

African tone phenomena

Edited by

Keith L. Snider

Virginia Beavon-Ham

Contemporary African Linguistics

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Keith L. Snider & Virginia Beavon-Ham (eds.). 2026. *African tone phenomena*
(Contemporary African Linguistics). Berlin: Language Science Press.

This title can be downloaded at:

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ISBN: `no digital ISBN`

`no print ISBNs!`

ISSN: 2511-7726

`no DOI`

Cover and concept of design: Ulrike Harbort

Fonts: Libertinus, Arimo, DejaVu Sans Mono

Typesetting software: Xe_{La}TeX

`redefine \publisherstreetaddress`

`redefine \publisherurl`

Storage and cataloguing done by `redefine \storageinstitution`

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Tribute to Mike Cahill (1954–2024)

Keith L. Snider

Dr. Michael Clark Cahill was born on February 7, 1954, in Albuquerque, New Mexico and passed away June 21, 2024, in Dallas, Texas, following a short, but valiant, fight with an aggressive form of cancer. I first pay tribute to Mike the scholar, with an overview of his numerous contributions to the linguistics world, and then to Mike, my friend and colleague.



Figure 1: Photo courtesy of Ginia Cahill

Mike was an outstanding linguist who made substantial contributions to the field of linguistics and to the field of African linguistics, in particular. For this, he was well-prepared. After graduating with a B.S. in Biochemistry from Iowa State University in 1977 and later teaching high school science and mathematics for five years in Iowa, Mike attended classes taught by the Summer Institute of Linguistics (later to become SIL International, and more recently SIL Global) at the University of North Dakota and then joined SIL in 1982. He graduated from The University of Texas at Arlington with an M.A. in Linguistics in 1985, and with a PhD in Linguistics from The Ohio State University in 1999. In the intervening years, Mike, and his wife, Ginia, lived in Yikpabongo, the main village in northern Ghana where the Konni [kma] language is spoken. While there, he analyzed Konni and worked with the Konni-speaking language community to develop an orthography and begin a Bible translation and literacy project. While

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in Ghana, Mike also served as the Linguistics Coordinator for the Ghana Institute of Linguistics Literacy and Bible Translation for three years. In 1993, the Cahill family returned to the U.S. to better help their second child, Laura, who has special needs.

After completing his PhD, Mike served in SIL's international administration headquarters in Dallas until he passed away. He was a Senior Linguistics Consultant for 13 years (2012–2024), he was SIL's International Linguistics Coordinator for eleven years (1999–2010), he was Editor-in-Chief of SIL's Global Publishing Services for six years (2010–2015), and finally, he was the Orthography Services Coordinator for nine years (2015–2024). He also taught phonetics and phonology courses intermittently for twelve years (2013–2024) at the Graduate Institute of Applied Linguistics in Dallas, TX (later to become Dallas International University) where he was an Adjunct Associate Professor.

Mike held a lifetime membership with the Linguistic Society of America and served on the LSA Committee on Endangered Languages and their Preservation as a member for 13 years and as Chairperson for one year. He was also a member of LSA's Ethics Committee for four years. He was a founding member of the Association of Contemporary African Linguistics (ACAL) and served as Secretary for ten years (2014–2024). As a regular attender of the LSA and ACAL meetings, he could usually be found staffing SIL's publications tables.

Mike's main research interests focused on tone, ATR vowel harmony, labial-velar stops, and in later years, orthography development. And of course, Konni, with numerous publications and conference presentations to his credit on these subjects. Two of his better-known publications were his 1999 PhD dissertation, *Aspects of the Morphology and Phonology of Konni*, published by SIL International in 2007, and *Developing Orthographies for Unwritten Languages*, co-edited with Keren Rice and published by SIL International in 2014. This latter publication grew out of an LSA symposium that he and Keren organized in 2011 on Developing Orthographies for Unwritten Languages. Finally, I wish to acknowledge one of Mike's papers that did not get written. As editors of the current volume, Virginia Beavon-Ham and I had invited Mike to contribute a paper on polar tones to this volume, which he had accepted to write. Regrettably, Mike passed away shortly after we received his abstract, and the paper was never written. It goes without saying that Mike's passing is a huge loss, not only to the African linguistics community, but to SIL and the greater linguistics world.

At a personal level, Mike was much more than a scholar. He was a son, a husband, a father, a colleague, and a friend. I first met Mike in the spring of 1983, shortly after he arrived to work with SIL in Ghana. We soon became, and

Tribute to Mike Cahill (1954–2024)

remained, good friends. Although physical distance over the years kept us separated most of the time, we stayed in regular touch by email. My wife, Ruth, and I were also new to Ghana in those days, having arrived there ourselves only one year earlier. For those of you who knew Mike, the characteristics that defined his life later were apparent from the beginning. He was such a friendly person. I am sure that Mike, like everyone, had his moments, but I can honestly say I never saw him truly angry. He was a positive-thinking person who tried to understand everyone's point of view.

A year before Mike showed up in Ghana to begin his linguistics career, a schoolteacher named Ginia arrived in Ghana. As an itinerant teacher, Ginia went around to the different villages where SIL personnel were working to help them home-school their children. Since Mike and Ginia were initially both based at the SIL headquarters in Tamale, it did not take long before it was noticed they had developed "an interest" in each other. And this revealed itself in several ways. For example, in those days, the Tamale office maintained single side-band radio communication each morning with all of us village teams to ensure that everyone was safe and doing well. At some point, the radio job was delegated to Mike, but it didn't take long before Ginia joined him. The laughter and giggling we heard each morning on the radio soon betrayed everything, and we were all delighted.

The following December (1983), Ruth and I invited Mike and Ginia to spend Christmas with us for a few days in our home in Kumindi, the Chumburung-speaking [ncu] village where we lived with our two small boys. While Mike was interested in everything to do with Chumburung and linguistics, as parents, we were also pleased to see him spend time chatting with our little boys, who really liked him. At one point, I recall the boys coming up to Mike with their pockets bulging, and their hands full of as many toads as they could possibly manage. It was toad season, and there were toads hopping everywhere. Mike exclaimed, "Oh, you have lots of frogs," to which our two-year old responded, "They're not fwogs, they're toads!" Mike loved children and was very patient with them, and they, in turn, loved him.

A day or two later, Ginia came up to Ruth and excitedly confided to her that that evening, Mike was going to propose to her and that she was going to say, "Yes," but that he didn't know her answer yet! And Ruth, of course, quickly told me! So, Ruth and I knew how the evening was going to turn out before Mike did. In any case, after a walk down by the river that evening, Mike and Ginia returned rather excited, and the rest is history, as they say. They married in Boxford, Massachusetts in July 1984, and went on to have three children, Deborah, Laura, and Stephen, and eventually four grandchildren. Two of Mike's children later

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followed his and Ginia's lead by also becoming involved in international and linguistic-related work, and I'm sure he was proud of them.

As mentioned above, Mike loved children. Over the years, I interacted with the Cahills on various occasions, including being an overnight guest in their home in Duncanville, Texas. Each time, I was deeply touched by the love, gentleness, and patience that Mike and Ginia bestowed on Laura as a child, and later as a young adult. It was a joy to watch Mike interact with his daughter so beautifully, and to see her return that affection so abundantly. Thank you, Mike and Ginia, for raising Deborah, Laura, and Stephen to be the beautiful people they are.

But the Cahill and Snider kids were not the only ones to profit from Mike's love for children. Beginning in 1998, Mike became involved as a leader with Boy Scouts of America Troop 141 in DeSoto, TX, when his son Stephen became a Cub Scout. Mike served first as a Den leader, then later as Scoutmaster, and Assistant Scoutmaster until his passing. He was awarded the Eagle District Lightning Award on three occasions, and he was a certified archery range master. The week before Mike passed away, he was still helping boys get advanced in their rankings. At Mike's memorial service, Don Heinen, one of Mike's fellow Scout leaders, concluded his eulogy by announcing the creation of the "Mike Cahill Memorial Campership" to be awarded to one deserving first year camper from Soaring Eagle District who would be afforded the opportunity to attend a Circle 10 Council summer camp.

Mike was a leader and a man of faith and strong convictions. He regularly attended church and led a "life group" at his church. A life group in this context is a small group of individuals who meet regularly, usually for Bible study and fellowship. He was an active member of the Atheist & Christian Book Club where he defended his faith, not only ably, but respectfully. He also did not hesitate to respectfully defend SIL against critics at linguistics conferences and elsewhere when warranted. Occasionally, Mike and I would share a room at linguistics conferences. On one of those occasions, a few of us went out to eat dinner at a restaurant one evening. Later, as we walked back to our hotel, we happened upon a brothel of some sort on our side of the street. Mike at once suggested we switch to the other side of the street, and he then led the way without waiting for consensus. Thank you, Mike, for your integrity and leadership.

In his role as SIL's International Linguistics Coordinator, Mike was my supervisor for eleven years, and I could only admire the skill with which he worked towards resolving interpersonal issues. He cared for people and treated everyone fairly, whether he agreed with them or not. As his daughter Deborah said in her eulogy, "He prioritized people and would stop to have a conversation with someone at the expense of whatever 'business' he had at hand." Whenever I think of

Tribute to Mike Cahill (1954–2024)

Mike, I think kind, gentle, respectful. The world has lost a gifted linguist, and we have all lost a wonderful friend.

Chapter 1

African tone phenomena in focus

Virginia Beavon-Ham

SIL Global

Keith L. Snider

SIL Global

This volume is designed to provide a single resource to help beginning linguists recognize and understand tone phenomena found in African languages. Accordingly, the content of articles is more descriptive than theoretical. The volume is divided into two parts, the first organized by particular tone phenomena, and the second organized by language groupings. The present chapter presents some of the challenges authors face in describing multiple tone height systems, including the lack of consensus on the labels given to the different tone and pitch heights. It also describes a lack of consensus among authors on how best to support tonal contrasts and discourages argumentation which relies on minimal tone sets that contain words that are not grammatically homogeneous. The remainder of this article consists of summaries of each chapter in the volume.

1 Introduction

Of all the languages spoken in Africa, it is hard to give a precise number of how many are tonal. This is not only because there is some disparity in how languages are counted, but also because many languages do not have documentation of their tone systems. That said, it is commonly understood that the Niger-Congo languages are almost all tonal (Yip 2002: 17), Nilo-Saharan languages are mostly tonal (Blum 2026 [this volume]), Khoisan languages are tonal, and some branches of Afroasiatic languages are tonal (Roberts 2026 [this volume]). This means that linguists describing African languages will be most likely confronted with the

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need to understand something about the tone system of that language. Yet this is a challenge because of the uniqueness of tone. Hyman (2011b) concludes that "...tone can do everything that segmental and metrical phonology can do, but that the reverse is not true" (p. 198). Moreover, when compared to other phonological properties, words like "ambiguous", "abstract", and "arbitrary" have been used to describe the complexity of tone (Hyman 2011b).

This volume is intended to equip linguists to produce more and better analyses of tone phenomena, particularly in some of the under-studied languages of Africa. Specifically, this volume seeks to quickly provide enough background so that linguists new to the topic can frame their analyses in the context of what is known. In addition, linguists with considerable experience in one language family's tone phenomena will discover new phenomena in exploring the descriptions of other language families.

The intent is to expose readers to a wide array of tone phenomena but also to limit the discussions specifically to the phenomena seen in Africa. The word "phenomena" is used because the discussions cover a variety of elements of tone description: tone inventories, presence or absence of floating tones, tone rules of various types, lexical and grammatical tone, etc.

To summarize, distinctives of this volume include: a) a collection of chapters that are all specifically relevant to the study of tone in African languages; b) a volume that is tailored to the needs of linguists beginning to delve into the study of tone in African languages; c) an examination of various language families in Africa and the tone phenomena seen in these families; and d) a single resource that quickly orients linguists to the breadth of topics that relate to African tone.

Each chapter is authored by a linguist or linguists who have experience in particular topics or language families related to African tone phenomena. The authors of the chapters in this volume employ primarily a phonology-driven approach to describing tone phenomena, although in many cases there exist laboratory studies exploring the finer details of the phonetic realization of these descriptions. While the authors come from different theoretical backgrounds, autosegmental theory is heavily employed throughout. In order to make the volume accessible to a broad audience, authors provide a clear description of the relevant phenomena and exploit theory only to an extent that enhances this description.

2 Scope of content

This volume is divided into two parts: the first consists of generalized descriptions of tone phenomena, and the second is an examination of tone phenomena

1 *African tone phenomena in focus*

that have been described in particular language groupings. The chapter on tone and music is a little unique in the scope of this volume, but will be of interest to anyone interested in the intersection of language prosody and the arts.

We recommend that readers not read this volume from front to back, but instead proceed directly to topics and language family descriptions that are of most immediate interest, circling back to other chapters as desired. Some topics are treated in multiple chapters, and this is natural in such a volume as readers may benefit from multiple authors' perspectives on the same topic. Following the tribute we have included to recognize the life of Michael Cahill, one of the intended authors, we begin with the first part of the volume. Here, we have ordered topics (apart from the present chapter) according to our impressionistic understanding of how commonly these tone phenomena are encountered across the languages of Africa. In the second part, chapters are ordered by language groupings, starting with Afroasiatic, then Nilo-Saharan, Niger-Congo Non-Bantu (proceeding from east to west geographically), and finally Niger-Congo Bantu.

This volume is not intended to explain how to analyze tone; rather, it is a tool that would complement such a book. There are a number of articles and books that address the process of tone discovery, including Hyman (2014), Marlo (2013), and Snider (2018). There are also excellent articles that outline general phenomena seen in African languages which would be of interest to readers of this volume, including Hyman (2001), Hyman (2007), Hyman (2011a), Hyman & Schuh (1974), and Odden (1995). Hyman (2011a) in particular addresses in depth the topic of tonal inventories, bearing in mind the possibility of tonal underspecification. In addition, the edited volumes of Downing & Rialland (2017) and Gussenhoven & Riad (2007), as well as the authored books by Wee (2019), and Yip (2002) provide excellent resources for the study of tone phenomena.

This volume brings together in one place not only comparative and typological descriptions of general tone phenomena but also overviews of the types of tone phenomena that have been observed in various language subdivisions of Africa. For linguists approaching the tone analysis of a single African language, this volume attempts to quickly orient them to what they might begin looking for—while also exposing them to the beautiful and sometimes dizzying diversity of what has already been described.

A gap in the coverage of the present volume is the scarcity of chapters describing Bantu languages. Only one chapter addresses tone in Bantu languages and this chapter focuses on Northwest Bantu. This was not by design; we had originally anticipated additional chapters that are not included in the final version of this volume. The topic of tone in Bantu languages, however, has been extensively covered elsewhere, including in: Downing (2011), Hyman (2025), Kisseberth &

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Odden (2003), Marlo (2013), Marlo & Odden (2022), and Odden & Bickmore (2014). Other gaps among dedicated chapters include: Atlantic, Kwa, Dogon, Adamawa, Ubangi, Kordofanian, Khoisan. However, the first half of the volume contains numerous examples from many languages found within these groupings.

3 Challenges in describing tone

3.1 Multiple tone height systems

In this section, we spotlight a challenge in the documentation of tone systems in Africa, which is the lack of consensus when representing four or more phonetic tone heights—both in terms of labeling and diacritic representation. The following examples illustrate this.

In this volume, there are references in the chapter on Nilo-Saharan languages to cases where an additional extra-high (X) level exists above the level of high in a X, H, M, L inventory (Blum 2026 [this volume]). In the discussion of Mande languages, extreme levels beyond H, M, L are referred to as super-high (SH) and extra-low (XL), and one example of an inventory of a language with four level heights is SH, H, L, XL (Konoshenko 2026 [this volume]). In yet a different chapter, super-low (sL) and super-high (sH) are terms used for extreme tone levels (Rolle & Bickmore 2026 [this volume]). A language cited in the chapter on Benue-Congo, also with four level heights, has the inventory extra-high, high, mid, low, while another has the inventory high, mid, low, extra-low (Blench 2026a [this volume]).

In each of these systems of labeling the four height levels, extra levels are designated with superlatives such as extra- or super-, and these are added to high or low. The choice of which extreme to choose for a fourth height may have to do with an analysis of the historical source of this level. For example, an extra-low tone height may be understood to have developed historically through lexicalization of the depressing effects of consonants. Similarly, an extra-high tone height may be understood to have developed historically through lexicalization of heights obtained because of tone raising rules Beavon-Ham & Pearce (2026 [this volume]).

In contrast with the options described above, another way that researchers sometimes refer to systems with four level tone heights is to maintain high and low as the extremes, and add two mids between the two. There is again some variety in the specifics of labeling. The mids are sometimes referred to as mid-low (m) and mid-high (M) (Russell & Sande 2026 [this volume]).

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The choice of having two mid tones is consistent with the Yip (1980) feature analysis of four heights being obtained from values related to two features: +/-Upper and +/-Raised. The higher of the two mids is +Upper, -Raised, and the lower is -Upper, +Raised. A complication for the description of a language with two mid tones is how to effectively use diacritics for marking the two levels of mid. (Russell & Sande 2026 [this volume]) provides one possible solution for this.

Related to this is how authors use the term register. The chapter on consonant-tone interaction by Beavon-Ham & Pearce refers to Volk (2013), who uses register in the sense of groupings of tone-bearing units into high and low register domains, depending on whether they are affected by depressor consonants (low register domains) or not (high register domains). Other chapters in this volume (e.g., Connell (2026 [this volume])) use register to distinguish register tone languages (i.e., languages with discrete level tones and contour tones composed of discrete level tones) from contour tone languages (i.e., languages with non-decomposable contour tones, such as those found mainly in Asia). Most chapters in this volume, however, use register in the sense of the lowered pitch levels that result from downdrift and downstep (cf. Snider (2026 [this volume])).

A further challenge which may influence decisions about labeling and representing tone heights is the presence of contour tones in many of these languages. The diacritic representation of contours which consist of any combination of the many options discussed above present some challenges for both writers and readers.

In conclusion, we are still some distance from achieving consensus on labeling and diacritic usage in describing tone systems of four (or more) tone heights. While it is not necessary (or even desirable) that a single convention be imposed, this lack of consensus may make comparative tone analysis difficult.

3.2 Establishing tonal contrast

A second challenge that lacks consensus among authors who document tone systems, including those authors in this volume, concerns how best to present data that support contrastive tone levels. When an author says a language has three or more tone levels, how can one be sure they are not conflating derived levels with underived levels? To avoid this problem, linguists strive to control for all factors that could potentially compromise phonological contrasts. And since the minimal pair concept epitomizes the principle of controlling for all possible factors that could affect the outcome, minimal pairs are typically seen as the gold standard for supporting tonal contrasts. Due, however, to the phenomenon of floating tone morphemes (cf. Rolle & Bickmore (2026 [this volume]), this volume; Rolle

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et al. (2026 [this volume]), this volume), minimal tone pairs and minimal tone contrast sets that ignore grammatical differences have the very real potential to compare (a) tones that are derived from interactions with tonal morphemes with (b) tones that are not derived. This, of course, undermines the *raison d'être* for employing minimal pairs and creates the potential to over-generate the number of perceived underderived contrastive levels.

Here is an example of what we mean. Ahoua (1986: 64) describes Baule [bci]¹ as having three contrastive tone levels, L, M, and H, and he supports them with this minimal triplet: bò 'bottom', bō 'bush,' and bó 'break'. Since these words are segmentally identical, there is nothing segmentally that could possibly affect any of the words in ways that do not affect all the words. However, while both 'bottom' and 'bush' are nouns, 'break' is a verb, and given the perplexing phonological behavior of the M tone, one must consider whether there are additional grammatical differences (latent historical noun class affixation?) that result in more phonetic levels than are truly contrastive. Mixing lexical and grammatical differences in minimal tone contrast sets increases the possibility of including any number of tonal morphemes that are not shared by all members of the contrast set. These tonal morphemes, in turn, can interact with underderived tones to generate derived tones.

Despite the compelling nature of the minimal triplet in Baule, it is telling that later publications (Ahoua 1996, Leben & Ahoua 1997) conclude that Baule has only two underderived tone levels, L and H. Thus, the minimal triplet in Ahoua (1986) is a classic case of overgeneration due to an over-reliance on minimal tone contrast sets that ignore grammatical differences. Now that the role of tonal morphemes in African languages is better understood, best practices in tone analysis suggest that if minimal tone contrast sets are employed, they should not only be segmentally identical, but also grammatically identical (Snider 2014, Snider 2018).

It is not always possible, of course, for all members of minimal tone pairs and contrast sets to be segmentally identical. In such cases (usually the norm), it is much better to employ near minimal tone pairs and contrast sets that have no grammatical differences than to employ segmentally identical sets that have grammatical differences. If all desired contrasts cannot be demonstrated within a single grammatical construction (segmental considerations aside), consider the strong possibility that what you are trying to present is a mix of derived tones with underderived tones, which is not desirable. While near minimal tone pairs that are grammatically homogeneous might not be as aesthetically pleasing as minimal pairs, they are less misleading and more convincing.

¹We employ ISO 639-3 codes as described in Eberhard et al. (2026) for each language cited.

4 Summaries and highlights: tonal phenomena

4.1 Tone spread and shift (Joshua A. G. Smolders)

In this chapter, Smolders (2026 [this volume]) proposes a typology of tone mobility. Tone mobility occurs when a surface tone is pronounced on one or more different tone-bearing units (TBUs) than the one(s) to which it is underlyingly associated. There are two types of mobility, spreading and shifting. In the case of spreading (the most common), the tone that is mobile continues to retain its association to the original TBU. In the case of shifting, the mobile tone loses its association to the original TBU. The proposed typology rests on four parameters: 1) the direction of movement: leftward or rightward, 2) the target of movement: bounded (one TBU) or unbounded (multiple TBUs), 3) which tone is mobile: H, L, or M, and 4) the type of movement: spread or shift. Based on a survey of some 100 languages from all the major phyla, Smolders provides examples of each possibility, as well as a chart that shows the percentages of languages attested to have each possibility.

Smolders then discusses interactions that arise from mobility. These include the creation of tonal contours, the delinking and deletion of target tones, and various things that can block movement such as morphological and syntactic boundaries, consonant types, and floating tones. Finally, he shows how given languages can employ more than one type of movement, and how tonal movement can occur lexically, postlexically, and historically.

4.2 Grammatical tone in African languages (Nicholas Rolle, Lee Bickmore)

In the first pages of Rolle & Bickmore (2026 [this volume]), the distinction between lexical tone and grammatical tone is established, and a definition is given for grammatical tone. The authors delineate a further subdivision between autonomous grammatical tone—in which the grammatical meaning is marked solely by the presence of a tone—and non-autonomous grammatical tone—in which grammatical tone marks a specific meaning when used in conjunction with another marker.

Grammatical tone types explored and categorized by Rolle & Bickmore are: tone addition, subtraction, replacement, shifting, exchanging, and spreading. Examples drawn from a number of language families show that floating grammatical tones can be found in singletons or in sequence. When a language has more than two levels of tonal height, grammatical tone can involve a scalar shift—for example, an upward shift of each tone height to the one above it.

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The authors show that grammatical tone can be prioritized over lexical tone. This is the case, for example, in several demonstrated instances of replacive tone on noun stems in the marking of plural. Grammatical tone can also be prioritized over grammatically meaningless boundary tones.

Examples from languages across Africa show many ways that grammatical tone changes can be independent of or dependent on other factors. Grammatical tone changes can operate without being restricted in their application by word boundaries. On the other hand, grammatical tone changes may need to be applied in compliance with a specific tone system requirement, such as the allowance of only one H tone per word. There may also be exceptional suspension of grammatical tone in unique circumstances.

4.3 Downdrift and downstep in African languages (Keith L. Snider)

In this chapter on downdrift and downstep (respectively also known in the literature as automatic and non-automatic downstep), [Snider \(2026 \[this volume\]\)](#) describes these phenomena as processes that lower the tonal register such that all tones are lowered that follow downstep junctures. While both phenomena are recursive and both lower the tonal register cumulatively, they are distinguished by whether they result in tones that are contrastive: downstepped tones are contrastive and downdrifted tones are not.

L tones are the most usual triggers of downdrift and downstep, and H tones are their most usual targets. However, L's can also downdrift and downstep M's, and floating H's can downstep L's. Moreover, adjacent H's can result in OCP violations that are resolved by downstep of the second H.

Snider also describes the phenomenon of double downstep of H tones, which, though rare, occurs in at least three distantly related language families: Western Nilotic, Grassfields Bantu, and Gur (Mabia) ([Angsongna et al. 2026](#)). When H tones undergo double downstep, the degree of lowering is twice that of normal downstep. Languages with double downstep also have instances of single downstep, which, when two instances happen to converge on a single tone-bearing unit due to morphological and syntactic arrangements, result in double downstep of the H.

Another rare downstep phenomenon Snider describes is total downstep. In languages with normal downstep, the downstepped H's are lower than the preceding H's but higher than the L's in HL sequences. However, in languages with total downstep, the downstepped H's are lowered to the level of the L's in HL sequences. He also notes that languages with total downstep do not have downdrift.

4.4 Floating tones (Nicholas Rolle, Larry M. Hyman, Florian Lionnet)

Assuming the theory of autosegmental phonology, Rolle et al. (2026 [this volume]) define floating tones as tones not associated to any tone-bearing units. Moreover, they are abstract, analytical constructs that cannot be heard, despite their effects being heard, but which, nevertheless, have the properties of associated tones. Floating tones are extremely useful for explaining several tonal effects such as 1) blocking other tones from spreading, 2) inducing downstep and upstep, and 3) preventing phrase-final L tones from falling. Also, many floating tones can be shown to have derived historically from associated tones.

While some floating tones co-occur with segmental material and explain contrasts between morphemes that are otherwise phonetically identical, others, called tonal morphemes, consist solely of floating tones that express grammatical meaning. Although tonal morphemes, in principle, could be lexical roots, in practice, they are confined to grammatical morphemes with inflectional and derivational meanings. Prosodically, tonal morphemes can pattern as affixes, clitics, or independent words.

The authors briefly describe theoretical alternatives to floating tones, but none of these are found to explain as broad a range of cross-linguistic effects as do floating tones.

4.5 Consonant-tone interaction (Virginia Beavon-Ham, Mary Pearce)

In Beavon-Ham & Pearce (2026 [this volume]), the relationship between consonant production and vowel tone is explored, looking at languages of Africa from both the Niger-Congo and Afroasiatic families as well as beyond. This relationship most commonly involves depressor consonants and lowering effects. There are examples of tone-lowering phenomena in Bantu languages, other languages of the non-Bantu Niger-Congo subfamilies, and Chadic languages—as well as occasionally elsewhere. Some low tone or lowering effects that are seen in the data include: lowering effects on adjacent vowels, blocking the spread of H tones, blocking the docking of floating H tones, and conditioning of H shift.

Certain observations of consonant-tone interaction highlight the difference between phonetics and phonology. While depressor consonants are most frequently sounds that are voiced obstruents, other sounds—such as voiceless glottal fricatives and implosives—do occasionally function as depressors (Bradshaw 1999, Odden 2007, Tang 2008, Volk 2013). In cases where the phonetic motivation for depressor activity is unclear, this presents challenges for attempts to formalize the nature of depressors (Maddieson 2003).

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Some languages in Africa also have raiser consonants. These have been shown to block the spread of L tone or to raise a L to M on an adjacent vowel.

Several of the languages discussed in Beavon-Ham & Pearce demonstrate diachronic progression in the relationship between consonant production and tone. In light of the possibility of diachronic change, the authors invite further research exploring how depressor and raiser categorization is generally maintained from one generation to the next.

4.6 Tone polarity (Douglas Pulleyblank)

Pulleyblank (2026 [this volume]) examines morphemes whose surface tone is systematically different from the tone of an adjacent element. In languages with two contrastive pitch levels (H and L), polar tones show a simple pattern: they surface as H next to L and as L next to H. However, the phenomenon becomes more complex in languages with three or more contrastive tone levels. In such cases, polar tone values are not always opposite to those of adjacent tones but might instead be determined by the natural tone class of the adjacent tone, or by whether the adjacent tone is peripheral or not.

Pulleyblank distinguishes between descriptively polar tones and analytically (true) polar tones. A tone is analytically polar when there is no principled reason to posit an underlying tonal specification that undergoes dissimilation. In such cases, the morpheme's surface tone is always different from the adjacent tone, and its underlying tonal value cannot be determined. Although some might analyze these morphemes as underlyingly toneless, Pulleyblank notes that some languages contrast true polar morphemes with other toneless morphemes whose surface tones are consistently identical with neighboring tones.

Cross-linguistically, polar tones are typically restricted to affixes and to specific lexically marked morphemes or morphological classes. Pulleyblank evaluates several theoretical approaches and argues that an Emergent Phonology framework provides the most unified account of the diverse polarity patterns observed.

4.7 Tone and music in Africa (Laura McPherson)

In this chapter, McPherson (2026 [this volume]) presents an overview of tonal textsetting, i.e., the influence of linguistic tone on vocal music, and an overview of the relationship between linguistic tone and musical surrogate languages (e.g., talking drums).

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When investigating tonal textsetting, researchers compare the direction of the linguistic tonal changes (rising, falling, level) with the direction of the comparable melodic changes of the surrogate languages and note whether the changes are parallel (both in same direction), oblique (one level, one moving), or opposing (the directions are opposite). Another factor in textsetting is whether the musical system encodes phonemic tone or surface tone. Moreover, the extent to which tone influences music differs across cultures. While Zulu [zul] demonstrates close matching between tone and music transitions (92% parallel and 97% non-opposing transitions), other cultures such as Tommo So [dto] mainly avoid opposing transitions, and others such as Shona [sna] pay little attention to tone.

Musical surrogate languages often transmit messages by encoding pitch changes, syllable length, and rhythm. However, since consonants and vowels are not encoded, there is often significant ambiguity. Ambiguity is often mitigated through replacing short words with lexicalized phrases, using rote formulas, relying on context, and employing previously established disambiguating phrases. Three surrogate languages are presented: the Sambla (Seenku) balafon [sos] the Igbo-oja flute [ibo], and the Yoruba bata drums [yor]. McPherson concludes that the interaction of tone and music remains understudied in African languages and that investigating the relationship between tone and music can help linguists analyze tone systems.

5 Summaries and highlights: language groupings

5.1 Tone in Cushitic languages (Christopher R. Green)

In this chapter on the Cushitic branch of Afroasiatic languages, [Green \(2026 \[this volume\]\)](#) provides an extensive literature review of Cushitic tone, accompanied by a helpful table (p.xx) titled, “Languages and data sources.” Green says that, by and large, Cushitic languages have “restricted” tone systems with “many, but not all properties of prototypical stress accent systems” (p. xx). In general, tonal contrasts in Cushitic languages are privative in nature, meaning that a single tone contrasts with its absence, and these contrasts are often found only on penultimate or final syllables. Syllables that do not host contrastive lexical tones are typically assigned predictable tones, often based on word shape. However, while contrastive tone in the lexicon does exist, contrastive grammatical tone is much more prevalent, including what Green calls, “subtractive tone,” i.e., the deletion of an existing tone.

Tonal alternations in Cushitic “are generally uncomplicated and mostly involve tone spreading, contour avoidance, non-finality, and, in some instances,

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tone (accent) polarity” (p. xx). One reason put forward to account for the simplicity of tonal alternations is the rudimentary tonal inventories of most Cushitic languages: H, or, in some cases, L, tones that contrast with their absence. Despite the presence of several descriptions of prosodic systems in Cushitic languages, these still lag far behind descriptions of other aspects of these languages, especially descriptions of morphology and syntax.

5.2 Tone in Chadic languages (James Roberts)

In this chapter on Chadic tone, [Roberts \(2026 \[this volume\]\)](#) surveys tone phenomena in the Chadic branch of Afroasiatic. Despite lexical tone having few minimal pairs and a low functional load in Chadic languages, the Chadic family is nevertheless the most tonal of the five Afroasiatic families: Semitic, Berber, Egyptian, Cushitic, and Chadic. Lexical tone systems in Chadic typically consist of two, or at most three, contrastive levels, with tone being more contrastive in nouns than in verbs, which are often toneless. Offsetting the low functional load of lexical tone in these languages is the relatively high degree of grammatical tone which, when coupled with toneless verbs, often makes verbal tone completely predictable once the grammatical categories are known.

The two most common tone phenomena across the four branches of Chadic are grammatical tone, which often marks tense, aspect, and mood; and consonant-tone interference (also known as consonant-tone interaction), whereby voiced obstruents often lower pitch and voiceless obstruents sometimes raise it. This interference can restrict tone patterns, generate additional tone levels, and interact with both lexical and grammatical tone. Other processes found in Chadic languages include tonal polarity, tone spreading, floating tones, and downstep, but these are much less frequent than grammatical tone and consonant-tone interference. With the low functional load of lexical tone, some Chadic languages resemble restricted accentual systems more than they do typical tone systems.

Given the low functional load of lexical tone and the fact that Chadic is the most tonal branch of the five Afroasiatic branches, it is unlikely that proto-Chadic was tonal. This raises the question of how Chadic languages became tonal. Since three of the four branches of Chadic do not border tonal languages, Roberts rejects the suggestion that Chadic languages acquired tone through contact with tonal languages. He instead believes that tonogenesis in these languages likely arose through language-internal mechanisms such as metrical prominence and consonant-tone effects becoming lexicalized.

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5.3 Tone in Nilo-Saharan (Mirella L. Blum)

In the chapter on Nilo-Saharan tone phenomena, [Blum \(2026 \[this volume\]\)](#) provides a broad summary of tone phenomena of a phylum whose status is still debated. Many Nilo-Saharan tone systems are not well documented and there is much variation between tone systems of this phylum.

Blum describes one of the interesting ways that tone displays itself in the morphology of Nilo-Saharan languages, which is that tone is involved in case-marking, although with considerable variety in the way this happens. For example, in Tugen [tuy], tone differences alone distinguish nominative from absolutive case, but benefactive and genitive cases are marked by segmental affixes ([Jerono 2013](#)). Case marking in Majang [mpe] involves a floating L tone which triggers non-automatic downstep ([Joswig 2019](#)).

Another interesting case from the Nilo-Saharan languages is that words derived from nouns in Kanuri [kau] are accompanied by a LH word-level tone pattern with the H appearing on the final syllable of the derived word ([Hutchison 1981](#)). Equally interesting is the case of tonal polarity described in the Kipsigis [sgc] language, where polarity is observed across an entire word ([Kouneli & Nie 2021](#)).

An unusual tonal phenomenon highlighted by Blum is that a HL fall in several of these languages has an unusual status. In Nuer [nus], H and HL fall tones are analyzed as being in complementary distribution. Moreover, H is associated with breathy voice and HL with modal voice ([Monich 2023](#)). In Shilluk [shk], a High Fall and a Late High Fall are each observed across three vowel lengths observed in the language, and the difference in timing across the vowel length is analyzed as representing a contrastive difference ([Remijsen & Ayoker 2014](#)).

5.4 Tone systems in the Bantoid languages (Roger Blench)

In this chapter, [Blench \(2026b \[this volume\]\)](#) notes that the 150–200 Bantoid languages (Niger-Congo), located on both sides of the Cameroon-Nigeria border, are not a single genetic family, but are rather a nexus of arguably twelve subgroups with significant diversity that nevertheless share many similarities with Narrow Bantu languages. Their tone systems range from having two contrastive tones, H and L, to systems having four and five contrastive tones, taking contour tones into account. Many of these languages have floating tones that resulted from the historical loss of segments which, in turn, sometimes trigger downstep and upstep. Grammatical tone, which often signals TAM distinctions on verbs and noun class distinctions on nouns, is also prevalent in Bantoid languages.

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Blench notes how older descriptions, on the one hand, sometimes tend to underestimate the number of tone contrasts due to improperly distinguishing them phonetically. On the other hand, older descriptions of languages with two tones and downstep also sometimes over-estimated the number of contrasts due to an inadequate understanding of downstep. In addition to describing how tone is sometimes represented orthographically in Bantoid languages, Blench describes the prevalence of tone surrogacy traditions that use drums, flutes, and slit-gongs that mimic the tone patterns of spoken speech to communicate messages.

5.5 Lexical and non-lexical tone in Mambiloid (Bruce Connell)

Connell (2026 [this volume]) examines the lexical, grammatical, and deictic functions of tone in Mambiloid, a subgroup of Bantoid languages (Niger-Congo) that straddle the Nigeria-Cameroon border. By employing minimal or near minimal tone contrast sets, all languages are demonstrated to have either three or four surface contrastive tone levels.²

After beginning with the general characteristics of tone in the Mambiloid subgroup, Connell provides more detailed overviews of tone functions in five languages: Ba Mambila [mcu], Sundani Kwanja [knp], Oumyari Wawa [www], Vute [vut], and Nizaa [sgi]. In these languages, tone can encode TAM distinctions, negation, associative and possessive marking, directional deixis in verbs (ventive vs. itive motion), as well as expressive meanings in ideophones. Unlike many other languages of Africa, downstep and downdrift are not manifestly present in Mambiloid. What *prima facie* might appear to be downstep typically turns out to be nothing more than a local lowering that at most affects two syllables. The chapter attributes tonal complexity in Mambiloid languages to the historical loss of noun class prefixes and the erosion of segmental material in suffixes, and it concludes with an implicit plea for more and better documentation of tone phenomena in Mambiloid.

5.6 Tone in Grassfields Bantu (Cameron Hamm, Rachel Robinson, and Pius W. Akumbu)

In this chapter, Hamm et al. (2026 [this volume]) present the internal sub-classification of the Grassfields Bantu (GB) languages (Niger-Congo) and then describe tone

²These levels, however, cannot be demonstrated using minimal contrast sets without mixing nouns and verbs. As Connell notes, in languages with four levels, all four levels are found only on nouns, with level 3 (second lowest) being rare. Similarly, verbs typically divide into only two groups, based on their tonal behaviour.

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in noun phrases and verb phrases, as well as clause-level and sentence-level tone phenomena. Historically at least, noun roots in GB languages were disyllabic and had four underlying tone patterns: */H/, */HL/, */LH/, and */L/. Today, however, noun roots in most GB languages are monosyllabic but still retain a four-way underlying tone contrast, though what this looks like differs from language to language. Some languages contrast H, M, a L that does not fall, and a L that falls phrase finally, which often correspond, respectively, to the historical patterns */H/, */HL/, */LH/, and */L/. In many cases, however, other languages do not have a M tone, but rather a downstepped H. Babanki [bbk] is exceptional in that it has both.

Contour tones are common, and floating tones are extremely common due primarily to the loss of the final TBU. While in some languages, /L/ is realized as low, /LH/ as rising, /HL/ as falling, and /H/ as high when preceded by a L-toned prefix, in most of the languages, things are not this straightforward. Noun class prefixes are typically /L/ in Eastern Grassfields Bantu, but /H/ in other subgroups; in some cases, the prefix /H/ is lowered to M due to a L boundary tone. Unlike noun roots, verb roots have only two patterns, /H/, and /L/. Inflectional suffixes often have inherent tones that are floating. GB languages have extensive noun class concord. Tonal processes common to GB languages include tone spreading, tone docking, tone merging, non-automatic downstep, polar tones, toneless morphemes, and floating tones, which occasionally give rise to double downstep.

5.7 Tonal systems in Benue-Congo and West Chadic A3 in Central Nigeria (Roger Blench)

Blench (2026a [this volume]) addresses the languages of Benue-Congo (Niger-Congo) that are outside of the Bantu and Bantoid branches. This chapter also includes (because of contact influence) West Chadic A3 languages. The language Boze [jer] provides interesting examples of sonorants, including /l/, /r/, /m/, /n/ and /ŋ/, which are tone-bearing in word-medial position; in such cases, no example is shown where the tone differs from that of the immediately preceding vowel (see Table 4 in the chapter).

In Rigwe [iri], a Plateau language, there are tonal changes related to the distinction between singular and plural (Table 8). Blench describes the raising of the prefix from extra-low to mid tone. Several other languages cited in this chapter that have plural and singular noun stems distinguished with changes in tone are Angan [kci] (Table 16) and Mbembe (Table 18, data cited from Richter 2014). In Mbembe [nza], a Central Jukunoid language, there are both tone and vowel differences between singular and plural noun stems.

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Also seen in the Rigwe language are TAM distinctions which involve tonal alternations both on the pronoun and on the verb. Similar complexity is seen in other languages, according to Blench. In Berom [bom], alternations both in vowel quality and tone of the first person singular pronoun distinguish between complex TAM categories.

Blench compares the complexity of tone systems of the Central Nigerian area to systems seen in the Mande area in Côte d'Ivoire and in the Grassfields of Cameroon, hypothesizing that the complexity comes from historical shortening of nominals with prefixes and CVCV roots.

5.8 Tone in Mande languages (Maria Konoshenko)

Konoshenko (2026 [this volume]) first discusses the inventories of tonal contrasts found in Mande languages (Niger-Congo). These range from two to five level tones—a broader range than we see in many of the chapters of this volume. That said, the distribution of these tones is not necessarily equally distributed; Seenku [sos], for example, has an inventory of SH, H, L, XL, but L has a 1.2% frequency in the lexicon (McPherson 2019: 90).

A few of the Mande languages are analyzed as having phonemic contours not decomposable as a sequence of two level tone heights. For certain other Mande languages, all combinations of level tones can be found as surface contours on bimoraic syllables.

The notion of tone melodies as summarized by Konoshenko refers to recurrent patterns of tone that are assigned at the foot or word level. Mande languages are where this type of tone melody was first noted, and data from several sources collated by Konoshenko illustrate what is meant by tone melodies. Syllable structure may affect melody association.

Mande languages mark TAMP distinctions with grammatical tones – a reality found in many other language families in Africa.³ There is also another function of grammatical tone marking in Mande languages, which is head-marking in compounds (Green & Konoshenko 2022). The final word of a compound can have tone neutralization of two types, which are described with multiple examples of each.

Konoshenko provides a comparison between tone rules in Mande and those documented in many other African languages. For example, H spread, while more common than M or L spread in Mande languages, is summarized by Konoshenko to be quite rare, and when it does exist, it is constrained to specific mor-

³TAMP is a common abbreviation for tense, aspect, mood, and polarity.

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phosyntactic contexts. The chapter also summarizes the research on intonation in Mande language – most of which has been carried out in the last ten years.

5.9 Tone in Kru languages (Katherine R. Russell, Hannah Sande)

Kru languages (Niger-Congo), described in [Russell & Sande \(2026 \[this volume\]\)](#), are complex in having at least three contrastive tone levels and often four. Interestingly, some of the Kru languages with three surface levels are analyzed as having four underlyingly contrastive tonal distinctions; two seemingly same surface levels in isolation (often mid-low and mid-high) can have distinctly different interactions with the tone of surrounding contexts. The authors cite Kagwé [gud], for example, which has nouns with level mid tone in their singular forms dividing into two different surface realizations in their plural forms ([Koopman & Sportiche 1982](#): 47–48).

Russell & Sande summarize the available research showing that rising contours are observed in all Kru languages, while falling contours are reported to be absent for some. This is counter to claims that rising contours are typologically the less common contours. Rising and falling contours can be restricted in certain languages to particular syllables within the word.

Grammatical tone has many different functions in Kru languages. For example, the authors discuss replacive tone in derivational processes in Guébie [gie]. Negation and aspect are frequently marked by grammatical tone, although there is considerable variety in the way grammatical and lexical tone interact. One of the most unusual instances of this is the scalar tone shift on the verb seen in Guébie imperfective marking ([Sande 2018](#)).

In some Kru languages (for example Grebo, discussed in [Newman 1986](#)), contour tones are analyzed not as a sequence of level tones, but as single tonal units—an analysis that is unusual for African languages. Russell & Sande discuss the arguments and implications of these findings.

5.10 Tone in Northwestern Bantu languages (Nadine Grimm)

[Grimm \(2026 \[this volume\]\)](#) begins with a clarification of the languages which will be included in the phylogentic category of northwestern Bantu (NWB) for the purposes of the tonal overview. Next, the author states that while many of the NWB languages (Niger-Congo) show features that are common for other Bantu languages (such as two levels of tone, contour tones, floating tones, downstep, and spreading), there are some things that may be more commonly seen in NWB than in other Bantu languages, such as having more than two phonemic

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distinctions, and having H vs. L systems rather than H vs. toneless. Furthermore, contour and floating tones in NWB are often linked to final vowel loss. A non-identical tone on a syllable that is lost can survive on the previous syllable as the second tonal unit of a contour tone; thus, contours tend to occur on final syllables.

Some NWB languages have underlying toneless TBUs in addition to H and L (Makasso 2012: 18). Grimm describes several observations that help distinguish L from toneless TBUs. These include a dichotomy between certain syllables that predictably surface as L and those that do not. An example provided by Grimm from Gyeli [gyi] is the leftward spread of H tone from the rightward edge of multi-syllable verbs. The first syllable of the verb is considered to have lexically assigned tone. H tone spreads leftward up to, but excluding, the first syllable of the verb.

Grammatical tone in TAMP encoding (a topic explored at length in Rolle & Bickmore (2026 [this volume])) is relatively frequently observed in NWB languages. Tone can also mark dependency of clauses. Some TAMP categories have a right-assigned H on the verb stem (sometimes labeled as a metatonic H) that is present when a verb is phrase medial rather than phrase final. Details vary regarding the specific conditions for the realization of this tone and whether its effects are also seen on the element following the verb, or whether (in some languages) an additional tone links the verb to its following element (often an object). Grimm provides numerous examples illustrating the variables to be considered.

Acknowledgements

As editors, we wish to thank the Pike Center for Integrative Scholarship for a grant that permitted us to obtain consultant help in the conversion to LaTeX of some of the chapters of this volume. We therefore thank Felix Kopecky, whose excellent and timely consultant services made the finalization of this volume achievable. Our gratitude also goes to Sebastian Nordhoff, whose responsiveness and care for detail greatly improved the quality of this publication.

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1 *African tone phenomena in focus*

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Chapter 2

Tone spread and shift

Joshua A. G. Smolders

SIL Global

Two of the most common tonal processes found in African languages are *spread* and *shift*, which are simply different expressions of the same phenomenon of tonal mobility. Both processes involve tones which were originally associated with one tone bearing unit (TBU) either forming new associations with neighbouring TBUs (i.e. spread) or entirely relocating to a different TBU (i.e. shift). In this chapter, a typology of tonal mobility is offered based on the major parameters of movement: direction (rightward vs. leftward), target (bounded vs. unbounded), mobile tone (H vs. L, or otherwise), and retention of the association to the TBU of origin (spread vs. shift). The different movement types are illustrated with examples from a variety of languages, and discussion is expanded to the complications that arise from movement, including tonal interactions (contour formation, delinking, deletion, assimilation, blocking, etc.), multiple kinds of movement applying in a single phonological derivation, lexical vs. postlexical movement, and historical movement.

1 Introduction

1.1 Tonal mobility

Tone in African languages tends to be “mobile”, which is to say that a given tone may be pronounced on a different tone-bearing unit (TBU) than the one to which it was originally associated (i.e. its lexical origin). Consider the following example from Bandi ([bza] a Mande language of Liberia). When the low-toned postposition /wà/ ‘on’ follows a low-toned noun such as /nikà/ ‘cow’, the resulting phrase is pronounced [nikà wà], as might be expected. When following a low-high or high-toned noun like /pèlé/ ‘house’ or /kéké/ ‘dog’, however, the result is [pèlé wá] and [kéké wá], respectively. How can we explain the change of

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/wà/ from low to high? It would seem unlikely that a new high tone (H) simply appeared out of thin air, unless there was evidence of a *tonal morpheme* intervening between the noun and postposition.¹ Rather, a reasonable hypothesis might be that the H of /pèlè/ and /kéké/ “moved” to the morpheme /wà/, as in example (1).²

- (1) Bandi (adapted from Mugele & Rodewald 1991: 112)
- kéké wà → kéké wá
 dog on
 ‘on a dog’

Using autosegmental theory to illustrate this movement, figure 1 shows that the H formed a new association to the TBU on its right, deleting the original low tone (L).³

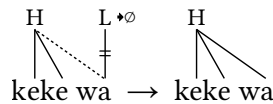


Figure 1: Rightward H spread in Bandi.

Now consider an example from Zulu ([zul] an Nguni language of South Africa, a.k.a. isiZulu). When a low-toned verb like /kù- limìsà/ ‘to help plough’ changes the infinitive prefix /kù-/ for a high-toned subject prefix like /bá-/ ‘we’, the result is pronounced [bàlimìsà], with a high tone on the penultimate syllable. Again, a reasonable hypothesis to explain this change is that the H of /bá-/ “moved” to the penultimate syllable, as shown in (2).⁴

- (2) Zulu (adapted from Downing 1990: 270)
- bá- lim -is -a → bàlimìsà
 3PL- plough -CAUS -FV
 ‘they help plough’

¹A tonal morpheme is a morpheme that consists entirely of a free-floating tone, without any lexical segments such as vowels or consonants. See Rolle et al. (2026 [this volume]).

²Here and throughout the paper, the TBU of a morpheme that is analysed as the lexical origin of the tone in focus (whether H or L or otherwise) is underlined in the data.

³In autosegmental notation, a solid line indicates an initial association between a tone and a TBU, a dashed line indicates a new association, a double-crossed line indicates a deleted association, →∅ indicates the deletion of a tone, and ∅→ indicates the insertion of a tone.

⁴Note that Zulu is analysed as having only H specified underlyingly (Downing 1990: 264), with any L being inserted by default at the end of the phonological derivation. Thus, underlying forms of low-toned morphemes are represented as unmarked.

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In autosegmental terms, the H of /bá/ first associated with a TBU to its right, then disassociated from its original TBU, and the resulting empty TBUs received a default L, as shown in figure 2.⁵

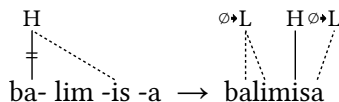


Figure 2: Rightward H shift in Zulu.

These examples illustrate two general types of tonal mobility common in African languages, known respectively as tone spread (example (1), figure 1), and tone shift (example (2), figure 2). I define these as follows:

- TONE SPREAD – when a tone forms a new association with one or more neighbouring tone-bearing units while retaining its previous associations.
- TONE SHIFT – when a tone forms a new association with a tone-bearing unit and loses its previous association.

As these definitions show, the key difference lies in whether the mobile tone retains the association to its original TBU. Kisseberth and Odden put it well in their chapter on tone in Bantu languages:

The difference between the two phenomena seems to be simply whether the H is realized just on the mora to which it has been extended (the ‘end-point’) or also on the original TBU and intervening moras. This unity is demonstrated by the fact that in some languages such as Xhosa (cf. Cassimjee and Kisseberth 1998) a structure may surface as shifting or spreading dependent on context. (Kisseberth & Odden 2006: 63)

Fundamentally, therefore, it can be said that the two phenomena are just different expressions of the same principle of tonal mobility.

1.2 A typology of spread and shift

To better understand tone spread and shift, it is necessary to first clarify the parameters of tonal mobility, then examine which movement types are attested and which are not. As to the parameters of movement, some pertinent questions include:

⁵This is a simplified representation. Downing (1990) actually analyses this as a series of rightward associations and left disassociations.

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- Which tones can move?
- Which direction do tones move?
- What is the target of the tonal movement?

I consider these in order below.

For African languages in general, the evidence is clear that both H and L can spread and shift, although the answer to which tones move will always be language specific. Of these two, H is far more likely to be the mobile tone; in many cases where L is mobile, H is also mobile, but the opposite cannot be said. As to other tones, some authors have offered examples of mid tone (M) movement (see §2.5), and some have even proposed moving contours (e.g. Newman 1986a), though this is comparatively rare since contours in African languages are usually analysed as *composite*, so any movement would be of the individual tones within the contour.⁶

As to direction, again this will depend on a given language, but there is good evidence that tones can move both to the *right* and *left*, with rightward movement being the more common (Hyman & Shuh 1974: 88). Of course, the terms *right* and *left* in this context refer to the axis of time, not spatial movement. Rightward moving tones are those which either draw out their pronunciation or are pronounced late in the spoken string, whereas leftward moving tones are anticipated or pronounced early in the spoken string.

Finally, the target of tonal movement can either be *bounded* or *unbounded*, sometimes also called *local* and *iterative* (Odden 2020: 11). Bounded movement means that a tone will spread or shift one TBU to the left or right. Conversely, unbounded movement means that a tone will spread or shift any number of TBUs until it hits some morphologically or prosodically important location (e.g. the end of a verb stem, or the penultimate mora), or is blocked by another tone or phonologically significant feature. While the evidence is not quite as strong as with the other parameters, bounded movement is generally more widely attested than unbounded.

With this understanding, a typology of tonal mobility in African languages can now be proposed. The various parameters of direction, target, tone, and type are organized in table 1. Each of the cells in *italics* gives a language in which a particular type of movement is attested, whereas empty cells represent

⁶Analysing a contour tone as *composite* means that, for example, a falling tone might be a HL sequence associated to the same TBU, and likewise a rising tone might be a LH sequence associated to the same TBU, rather than *unary* tones F and R, respectively.

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unattested types of movement. Using this typology, one can consistently classify any tonal movement observed in a specific language (e.g. the first cell indicates that Ikorovere Makua has *rightward bounded H spread*). One can also draw a clear generalization about the relative frequency of these parameters of movement (see §1.3): rightward movement is more common than leftward, bounded movement is more common than unbounded, H movement is more common than L (and other tones), and spread is more common than shift.⁷

Table 1: A typology of tonal mobility.

direction	target	tone	spread	shift
rightward	bounded	H	<i>Ikorovere Makua</i>	<i>Dana</i>
leftward	bounded	H	<i>Konni</i>	<i>Runyankore</i>
rightward	bounded	L	<i>Buli</i>	<i>Fur</i>
leftward	bounded	L	<i>Taita</i>	<i>Lendu</i>
rightward	unbounded	H	<i>N. Sotho</i>	<i>siSwati</i>
leftward	unbounded	H	<i>Ekegusii</i>	-
rightward	unbounded	L	<i>Ruwund</i>	-
leftward	unbounded	L	<i>Hausa</i>	-

While this typology is a useful place to begin, when getting into the particulars of a language things become more complicated. Like any other phonological process, tone movement does not occur in isolation, and so a seemingly simple movement may have a great variety of realizations depending on the phonological context in which it takes place. Therefore, some additional questions which will be investigated in this chapter include:

- What kinds of tonal interactions occur when tones move?
- When in the phonology does tone movement occur?

⁷It should be noted that this typology accounts only for surface phenomena, but particular theories will inevitably analyse and categorize things differently. For example, for my purposes the phenomenon of tone “doubling” or “copying” is treated the same as “bounded spread”, however under certain theoretical considerations there may be a meaningful distinction between the two. Even in autosegmental theory a sequence of two high-toned syllables can be represented as a single H associated to two TBUs or two distinct H tones associated to adjacent TBUs, though in general it is held that the latter is disallowed and so “fixed” by a phonological process such as merging the two tones. Other theories also lump bounded and unbounded movement together and analyse the phenomena as a matter of fixing boundaries of a tonal domain, whether local or distant (David Odden, personal communication).

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In the remainder of this chapter, I first give examples of the various types of movement, analysed and illustrated using autosegmental theory (§2). I then examine how moving tones can interact, including contour formation, delinking, deletion, assimilation, and blocked movement (§3). Finally, I look at when in the phonology tone movement may take place, giving an example of multiple movements occurring to produce a single surface form, looking at lexical versus post-lexical movement, and discussing historical (i.e. unproductive) movement (§4).

1.3 Typological corpus

Generalizations made about the typology presented in this chapter are based on my own survey of the descriptive literature of specific tone systems in Africa (Smolders 2025), covering 100 languages from 32 countries, representing 28 language families (including 2 isolates) from all the major phyla: Niger-Congo (68 languages, 38 of which were Bantu), Nilo-Saharan (15), Afro-Asiatic (13), Khoisan (2), plus 1 complete isolate and 1 creole. Due to the size of the corpus, I was unable to verify most analyses of movement, thus an attestation of a particular movement is only a claim that it is described as such by the references given.⁸ Table 2 below summarizes the results of this survey, first giving the percentage of languages in the corpus attesting a particular type of movement without reference to which tones are moving, and then breaking down attestations by moving tone H, M, and L.

Table 2: Frequency of movement types by % of languages with attestations based on a survey of 100 language descriptions (Smolders 2025).

	LBSp	RBSp	LUSp	RUSp	LBSh	RBSH	LUSh	RUSh
Total	18%	56%	9%	19%	7%	20%	0%	8%
H	14%	46%	8%	14%	7%	18%	0%	8%
M	1%	2%	0%	2%	0%	1%	0%	0%
L	8%	27%	2%	8%	2%	2%	0%	0%

⁸In cases where authors’ terminology differed from mine, I relabelled phenomena accordingly. So, what I call *shift* a given author might call *spread*, or what I would analyse as *leftward bounded H spread* causing M formation, an author might identify as L raising before a H. In a few select cases I also recorded an attestation where the phenomena were clear in the examples but not described by the author *per se*.

2 Examples of spread and shift

2.1 Bounded spread

I begin this overview of tonal mobility types with bounded spreading, starting with the most widely-attested type, *rightward bounded H spread*. A prime example of this can be found in the Ikorovere dialect of Makua ([vmw] a Bantu language of Mozambique, a.k.a Makhuwa). In this language, in the negative perfect a H tone is assigned to the second TBU of the verb stem and spreads one TBU to the right, whether the target is a different morpheme as in (3a), or within the verb as in (3b), illustrated in figure 3.

- (3) Ikorovere Makua (adapted from Cheng & Kisseberth 1980: 30 and Odden 2020: 12)

- a. a- ki- lokóth -al -e → àkìlòkòt^hàlè
 NEG 1SG pick.up -TA -FV
 ‘I have not picked up.’
- b. a- ki- maálih -al -e → àkìmàálihàlè
 NEG 1SG silence -TA -FV
 ‘I have not silenced.’

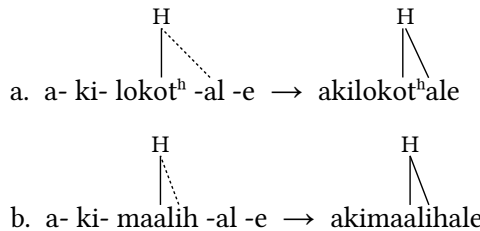


Figure 3: Rightward bounded H spread in Makua.

In both of these examples, the H forms a new association to the TBU immediately to the right of its origin. After this spread, all other TBUs are assigned L by default.⁹

This type of spreading can also occur in the opposite direction, as in Konni ([kma] a Gur (Mabia) language of Ghana, a.k.a Konni, Koma, Komung), where *leftward bounded H spread* can be observed. In example (4a) below, the underlying H tone of the definite suffix /-há/ spreads leftward onto the preceding noun

⁹For simplicity, I do not always depict default tone insertion in the autosegmental illustrations, as here.

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class 1 plural suffix /-à/, which is itself attached to the H noun root /nú/ ‘chest’.¹⁰ As illustrated in figure 4, the spreading H delinks the L, causing it to float, the effect of which is non-automatic downstep of the suffix H (cf. Snider 2026 [this volume]).¹¹ For context, (4b) shows the plural form of the noun without the definite suffix.

(4) Konni (adapted from Cahill 2004: 6)

- a. núr -à -há → nú¹ráhá
chest -C1.PL -DEF
‘the chests’
- b. núr -à → núrà
chest -C1.PL
‘chests’

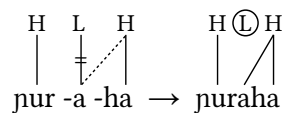


Figure 4: Leftward bounded H spread in Konni.

An example of *rightward bounded L spread* can be found in Buli ([bwu] a Gur language of Ghana). Although there are various effects that this L tone spreading has depending on the neighbouring tones (see §3), for clarity I discuss only the simplest context. In (5), when a low-toned noun /kò/ ‘father’ is followed by two morphemes with H tones (in this case the high-toned noun /bíík/ ‘child’ and the high-toned definite suffix /-ká/), the L will spread to the next TBU and displace the originally associated H, which then merges with the following H.¹²

(5) Buli (adapted from Schwarz 2003: 126)

- kò -bíí -ká → kòbíìká
father.c1/2 -child.c5/6 -DEF
‘the patrilineal relative’

¹⁰Cahill (2004) analyses the c1.PL suffix as having a polar tone, so it is L by virtue of the preceding H root. See Pulleyblank (2026 [this volume]) for more on polar tones.

¹¹In the autosegmental representation the floating tone is circled, and in the data downstep is represented by the ¹ symbol. See Snider (2026 [this volume]) for more on automatic and non-automatic downstep.

¹²Note that Schwarz analyses the TBU in Buli as the syllable, thus the L spread is still analysed as bounded even when it is to a bimoraic syllable. Also note that in the plural, the vowel of the noun root is short. It is possible to analyse the TBU as the mora and still maintain that this is bounded spread by proposing a lengthening process that occurs after tone spread.

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When this exact construction is made plural by adding the toneless suffix /-sa/, the L again spreads rightward only one TBU displacing the H, which then associates to this toneless TBU and merges with the following H, as seen in (6), illustrated in figure 5.

- (6) Buli (adapted from Schwarz 2003: 126)
 kò -bí -sa -ŋá → kòbìsàŋá
 father.c1/2 -child.c5/6 -PL -DEF
 ‘the patrilineal relatives’

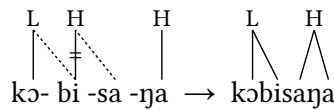


Figure 5: Rightward bounded L spread in Buli.

The final type of bounded spread, *leftward bounded L spread*, is a relatively rare phenomenon, but it can be observed in a particular context in Taita ([dav] a Bantu language of Kenya, a.k.a. Dabida). For verbs with an underlying H tone that form a three-syllable stem (i.e. the verb root plus any suffixes), when the final TBU is at the end of a phrase, a boundary L tone associates to it and then spreads one TBU to the left. In example (7), the stem of ‘to see each other’ is given after the H tone of the verb has spread to the penultimate syllable. Since the final TBU remains empty, it receives the L boundary tone. This L then spreads leftwards one TBU, to the second mora of the long vowel in /-aan/, resulting in a falling tone. This process is illustrated in figure 6.

- (7) Taita (adapted from Odden 2001: 82)
 ku- bón -áán -a -∅ → kùbónáànà
 INF- see -RECIP -FV -BT
 ‘to see each other’

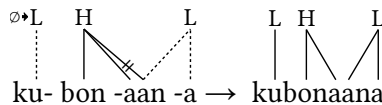


Figure 6: Leftward bounded L spread in Taita.

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2.2 Unbounded spread

Next I look at examples of unbounded spreading, where the moving tone continues to form new associations in a particular direction until it reaches an important morphological or prosodic target.

In the northwestern dialect of Northern Sotho ([nso] a Bantu language of South Africa) a robust pattern of *rightward unbounded H spread* can be observed. For any verb phrase containing a single H, that H will spread rightward, forming new associations with every TBU until it reaches the penultimate syllable, a well-known point of prosodic prominence for Bantu languages. This occurs within a simple verb stem (8a), as well as across multiple toneless suffixes (8b), and even across word boundaries (8c). This last example is illustrated in figure 7.

(8) Northern Sotho (adapted from [Zerbian 2006](#): 149–150)

- a. go khúrumetš-a → go khúrúmétša
 INF cover -FV
 ‘to cover’
- b. go rék -el -an -a → go rékélána
 INF buy -BEN -RECIP -FV
 ‘to buy for each other’
- c. go rēm -a mohlare → go rémá móhlare
 INF split -FV tree
 ‘to split a tree’

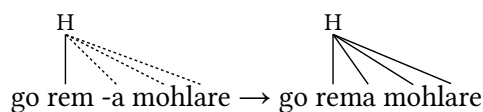


Figure 7: Rightward unbounded H spread in Northern Sotho.

An example of *rightward unbounded L spread* can be seen in Ruwund ([rnd] a Narrow Bantu language of the Democratic Republic of Congo, a.k.a Ruund). Unlike most other two-tone languages, in Ruwund the L is the marked tone and H is default ([Nash 1994](#): 224). In verb phrases, certain prefixes like the third person singular subject /wù-/ have a lexical L which spreads across any number of morphemes to the edge of the prefixal domain (indicated below with a bracket), after which toneless TBUs receive a default H, as seen in (9a), illustrated in figure

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8. For context, in (9b) the same construction is given but with the toneless first person subject prefix /ni/. In this case all TBUs receive a default H.¹³

(9) Ruwund (adapted from Nash 1994: 240–241, 251)

- a. wù- a- ma- mu- [fut -ol -in -a →
 3SG.SBJ- PST1- 3PL.OBJ- 3SG.OBJ- pay -INTS -APPL -FV
 wàmàmùfútóliná

‘She paid him a lot for them.’

- b. ni- a- ma- mu- [fut -ol -in -a →
 1SG.SBJ- PST1- 3PL.OBJ- 3SG.OBJ- pay -INTS -APPL -FV
 námámùfútóliná

‘I paid him a lot for them.’

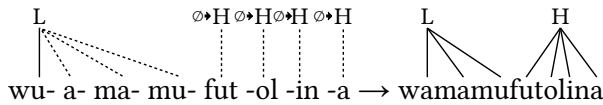


Figure 8: Rightward unbounded L spread in Ruwund.

Looking now at *leftward unbounded H spread*, an example can be found in Ekegusii ([guz] a Bantu language of Kenya). Here, in certain grammatical constructions (e.g. the habitual) a H tone is suffixed to the verb stem (cf. Rolle & Bickmore 2026 [this volume]).¹⁴ As long as the initial TBU of the verb stem does not have an association with a H outside the stem, this H tone will associate to the final vowel and spread leftward to the last non-initial TBU of the stem.¹⁵ So for example, in (10a) the H associates to the final vowel /-a/, then spreads leftward to the second TBU of the stem (the stem boundary is indicated by a bracket). This process is illustrated in figure 9. In (10b) the same pattern occurs, but since the verb has a lexical H tone, it merges with it after spreading. In (10c) the spreading

¹³Note that the vowels of both /wù-/ and /ni-/ coalesce with the tense prefix /a-/.

¹⁴It would be possible to analyse this as a H which associates to the second TBU of the stem and spreads rightward to the end of the word, but data in which the spread is blocked shows that this H indeed initially associates to the final TBU.

¹⁵Ekegusii also has rightward bounded H spread of prefixes. Sometimes this spreading will occur across the stem boundary, and sometimes it will land adjacent to the stem boundary and merge with a H on the root. This is a good example of how many languages attest multiple types of movement that can interact with each other.

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does not occur because the first TBU of the stem is associated to a H outside the stem.

(10) Ekegusii (adapted from Bickmore 1999: 115–117)

- a. tó- raa [timok -er -á → tóráà timókérá
 3PL- SBJ rest -APPL -FV
 ‘We will rest (appl).’
- b. tó- raa [tákun -er -á → tóráà tákúnérá
 3PL- SBJ chew -APPL -FV
 ‘We will chew (appl).’
- c. tw- áá- [timok -er -á → twáátimòkèrá
 3PL- PST rest -APPL -FV
 ‘We just rested (appl).’

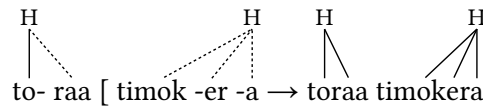


Figure 9: Leftward unbounded H spread in Ekegusii.

The last possible type of spread, *leftward unbounded L spread*, is evidently rare. I found only one possible attestation in my corpus, from Hausa ([hau] a Chadic language of Nigeria and Niger), where so called “tone-integrating” suffixes spread their tone (whether L or H) onto the stem, overriding any lexical tone. When such a suffix has a tone pattern beginning with a L, as seen with the plural /-àikúú/, it spreads leftward until the left edge of the stem, as in 11, illustrated in 10.¹⁶

(11) Hausa (adapted from Newman 1986b: 252)

- ráánáá -àikúú → ràànàikúú
 day -PL
 ‘days.’

¹⁶Whether this is bounded or unbounded spread depends on whether the TBU in Hausa is analysed as the syllable or the mora, and whether spreading occurs before or after vowel coalescence.

2 Tone spread and shift

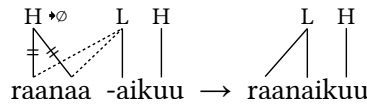


Figure 10: Leftward unbounded L spread in Hausa.

2.3 Bounded shift

The next major type of movement in this overview is bounded shift. As a reminder, the primary difference between tone spread and tone shift is whether the tone retains its association to its TBU of origin.

Dana (currently classified as a dialect of Opo, [lgn] a Koman language of South Sudan and Ethiopia, a.k.a. Burun-Koma) provides a good example of *rightward bounded H shift*. In associative compounds (i.e. nouns formed with the construction N_1 -ASC N_2) the associative marker /-ma/ is underlyingly H. This H, however, is never realized on the morpheme itself. Instead, it shifts rightward to the first TBU of N_2 , and the /-ma/ itself gets a default L.¹⁷ Thus in (12a) the H of /-ma/ associates to the first TBU of the low-toned N_2 /dɪnc'à/ displacing the underlying L. Likewise in (12b) the H associates to the only TBU of the low-toned N_2 /dɛ/ and forms a falling contour. (12a) is illustrated in figure 11.

(12) Dana (Smolders personal data, from John David Amune).

- a. wàc'à -má dɪnc'à → wàc'àmádɪnc'à
 fish -ASC skin
 'skinfish'
- b. dɪnc'à -má dɛ → dɪnc'àmádɛ
 skin -ASC eye
 'eyelid'

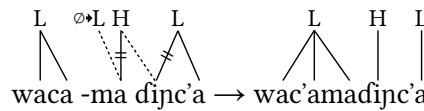


Figure 11: Rightward bounded H shift in Dana.

¹⁷ Another potential analysis is that the morpheme has a lexical LH pattern, and the L retains its association to /-ma/ whereas the H shifts. Based on other data, this morpheme is not the target of L spread from a preceding morpheme, though in the examples here there is a preceding L which merges with the default.

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Leftward bounded H shift can be found in Runyankore ([nyn] a Bantu language of Uganda, a.k.a. Nyankore). Here, as a general pattern a H will shift from an utterance final TBU onto the penultimate TBU. Thus, in (13a) the lexical H of the adjective /bí/ ‘bad’ shifts to the toneless class prefix /mu-/, illustrated in figure 12. In (13b) the H of the verb stem /rí/ ‘eat’, after it has coalesced with the final vowel /-a/, shifts onto the toneless distant past tense prefix /ka-/. As is the case with many of the examples given so far, after H movements are resolved toneless TBUs receive a default L.¹⁸

(13) Runyankore (adapted from Hyman 2022: 39)

- a. mu- bí → múbí
 c1- bad
 ‘bad person’
- b. tu- ka- rí -a → tùkárýà
 1PL- PST3- eat -FV
 ‘We ate.’

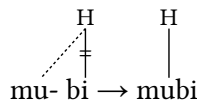


Figure 12: Leftward bounded H shift in Runyankore.

Rightward bounded L shift was attested in only two languages in my survey, one being a case of displaced tones (discussed in §2.5), and the other coming from Fur ([fvr] a Fur language of Sudan). Here, underlyingly HL monosyllabic nouns and underlying LHL bisyllabic nouns will form a falling contour on the last TBU of the singular stem, but when the toneless plural suffix /-a/ is attached as in (14a) and (14b), the final L tone shifts to it, decomposing the contour. This is illustrated for /kàràb -a/ in figure 13. Interestingly, the H of a LH contour under the same conditions will spread instead of shift.

(14) Fur (adapted from Waag & Kutsch Lojenga 2004: 19–20)

- a. púl -a → púlà
 groundnut -PL
 ‘groundnuts’

¹⁸This is not to say that Runyankore has no overt Ls, however, since Hyman analyses the motivation for the leftward H shift as the presence of an utterance final boundary L (Hyman 2022: 41).

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- b. kàrāb -a → kàrábà
 animal -PL
 ‘animals’

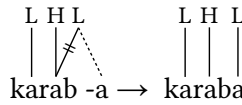


Figure 13: Rightward bounded L shift in Fur.

Leftward bounded L shift was only attested in two languages in my survey, and in each case H also shifted under the same conditions. One example comes from Lendu ([led] a Central Sudanic language of the Democratic Republic of Congo), where monosyllabic nouns may take the suffix /-í/, as in /tʃò-í/ → [tʃòí] ‘bees’ and /nò-í/ → [nòí] ‘female slave’. However, for monosyllabic nouns beginning with a subset of the consonant inventory (s, z, $\widehat{ts}$, $\widehat{dz}$, r, $\mathfrak{r}$), the root vowel elides and its tone shifts onto the preceding consonant, as seen with L in (15a) and figure 14, and with H in (15b).¹⁹

(15) Lendu (adapted from Trifković 1977: 123–124)

- a. sò -í → sì
 grass.sp -SFX
 ‘variety of grass’
 b. rí -í → rí
 grass.sp -SFX
 ‘ghost’

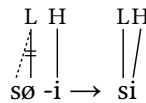


Figure 14: Leftward bounded L shift in Lendu.

¹⁹Trifković makes a distinction between an onset consonant that is functioning as a TBU and one that is not: čv vs. čv̌, é̌v vs. čv̌, and čv̌ vs. čv̌. I do not know how this is produced or perceived phonetically, especially with the voiceless consonants, and she gives no explanation.

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2.4 Unbounded shift

The final type of tonal movement I look at is unbounded shift, of which there is only one attested subtype identified in African languages so far: *rightward unbounded H shift*. An example of this can be found in siSwati ([ssw] a Narrow Bantu language of Eswatini, a.k.a. Swati). In this language, any single H originating within the phonological word of a verb will shift to a particular prosodic target, either the antepenultimate TBU if the word is phrase-final, or the penultimate TBU if not phrase-final. Thus, in (16a) the H of the verb /búlal/ shifts two TBUs to the right, to the antepenultimate position (determined before penultimate lengthening), associating to the suffix /-is/. In (16b) that same H only shifts one TBU to the right, targeting the penultimate position, which happens to be the second syllable of the verb root. Likewise the H of the prefix /bá-/ in (16c) shifts past all prefixes and the toneless verb root /łagul/ to the third-to-last TBU phrase-finally, but when non-final in (16d) it targets the second-to-last TBU, as illustrated respectively in figure 15.

(16) siSwati (adapted from Malambe 2017: 324)

- a. ni- ya- búlal -is -an -a → nìyàbùlálísàànà
1SG- PROG- kill -CAUS -RECIP -FV
'You (pl.) are causing to kill one another.'
- b. ni- tawu- búlal -a Siphò → nìtawùbùlálà Siphò
1SG- FUT- kill -FV Siphò
'You (pl.) will kill Siphò.'
- c. bá- ya- łagul -is -an -a → bàyàłàgùlísàànà
C2.SBJ- PROG- weed -CAUS -RECIP -FV
'They are causing to weed one another.'
- d. bá- to- nits^h -is -an -a Siphò → bàtònits^hisàànà Siphò
C2.SBJ- FUT- drink -CAUS -RECIP -FV Siphò
'You (pl.) will cause Siphò to drink together.'

2.5 Other moving tones and tone features

So far in this section I have only used instances of H and L tones moving to illustrate the typology presented in table 1. It is possible, however, for other tones to move, though such phenomena are comparatively less widely attested (a fact likely related to the inverse correlation between the number of tone levels in a language and tonal mobility). For example, Lesage (2020) describes both

2 Tone spread and shift

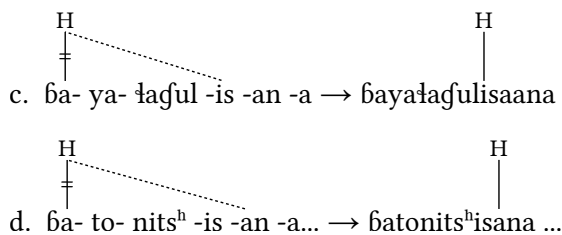


Figure 15: Rightward unbounded H shift in siSwati.

rightward bounded H spread and *rightward bounded M spread* in Kam ([kdx] an Adamawa language of Nigeria, a.k.a. ànwòm). A spreading H will displace any L or M associated to the target TBU (see §3.2 for more on delinking), but a spreading M will only displace a L tone. This supports the universal hierarchy of tone spreading proposed by Hyman & Shuh (1974: 89) that “if tone spreading takes place from a mid tone into a following low, then tone spreading also takes place from a high tone into a following low tone in that language.” Lesage (2020) also describes *rightward bounded M and L shift* in Kam, but only in the context of displacement by another tone spreading.

Additionally, it is important to note that under most theoretical frameworks the composition of tones are not as simple as unary H, M, and L. So, for example Yip (2002: 42–43) defines four tone levels via a “register” feature [+/- Upper] and a “tone” feature [+/-high]. Similarly, Snider (2020) conceptualizes an autosegmental structure of tone which has both a tonal tier (with features H and L) and a register tier (with features h and l). Under such theoretical frameworks, features or nodes other than the simple tone may spread or shift to account for various phenomena. Covering these possibilities is beyond the scope of this chapter (with the exception of figure 22 in §3.3), but the reader should nevertheless be aware that when linguists describe patterns of tone movement, they may be utilizing such frameworks.

3 Tonal interactions arising from mobility

3.1 Contour formation

In the previous section, to whatever extent possible, I presented uncomplicated examples of the tonal movement types. In reality, however, moving tones inevitably interact with their environment, producing more complex surface forms. In this section, I demonstrate some of the phenomena that can arise as a result.

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A very common result of moving tones is contour formation. Here a tone may spread or shift to a TBU that already has a different tone associated to it. If the phonology permits, both tones can simultaneously occupy the same TBU and form a contour tone (see footnote 6 and example 12b). Whether contour formation occurs, and whether those contours may be falling or rising or both, are parameters unique to each language. As Hyman & Shuh (1974) noted:

... there is apparently no implicational relationship between spreading from a lower to a higher syllable to give rising tones and spreading from a higher to a lower syllable to give falling tones. Nupe has created rising tones but not falling tones (apart from a few exceptions deriving from the loss of final syllables; cf. George 1970). On the other hand, Aboh Igbo and Ngizim have falling tones (from H-L) but do not permit rising tones. (Ngizim does allow rising tones under limited conditions, but they never result from tone spreading.) (Hyman & Shuh 1974: 10)

A good example of contour formation can be seen in Yemba ([ybb] a Grassfields Bantu language of Cameroon, a.k.a. Bamileke-Dschang) (cf. Hamm et al. 2026 [this volume]). In the associative construction seen in (17), the L tone of the class prefix /mǎ-/ will spread rightward onto the second noun. Given that the lexical tone of N_2 /tsǒŋ/ is H, both the L and H end up associated to a single TBU, forming a rising contour, as illustrated in figure 16.²⁰

- (17) Yemba (adapted from Hyman 1985)
 è- fǒ è- mǎ- tsǒŋ → èfǒ mǎtsǒŋ
 c1- chief c1.ASC- c2- thief
 ‘chief of thieves’

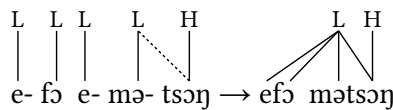


Figure 16: Rising contour formation in Yemba.

Contours may also result from spread only applying to one mora of a heavy syllable. In Southern Sotho ([sot] a Southern Bantu language of Lesotho) there is *rightward bounded H spread*, as well as penultimate vowel lengthening which

²⁰Note in this data that the c1.ASC marker coalesces with the vowel of the preceding noun, and that at the end of the phonological derivation all adjacent L tones merge.

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applies before this tonal movement. Since the TBU in this language is the mora, three different surface forms of a given high-toned monosyllabic verb root can be observed. First, when lengthening does not affect the syllables containing the origin or target TBUs, the H tone distributes across two syllables, as seen in (18a). Second, when the syllable containing the TBU *of origin* is subject to penultimate lengthening, the spread will associate to the newly inserted mora, resulting in a long level H tone as in (18b). Finally, when the syllable containing the *target* TBU is subject to penultimate lengthening, the H tone will spread only to the first mora, resulting in a long vowel with a falling tone as in (18c), illustrated in figure 17. In all cases, at the end of the phonological derivation empty TBUs receive a default L tone.

(18) Southern Sotho (Smolders personal data, from Bohang Matsumunyani)

- a. k'ɪ- a- rúṭ -a ba- na → k'ìàrúṭá bààná
1SG- PROG- teach -FV C2- child
'I am teaching the children.'
- b. k'ɪ- a- rúṭ -a → k'ìàrúṭà
1SG- PROG- teach -FV
'I am teaching.'
- c. k'ɪ- a- rúṭ -is -a → k'ìàrúṭiisà
1SG- PROG- teach -CAUSE -FV
'I am causing to teach.'

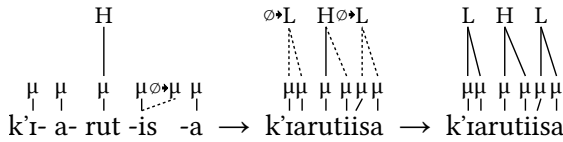


Figure 17: Contour formation in Southern Sotho.

3.2 Delinking and deletion

If a language disallows two tones to be associated to the same TBU (in at least some contexts), a moving tone may cause the originally associated tone to delink (i.e. break its association). A clear example of this can be seen in Sezo ([sze] an Omotic language of Ethiopia, a.k.a. Seze), where there is *rightward bounded H spread* from prefixes to a following noun. If the first syllable of the noun is bimoraic and bears a L, the H spread causes the L to delink from the first mora,

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resulting in a falling contour, as seen in (19a). If, however, the first syllable of the noun is monomoraic, the H spread causes the L to delink and float, resulting in non-automatic downstep of a following high, as seen in (19b).²¹

(19) Sezo (adapted from Desta 2016: 76)

- a. jé- tààlí → jétààlí
DEF- chief
'the chief'
- b. jé- gizzí → jégízí¹
DEF- money
'the money'

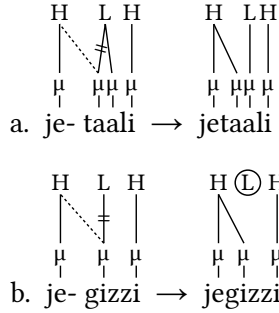


Figure 18: L delinking in Sezo.

In some cases a tone that is delinked by the movement of another tone may simply delete, rather than persisting as a floating tone or retaining an association to another TBU. This is the case with *rightward bounded H spread* onto the inalienable possessive suffix in Laal ([gdm] a language isolate of Chad). In (20a) and (20b) it can be seen that the H tone spreads to the suffix /-àr/ causing the underlying L to simply disappear, neither forming a contour nor floating. This is illustrated in figure 19. The existence of that L can be seen in (20c) where the spreading occurs only to the second mora of the long vowel in the noun root itself, never reaching the suffix. If the suffix were toneless, we would expect a M (which Lionnet (2022: 286) analyses as “weak” or “default”), but since it surfaces with a L, it is analysed as underlying.

²¹In Desta (2016) the data in (19a) also indicates the final H is downstepped. However, since there is an explicit (non-floating) L tone preceding it, I assume this is the result of automatic downstep, and so I have removed the ¹ from the data.

(20) Laal (adapted from [Lionnet 2022](#): 283)

- a. tím -àr → t_ímár
hand -INPOSS
'his hand'
- b. ság -àr → s_ágár
brother.in.law -INPOSS
'his brother-in-law'
- c. wúur -àr → w_úúràr
thigh -INPOSS
'his thigh'

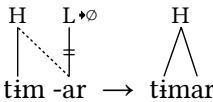


Figure 19: L deletion in Laal.

3.3 Assimilation

For the purposes of this chapter, I define “assimilation” as tonal movement which causes a tone on the target TBU to take on some property of the moving tone without delinking or forming a contour.²² In the literature on tonal phenomena, however, the term “assimilation” has been used in various ways, and one might argue that all tone spread is in fact a kind of assimilatory process ([Hyman & Shuh 1974](#): 88). Thus, for example [Yip \(2007: 232\)](#) describes data from [Akinlabi & Liberman \(2000\)](#) as “assimilation”, whereas I would refer to it as “contour formation resulting from bounded spreading” (i.e. the patterns HL → HF, and LH → LR). Likewise [Rolle \(2018: 19–20\)](#) describes what I call “spreading with contour formation” as “horizontal assimilation”, and what I am calling “assimilation” as “vertical assimilation”.

The clearest example of assimilation as I define it is M formation arising from the spread or shift of a H to a L target, or a L to a H target. For example, in Kyak ([bka] an Adamawa language of Nigeria), H can spread either leftward or

²²I am using “property” here in a non-technical way, though particular theoretical frameworks will undoubtedly want to explain such phenomena as the spread of a particular feature or node in the structure of a tone.

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rightward to a neighbouring L to form a “phonetic mid” (Harley 2020: 379).²³ In (21) the underlyingly L word /kàp/ ‘tree’ is shown in two contexts modified by a quantifier. In (21a), when followed by another underlying L word, nothing remarkable occurs, but in (21b), when followed by the underlyingly H word /k^wín/, the H spreads leftward, raising /kàp/ to a M tone, as illustrated in figure 20.

(21) Kyak (adapted from Harley 2020: 387)

- a. kàp tè → kàp tè
tree three
‘three trees’
- b. kàp k^wín → k̄ap k^wín
tree one
‘one tree’

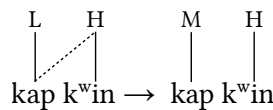


Figure 20: Assimilation of L to M in Kyak.

A very interesting case of tone assimilation can be found in Bafut ([bfd] a Narrow Grassfields Bantu language of Cameroon), where there is bidirectional assimilation, with both H lowering after a L and L raising before a H.²⁴ When noun stems are formed, if the class prefix is L and the noun root has an initial H, the prefix L is raised and the H of the noun root is lowered, but not to the point that they are on the same phonetic level. This can be seen in (22), where raising and lowering are represented by (↑) and (↓), respectively, following Chumbow & Neba (2011). The raising of the L is confirmed when the noun appears after another L in context, as in (23).

(22) Bafut (adapted from Chumbow & Neba 2011: 45)

²³Pending further research, Harley does not explain the exact conditions for this spread, however from the data it appears that it only occurs in certain environments, such as across morpheme boundaries within a noun phrase, as shown here. There are other contexts in which L remains phonetically low both preceding and following a H. Harley also describes downstep as a separate process.

²⁴Note that this H lowering is not simply automatic downstep, since subsequent H tones are not also lowered. Bafut has non-automatic downstep caused by a floating L, but not automatic downstep (Chumbow & Neba 2011: 44). See Snider (2026 [this volume]) for more on downstep.

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- a. bi- ləŋə → ↑bi↓ləŋə
 c2- horse
 ‘horses’
- b. à- báâ → ↑à↓báâ
 c2- fufu
 ‘fufu’

- (23) Bafut (adapted from Chumbow & Neba 2011: 48)
 nìbàʔàlìm bi- síŋ → nìbàʔàlìm ↑bi↓síŋ
 Nebalum c2- bird
 ‘Nebalum [proper name] of birds’

What makes this case particularly interesting is that it cannot be accounted for using a basic autosegmental representation, since one would have to posit crossing lines of association, as shown in figure 21. To solve this, Chumbow & Neba (2011) use Register Tier Theory or RTT (Snider 1999, 2020) to propose leftward shift of a low register feature (l) to the H, followed by the insertion of a default high register feature (h). While it is not in the scope of this chapter to explain individual theories and their notation in detail, a representation is given in figure 22 for the reader’s consideration.

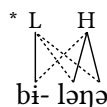


Figure 21: Invalid account of bidirectional assimilation in Bafut.

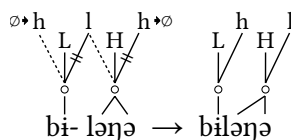


Figure 22: RTT account of bidirectional assimilation in Bafut.

3.4 Blocked movement

Even when a robust pattern of tone spread or shift is identified there may be exceptions where the movement behaves differently (e.g. unbounded spread appears to be bounded) or does not occur at all. Such cases can often be explained as blocked movement arising from the tonal environment.

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For example, tone movement may be blocked by a constraint against two tones of the same value being associated to adjacent TBUs.²⁵ In other words, if there are two distinct H tones or L tones next to each other, a language may prefer to break them up with an intervening tone of a different value. In Thetogovela (a dialect of Moro, [mor] a Kordofanian language of Sudan) there is *rightward unbounded H spread* from the locative prefix /é-/ onto nouns, as seen in (24a). If there is a H in the noun's lexical tone pattern, however, the prefix H will only spread onto a TBU that is not adjacent to the noun's lexical H. Thus, in (24b) the H of /é-/ spreads onto the first TBU of /ðəŋəlá/ but not the second, because the second is adjacent to the root H. In (24c) no spreading occurs at all because the first TBU is adjacent to the root H. Finally, in (24d) the two H tones do end up associated to adjacent TBUs, but only because preservation of tones in this language appears to take priority over avoidance of adjacent identical tones.

(24) Thetogovela (adapted from [Rose & Piccinini 2016: 21](#))

- a. é- ðaba → éðábá
LOC- cloud
'at cloud'
- b. é- ðəŋəlá → éðəŋəlá
LOC- tongue
'at tongue'
- c. é- ðabará → éðabará
LOC- cotton
'at cotton'
- d. é- ðáŋála → éðáŋála
LOC- sheep
'at sheep'

Another case of this can be found in Bamana ([bam] a Mande language of Mali and Côte d'Ivoire, a.k.a Bambara, Bamanankan), where there is *rightward unbounded L spread* (cf. [Konoshenko 2026 \[this volume\]](#)).²⁶ When a particular

²⁵This ban on adjacent identical tones is usually attributed to the Obligatory Contour Principle (OCP, see [Leben 1973](#) and [Yip 2002](#)). The OCP is generally understood to ban identical adjacent tones on the autosegmental level, not merely to adjacent associations at the segmental level. Thus in the data given for Thetogovela, it is assumed that a default L is inserted between the two H tones in (24b) and (24c), and associates to the rightmost available TBU. In (24d) it is assumed that the two H tones merge.

²⁶Unlike most other two-toned languages, in Bamana the L is the marked tone ([Vydrin 2016: 86](#)).

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TBU has an underlying L, and there are no other morphemes with underlying L tones within the word, that L will spread rightward until the end of the word, as in (25).

- (25) Bamana (adapted from [Vydrin 2016](#): 87)
- | | | |
|------------------|---|---------|
| gèlɛnya | → | gèlɛnyà |
| grow.difficult | | |
| ‘grow difficult’ | | |

This L spread is blocked, however, when there are two L tones in adjacent tonal domains (indicated here by a bracket).²⁷ In such cases, a H is inserted into the first domain to break up the sequential L tones. Thus in (26a), where the two L domains consist of a single syllable, a H is inserted and associates to the sole TBU of the first domain, forming a rising contour tone. Likewise in (26b) where the two L domains each have two syllables, a H is inserted onto the second TBU of the first domain and spread is blocked (evidenced by the lack of the contour). L tone spread proceeds as normal in the second domain.

- (26) Bamana (adapted from [Vydrin 2016](#): 87)
- | | |
|--------------|--------------------|
| a. mǎ | [dòn → mǎdòn |
| approach | |
| ‘approach’ | |
| b. kùnnà | [sìrì → kùnnàsìrì |
| be.ashamed | |
| ‘be ashamed’ | |

Tone movement can also be blocked by floating tones. In Kenyang ([ken] a Southern Bantoid language of Cameroon), the H of a subject prefix will spread to the verb if possible (cf. [Blench 2026 \[this volume\]](#)). Thus in (27a), in the perfective form, the H of the 3PL prefix passes over the zero PFV marker and associates to the verb /kà/, forming a falling contour tone. In the imperfective, however, there is a floating L IPFV marker between the prefix and the verb which blocks this spread, as seen in (27b).

- (27) Kenyang (adapted from [Snider 2022](#): 3)

²⁷[Vydrin \(2016: 86\)](#) defines a tonal domain as “[a] sequence consisting of a dominant syllable and the recessive syllables to its right (within a word or a tonally compact group)...”.

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- a. bá- Ø- kò → bákô
3PL- PFV- walk
'they walked'
- b. bá- ò- kò → bákò
3PL- IPFV- walk
'they are walking'

Movement may also fail to occur by lexical exception, where the conditions for blocking spread or shift existed historically but have since disappeared, or where the target of spread or shift has elided. This is found in Shambaa ([ksb] a Northeast Bantu language of Tanzania), where there is *rightward bounded H spread* from the penultimate syllable to the next TBU (i.e. the final syllable), as seen in (28a). A certain number of stems break this general rule, however, such as (28b) and (28c). This can be explained by the fact that these verbs historically had long penultimate vowels, so the target of the bounded spread used to be the second mora of the long vowel. This mora has since elided, but the spread has not reapplied to the new prosodic structure.

(28) Shambaa (adapted from Kisseberth & Odden 2006: 61)

- a. ku- kóm -a → kùkómá
INF- kill -FV
'to kill'
- b. ku- lál -a → kùlálà
INF- sleep -FV
'to sleep'
- c. ku- vyál -a → kùvyálà
INF- give.birth -FV
'to give birth'

Finally, movement may be blocked by depressor or non-depressor consonants (see Beavon-Ham & Pearce (2026 [this volume]) for more on consonant tone interaction). These are natural classes of consonants in a language, the former usually consisting of voiced obstruents and the latter of voiceless obstruents, sonorants, and implosives (Snider 2018: 44). In Tsonga ([tso] a Southern Bantu language of South Africa, Mozambique, and Zambia) there is *rightward bounded H spread*, as seen in (28) where the H tone of the copula /(h)í/ associates to the first TBU of a following noun, in this instance forming a falling contour tone.

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- (29) Tsonga (adapted from Baumbach 1987 as found in Seunghun 2008: 21–22)

í kòmbé → í kòmbé

COP impala.SP

‘it is an ousted impala’

If, however, the noun begins with a depressor consonant (i.e. voiced stops, breathy consonants, and aspirated consonants in Tsonga), such spreading is blocked, as seen in (30).

- (30) Tsonga (adapted from Baumbach 1987 as found in Seunghun 2008: 21–22)

a. í ĩwàrí → í ĩwàrí

COP partridge

‘it is a partridge’

b. í bùfá → í bùfá

COP hen.SP

‘it is a hen that eats its own eggs’

c. í tsʰùrí → í tsʰùrí

COP mortar

‘it is a [sic] mortar’

In Ngizim ([ngi] a Chadic language of Nigeria), *rightward bounded H spread* is blocked by voiced obstruents, and conversely *rightward bounded L spread* is blocked by any other consonant (cf. Roberts 2026 [this volume]). Thus in (31a) the H of the 1SG.PFV marker /ná-/ passes over the implosive /d/ and spreads to the first TBU of the verb root, making a falling contour. In (31b), however, this H spread is blocked by the voiced obstruent /b/. Conversely, in (32a) the L of the subjunctive prefix /à-/ spreads past the voiced obstruent /b/ onto the first TBU of the verb, delinking the lexical H, but in (32b) this spreading is blocked by the voiceless obstruent /k/.

- (31) Ngizim (adapted from Schuh 1971 as found in Seunghun 2008: 23)

a. ná- dānkó -w → ná dānkú

1SG.PFV sew -PFV

‘I sewed (it)’

b. ná- bàkó -w → ná bàkú

1SG.PFV burn -PFV

‘I burned (it)’

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(32) Ngizim (adapted from Schuh 1971: 54)

- a. à dábǵǵ -í → à dàbǵí
 SBJV bury IMP.SG
 ‘bury!’
- b. à káásǵ -í → à káási
 SBJV sweep IMP.SG
 ‘sweep!’

4 Analysing movement

4.1 Multiple movements and interactions

So far in this chapter I have isolated examples of spread and shift and the resulting interactions with their environments. For a given language, however, one may observe multiple different kinds of movement and interactions, sometimes all applying together to explain a single surface form. To illustrate this I discuss the case of Ci-Ruri (a dialect of Kwaya, [kya] a Bantu language of Tanzania) where several types of shift and spread occur, along with shift blocking, tone deletion, and both rising and falling contour formation.²⁸

The first phenomenon to note in Ci-Ruri is that of *rightward bounded H shift*. This can be seen when a verb has a lexical H tone, as in (33a) where the H of /tém/ shifts to the following toneless applicative marker, or likewise when an object marker has a H tone, as in (33b) where the H of /mú-/ shifts to the following toneless verb stem.

(33) Ci-Ruri (adapted from Massamba 2010, noun class information from Odom 2016)

- a. o- ku- t_ém- er -a → òkùt_ém_érà
 PP- INF- cut -APPL -FV
 ‘to cut (appl.)’
- b. o- ku- m_ú- gur -a → òkù_mù_gú_rà
 PP- INF- C1.OBJ- buy -FV
 ‘to buy him/her’

²⁸Note that while the data for this section comes from Massamba (2010), I do not attempt to reconstruct his analysis of accent formation and rules. Rather, I describe the patterns using the typology and phenomena already established in this chapter.

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There are constraints operating on this shift, however. If the shift would cause the H tone to associate to the final TBU of the word, it is blocked, as shown in (34a). Further, if both the object prefix and the verb root have H lexical tones, the verb's tone is deleted, as shown in (34b) (it is determined that the verb tone deletes because otherwise one would expect the H to shift to the applicative suffix).

(34) Ci-Ruri (adapted from Massamba 2010, noun class information from Odom 2016)

- a. o- ku- tém-a → òkùtémà
 PP- INF- cut -FV
 'to cut'
- b. o- ku- mú- tém-er -a → òkùmùtémèrà
 PP- INF- C1.OBJ- cut -APPL -FV
 'to cut (appl.)'

In addition to this, if there is a second object marker with a H tone, as in (35), that H does not delete like the verb's tone, but rather shifts (*rightward unbounded*) to the ultimate TBU, creating a falling contour with the final L tone.²⁹ After this shift, the H then undergoes *leftward bounded spread* to the penultimate TBU, creating a rising contour. This same pattern holds for both verbs with a lexical H (35a) and toneless verbs (35b) since the verb tone deletion still applies whether there is one or two object prefix H tones.

(35) Ci-Ruri (adapted from Massamba 2010, noun class information from Odom 2016)

- a. o- ku- βá- lí- gur -ir -a → òkùβàlígùrírâ
 PP- INF- C2.OBJ- C5.OBJ- cut -APPL -FV
 'to cut it for them'
- b. o- ku- βá- lí- tém-er -a → òkùβàlítémèrà
 PP- INF- C2.OBJ- C5.OBJ- cut -APPL -FV
 'to buy it for them'

To bring this all together, I illustrate the various processes required to derive the surface form of (35b) in figure 23: (1) verb tone deletion, (2) rightward unbounded H shift and falling contour formation, (3) rightward bounded H shift, (4) default L insertion along with leftward bounded H spread and rising contour formation, (5) surface realization.

²⁹Massamba (2010) argues that this L is generated by the accent. I rather suspect it is a boundary tone. For the purpose of this description I simply note that it is there.

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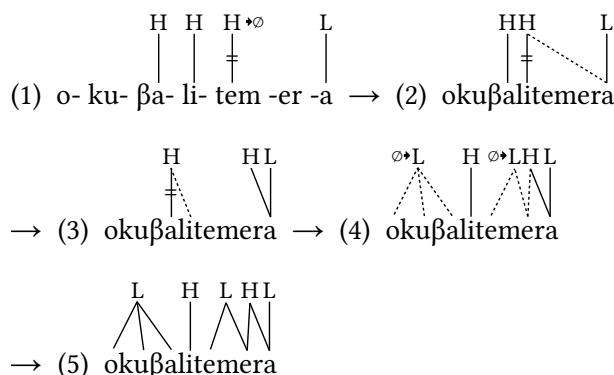


Figure 23: Complex tone movement in Ci-Ruri.

4.2 Lexical versus postlexical movement

The example of Ci-Ruri raises the question of *when* exactly in the phonological derivation tone movement may occur. For Ci-Ruri at least, certain types of movement must apply before others in order to support the analysis. Similarly, in other languages it seems that certain tonal movements must occur in a fixed order relative to other phonological processes like vowel lengthening—whether *before*, as seen with S. Sotho in (18), or *after* as seen with Buli in (5) (n.b. footnote 12). The long answer, of course, is that a given analysis will depend on the theoretical framework being employed, but for the purposes of this chapter I assume that tone movement can be either lexical (phonemic) or postlexical (phonetic). The importance of determining whether an instance of tonal movement is lexical or postlexical not only has implications for phonological analysis (e.g. whether a M or falling tone is contrastive or not), but also for orthography development (for which I refer the reader to works such as Snider 2018 chapter 4, and Cahill 2019).

Snider (2018) provides a helpful distillation of criteria proposed by various authors, which I in turn have summarized in table 3. Using these criteria, the examples given in this chapter so far can be re-examined to determine their status. For example, *rightward bounded L shift* in the Dana associative construction, as seen in (12), meets criteria (c) for being a lexical process, since it refers to the morpheme boundary of the associative construction. Likewise, *rightward bounded H spread* in Shambaa (28) must be a lexical process, since it has genuine lexical exceptions. In contrast, *rightward unbounded H spread* in N. Sotho (8) meets criteria (b) for being a postlexical process by applying across word boundaries, specifically in (8c). Also *leftward bounded H spread* in Kyak, which causes the assimilation of L to M as shown in (21), meets criteria (a) for being a post-lexical process,

since the output M is in complementary distribution with L, and therefore not contrastive.

Table 3: Determining lexical and postlexical tonal processes, adapted from Snider (2018: 95).

Lexical <i>a process must be lexical if...</i>	Postlexical <i>a process must be postlexical if...</i>
a. There are genuine exceptions (i.e. single words or morphemes, classes of words or morphemes).	a. The output is non-contrastive.
b. The process lacks phonetic motivation.	b. The process occurs across a word boundary.
c. It is necessary to refer to a morpheme boundary.	c. Only occurs at the beginning or end of a phrase.
d. Linguistically naïve speakers are aware of the effect.	d. Linguistically naïve speakers are unaware of the effect.
	e. The output is gradient.

4.3 Historical lexical movement

One final phenomenon to address in relation to tonal movement is *historical movement*, where spread or shift occurred earlier in a language’s development but is no longer productive. Its traces remain in the phonology, appearing as lexical exceptions to general tone patterns, or unexpected patterns given data from related languages, or some other way. An example of this has already been given from Shambaa (28), but perhaps the most famous case comes from Gikuyu ([kik] a Narrow Bantu language of Kenya, a.k.a. Kikuyu). Consider the data in table 4, which compares the Proto-Bantu forms of various different morphemes with four closely related languages–Tharaka [thk], Kamba [kam] (a.k.a. Kikamba), Mwimbi [mws] (a.k.a. Mwimbi-Muthambi), and finally Gikuyu.³⁰

These data reveal that tone patterns in the three other languages faithfully correspond to the reconstructed tone of Proto-Bantu, but that in Gikuyu all tones in the pattern have shifted one mora to the right. For morphemes with a proto H

³⁰I have adapted these data to use the contemporary standard notation of extra-high (ḙ) and extra-low (ḙ), except for the two cases of falling tones (ḙ) which should be read as extra-high to extra-low falling.

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Table 4: Historical tone shift in Gikuyu (adapted from Clements & Ford 1979: 187).

gloss	Proto-Bantu	Tharaka	Kamba	Mwimbi	Gikuyu
‘-poss’	*-yé ^{né}	-é ^{né}	-é ^{né}	-é ^{né}	-è ^{né}
‘big’	*-né ^{nè}	-né ^{nè}	-né ^{nê}	-né ^{ně}	-nè ^{né}
‘tree’	*-tí	mò-té	mò-té	mò-té	mò-tě
‘bushland’	*-cà ^{ká}	kè-ðà ^{ká}	kè-ðè ^{ká}	kè-ðà ^{ká}	yè-ðà ^{kă}
‘bamboo’	*-dà ^{ŋgí}	mò-rà ^{ŋgí}	mò-à ^{ŋgí}	mò-là ^{ŋgí}	mò-rà ^{ŋgĩ}
‘charcoal’	*-ká ^{dà}	è-kà ^{rà}	è-kă ^â	è-kà ^{lă}	ì-kà ^{râ}

pattern, such as the possessive suffix /-^{nyé}/, the result in Gikuyu is LH /è^{né}/, with the H shifting right and the first TBU receiving a default L. For morphemes with a HL pattern like /-^{nenè}/, the H shifts right and the final L gets “bumped” off the morpheme entirely, becoming a floating L and causing non-automatic downstep of a following word. The first TBU again receives a default L. With the nouns, Clements & Ford (1979) analysed that all tones, including those of the noun class prefixes (which are all L in these data), first shift one TBU to the right, then spread left if possible. Thus for a noun like /mò-tě/, the L of the prefix first shifted right to the noun root then spread left to the prefix, whereas the H of the root shifted to nowhere and then associated left again to the root to form a rising contour. In a noun like /ì-kà^{râ}/ with a HL root pattern, however, the final L shifted to nowhere but was disallowed from associating left again (likely due to a constraint against falling tones), and instead floated, causing non-automatic downstep. These latter two examples are illustrated in figure 24 .

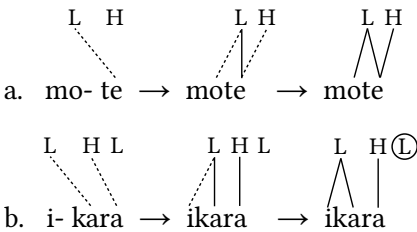


Figure 24: Historical tone shift in Gikuyu nouns.

The truly incredible thing about this shift is that it does not merely describe a few irregular morphemes in the language, but is rather representative of hundreds of morphemes (Clements & Ford 1979: 188), evidencing that it was once a

broadly productive pattern.³¹

5 Conclusion

In this chapter I have offered a typology of tone movement using the parameters of the *direction* of movement (rightward vs. leftward), *target* (bounded vs. unbounded), *active tone* (H vs. L, or otherwise), and whether the tone retains its association with the TBU of origin (spread vs. shift). Based on a survey of 100 African languages from 28 language families, it was shown that the resulting types are not equally attested (table 2), the most common being *rightward bounded H spread*, and the least common being three unattested types: *leftward unbounded H shift*, and both *leftward* and *rightward unbounded L shift*. In §2 examples from various languages were used to illustrate each attested type. In §3 I demonstrated the various phenomena that can occur as a result of tones moving, such as merging, insertion, deletion, displacement, contour formation, and blocked movement. Finally in §4 I showed how in a given language one might encounter multiple different kinds of movement, perhaps occurring at different stages in the phonological derivation (depending on the theoretical framework being employed); one might even find evidence of historical movement which is no longer productive.

³¹Clements & Ford (1979) in fact note that the process was still active in nouns until the recent past at the time of writing, evidenced by the fact that English borrowed terms displayed the same shift when compared to sister languages, such as /òβísi/ 'office' in Tharaka being /òβísi/ in Gikuyu.

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Abbreviations

↑	raised	LBSp	leftward bounded spread
↓	lowered	LOC	locative
1	first person	LUSp	leftward unbounded shift
3	third person	LUSp	leftward unbounded spread
APPL	applicative	M	mid tone
ASC	associative	N ₁	noun 1
BEN	benefactive	N ₂	noun 2
BT	boundary tone	NEG	negative
c1	class 1	OBJ	object
c1/2	class 1 and 2	PFV	perfective
c2	class 2	PL	plural
c5	class 5	PP	pre-prefix
c5/6	class 5 and 6	PROG	progressive
CAUS	causative	PST	past
COP	copula	PST1	near past
CP	class prefix	PST3	distant past
DEF	definite	R	rising tone
F	falling tone	RBSp	rightward bounded shift
FUT	future	RBSp	rightward bounded spread
FV	final vowel	RTT	Register Tier Theory
H	high tone	RUSp	rightward unbounded shift
h	high register	RUSp	rightward unbounded spread
IMP	imperative	RECIP	reciprocal
INF	infinitive	SBJ	subject
INPOSS	inalienable possession	SBJV	subjunctive
INTS	intensifier	SFX	undefined suffix
IPFV	imperfective	SG	singular
L	low tone	SP	specific/species
l	low register	TA	tense aspect
LBSp	leftward bounded shift	TBU	tone bearing unit

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Chapter 3

Grammatical tone in African languages

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In this chapter, we examine grammatical tone data from approximately thirty different African languages from all over the continent, organized for convenience around the four Greenbergian phyla Niger-Congo, Nilo-Saharan, Afro-Asiatic, and Khoisan. We start by establishing the key differences between lexical and grammatical tone, and then show that any inflectional or derivational property which can be expressed by segmental morphemes can also be expressed by tone. One important patterning is that grammatical tone on its own might signal the grammatical property being conveyed, while in other cases it might be one of several co-exponents of the grammatical expression. We also lay out the distinct ways grammatical tone can be implemented, e.g. addition, subtraction, replacement, shifting, exchanging, spreading, among others. In highlighting the myriad grammatical properties that grammatical tone is used to express, we emphasize that accurate documentation and description is absolutely indispensable for a full understanding of most African languages.

1 Introduction

This chapter will define, describe and illustrate grammatical tone in a number of African languages. Admittedly, this is a topic that strikes fear in the heart of many researchers. And not without cause. Hyman (2011) noted that “since some tonal phenomena have no segmental or stress analogues, anyone who is interested in the outer limits of what is possible in phonology would thus be well-served

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to understand how tone systems work”, and grammatical tone is exemplary of such ‘outer limits’. Along similar lines, Kisseberth & Odden (2003: 70) in their discussion on Bantu remarked that some of its tone systems reveal “patterns of extraordinary complexity”, of which grammatical tone is part-and-parcel. In this paper we hope to make at least some progress in demystifying grammatical tone, by carefully defining what it is and providing enough examples so that current African language researchers become more confident in identifying and describing such patterns in the languages they work on.

In this first section, we discuss the difference between lexical and grammatical tone. We also describe the kinds of things that grammatical tone is used for as well as the range of ways in which grammatical tone can be employed. Grammatical tone will then be illustrated in various African languages, organized based on Greenbergian language phylum: Niger-Congo (Section 2), Nilo-Saharan (Section 3), Afro-Asiatic (Section 4), and finally Khoisan (Section 5).

We note here that in a paper of this limited length, it is obviously impossible to document every claim of a grammatical tone operating in some African language. We have therefore endeavored to include the most interesting and instructive cases which illustrate the effects of grammatical tone. In some language families there are relatively few reliable descriptions, while in others there is an abundance of them. In the latter groups, while ultimately choosing which specific languages to highlight is by definition subjective, we’re confident that the range and depth of the examples presented will accomplish the goal of familiarizing the reader with the many ways in which grammatical tone is used in African languages.

Throughout this chapter, we shall largely employ diacritics on the vowels to represent vowels, following the International Phonetic Alphabet conventions (i.e. à = super-low/sL, à = low/L, ā = mid/M, á = high/H, and ǎ = super-high/sH, and various contours formed from these). We note overtly when we deviate from this transcription practice. In cases where we provide no tone diacritic, this should be interpreted as a tonally unmarked syllable (i.e. toneless), which gets its phonetic pitch value by default (usually, surfacing as low).

1.1 Lexical vs. grammatical tone

We can separate tonal function into two broad categories: lexical versus grammatical. Lexical tone is part of the underlying form of a morpheme, on par with nasality, vowel height, sonority and other contrastive phonological features. This is most easily illustrated with minimal pairs, such as the infinitival forms in the

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Bantu language Setswana (ISO 639-3 code [tsn] – Botswana), as provided in their phonetic forms in Table 1.

Table 1: Setswana tonal minimal pairs on verbs (Jones et al. 2018)

Low-toned roots		High-toned roots	
[gò-dùm-à]	‘to roar’	[gò-dúm-à]	‘to spray with insecticide’
[gò-sèlw-à]	‘to pick up’	[gò-sélw-à]	‘to oversleep’
[gò-bàk-à]	‘to praise in song’	[gò-bák-à]	‘to bake bread’

Each member of the three pairs of forms above are segmentally identical – the only thing that distinguishes them is tone. The infinitival prefix *gò-* always surfaces as low, as does the Bantu final vowel *-à*. The tonal difference is situated solely within the root. Given the fact that the roots are segmentally identical, it is impossible to predict the tone on the basis of any phonological aspect of the root, and we therefore may conclude that tone is contrastive on verb roots in this language. Looking deeper at the language, one must decide how to interpret the phonological status of this two-way contrast, i.e. as H vs. L (e.g. /sélw-/ ‘oversleep’ vs. /sèlw-/ ‘pick up’), as H vs. toneless ($\emptyset$) (e.g. /sélw-/ vs. /selw-/), or perhaps as $\emptyset$ vs. L (e.g. /selw-/ vs. /sèlw-/). See Rolle et al. (2026 [this volume]) for discussion on positing toneless syllables.

Moreover, if one has only the data in Table 1, it might not be clear whether the High tone in Setswana should be pre-associated to its vowel or merely be sponsored by it and come in ‘floating’. We maintain that this is a separate analytic issue. What is clear is that a High tone must be part of the root meaning ‘oversleep’, and thus it is lexical. Depending on the tonal patterns of these forms in their simple and derived instantiations, the lexical High might be argued to be pre-linked or floating. See Rolle et al. (2026 [this volume]) for an overview of ‘floating tones’.

The other thing to emphasize here is that lexical tones can potentially be present in any type of root, as well as in affixes. Lexical tone is a property not just of verbs (as in Table 1), but can be a property of any part of speech, including nouns, adjectives, adverbs, pronouns, clitics, prefixes, suffixes, and so on. A comparable example of tonal minimal pairs in Setswana nouns is in Table 2 (note the noun class prefix before the noun root in the latter two rows).

That lexical tone is not limited to roots is demonstrated below, with a three-way minimal pair of prepositions from Aghem ([agq] – Grassfields: Cameroon) in Table 3 and a two-way minimal pair of derivational suffixes in Herero ([her] –

Table 2: Setswana tonal minimal pairs on nouns (Jones et al. 2018)

Low-toned roots		High-toned roots	
[thàkà]	‘pupil (of eye)’	[thákà]	‘friend, peer’
[mà-fùlò]	‘pastures’	[mà-fúlò]	‘foam, froth’
[mà-bòkò]	‘praise poems’	[mà-bòkó]	‘brains’

Bantu: Namibia) in Table 4. (Note the presence of a downstep [†] in Table 3 which indicates the lowering of a high tone’s register, but not as low as a true Low tone.)

Table 3: Aghem prepositions (Hyman 2010)

High-toned locative	á ‘in’	á [†] ndúghó	‘in the house’
Falling-toned reciprocal	â ‘to’	â bvá [†] tó	‘to the dogs’
Low-toned comitative	à ‘with’	à bvá [†] tó	‘with the dogs’

Table 4: Herero suffixes (Hyman 2018, citing Wilhelm Möhlig, p.c.)

Toneless stative suffix	-ak	zúv-ak-a	‘be heard’
High-toned neutro-passive suffix	-ák	zúv-ák-a	‘be hearable’

In these tables, the lexical tone is part of the underlying representation of an individual morpheme and appears on that morpheme. It does not appear in all comparable parts of speech (e.g. all nouns, or all suffixes), and the tone is not extractable and does not mean anything on its own (i.e. in Aghem, Low tone does not combine with other prepositions with a meaning ‘comitative’). This contrasts with the subject of this chapter, namely grammatical tone. Here we adopt a definition of grammatical tone as a productive tone alternation triggered by a particular grammatical context (rather than purely a phonological context), which functions to express the meaning of that grammatical context.

Let us briefly consider the parts of this definition. First, the alternation must be productive – that is, it cannot be exhibited by only a few morphemes, but rather targets an unbounded class. If a grammatical tone expressing Remote Future affects one verb stem, it should affect all of them (all else being equal). Second, the tone alternation cannot be due to a language-wide rule which applies whenever a certain phonological contexts is met, e.g. a predictable phonological rule

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inserting a H tone between two any two L tones, regardless of morphological context. Such a process is more akin to segmental epenthesis rule, and likewise grammatical tone is more akin to segmental affixation. Third, grammatical tone signals meaning – that is, it is part of the obligatory morphological expression of a linguistic category. One can forego grammatical tone as much as one can forego deriving an agentive noun or inflecting a finite verb: almost never, and only under highly specific circumstances.

1.2 Types of grammatical tone

How exactly is grammatical tone added to or implemented on the base when it is found? The first important point to make is that very often it’s not just the presence of a grammatical tone (or sequence of tones) on its own which expresses some grammatical meaning. In situations where that is the case, we will refer to that as ‘autonomous grammatical tone’ (also referred to in the literature as ‘tonal morphs’ or ‘morphemes’). But often grammatical tone is one of several co-exponents of some grammatical expression, and here we refer to it as ‘non-autonomous grammatical tone’. For instance, some languages might express the Potential mood by a combination of a segmental verbal prefix and a grammatical tone suffix, others by tone alone.

A partial taxonomy of grammatical tone patterns includes several different types, such as tone addition, subtraction, replacement, shifting, exchanging, and spreading. We provide a very brief description of these here, as they will be illustrated more fully in Sections 2-5 to come (cf. also Smolders (2026 [this volume])).

Tone Addition involves the insertion of a grammatical tone on a specific Tone Bearing Unit (TBU). This is the case in Makonde ([kde] – Tanzania) in Table 5, where various inflectional contexts add a grammatical tone pattern onto the penultimate TBU, either a Low ① in the hypothetical (HYP), a High ② in the consecutive (CONS), a ①② in the future (FUT), or a ①②① in the remote past (RPST).

Table 5: Makonde additive grammatical tone (Manus 2014: 267–268)

Low-toned roots		High-toned roots		
HYP	káléká ngu-①-takatukil-e	→	... ngu-takatuk[ìì]le	‘if I stood up’
CONS	ni-ka②-takatukil-a	→	ni-ka-takatuk[íí]la	‘and I stood up’
FUT	ni-nda①②-takatukil-a	→	ni-nda-takatuk[íì]la	‘I will stand up’
RPST	ní-ndi①②①-takatukil-a	→	ní-ndí-takatuk[ìì]la	‘I stood up’

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Note that in Table 5, we denote the grammatical tone in a circle, a convention which we will continue this point forward. The circled tone is generally reserved for ‘floating tones’ (introduced above), which may be intuitive for some grammatical tone patterns but much less so for others. We use the circle convention for convenience in this paper, and acknowledge that other formalizations/representations are possible for many of the grammatical tone patterns we highlight below.

Moving on, when some grammatical expression involves the deletion of some lexical tone this is a case of Tone Subtraction. For instance, the addition of some segmental suffix might delete all of the tones within the root to which it attaches, leaving other productive tonal processes to value the TBUs which lost their tonal specification in this process (such as late default fill-in rules). Consider the Ijoid language Nkoroo ([*nkx*] – Nigeria). In Noun-Noun compounds, the underlying tones of the second noun (the head noun) delete, as represented below in the intermediate forms in Table 6. How the tones on the second noun ultimately surface depends on the first noun’s underlying tones. If the first noun ends in H then the second noun surfaces [HL], but if the first noun ends in L then the second noun surfaces as [LL]. This can straightforwardly be understood as tone spreading to the toneless second noun, plus default low on all non-initial vowels.

Table 6: Nkoroo subtractive grammatical tone in Noun-Noun compounds (Akinlabi et al. 2003)

Noun 1		Noun 2		Intermediate		Surface	
bíó	‘leg’	fúró	‘belly’	→	bíó	furo	→ [bíó fúró] ‘thigh’
ánáná	‘sheep’	tòkú	‘child’	→	ánáná	tòku	→ [ánáná tòkù] ‘lamb’
ḃòḃkò	‘chicken’	tòkú	‘child’	→	ḃòḃkò	tòku	→ [ḃòḃkò tòkù] ‘chick’
wàrí	‘house’	tòkú	‘child’	→	wàrí	tòku	→ [wàrí tòkù] ‘servant’

Relatedly, Tonal Replacement occurs when existing lexical tones are not simply deleted, but systematically replaced by some grammatical melody. Unlike in Nkoroo, in some languages where underlying tones are deleted, a fixed grammatical tone provides the replacement, regardless of the underlying tone of triggering environment. In a related Ijoid language Kalabari ([*ijn*] – Nigeria), the tones of

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the noun are deleted and replaced by a uniform ①② pattern in the context of a proximate demonstrative ‘this’, shown in Table 7.

Table 7: Kalabari replacive ①② tone (Harry & Hyman 2014)

HH	námá	‘animal’	→	mí námá	‘this animal’
LL	pùlò	‘oil’	→	mí pùló	‘this oil’
HL	bélè	‘light’	→	mí bèlé	‘this light’
LH	gàrí	‘garri flour’	→	mí gàrí	‘this garri flour’
H’H	ḃá’rá	‘hand’	→	mí ḃàrá	‘this hand’

Next, Tone Shifting involves the shifting of an underlying (lexical) tone to a different TBU within the same morpheme. For instance, a particular suffix might induce the shift of a root-initial tone to some other TBU. Crucially, it would need to be demonstrated that this shift was not a general rule (which would be triggered by any suffix) nor can it be phonologically predictable (e.g. occurring when the tone was followed by some number of TBUs). Tone Shifting is found in the Bantu language Tachoni ([Its] – Kenya), where in the imperative, toneless stems remain toneless but stems with initial high tone shift this high to the penult (which sometimes appears additionally one TBU to the left). This is shown in Table 8.

Table 8: Tachoni shifting tone in imperative (Odden 2009)

Toneless stems			H-toned stems		
chiingila	→	[chiingila] ‘carry for!’	káraanga	→	[karáánga] ‘fry!’
beechakala	→	[beechakala] ‘belch!’	bótooxanila	→	[botooxánila] ‘go around for!’

The last kind of grammatical tone and related to the last one is Exchanging Tone. One such case would be a stem with an underlying H losing it (like subtractive tone) but a stem without any underlying tone gaining a H (like additive tone). Exchanging tone can also be seen in Tachoni, in its negative imperative. Simplifying a bit, the data in Table show that toneless stems are assigned a H which appears after the prefix *xa-*, whereas H-toned stems surface toneless with neither underlying tone nor any grammatical tone (what Odden calls ‘mutual annihilation’).

Table 9: Tachoni exchanging tone in negative imperative

Toneless stems		H-toned stems	
bala	→ [o-xa-bála] ‘don’t count!’	lúma	→ [o-xa-luma] ‘don’t bite!’
beechakala	→ [o-xa-bééchakala] ‘don’t belch!’	káraanga	→ [o-xa-karaanga] ‘don’t fry!’

Other grammatical tone patterns exist as well, such as grammatically-induced spreading operations (where no general spreading rule would account for the pattern), or other more phonologically natural processes (e.g. assimilation, dissimilation, raising, lowering, plateauing, etc.). We point out, however, that natural tone rules which have highly specific prosodic or (morpho)syntactic conditioning are frequently excluded from grammatical tone discussion. For instance, if a language has a rule which deletes a High tone before another High tone, but only within a stem, or alternatively within a phonological phrase. We do not classify such cases as grammatical tone, as the pattern resulting from this change signals no change in meaning.

1.3 What is grammatical tone used for?

Given the definition and range of grammatical tone patterns in the previous subsection, what can grammatical tone be used for? One striking way to respond to this would be: any inflectional or derivational process which a segmental morpheme can express, tone too can express! Perhaps Hyman (2011) said this best: “I conclude that tone can do everything that segmental and metrical phonology can do, but that the reverse is not true”. This appears to hold, and no strict gaps are known where an inflectional or derivational process can only be expressed by segments and never tone. However, it may very well be the case that grammatical tone is used to express some inflectional and derivational properties more than others, e.g. verbs commonly inflecting for tense/aspect/mood via grammatical tone, or nouns commonly marked by grammatical tone in compounds/possessives/associative constructions. At the same time, we also find grammatical tone commonly used syntactically, for example, in syntactic agreement between a head and modifier or complement. In Sections 2-5 below, we will describe specific cases of grammatical tone in a number of African languages which illustrate how it is used.

2 Niger-Congo

The Niger-Congo phylum is one of the largest in the world, and even if one excludes its more controversial branches (e.g. Ijoid, Mande, Kordofanian, among a few others), it still constitutes one the largest families in the world. Grammatical tone is rampant in Niger-Congo, and it is hard to find languages within which grammatical tone does not play a major role. We shall go through several major sub-branches and discuss common grammatical tone patterns (though for considerations of space, we simply cannot explore each and every sub-branch, and leave out discussion of Adamawa, Ubangi, Gbaya, Senoufo, Dogon, and various ‘Atlantic’ families).

2.1 Benue-Congo

Perhaps the most famous sub-branch of Niger-Congo is Benue-Congo, no doubt due to the presence of its expansive sub-sub-branch Bantu. We have seen several cases of Bantu already, such as Makonde in Table 5 above where a particular grammatical tone pattern is part of the expression of verb inflection. Such patterns are omnipresent in Bantu, and can form some of the most complex tonal patterns known to linguistics.

Consider the Bantu language Kuria ([kuj] – Kenya), which has become well-known for its ‘counting effects’ in grammatical tone assignment (Paster 2019). Table 10 provides a mini-paradigm involving the prefix *to-* 1p ‘we’ with the verb root *kirigiit-* ‘scrub’ in various inflected contexts. Depending on the grammatical context, a grammatical ④ tone is systematically assigned to the first, second, third, or fourth mora of the stem (denoted in curly brackets), and even to the first and fourth simultaneously. Note that the forms show the effects of general tone spreading rules. Where the grammatical tone is assigned originally is the underlined TBU of the stem.

Table 10: Kuria counting effects (Mwita 2008)

④ ₁	Habitual past	n-to-ogá-{ <u>kirigiit</u> -íre}	‘we used to scrub (then)’
④ ₂	Past prog. (foc.)	n-to-oga-{ <u>kirigiit</u> -íre}	‘(indeed) we have scrubbed’
④ ₃	Remote fut. (foc.)	n-to-re-{ <u>kirigiit</u> -a}	‘we will scrub (then)’
④ ₄	Inceptive	to-ra-{ <u>kirigiit</u> -a}	‘we are about to scrub’
④ ₁₊₄	Narrative past	to-gá-{ <u>kirigiit</u> -a}	‘(and) we scrubbed’

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If we assume that grammatical tone generation and docking both take place in the morphology, we might expect that these processes should be confined to the morphological word. While this is true for many languages, there is a variety of Kuria which convincingly shows that grammatical tone assignment must be done at the level of the phrase instead of the word. Marlo et al. (2014) show that in the Buguumbe dialect of Kuria, speakers realize some verbal grammatical High tones within the *following* word. For example, the Inceptive targets the fourth mora of the stem, and when the stem has four morae then the grammatical ③ docks onto that stem-final fourth mora (as in the first row of Table 11). But when the stem is shorter than four morae, the ③ docks within the following word, on whatever is the fourth mora from the left edge of the verb stem.

Table 11: Kuria inceptive assigns grammatical tone into next word

to-ra-{karaang-á éyétóókè	‘we are about to fry a banana’
to-ra-{sukur-a éyétóókè	‘we are about to rub a banana’
to-ra-{rom-a eyétóókè	‘we are about to bite a banana’
to-ra-{ry-a eyetóókè	‘we are about to eat a banana’

Let us take this opportunity to situate Kuria in a broader Bantu context. While grammatical tones in Kuria target one or more of the first four TBUs of verb stem, additional targets are found in other Bantu languages. Example (1) summarizes the docking targets found in the more than two dozen languages as described in Odden & Bickmore (2014), a special volume of *Africana Linguistica*. (All references within these examples are from that volume – see that source for referential codes and geographic locations.)

- (1) Position in stem targeted by grammatical tones across Bantu
- a. 1st: Jita (Downing 2014), Kuria (Marlo et al. 2014), Nyala-West (Ebarb et al. 2014), Orungu (Maniacky & Ambouroue 2014), Simakonde (Manus 2014), Wanga (Ebarb et al. 2014)
 - b. 2nd: Cilungu (Bickmore 2014), Copperbelt Bemba (Bickmore 2014), Emakhuwa (Kisseberth & Guérois 2014), Fipa (Bickmore 2014), Idhakho (Ebarb et al. 2014), Kamba (Roberts-Kohn 2014), Kuria (Marlo et al. 2014), Namwanga (Bickmore 2014), Totela (Crane 2014)
 - c. 3rd: Emakhuwa (Kisseberth & Guérois 2014), Kuria (Marlo et al. 2014)
 - d. 4th: Kuria (Marlo et al. 2014)
 - e. Antepenult: Bena (Morrison 2014)

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- f. **Penult:** Bena (Morrison 2014), Emakhuwa (Kisseberth & Guérois 2014), Jita (Downing 2014), Kamba (Roberts-Kohno 2014), Simakonde (Manus 2014)
- g. **Final:** Basaa (Makasso 2014), Bakweri (Marlo & Odden 2014), Cilungu (Bickmore 2014), Copperbelt Bemba (Bickmore 2014), Emakhuwa, Fipa (Bickmore 2014), Jita (Downing 2014), Kamba (Roberts-Kohno 2014), Kwanyama (Halme-Berneking 2014), Mbadja (Halme-Berneking 2014), Namwanga (Bickmore 2014), Nyala-West (Ebarb et al. 2014), Orungu (Maniacky & Ambouroue 2014), Totela (Crane 2014), Wanga (Ebarb et al. 2014)

In addition to those listed above, Nyala-West has a pattern which targets the first mora of the second stem syllable, and Wanga and Kifuliiru have patterns which target the leftmost free TBU.

That same volume details the range of tones and combinations of tone within each language (as analyzed by the authors) which are used as grammatical tone melodies. Example (2) summarizes this for the languages examined above. (Note that sL= super-low and sH= super-high).

(2) Grammatical tone melodies across Bantu

- a. **Just H:** Basaa (Makasso 2014), Bena (Morrison 2014), Emakhuwa (Kisseberth & Guérois 2014), Ecuwabo (Kisseberth & Guérois 2014), Jita (Downing 2014), Kuria (Marlo et al. 2014), Cilungu (Bickmore 2014), Idhako (Ebarb et al. 2014), Nyala (Ebarb et al. 2014), Wanga (Ebarb et al. 2014), Makwe (Devos 2014), Mbadja (Halme-Berneking 2014), Shona (Odden 2014), Totela (Crane 2014)
- b. **H, L:** Orungu (Maniacky & Ambouroue 2014)
- c. **H, LH:** Dawida (Philippson 2014)
- d. **H, L, LH:** Dibole (Leitch 2014)
- e. **H, L, LH, HL:** Bakweri (Marlo & Odden 2014)
- f. **H, L, sH, sL, H-sL:** Kamba (Roberts-Kohno 2014)
- g. **H, L, LH, HL, HH, LL:** Kifuliiru (Van Otterloo 2014)
- h. **LH, LHL:** Lulamogi (Hyman 2014)
- i. **H, L, LH, LHL:** Simakonde (Manus 2014)

Bantu languages thus differ in terms of their inventory of grammatical tones and where these grammatical tones associate (cf. Grimm 2026 [this volume]).

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One of the languages found above, Orungu ([mye] – Gabon), shows that languages can also differ as to the behavior of grammatical tone akin to the divisions we made in Section 1.2 (e.g. additive, subtractive, replacive, *etc.*). Most grammatical ⑩ tones affect their local environment, but one sponsored by the negative prefix à- NEG is different: it is a replacive pattern which spreads to all TBUs to its right. A remarkable aspect of this pattern is that its scope is unbounded: the grammatical tone replaces the underlying tone even of whole words to its right, until a clause boundary. A data set is provided in Example (3).

(3) Orungu replacive grammatical tone (Ambouroue 2007, Maniacky & Ambouroue 2014)

- a. [à-yól-á]
à⑩-yól-a
NEG-buy-FV
'don't buy!'
- b. [à-yómb-íz-ó-ní]
à⑩-yómb-íz-ò-nì
NEG-chase-CAUS-PASS-PL
'don't be chased away!'
- c. [à-ríy' áwán' áŋká yó 'ŋkóló]
à⑩-ríy-à àwáná áŋká yó ìŋkòlò
NEG-leave-FV children alone P tonight
'do not leave the children alone tonight'

Bantu is far from the only sub-family in Benue-Congo which exhibits grammatical tone. In non-Bantu languages westward in Nigeria, such tonal effects are also found but often without the characteristic 'flare' of Bantu verb inflection (cf. Blench 2026a [this volume]). In Yoruba ([yor]), a sizable literature has emerged on the function of the so-called subject-marking 'High Tone Syllable' (HTS) which appears between the subject and predicate in many (if not most) inflectional environments (Awobuluyi 1975, Bisang & Sonaiya 1999, *inter alia*). The HTS is realized either as a copy of the preceding vowel (i.e. the last vowel of the subject) or it replaces the final tone of the subject. For example, /bābā/ 'father' becomes either [bābā á] or [bābá] in the first row of Table 12.

Moreover, in the fellow Nigerian language Igbo ([ibo]) a low tone appears at the subject-predicate boundary to indicate a simple polar question, i.e. ⑩ Q. This grammatical tone associates to the prefix on the verb but then also spreads backwards onto any contiguous string of high tones on the subject, shown in example (4).

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Table 12: Yoruba grammatical ⑤ at subject-predicate boundary (Awobuluyi 1975)

bābā ⑤ lẹ	→	[bābā á lẹ] ~ [bābá lẹ]	‘Father went’
wọ̀n ⑤ lẹ	→	[wọ̀n ọ̀n lẹ] ~ [wọ̀n lẹ]	‘they went’
ògèdè ⑤ wọ̀n	→	[ògèdè é wọ̀n] ~ [ògèdẹ̀ wọ̀n]	‘bananas are expensive’

(4) Igbo polar question ① subject-predicate boundary (Clark 1990)

- a. [hà àlá'ɣí]
há ① á-'lá'ɣí
they Q PRF-leave\NEG
‘didn’t they leave?’
- b. [ùwè ọ̀chà ọ̀-dị n-ébè à]
ùwé ọ̀chá ① ọ̀-dị n-ébè à
shirt clean Q PRF-is in-place this
‘Is there a clean shirt here?’

Grammatical tone patterns could also easily be shown for the other Benue-Congo families (e.g. Kainji, Edoid, Platoid, Nupoid, Delta Cross, Jukunoid, or non-Bantu Bantoid such as Grassfields) (cf. [Blench 2026b \[this volume\]](#)).

2.2 Kwa

Another Niger-Congo family is Kwa, containing languages related to Akan/Twi and Ewe, among others. In the Gã language ([gaa] – Ghana), grammatical tones appear after the subject marker at the left edge of the verb. The Habitual sponsors a floating ① which docks to the preceding subject marker if it is toneless (all but the first row of Table 13) but results in downstep ¹ if the subject marker is high-toned (the first row).

In contrast, inflection from the Subordinate grammatical ⑤ and the Simple Past grammatical ① are different: they always dock to the preceding subject marker and neutralize any underlying tone difference (the H vs. Ø distinction). These patterns are in Table 14. Observing the difference between the Habitual and the Simple Past specifically, this demonstrates that the value of the grammatical tone is not enough to predict behavior.

In another Kwa language, Saxwe ([sxw] – Benin), we find evidence of grammatical tones being prioritized for survival over tones with no meaning such as boundary tones. Clauses by default are realized with a meaningless L% boundary

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Table 13: Gã floating tone from habitual (Paster 2003)

Subject		Habitual ‘X sing(s)’		
1S	mí-	mí- [Ⓛ] -lá-ɔ	→	mí- [Ⓛ] -lá-à
2S	o-	o- [Ⓛ] -lá-ɔ	→	ò-lá-à
3S	e-	e- [Ⓛ] -lá-ɔ	→	è-lá-à
1P	wɔ-	wɔ- [Ⓛ] -lá-ɔ	→	wò-lá-à
2P	nyě-	nyě- [Ⓛ] -lá-ɔ	→	nyè-lá-à
3P	amě-	amě- [Ⓛ] -lá-ɔ	→	àmè-lá-à

Table 14: Gã neutralization of subject agreement by inner inflection

Subject		Subordinate ‘that X sing’			Simple past ‘X cultivated’		
1S	mǎ-/mí-	mǎ- [Ⓛ] -lá	→	mǎ-lá	mí- [Ⓛ] -dú	→	mǐ-dú
2S	o-	o- [Ⓛ] -lá	→	ó-lá	o- [Ⓛ] -dú	→	ò-dú
3S	e-	e- [Ⓛ] -lá	→	é-lá	e- [Ⓛ] -dú	→	è-dú
1P	wɔ-	wɔ- [Ⓛ] -lá	→	wó-lá	wɔ- [Ⓛ] -dú	→	wò-dú
2P	nyě-	nyě- [Ⓛ] -lá	→	nyé-lá	nyě- [Ⓛ] -dú	→	nyè-dú
3P	amě-	amě- [Ⓛ] -lá	→	àmé-lá	amě- [Ⓛ] -dú	→	àmè-dú

tone (BT) at the right edge of a clause (though a small number of contexts have a H% boundary tone, e.g. certain clauses with a syntactic gap). We exemplify the surface forms when a L% boundary tone is present in Example (5), with both final high-toned and mid-toned words. We see that only the mid-toned verb /sē/ changes, becoming a falling mid-low tone [sè] – high tones remain unaltered.

(5) Saxwe clause-final boundary tone L% (Beavon-Ham 2019)

- a. [ōxē sǐ]
 ōxē sǐ L%
 bird leave BT
 ‘a bird left’
- b. [ōxē sè]
 ōxē sē L%
 bird hear BT
 ‘a bird heard’

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In contrast, there also exists a clause-final grammatical ④ which is part of the expression of negation (along with the pre-verbal particle ǝ NEG). Unlike with the boundary tone, with negative grammatical tone both underlying mid and high are affected, i.e. /sǝ/ becomes [sǝ̃] (6a) just as /nǝ/ becomes [nǝ̃] (6b).

(6) Saxwe clause-final grammatical ④ expressing negation

- a. [kōfí ǝ̃ sǝ̃]
 kōfí ǝ̃ sǝ̃ ④
 Kofi NEG leave NEG
 ‘Kofi did not leave’
- b. [ōsǝ̃ ǝ̃ gbè nǝ̃]
 ōsǝ̃ ǝ̃ ④ ④ gbè nǝ̃ ④
 horse NEG IRR IPFV refuse IPFV NEG
 ‘the horse will not refuse’

2.3 Kru

To the west of the Kwa family are the Kru languages (sometimes not even included in Niger-Congo, e.g. in Glottolog – Hammarström et al. 2025). Exemplary of grammatical tone in Kru is Guébie ([gie] – Côte d’Ivoire) (cf. Russell & Sande 2026 [this volume]). Negation is expressed as a high grammatical tone ④ which appears between the subject and predicate (Guébie has four pitch heights which Sande transcribes numerically with 4, 3, 2, and 1, where 4 is the highest and 1 the lowest). Example (7) shows that the subject /ja³¹/ ‘coconuts’ with an underlying falling contour surfaces as [ja³¹⁴] with a dipping contour after the floating ④ docks in the negative.

(7) Guébie grammatical ④ marking negation on subjects (Sande 2022)

- a. [ja³¹ na⁴nɛ⁴]
 ja³¹ na⁴nɛ⁴
 coconuts be.good
 ‘coconuts are good’
- b. [ja³¹⁴ na⁴nɛ⁴]
 ja³¹ ④ na⁴nɛ⁴
 coconuts NEG be.good
 ‘coconuts are not good’

A complete set of patterns are provided in Table 15, which demonstrates the straightforward addition of the negative tone regardless of which tone the subject ends in. This set also demonstrates the absence of neutralization, as even

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words with underlying /4/ tone show lengthening in the context of negative ④ (e.g. /ɲu⁴/ ‘water’ surfaces as [ɲu⁴u⁴]).

Table 15: Guébie summary of grammatical ④ NEG (Sande 2022)

Underlying	Under negation
/...1/	→ ...14
/...2/	→ ...24
/...3/	→ ...34
/...4/	→ ...44
/...23/	→ ...234
/...24/	→ ...244
/...41/	→ ...414
/...42/	→ ...424
/...31/	→ ...314
/...32/	→ ...324

Even more intriguing is the marking in imperfective which shows a scalar shift. In imperfective contexts, the initial tone of the verb lowers one level in pitch, e.g. /li²ʔe³/ ‘dine’ (8a) surfaces as [li¹ʔe³] in imperfective contexts (8b). In the latter, we gloss this simply as a morpheme ① IPFV, standing for the scalar tone effect.

(8) Guébie imperfective scalar shift (Sande 2018)

- a. [ɔ³ li²ʔe³]
 ɔ³ li²ʔe³
 3S.HUM dine
 ‘(s)he dined’
- b. [ɔ³ li¹ʔe³]
 ɔ³ ① li²ʔe³
 3S.HUM IPFV dine
 ‘(s)he is dining’

The Guébie scalar shift shows a unique property, as well: if the verb is underlying /1/ and is thus already at the bottom of the pitch scale, it forces the preceding word to raise, i.e. the final tone of the subject shifts upward, e.g. 3SG $\epsilon^3 \rightarrow \epsilon^4$ in Table 16. Remarkably, if the final subject’s tone is already at the top of the scale /4/ then a derived super-high tone surfaces [5] which is otherwise not used in the language.

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Table 16: Guébie scalar shift affects subject with /1/ verbs (Sande 2018)

$\varepsilon^3 \text{ } \text{ɓɔ}^1$	→	$\varepsilon^3 \text{ } \text{ɓɔ}^1$	‘it withered’
$\varepsilon^3 \text{ } \text{Ⓢ} \text{ } \text{ɓɔ}^1$	→	$\varepsilon^4 \text{ } \text{ɓɔ}^1$	‘it withers’ (Imperfective)
$e^4 \text{ } \text{pa}^1$	→	$e^4 \text{ } \text{pa}^1$	‘I ran’
$e^4 \text{ } \text{Ⓢ} \text{ } \text{pa}^1$	→	$e^5 \text{ } \text{pa}^1$	‘I run’ (Imperfective)

2.4 Gur (Mabia)

To the general north of the Kwa and Kru area is the Gur family of Niger-Congo, where grammatical tone again is omnipresent (Angsongna et al. 2026). In one major Gur language, Dagbani ([dag] – Ghana), there exist the typical grammatical tone patterns found across Africa, such as marking noun compounds and verb inflection involve negation, tense, and aspect. Dagbani contrasts underlying low-toned roots (e.g. *pìì* ‘choose’) versus high-toned roots (e.g. *gbáá* ‘catch’), whose tones are altered in context. The Present Perfective replaces this distinction with a uniform Ⓢ(Ⓢ) pattern (Table 17), while the negative equivalent replaces it with just Ⓢ (note orthogonal downstepping and spreading rules). In contrast, other contexts do not neutralize underlying verb tone. They either preserve the tone pattern on the verb itself (e.g. the Relative Present Perfective in the third row in Table 17), or effects of the underlying tone are felt on the surrounding environment (e.g. downstep vs. spreading on the object in the last row with the Future). Such data also show that whether inflection is at the matrix clause level or within an embedded clause can affect grammatical tone assignment, found here in Dagbani but more robustly throughout Bantu.

In addition to sensitivities to tense, aspect, negation, and clause-type, the linguistic category mood can also dictate grammatical tone patterns. This is seen in another Gur language Turka ([tuz] – Burkina Faso), where a grammatical Ⓢ docks onto the subject to express the subjunctive (SjTV). In this language, the Ⓢ only docks to the right edge of the subject and cannot extend to other syllables or words past this position.

- (9) Turka subjunctive Ⓢ marking on subject (Monich 2020, citing Suggett 2005)

- a. [wú¹kú]
 wù¹ Ⓢ¹ kú¹
 3SG SjTV die\SjTV
 ‘he would die’ (Cf. indicative: [wù¹kù¹] ‘he dies’)

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Table 17: Dabgani tone inflection on verbs, with object *kòdú* ‘banana’
(Hyman & Olawsky 2004)

Inflection	Low-toned roots	High-toned roots
Present	ò pìi kó ¹ dú	ò gbàà kó ¹ dú
perfective	‘he has chosen a banana’	‘he has caught a banana’
Negative pres.	ò bá píi kòdú	ò bá gbáá kòdú
perfective	‘he hasn’t chosen a banana’	‘he hasn’t caught a banana’
Relative pres.	ṇùn pìi kòdú	ṇùn gbáá kòdú
perfective	‘...who has chosen a banana’	‘...who has caught a banana’
Future	ò-n píi kó ¹ dú	ò-n gbáá kòdú
	‘he will choose a banana’	‘he will catch a banana’

- b. [wù kúúgú ¹cá]
 wù kúúgú ① ¹cá
 3SG death SJTV difficult\SJTV
 ‘may his death be difficult’

2.5 Mande

The Mande family is only controversially included in Niger-Congo, but it exemplifies the same breadth of grammatical tone patterns as the core of the phylum. We can exemplify this with Mande languages located in different areas of the Mande spread zone: Vai ([vai] – Liberia), Soninke ([snk] – Mali), and Seenku ([sos] – Burkina Faso). For further overviews of Mande grammatical tone, see Green (2018), Konoshenko (2018, 2022), *inter alia* and Konoshenko (2026 [this volume]).

First, the Vai language of Liberia shows grammatical tone in both its nominal and verbal system (Welmers 1976). Nouns contrast for tone, such as the minimal triplet in Table 18 showing H vs. L vs. LH (rising tone). To mark the definite versions of these nouns, one adds an autonomous ①② grammatical tone sequence to the right edge (thus, /làá/ ‘clay pot’ becomes [lǎǎ] ‘the clay pot’ with two rising tones).

Moreover, on verbs one sees a replacive pattern in the imperative (IMP), where the underlying tone of the verb is replaced by a uniform low tone (10). One exception to this is the motion verb *táá* ‘go’ (10d) which maintains its underlying high tone in the imperative (it is ‘indomitable’ in the terms of Rolle 2018). Such

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Table 18: Vai grammatical tone marking definiteness on nouns
(Welmers 1976)

H	láá	‘plankfish’	→	HR	láǎ	‘the plankfish’
L	làà	‘paddle’	→	LR	làǎ	‘the paddle’
LH	làá	‘clay pot’	→	RR	làǎ	‘the clay pot’

patterns show that grammatical tone may be suspended on a case-by-case basis, sometimes word-by-word.

(10) Vai replacive grammatical tone marking imperative (Welmers 1976)

- a. [mú kùnè]
mú ① kúné
us IMP wake
‘wake us up!’
- b. [kǒŋ tiè]
kǒŋ ① tié
stick IMP cut
‘cut sticks!’
- c. [kǒŋě lèsì]
kǒŋ-ě ① lési
stick-DEF IMP catch
‘catch the stick!’
- d. [í táá] (Cf. *[í tàà])
í Ø táá
you IMP go
‘go!’

Moving northward, Soninke shows grammatical tone changes when multiple words are placed in a compound, where the first word is the modifier and the second element is the head (Creissels 2016), a common pattern in (Western) Mande. In such constructions, the first word loses all tone but its first toneme which then spreads to the end of that word (the second word – the head noun – is only minimally altered tonally). Underlying H-tone initial words of various shapes (e.g. H, HL, HHL, HLF) surface as all-high (the top four rows of Table 19), while L-initial words surface as all-low (the bottom five rows). One might view this as a grammatically-induced spreading operation, rather than the addition of any special (floating) tone (cf. Rolle et al. 2026 [this volume]).

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Table 19: Soninke grammatically-induced spreading operation of the first word in a compound (Creissels 2016)

Change	Underlying		Compound	Meaning
H→H	tááxú hô	→	tááxú-hô	‘seat’ (lit. sit-thing)
HL→H	yúgò séntààdê	→	yúgú-n-céntààdê	‘man’s comb’ (lit. man-comb)
HHL→H	líhíndì màsî	→	líhíndí-mási	‘sewing machine’ (lit. sew-machine)
HLF→H	bánbàrà-llènmê	→	bánbárá-llènmê	‘young Bambara’ (lit. Bambara-DIM)
L→L	sì hàtê	→	sii-hàtê	‘horse skin’
LH→L	sèhé-llénmà	→	sèhè-llénmà	‘interlocutor’ (lit. speak-DIM)
LF→L	kìidê táxàyê	→	kiiidì-táxàyê	‘Baobab condiment’
LHL→L	tùbáábù qálàsî	→	tùbààbù-qálàsî	‘European money’
LLF→L	sòònìnkê qánnè	→	sòòninkà-n-qánnè	‘Soninke language’

Finally, moving eastward into Burkina Faso, the Seenku language demonstrates a scalar shift akin to the Guébie one in (8), though much more reduced. To form plurals, in addition to suffixation, a noun’s super-low (sL) tones become low (L), high tones (H) become super-high (sH), and super-high tones remain as such (there are incidentally no underlying L-toned nouns in Seenku). This is demonstrated in Table 20 (as with the Guébie example, the circled ① stands for the tone process).

Table 20: Seenku scalar shift in plural formation (McPherson 2017)

Change	Example		Plural form	Meaning
sL→L	jũ-①-i	→	jùi	‘hills’
H→sH	bí-①-i	→	bǐ	‘goats’
sH→sH	sũ-①-i	→	súi	‘antelopes’

3 Nilo-Saharan

Next, let us examine another Greenbergian phylum, Nilo-Saharan. Although this has not been proven to be a genealogical unit *per se* (e.g. Glottolog does not treat it as such – Hammarström et al. 2025), its various families all show robust grammatical tone patterns. We shall examine some of its most recognizable families, namely Nilotic, Central Sudanic, and Nubian. For space reasons, we do not discuss Saharan, Songhay, Maban, Fur, Ik, Surmic, or various isolates (e.g. Kunama).

3.1 Nilotic

Nilotic languages are known for complex tonal alternations on nouns, typically understood as a reflex of their grammatical case system (cf. Blum 2026 [this volume]). Consider the Nilotic language Tugen ([tuy] – Kenya). There are at least 15 distinct alternation patterns when comparing absolutive case (the elsewhere form of the noun) to its nominative case form. Representative nouns are provided in Table 21.

Table 21: Tugen grammatical tone expressing case (Jerono 2013)

Absolutive	Tone		Nominative	Tone	Meaning
kibùkàndiì	L	→	kibùkándií	LH	‘guitar’
chèèp-ò	L	→	chéép-ò	HL	‘daughter of’
kwááriík	H	→	kwáàrìik	HL	‘tendon’
móíta	H	→	mòità	L	‘calf’
káályááng’ík	H	→	kààlyááng’ík	LH	‘flies’
mùrsiík	LH	→	mùrsiik	L	‘sour milk’
chèèmòòsíí	LH	→	chéémóósíí	H	‘ogre’
káámèè	HL	→	kààmèè	L	‘mother’
cháálwòòk	HL	→	cháálwóók	H	‘wrongdoing’
séèsèè	HL	→	séèsée	HLH	‘dog’
láàkwéé	HLH	→	láákwéé	H	‘child’
òinóósyék	HLH	→	òinòòsyèk	HL	‘rivers’
ùsíí	LHL	→	úsíí	H	‘thread’
bèèlyóóndé	LHLH	→	béèlyóóndé	H	‘elephant’
kíkóòmbéé	LHLH	→	kíkóòmbéé	HLH	‘cup’

Tone marking grammatical case is widespread within the Nilotic family of Northeast Africa, also including Maasai (Tucker & Mpaayei 1955), Datooga

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(Kießling 2007), Kipsigis (Toweett 1979, Kouneli & Nie 2021), and Jumjum (Andersen 2004). In this last language Jumjum ([jum] – Sudan), the underlying tone of nouns is replaced by ① in the so-called ‘antigenitive’ case (which marks the non-possessor noun in a possessive construction), illustrated in Table 22.

Table 22: Jumjum antigenitive case marking (Andersen 2004)

Absolutive	Tone		Antigenitive	Tone	Meaning
ḍéɛŋ	H	→	ḍèɛŋ	L	‘cow’
yîm	HL	→	yîm	L	‘giraffe’
kùun	L	→	kùun	L	‘thorn’
dîir-gá	HH	→	dîir-gà	LL	‘birds’
cìcàm	HL	→	cìcàm	LL	‘knife’
càw-ná	LH	→	càw-nà	LL	‘arrow’
mín-nèní	HLH	→	mìn-nèni	LLL	‘child’
bòròŋ-gó	LHH	→	bòròŋ-gò	LLL	‘clothes’

It is not just nouns but also verbs which can display complex tonal alternations. This is seen prominently in Dinka ([dib] – South Sudan). Here, grammatical tone may result in complex fusional patterns where derivational and inflectional meaning are melded together. In the Agar dialect (Andersen 1992), derivational categories expressing directionality (e.g. the Centripetal CP) and valency (e.g. the Benefactive BEN, the Antipassive AP) are interwoven with inflectional categories of finiteness, person, number, (non-)topicality, and voice (e.g. Passive PASS). Verbs contrast for tone underlyingly (H vs. L), and this can be preserved or replaced depending on the precise features being expressed. The tonal patterns of various inflected-derived cells are provided in Table 23, showing only the patterns on CVC roots. Surface patterns are always high (H), low (L), or falling (F) (there are other (supra)segmental on the root, not shown).

The grammatical tone patterns of Table 23 Dinka evidence two morphological ‘layers’, which following the Dinka tradition we call ‘In’ (for an ‘inner’ layer) and ‘Out’ (for an ‘outer’ layer). In the inner layer, the pattern is governed largely by derivational categories (at the top), as reflected by the uniform tone values in the columns. In contrast, in the outer layer the pattern is governed by inflectional categories (at the left), reflected by the uniform rows.

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Table 23: Dinka layering of inflection-derivation grammatical tone (Andersen 1992, Trommer 2011)

		H-toned CVC roots					L-toned CVC roots				
		Ø	BEN	CP	AP	BEN.AP	Ø	BEN	CP	AP	BEN.AP
In	FIN	L	H	L	F	F	L	F	L	F	F
	1SG/3SG	L	H	L	F	F	L	F	L	F	F
	PL	H	H	L	F	F	H	F	L	F	F
	NFIN	F	H	L	L	F	L	F	L	F	F
Out	NTOP.SBJ	H	H	H	H	H	H	H	H	H	H
	PASS.CT	F	F	F	F	F	F	F	F	F	F
	PASS	H	F	F	F	F	H	F	F	F	F
	2SG	L	L	H	L	L	L	L	H	L	L

3.2 Central Sudanic

Another Nilo-Saharan family with robust grammatical tone is Central Sudanic. In Ngiti ([niy] – The Democratic Republic of the Congo), verbs are inflected both for tense/aspect/mood as well as subject agreement. In Table 24, the intransitive verb *ikpè* ‘cough’ appears in various inflection contexts which systematically alter its tone (with or without additional affixation). The basic inflected form of the stem is provided in this table under ‘Basic’, which is the form of the verb without subject agreement morphology. Depending on the exact inflection, the stem bears various tonal patterns (viz. MH-L, HL-M, LL-L, or ML-M). This is the first inflectional layer.

To the right of these basic stems are forms which bear subject agreement prefixes, here with third singular *k-* 3SG ‘he/she’ which adds a grammatical ①. At this second layer, when grammatical tone is added it creates rising contour tones, maintaining the basic inflected stem’s tone when possible. (Note that this example places the subject agreement forms in the negative with an independent word *inzá* ~ *īnzá* NEG – these forms better illustrate the relevant tonal patterns. The translations provided are approximate based on the description in Kutsch Lojenga (1994).)

In the related Sara-Bongo-Bagirmi family, comparable inflectional tone shows a cross-linguistically rare pattern: sensitivity to the initial segment of the target. We can illustrate this with the Eastern Sara language Nar ([mwm] – Chad). Table 25 shows person/number affixes in an unmarked tense/aspect, translated

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Table 24: Ngiti layering of grammatical tone (Kutsch Lojenga 1994)

Inflection	Tone	Basic	With additional tone layer (NEG + <i>k</i> Ⓢ- 3SG)	
Past hab.	MH-L	ìkpè-nà	ìnzì k-ìkpè-nà	‘(s)he didn’t use to cough’
Past cond.	HL-M	ìkpè-nā	ìnzì k-ìkpè-nā	‘if (s)he hadn’t coughed’
Recent past	LL-L	ìkpè-nà	ìnzà k-ìkpè-nà	‘(s)he didn’t cough’
Future	ML-M	ìkpè-nā	ìnzì k-ìkpè-nā	‘(s)he won’t cough’

as simple present. With consonant-initial roots (e.g. C-initial /tógō/ ‘wash’), the tone of the root does not alter in these contexts. In contrast, vowel-initial roots show grammatical tone changes (e.g. V-initial /ígì/ ‘sell’, with the same underlying tones). The most telling forms are the third person, which are prefixless and surface with underlying HM with C-initial roots (i.e. [tógō] in this table), but as LL bearing grammatical tone expressing subject agreement with V-initial roots (i.e. [ígì]).

Table 25: Nar consonant-initial roots do not take grammatical tone but vowel-initial roots do (Keegan 2012)

	Tone	C-initial	Translation	Tone	V-initial	Translation
1S	M-HM	m̄-tógō	‘I wash’	HM	m-ígì	‘I sell’
2S	M-HM	ī-tógō	‘you wash’	HM	ígì	‘you sell’
3S	HM	tógō	‘(s)he washes’	LL	ìgì	‘(s)he sells’
1P	L-HM	j̄-tógō	‘we wash’	LM	j-ìgì	‘we sell’
2P	M-HM-M	ī-tógō-ī	‘you wash’	H-M	íg-ī	‘you sell’
3P	HM-M	tógō-n̄	‘they wash’	LL-M	ìgì-n̄	‘they sell’

Keegan shows that this distribution applies to monosyllabic roots as well (e.g. /ɔ̄r/ ‘pull’ becomes [ɔ̄r], but counterpart C-initial monosyllabic roots do not change), showing one cannot simply explain away these facts through prosodic footing or frozen morphology.

3.3 Nubian

While far smaller and less thoroughly documented than the Nilotic and Central Sudanic families, the Nubian family too shows grammatical tone, especially in

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marking nouns (albeit usually of the non-autonomous type whereby the tone co-occurs with overt affixation). In Nobiin ([fia] – Egypt), nouns in isolation bear underlying tone, typically a single high span at the right edge, or an all low pattern. To indicate plurality, these nouns are marked by a suffix, idiosyncratically selected by particular roots (e.g. *-ìì*, *-ncìì*, *-ríí*, and *-kúú*, among others). Regardless of which suffix is selected, in all contexts the underlying tone of the root is neutralized to low, shown in Table 26. This affects Nubian-derived vocabulary as well as that which is Arabic-derived (e.g. /áádèm/ → [ààdèm-ríí] ‘people’).

Table 26: Nobiin neutralizing tone in plural (Werner 1987)

Underlying	Example	Plural form
H	ág	→ àg-ìì ‘mouths’
	éen	→ èèn-cìì ‘women’
	kám	→ kàm-ríí ‘camels’
	áy	→ ày-kúú ‘hearts’
L	gììd	→ gììd-ìì ‘grasses’
	tùù	→ tùù-ncìì ‘bellies’
	kòòg	→ kòòg-ríí ‘ravens’
	dir	→ dir-kúú ‘sandals’
HH	féntí	→ fènt-ìì ‘dates’
	úmbúú	→ ùmbùù-ncìì ‘palm stems’
	ájáw	→ àjàw-ríí ‘sweets’
	kúrsí	→ kùrsì-gúú ‘chairs’
LH	kàdíís	→ kàdiis-ìì ‘cats’
	àrrée	→ àrrèè-ncìì ‘waterfalls’
	mèrgír	→ mèrgìr-íí ‘belts’
	gàttí	→ gàttì-gúú ‘odors’
LHH	àgíssí	→ àgiss-ìì ‘fishes’
	shìgírtí	→ shìgìrtìì-ncìì ‘hairs’
	kòkóttí	→ kòkòttì-ríí ‘muds’
	bèltááyíb	→ bèltààyìb-úú ‘earrings’
LLH	èskàléé	→ èskàlèè-ncìì ‘waterwheels’
	gàngàréé	→ gàngàrèè-gúú ‘spikes’

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4 Afro-Asiatic

Next consider the Afro-Asiatic phylum. While its most famous families are non-tonal (i.e. Semitic, Berber, and Egyptian), those branches which have tone show robust grammatical tone patterns. We shall discuss key examples in families Chadic, Cushitic, and Omotic (the last being most likely several independent families unrelated to the rest of Afro-Asiatic, but within its orbit) (cf. [Roberts 2026 \[this volume\]](#) and [Green 2026 \[this volume\]](#)).

4.1 Chadic

The most populous Chadic language is Hausa ([hau] – Nigeria), spoken by tens of millions, and this language shows rampant grammatical tone effects in its nominal and verbal system. Like the Nobiin facts in [Table 26](#) above, plural formation systematically replaces the tone of the base noun. But unlike in Nobiin, the exact grammatical tone pattern varies according to the precise suffix. [Table 27](#) shows that the plural suffixes *-óóCíí* (where ‘C’ is a copy-consonant of the root) and *-únàà* assign a replacive ⑥ tone to the root, while the suffix *-áí* assigns replacive ⑦.

Table 27: Hausa plural suffixes assigning either ⑥ or ⑦ replacive tone ([Newman 1986, 2000](#), [Inkelas 1998](#))

Replacive ⑥	gyàlè	→	gyál-óólíí	‘shawls
	óófíis	→	óófíis-óófíí	‘office’ (< English)
	kààsúwá	→	káásúy-óóyíí	‘markets’
	tààtsúúniyáá	→	táàtsúúniy-óóyíí	‘birds’
	hántàà	→	hánt-únàà	‘livers’
	rììgáá	→	ríg-únàà	‘gowns’
	àkààwú	→	ákáaw-únàà	‘clerks’
	àgóógó	→	ágóóg-únàà	‘clocks’
Replacive ⑦	wàkíìlìì	→	wàkìil-áí	‘representatives’
	dáálibíí	→	dààlib-áí	‘student’
	àlmákàfíí	→	àlmàkàs-áí	‘scissors’ (< Arabic)
	máḃúḃḃúgáá	→	màḃùḃḃùg-áí	‘water springs’

Within the verbal system, replacive grammatical tone also plays a major role in derivational and inflectional morphology. One example is the imperative, which

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Table 28: Hausa imperative expressed by replacive ①② (Newman 2000, Trommer 2011)

LH	fitá	→	LH	fitá	‘go out!’
HL	tá:ǰi	→	LH	tà:ǰi	‘get up!’
HH	kwà:ná	→	LH	kwà:ná	‘spend the night!’
LHL	tàimákà:	→	LLH	tàimàkí	‘help!’
HLH	kàràntá:	→	LLH	kàràntá:	‘read!’
HHH	nànnè:mó:	→	LLH	nànnè:mó:	‘seek (them) repeatedly!’

overwrites any lexical tone with a replacive ①② pattern, an autonomous grammatical tone pattern. This is exemplified in Table 28.

At the same time, not all grammatical tone is replacive in Hausa (Inkelas 1998). One is the definite clitic =①n ~ =①ĩ ‘the’ which assigns an additive ① tone to the target of grammatical tone (on the distribution of the two allomorphs, see Newman 2000). This specifically does not result in replacement of the entire word’s tones, but typically results in a falling contour.

Table 29: Hausa additive ① sponsored by definite clitic (Newman 2000)

bàká:=①n	→	bàkân	‘the bow’
ùngùlú:=①n	→	ùngùlûn	‘the vulture’
mànò:má:=①n	→	mànò:mân	‘the farmers’
gó:ǰíyó:yí:=①n	→	gó:ǰíyó:yîn	‘the mares’
mó:tà:=①ĩ	→	mó:tàĩ	‘the car’

4.2 Cushitic

While less robustly tonal than Chadic, tonal Cushitic languages also show grammatical tone patterns, typically which comply with pre-existing aspects of their tonology (e.g. allowing only one H per word). One famous case is the correlation between the position of high tone and gender and number in Somali ([som] – Somalia), shown in Table 30. A general correlation in the language is that male and/or singular nouns have a H on the penultimate mora, while counterpart female and/or plural nouns shift the H to the final mora.

On these patterns in Somali (and also the Cushitic language Rendille), Mous (2021) cites Oomen (1981) on their origin via a former suffix, e.g. a (reconstructed)

Table 30: Somali marking gender/plural *via* tone shift (Hyman 1981)

Male and/or Singular		Female and/or Plural	
náil	‘male lamb’	naíl	‘female lamb’
ínan	‘son, boy’	inán	‘daughter, girl’
nácas	‘stupid man’	nacás	‘stupid woman’
qaálin	‘young male camel’	qaalín	‘young female camel’
daméer	‘he-donkey’	dameér	‘she-donkey’
caasáan	‘young he-goat’	ceesaán	‘young she-goat’
éi	‘dog’	eí	‘dogs’
túug	‘thief’	tuúg	‘thieves’
kálax	‘ladle’	kaláx	‘ladles’
bállí	‘reservoir’	ballí	‘resevoirs’
doofaar	‘pig’	doofaár	‘pigs’
Soomaáli	‘Somali man’	Soomaali	‘Somali people’

feminine suffix **-et* which shifted default high tone assignment and then was subsequently lost. See Mous as well for wider discussion of grammatical tone in Cushitic.

4.3 Omotic

Also generally included under the umbrella of Afro-Asiatic are the Omotic languages (split up in Glottolog into genealogically unrelated families North Omotic, South Omotic, Blue Nile Mao, and Dizoid – Hammarström et al. 2025). Despite their questionable status as Afro-Asiatic, they are still often described using terms from Afro-Asiatic and Semitic linguistics. For example, in Ganza ([gza] – Ethiopia) when roots are modified by certain morphological environments (e.g. by a possessor) they appear in a ‘construct state’ (e.g. Frajzyngier & Shay 2012: 533 for some background in Afro-Asiatic). In Ganza, the changes from the underlying tones can be called ‘construct tone’. For example, in the construct state of lexical roots underlying tone is replaced by a construct tone pattern which is only partially phonologically predictable. Table 31 shows that construct tone reduces tonal contrast to three shapes: L, HL, or H¹ (note that the downstep ¹ and upstep ¹ symbols indicate changes on the following word).

For nouns and adjectives, construct tone is triggered by most modification. However, with verbs construct tone is triggered by what appears to be an id-

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Table 31: Ganza underlying vs. grammatical ‘construct tone’ (Smolders 2016)

	Underlying			Construct		Meaning
Pattern 1	HH	tákú	→	LL	tàkù	‘run’ (PL)
	LL	kùrù	→	LL	kùrù	‘genet’
	HH ^l	t’úlá ^l	→	LL	t’ùlà	‘dikdik’
	LH	p’òfó	→	LL	p’òfò	‘chase’
	LH ^l	kjàlá ^l	→	LL	kjàlà	‘colobus’
Pattern 2	HL	tájà	→	HL	tájà	‘sojourn’
	H ^l H	kí’á	→	HL	kí’à	‘catch’
Pattern 3	HH ^l	sásó ^l	→	HH ^l	sásó ^l	‘monkey’
	LL ^l	wàṅà ^l	→	HH ^l	wàṅá ^l	‘chicken’
	H ^l H	ǰí’ní	→	HH ^l	ǰíní ^l	‘work’

iosyncratic set of contexts. For example, the verbal clitic =*na* conditions the construct form of the verb /p’áp’í/ ‘gather’ (11a, where it surfaces with [L] construct tones), while the clitic =*bo* preserves the underlying high tones (11b). These two clitics are translated identically, and throughout Smolders’ work appear as the default inflectional forms conveying basic present/past tense (depending on the verb’s inherent semantics).

- (11) Ganza idiosyncrasy of construct tone triggers (e.g. =*na*) and non-triggers (e.g. =*bo*) (Smolders 2016)
- a. [hàgá p’àbìnà]
hà^l=ga p’áp’í=na
AFF=3M.SBJ gather\CNST=CL
‘he gathers’
- b. [hàgá p’ábìbô]
hà^l=ga p’áp’í=bo
AFF=3M.SBJ gather=CL
‘he gathers’

5 Khoisan

Our last Greenbergian phylum is Khoisan, which at this point has been demonstrated as not forming a valid language family, but rather five (Güldemann &

Fehn 2014): three families Khoe-Kwadi, Kx’a, Tuu, and two isolates Hadza and Sandawe. We provide only a brief discussion of Khoisan, leaving out entirely Kx’a, Tuu, and Hadza. Khoisan languages generally lack the grammatical tone flourish of other African family stocks in terms of nominal or verbal inflection (especially compared to nearby Bantu languages). Still, grammatical tone can be identified, most clearly with types of derivational morphology.

5.1 Khoe-Kwadi

The Khoe-Kwadi family contains the largest and perhaps best documented Khoisan language, Khoekhoe a.k.a. Khoekhoegowab or Nama/Damara ([naq] – Namibia). Its tone system has been extensively studied for over 100 years (Haacke 1999, Haacke 2008, Brugman 2009, Kusmer 2020), and from this literature there is consensus that Khoekhoe has four pitch heights which combine to form six primary melodies distributed across a bimoraic lexical stem. A six-way minimal pair with an approximate citation pronunciation is provided in Table 32, to which we add an informal label for each melody and our working phonological analysis of the underlying tones (note 4=highest á, 3=high á, 2=low à, and 1=lowest ä, and that post-vocalic /m/ is moraic and tone-bearing).

Table 32: Khoekhoe minimal pair of six melodies distributed across bimoraic stem (Haacke 2008)

Label	Approximate pitch	Underlying	Example	Meaning
‘Top’	[44] ~ [43]	/4/	!óǀms	‘fist’
‘Higher-Rising’	[24] ~ [34]	/24/	!òǀms	‘pollard’
‘Intermediate’	[33] ~ [32]	/3/	!óǀm	‘coagulate’
‘Level’	[22]	/2/	!òǀm	‘force out’
‘Bottom’	[11] ~ [12]	/1/	!ǀǀm	‘push’
‘Lower-Rising’	[12] ~ [13]	/12/	!ǀǀms	‘udder’

Khoekhoe shows several ‘flip-flop’ rules (a term taken from the Chinese tone literature – e.g. Wang 1967), which appear in certain derivational contexts. One is weak flip-flop, which is triggered across numerous affixal and compound constructions. One productive trigger is the applicative suffix -bǎ , which adds an argument to the verb structure. An example set is provided in Table 33, which shows that the six underlying melodies form three pairs which neutralize their distinction: /4/ and /24/ both become 24, /3/ and /2/ become 2, and /1/ and /12/ become 12.

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Table 33: Khoekhoe weak flip-flop triggered by applicative -*bǎ* (Haacke 1999)

Pair	Tone change			Example		Applicative form		
Pair 1	/4/	→	24	kǔrú	‘make’	→	kùrú-bǎ	‘make for’
	/24/	→	24	!nàrí	‘steal’	→	!nàrí-bǎ	‘steal for’
Pair 2	/3/	→	2	óá	‘return’	→	òá-bǎ	‘return for’
	/2/	→	2	sàrì	‘visit’	→	sàrì-bǎ	‘visit for’
Pair 3	/1/	→	12	!nǎrì	‘drive’	→	!nǎrì-bǎ	‘drive for’
	/12/	→	12	ǔrì	‘jump’	→	ǔrì-bǎ	‘jump for’

The other type is strong flip-flop. This appears only sporadically, but is systematically triggered by causative reduplication, whereby causation is expressed by two parts: a suffixed reduplicant which bears a fixed /21/ tone (the second copy) and a tone change on the root (the first copy). The strong flip-flop tone change involves the same pairs as its weak counterpart, but with tonal exchange rather than neutralization. In the first pair in Table 34, underlying /4/ becomes 24 while underlying /24/ becomes 4. Equally, /3/ becomes 2 but /2/ becomes 3, and /1/ becomes 12 but /12/ becomes 1.

Table 34: Khoekhoe strong flip-flop (tonal exchange) triggered by causative reduplication (Brugman 2009)

Pair	Tone change			Example		Causative form		
Pair 1	/4/	→	24	!nǒm̃	‘smile’	→	!nòm̃-!nòm̃	‘make smile’
	/24/	→	4	!nòǒ	‘measure’	→	!nǒǒ-!nòǒ	‘estimate’
Pair 2	/3/	→	2	sóm̃s	‘shade’	→	sòm̃-sòm̃	‘make shady’
	/2/	→	3	kòñ	‘move’	→	kóñ-kòñ	‘move to and fro’
Pair 3	/1/	→	12	!nǎm̃	‘love’	→	!nǎm̃-!nǎm̃	‘inspire to love’
	/12/	→	1	!nǔwù	‘short’	→	!nǔwù-!nǔwù	‘shorten’

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What unifies these flip-flop rules is that both reference a /2/ tone. In weak flip-flop (Table 33), a grammatical ② is either added to the stem (positioned at whichever edge creates a rising contour) or it overwrites stem tone (as with the ‘Intermediate’ /3/ melody). This results in the neutralization of the three paired melodies. In contrast, in strong flip-flop (Table 34) grammatical ② only appears on stems which lack 2 underlying, but 2 is deleted if one is already there (hence, exchange). Comparable grammatical flip-flop patterns are found in related Khoe-Kwadi languages, e.g. G|ui (Nakagawa 2006).

5.2 Sandawe

Last but not least is the Sandawe language ([sad] – Tanzania), of unclear affiliation to other Khoisan languages (but perhaps related to Khoe-Kwadi). In the Sandawe literature, a process of ‘grammatically-conditioned tone lowering’ is described (e.g. Hunziker et al. 2008, Eaton 2010, *inter alia*), whereby the tones of the second word of certain noun phrases systematically lower. This is shown with various possessive constructions, compounds, and [NOUN ADJECTIVE] constructions in Example (12). In these data, the first word’s tones remain intact while the second’s lower.

Despite the name of this ‘tone-lowering’ operation, both Hunziker et al. (2008) and Eaton (2010) show that the underlying tone contrast of the target is still phonetically detectable, and therefore not completely neutralizing. If the second word ends in an underlying low tone – whether simply low (12a) or a falling contour (12b-12c) – then the final syllable bears a super-low tone (more accurately, a low-to-super-low contour where the pitch falls to the lowest point in a speaker’s pitch range). In contrast, if the second word ends in high underlyingly (whether a contour or not), then the final syllable lowers less, only to a flat level-low pitch (12d-12f).

- (12) Sandawe grammatically conditioned tone-lowering, but without complete neutralization (Hunziker et al. 2008, Eaton 2010)

- a. [hèwé hùmbù]
hèwé hùmbù
he cow
‘his cow’
- b. [lʰàtʃʰú k’àn̩kàrà]
lʰàtʃʰú k’àn̩kàrà
lion black
‘black lion’

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- c. [kǒ:ŋgó: bǒ:k'ò]
 kǒ:ŋgó: bǒ:k'ò
 Gkoongoo spoon
 'Gkoongoo's spoon'
- d. [gélé màtò]
 gélé mǎtó
 Gele gourd
 'Gele's gourd'
- e. [lʰàtʃʰú ts'wǎ:]
 lʰàtʃʰú ts'wǎ:
 lion tail
 'a lion's tail'
- f. [hàpú k'àmè]
 hàpú k'ámé
 you beer
 'your beer'

The maintenance of this contrast may point to the role of phonological phrasing in accounting for these tone-lowering effects. Interestingly, a similar process is found in tone sandhi in Khoekhoe, whereby non-initial words change to a special 'sandhi' form (separate and distinct from the flip-flop rules of Tables 33 and 34 above). See Kusmer (2020) for details.

6 Conclusion

In this chapter, we examined grammatical tone data from approximately thirty different African languages from all over the continent, and made reference to properties of several dozen others. We defined and explored the differences between lexical and grammatical tone and ultimately showed that they are both robustly attested in languages of every phylum. In terms of the range of grammatical properties which can be conveyed using grammatical tone, we suggest that any inflectional or derivational property which can be expressed by segmental morphemes can also be expressed by tone. We showed that in some cases a grammatical tone on its own might signal the grammatical property being conveyed, while in other cases it might be one of several co-exponents of the grammatical expression. We observed that the way grammatical tone is actually implemented takes a variety of forms (e.g. addition, subtraction, replacement, shifting,

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exchanging, and spreading) and these were all exemplified in various African languages.

Because of its suprasegmental nature, tone can be quite challenging to describe and accurately document. This is true of lexical tone, and even more so with regard to grammatical tone. But given the myriad grammatical properties that it expresses in whole or part, as illustrated in the sections above, its documentation and description is absolutely indispensable for a full understanding of such data. Careful and intense research on African tone languages has contributed substantially to our general understanding of grammatical tone, and we expect ongoing such research will deepen this even further.

Symbols and abbreviations

∅	toneless	ISO	International Organization for Standardization
Ⓣ	floating/grammatical tone		
%	boundary tone	L	low tone
↓	downstep	M	mid tone
↑	upstep	sH	super-high tone
C	consonant	sL	super-low tone
F	falling tone	TBU	tone-bearing unit
H	high tone	V	vowel
HTS	High Tone Syllable (Yoruba)		

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Glosses

1	first person	HUM	human
2	second person	HYP	hypothetical
3	third person	IMP	imperative
AFF	affirmative	IPFV	imperfective
AP	antipassive	IRR	irrealis
BEN	benefactive	M	male
BT	boundary tone	NTOP	non-topic
CAUS	causative	NEG	negative
CL	clitic	P	preposition
CNST	construct	PASS	passive
CONS	consecutive	PL	plural
CP	centripetal	PRF	perfect
CT	circumstantial	Q	question
DEF	definite	RPST	remote past
DIM	diminutive	SBJ	subject
FUT	future	SG	singular
FV	final vowel (Bantu)	SJTV	subjunctive

Acknowledgements

We would like to thank our department colleagues, the anonymous reviewers, the editors of this volume, and the dedicated staff at Language Science Press (especially Sebastian Nordhoff). All errors are our own.

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Chapter 4

Tonal downdrift and downstep in African languages

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Downdrift and downstep are phonological processes that lower the tonal register in such a manner that all tones that follow the downstep juncture are just that much lower than they otherwise would have been. Tones that follow downdrifted and downstepped tones maintain their contrastive pitch heights relative to each other, albeit on the new, lower pitch register. Downdrift and downstep can also occur multiple times in the same utterance (i.e., they can be recursive), and their effects are cumulative so that each occurrence of register lowering results in an ever-lower setting of the pitch “ceiling.” Although downdrift and downstep are most often described as targeting H tones, they can also target M and L tones. The present work describes the various downstep scenarios found in African languages, including total downstep and double downstep, that can ensue when different triggers interact with different targets.

1 Introduction

Downdrift and downstep are phonological processes that lower the tonal register in such a manner that a new (lower) “ceiling” is established for all tones that follow the downstep juncture. Tones that follow downdrifted and downstepped tones, in turn, maintain their contrastive pitch heights relative to each other, albeit on the new lower register. This downward shift in register is similar to a singer changing a song to a lower key. Downdrift and downstep can also occur multiple times in the same utterance (i.e., they can be recursive), and their effects are cumulative so that each occurrence of register lowering results in setting a new, lower, pitch ceiling. It is also the case that the lowering effect is exclusive of

Keith L. Snider. 2026. Tonal downdrift and downstep in African languages. In Keith L. Snider & Virginia Beavon-Ham (eds.), *African tone phenomena*, 103–137. Berlin: Language Science Press. DOI: ??

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whatever lowering may be attributable to intonational effects such as declination (see below) and to whatever raising may be attributable to H tones being raised before L tones.

Each of these characteristics of downstep is dealt with in detail below. However, since the terminology that surrounds downstep and other pitch-lowering phenomena is often confusing, due to the same terms sometimes being used to mean different things by different authors, I first clarify the terms employed in this paper.

The earliest description of downstep as a phenomenon seems to be Christaller's (1875) *Grammar of the Asante and Fante language*, although the earliest use of the actual term DOWNSTEP, is probably F. D. D. Winston's (1960: 189) description of the "mid tone" in Efik [efi]¹ (Benue-Congo, Nigeria) as "a unit of 'downstep', which may be symbolized by a superior exclamation mark, contrasting with its absence."

There are, however, two types of downstep: one that is completely predictable from the surface contexts (i.e., its effects are non-contrastive), often called DOWN-DRIFT, and one that is not predictable (i.e., its effects are contrastive), often just called DOWNSTEP.² Downdrift, as it is normally thought of, applies to alternating sequences of H's and overt L's (e.g., H¹LH¹LH) such that whenever a L follows a H, a new downstep juncture is created.³ As such, it is, of course, completely predictable.

Downstep, on the other hand, is not predictable from surface contexts. In most cases, it occurs between sequences of like tones, usually H tones (e.g., H¹H), where the downstepped tones contrast with their non-downstepped counterparts (e.g., H¹H contrasts with HH). To account for downstep, linguists often posit a FLOATING (latent) L tone at the downstep juncture.

Floating tones, often represented in parentheses as (L) or (H) in this paper, are tones not associated with TONE-BEARING UNITS (TBUs) in autosegmental representations (cf. Rolle et al. 2026 [this volume]). Often shown to result from the loss of their TBU historically, floating tones are extremely useful in helping to explain certain tonal phenomena clearly and in an intuitively satisfying manner. Throughout this paper, I therefore employ the notion of floating tones when doing so enhances clarity.

¹At the first mention of a language name, its ISO 639-3 code (Eberhard et al. 2024), language family, and country where spoken are provided.

²See Welmers (1973: 93) for early usage of these terms.

³Whether the downstep juncture in downdrift environments occurs between H and L (as assumed in this paper), or between L and H (as was probably the original intuition) remains an open question and is inconsequential for this paper (see also Snider 2020).

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Unfortunately, the term downdrift is also used by some (e.g., [Hombert 1974](#): 172, fn. 6) to refer to DECLINATION. Declination is the gradual phonetic lowering of pitch that occurs over the course of an utterance, even when all tones are identical (see Figure 1).

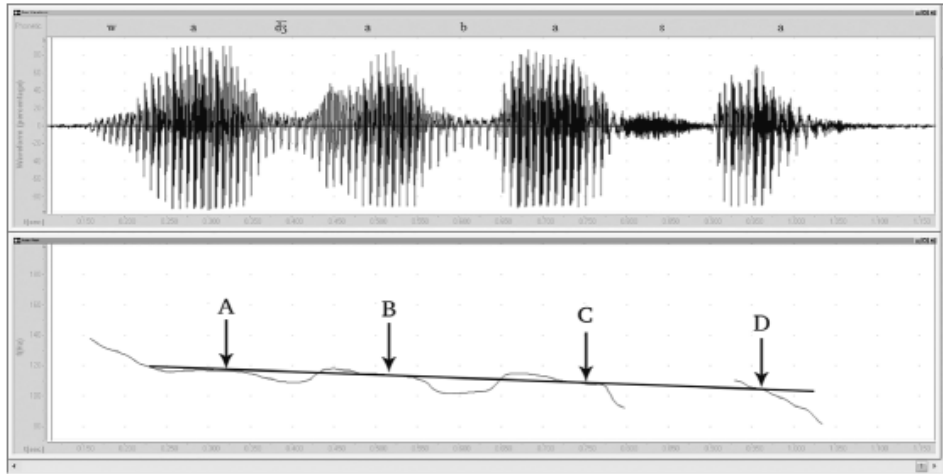


Figure 1: Declination of all L tones in Chumburung ([Snider 2018](#): 73).

As may be seen in Figure 1, the pitches in this all-L utterance (*wàdʒà bàsà* ‘cloth’s needle’) descend at a remarkably constant rate: A-116.4 Hz, B-113.6 Hz, C-109.1 Hz, and D-104.9 Hz ([Snider 2018](#): 70). Declination is a purely low-level phonetic phenomenon, but it needs to be taken into consideration when comparing acoustic measurements of different pitch heights to determine the degree of lowering attributable to declination and to ensure that the degrees of lowering attributed to downdrift and downstep are not, in fact, solely attributable to declination.

Recognizing that several characteristics of downdrift are shared with downstep, Stewart (1965) coined the terms AUTOMATIC DOWNSTEP and NON-AUTOMATIC DOWNSTEP to refer to downdrift and downstep, respectively. He included the word downstep in both terms to acknowledge their shared characteristics (i.e., that both were register lowerings triggered by L tones) and to communicate his claim that downdrift and downstep were essentially the same phenomenon. Based mainly on audio impressions in those days, overt L tones were seen as triggering the downward shifting of the tonal register to the same degree as that triggered by floating L tones (e.g., HLH [– –] = H(L)H [– –]). Wishing also, however, to recognize their distinctive natures (i.e., that downdrift is automatically

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triggered by overt L tones, and downstep non-automatically triggered by floating L tones),⁴ Stewart called downdrift *AUTOMATIC DOWNSTEP* and downstep *NON-AUTOMATIC DOWNSTEP*.⁵ Although Stewart's terms have gained some currency in the literature (including in many of my own works), it is also the case they have not completely supplanted the original terms downdrift and downstep.

Since the coining of Stewart's terms, however, additional examples of downstep have come to light that now call into question the usefulness of the terms *automatic* and *non-automatic* downstep (see Section 5). Accordingly, in this paper, I am reverting to the use of *DOWNDRIFT* instead of *AUTOMATIC DOWNSTEP* and to *DOWNSTEP* instead of *NON-AUTOMATIC DOWNSTEP*. In the discussion in Section 5, I conclude that what unites downdrift and downstep is not that they are necessarily the same phenomenon (i.e., that they necessarily have the same degrees of lowering), but that both lower the pitch register and both can apply recursively. And I conclude that what distinguishes them is whether their outputs are contrastive. If the output is contrastive, it is downstep; if it is not contrastive, it is downdrift.

Because downdrift is totally predictable from the surface context (i.e., the surrounding surface tones), it is normally not indicated in phonetic transcriptions. Rather, a brief description of downdrift sometimes appears near the beginning of papers, with a statement along the lines of, "Following common practice, downdrift is not transcribed in surface representations." On the other hand, downstep (viz. non-automatic downstep) is not predictable from the surrounding surface tones and needs to be represented in phonetic transcriptions. Prior to the widespread use of personal computers, authors were mainly limited to the symbols found on typewriters, so a super-scripted exclamation mark ([!]) at the downstep juncture came to represent downstep. Some older publications also used a straight apostrophe to represent downstep. More recently however, a super-scripted down-arrow ([↓]), recognized by the International Phonetic Association

⁴Stewart, of course, did not use the term "floating tone," since that term was introduced sometime later. Instead, he referred to what is now called a floating tone as "a zero tone-bearing unit" (Stewart 1965: 20–21) and, following Winston (1960), as "a phoneme of downstep" (Stewart 1965: 8). The earliest use of the term "floating tone" of which I am aware is Trutenau (1972: 88–89, fn 11) in which, with reference to Ga [gaa] (Kwa, Ghana), he refers to "a preceding *EXTRASEGMENTAL* ('floating') tone," and later in the same footnote, "its original tone is retained while the 'floating' low tone is copied." Although the publication date of the paper is 1972, the editor has placed at the end, "Received April 1970." My thanks to Larry Hyman (personal communication) for drawing this reference to my attention.

⁵Stewart confirmed this reasoning to me during a conversation we had at some point during my years of graduate study at Leiden University.

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(IPA), has mainly replaced these older symbols.⁶ Here are how these symbols appear in linguistic literature.

- (1) Representations of downstep in Bamileke-Dschang⁷ [ybb] (Grassfields Bantu, Cameroon)
 - a. *ɲká 'sáɲ* ‘monkey of bird’ (Stewart 1981: 120)
 - b. *ɲdzà 'ɲdzwì* ‘the axe of the leopard’ (Hyman & Tadadjeu 1976: 93)
 - c. *á 'vá* ‘he/she/it came’ (Leben 2018: 4)

The remainder of this paper is structured as follows: In Section 2 I discuss how H tones may be downdrifted by overt L’s and M’s and be downstepped by floating L’s. This covers canonical downstep, as well as TOTAL DOWNSTEP of H’s, DOUBLE DOWNSTEP of H’s, and downdrift of H’s that narrow the pitch register. In Section 3 I describe how overt and floating L’s can respectively trigger downdrift and downstep of M tones, and in Section 4 I show how floating H’s can trigger downstep of L tones. In Section 5, I review the pertinent facts about downdrift and downstep, I discuss the usefulness of the terms automatic downstep and non-automatic downstep, and I suggest further avenues of research needed in future studies of downstep.

2 Downdrift and downstep of H’s

There are three triggers known to downdrift and downstep H tones: L tones can trigger both downdrift and downstep of H’s, overt M tones can trigger downdrift of H’s, and adjacent H tones can trigger downstep of the second H.

2.1 Downdrift and downstep of H’s triggered by L’s

There are two types of L triggers: overt L tones that trigger downdrift, and floating L tones that trigger downstep.

⁶The opposite phenomenon, UPSTEP (not treated in this paper), is represented with the up-arrow (viz. [↑]).

⁷Although the linguistic literature refers to this language as Bamileke-Dschang, or simply Dschang (i.e., the dialect of Bamileke spoken in and around the city of Dschang), its autoglossonym is Yemba.

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2.1.1 Canonical downdrift and downstep

I begin with canonical downdrift, triggered by overt L tones interspersed between H tones or sequences of H's. Here is a typical example of canonical downdrift, adapted from an acoustic study of downstep in Bimoba [bim] (Gur (Mabia),⁸ Ghana).

(2) Downdrift in Bimoba (Snider 1998: 87, 98)

[	—	—	—		—	—	—	—		—	—	]
pòrùk	món	gòrí					pórúk	bòn		bík		
spider	red	look.at.IPFV	spider				black	small				
‘red spider is looking at small black spider’												

We know that (2) is an example of downdrift because all tones that follow the downstep junctures, indicated by vertical strokes, are lower than they otherwise would have been. Compare the level of the initial L-toned *pòrùk* (before the first downstep juncture) with that of L-toned *gò* (after the first downstep juncture), and the level of H-toned *món* (before the first downstep juncture) with that of H-toned *rí* (after the first downstep juncture).⁹ We also know it is downdrift rather than the creation of discrete third and fourth levels because it is recursive, with the LH pitch pattern successively lowered between the vertical strokes. Finally, we know it is downdrift because its occurrence is non-contrastive in that it occurs at each instance of H followed by L (i.e., H^L never contrasts with HL in the same downdrift span).

Of course, the question remains whether the lowerings in (2) might be solely attributable to declination. To test this, Snider (1998) compares acoustic measurements of the utterance in (2) with all H-toned and all L-toned utterances of similar lengths to determine the degree of total lowering in (2) that can be attributed solely to declination.

The figures in Table 1, extrapolated from Snider (1998), clearly reveal that the lowerings in (2) cannot be attributed solely to declination.¹⁰

⁸Gur is also known in some recent literature as Mabia. Throughout this paper, I refer to the language family names with the names that appear in the sources I am citing, or, if non-existent, the name as it appears in the Ethnologue.

⁹The observant reader will notice in (2) that whereas the first instance of *pòrùk* ‘spider’ is all L-toned, the second instance is all H-toned. This is because the second instance has undergone H-Spread from the H tone of the preceding verb *gòrí* ‘look at’. While the language permits H's to spread from verb to object, it does not permit them to spread from subject to verb (i.e., from the H tone of *món* ‘red’ to the L tone of *gòrí* ‘look at’).

¹⁰Cross-linguistically, H tones are typically raised before L's, floating or otherwise. To com-

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Table 1: Downdrift and declination measurements in Bimoba (Snider 1998)

Lowering of final tones attributable ...	(Hz./syll.)	
	H tones	L tones
to downdrift and declination	11.7	4.9
solely to declination	1.0	2.8
solely to downdrift	10.7	2.1

Here is a second example of downdrift, from Chumburung [ncu] (Kwa, Ghana).

(3) Downdrift in Chumburung (Snider 2007: 4)

[	-	-		-		-	-		-		-	-	]
nàná			mí			nàná			mí			nàná	
grandfather			3POS			grandfather			3POS			grandfather	
'grandfather's grandfather's grandfather'													

As may be seen in (3), the downdrift effect is recursive and totally predictable (non-contrastive).

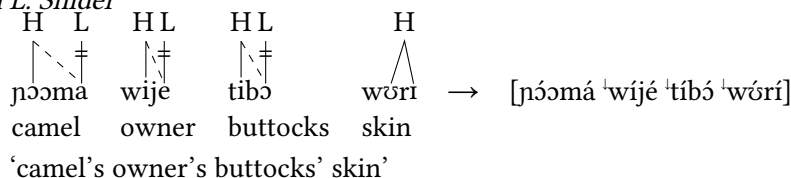
While downdrift is triggered by overt L tones, downstep of H tones is triggered by floating L tones, as mentioned above. Floating L's can be either derived or underlying. Derived floating L tones can arise from a L tone losing its tone-bearing support due to phonological processes such as vowel deletion or vowel coalescence, to the spreading of a H with delinking of an adjacent (usually a following) L, or to the default insertion of floating L's onto toneless TBUs. Underlying floating L tones often occur on the right edges of segmental lexical or grammatical morphemes, or they occur alone as grammatical tonal morphemes. I discuss each, in turn.

Here is an example, again from Chumburung, of downstep triggered by derived floating L's that have been delinked from their TBUs due to the rightward spreading of H-tones (cf. Smolders 2026 [this volume]).

(4) Downstep in Chumburung due to H-spread (personal field notes)

pensate for this, these figures avoid comparing measurements of the final H in (2), which is not followed by L, with measurements of the preceding H, which (like the H before it) is followed by L.

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Example (4) demonstrates how H tones in Chumburung spread rightward (viz. perseveratively) within words to the right edges of the words and delink the L tones from their TBUs, as indicated by the chop marks across the association lines of the L tones. As indicated by the downstep arrows, the H tones to the right of the now floating L tones are downstepped, together with all tones that follow them. This example demonstrates that downstep in Chumburung is clearly recursive and affects all following tones.

Next, we examine another case of H downstep in Chumburung in which the floating L that triggers the downstep has lost its TBU due to the fusion of its vowel with the vowel that follows it. Here and elsewhere, the ghost-like dotted circle symbol $\circ$ serves as a placeholder for diacritic marks that represent floating tones. Although it represents a synchronically deleted TBU in this example, this is not necessarily always the case, as in (10) and (11) below, where it serves as a placeholder for a grammatical tone (cf. Rolle & Bickmore 2026 [this volume]).

- (5) Downstep in Chumburung due to vowel coalescence (Snider 1989: 225)
 kúri ɔ́ɔ → kúr $\circ$ ɔ́ɔ → [kúr $\circ$ ɔ́ɔ]
 husband stranger
 ‘husband’s stranger (i.e., guest)’

In (5), the final underlyingly L-toned vowel of ‘husband’ is deleted whenever it precedes another word that begins with a vowel. This leaves its L tone floating, which, in turn, downsteps the following H tones of ɔ́ɔ ‘stranger’. (In isolation, kúri is realized with a H-falling pitch, due to lexical spreading of the preceding H tone.)

These final examples of downstep attributable to derived floating L’s involve floating L’s that are assigned by default to toneless TBUs in Tiriki [ida] (Bantu, Kenya, Paster & Kim 2011). Tiriki has an underlying three-way tonal contrast between L, $\emptyset$, and H. I first demonstrate contrast between H and toneless TBUs in (6).

- (6) Contrast between H and $\emptyset$ in Tiriki (adapted from Paster & Kim 2011: 80)
 a. /mu-limi/ → [mù-limì] ‘farmer’ /va-limi/ → [và-limì] ‘farmers’
 b. /mu-xáli/ → [mù-xàlì] ‘wife’ /va-xáli/ → [và-xàlì] ‘wives’

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In (6a) the root /limi/ ‘husband’ is underlyingly toneless, as are the singular and plural noun class prefixes /mu-/ and /va-/. This results in all TBUs being realized with default surface L tones. In (6b), while the root /xáli/ ‘wife’ begins with an underlying H tone, its singular and plural noun class prefixes, being the same prefixes as in (6a), are underlyingly toneless. Notice how the H tone of /xáli/ spreads leftward onto the toneless prefixes of ‘wife’ and ‘wives’ so they are realized with surface H’s. Also notice how the final syllable of /xáli/, which does not undergo leftward spreading of the H, is realized with a default L tone. In (7), one observes that the leftward spreading of H is unbounded (i.e., it spreads to more than one TBU).

(7) Unbounded leftward spreading of H (adapted from Paster & Kim 2011: 80)

- a. /mu-limi/ → [mùlimi] ‘farmer’
- b. /mu-limi mu-láhi → [mú-límí mú-láhi] ‘good farmer’

Notice how in (7b), the underlying H of /láhi/ ‘good’ spreads unblocked to the very left edge of the phrase. This results in all TBUs in the phrase being realized with surface H’s except for the final TBU on the right edge, which is not subject to the leftward spreading of H. Next, look at the examples in (8), which demonstrate contrast between L and toneless TBUs.

(8) Contrast between L and ∅ in Tiriki (adapted from Paster & Kim 2011: 81)

- a. /và-lix-a-molom-el-e mu-limi/ → [và-lixà-mòlòm-èl-è mù-limi]
‘they will speak for a farmer (distant future)’
- b. /và-lix-a-molom-el-e mu-lína/ → [và-lixá-mólóm-él-é mú-línà]
‘they will speak for a friend (distant future)’

In (8a), all TBUs are underlyingly toneless except for the subject prefix /và-/, which is underlyingly L. Since there are no H tones to spread leftward in this example, the toneless TBUs are all realized with surface L’s and are identical in pitch to that of the underlying L-toned TBU. In (8b), however, the underlying H of /lína/ ‘friend’ spreads leftward like the underlying H of /láhi/ ‘good’ in (7b). This time, however, it is blocked from spreading to the very left edge of the phrase by the underlying L of /và-/. It is at this point that the contrast between L and ∅ reveals itself: While all the toneless TBUs undergo the leftward spreading of H, the underlyingly L-toned subject prefix /và-/ does not and is instead realized with surface L. Finally, consider the examples in (9), which demonstrate downstep in Tiriki.

(9) Downstep in Tiriki (adapted from Paster & Kim 2011: 82)

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- a. /và-à-mú-molom-eel-e mu-limi/ → [và-à-mú-mòlòm-èèl-è mù-lìmi]
‘they have just spoken for a farmer’
- b. /và-à-mú-molom-eel-e mu-línà / → [và-à-mú-‘mólóm-éél-é mú-línà]
‘they have just spoken for a friend’

In (9a), all TBUs are toneless except for the underlyingly L-toned /và-/ and /à-/, and the underlyingly H-toned /mú-/ prefixed to the verb root /molom/ ‘speak’. Since there are no H tones to the right of any toneless TBUs, all toneless TBUs in (9a) are realized with surface default L’s. In (9b), however, the underlying H tone of /línà/ ‘friend’ spreads leftward until it is blocked by the underlying H tone of /mú-/. Notice, however, that downstep occurs between the underlying H tone of /mú-/, and the surface H tones of [molom]. Why? Paster & Kim analyze downstep in Tiriki as triggered by default L tones that are inserted between underlying H tones any time there is at least one intervening TBU that is underlyingly toneless.

The floating L’s that trigger downstep of H’s in these next examples are not derived but underlying. In Kenyang [ken] (Wide Bantu, Cameroon), the trigger for downstep in (10) is a floating L-toned imperfective marker. Historically, grammatical floating tones are sometimes created by the historic loss of their tone-bearing support (e.g., vowels). In such cases, when the TBUs of grammatical morphemes are deleted, leaving their tones intact, these now floating grammatical tones are referred to as TONAL MORPHEMES. The verb complex in (10) consists of a subject prefix, followed by a tense/aspect/mood marker, followed by the verb root.

- (10) Downstep in Kenyang due to floating L-toned imperfective aspect marker (Eyongetah 2017)
 - a. /bá-Ø-pá/ → [bápá] ‘they spat’
3PL-PFV-spit
 - b. /bá-ò-pá/ → [bá’pá] ‘they are spitting’
3PL-IPFV-spit

In Kenyang, perfective aspect is unmarked, indicated by the NULL symbol, with the result that the two H tones in (10a) are phonetically realized H. Imperfective aspect, however, is marked solely by a floating L that occurs between the H-toned 3P subject prefix *bá-* and the H-toned verb root *pá* ‘spit’ in (10b). In this case, the floating L manifests its presence by downstepping the following H tone, as in *bá’pá* ‘they are spitting’.

These next examples of downstepped H tones in Bamileke-Dschang also involve downstep triggered by underlying grammatical floating L’s.

- (11) Downstep in Bamileke-Dschang (Hyman 1985b: 68)

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/sǎŋ ò sǎŋ ò sǎŋ / → [sǎŋ ˈsǎŋ ˈsǎŋ] ‘bird of bird of bird’
bird AM9 bird AM9 bird

In Dschang, most nouns begin with a noun class prefix, abbreviated in the glossing with the numbers of their classes (e.g. c7). The ASSOCIATIVE CONSTRUCTION, which often translates as ‘noun1 of noun2’, consists of the head noun (noun1) followed by the ASSOCIATIVE MARKER (AM), which is concordant with the class of the head noun. In slow speech, the AM is realized as *è* in classes 1 and 9, *á* in class 7, and *é* in all other classes. In normal speech, however, the segmental vowel is usually dropped, leaving behind its (now floating) tone. For classes 1 and 9, the AM in normal speech is (L), and for all other classes it is (H). This, in turn, is followed by the subordinate noun (noun2). The subordinate noun consists of its own noun class prefix, if any, followed by its stem.

In each of the examples above, the downstepped H immediately follows the floating L trigger, and this is normal, also for Dschang, as may be seen in (11). A comparison of (11) with (10b) quickly demonstrates that H tones undergo downstep when preceded by the sequence H(L). (11) also demonstrates that downstep in Dschang is recursive.

However, in this next example set, notice how in (12a), the floating L that triggers the downstep does not occur before (i.e., to the left) of the downstepped H, as it does normally in Dschang and in most other languages. Rather, the floating L trigger in (12a) immediately follows the downstepped H, and this is unusual. When this occurs in Dschang, as is the case in (12a), the floating L must be part of the stem tone pattern to its left, and there must be a L tone before that stem. If these conditions are not met, the downstepped H occurs at the location of the floating L, as it does elsewhere in Dschang.

(12) Downstep to the left of the floating L (Hyman 1979a: 11)

- a. /à-páò/ c7-taro → [àˈpá] ‘taro’
- b. /à-pá/ c7-lid → [àpá] ‘lid’

Looking at (12a), a first question that arises concerns whether this truly is downstep, as opposed to downdrift, in which all H tones that follow L tones are automatically downdrifted? Downdrift, however, does not occur in Dschang. This is readily confirmed by comparing (12a) with (12b), where it may be seen that the H that follows the L in (12b) *ápá* ‘lid’ does not undergo downdrift, as it does in so many other languages. In this respect, it contrasts with the downstepped H in (12a) *àˈpá* ‘taro’. In other words, the trigger for downstep in *àˈpá* is not the overt L that precedes the downstep juncture, but rather the floating L that follows it.

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A second question, however, also comes to mind, questioning whether the downstepped H might not actually be a M tone? This question, however, is also easily dispelled by the examples in (13) and (14).

- (13) Following tone affected by downstep (Hyman 1979a: 11)

/à-páò ó sɛŋ → [à[˥]pá sɛŋ]

c7-taro AM7 bird

‘taro of bird’

If the downstepped H of ‘taro’ were truly M, then ‘taro of bird’ in (13) would be realized *[àpā sɛŋ] instead of [à[˥]pá sɛŋ]. Since the lowering effect of downstep affects all tones that follow the point of downstep, that sɛŋ is downstepped to the same pitch level as pá demonstrates that pá is a downstepped H, and not a M. Further evidence that words ending in H(L) in Dschang are downstepped H’s, as opposed to M’s, may be seen by the presence of recursion in (14).

- (14) Recursive effect of downstep (Hyman & Tadadjeu 1976: 98)

/à-púò ó ò-móò → à[˥]pú [˥]mó

c7-arm AM7 C1-child

‘arm of child’

In (14), the roots for ‘arm’ and ‘child’ are respectively preceded by an overt L and a floating L, and both are followed by floating L’s. In the case of /à-púò/ ‘arm’, the downstep is effected by the following floating L on its right edge (cf. [à[˥]pá] ‘taro’ vs. [àpá] ‘lid’ in (13a) and (13b), respectively). If the downstepped H’s in (14) were truly adjacent M’s, then ‘arm of child’ would be realized *[àpū mō] instead of [à[˥]pú [˥]mó]. Since the lowering effect of downstep is recursive however, that the pitch of mó is downstepped relative to the pitch of pú demonstrates that [˥]pú and [˥]mó are downstepped H’s, and not M’s.

Not all H tones that are lowered after floating L’s, however, are instances of downstep. In Kom [bkm] (Grassfields Bantu, Cameroon), when a H tone follows a floating L in examples like (15), the H is not downstepped but is rather lowered to M.

- (15) H lowered to M in Kom [bkm] (adapted from Hyman 2005: 321)

/ó-sòyín sé tè-fòyn/ → sóyòín sé tèfòyn → [sòyñ sé tèfòyn]

c10-friend c10AM c13-chief

‘friends of chiefs’

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In (15), the prefix of ‘friend’ is a floating H tone that spreads onto the stem /sòyín/ and sets the L tone of that stem floating. This floating L, in turn, lowers the following H of /sòyín/ to M. We know, however, that this lowering is not downstep because it affects only the first H tone after the floating L. As may be seen in (15), the H-toned associative marker *sé* that follows the lowered H is not itself lowered to M but is realized at the same phonetic pitch as the H that precedes the floating L. This indicates that the lowering of H after the floating L is just that, a local lowering of H to M and not downstep of the H.¹¹

To conclude this discussion of canonical downstepped H’s triggered by L tones, most instances of H’s that undergo downstep are triggered by floating L’s that occur before the downstepped H’s. In Dschang, the downstepped H can also occur to the left of the floating L that triggers the downstep. In this case, the floating L trigger must be part of the tone pattern of the stem on the left that undergoes the downstep. Also, the H of the downstepped stem must be preceded by a L tone. If not, downstep in Dschang only occurs to the right of the floating L. Although downstep of H tones triggered by floating L’s is widespread across Africa, I am currently not aware of any cases outside of the Grassfields Bantu area of Cameroon in which the downstepped H can also occur to the left of a floating L (cf. Hamm et al. 2026 [this volume]).

In these next sections, I discuss three non-canonical downstep phenomena: total downstep, double downstep, and narrowing of the pitch register.

2.1.2 Total downstep

I now discuss what is known as TOTAL DOWNSTEP. The earliest use of this term of which I am aware is Meeussen (1970), who probably coined it: “There seems to be a phenomenon which could be called ‘total downstep’, in which a high is not lowered just a little, but all the way to the next lower register” (Meeussen 1970: 270). Languages reported to have total downstep include Kikuyu [kik] (Bantu, Kenya, Clements & Ford 1979, 1981) and Izi [izi] (Benue-Congo, Nigeria, Meier & Bendor-Samuel 1974, Meier et al. 1975).¹² Here is an example of total downstep in Kikuyu, adapted from Clements & Ford (1981).

¹¹Although not pertinent to the present concerns, the underlying H tone of /sé/ spreads onto the L-toned prefix of /tè-fòyn/ ‘chiefs’ where it joins with that L tone to create a high-falling pitch.

¹²Clements (1981: 88) also describes Riccitelli’s (1965) presentation of tone in Bwamu [box] (Gur, Burkina Faso and Mali, Riccitelli 1965) as demonstrating TOTAL DOWNDRIFT, in which the downdrifted H’s are lowered to the same pitch level as the preceding overt L’s.

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- (16) Total downstep in Kikuyu (Clements & Ford 1981: 315)

$$\left[\begin{array}{ccccccc} - & - & & - & - & - & - \end{array} \right]$$
 áhèiré moáyáhìná 'njátá
 he-gave weakling star
 'he gave weakling a star'

In (16), one observes there is no downdrift (i.e., H's that follow L's are realized at the same phonetic pitch as H's that precede L's), and the downstepped H's of *njátá* 'star' are realized at the same phonetic level as the L that precedes the final H of *moáyáhìná* 'weakling'.¹³

Clements (1981) describes total downstep and total upstep (not discussed in this paper) systems as follows:

A few terracing systems are known in which downstep or upstep is “total” rather than partial (Meeussen 1970). In these systems, the drop (or rise) determined by downstep or upstep involves the full interval separating the adjacent tone levels. A further characteristic of systems of this type is that they appear not to have systematic downdrift.” Clements (1981: 87–88)

Clements & Ford (1981) describe total downstep in Kikuyu like this:

Phonetically, a single downstep is equivalent to the interval between an initial low and high tone. Thus, a low tone followed by a downstepped high tone (L!H) is phonetically identical to a low tone followed by a low tone (LL), and a high tone followed by a downstepped high tone (H!H) is phonetically equivalent to a high tone followed by a low tone (HL). (Clements & Ford 1981: 315)

In other words, the interval between a H and a downstepped H is significantly greater in languages with total downstep than it is in languages with partial downstep.

Chumburung, and other languages with PARTIAL DOWNSTEP¹⁴ systems like it, are interesting because in one respect, they fit part of Meeussen's definition of total downstep systems in that downstepped H's are realized at the level of preceding L's. Here are examples of downstep and downdrift in Chumburung

¹³For an interesting theoretical account of total downstep, see Manfredi's (1993) prosodic government approach.

¹⁴Clements (1981: 88) may have been the first to use the term PARTIAL DOWNSTEP to distinguish regular downstep from total downstep.

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that could lead to possible confusion about total downstep in languages like these.

(17) Downstep and downdrift in Chumburung

a. Downstep (Snider 2020: 143)

$\left[\begin{array}{cc} - & - \end{array} \right]$
 kitée ¹dápó
 feather hawk
 ‘feather’s hawk’

b. Downdrift (repeated from (3))

$\left[\begin{array}{ccccc} - & - & - & - & - \end{array} \right]$
 [nàná mí nàná mí nàná]
 grandfather 3POS grandfather 3POS grandfather
 ‘grandfather’s grandfather’s grandfather’

In (17a), one notices that the downstepped H of *dá* is realized at the same phonetic pitch as the preceding L of *ki*, which is similar to Kikuyu. It is also the case that the downdrifted H of the 2nd bold *ná* in (17b) is realized at the same phonetic pitch as the preceding L of the first bold *nà*. However, while the Chumburung partial downstep of (17a) might, at first glance, look like the Kikuyu total downstep of (16), downstep in languages like Chumburung is clearly not the canonical total downstep described by Meeussen (1970) and Clements & Ford (1981). First, total downstep languages do not have downdrift, as does Chumburung, and this is important (see discussion immediately below). Second, whereas the [H¹H] drop in total downstep languages is equivalent to that of [HL], the [H¹H] drop in Chumburung is about half that of the [HL] drop, as may be seen by comparing (17a) with (17b).

Chumburung, however, does fit the description of what Stewart (1993) calls COVERT TOTAL DOWNSTEP languages. Consider the following:

In this paper, I distinguish between overt total downstep languages, which are immediately recognizable as such by Meeussen’s criterion, and covert total downstep languages such as Akan, which Meeussen does not recognize as total downstep languages but which meet his criterion as soon as they are treated as having automatic downstep in the context [H_L] rather than in the context [L_H]: in the overt total downstep languages the [H¹H] drop is patently the same size as the [HL] drop, but in languages such as

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Akan [and Chumburung] there is no HL drop[,] as [H¹L] occurs to the exclusion of [HL], and the [H¹L] drop, is as one would expect, twice the size of the [H¹H] drop. (Stewart 1993: 185)

If one accepts Stewart's overt/covert distinction and his criteria for covert total downstep, then Chumburung and many other partial downstep languages in Africa satisfy that criteria. Essentially, this means that the only difference between total downstep in languages like Kikuyu, and partial downstep in languages like Chumburung and Akan, where the first and last pitches of LH¹H sequences are equal, is that the latter have downdrift and the former do not. However, I'm not really sure how useful this distinction is. When linguists speak of total downstep, they are typically not thinking in Stewart's terms. Rather, total downstep for most linguists means canonical total downstep, as described by Meeussen and Clements & Ford, and I would be loathe to label partial downstep languages like Chumburung and Akan as covert total downstep languages.

2.1.3 Double downstep

While canonical downstep of H tones is widespread and generally well documented in African languages, a much rarer and less studied type of downstep is DOUBLE DOWNSTEP. All six languages known to me with reported instances of double downstep in Africa are also reported to have single canonical downstep, with the double downstep drops being twice that of their single counterparts. Despite the phenomenon being quite rare, its distribution nevertheless appears to be rather widespread, from the Gur family in Burkina Faso (Angsongna et al. 2026) to Grassfields Bantu in Cameroon to Western Nilotic in Uganda (cf. Blum 2026 [this volume]). Then there are two Oceanic languages in New Caledonia (non-African) also reported to have double downstep. All languages known to me that are attested to have double downstep appear in Table 2, with references.

A review of these descriptions reveals that double downstep in all cases occurs when the outputs of two independent downstep processes converge in the same environment. In all these languages except Northern Toussian, Drubea, and Numèè, these two independent downstep processes produce the derived sequence H (LHL) H (i.e., a sequence of two overt H's separated by the floating tone sequence LHL).

Consider the following discussion and examples, as I have interpreted Hieda's (2010) description and analysis of double downstep in Kumam. The underlying tone patterns of the different morphemes for the Kumam examples below appear in (18), followed by an example of single downstep in (19).

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Table 2: Languages reported to have double downstep of H tones

Family/Language	Country	Reference
Gur		
Northern Toussian [tsp]	Burkina Faso	Struthers-Young 2025
Grassfields Bantu		
Bamileke-Dschang [ybb]	Cameroon	Hyman & Tadadjeu 1976
Bamileke-Meduma [byv]	Cameroon	Voorhoeve 1971
Chuffie' [bfj]	Cameroon	Hamm 2024
Western Nilotic		
Kumam [kdi]	Uganda	Hieda 2010
Acooli [ach]	Uganda	Hieda 2011
Oceanic		
Drubea [duf]	New Caledonia	Lionnet 2025
Numèè [kdk]	New Caledonia	Lionnet 2025

(18) Underlying tones in Kumam

/ɛ̃/=	3SG (subject)
nɛk	'kill'
(H)	Indicative (IND)
(L)	Perfective aspect (PFV)
Toneless	Imperfective aspect (IPFV)
-ɔ̌	Transitive formative (TR)
-á	1SG (object)

(19) Single downstep in Kumam (Hieda 2010: 23)

LH(H) L H		L H L H		L H(L)H	
∕	∕	∕	∕	∕	
/ɛ̃=nɛk-ɔ̌-a/	→	ɛ̃=nɛk-ɔ̌-a	→	ɛ̃=nɛk-a	[ɛ̃nɛ́ká]
3SG=IPFV:IND:kill-TR-1SG					
‘He kills me.’					

In (19), the floating H grammatical tone of the indicative marker associates to the toneless root /nɛk/, and it absorbs the H of the rising tone of the 3sg subject marker /ɛ̃/. A segmental hiatus resolution process deletes the vowel of the TRANSITIVE FORMATIVE marker -ɔ̌ when it is adjacent to the 1SG object suffix /-á/, leaving the TR marker's L tone floating. This floating L, in turn, triggers

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downstep of the final H of the 1SG object marker /-á/. In this respect, it is typical of canonical downstep in many other languages.

Next look at an example of double downstep in Kumam.

- (20) Double downstep in Kumam (Hieda 2010: 23)

$$\begin{array}{ccccc} \text{LH(LH)L H} & & \text{L H(LH)L H} & & \text{L H(LHL)H} \\ \text{/}\epsilon\text{=n}\epsilon\text{k-}\grave{\text{o}}\text{-a/} & \rightarrow & \epsilon\text{=n}\epsilon\text{k-}\grave{\text{o}}\text{-a} & \rightarrow & \epsilon\text{=n}\epsilon\text{k-a} \quad [\epsilon\text{n}\epsilon^{\text{H}}\text{ká}] \\ \text{3SG=PFV:IND:kill-TR-1SG} & & & & \\ \text{'He killed me.'} & & & & \end{array}$$

Morphologically, the only difference between (19) and (20) is the change from IMPERFECTIVE ASPECT (toneless) in (19) to PERFECTIVE ASPECT (floating L grammatical tone) in (20). In (20), the H of the 3SG subject prefix /ε=/ shifts rightward to the toneless root /nek/, leaving the L of the underlying /LH/ pattern still associated to the prefix. The floating L perfective aspect marker and the floating H indicative marker remain floating. After hiatus resolution deletes the vowel of the TR marker -ǝ, this adds the now floating L of the TR to the already floating LH tones of the perfective and indicative markers, which, in turn, trigger double downstep of the H tone of the 1SG object suffix /-á/. As may be seen, this combination of H (LHL) H tones would theoretically result in downstep of the bold floating H followed by a second downstep of the final overt H.

As for Northern Toussian, Struthers-Young (2025) also argues that double downstep in this language is the result of two independent instances of single downsteps. The first instance is triggered by a floating L grammatical tone that precedes the verb, and which occurs when a verb is “absent [a] preverbal internal argument” (Struthers-Young 2025: §3.2.1). Verbs that lack preverbal internal arguments are glossed APVIA. Here is an example.

- (21) Grammatically conditioned downstep in Northern Toussian (adapted from Struthers-Young 2025: Ex. 32b)

$$\begin{array}{ccc} \text{sú} & \text{ká} & \grave{\text{o}}\text{já} \\ \text{father NEG watch.APVIA} & \rightarrow & \text{sú} \text{ká}^{\text{H}}\text{já} \\ \text{'Father did not watch.'} & & \end{array}$$

The second instance of single downstep is prosodically conditioned and occurs when a M or H tone follows a M that is positioned at the right edge of a phonological phrase. Here is an example in which a H tone follows the M-toned 1s marker *pǝ*, which occurs at the right edge of its phonological phrase (placed in parentheses).

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- (22) Prosodically conditioned downstep in Northern Toussian (adapted from [Struthers-Young 2025](#): Ex. 1)

(sú p̃) n=ʔá m̃=púñ tǎ
father 1s IPFV=watch 1SG=mother place

‘When father is going to watch at my mother’s house’

The verbal imperfective marker *n=*, which precedes the verb ‘watch’, is toneless and has no effect on the surface tone of the verb. When these two instances of independent downsteps (due to floating grammatical L and to prosodically conditioned downstep) converge on a single TBU, double downstep ensues, as may be seen in (23). Again, the phonological phrase of the M toned word that triggers the prosodically conditioned downstep is placed in parentheses.

- (23) Double downstep in Northern Toussian (adapted from [Struthers-Young 2025](#): Ex. 40)

(pē) n=ʔá m̃=púñ tǎ → pē n=ʔá m̃=púñ tǎ
husband IPFV=watch.APVIA 1SG=mother place

‘The husband watched at my mother’s house.’

Unlike the other five languages in Africa with double downstep, however, double downstep in Northern Toussian only occurs after M’s (viz. [M^uH]) due to the condition stated above that the prosodically conditioned downstep only occurs after M tones.

2.1.4 Downdrift of H’s that narrow the pitch register

These next examples of downdrift are also not canonical. While canonical downdrift lowers all H and L tones that follow the point of downstep, these examples demonstrate downdrift that recursively lowers the H tones, but leaves the L tones at their original pitch levels. This ever lowering of H’s but not L’s results in a narrowing of the pitch register. Arguably, this is not a lowering of the tonal register since L’s are not affected in this phenomenon. It is therefore questionable whether this phenomenon should be called downdrift. Regardless of whether it should be called downdrift, however, I include it here because it is recursive. These next two examples are illustrative.

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- (24) Register narrowing in Southern Nuni (Kurrle 1988: 30)

$$\left[\begin{array}{ccccc} - & - & - & - & - \end{array} \right]$$

/ná yà wá mà wá/ → na ya wa ma wa
 we FUT will hit her
 ‘We will hit her.’

Kurrle (1988: 30) claims that H tones downdrift after L tones in Southern Nuni [nnw] (Gur, Burkina Faso) and supports this with the example in (24). Although her prose description does not specifically say that the H’s descend and the L’s do not, the diagram she provides for this example clearly shows this happening, as shown in (24). Here is a second example of this phenomenon from Nawdm.

- (25) Register narrowing in Nawdm (Nicole 1980: 135)

$$\left[\begin{array}{cccc} - & - & - & - \end{array} \right]$$

/ʔàbá ʔà lógè/ → ʔaba ʔa loge
 3SG be 3SG take
 ‘he will take’

As one can see in (25), with the H’s descending and the L’s remaining unchanged, the pitch range continues to narrow as the utterance progresses. In this case, Nicole (1980) specifically claims in prose that the L’s do not descend, which makes the Nawdm case for this phenomenon slightly stronger than that for Southern Nuni. Although documentation of lowerings attributed to downdrift in which the L’s do not also descend are scarce, Larry Hyman (personal communication, 4/15/2025) provides additional support for the phenomenon with reference to Igbo [ibo] (Benue-Congo, Nigeria): “When I studied Igbo decades ago, it appeared that the L’s did not lower in downdrift. I know there’s been subsequent work, but I did not perceive the lowering, since the lowering of H’s was much more perceptible.”

2.2 Downdrift of H’s triggered by M’s

Mid tones are also documented to trigger downdrift of H’s, although this is at least unusual. Languages in which this is reported include: Bimoba (Snider 1998), Yala of Ikom [yba] (Benue-Congo, Nigeria), and Northern Toussian (Struthers-Young 2025). Here is an example from Yala of Ikom.

M’s in Yala downdrift following H’s to an intermediate level between that of M’s and H’s, similar to the downdrift of H’s by interspersed L’s.

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- (26) Downdrift of H's triggered by L's and M's in Yala of Ikom (Armstrong 1968: 51)

$$\begin{array}{c} \left[\begin{array}{ccccccc} & - & - & & - & - & - \\ - & & & - & & - & - \end{array} \right] \\ /àwá fěp sē rójōré/ \rightarrow [àwá fěp sē rójōré] \\ \text{'all of them (habitually) ate food'}$$

In (26), the L tone downdrifts everything to its right, including the following H and the first M, which is lowered to the level of the triggering L. Moving on to the rest of the utterance, *sē rójōré*, it has the sequence MHMH. Here, one observes the downdrift effect one typically finds when H's are interspersed with L's. Notice that downdrift in this example results not only in the second M being downdrifted relative to the first M, but the second H is also downdrifted relative to the first H. The effect is also recursive in that the MH pattern is repeated. Here is another example of M tones downdrifting H's, this time from Bimoba. In this case, however, the downdrift effect is total in that the H's are downdrifted to the level of the preceding M's.

- (27) Downdrift of H's triggered by M's in Bimoba (personal field notes)

$$\begin{array}{c} \left[\begin{array}{ccc} - & - & - \end{array} \right] \\ /tō-ók \quad mǎ-tí \quad tō-ók/ \rightarrow [took \text{ mari } took] \\ \text{colleague-NC squash-IPFV colleague- NC} \\ \text{'colleague squashes colleague'}$$

In (27), each instance of H is downdrifted to the level of the preceding M tone. That this is downdrift may be seen in that the effect is not local (i.e., M's that follow downdrifted H's maintain their relative pitch-height relationships with the preceding downdrifted H's). The downdrift effect is also clearly recursive in that it is repeated twice after the initial instance.

2.3 Downstep of H's triggered by OCP violations

Not only can adjacent L's trigger the downstep of H's, but in some languages, independent adjacent H's can also trigger it. While not as common as the downstep of H's triggered by L's, this phenomenon is nevertheless relatively widespread, especially in Bantu languages, and it is documented to occur in other language families as well (e.g., Gur). Some of the languages reported to have it include: Supyire [spp] (Gur, Mali, Carlson 1983), Tiriki (Paster & Kim 2011), Shambala [ksb] (Bantu, Tanzania, Odden 1982), and Namwanga [mwn] (Bantu, Zambia,

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Bickmore 2000). Whenever two independent H's occur in juxtaposition, a violation of the OBLIGATORY CONTOUR PRINCIPLE (OCP) occurs. The OCP, originally proposed by Leben (1973), is a general principle, which, when applied to tone, prohibits identical tones from being adjacent on the tonal tier.¹⁵ Languages employ various strategies for resolving OCP tonal violations. With respect to adjacent H tones, these include: a) fusion of the two H's (i.e., two underlying H's linked to a single TBU being realized as a single H linked to a single TBU), b) deletion of one or the other of the two H's, with replacement of the deleted H by a default L tone (e.g., Meeussen's Rule),¹⁶ and c) the insertion of a floating L between the two H's. This is analogous to the insertion of an epenthetic vowel to break up an illegal consonant cluster in segmental phonology.

Following Paster & Kim (2011), I suggest that due to the strong correlation between floating L's and downstep cross-linguistically, anytime an OCP violation triggers downstep of H tones, there is a process that inserts a floating L between the two H's to resolve the violation. This L tone, in turn, triggers downstep of the following H.¹⁷

Whether downstep in such cases is triggered by insertion of a floating L or by some other means, however, is irrelevant for the present concerns. What is important for field workers analyzing tone languages is that downstep of H tones can be triggered by adjacent independent H tones, with no external evidence for intervening floating L's. External evidence for intervening floating L tones could be words that lack L tones in downstepping environments, but which bear L tones in other environments, or which can be demonstrated to have borne L tones historically.

As in some other Bantu languages, downstep in Shambala occurs between adjacent independent H's. Consider the examples in (28) and (29).

¹⁵ Archangeli & Pulleyblank (1986: 30–31) recognizes two types of OCP restrictions: a) skeleton sensitive restrictions, in which adjacent like tones are prohibited from occurring on adjacent TBUs, as in * $\acute{x} \acute{x}$, and b) skeleton insensitive restrictions, in which adjacent like tones are prohibited from occurring on non-adjacent TBUs separated by toneless TBUs within a specified domain, as in * $\acute{x} x \acute{x}$. See also Suzuki (1998) and Hyman (2012: 27).

¹⁶ Meeussen's Rule, often stated as $HH \rightarrow HL$, is named after A.E. Meeussen, who was the first to notice that stem initial H tones in Tonga verbs were realized L when they followed certain H-toned clitics (Meeussen 1963).

¹⁷ An anonymous reviewer has asked how downstep should be accounted for in languages in which L tones do not exist underlyingly (i.e., the tonal contrast is between H and $\emptyset$)? This is a great question, however, responding to it properly would take me beyond the constraint imposed on papers in this volume that they exploit theory only minimally. Suffice it to say that following Paster & Kim (2011), I assume that default L tones can be inserted into structures at early and different stages of phonological derivations.

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- (28) Downstep between H's in Shambala (Odden 1982: 187)

kúi 'dog'
/ní kúi/ → [ní 'kúi] 'it is a dog'

However, when this happens, the two H's must be independent tones, not one H linked to two TBUs. When a single H is linked to two or more TBUs, downstep is not triggered, as may be seen in (29a).

- (29) Lack of downstep between single H linked to two TBUs in Shambala (Odden 1982: 187)

- a. nwáná 'child'
- b. dú 'only'
- c. nwáná 'dú 'only a child'

In (29a), the two [H] pitches of *nwáná* 'child' are the surface representations of a single /H/ that has spread onto the following toneless TBU with the result that downstep is not triggered on the second TBU. As for the H tone of *dú* 'only', it is separate from that of the doubly linked H in *nwáná* 'child' with the result that in (29c), downstep is only triggered between *nwáná* and *dú*.

When dealing with languages like Shambala, it is important to remember that tones can spread, not only within morphemes and words, but also across word boundaries. Accordingly, when a H tone spreads from one morpheme or word to another, this still counts as a single H tone.

3 Downdrift and downstep of M's, triggered by L's

In several languages with three contrastive pitch levels, L tones are reported to trigger the downdrift and downstep of M's, as may be seen in the example of downdrift from Yala of Ikom and the examples of downstep from Ngamambo [nbv] (Benue-Congo, Cameroon, Hyman 1986) and Gokana [gkn] (Benue-Congo, Nigeria, Hyman 1985a) (cf. Blench 2026 [this volume]).

At this point, I am not aware of triggers other than L's and the prosodically triggered downstep of M's documented for Northern Toussian (Struthers-Young 2025) that cause M tones to be downdrifted or downstepped. Examples of downdrift and downstep appear in (30–32).

- (30) Downdrift of M's triggered by L's in Yala of Ikom (Armstrong 1968: 51)

$\left[\begin{array}{ccc} - & - & - \end{array} \right]$
[ìjěná yín]
'his jaw'

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It is clear from this example that the downdrifting of M's in Yala results not only in the second L being downdrifted relative to the first L, but the second M also being downdrifted relative to the first M (i.e. the downdrift is recursive).

Next, is an example that demonstrates contrast between HM and H'M in Ngamambo, followed by an example of a downstepped M triggered by a floating L in Gokana.

- (31) Contrast between M and 'M in Ngamambo (Hyman 1986: 123)

ā-sīŋ á rǎ-kōn 'place of the beds'

ā-sīŋ á rǎ-bēi 'place of the knives'

In (31), the M and the downstepped M are realized at different pitch heights.

- (32) Downstep of M in Gokana (Hyman 1985a: 115)

/àé mōn gè ò níʔeí/ → àé mōn ògè níʔeí → [àé mōn ˈgè níʔeí]

3SG see knife JUNCTURE today

'He will see a knife today.'

As demonstrated in (32), the surface M tone of gè 'knife' is downstepped by a preceding floating L. As discussed in Hyman (1985a), the underlying L tone of /gè/ is set floating by the leftward docking of a junctural M tone onto /gè/. The now floating L, in turn, downsteps the now M of gè.

Unfortunately, the evidence for downstepped M's is not as robust as for downstepped H's. There are two reasons for this. First, there are significantly fewer languages with three contrastive tone levels than with two. So, while all tone languages have H tones that could potentially undergo downstep, fewer languages have M tones that could potentially undergo downstep. Accordingly, there are significantly fewer descriptions of downstepped M's possible than of downstepped H's. Second, demonstrating the recursive nature of downstep was not necessarily an expectation for claims of downstep when some of the earlier descriptions of downstep were written. Moreover, some of these descriptions are focussed on things other than downstep, so data helpful to the issues at hand do not necessarily appear in all the descriptions. Nevertheless, it is hoped that in the future, the recursive nature of downstep will be demonstrated for downstepped M tones.

4 Downstep of L's, triggered by floating H's

That H and M tones could be downdrifted and downstepped to lower levels by overt and floating L's is understandable. However, that L tones could be downstepped to even lower levels by floating H's would seem to be counter intuitive,

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a notion supported by the relative rarity of the phenomenon.¹⁸ Despite its being relatively rare, the downstep of L tones is nevertheless reported for several languages, not all of which are necessarily closely related. The phenomenon was first documented in Bamileke-Dschang in Hyman & Tadadjeu (1976), and more recently in several other Bamileke languages spoken in Cameroon, including Ngyemboon [nnh] (Anderson 1983) and Ghomala' [bbj] (Nissim 1981). It has also been documented in Nawdm [nmz] (Gur, Togo, Nicole 1980); and the Nandi dialect of Kalenjin [kln] (Nilotic, Kenya, Creider 1982). For most of these languages, the downstep is either claimed to be triggered by floating H's (Anderson 1983 for Ngyemboon, Hyman 1984 for Kalenjin), or is at least amenable to such claims (Nicole 1980 for Nawdm). At this point, I am not aware of any languages with downstepped L tones in which the downsteps are attributed to anything other than floating H tones.

In an analysis of downstepped L tones in Bamileke-Dschang, based mainly on the data of Hyman (1985b) and which adopts Hymans' underlying forms, Snider (2020) claims that all instances of downstepped L tones in Hyman's data are attributable to either underlying floating H's, as in (33), or to derived floating H's, as in (34). In (33), all underlying tones are L except for the lexical floating H's on the right edges of the different instances of *ndzàŋ* 'axe' and *kàŋ* 'squirrel'. These examples demonstrate unequivocally the potential for successive downsteps of L in Bamileke-Dschang.

(33) Downstepped L due to underlying floating H in Dschang (Snider 2020: 128)¹⁹

- a. /n̄-dzàŋ́ è n̄-dzàŋ́ è n̄-dzàŋ́ è n̄-dzàŋ́/ →
 NC-axe AM NC-axe AM NC-axe AM NC-axe
 [n̄dzàŋ́ n̄dzàŋ́ n̄dzàŋ́ n̄dzàŋ́]
 'axe of axe of axe of axe'
- b. /kàŋ́ è kàŋ́ è kàŋ́ è kàŋ́/ →
 squirrel AM squirrel AM squirrel AM squirrel
 [kàŋ́ kàŋ́ kàŋ́ kàŋ́]
 'squirrel of squirrel of squirrel of squirrel'

In Dschang, the vowel of the associative marker (è) is often deleted in regular speech, as it is in these examples and in (34), below. However, since the AM is

¹⁸ An anonymous reviewer, however, has suggested that since the well-known phenomenon of H's being raised before L's (cf. footnote 10, Section 2.1.1) increases the pitch difference between H's and following L's, the downstepping of L's after floating H's may not be all that unnatural.

¹⁹ I elicited and recorded these utterances from the late Professor Maurice Tadadjeu, the original source of the data in Hyman & Tadadjeu (1976).

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underlyingly L-toned, like all the other tones in the utterance except for the floating H's, the deletion of its vowel does not affect the pronunciation of any surface tones. As one can see at each instance of a floating H, the L's that immediately follow it are downstepped relative to those that immediately precede it. One can also see in both examples that the downstep is recursive and that each instance of downstep in (33a) affects more than just one TBU.

Now look at this next example of downstepped L in Dschang, this time triggered by a derived floating H. As in (33), all underlying tones are L except for one, in this case the H tone of /dɔŋ/.

(34) Downstepped L due to derived floating H in Dschang (Snider 2020: 130)

L	H (L)	L	L	L	
	↑				
n-dɔŋ	e	mən-dzwi	→	[n'dɔŋ mɛndzwi]	
NC-horn	AM	NC-leopard			
'horn of leopard'					

In (34), the floating L on the right edge of 'horn' spreads leftward onto *dɔŋ*, delinking the /H/ from *dɔŋ* and setting it floating.²⁰ As in (33), there is now a floating H trapped between two associated L's, which triggers downstep of the second L and all following tones. Although this example does not demonstrate the recursive nature of downstep, it does demonstrate how the tones following the downstepped H are also downstepped, as evidenced by all L's that follow the downstep juncture being realized at the same phonetic pitch level.

5 Discussion

Here is a review of the pertinent facts about downdrift and downstep. To be considered downdrift or downstep, a pitch lowering phenomenon must lower the tonal register in such a manner that *all* tones that follow the downstep juncture are demonstrably lower than they otherwise would have been (i.e., they are pitch register lowering phenomena). Also, the phenomenon must be demonstrably recursive (i.e., the lowering must be repeatable in the same utterance), and the effect must be cumulative, with each instance of downstep resulting in a new, ever lower, register. When these conditions are met in a single utterance, there is

²⁰The leftward docking of the floating L tone, as formulated in (34), is an over-simplification of two independent rules in Snider (2020). This complication is ignored in this example since both accounts (i.e., Snider 2020 and the present work) end identically with the H tone set floating between the two associated L's.

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a downward “terracing” effect. Cross-linguistically, all tones can undergo downstep.

Downdrift and downstep of H tones can respectively be triggered by overt and floating L’s, by M’s, and by OCP violations between adjacent H tones. Canonical downdrift is triggered by overt L’s, and the phenomenon is usually called either downdrift or automatic downstep. Canonical downstep is triggered by floating L’s, and it is usually called either downstep or non-automatic downstep. This paper employs the more traditional terms, downdrift and downstep.

Although most triggering L’s occur before (i.e., to the left of) downstepped H’s in Bamileke-Dschang, they can also occur to the right of the first downstepped H, with the effect progressing rightward from that downstepped H. The downstepping of H tones by L tones is the most common form of downstep, and it is widespread across Africa. In a small number of languages with downstep, the degree of downstep is total such that the phonetic pitch lowering of H’H is equivalent to that of HL. This is called total downstep. Also in a small number of languages with downstep, double downstep of H occurs when two downstep environments converge in such a manner that the floating sequence (LHL) occurs between two H’s, thereby triggering a drop in pitch that is twice that of normal downstep. There are also languages in which downdrift recursively lowers H tones, but leaves L tones untouched, thereby progressively narrowing the pitch register between H and L tones.

M tones can be downstepped by preceding L’s. However, despite seemingly being natural enough, evidence that it is recursive is not as robust as one might expect, and the phenomenon itself does not appear to be overly widespread, although it might be more prevalent than is currently recognized.

Although downstep of L tones is not common, languages that exhibit this phenomenon are not confined to a single language family, nor are they necessarily near each other geographically. So far, the evidence points to L tones being downstepped solely by floating H tones. The following quotation, with reference to the Nandi dialect of Kalenjin, is illustrative:

This floating H is, I claim, the trigger for the downstepping of the second L. If this is correct, we not only have further evidence for considering most cases of downstep to be triggered by a preceding floating tone..., but we also see evidence that the trigger of a downstepped L may be a floating H. (Hyman 1984: 674–675)

For those interested in pursuing the phenomenon of downstepped L tones from a theoretical point of view, the interested reader is referred to chapters 3

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and 7 of Snider (2020) where a typology of downstep triggers is presented within the model of Register Tier Theory.

In the discussion of downstep terminology in Section 1, I call into question the usefulness of the terms automatic and non-automatic downstep, to which I had formerly ascribed. These labels, of course, would make sense if all examples of downstep were as Stewart originally thought them to be: Automatic downstep and non-automatic downstep are basically the same phenomenon, they necessarily result in the same degrees of lowering, and if a language has non-automatic downstep, it must also have automatic downstep.

There are two problems, however, with the notion that downdrift and downstep are a single phenomenon (Stewart's reason for joining the two with the word "downstep"). The first lies in Stewart's implied assumption (Stewart 1965, 1971, 1981, 1983) that the degree of lowering effected by each is equivalent. And for the record, phonetic equivalence between the two is indeed demonstrated to hold for the following languages: Akan [aka] (Kwa, Ghana, Genzel & Kügler 2011), Bimoba (Snider 1998), Chumburung (Snider 2007), and Igbo (Laniran 1992).²¹ However, it is also demonstrated not to hold for other languages (e.g., Abron [abr] (Kwa, Ghana) and Adioukrou [adj] (Kwa, Ivory Coast), Leben & Ahoua (2017). Leben (2018: 3, fn 3) also notes that in an instrumental study of Dagara-Wulè [dgi] (Gur, Burkina Faso), Rialland & Somé (2011) found downdrift and downstep to differ in their patterns of lowering in that language. So, to the extent that the cited works are correct, whether the effects of downdrift and downstep are phonetically equivalent is language specific, and not universal. This lack of universality undermines the notion that automatic and non-automatic downstep are truly the same phenomenon in all languages, which, in turn, casts doubt on the usefulness of Stewart's including the term downstep in each term.

A second problem with Stewart's understanding that downdrift and downstep are essentially the same phenomenon concerns Stewart's original conclusion that languages with downstep must necessarily have downdrift as well (Stewart 1971: 195). If that conclusion were true, then it would clearly support the notion that the phenomena are essentially one. However, more recent studies demonstrate this conclusion to have been premature, as Stewart also recognized later (Stewart 1983: 68). As Hyman (1979a) states:

The result is we now have languages representing all four possible combinations of downstep and downdrift: (1) downstep and downdrift (Igbo, Akan);

²¹Contrary to Genzel & Kügler's (2011) claim for Akan, Dolphyne (1994) had earlier concluded that the phonetic effects of downdrift and downstep were not the same in Akan, and contrary to Laniran's (1992) claim for Igbo, Liberman et al. (1992) conclude they are not the same for Igbo.

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(2) downstep without downdrift (Dschang); (3) downdrift without downstep (Hausa); and (4) no downstep and no downdrift (Haya). (Hyman 1979a: 26, fn 19)

Given that the existence of downstep is not dependent on the existence of downdrift, this again undermines the notion that downdrift and downstep are the same phenomenon in all languages, which, in turn, again casts doubt on the usefulness of Stewart’s terms, cross-linguistically.

So, where does this leave us? I propose that what unites the two phenomena is that both are pitch register lowering phenomena that apply recursively, and what distinguishes them is whether their outputs are contrastive (i.e., not predictable from surface structures). This is summed up in Table 3.

Table 3: Characteristics of downdrift and downstep

	What distinguishes them	What unites them
Downdrift	Non-contrastive outputs	Both lower the pitch register,
Downstep	Contrastive outputs	Both apply recursively

While much is known about downdrift and downstep, much also remains unknown, so it is hoped that future research will answer some of the remaining questions. For example, while adjacent H’s are known to trigger downstep of the second H (e.g., Shambala) do adjacent M’s ever trigger downstep of the second M, or do adjacent L’s ever trigger downstep of the second L? If not, why not? Assuming Register Tier Theory, Snider (2020) proposes that H’s are intrinsically linked to H registers and L’s to L registers. He suggests that any tone can be downstepped by any other intervening floating tone, providing the intervening tone has a register value opposite to that of the downstepped tone (e.g., H(L)H → H^lH and L(H)L → L^lL). He also proposes there are two kinds of M tones: H’s with L registers and L’s with H registers. If true, are there M tones with register values opposite to L tones that could result in L(M)L → L^lL? And are there H tones with register values opposite to M tones that could result in M(H)M → M^lM? Again, if not, why not?

Much also remains to be discovered through instrumental studies of downdrift and downstep that investigate the equivalence between different phonetic pitch levels. For example, there are a number of claims, based on auditory impressions only, that downstepped H’s in some languages are phonetically equivalent to M’s. These include Supyire (Carlson 1983), Babanki [bbk] (Grassfields Bantu,

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Cameroon, Hyman 1979b), and Kagoro [kcg] (Plateau, Nigeria, Van der Kolk 1992). Unfortunately, very few claims of this nature are supported with instrumental data. Similarly, although downstepped H's and M's have different phonological behaviors, more instrumental studies are still needed to further support (or to debunk) claims of phonetic equivalence between them.

Acknowledgments

I wish to thank the following people for their generous help with this paper. Larry Hyman and Will Leben were most helpful with extensive email discussions on downstep in the lead-up to writing the first draft and then later with their comments on an early draft. Later still, Larry, who chose not to remain anonymous, also provided a helpful formal review. I'm also grateful for email interactions with the following people: Bruce Connell on total downstep, David Odden on downstep triggered by OCP violations in Shambala, and Anthony Struthers-Young on double downstep in Northern Toussian. Finally, I wish to thank the second reviewer (anonymous) for their comments. Thanks to the excellent help I received, I have only myself to blame for any remaining shortcomings.

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Chapter 5

Floating tones

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Floating tones are tones that are not associated with any tone-bearing unit (e.g., segmental material), and they have been used widely in the analysis of a variety of tone systems since the second half of the 20th century. Floating tones are inherently abstract and do not constitute an empirical object (one cannot hear a floating tone). Rather they are an analytical device, one that we show is indispensable for tonal analysis. This chapter gives an overview of floating tones, their nature, and the variety of tonal phenomena they have been used to account for (downstep, blocking effects, etc.). While most of our examples come from African languages (primarily from the Niger-Congo stock, or those within its orbit), our discussion should be relevant to tone languages in general.

1 Introduction

Tone languages can be defined as those where “an indication of pitch enters into the lexical realization of at least some morphemes” (Hyman 2001: 1368; see also Welmers 1959, 1973), which constitute roughly half the world’s languages and 80% of the languages of Africa (Maddieson 2022, 2023).¹ Tone languages typically

¹Maddieson bases his estimate of 46.7% on 6,520 languages listed in the *Ethnologue* (Eberhard et al. 2023) and other sources. Our 80% African calculation is based on Maddieson’s Table 2 which

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possess minimal pairs such as in (1) from the Mande language Vai [vai] (Welmers 1976: 31) (cf. Konoshenko 2026 [this volume]). Here, the same segmental sequence *laa* expresses three distinct nouns depending on whether it is high (H), low (L), or rising (rendered LH). Note that *aa* stands for a long vowel, i.e., the vowel /a/ linked to two morae, that is, two Tone Bearing Units (TBUs).

- (1) Vai – Tonal minimal triplet in nouns
- a. High H *láá* [lá:] ‘a plankfish’
 - b. Low L *làà* [là:] ‘a paddle’
 - c. Rising LH *làá* [lǎ:] ‘a clay pot’

Perhaps the most insightful development in the analysis of tone is separating segments and tones onto separate ‘tiers’, as developed in Goldsmith’s (1976) Autosegmental Phonology. In this framework, the minimal triplet in (1) would be rendered as in (2) where phonological elements on the tonal tier are connected via ‘association lines’ to TBUs, here conflated with segmental representations on the segmental tier.

- (2) Vai – Autosegmental diagrams
- | | | | |
|----------|----------|----------|----------------|
| H | L | L H | tonal tier |
| | | | |
| a. l a a | b. l a a | c. l a a | segmental tier |

Separating the tonal and segmental tiers allows one to easily model one-to-one associations (as in 2c), and also many-to-one associations where an element on one tier associates to multiple elements on another tier (2a-2b). Also, because the tiers are independent, it is possible for elements to be present on one tier without any elements occurring on the other. Thus, representations of underlying morphemes as well as surface forms may contain ‘toneless’ segments which are not associated to a tone, or tones which are not associated to a segment. The latter constitutes what is known as ‘floating tone’.

Floating tones can be demonstrated with the definite versions of these Vai nouns, shown in (3). Descriptively, this involves the addition of a rising tone at the right edge of the noun.

indicates the number of tonal languages in Niger-Congo (1,458 out of 1,551), Afroasiatic (252 out of 384), and Nilo-Saharan (180 out of 207), to which we added 27 Khoisan languages occurring in the *Ethnologue*.

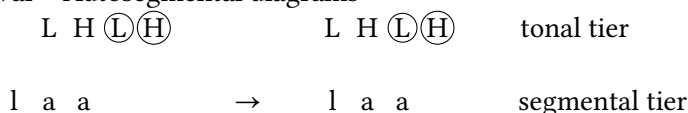
5 Floating tones

(3) Vai – Definite forms of nouns

- | | | | | | |
|----|-----------|------------|---------------|------------|-----------------|
| a. | High H | <i>láá</i> | ‘a plankfish’ | <i>lǎǎ</i> | ‘the plankfish’ |
| b. | Low L | <i>làà</i> | ‘a paddle’ | <i>lǎǎ</i> | ‘the paddle’ |
| c. | Rising LH | <i>làá</i> | ‘a clay pot’ | <i>lǎǎ</i> | ‘the clay pot’ |

We might think of the definite marker as a floating rising tone morpheme, which must associate to the segmental tier since it is not pre-associated to anything. A mapping from the underlying form to the surface form for [lǎǎ] ‘the clay pot’ is provided in (4). We follow the common convention of circling floating tones (e.g., [Pulleyblank 1986](#), [Yip 2002](#)), thus the floating rise is represented as a $\textcircled{\text{L}}\textcircled{\text{H}}$ sequence.²

(4) Vai – Autosegmental diagrams



In what follows, we shall discuss cases akin to Vai where floating tones are part of the underlying representation of morphemes which usually associate in the course of a phonological derivation. However, there are also cases where they have been posited to remain floating either as a trigger (e.g., of downstep) or a blocker (e.g., of tone spreading), among other effects (cf. [Snider 2026 \[this volume\]](#)). We shall collectively refer to these as ‘floating tone effects’, and demonstrate how floating tones have proven useful in individual instances, both for underlying and derived floating tones.

The outline of our chapter is as follows. Section 2 discusses floating tones as part of morphemes or as independent morphemes themselves, which we refer to as their ‘morphological status’. Section 3 demonstrates the use of floating tones as an analytic device to account for certain tonal behavior, which we collectively call ‘floating tone effects’. This includes blocking effects, downstep and upstep/raising, and contrastive final level L tones. Section 4 discusses certain analytical issues which come up when one adopts floating tones, specifically related to the inventory of floating tones, and issues of locality, prosodic category, directionality, and markedness. Section 5 provides several analytical alternatives to floating tones, and briefly assesses their respective merits. Section 6 provides our conclusion.

²Although the $\textcircled{\text{L}}\textcircled{\text{H}}$ input is no longer floating in the output, we leave the circles around the tones for identificational purposes.

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2 Morphological status

Given autosegmental architecture, the only difference between floating tones and non-floating tones is that the former are unassociated to a host. As such, we expect floating tones to have (roughly) the same morphological status as their associated counterparts. In this section, we show how floating tones may be the sole expression of the morpheme, referred to as ‘tonal morphemes’ (Section 2.1), or accompany the segmental components of a morpheme, where they act as part of the expression of the morpheme’s meaning (Section 2.2).

2.1 Tonal morphemes

Cases where tone alone expresses linguistic meaning are referred to as ‘tonal morphemes’. We have seen this already in Vai (3–4), where the marker of definiteness was said to be a floating tone sequence $\textcircled{\text{L}}\textcircled{\text{H}}$ which docks to the right edge of a noun. Such tonal morphemes are capable of a full range of functions on par with their segmental counterparts, marking both derivational meanings (e.g., changes to part-of-speech, valency, diminutivization, *inter alia*) and inflectional meanings (e.g., case, number, gender/class, agreement/concord, finiteness, tense/aspect/mood, polarity/negation, realis, focus/topic, *inter alia*). In fact, the only type of tonal morpheme we have not seen is that of a lexical root. One might, however, envision such a ‘root tonal morpheme’, e.g., a floating tone sequence $\textcircled{\text{L}}\textcircled{\text{H}}$ meaning ‘eat’ which does not co-occur with any segments, whose tone is realized on neighboring affixes (comparable to segmentally null roots found in certain highly synthetic languages, e.g., Menominee [mez] – Bloomfield 1962: 63).

Tonal morphemes can often be shown to derive from associated tones (but certainly not always), where material on the segmental tier has eroded but tones on the tonal tier remain (the so-called ‘stability’ of tone – Goldsmith 1976, Yip 2002). This may result in allomorphs in synchronic systems. A straightforward example comes from the language isolate Laal [gdm] (Lionnet 2021), which has a series of connective markers which agree with the head of the noun phrase in gender and number (5a). All of these consist of a H-toned CV syllable which may appear as such (first example in 5b–d). In all gender/number combinations except masculine and feminine singular, the connective also has a floating $\textcircled{\text{H}}$ allomorph (second example in 5b–d). The realization of this allomorph on a monosyllabic word results in a rising tone in (5c). As seen in (5d), if the preceding tone is already H then the $\textcircled{\text{H}}$ tonal allomorph does not have a discernible effect on the surface.

- (5) Laal – Connective morphemes show tonal allomorphs (data from Lionnet’s field notes)

- | | | | | | | | |
|----|-----------|-------------|--|--|--|--|--|
| a. | <i>já</i> | MASC.SG | | | | | |
| | <i>jí</i> | FEM.SG | | | | | |
| | <i>má</i> | NEUT.SG | | | | | |
| | <i>yá</i> | NEUT.PL | | | | | |
| | <i>yí</i> | PL, NHUM.SG | | | | | |
-
- | | | | | | | | | | |
|----|--------------|-----------|------------|---|--------------|-----|------------|---|------------------|
| b. | <i>nyàw</i> | <i>yí</i> | <i>dēē</i> | ~ | <i>nyàw</i> | Ⓜ | <i>dēē</i> | → | <i>nyǎw dēē</i> |
| | house | CON | at:me | | house | CON | at:me | | ‘my house’ |
| c. | <i>dōrūm</i> | <i>yí</i> | <i>dēē</i> | ~ | <i>dōrūm</i> | Ⓜ | <i>dēē</i> | → | <i>dōrúm dēē</i> |
| | rope | CON | at:me | | rope | CON | at:me | | ‘my rope’ |
| d. | <i>wúrá</i> | <i>yí</i> | <i>dēē</i> | ~ | <i>wúrá</i> | Ⓜ | <i>dēē</i> | → | <i>wúrá dēē</i> |
| | things | CON | at:me | | things | CON | at:me | | ‘my things’ |

Markers akin to Laal connectives (also known as ‘associative markers’) are commonly expressed as tonal morphemes, where they act to ‘link’ or ‘associate’ a head noun with either a modifier (often a modifying noun, as in compounds and possession). Examples are pervasive across the African continent (Cahill 2000), including Benue-Congo (e.g., Epira [igb] – Adiva 1989: 65–66), Gur (Mabia) (e.g., Dagbani [dag] – Olawsky 1999: 205ff.), Mande (Green 2018), Kru (Marchese 1979: 77), Chadic (Schuh 2017: 141), Nubian (e.g., Midob [mei] – Werner 1993: 24, and Khoe-Kwadi (e.g., G|ui [gwj] – Nakagawa 2006: 60–80) (cf. Blench 2026 [this volume], Russell & Sande 2026 [this volume], and Roberts 2026 [this volume]).³

Tonal morphemes may be relatively simple like the Ⓜ tonal allomorph in Laal, but sequences of floating tones exist as well. Consider the language isolate Kunama [kun] (Connell et al. 2000), which has a basic high/mid/low (H/M/L) distinction forming various contours (falling $\widehat{HM}$, $\widehat{ML}$, $\widehat{HL}$, and rising $\widehat{MH}$). Compounds and possessive constructions are expressed by juxtaposing two nouns which are linked via a floating tone sequence $\widehat{L}\widehat{M}\widehat{H}$, e.g., in (6) where we gloss it as LINK.

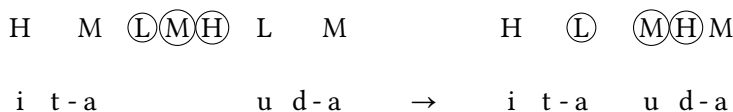
³Outside of Africa, such tonal morphemes are common in noun phrases too, e.g., in Norwegian [nor] (Wetterlin 2010) in Europe, Metnyo Ambel [wgo] (Arnold 2018) and Fore [for] (Scott 1978) in the Pacific, Kuki-Thaadow [tcz] (Hyman 2010: 43) in SE Asia, among others.

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- (6) Kunama – Tonal morpheme as floating tone sequence
- | | | | | | |
|----|---------------|--|----------------|---|-----------------------|
| a. | <i>ít-ā</i> | $\textcircled{\text{L}}\textcircled{\text{M}}\textcircled{\text{H}}$ | <i>úd-ā</i> | → | <i>ít-à úd-ā</i> |
| | house-SG | LINK | door-SG | | ‘house’s door’ |
| b. | <i>àbìn-ā</i> | $\textcircled{\text{L}}\textcircled{\text{M}}\textcircled{\text{H}}$ | <i>úkùn-ā</i> | → | <i>àbìn-à úkùn-ā</i> |
| | elephant-SG | LINK | ear-SG | | ‘elephant’s ear’ |
| c. | <i>kīīš-ā</i> | $\textcircled{\text{L}}\textcircled{\text{M}}\textcircled{\text{H}}$ | <i>āsāṁg-ā</i> | → | <i>kīīš-à āsāṁg-ā</i> |
| | girl-SG | LINK | head-SG | | ‘girl’s head’ |

In these examples, the $\textcircled{\text{L}}$ tone delinks the M of the *-ā* singular suffix of the first (modifying) noun, while the $\textcircled{\text{M}}\textcircled{\text{H}}$ portion docks to the second noun (the head noun). This creates a $\widehat{\text{HM}}$ contour tone on the suffix of in (6a). In contrast, in (6b–c), there is enough segmental material for both $\textcircled{\text{M}}$ and $\textcircled{\text{H}}$ to be realized on the stem itself, the underlying stem tones being deleted to make room for the floating tones (note that the change of H to M on [ŋ] in /āsāṁg-/ → [āsāṁg-] in (6c) is caused by an unrelated tonal process). A simple autosegmental representation of (6a) is provided in (7).

- (7) Kunama – Autosegmental diagram of [ít-à úd-ā] ‘house’s door’



For decades, tonal morphemes have also been the de facto standard in another prosodic area: intonation, especially that of the Autosegmental-Metrical framework (e.g., [Pierrehumbert 1980](#), [Ladd 2008](#)). Under this perspective, one could interpret the rising tone to mark yes/no questions in (Standard) English as a kind of floating tone sequence, since it enters the phonological derivation unassociated to any specific element on the segmental tier. Even many tone languages have such intonational tonal morphemes to mark specific types of clauses, e.g., the ‘African lax question prosody’ whereby utterance-final L tones commonly indicate yes/no questions especially in the tonally-rich middle belt of Africa ([Rialland 2009](#)).

2.2 Floating tone as part of morphemes

There is also evidence that floating tones can form part of the underlying representation of morphemes, where they accompany segmental material. Consider another Mande language, Guinean Kpelle [gkp]. In this language, nouns pronounced with all L tones form two classes, exemplified by *yàlà* ‘lion’ and *gbònò*

5 Floating tones

‘ring’ in (8). The latter has no unexpected tonal impacts on its environment, while the former systematically adds a H tone to the following morpheme, both within and across words (Konoshenko 2008: 24).

(8) Guinean Kpelle – Distinct tonal behaviors of L-toned nouns

- a. *yàlà* ‘lion’ *yàlà-yàà* ‘lions’ *ń yàlà kàà* ‘I saw a lion’
 b. *gbònò* ‘ring’ *gbònò-yàà* ‘rings’ *ń gbònò kàà* ‘I saw a ring’

A straightforward way to capture the above difference is to posit a floating tone in the underlying representation of /*yàlà*($\textcircled{\text{H}}$)/ ‘lion’. This ($\textcircled{\text{H}}$) associates to a following L tone syllable if one exists but is otherwise deleted. Autosegmental diagrams of the plural forms with -*yàà* are in (9).

(9) Guinean Kpelle – Autosegmental diagrams of the plural forms with -*yàà*

- | | | |
|--|---|-------------------------------------|
| L ($\textcircled{\text{H}}$) L | | L ($\textcircled{\text{H}}$) L |
| a. y a l a - y a a | → | y a l a - y a a |
| L L | | L L |
| | | |
| b. gb o n o - y a a | → | gb o n o - y a a |

Konoshenko (2008: 26–27) shows that the *yàlà*-type is realized with M rather than L tone in closely related varieties – e.g., *yālā* ‘lion’ in Liberian Kpelle [xpe] contrasting against *gbònò* ‘ring’ – and posits that this difference derives historically from a *cvc̣ṿ vs. *cvc̣ṿ contrast (cf. Hyman 2003). This represents a running theme in floating tones: they diachronically derive from previously associated tones which become dislodged from their original position.

Moreover, in many languages the same sequence of tones can contrast depending on whether they are floating or associated in the underlying representation. A three-way contrast is found in the Grassfields Bantu language Bamileke-Ghomala’ [bbj] (Nissim 1981) where nouns may have a LH sequence pre-associated to the segmental tier (10a), or can be a L followed by a floating ($\textcircled{\text{H}}$) (10b) or a H preceded by a floating ($\textcircled{\text{L}}$) (10c). These three patterns contrast against a simple linked H tone (10d). (Note that the $\downarrow$ indicates downstep, and the degree sign ° indicates a non-falling L tone before pause.)

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(10) Bamileke-Ghomala' – Contrast between associated and floating tones

(Nissim 1981: 153, 156, 249)

- | | | | | | | | |
|----|---------------|-------------------------|---|-------------------------|----------|------------------------------------|----------------------|
| a. | LH | <i>bvǎ</i> | → | <i>bvǎ</i> | 'dog' | <i>thá</i> [↓] <i>bvǎ</i> | 'head of the dog' |
| b. | L(<u>H</u>) | <i>bàp</i> (<u>H</u>) | → | <i>bàp</i> [°] | 'animal' | <i>thá</i> <i>bàp</i> [°] | 'head of the animal' |
| c. | (<u>L</u>)H | (<u>L</u>) <i>dhá</i> | → | <i>dhá</i> | 'spouse' | <i>thá</i> [↓] <i>dhá</i> | 'head of the spouse' |
| d. | H | <i>fá</i> | → | <i>fá</i> | 'parent' | <i>thá</i> <i>fá</i> | 'head of the parent' |

For maximum clarity, we provide autosegmental diagrams for these four contrasts, in (11).

(11) Bamileke-Ghomala' – Autosegmental diagrams of floating tone contrasts

- | | | | | | | | |
|----|--------------------|----|----------------------------|----|--------------------|----|-------------------|
| | L H | | L (<u>H</u>) | | (<u>L</u>) H | | H |
| a. | <i>bv</i> <i>ǎ</i> | b. | <i>b</i> <i>a</i> <i>p</i> | c. | <i>dh</i> <i>ǎ</i> | d. | <i>f</i> <i>á</i> |

The exact evidence for these floating tones involves both their surface pronunciation and the tonal effects they have on their environment. We turn to this in §3.

Just as floating tone can be part of the representation of lexical morphemes, it can also co-accompany grammatical morphemes, e.g., affixes and clitics. In Chichewa [nya] (Hyman & Mtenje 1999), there are two otherwise homophonous derivational suffixes which appear in the same post-verbal field: toneless causative *-its* versus intensive *-its*(H) with a floating tone, in (12).⁴

(12) Chichewa – Homophonous suffixes with and without (H) Hyman & Mtenje 1999: 95, 98

- | | | | |
|----|---|---|---------------------------------------|
| a. | No derivational suffixation: | | |
| | <i>thandiz-a</i> | → | <i>thandiz-a</i> 'help!' |
| b. | Causative suffixation: | | |
| | <i>thandiz-its-a</i> | → | <i>thandiz-its-a</i> 'cause to help!' |
| c. | Intensive suffixation: | | |
| | <i>thandiz-its</i> (<u>H</u>)- <i>a</i> | → | <i>thandiz-its-á</i> 'help a lot!' |

Such grammatical floating tones are very common in the derivational and inflectional morphology of African (and other) languages.

In languages with floating tones, typically they only appear on a minority of morphemes, whereas most morphemes do not have floating tones. However, in

⁴These forms are further subject to penultimate lengthening, H tone anticipation, and default L insertion (cf. Hyman & Mtenje 1999: 128, endnote 7 and references therein; for a recent description and analysis of the phonology of Chichewa, see Downing & Mtenje 2017).

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the Ijoid language Izon [ijc], [Rolle \(2021\)](#) posits that virtually all lexical items sponsor floating tones, which dock to the following morpheme (and frequently override that morpheme's tone). Representative examples of the 'ubiquity' of floating tone are in (13), which show four different patterns on nouns: either a $\textcircled{\text{L}}$ or $\textcircled{\text{H}}$ floating tone alone (13a–b), or a $\textcircled{\text{L}}\textcircled{\text{H}}$ or $\textcircled{\text{H}}\textcircled{\text{L}}$ sequence (13c–d) (association of these floating tones is sometimes vacuous, as in 13c).

- (13) Izon – Representative examples of ubiquitous floating tone on virtually all lexical items

- | | | | | | | |
|----|--|--|-------------|---|------------|-------------|
| a. | Ends in $\textcircled{\text{L}}$ | <i>wún</i> $\textcircled{\text{L}}$ | <i>ẹ́rì</i> | → | [wún ẹ̀rì] | 'see sand!' |
| b. | Ends in $\textcircled{\text{H}}$ | <i>fù</i> $\textcircled{\text{H}}$ | <i>ẹ́rì</i> | → | [fù ẹ́rì] | 'see salt!' |
| c. | Ends in $\textcircled{\text{L}}\textcircled{\text{H}}$ | <i>tà</i> $\textcircled{\text{L}}\textcircled{\text{H}}$ | <i>ẹ́rì</i> | → | [tà ẹ́rì] | 'see wife!' |
| d. | Ends in $\textcircled{\text{H}}\textcircled{\text{L}}$ | <i>wò</i> $\textcircled{\text{H}}\textcircled{\text{L}}$ | <i>ẹ́rì</i> | → | [wò ẹ́rì] | 'see him!' |

Another potential case of ubiquitous floating tone concerns languages with stem or word 'tone melodies' which predictably map onto the corresponding tone-bearing units (see [Hyman 2003](#) for a brief history of the concept of 'tone melody' in African linguistics). Mende [men] is perhaps the most famous language in this regard ([Leben 1973](#), [Leben 1978](#)), but a clearer case of predictable tone association (with fewer counterexamples) comes from the Bantu language Kukuya [kkw] ([Paulian 1975](#), [Hyman 1987](#)). This language has exactly five tone melodies L, H, LH, HL, and LHL. These map onto the stem constituent in a predictable fashion, shown in (14) (association ignores prefixes, which are outside of the stem).

- (14) Kukuya – Five tone melodies (on stem)

- | | | | |
|----|-----|-------------------|---------------------------|
| a. | L | <i>(kì-)bàà</i> | <i>(kì-)bàlàgà</i> |
| | | 'jealousy' | 'to change route' |
| b. | H | <i>(mà-)báá</i> | <i>(lì-)bálágá</i> |
| | | 'cheeks' | 'fence' |
| c. | LH | <i>(mò-)sàá</i> | <i>m^wàràgí</i> |
| | | 'seed necklace' | 'younger brother' |
| d. | HL | <i>(kì-)káà</i> | <i>(kì-)kàràgà</i> |
| | | 'to grill' | 'to be entangled' |
| e. | LHL | <i>(ndé-)kàây</i> | <i>(ndé-)kàlógí</i> |
| | | 'he loses weight' | 'he turns around' |

In such a language, words with the LH melody such as *m^wàràgí* 'younger brother' have only one association possibility (trisyllabic L.L.H) and thus cannot

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contrast against words of the shapes L.H.H, L.L.LH, etc. Tonal melodies are not traditionally analyzed as floating tones, because their association is predictable and systematic, and the trivial step of association is unnecessary to include in the underlying representation. However, strictly speaking, if tonal melodies are not associated in underlying representations, this is equivalent to being floating tones, shown in (15) (cf. also McPherson 2022).

(15) Kukuya – Predictable association of tone melody as ‘floating tones’

$$\begin{array}{ccc} \textcircled{\text{L}} \textcircled{\text{H}} & & \textcircled{\text{L}} \quad \textcircled{\text{H}} \\ m^w a r \text{ ə } g i & \rightarrow & m^w a r \text{ ə } g i \end{array}$$

3 Floating tone effects

As has been established, the primary reason for proposing floating tones is to account for tonal patterns which are conditioned by the presence of a morpheme and are not predicted based simply on the tones that are realized on that morpheme or any other morpheme around it. Such patterns we collectively refer to as ‘floating tone effects’, the empirical substance motivating a floating tone analysis.

The most straightforward of these is the presence of a tonal alternation on a neighboring morpheme, e.g., the fact that Guinean Kpelle *yàlà* ‘lion’ causes a following L to become H as in *yàlà-yàà* ‘lions’ (cf. *gbònò-yàà* ‘rings’ in example 8 above). In this section, we shall show additional floating tone effects in three interrelated patterns: blocking effects, downstep and upstep/raising (i.e., register effects), and final level Ls.

3.1 Blocking effects

A widespread property of tone is its ability to spread onto neighboring environments, e.g., a H spreading one position to the right (cf. Smolders 2026 [this volume]). When two tonally identical morphemes should, according to the rules of the language, spread their H tone, but only one of them does, one common analysis is to posit that there is a covert floating tone which ‘blocks’ spreading.

Such effects are common in the Grassfields languages of Cameroon (cf. Hamm et al. 2026 [this volume]). For example, in Aghem [agq] (Hyman 1979, 2003), H-toned nouns can be split into different groups based on their tonal effects, though they are pronounced identically in isolation. In one type of H words, the

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H spreads onto a following L tone word (16a), whereas in another type of H words there is no tone spreading (16b).⁵

(16) Aghem – Two types of H-toned nouns

- a. *kí-fé* ‘leg’ (*kí-*)*fé* *kíà* ‘your (SG) leg’
 b. *kí-wó* ‘hand’ (*kí-*)*wó* *kíà* ‘your (SG) hand’

This contrast falls out naturally in an analysis with floating tones: the first type ends in a H which spreads one position to the right (17a), but the second type has a final floating $\textcircled{L}$ which blocks this spreading operation (17b). (Additional evidence for the presence of this floating $\textcircled{L}$ is given in the next section.)

(17) Aghem – Blocking effect from floating tone

- | | | | | | | |
|----|---|---------------------|---|-------|---|-----------------------|
| H | H | L | | H | H | L |
| | | | | | | |
| a. | k | i | - | f | e | k i a |
| | → | | | (k i) | - | f e k i a |
| | | | | | | |
| | H | H $\textcircled{L}$ | L | | H | H $\textcircled{L}$ L |
| | | | | | | |
| b. | k | i | - | w | o | k i a |
| | → | | | (k i) | - | w o k i a |

3.2 Downstep and upstep/raising

Another common motivation for floating tones is to explain otherwise unpredicted register changes which lower or raise the pitch range, i.e., ‘downstep’ and ‘upstep’.⁶ We can demonstrate downstep using the Aghem data from Section 3.1: /*kí-fé*/ ‘leg’ vs. /*kí-wó* $\textcircled{L}$ / ‘hand’ with a floating tone. When these nouns appear before a H-toned modifier such as *kín* ‘this’ (agreeing in noun class), they again show different tonal patterns. Without a floating tone, both the noun and the modifier are realized at the same H tone level (18a), whereas when there is a floating $\textcircled{L}$ the H tone of the modifier is realized within a lower register, indicated by the downstep mark ↓ (18b).

⁵In this example the two nouns have an underlying /H/ prefix which is deleted before most modifiers. See the original sources for details.

⁶Note that downstep and upstep do not always result from the effect of a floating tone, but may also be caused by associated tones, as in the automatic downstepping (‘downdrift’) of a H tone by a preceding associated L tone, or cases where a H is automatically downstepped after another H (cf. Stewart 1965, 1983; Connell 2011; Snider, this volume, among others). Downstep may even be a phonological primitive present in underlying representation, independent of tone, e.g., in the African language Baga Pukur [bcg] (Rochant 2023; cf. also Lionnet 2022, 2025, on the Oceanic languages Paici [pri], Drubea [duf], and Numèè [kdk] of New Caledonia).

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(18) Aghem – Two types of H-toned nouns

- a. /kí-fé kín/ → (kí-)fé kín ‘this leg’
 b. /kí-wó $\textcircled{L}$ kín/ → (kí-)wó $\downarrow$ kín ‘this hand’

As emphasized by Hyman (2003: 160), by positing a single floating tone we can simultaneously account for both the downstep pattern in (18b) and the blocking effect in (16b). That there are two pieces of evidence of a floating $\textcircled{L}$ at the end of the root /wó $\textcircled{L}$ / shows that the analysis is not circular or ad hoc. We will return to this argument in Section 5, where we discuss alternatives.

An interesting case of floating tones in sequence is found in Bamileke-Medumba [byv], another Grassfields language (Voorhoeve 1971). In Medumba, nouns which are pronounced as H in isolation can correspond to different underlying forms: a simple underlying /H/ (19a), a H with a preceding floating tone / $\textcircled{L}$ H/ (19b), and a H with a preceding and following floating tone / $\textcircled{L}$ H $\textcircled{L}$ / (19c). They are differentiated when they combine in a ‘noun of noun’ genitive construction linked only by a floating $\textcircled{H}$ tonal morpheme (another example of this linking function, cf. Section 2.1). When there is only a H tone on both sides of the tonal morpheme $\textcircled{H}$ (19d), the result is a level H tone across the phrase (the initial floating tone of $\textcircled{L}$ tí ‘tree’ has no effect here). In contrast, the final floating tone of $\textcircled{L}$ yú $\textcircled{L}$ causes downstep of any H tone which follows, in (19e). What is remarkable is the case of (19f), where the tonal morpheme $\textcircled{H}$ is flanked by $\textcircled{L}$ tones: the second noun ‘child’ is realized on a double downstepped level, a greater interval than a single downstep.

(19) Medumba – Effects of multiple floating tones

- a. /sán/ → sán ‘bird’
 b. / $\textcircled{L}$ tí/ → tí ‘tree’
 c. / $\textcircled{L}$ yú $\textcircled{L}$ / → yú ‘thing’
 / $\textcircled{L}$ mén $\textcircled{L}$ / → mén ‘child’
 d. / $\textcircled{L}$ tí + $\textcircled{H}$ + sán/ → tí sán ‘tree of bird’
 e. / $\textcircled{L}$ yú $\textcircled{L}$ + $\textcircled{H}$ + sán/ → yú $\downarrow$ sán ‘thing of bird’
 f. / $\textcircled{L}$ yú $\textcircled{L}$ + $\textcircled{H}$ + $\textcircled{L}$ mén $\textcircled{L}$ / → yú $\downarrow\downarrow$ mén ‘thing of child’

One can understand this as the $\textcircled{L}$ before the second noun causing the first downstep and the $\textcircled{L}$ before the tonal morpheme $\textcircled{H}$ as causing the second downstep, resulting in a H $\downarrow\textcircled{H}$ $\downarrow$ H sequence. The downsteps then ‘combine’ to form the [H $\downarrow\downarrow$ H] surface form.⁷

⁷Another option for (19f) is to elongate the vowel of ‘thing’ and produce a falling contour with successive downsteps, i.e., yú $\downarrow$ yú $\downarrow$ mén (Voorhoeve 1971: 48).

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Although floating L tones often trigger downstep, they occasionally instead condition a preceding ‘upstep’ whereby the register is raised rather than lowered. One example comes from Ebira [igb] (Scholz 1976), a language with a three-way H/M/L distinction. In this language, H tones (including sequences of H tones) are automatically raised to super-high (S) when they precede a L tone, both within words (20a) and across words (20b).⁸

(20) Ebira – Change of H to super-high (S) before a L-toned syllable (Scholz 1976: 67)

- | | | | | |
|----|--------------|---|--------------|-----------------------------|
| a. | /àgùgù/ | → | [àgùgù] | ‘crocodile’ |
| b. | /ínómí nìní/ | → | [ínómí nìní] | ‘birds’ (i.e., bird PLURAL) |

This upstep/raising process applies opaquely in cases where there is a latent L tone which does not appear on the surface. Positing a floating ① in these contexts unifies the triggering environments: the change to S applies before any kind of L tone on the tonal tier (whether associated or not). The floating ① may be present in the underlying representation (where it minimally contrasts in the same position with words without floating tone – 21a and 21b), or it may be a derived floating ① which comes from a pre-associated L tone which becomes delinked due to operations like H tone spreading (21c) and vowel elision (21d).

(21) Ebira – Change of H to S before floating ① (Scholz 1976: 41,63,68,71,72,82)

- | | | | | |
|----|------------|---------|-----------|-----------------------|
| a. | /hámá wá/ | → | [hámá wá] | ‘imitate them!’ |
| | /cémá① wá/ | → | [cěmǎ wá] | ‘lift them!’ |
| b. | /ōzí émī/ | → | [ōzí émī] | ‘my son’ |
| | /ōzí ①yọ/ | → | [ōzí yọ] | ‘how is the child?’ |
| c. | /ó-wèyí/ → | ó-wé①yí | → | [ǒwéyí] ‘a small one’ |
| d. | /dọ òré/ → | d-ó①ré | → | [dǒré] ‘get glass’ |

3.3 Final level L tone

A final type of floating tone effect comes from languages which contrast two different types of pre-pausal L tones: one L which descends to the bottom of one’s pitch range (transcribed simply L) versus another L which has a level pitch and does not descend as such (transcribed as L° with a degree symbol). Under a

⁸Given that the tones following the raised tone go unaffected by this raising process (as in 20b), it may be more appropriate to simply call this ‘H-raising’ rather than ‘upstep’. Our main point remains, however: this same process is triggered equally by associated and floating L tones.

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floating tone analysis, the level L° is due to a final floating tone ($\textcircled{H}$ or $\textcircled{M}$) which blocks the utterance-final fall of the ordinary L.

Yet another Grassfields language can demonstrate this pattern, this time from Bamileke-Fe’fe’ [fmp] (Hyman 1972) with a basic L/M/H contrast. Example (22a) demonstrates a contrast between two types of Ls. The noun *pàà* ‘wing’ in isolation is pronounced with descending pitch before pause, a cross-linguistically widespread process of final downgliding. In contrast, the noun *mbàà* ‘meat’ has the low level pitch without this downgliding. In context (22b), we see that ‘wing’ stays L when followed by the proximal demonstrative *lè* ‘this’, while ‘meat’ is realized with a $\widehat{LM}$ rising contour.

(22) Fe’fe’ – Two types of floating Ls (Hyman 1972: 131–139)

- a. *pàà* ‘wing’ *mbàà* ‘meat’
- b. *pàà* *lè* ‘this wing’ *mbàà* *lè* ‘this meat’

We can account for this contrast by positing a final floating $\textcircled{M}$ on ‘meat’. When followed by a L tone, this $\textcircled{M}$ docks to the preceding word resulting in a contour, but utterance-finally it is not realized, and simply blocks the application of final L ‘downgliding’ (i.e., only L’s which are absolutely final on the tonal tier undergo this process). Although $\textcircled{M}$ does not associate per se, it still affects the surface pronunciation, as shown in the autosegmental representations in (23).

(23) Fe’fe’ – Floating $\textcircled{M}$ blocks final L ‘downgliding’

- | | | |
|-----------------------|---|-----------------------|
| L $\textcircled{M}$ | | L $\textcircled{M}$ |
| a. mb a a # | → | mb a a ° # |
| L $\textcircled{M}$ L | | L $\textcircled{M}$ L |
| b. mb a a l e | → | mb a a l e |

4 Typological parameters

Once floating tones are assumed, they become a phonological object which we can evaluate like any other less-abstract object. In this section, we explore several typological issues which arise with floating tones, having to do with inventory (what tones can float and in which contexts), locality (how locally does the floating tone associate), prosodic status (can tonal morphemes pattern as affixes, cli-

tics, or even independent words), directionality (do floating tones dock leftwards or rightwards), and markedness (what phonological conditions favor/disfavor tone docking).

4.1 Inventory

Thus far, we have seen many different kinds of floating tones. This includes various qualities such as floating $\textcircled{\text{H}}$'s, $\textcircled{\text{L}}$'s, and even $\textcircled{\text{M}}$'s, appearing in various combinations to form a complex floating tone sequence such as $\textcircled{\text{H}}\textcircled{\text{L}}$ in Izon (13), $\textcircled{\text{L}}\textcircled{\text{H}}\textcircled{\text{M}}$ in Kunama (6), etc. At the same time, we saw that floating tones may appear on their own as 'tonal morphemes' or can come before or after a segmental sponsor (sometimes both – Medumba in 19c). We can thus ask: are there any limits on a language's inventory of floating tones?

One possibility is that only tones which can be associated can be floating. In other words, if we find floating $\textcircled{\text{H}}$ underlyingly, we will also find underlying associated H. While tempting, there are in fact analyses which posit that certain tone qualities must come in floating. We call these 'inherently floating tones'. This is how Daly & M. (2007) characterize the contrast between H and L in the Mexican language Peñoles Mixtec [mil], where these two tones systematically behave differently. To account for this asymmetry, Daly and Hyman propose to analyze L tones as inherently floating – i.e., all H tones are underlyingly linked to a segment, while all L tones enter the derivation floating. While this remains an analytical possibility, this has not been proposed to account for any African tonal language to date.

The closest cases of inherently floating tone we can think of in Africa come from grammatical tone patterns in three languages (cf. Rolle & Bickmore 2026 [this volume]). One is Fe'Fe', already seen in Section 3.3 above, which has a lexical M vs. L distinction, but an additional H tone contrast is found only in grammatical contexts. For example, *vhū* 'ashes', *mōh* 'fire', and *mūū* 'child' are underlyingly M, but are replaced by a tonal morpheme $\textcircled{\text{H}}$ in connective constructions, e.g., *vhú mōh* 'ashes of fire', *vhú mōh mūū* 'ashes of fire of child' (Hyman 1976: 127–128). If this tonal morpheme is interpreted as floating, then this would entail that all H tonemes come in floating. Two additional examples are Kisi [kss] (Childs 1995) and Igala [igl] (Ejeba 2016), which both have a super-high (S) tone that occurs only in certain grammatical contexts, primarily negation. In Kisi, negation is expressed as a change to S on the verb plus a final particle *lé* NEG (24), while in Igala negated clauses the tone on subject clitics (scl) becomes S co-accompanied by the clitic *=ń* NEG (25).⁹

⁹A reviewer brings up an alternative that Igala S is derived from a floating low (comparable

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- (24) Kisi – Super-high tone (S) restricted to grammatical tone patterns (Childs 1995: 49)

a.	à	dàtá	yá	b.	à	dàtǎ	yá	lé
	you	condemn	me		you	condemn.NEG	me	NEG
	‘You condemned me.’				‘You didn’t condemn me.’			

- (25) Igala – S toneme restricted to negation (Ejeba 2016: 326)

òdʒòrú=ǔ	dʒē	èg ^w à=ń
Ojoru=NEG.SCL	eat	beans=NEG
‘Ojoru did not eat beans.’		

Another potential type of restriction is not whether certain tones can float, but whether they are permitted to float if they are adjacent to an associated identical tone. For example, one could imagine a language with a ban on $H(\bar{H})$ (a ban in line with the ‘Obligatory Contour Principle’). While most of the examples we have encountered incidentally comply with such a restriction, we cannot hold this to be a true universal. Indeed, we have seen comparable sequences, e.g., in the Izon word $tà(\bar{L})(\bar{H})$ ‘wife’ (13c), whereby a linked L is followed by a floating $\bar{L}$ (see Rolle 2021 for the original arguments). Another case comes from Dagbani (Hyman 1993: 242), where monosyllabic H nouns are interpreted as having a floating $(\bar{H})$ suffix in order to account for an unexpected limitation in a spreading rule, e.g., $bá-(\bar{H})$ ‘father’.

In total, while it is certainly the case that an individual language may impose a restriction on its inventory of floating tones or the types of environments they may appear adjacent to, with rare exceptions, floating tones are usually a subset of associated tones.

4.2 Locality

In the examples we have seen thus far, floating tone has a local effect: it docks to an adjacent syllable/morpheme, it triggers downstep or raising of an adjacent tone, or it blocks an adjacent tone from triggering a tonal operation (e.g., spreading, or final L ‘downgliding’). In general, floating tones typically target local environments. This is only a strong tendency, however, as non-local floating tone patterns are attested as well (depending on one’s interpretation), usually under specific circumstances.

to Ebira in 21). Regardless, the Igala data demonstrates our main point about tonal inventories, irrespective of the chosen analysis.

Non-local linking of floating tones can be the result of ‘prominence-based association’. In this type, a floating tone associates to a phonologically-prominent target in the relevant domain, such as to stress or to a particular boundary (or a fixed position relevant to that boundary). In order to qualify as prominence-based association, we must find evidence independent from floating tone association for this position being prominent. One example comes from the Bantu language Zanzibar Simakonde [kde] (Manus 2014), hereafter simply Makonde. As is common in Eastern Bantu, the penultimate position of the phrase can be identified as prominent since it undergoes penultimate vowel lengthening. Manus demonstrates that pre-verbal tense/aspect/mood (TAM) prefixes can co-vary with one of four floating tone patterns: $\underline{\text{L}}$, $\underline{\text{H}}$, $\underline{\text{L}}\underline{\text{H}}$, or $\underline{\text{L}}\underline{\text{H}}\underline{\text{L}}$. While the exact pattern is idiosyncratic to the TAM prefix, where they appear is predictable: they are always realized on the penultimate syllable, the position of prominence. In the examples in (26), the penultimate position shows both penultimate lengthening and floating tone association (data from Manus 2014: 267–268). The origin of the floating tones and its surface position are underlined.¹⁰

(26) Makonde – Prominence-based association of floating tones

- | | | | |
|----|-------------------|--|---------------------|
| a. | CONSECUTIVE | /ni- <u>ka</u> $\underline{\text{H}}$ - | takatukil-a/ |
| | ‘and I stood up’ | ni- <u>ka</u> - | takatuk <u>i</u> la |
| b. | FUTURE | /ni- <u>nda</u> $\underline{\text{L}}\underline{\text{H}}$ - | takatukil-a/ |
| | ‘I will stand up’ | ni- <u>nda</u> - | takatuk <u>i</u> la |
| c. | REMOTE PAST | /ní- <u>ndi</u> $\underline{\text{L}}\underline{\text{H}}\underline{\text{L}}$ - | takatukil-a/ |
| | ‘I stood up’ | ní- <u>ndi</u> - | takatuk <u>i</u> la |
| | 1SG- TAM- | | stand.up-INFL |

Of course, as with all uses of the floating tone concept, it is debatable whether floating tones are the best way to capture these data (as opposed to the tones being pre-linked to a specific affix and then delink and move to the penult). However, our main point remains: the segmental component of the TAM category appears non-local to the tonal component.

Not all non-local floating tone association is due to prominence, however. Consider another Bantu language, Luganda [lug] (Hyman & Katamba 2010), which contrasts H vs. L vs. Ø (toneless) on syllables/morphemes. Consider the example in (27) (in the underlying forms in this example, we denote H with an acute

¹⁰Note that the H tone on the 1SG prefix *ní-* in (26c) which spreads onto the following syllable constitutes a separate type of grammatical process, independent of those floating tones attracted to the penultimate position. See Manus (2014: 268ff.) for details.

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accent on its host vowel, L with a grave accent, and Ø as unmarked).¹¹

(27) Luganda - Non-local tone association across clitics (Hyman & Katamba 2010: 95)

- | | | |
|----|---|-------------------------------|
| a. | /te-y-a-túùk-à/ | (output of lexical phonology) |
| | tè-yà-túùk-à | (surface output) |
| | NEG-3SG-arrive-FV | |
| | ‘s/he didn’t arrive’ | |
| b. | /te-y-a-túùk-à | (H)bùli lunákù/ |
| | tè-yà-túùk-à | bùli lùnákù |
| | NEG-3SG-arrive-FV | every day |
| | ‘s/he didn’t arrive every day’ | |
| c. | /te-y-a-túùk-à | na=byaa=(H)bùli mu-limi/ |
| | tè-yà-túùk-à | nà=byàà=bùli mù-límí |
| | NEG-3SG-arrive-FV | with=those.of=every CL-farmer |
| | ‘s/he didn’t arrive with those of every farmer’ | |
| d. | /te-y-a-túùk-à | na=byaa=mu-limi/ |
| | tè-yà-túùk-à | nà=byàà=mù-límí |
| | NEG-3SG-arrive-FV | with=those.of=CL-farmer |
| | ‘s/he didn’t arrive with those of the farmer’ | |

Some words have an initial floating (H), such as (H)bùli ‘every’ which appears initial within its noun phrase. While the verb occurs with final L in (27a), the floating (H) of ‘every’ is realized on the final syllable of the verb in (27b). (We underline the relevant positions in the example.) Example (27c) demonstrates that even when the trigger (H)bùli is separated by one or more proclitics (here *na*= ‘with’ and *byaa*= ‘those of’), it still assigns its tone non-locally to the final position of the verb. Example (27d) shows that these clitics do not otherwise result in a H tone on the verb.

An even more extreme example of non-prominence-based non-local association comes from the counting rules of Kuria [kuj] (Mwita 2008, Marlo et al. 2015). As in Makonde in (26) above, inflectional prefixes idiosyncratically sponsor one of several floating tone patterns. However, unlike in Makonde these do not associate to the same position; rather, each of them targets a different position within the verbal stem. Depending on the grammatical context, a floating tone may be assigned to the first ((H)¹ in 28b), second ((H)² in 28c), third ((H)³ in 28d), of fourth

¹¹We refer the reader to the original source for details on how toneless syllables get tone on the surface (e.g., involving spreading, boundary tones, among other factors).

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mora of the stem ($\textcircled{H}^4$ in 28e), and even to the first and fourth simultaneously ($\textcircled{H}^1\textcircled{H}^4$ in 28f). An example set is in (28) using the verb *kirigiit*- ‘scrub’, which forms a stem constituent labeled ‘ Σ ’. As above, the position that floating tones associate to is underlined.¹²

(28) Kuria – Contrastive association of floating tones (Mwita 2008: 305–336)

- a. $\emptyset$ – HORTATORY IMPERATIVE
/to-tá-(kirigiit-a) Σ / → to-tá-(kirigiit-a) Σ
‘let us scrub’
- b. $\textcircled{H}^1$ – HABITUAL PAST
/n-to-ogá $\textcircled{H}^1$ -(kirigiit-ire) Σ / → n-to-ogá-(kírigiit-ire) Σ
‘we used to scrub (then)’
- c. $\textcircled{H}^2$ – PAST PROGRESSIVE (FOCUS)
/n-to-oga $\textcircled{H}^2$ -(kirigiit-ire) Σ / → n-to-oga-(kírigiit-ire) Σ
‘(indeed) we have scrubbed (today)’
- d. $\textcircled{H}^3$ – REMOSTE FUTURE (FOCUS)
/n-to-re $\textcircled{H}^3$ -(kirigiit-a) Σ / → n-to-re-(kirigíit-a) Σ
‘we will scrub (then)’
- e. $\textcircled{H}^4$ – INCEPTIVE
/to-ra $\textcircled{H}^4$ -(kirigiit-a) Σ / → to-ra-(kirigíit-a) Σ
‘we are about to scrub’
- f. $\textcircled{H}^1\textcircled{H}^4$ – NARRATIVE PAST
/to-gá $\textcircled{H}^1\textcircled{H}^4$ -(kirigiit-a) Σ / → to-gá-(kírigíit-a) Σ
‘(and) we scrubbed’

As extensively argued in Marlo et al. (2015), there is no evidence that any of these positions is particularly prominent in Kuria (and some are rare or otherwise unattested cross-linguistically, i.e., $\textcircled{H}^3$ and $\textcircled{H}^4$).

In the above non-local patterns, there is no tone intervening between the origin of the floating tone and its final surface position. Cases of an associated tone being transparent to non-local association are extremely rare. One case outside of Africa is the Mexican language San Miguel el Grande Mixtec [mig] (Mak 1950, Zimmermann 1950) whereby a floating $\textcircled{H}$ which normally docks locally skips over the first tone under exceptional conditions (e.g., $\textcircled{H}$ + ML → M $\textcircled{H}$). However, in most examples in Africa (as worldwide), non-locality requires that there be no intervening tones.¹³

¹²Note that the actual surface forms in Kuria differ slightly from what is provided in (28), due to orthogonal tone spreading rules, not shown.

¹³One possible instance of non-local association over an intervening tone is in Tommo So [dto]

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4.3 Prosodic status

Segmental grammatical morphemes are often catalogued as to how tightly they ‘cohere’ with their environment, what we call their ‘prosodic status’. Morphemes may be tightly bound to their base as affixes, loosely bound as clitics, or not bound at all as independent function words. If ‘tonal morphemes’ as defined in Section 2.1 are just like regular segmental morphemes, only without any segmental content, then we should expect them to demonstrate the same variety of prosodic statuses. Such issues of prosodic status are unfortunately rarely discussed with respect to tonal morphemes and floating tones in general.

One overt attempt at assessing different kinds of tonal morphemes explicitly with these prosodic categories in mind is Van de Velde’s (2008, 2009) analysis of Eton [eto]. Van de Velde starts from a simple logical extension: since tonal rules are sensitive to different kinds of boundaries (e.g., affix, clitic, and word), the tonal behavior of floating tones should reveal the prosodic status (affix, clitic, or word) of the tonal morphemes they are part of. That is, it should not matter whether the tones that undergo tonal rules are underlyingly associated with segments or not, and we should be able to find affixes, clitics, and independent grammatical words that consist only of tonal material. What Van de Velde shows is that floating tones do indeed have three distinct sets of patterns, which correspond exactly to the patterns found with tones unequivocally associated with segmental affixes, clitics, and independent grammatical words.

To illustrate the first type, let us look at the augment prefix (used on nouns in certain contexts). This prefix has several allomorphs based on the noun class of the noun to which they attach, some of which consist of a vowel and a H tone, as in class 7 /i-/ in (29a’, b’), while others consist only of a floating $\textcircled{\text{H}}$ tone, as is the case for nouns of class 5 illustrated in (29a, b). As can be seen in (29a, b), the floating $\textcircled{\text{H}}$ of the augment docks to a following L-toned syllable and thereafter spreads rightward onto all following L-toned syllables to form a HL contour tone with the last L in the sequence. This directly mirrors the behavior of the H tone from the segmentally-overt allomorphs of the augment prefix in (29a’, b’).

(McPherson 2014, McPherson & Heath 2016), though the original sources expressly do not adopt a floating tone analysis for the relevant tonal patterns. See these sources and Rolle (2018: 30–31, 81–82) for details.

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- (29) Eton – Tonal morpheme as a prefix / $\textcircled{\text{H}}$ -/
- | | | | | |
|-----|-------------------------------------|---------|---|--------------------------------|
| a. | / $\textcircled{\text{H}}$ -è-dʒɔŋ/ | dì | → | <u>é</u> -dʒɔŋ dì |
| | AUG5-CL5-hole | CL5.DEM | | ‘this hole’ |
| a’. | /í-bì-sínà / | bí | → | <u>í</u> - <u>bí</u> -sínà bí |
| | AUG7-CL7-hole | CL7.DEM | | ‘these cats’ |
| b. | / $\textcircled{\text{H}}$ -è-dʒɔŋ/ | dì | → | <u>é</u> -dʒɔŋ dì |
| | AUG5-CL5-clan | CL5.DEM | | ‘this clan’ |
| b’. | /í-bì-lòlì / | bí | → | <u>í</u> - <u>bí</u> -lwàlì bí |
| | AUG7-CL5-clan | CL7.DEM | | ‘these ducks’ |

In contrast, consider two other tonal morphemes: those that act as a connector CON= in a ‘noun of noun’ construction (30), and those that act as a locative preposition (30). These are analyzed as a proclitic and as a word, respectively. Both have several allomorphs, consisting of either a (C)V syllable with an associated H tone (examples a’ and b’ in 30 and 31), or only a floating $\textcircled{\text{H}}$ tone (examples a and b). The floating $\textcircled{\text{H}}$ allomorph of each of these two morphemes shows distinct behavior compared to the prefixal $\textcircled{\text{H}}$ seen in (29a–b). Firstly, while in both cases, the $\textcircled{\text{H}}$ is realized on the following position (on the class prefix in all examples in 30 and 31), it does not spread any further after association. Compare the falling tone in the surface form of the root in (29b) above to the L tone on the root in (30b) and (31b) below. Independent word tonal morphemes further condition downstep when they are followed by another H tone (31a), which is not seen with either prefixal $\textcircled{\text{H}}$ (29a) or proclitic $\textcircled{\text{H}}$ (30a). In each of these cases, the behavior of the tonal morpheme mirrors the tonal behavior of its segmental allomorph.

- (30) Eton – Tonal morpheme as a proclitic / $\textcircled{\text{H}}$ =/
- | | | | | |
|-----|-------------|------------------------------------|---|-----------------------------|
| a. | /bj-ǎ | $\textcircled{\text{H}}$ =bì-sínà/ | → | bj-ǎ <u>bí</u> -sínà |
| | CL8-claw | CL8.CON=CL8-cat | | ‘the claws of the cats’ |
| a’. | /bj-ǎ | bí=ì-sínà / | → | bj-ǎ b= <u>í</u> -sínà |
| | CL8-claw | CL8.CON=CL8-cat | | ‘the claws of a cat’ |
| b. | /mà-vùl | $\textcircled{\text{H}}$ =bì-lòlì/ | → | mà-vùl <u>bí</u> -lwàlì |
| | CL6-feather | CL6.CON=CL7-duck | | ‘the feathers of the ducks’ |
| b’. | /mà-vùl | má=ì-lòlì/ | → | mà-vùl m= <u>í</u> -lwàlì |
| | CL6-feather | CL6.CON=CL7-duck | | ‘the feathers of a duck’ |

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- (31) Eton – Tonal morpheme as an independent word (H)
- a. / $\textcircled{\text{H}}$ è-dʒóŋ/ → $\underline{\acute{\epsilon}}\text{-}\downarrow dʒóŋ$
LOC CL5-hole ‘in the hole’
- a’. /á mǝ-dʒóŋ/ → á $\underline{mǝ}\text{-}\downarrow dʒóŋ$
LOC CL6-hole ‘in the holes’
- b. / $\textcircled{\text{H}}$ è-dʒǝŋ/ → $\underline{\acute{\epsilon}}\text{-}dʒǝŋ$
LOC CL5-clan ‘in the clan’
- b’. /á mǝ-dʒǝŋ/ → á $\underline{mǝ}\text{-}dʒǝŋ$
LOC CL6-clan ‘in the clans’

We summarize the three sets of tonal patterns in Table 1.

Table 1: Three types of tonal morphemes in Eton

		L.L	L.H	Spreading after association	Downstep
Prefix	/ $\textcircled{\text{H}}$ -/	[$\textcircled{\text{H}}$. $\textcircled{\text{H}}$ L]	[$\textcircled{\text{H}}$.H]	Yes	No
Proclitic	/ $\textcircled{\text{H}}$ =/	[$\textcircled{\text{H}}$.L]	[$\textcircled{\text{H}}$.H]	No	No
Word	/ $\textcircled{\text{H}}$ /	[$\textcircled{\text{H}}$.L]	[$\textcircled{\text{H}}$. $\downarrow$ H]	No	Yes

Given the parallels with their segmental counterparts, extending the affix/clitic/word distinction to tonal morphemes appears to be warranted for Eton. However, there exist other cases where floating tonal morphemes show distinct tonal behavior (e.g., triggering vs. not triggering spreading), but where a comparable prosodic categorization as in Eton is without support (we will turn to one of these shortly in Section 4.5 from Giphende). We thus emphasize that just as analyzing tonal morphemes as floating tones is an analysis, so is their prosodic status and the constituents they form.

4.4 Directionality

Another parameter of floating tones is directionality: A floating tone may dock to the right or to the left of its postulated origin depending on a number of factors. The easiest way to see directionality is to show how it applies differently in different varieties of a language. Consider the case of Igbo where a floating $\textcircled{\text{H}}$ tone functions to link two nouns in a ‘noun of noun’ construction (Hyman

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& Schuh 1974; see also Section 2.1 above). In Central Igbo [ibo], the $\widehat{H}$ tonal morpheme docks leftward to the end of the first noun (32a), while in Aboh Igbo [ukw] it docks rightward to the beginning of the second noun (32b). This is consistent with Aboh having regular H tone spreading (*ézi* → [ézi] ‘pig’), which Central Igbo does not (cf. [ézi]).

- (32) Igbo – Directionality of association of tonal morpheme
- a. Central Igbo: *àgbà* + $\widehat{H}$ + *èñwè* → *àgbà* *èñwè* ‘jaw of monkey’
 - b. Aboh Igbo: *ègbà* + $\widehat{H}$ + *èñwè* → *ègbà* *èñwè* ‘jaw of monkey’

The exact conditions on directionality vary greatly from language to language, and even family to family. When generalizing over floating tone directionality across Eastern Grassfields Bantu, Hyman & Tadadjeu (1976: 62) find several general criteria, summarized in (33).

- (33) Eastern Grassfields Bantu – Criteria for floating tone directionality:
- a. Realization: Associate in the direction that has greater tonal effect on surface
e.g., for an input L $\widehat{H}$ H, associating to the left would have a greater effect than associating to the right where the output would be surface-identical to L H
 - b. Boundary strength: Do not associate in a direction that crosses certain boundaries
e.g., for an input L $\widehat{H}$ [L (where ‘[’ indicates a stem boundary), associating to the left would not cross the stem boundary while associating to the right would
 - c. Naturalness: Associate in the direction of a more phonologically natural outcome
e.g., for an input $\widehat{HL}$ $\widehat{H}$ L involving a falling $\widehat{HL}$ contour, associating to the left would create a marked $\widehat{HLH}$ falling-rising contour which is worse than associating to the right creating two falling contours

4.5 Markedness

In short, a number of factors are potentially involved, and the way floating tones dock can only be partially predicted. This leads into our last parameter: phonological markedness.

One of the great insights of phonology is that sequences of segments, tones, and their combinations can be assessed as to whether they are relatively ‘marked’

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or ‘unmarked’ in terms of complexity or frequency, e.g., a word *spear* in English is less marked than *sphere* with an unusual [sf] onset. Particularly marked structures can be actively avoided in languages and structures with floating tones are no exception.

Consider the case of Ik [ikx], a Kuliak language of Uganda with a H vs. L tonal contrast (Schrock 2014: 121, 598). This language shows depressor consonant effects, whereby the pitch lowering effects of voiced obstruents has been phonologized (cf. Beavon-Ham & Pearce 2026 [this volume]). Consider the association of a floating $\textcircled{\text{H}}$ which appears with the subordinating conjunction clitic *na=* CONJ ‘if/when’. The floating tone associates to the syllable immediately following *na=* if that syllable begins with a sonorant or voiceless obstruent (i.e., non-depressors, 34a–b), but falls on *na=* itself if the following syllable begins with a voiced obstruent (i.e., a depressor, 34c) (L tones are not transcribed in the examples).

(34) Ik – Effect of depressor consonants on floating tone association

- a. /na $\textcircled{\text{H}}$ =ɲk-ese dád-a dakú-gwarí-oó/ → *na=ɲk-ese ...*
 CONJ=eat-SPS honey-NOM tree-top-ABL
 ‘if honey is eaten from the tree top’
- b. /na $\textcircled{\text{H}}$ =kək-ese ... → *na=kók-ese...*
 CONJ=close-SPS ‘if (SUBJECT) is tied...’
- c. /na $\textcircled{\text{H}}$ =gon-ese ... → *ná=gon-ese...*
 CONJ=look-SPS ‘if (SUBJECT) is looked at...’

In Ik, a floating tone docks rightward except if it would create a marked syllable (i.e., a H tone with a depressor consonant), in which case it docks leftward.¹⁴

It is just as common, however, to find floating tones insensitive to depressor consonant effects. And just as we saw in §4.3, floating tones even in the same language may not be equally sensitive to markedness conditions. One illustrative example comes from the Bantu language Giphende [pem] (Hyman 2017: 113–114). In Giphende, nouns have an underlying tone which is systematically changed depending on its syntactic position, e.g., /gì-kómbó/ ‘broom’ with underlying /L-H.H/ (where /gì-/ is a noun class prefix) is pronounced [gì-^lkómbó] when used as a syntactic object, [gì-kómbó] as a focused object or in a genitive, and [gì-kómbó] in predicative position (‘it’s a ’). Hyman divides these into four ‘tonal

¹⁴That the floating $\textcircled{\text{H}}$ tone can dock to the morpheme *na=* which introduces it is an instance of ‘self-association’. In many languages, self-association is itself marked and actively avoided (a markedness constraint which goes by various names in the literature – see McPherson 2014: 89, and Rolle 2018: 258 for discussion). Another kind of markedness is thus whether or not a floating tone can dock to the morpheme which sponsors it.

cases’ in (36). We label these as FOCUS.OBJECT FOC), GENITIVE (GEN), ACCUSATIVE (ACC), and PREDICATIVE (PRED) for our purposes, realized as the tonal morphemes labeled $\textcircled{H}_F$, $\textcircled{H}_G$, $\textcircled{H}_A$, and $\textcircled{H}_P$ respectively.¹⁵

These four morphemes can be assessed as to which conditions allow them to dock, and when they dock what their behavior is afterwards. As per the first aspect, FOCUS.OBJECT $\textcircled{H}_F$ only docks to all L-toned nouns, GENITIVE $\textcircled{H}_G$ docks as long as it would not be adjacent to a H, while ACCUSATIVE $\textcircled{H}_A$ and PREDICATIVE $\textcircled{H}_P$ always dock. And as per the second aspect, $\textcircled{H}_F$ systematically spreads in its only context (with all-L nouns), $\textcircled{H}_G$ and $\textcircled{H}_P$ only spread in this all-L context but otherwise do not spread. Finally, accusative $\textcircled{H}_A$ always spreads over L tones up to the next H, which it causes to downstep. This is summarized in Table 2.

Table 2: Tonal case in Giphende

Underlying tone of noun	Case $\textcircled{H}_F$ FOC	$\textcircled{H}_G$ GEN	$\textcircled{H}_A$ ACC	$\textcircled{H}_P$ PRED
L-L.L	$\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}.L$
L-L.L.L	$\textcircled{H}\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}\textcircled{H}.L$
L-L.H	L-L.H	$\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}.\downarrow H$	$\textcircled{H}\textcircled{H}.L$
L-L.H.L	L-L.H.L	$\textcircled{H}\textcircled{H}.L$	$\textcircled{H}\textcircled{H}.\downarrow H.L$	$\textcircled{H}\textcircled{H}.L$
L-H.L	L-H.L	L-H.L	$\textcircled{H}\textcircled{H}.\downarrow H.L$	$\textcircled{H}\textcircled{H}.L$
L-H.H	L-H.H	L-H.H	$\textcircled{H}\textcircled{H}.\downarrow H.H$	$\textcircled{H}\textcircled{H}.H$
L-H.H.L	L-H.H.L	L-H.H.L	$\textcircled{H}\textcircled{H}.\downarrow H.H.L$	$\textcircled{H}\textcircled{H}.H.L$
L-H.H.H	L-H.H.H	L-H.H.H	$\textcircled{H}\textcircled{H}.\downarrow H.H.H$	$\textcircled{H}\textcircled{H}.H.H$

We can understand these Giphende patterns as different strategies for avoiding two different H tones on the tonal tier within the same domain (in formal terms, avoiding a violation of the ‘Obligatory Contour Principle’). $\textcircled{H}_F$ FOC is the strictest and only associates if there is no other H in the word, $\textcircled{H}_G$ GEN is less strict in that it can dock as long it would not be adjacent to another H, whereas $\textcircled{H}_A$ ACC and $\textcircled{H}_P$ PRED are the least strict in allowing the floating tone to dock in any context (with idiosyncratic spreading and downstep effects thereafter to solve any violation of the ‘Obligatory Contour Principle’). Sensitivity to the above and other markedness considerations is thus not a single parameter which can be set for an entire language, but rather is calibrated for each individual floating tone or tonal morpheme.

¹⁵See the original source (Hyman 2017) for details on the exact distribution of these tonal morphemes, which depend on factors such as negation, dislocation, relative clausehood, etc.

5 Alternatives to floating tones

We have presented a plethora of examples where floating tones have proven to be useful in accounting for otherwise unexplained tonal patterns, and in the last section we showed how they can be parameterized along several phonological dimensions, just like less abstract phonological objects such as segments or associated tones. In this section, we shall explore four families of alternatives to floating tones: 1) positing a distinct string of pre-associated tones, 2) endowing a tone with a diacritic, 3) adjusting the ‘strength’ or ‘gradience’ of a tone, or 4) assuming universal allomorphy.

To demonstrate each of these alternatives, let us return to the Ghomala’ data from Section Section 2.2 (Nissim 1981: 150, 153). In Table 3, we provide a six-way tonal contrast using only H and L tonemes. Under a floating tone approach, there is a critical contrast between LH vs. L(H) vs. (L)H, which contrast against associated H, L, and HL.

Table 3: Floating tones and their alternatives (numbered 1 through 4) in Ghomala’

	[fá] ‘parent’	[tà] ‘pot’	[bûə] ‘madman’	[bvǔ] ‘dog’	[bàp°] ‘animal’	[dhó] ‘spouse’
Floating tone analysis	H /fá/	L /tà/	HL /bûə/	LH /bvǔ/	L(H) /bàp(H)/	(L)H /(L)dhó/
1) Different Tones	H fá	L tà	HL bûə	LH bvǔ	M bāp	MH dhə̃
2) Diacritic	H1 fá	L1 tà	HL bûə	LH bvǔ	L2 bàp	H2 dhó
3) Strength	H _[1.0] fá	L _[0.5] tà	HL bûə	LH bvǔ	L _[1.0] bàp	H _[0.5] dhó
4) Allomorphy	/fá/	/tà/ /ᵀtà/	/bûə/	/bvǔ/ /ᵀbvú/	/bàp°/ /ᵀbàp°/	/dhó/ /ᵀdhó/

For Ghomala’, floating tones were posited to capture effects not observed with their counterparts without floating tones, e.g., the contrast between /fá/ ‘parent’ and /(L)dhó/ ‘spouse’ in (35) within ‘noun of noun’ constructions, where the floating (L) of the latter triggers downstep in (35b).

(35) Ghomala' – Floating tones posited to capture downstep in 'noun of noun' construction

- a. thó fá → thó fá 'the head of the parent'
 b. thó $\textcircled{L}$ dhó → thó $\downarrow$ dhó 'the head of the spouse'

Under the first alternative analysis in Table 3, words with erstwhile floating tones actually constitute different tones or tone sequences, i.e., the form for 'animal' could be /bāp/ with a M tone rather than /bāp $\textcircled{H}$ / with a floating tone, while 'spouse' could be /dhə $\textcircled{L}$ / with a MH contour tone rather than / $\textcircled{L}$ dhó/. This would avoid floating tones, but necessitate both more tone heights (a H vs. L. vs. M contrast) as well as particular realizational rules which map /MH/ to surface [H] in isolation but as [$\downarrow$ H] after a H tone.

Under the second alternative, we would posit two qualitatively different types of H and L tones endowed with a diacritic, e.g., H vs. H2 and L1 vs. L2. Tonal rules would then be sensitive to this distinction, such that the trigger or target of a rule could target both or only one. For example, H1 would not downstep after another H tone, but H2 would downstep (as in 35b above). As argued in Hyman (2003), what this diacritic solution misses is the fact that floating tone effects act 'as if' a tone was there, i.e., we posit an abstract / $\textcircled{L}$ / or / $\textcircled{H}$ / because it patterns like a concrete /L/ or /H/. There is nothing explicit like this in the diacritic approach.¹⁶

In the third, from the Gradient Symbolic Representation literature (Smolensky & Goldrick 2016, Zimmermann 2018, 2019, Hsu 2022), floating tone effects would involve qualitatively identical tonemes without any diacritics, but they would bear different 'weights' or 'strengths'. For example, the H tone on 'parent' would be a 'strong' H tone with a weight of [1.0] (i.e., /fá_[1.0]/) while the H on 'spouse' would be a 'weak' one with a weight of perhaps [0.5] (i.e., /dhó_[0.5]/). The former tones would not undergo any downstep when they follow another H (i.e., they are strong enough to resist any tone change), while the latter tones would show downstep (i.e., they are not strong enough to resist tone change). The main challenge of this approach is explicitly connecting the 'strength' of a tone to the behavior it shows—and, again, accounting for the parallel behavior of $\textcircled{L}$ s and Ls.

Finally, the last and alternative to floating tones posits pure allomorphy. One recent proposal is the Emergent Phonology model of Archangeli & Pulleyblank (2022) which eschews underlying representations for 'morph sets', essentially turning all phonological alternations into a matter of allomorph selection. To

¹⁶For further argumentation against diacritic approaches, see, *inter alia*, Clements & Ford (1977, 1979), summarized in Lionnet & Hyman (2018).

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demonstrate this model, consider the Yoruba [yor] data in (36). Such data are conventionally understood as the result of a productive and language-general rule whereby the H in L-H sequences becomes rising due to L tone spread, i.e., the verb in the example has one underlying form /rí/ which surfaces as [rí] or [rĩ] depending on whether tonal spreading takes place. In contrast, under Emergent Phonology this verb would consist of the morph set /rí/, /rĩ/ where neither is the underlying representation of the other. The allomorph /rí/ is selected after a H-toned word whereas the allomorph /rĩ/ is selected after a L-toned word.

- (36) Yoruba – Tonal alternations as stored allomorphs in a morph set {rí, rĩ}
 a. ó rí ‘he/she/it saw’ b. kò rĩ ‘he/she/it did not see’

Emergent Phonology is a self-described ‘surface-to-surface model’, which has no conceptual place (or need) for floating tones (39). The contrastive floating tone effects in (41) would be captured by distinct morph sets, e.g., the ‘parent’ type we analyzed as /H/ would constitute a morph set only consisting of H-toned {fá}, while the ‘spouse’ type we analyzed as / $\textcircled{\text{L}}$ H/ would constitute a larger morph set {dhá, $\downarrow$ dhá}, and so on.

In comparing these five possible analyses, one can see that floating tones, contrary to the other four alternatives, are able to capture multiple tonal phenomena simultaneously. This gives floating tones a clear advantage over the alternatives, and explains why they have been part and parcel of tonological work (especially in Africa). This was seen in Aghem (Section 3) where we posited a contrast between /kí-fé/ ‘leg’ with no floating tone versus /kí-wó $\textcircled{\text{L}}$ / ‘hand’. The floating $\textcircled{\text{L}}$ conditions downstep on a following H tone as well as blocking H tone spread onto a following L tone. Crucially, we know independently that an overt (associated and realized) L tone has the exact same effects. Our contention is that any adequate analysis should capture both of these patterns using the same machinery.

Moreover, one of the most intuitive motivations for floating tones concerns the very existence of tonal morphemes. Whether one adopts different tones, quantitative diacritics, or qualitative strength, tonal morphemes (and hence floating tones) exist in underlying representations. Of the alternatives, only the Emergent Phonology model could capture floating tone effects without literal tonal morphemes, by assuming one morph in a neutral context (e.g., /tâ/) and another in a specific grammatical context (e.g., /tǎ/), which would correspond to a floating tone effect. We will not further evaluate the merits and shortcomings of Emergent Phonology in this paper, but merely point out that if it is adopted, the phonological module would look so radically different that even entertaining the idea

of floating tones becomes illogical. Although we disprefer these alternatives for capturing floating tone effects, they may prove successful in individual cases. That is, one may wish to posit distinct pre-associated tone patterns rather than floating tones for an individual tonal effect, and stored allomorphy rather than a tonal operation involving association of floating tones to account for some other tonal effect.

6 Summary

In this chapter, we have overviewed the use and function of floating tones, demonstrating their integral part in both the description and formal analysis of African languages. We showed that floating tones can be part of morphemes or act as independent morphemes themselves. Their function as an analytic device is to account for certain otherwise unpredicted tonal behavior, what we called ‘floating tone effects’. Examples include tone patterns on surrounding morphemes, as well as blocking effects, downstep and upstep/raising, and contrastive final level low tones. Once floating tones are treated as *bona fide* phonological objects, a number of analytic issues arise. Some of those include its inventory in a language, as well as locality, prosodic status, directionality, and markedness. Throughout this chapter we emphasized that floating tones are an analysis with several alternative analyses available, although none was found to have the range and flexibility of the floating tone construct.

In conclusion, abstract floating tones and their function as purely tonal morphemes are parallel to their less abstract segmental counterparts in virtually all details. They can occur as an affix or as a clitic, depending on whether it comes in at the word or phrase level, and may either be added outside the tone of its base or host, or it may replace it entirely. A floating tone may be the sole exponent of a morphological expression, or it may co-occur with another morph in a type of bipartite construction, e.g. a tense designation defined as a tonal prefix plus a segmental suffix. Ultimately, the most important take-away is that if one wishes to eschew floating tones entirely, any alternative must capture the full range of floating tone effects which we have outlined in this chapter.

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Abbreviations

AUG	augment (followed by class number when relevant)
CL	noun class (followed by class number when relevant)
CON	connective
CONJ	conjunction
FV	final vowel
NHUM	non-human
PRED	predicative
SCL	subject clitic (in Igala)
SPS	sequential impersonal passive (in Ik)

Acknowledgements

The authors wish to thank two anonymous reviewers as well as the editors of this volume for very useful feedback.

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Chapter 6

Consonant-tone interaction

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Consonant production and vowel tone are phonologically linked in diverse ways in a number of languages of Africa, from both the Niger-Congo and Afroasiatic families as well as beyond. We look at depressor consonants and lowering effects, followed by an examination of raiser consonants. We find variety not only in the types of consonants that have an interaction with tone, but also in the kinds of effects observed from this interaction. We also see diachronic progression in the relationship between consonant production and tone for certain languages.

1 Introduction

In a number of languages of Africa, from both the Niger-Congo and Afroasiatic families as well as beyond, there is interaction between consonants and tone. We find variety not only in the types of consonants that have an interaction with tone, but also in the kinds of effects observed from this interaction.

Consonant-tone interactions in African languages illustrate the distinction between phonetics and phonology. We look first in Section 2 at the phonetics of consonant co-articulation effects on an adjacent vowel before moving to the description of phonologized consonant-tone effects. Because lowering effects are the most common, we begin in Section 3 describing ways that depressors (consonants that have a lowering effect) are seen to affect tonal processes. We then examine in Section 4 the raising effects of certain consonants, looking at tonal processes which are affected by raiser consonants. In Section 5 we show which

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language families of Africa have generally been shown to have consonant-tone interaction.

There is very little documentation of the reverse direction of influence—tone having an effect on consonants. A synchronic example of such a case is given in Section 6. Next, we illustrate in Section 7 some examples of diachronic change in the relationship between consonants and tone. Finally, we conclude in Section 8 with some reflections about the trends we see in consonant-tone interaction systems in Africa.

2 Consonants and fundamental frequency of vowels

Universally, some consonants exert a phonetic influence on the fundamental frequency of adjacent vowels. Fundamental frequency is a measure of the lowest frequency of a complex sound wave. Fundamental frequency, often abbreviated as F0, is a physical property; pitch, on the other hand, is an auditory property allowing a listener to perceive a sound on a scale from low to high (Ladefoged & Johnson 2011: 24). As fundamental frequency increases, so does pitch.

Phonetic co-articulation effects between obstruents and vowels have been well documented in languages that are tonal as well as those that are not. In the context of a voiceless obstruent, the following vowel typically has an initial raised F0 pattern, whereas voiced obstruents tend to cause lower initial vowel F0 (Hombert et al. 1979; Kingston & Diehl 1994; Kirby & Ladd 2016; Gao & Kirby 2024).

Halle & Stevens (1971) relate the lowering and raising effects of voiced and voiceless obstruents to slack and stiff vocal cords, respectively. Research efforts continue to refine our understanding of the interplay of muscles and other laryngeal anatomy involved in regulating voicing and F0. Complex interactions between the thyroarytenoid and cricothyroid muscles of the larynx regulate vocal cord stiffness, resulting in F0 raising and lowering during vowel production (Jiang et al. 2024). In addition to the interactions of muscles, the width of the glottal opening is another factor that may affect F0 (Chhetri et al. 2014).

Consonant-tone interaction is frequently seen as phonologized patterns originating from phonetic co-articulation effects (Michaud & Sands 2020). Phonetic effects on F0 clearly do not necessarily result in phonological consonant-tone interaction in tone languages. Hombert (1977) examines F0 in Yoruba, a surface 3-level tone language. For the Yoruba speakers studied, as the voiceless obstruents [k] and [g] transition to the following vowel, the consonant-related effect on fundamental frequency is no longer observable somewhere between 40 and 60 msec into the vowel in most cases. Given that vowels produced by these speakers lasted approximately 200 msec, this is roughly 1/4 of the vowel duration. In

tone descriptions, it is helpful to clarify in discussions of consonant-tone interaction whether interactions which are measurable through acoustic measurement impact the hearer's perception of categories of pitch. In the case of Yoruba, the consonant-related effects on fundamental frequency do not influence tone perception in a phonological sense.

We see in Section 3.5 and Section 4.3 that from a synchronic standpoint, not all consonant-tone interaction can be shown to be phonetically motivated by voicing contrasts (Downing 2009; Michaud & Sands 2020; Maddieson & Sands 2022).

3 Depressor consonants and lowering effects

Lowering effects are the most widely documented of the consonant-influenced effects in the African literature. Consonants that have a lowering effect on tone are generally labeled as depressor consonants or depressors. The most frequently cited category of depressor consonants is that of voiced obstruents, although Section 3.5 shows there are other consonants that have a lowering effect on tone.

3.1 Lowering effects on adjacent vowels

In some languages, depressor consonants lower the tone of the immediately following vowel, either in some or all grammatical contexts. This type of interaction typically creates asymmetries in the lexicon where surface tone patterns are in complementary distribution and conditioned by the presence or absence of a depressor. Even in languages that have very little tonal sandhi, this type of lowering effect may be observed.

The many language variants within the Gbe continuum (spoken in Ghana, Togo and Benin) provide examples of this type of interaction. Bole-Richard (1983) analyzes Gen (also known as Gengbe) as having an underlying /H, L/ system. The examples in Table 1 show underlying representations (UR) and their surface realizations in monomorphemic nouns. Nouns are divided into those that have a depressor consonant (D) and those that do not (non-D).

In Gen, initial vowels on nouns have L tone.¹ We see in Table 1 that for the /L.L/ underlying tone pattern, both nouns with depressors and non-depressors alike have a L.L realization. However, an underlying /H/ following a voiced obstruent

¹These initial vowels in Gbe languages are generally understood to be developed from historic noun class prefixes (Good 2012).

Table 1: Gen (Gbe; Bole-Richard 1983: 107–110)

UR	D		non-D	
/L.H/	ègǎ	‘chief’	ètɔ	‘father’
	édɔ	‘work’	àlɔ	‘cheek’
	àbɔ	‘arm’	èmɔ	‘path’
	àdũ	‘tooth’	ènú	‘thing’
/L.L/	ègà	‘metal’	ètɔ	‘stream’
	édò	‘sickness’	àlɔ	‘hand’
	èvi	‘red kola’	èmò	‘machine’

is realized LH.² Looking at these data prior to analysis, what one would observe is that following depressors, L.LH and L.L surface patterns are observed; following non-depressors, L.H and L.L surface patterns are observed.

One phonological explanation that has been given for data such as the Gen data is that depressors condition the partial spread of L tone from the initial vowel to the following H (cf. Smolders 2026 [this volume]). This is the analysis discussed for Nupe in Hyman (1973). Another proposal for this type of phenomenon is described in Bradshaw (1999) and involves a single [L/voice] feature that can be shared by both consonant and vowel; associated to the vowel, this feature is realized as low tone. A third interpretation (similar to the previous) is that the depressor consonant inserts L tone. This type of proposal is given for Ikalanga in Hyman & Mathangwane (1998), where a depressor consonant conditions L tone insertion on the mora.

In the Gbe variety Fon (also known as Fɔ̀ngbe), monomorphemic nouns can have either a V.C(C)V or C(C)V syllable profile—the latter missing an initial vowel. Lefebvre & Brousseau (2002) describe Fon as having a binary underlying /H, L/ tonal contrast. Disregarding the initial vowel (which is always L), nouns are either /H/ or /L/. Table 2 captures the noun tone paradigm in Fon.

Here again we see certain consonants conditioning rising surface tone patterns. However, unlike in Gen, sonorants in Fon pattern with voiced obstruents in having a depressor effect; a vowel that follows a sonorant has underlying /H/ realized as LH. Brousseau (1993) concludes that both sonorants and voiced obstruents insert L tone in Fon.

²Bole-Richard also mentions six exceptional forms that have a rising contour following a voiceless obstruent or sonorant. These are [ètǎ] ‘head’, [àtǎ] ‘chief’, [ètò] ‘mortar’, [ènǔ] ‘mouth’, [àyǐ] ‘beans’, and [àyǎ] ‘wind’.

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Table 2: Fon (Gbe; Brousseau 1993: 8, 12, 13, 17, Lefebvre & Brousseau 2002: 20, 26, 48)

UR	D		non-D	
/L.H/	àvǔ	‘dog’	àsǔ	‘crab’
	àdǐ	‘soap’		
	àǎǎ	‘bridge’		
	àǐǐ	‘liver’		
/L.L/	àzǔ	‘disease’	àsì	‘wife’
	àwù	‘clothes’		
/H/	vǐ	‘child’	xú	‘bone’
	dě	‘tongue’		
	mǎ	‘leafy vegetable’		
	lǒ	‘crocodile’		
/L/	dǎ	‘snake’	xò	‘building’

We have seen that L insertion (or spreading) on an adjacent vowel can result in rising contours on a single syllable; this is not always the case. Another example of the lowering effect of depressor consonants can be seen in Suma (Adamawa) verbs in the present tense (Bradshaw 1995). Verbs in Suma are underlyingly toneless and their surface tone is attributable to tonal morphemes that indicate tense or word class. For instance, the verb ‘look for’ is realized [kírí] in the present, [kírā] in the past, and [kiri] in the neutral tense (Bradshaw 1995: 258). Bradshaw demonstrates that /H/ verbs (those in the present tense) have a H or LH pattern depending on whether they begin with a depressor or a non-depressor. Depressors include the sounds [b, d, g, gb, v, z], all of which are contrastively voiced. Verbs in the present tense are given in Table 3, where verbs have depressors (D) or non-depressors (non-D) in word-initial position.

Note here that we see a LH contour on one-syllable verbs. However, on multisyllabic verbs, we see L on the first syllable and H on the following syllables. Bradshaw notes that this lowering effect is not seen for verbs in the past, which have M grammatical tone (cf. Rolle & Bickmore 2026 [this volume]). For the past, surface tone is M regardless of the consonants found in the verb (Bradshaw 1995: 264).

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Table 3: Suma (Adamawa-Ubangi; [Bradshaw 1995](#): 263)

D		non-D	
bõm	‘be blind’	búk	‘applaud’
dĩk	‘be sonorous’	dǎŋ	‘mount’
gǎy	‘reprimand’	ée	‘leave behind’
gbǎk	‘borrow’	fódĩ	‘stir briskly’
vǎy	‘bet’	kírí	‘look for’
zǎfi	‘sneeze’	nóy	‘boil’
bùsí	‘be bland’	rém	‘be able to’
díkíli	‘tickle’	yári	‘unravel’
gòbí	‘twist’	ndángí	‘boom’
zíkídĩ	‘delay’	níkírí	‘exaggerate’

3.2 Blocking the spread of H tone

The Mijikenda, Shona, and Nguni subgroups of Bantu languages provide numerous examples of how depressor consonants block H spread. In Ikalanga ([Hyman & Mathangwane 1998](#)), depressor consonants block H spread across word boundaries. We can compare 1a to 1b and 2a to 2b. In these examples, H is spread across the word boundary onto a class prefix except when there is a depressor consonant to block the spread (seen here in the b examples).³

- (1) Ikalanga (Shona subgroup; [Hyman & Mathangwane 1998](#): 204–205)
 - a. ku-túm-á cǐ-pó ‘to send a gift’ túm-á ‘send’; ci- CL7; pó ‘gift’
 - b. ku-túm-á zvii-pó ‘to send gifts’ túm-á ‘send’; zvi- CL8; pó ‘gift’
- (2) Ikalanga (Shona subgroup; [Hyman & Mathangwane 1998](#): 204–205)
 - a. ku-c-á cǐ-pó ‘to fear a gift’ c-á ‘fear’; ci- CL7; pó ‘gift’
 - b. ku-c-á zvii-pó ‘to fear gifts’ c-á ‘fear’; zvi- CL8; pó ‘gift’

In 1a and 2a, H tone spread is not blocked by the onset of the class 7 prefix *ci-*. However, H tone spread is blocked by the depressor onset of the class 8 prefix *zvi-* in 1b and 2b.

³Researchers of Bantu languages follow a convention of leaving L tone unmarked; we maintain this convention when citing Bantu data (cf. [Grimm 2026 \[this volume\]](#)).

3.3 Blocking the docking of floating H tone

Depressors can also have an effect of blocking the docking of a floating H tone (cf. [Rolle et al. 2026 \[this volume\]](#)). This is the case in Gbaya bokota, an Ubangian language of Central African Republic ([Bradshaw 1999](#): 30–31). In this language, depressors (defined as voiced obstruents) block the docking of H tone which marks the associative construction. This floating H tone is located between the head noun to the left and the modifier noun to the right.

In the absence of a depressor, when the modifier noun is /L/ on the first tone-bearing unit, we see in [Table 4](#) that the associative floating H is able to dock to this noun, resulting in surface H on the first tone-bearing unit. (Note that prenasalized stops are not depressors in Gbaya bokota.) However, when the modifier noun has a depressor onset, the floating H of the associative construction is unable to dock. This is seen in [Table 5](#).⁴

Table 4: Gbaya bokota (Ubangi; [Bradshaw 1999](#): 30–31)

head + modifier	associative	gloss
wò + rì	wō rí	thirst for water
wò + mbòrō	wō mbórō	hunger for monkey
gbàà + mbòŋgò	gbàà mbóŋgò	corn seed
ēē + mbòrō	ēē mbórō	monkey paws

Table 5: Gbaya bokota (Ubangi; [Bradshaw 1999](#): 30–31)

head + modifier	associative	gloss
wò + gòrē	wō gòrē	hunger for chicken
wò + dùwà	wō dùwà	hunger for goat
gbàà + zàwà	gbàà zàwà	peanut seed
ēē + dàwà	ēē dàwà	lion paws

⁴[Bradshaw \(1999\)](#) does not explain the M on the first noun. There appears to be a process of L raising to M on the vowel preceding the associative floating H.

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3.4 Conditioning of H shift

Bantu languages have processes of H tone shifting (cf. Smolders 2026 [this volume]) where a H tone may shift over one or many TBUs across a word from a point of origin to a targeted syllable (Marlo & Odden 2022). Depressor consonants can affect this long distance shifting. In many languages where depressors condition H shift, this conditioning can include one or both of: a) the blocking of H docking, and b) the insertion of L to an adjacent vowel. Examples of the conditioning of H shift can be found in the Mijikenda and Nguni languages.

In Siswati (Nguni), H shift targets the antepenultimate syllable. However, when the antepenultimate syllable has a depressor onset, H shifts one more space to the right, to the penultimate syllable. This is seen in 3b and c, where both [v] and [mb] are depressors.

- (3) Siswati (Nguni; Cassimjee & Kisseberth 1992: 27, 28)

a. /kú-hlokolota/ ‘to prod’	ku-hlokólo:ta
b. /kú-vikela/ ‘to doge [sic]’	ku-vikê:la
c. /kú-vimbetela/ ‘to trap’	ku-vimbetê:la

When both the antepenultimate and penultimate syllables have a depressor, the antepenultimate receives H (Odden 2010). However, the depressor onset plays a role in the creation of a rising tone on that syllable. This is seen in 4.

- (4) Siswati (Nguni; Odden 2010: 93)
/kú-gezela/ ‘to bathe for’ ku-gěze:la

Another interesting case of depressors conditioning H shift is found in Chikauma (Mijikenda), described by Cassimjee & Kisseberth (1992) and shown in 5.

- (5) Chikauma (Mijikenda; Cassimjee & Kisseberth 1992: 32–35, cited in Hyman 2025: 58)
- | | |
|------------------|------------------------|
| /ú-a-zi-fugul-a/ | ‘s/he is untying them’ |
| w-á-zí-fúgúl-a/ | H spread |
| w-á-zi-fúgǔ:l-a | L insertion |

In 5, H is spread rightward from the first syllable and the target is the penultimate syllable. The output has a low tone on the syllable [zi-] because the depressor [z] inserts L tone (followed by delinking of H). While the depressor [g] does

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not prevent association of H to its targeted penultimate syllable, this phonetically lengthened syllable is realized with a rising contour.⁵

We see in the final realization in 5 that each depressor consonant has a H TBU preceding it—which Cassimjee and Kisseberth label as “spurious” Hs left behind from the spread of H from its point of origin to the penultimate syllable.

Looking at these examples, the challenge is to account for the fact that H shift is transparent to the presence of depressors—meaning that H tone shifts to its target syllable without being blocked by any depressor-induced L tone in the intervening syllables. This must be explained in conjunction with the fact that L insertion is the proposed means by which rising surface tones are created. A number of studies (among them Cassimjee & Kisseberth 1992; Hyman & Mathangwane 1998; Hyman 2014; Bradshaw 1999; Downing 2011; Downing 2017; Volk 2013) discuss the theoretical challenges of explaining this transparency and opacity to depressors.

A last example of depressors conditioning H shift comes from Giryama (Mijikenda), looking specifically at phonological words which have two underlying H tones. In Giryama, H tone shift targets the penultimate syllable. When one H tone is in the verb complex, it is realized on this target syllable regardless of whether it originates from the verb stem (as in 6a and 6b) or from a prefix (as in 6c and 6d, which include the third singular [á-] in the present tense). Note that in 6d, the presence of depressors in the syllables between the point of origin and the target does not affect this H shift. (In this language, [β] is not a depressor.)

(6) Giryama (Mijikenda; Volk 2013: 389–392)

- | | |
|--|------------------|
| a. /ku-βáβukira/ ‘to be searchable’ | ku-βaβukí:ra |
| b. /ku-bámahula/ ‘to wrench apart’ | ku-bamahũ:la |
| c. /á-na-mala/ ‘s/he is wanting’ | a-na-mǎ:la |
| d. /á-na-zazigirika/ ‘s/he is willing to play’ | a-na-zazigirí:ka |

When both the prefix and the verb have underlying H tones, the second of the two H tones shifts to the penultimate syllable, just as in 6. This is seen in all of the examples of 7. Where consonant-tone interaction comes to play is in the realization of the first H. In the absence of a depressor, this tone shifts to the first syllable of the verb stem (seen in 7a and 7b). However, if that syllable has a depressor onset, the H is realized instead on the last syllable before the stem (7c and 7d). (In these examples, [á-] is the third singular prefix and [na-] is the present tense prefix.)

⁵The form given here follows that of (Hyman 2025), however notation of lengthening is adapted to match other Bantu data in this chapter.

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(7) Giryama (Mijikenda; Volk 2013: 391)

- | | |
|------------------------------------|-----------------|
| a. /á-na-βáβukira/ ‘be searchable’ | a-na-βáβukĩ:ra |
| b. /á-na-fútama/ ‘squat’ | a-na-fútă:ma |
| c. /á-na-bámbahula/ ‘wrench apart’ | a-ná-bambahũ:la |
| d. /á-na-dáŋgula/ ‘persevere’ | a-ná-daŋgũ:la |

3.5 Diverse phonetic properties of depressors

While voiced obstruents are the most common depressors, languages vary in the categories of sound that are treated as depressors. For example, in the Chadic language Zina Kotoko (Odden 2007), the F0 of the vowel immediately following an implosive is slightly raised in a manner similar to the raising observed in the environment of a voiceless obstruent. Yet despite this phonetic reality, implosives operate as depressors in the phonology of this language. In Zina Kotoko, implosives are among the consonants that condition a M to L lowering rule in the recent past tense; voiced obstruents, sonorants, and the glottal stop also condition this same lowering rule. The phonological categorization of implosives as depressors in Zina Kotoko highlights the difference between phonetics and phonology (Odden 2007).

Tang (2008) and Bradshaw (1999) both include surveys of consonant-tone interaction which show that, in addition to voiced obstruents, the voiceless glottal fricative is one of the most frequent additions to the list of depressor consonants, with sonorants and implosives also occasionally having an affinity for lower tone. More recently, Mathes (2015) argues that Tsua, a Khoisan language, includes among its depressors voiced obstruents, the glottal fricative /h/, and aspirated obstruents.

In Giryama, Volk (2013: 392) argues that the sounds [h], [f], and [ð] have depressor and non-depressor counterparts. This challenges the relationship between consonant voicing and low tone that is most commonly observed for depressors. This difference in Giryama between depressor and non-depressor fricatives is seen in H tone shift when there are two H tones in the phonological word. In 7, we saw that that first of two underlying Hs in Giryama targets the first syllable of the verb stem. However, if a depressor is in the onset of that target syllable, then H is realized instead on the preceding syllable. We now look at further examples in 8, 9, and 10, where the first two verbs have non-depressor [h], [f], and [ð] and the other verbs have depressor [h], [f], and [ð].

(8) Giryama (Mijikenda; Volk 2013: 392)

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- | | |
|-----------------------------------|----------------|
| a. /á-na-hóhola/ ‘bend’ | a-na-hóhǒ:la |
| b. /á-na-háhisa/ ‘cause to vomit’ | a-na-háhǐ:sa |
| c. /á-na-hóhosa/ ‘pervert’ | a-ná-hohǒ:sa |
| d. /á-na-héndzeza/ ‘consent’ | a-ná-hendzě:za |

(9) Giriyama (Mijikenda; Volk 2013: 392)

- | | |
|----------------------------------|-----------------|
| a. /á-na-fútama/ ‘squat’ | a-na-fútǎ:ma |
| b. /á-na-fwéðeðeka/ ‘be ashamed’ | a-na-fwéðeðě:ka |
| c. /á-na-fúfula/ ‘revive’ | a-ná-fufǔ:la |
| d. /á-na-fínikira/ ‘cover’ | a-ná-finikǐ:ra |

(10) Giriyama (Mijikenda; Volk 2013: 392)

- | | |
|---|---------------|
| a. /á-na-ðéðedza/ ‘coax’ | a-na-ðéðě:dza |
| b. /á-na-ðúnyuka/ ‘become contused’ | a-na-ðúnyǔ:ka |
| c. /á-na-ðyóma/ ‘speak with foreign accent’ | a-ná-ðyǒ:ma |

Volk (2013) concludes that depressors in Giriyama do not insert L tone, but rather they lower register. Note that Volk uses the terminology of register in the sense of a division into high and low register domains, depending on whether tones are affected by depressor consonants or not. This is in contrast with the lowered pitch levels that result from downdrift and downstep that are often referred to as having a lowered register (cf. Snider (2026 [this volume])). It is not in the scope of this paper to discuss tonal register, but see Volk (2013) for the description of this register analysis for Giriyama.⁶

Languages of the Nguni and Shona subgroups also have depressors that are not voiced. Downing & Gick (2005) show that Nambya (Shona) has a depressor [f] and a non-depressor [f], both voiceless. They also show that Botswana Kalang’a (also Shona) shows lowering behavior following the depressor aspirated stops [p^h] and [t^h]. This lowering is not seen following the non-depressor aspirated voiceless stops [p^h] and [t^h]. To explain depressor lowering effects in the absence of voicing, Downing (2017: 128), employing an Optimality Theoretic framework, proposes the constraint stated in 11.

- (11) *(DEPRESS, H): If depressor syllable, then not High tone.

⁶On a separate note, it is interesting that the underlying lexical assignment of L tone does not appear to be a hypothesis under consideration for cases like Giriyama, perhaps because Narrow Bantu languages in this area typically view surface L as unspecified in the underlying tone system.

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Rycroft (1980: 125–126) posits that “breathy-voice phonation” rather than voicing might be the phonetic correlate to depressors in Nguni languages. Traill *et al.* (1987), however, shows that Zulu (Nguni) vowels lowered by depressors do not exhibit breathiness. Phonetic studies of Zulu by Chen & Downing (2011) demonstrate that depressors lower not just the surface realizations of H vowels, but also those of L vowels—showing that even in the phonetic realization of the depressor effect on vowel tone there can be some variability. Traill (1990) claims that voiceless obstruents can have a depression effect in Siswati (also Nguni).

This summary shows that there is quite a diverse class of sounds that function phonologically as depressors. There is also some variability in the manner that depressors exert their lowering effect on pitch. In examination of the phonetic variability of depressors in Bantu, Maddieson (2003) concludes that depression represents “a special laryngeal posture consistent with very low pitch co-produced with the consonant it is associated with” (p. 30). In summary, while the prototypical depressor is a voiced obstruent, there are other sounds that function phonologically as depressors—including some sounds where there is unclear phonetic motivation for this phonological function.

4 Interactions that involve raiser consonants

While tone depression is far more commonly attested in African languages, tone raising is also a documented possibility. Tone raising is observed in Chadic languages (cf. Roberts 2026 [this volume]).

4.1 Raising of L on the adjacent vowel

An interesting case of raising is found in Musey, which has three level tones (Shryock 1995). Consonants are divided into categories of what Shryock labels as high consonants (raisers) and low consonants (non-raisers). Raisers include voiceless obstruents, sonorants, and implosives. Tone is associated right to left in nominal stems, and for /L/ stems with a raiser consonant in the onset of a syllable unassigned for tone, the noun receives M tone; otherwise it receives L. This is seen in comparing 12a and 12b.

(12) Musey (Chadic; Shryock 1995: 6)

- | | |
|--------------------|-------|
| a. /kulùf/ ‘fish’ | kùlùf |
| b. /buzùr/ ‘blood’ | bùzùr |

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We see this raising effect nicely illustrated in ideophones in Musey. Underlyingly, an ideophone may only have /M/ or /L/, but /L/ is raised to M on the initial syllable of the word following the group of raisers (Roberts & Soulokadi 2019). This is seen in Table 6. Note that the raiser has no effect on M.

Table 6: Musey (Chadic; Roberts & Soulokadi 2019: 219)

UR	non-R		R	
/L/	bùv	penetration of needle	kīm	resonance of a loud sound
	mbilid	seriousness of a lie	mīnìd	force of a hit against s.th.
	gàrnàg	thoroughly burnt state	kōrògdòg	sensation of bodily weakness
/M/	bīr	severity of a beating	kīg	action of stabbing
	mbēlēŋ	smallness of an opening	mārāɫ	suddenness of flight
	gēwrēŋ	smallness of an object	kīrmīdīŋ	intense darkness

(13) Musey (Chadic; Roberts & Soulokadi 2019: 222)

- a.

lērlēr

running of nose mucus
- b.

cēdēvcēdèv

action of eating

Note that in these analyses of Musey, the group of non-raisers is not described as having a depressor effect because the presence of a non-raiser has no lowering effect on M.

It is possible that a language may also have three categories of consonants—depressors, raisers, and a third category of consonants that are neutral to tone effects. A three-way categorization of consonants is shown in the discussion of Kera in Section 4.3.

4.2 Blocking of L spread

Besides raising the tone on the adjacent vowel, raiser consonants can also block L spread; this is seen in Bade (Tang 2008: 65–67, citing data from Schuh 2002). In Bade, L spreads rightward to a H found at the right boundary of a word, delinking the H; this happens in the environment where there is a H following the word boundary. This can be seen in Table 7, where L spread results in the syllable before the word boundary being L.

Table 7: Bade (Chadic; Tang 2008: 66)

underlying	surface	gloss
/dʒə.ɖǵá kó:rón/	dʒə.ɖǵə kó:rón	we followed a donkey
/dʒə kèrə kó:rón/	dʒə kèrə kó:rón	we stole a donkey
/dʒə tàǵá kó:rón/	dʒə tàǵə kó:rón	we released a donkey
/kàyá:n póm/	kàyà:n póm	not a squirrel

Table 8: Bade (Chadic; Tang 2008: 67)

underlying	surface	gloss
/dʒə dèpsá kó:rón/	dʒə dèpsá kó:rón	we hid a donkey
/dʒə gǵá kó:rón/	dʒə gǵá kó:rón	we caught a donkey
/tǵká:n póm/	tǵká:n póm	not an orphan
/mǵén póm/	mǵén póm	not death

However, the presence of a raiser (defined as a voiceless obstruent) prevents L spread from happening. This can be seen in Table 8.

Tang (2008: 67–68) makes the claim that this is best analyzed as the blocking of L spread by raiser consonants rather than the conditioning of L spread by depressor consonants. If the latter analysis were proposed, the category of depressor consonants would need to include implosives and sonorants.

4.3 Diverse phonetic properties of raisers

In Section 4.1, we discussed the effect that raiser consonants have on L, raising it to M. In a detailed study of Musey, Shryock (1995: 6, 9, 20, 84) concludes that obstruent pairs consisting of raisers and non-raisers (such as [k] and [g]) have undergone a historical development in which Proto-Chadic voicing contrast has largely been lost. Instead, a more subtle phonetic contrast exists to differentiate these phonologically contrastive sounds; this phonetic contrast is described for raisers (labeled as high consonants) as being a combination of greater longitudinal vocal fold tension and greater subglottal pressure (p. 84).

Loss of voicing contrast is shown for Kera, another Chadic language, with the analysis of this language changing over the course of years of research (Ebert 1979; Pearce 1999; Pearce 2013b). Pearce (1999) began with the analysis that nouns

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are divided into three groupings: those that begin with depressor consonants (D), those that begin with raiser consonants (R), and those that begin with sonorants. This is seen in Table 9. Only syllables with raiser consonant or sonorant onsets have surface M realizations and there is no H realization for syllables with depressor onsets.

Table 9: Kera (Chadic; data from [Pearce 2005](#): 64–66)

Surface	D		R		son	
H			káasáw	‘millet’	móyán	‘river’
M			pāatāl	‘needle’	māanī	‘co-wife’
MH			tāatá	‘big jar’	māahúr	‘flute’
HM			tāasā	‘cup’	máalāŋ	‘bird of prey’
L	dàagà	‘mat’			mòlòorò	‘saliva’
LH	dàktéláw	‘bird’			hùlúm	‘man’
HL			táabùl	‘table’	mánhòr	‘ten’

[Pearce \(2005\)](#) found that differences that had to do with consonant-tone interaction were fluctuating over time and geographic location. This led to the observation that in some locations, the difference in phonetic manifestation between depressor and raiser consonants was voice onset time (VOT) rather than voicing, with increased VOT co-occurring with higher pitch ([Pearce 2005](#): 79). Furthermore, the analysis concluded that for some speakers, tonal contrast was being phonologized; this is discussed further in Section 7.3.

These observations of phonetic diversity in systems with raiser consonants are not unlike the discussion in Section 3.5 of the phonetic variability of depressors in Bantu. Phonologically distinctive sounds that have an effect on tone sometimes have phonetic manifestations other than the presence or absence of voicing.

5 Languages exhibiting consonant-tone interaction

It is helpful to know where consonant-tone interaction has been identified in Africa. Both [Bradshaw \(1999\)](#) and [Tang \(2008\)](#) have surveys of languages documented as having consonant-tone interaction. While these lists are not exhaustive (research published since these studies has continued to identify more cases), they are indicative of where one might expect to find this phenomenon in Africa.

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Table 10 through Table 11 are compilations of both lists without independent examination of the criteria that determined inclusion in these lists. These tables include only African languages; Tang’s survey includes languages outside of Africa. Language family information has been added from information provided in the Ethnologue when necessary (Eberhard et al. 2024). In these tables, there is a reference to whether the active consonants concerned were voiced obstruents (vcd), voiceless obstruents (vcless), sonorant (son), implosive (implos), sequences of nasal+consonant (NC), or whether aspiration (aspir) or other specific sounds such as the glottal stop were involved.

Generally, we see that Narrow Bantu languages and Chadic languages are well represented in these inventories; Table 10 lists Narrow Bantu languages and Table 12 lists Chadic languages. In addition, there are some Kwa, Kru and Adamawa-Ubangian languages, as well as a few other Niger-Congo subdivisions as shown in Table 11.

Table 10: Narrow Bantu (various subgroups; from Bradshaw 1999: 6(B) and Tang 2008: 25–26(T))

language	consonants	survey
Botswana Kalanga’a	vcd; aspir	T
Digo, Chirihe, Chikauma	vcd	B; T
Ikalanga	vcd	B
Isixhosa	vcd; NC; implos	B; T
Makaa	vcd	B; T
Nambya	vcd; depressor /f/	T
Siswati	vcd; NC; slack/lax	B; T
Yaka	vcd	B; T
Zulu	slack/lax	T

In Table 10, we see good representation of the three subgroups which are Mijikenda (Digo, Chirihe, Chikauma), Shona (Ikalanga and Nambya), and Nguni (Isixhosa, Siswati, Zulu). Studies since Bradshaw (1999) and Tang (2008) include additional descriptions of consonant-tone interaction in Mijikenda (Duruma, Giryama, Kambe: Volk 2013) and in Nguni (Phuthi: Donnelly 2009). In addition, depressor effects have been described in Malawian CiTonga (Mkochi et al. 2022).

We can summarize by saying that many branches of the Niger-Congo family are represented in Table 10 and Table 11. From the Afroasiatic family, it is primarily Chadic languages that have been documented to have consonant-tone interaction.

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Table 11: Various (Adamawa-Ubangi, Gur (Mabia), Kordofanian, Kru, Kwa, Nupoid; from Bradshaw 1999: 6 and Tang 2008: 25–26)

family	language	consonants	survey
Ubangi	Suma	vcd	B; T
Ubangi	Gbaya bokota	vcd	B; T
Gur	Dagara-wule	vcd	B; T
Kordofanian	Moro	?	T
Kru	Bassa	vcd; vcless; son; implos; /h/	T
Kru	Kuwaa	vcd	T
Kwa	Ewe	vcd; son	B; T
Kwa	Siya	vcd	B
Kwa	Ebrié	unclear	B; T
Kwa	Tuwuli	?	T
Nupoid	Nupe	vcd; vcless; son	B; T

Table 12: Chadic (various subgroups; from Bradshaw 1999: 6 and Tang 2008: 25–26)

language	consonants	survey
Bade	vcd; vcless; son	B; T
Bolanci	vcd	B
Bole	vcd	T
Kotoko	vcd; son; implos; ?	T
Lamang	vcd	T
Masa	vcd; vcless; implos	T
Miya	vcd	B; T
Mulwi	vcd	B; T
Musey	vcd; vcless; son; implos	T
Ngizim	vcd; vcless; implos	T
Ouldeme	vcd; NC	B
Sayanci	vcd	T

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These tables are indicators of general trends regarding the extent of consonant-tone interaction in African languages. It is expected that the list of subfamilies where consonant-tone interaction is attested will continue to increase as research is furthered. Mathes (2015), for instance, describes consonant-tone interaction in a Khoisan language.

6 Tones that have an effect on consonants

The large majority of examples from crosslinguistic studies of African languages show consonants having an effect on tone rather than the reverse—tones having an effect on consonants (Bradshaw 1999, Tang 2008). Hyman (1973) and Hyman & Shuh (1974) take a strong position on this, arguing that tone does not affect consonants. In ensuing publications on this topic (including Maddieson 1977), the synchronic phonological effects of tone on consonants are illustrated primarily from non-African languages.

As an example of diachronic change, Maddieson (1977), citing Dwyer (1974) describes Kpelle as having different processes depending on whether prefixed nasals have H or L tone (seen in 14).

(14) Kpelle (Mande; Maddieson 1977: 95)

- a. *p → p
- b. *ń-p → mb
- c. *̀n-p → bb

In synchronic phonology, we find a purported example of tone having a phonological interaction with consonants in Orungu (Myene), a Bantu language spoken in Gabon (Maniacky & Ambouroue 2014). In this language, there are consonant alternations—each alternating pair including both a “strong” consonant and a “weak” consonant. The initial consonant of a verb root is conditioned by tone on the verb. The initial consonant is realized with the weak consonant when the syllable is associated to L tone (part of the tense and aspect marking). It is realized with a strong consonant when the syllable is associated to H tone. Sometimes the weak/strong alternation involves a voiced/voiceless pair. In other pairs, the weak consonant is a sonorant that is pronounced in the same place of articulation as its corresponding strong consonant, an implosive. Maniacky & Ambouroue (2014) note that verb root initial consonants that are nasals do not have alternations such as this.

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These consonant (C) alternations can be seen in Table 13, where L tone marks the remote past and H tone marks the present. An example of the underlying representation is given in 15.

Table 13: Orungu (Myene; data from Maniacky & Ambouroue 2014: 245)

weak C	strong C	L tone remote past	H tone present	gloss
β	p	à[βíkí	è[píkà	‘to grill’
v	f	à[vèlí	è[félà	‘to put’
r	t	à[rèní	è[ténà	‘to cut’
l	ɸ	à[lèní	è[dénà	‘to cry’
z	s	à[zíkí	è[síkà	‘to push’
ɣ	k	à[ɣílí	è[kílà	‘to grate’
y	ɸ ^y	à[yèní	è[ɸyénà	‘to see’
w	b	à[wèlí	è[bélà	‘to want’

- (15) Orungu (Myene; data from Maniacky & Ambouroue 2014: 245)
- à[βíkí /à-à-βík-í^H/ ‘s/he grilled’
- è[píkà /à-è-βí^Hk-à/ ‘s/he grills’

Maniacky & Ambouroue (2014) argue that these alternations are seen for root-initial consonants of all verbal forms with the exception of infinitives, which are underlyingly L but have strong consonants. The authors argue that this is because infinitives behave partly as nominals.

The posited effect of tone on consonants in Orungu is only in root-initial position of the verb; in other positions, tone does not cause alternation of consonants. While the claim is that tone affects consonants, there does seem to be some circularity in the conclusions drawn in Maniacky & Ambouroue (2014: 245, 253, 255) that underlying tone can be inferred from the initial consonant seen in certain forms. Thus when H tone is found in a syllable with a weak consonant, this is explained as the result of a preceding H or prefixed floating H (represented as _H) associating to a vowel with underlying L tone. This is seen in 16.⁷

⁷According to Maniacky & Ambouroue (2014: 253), vowels without tone marking represent cases where underlying tone cannot be determined.

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- (16) Orungu (Myene; data from Maniacky & Ambouroue 2014: 245, 253, 255)
- | | | |
|------------|------------------------------|--------------------|
| àzwá'[βíká | /àzó=a- _H -βík-a/ | 'let us not grill' |
| è[zélé | /à-é- _H -zèl-e/ | 's/he is not' |
| wíyá[βíkè | UR not given | 'that they grill' |

The Orungu data are typologically unusual and pose interesting analytical challenges. This phenomenon is clearly related to morphology. Two changes are observed when comparing the remote past to the present—both a consonant change from weak to strong and a tone change from L to H. However, as we see in 16, there can be surface forms with weak consonants and H. Moreover, infinitive forms have strong consonants and L.

It is possible that the present tense is marked by an initial strong consonant as well as H tone—the two being unrelated. This could be true if, for example, the present prefix [è-] developed from a historic prefix such as *éN-, which would condition strengthening of the root-initial consonant before subsequent diachronic loss of the nasal. The question is whether this is a phonologically productive relationship where tone affects the consonant, or whether there is in some verbal forms a synchronic lexical co-occurrence due to historic factors. This language merits further examination, especially in light of the apparent world-wide rarity of cases where tone affects consonants. Further research from Africa showing clear examples of tone affecting consonants would be invaluable.⁸

7 Diachronic change and consonant-tone interaction

Languages are constantly in flux and speakers may have differences in the combination of phonetic cues that are used to differentiate phonemic categories. Mechanisms of historic change such as reanalysis and analogy apply to tone phenomena. In this section, we look at examples of diachronic change in the relationship between consonants and tone.

7.1 Lexicalization of pitch contrast effects

In Lamang nouns, it is a general observation that depressors co-occur with L tone while non-depressors co-occur with H tone (Wolff 1987). Depressors are defined as voiced obstruents, while non-depressors include sonorants, glottalized and voiceless obstruents. An example of predictable tone in Lamang nouns is seen in 17.

⁸We would like to thank our reviewers for their comments which led to this detailed discussion of the Orungu data.

- (17) Lamang (Chadic; Wolff 1987: 200)
- a. tlá

b. ghwà
- cow

mountain

These co-occurrence patterns do not hold elsewhere in the phonology. In ideophones and other parts of speech, tone is independent of syllable onset type. For example, tone plays a significant role in verbal tense and aspect. Grammatical tone is contrastive and does not follow any pattern of consonant-tone interaction such as that seen in nouns (Wolff 1987: 201).

Wolff (1987) describes this as a pitch/tone system, where diachronic processes have enhanced pitch contrasts on nouns, yielding H in a syllable with a non-depressor consonant and L in a syllable with a depressor consonant. In this example of Lamang nouns, H is not raised from a lower underlying level, nor is L lowered from an underlying raised level in the synchronic phonology.

7.2 Lexicalization of lowering effects

The Gbe language Saxwe (also known as Saxwɛgbe) presents a more complex case than that described for the Gbe languages Gen and Fon in Section 3.1. There is alternation of surface tone pattern realizations when comparing nouns with voiced obstruents to nouns with voiceless obstruents. This can be seen in Table 14.⁹

Table 14: Saxwe (Gbe; data from Beavon-Ham 2019: 64, 96–98, 331–332)

UR	vcd except b/d	vcless	son
/Ø.H/	–	ōsó āká	‘horse’ ‘coal’
/Ø.Ø/	–	ōxē āfō	‘bird’ ‘leg’
/Ø.LH/	ōvũ ōgbō	–	‘dog’ ‘goat’
/Ø.L/	āvò ōdà	–	‘cloth’ ‘snake’

⁹In this table, we see the three-way underlying distinction in Saxwe represented as /H, Ø, L/; this is a reanalysis of Beavon-Ham (2019), which has the underlying distinction as /H, M, L/.

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In Table 14, we see complementary distribution in the paradigm of noun tone patterns when we examine the columns of voiced obstruents and voiceless obstruents.¹⁰ Voiced obstruents are immediately followed by L in their noun tone realization, while voiceless obstruents are never immediately followed by L in their noun tone realization. Looking at these data alone, one might hypothesize that voiced obstruents have a lowering effect on tone by inserting L tone.

However, nouns containing sonorants are followed by surface L, M, or H; they display a range of tone patterns, including those associated with voiced as well as voiceless obstruents. The sounds [b] and [d] pattern with sonorants in being followed by surface L, M, or H. This is shown in Table 15.

Table 15: Saxwe (Gbe; data from Beavon-Ham 2019: 63, 96–97)

UR	b	d
/Ø.H/	ābó ‘arm’ ōbó ‘amulet’	ōdā ‘silence’
/Ø.Ø/	ābî ‘wound’ ābā ‘forked branch’	ōdâ ‘hair’
/Ø.LH/	–	–
/Ø.L/	ābò ‘cooked beans’ ōbà ‘manioc dish’	ōdò ‘fishing net’

We therefore cannot conclude without complications that in the synchronic phonology that surface L is due to the presence of voiced obstruents.¹¹ The conclusion drawn in Beavon-Ham (2019) is that the historic phonological relationship between L and voiced obstruents has become lexicalized in the language, meaning that L tone is now assigned in the speaker’s mental lexicon rather than obtained as an output of the phonology. Inconsistent consonant-tone interactions in grammatical morphemes (including L on syllables with no consonant onset) is further indication that L has become lexicalized as a distinct underlying tone level rather being a surface output of the phonology (Beavon-Ham 2019). Language contact with neighboring Yoruboid languages that have a three-way tonal distinction may have influenced this change.

¹⁰There are other minor noun tone patterns not shown in this table that can be observed in Saxwe, discussed in Beavon-Ham (2019).

¹¹A reviewer points out that from a diachronic standpoint, this circumstance may be due to a change such as *b and *mb merging as [b].

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The lexicalization of consonant-tone interaction may also have been influenced by change related to the sounds [b] and [d] and a resulting ambiguity as to whether these sounds pattern with depressors or non-depressors. These two sounds are purported to come from pre-Gbe implosives *ɓ and *ɗ (Stewart 1989: 240–241). Moreover, [d] has no voiceless counterpart and the voiceless [p] is marginal in the phonology of the language. These reasons might explain a historic categorization of nouns with present-day [b] and [d] as belonging to the category of nouns containing non-depressors. Given this background, it becomes more understandable why nouns with [b] and [d] have tone realizations that pattern with both voiced and voiceless obstruents in the synchronic phonology.

7.3 Diachronic change in Kera

Another case of diachronic change is seen in Kera (Chadic). Pearce (2005) found that differences that had to do with consonant-tone interaction were fluctuating over time and geographic location.

Pearce (1999) provides the analysis that nouns are divided into three groupings: those that begin with depressor consonants (D), those that begin with raiser consonants (R), and those that begin with sonorants. This has an effect on tone; only syllables with raiser consonants or sonorant onsets have surface M realizations and there is no H realization for syllables with a depressor onset. This was described in Section 4.3.

In Pearce (2005), the observation is made that for certain people, consonant-tone interactions are represented differently than previously analyzed. For certain Kera speakers, tone is phonemic while consonant VOT cues enhances the tone (Pearce 2005: 79). VOT value is realized on a gradient based on the pitch. The higher the pitch, the later the voicing begins.

However, in addition to this observed increase in tonal contrast for some speakers, Kera also shows a different diachronic progression for specific subpopulations. For female Kera speakers living in urban areas, there is diachronic loss of tonal contrast and strengthening of voicing contrast (Pearce 2013a; Pearce 2013b). This subsection of speakers has not maintained a three-way tonal contrast throughout the language. French is the language of education and integration into urban life. For these female speakers, their speech when using Kera is influenced by French voicing and lack of tone. The result for these female urban speakers is that pitch differences are minimized except for minimal pairs, but the consonant voicing contrast is strengthened. This results in a tone system that distinguishes two contrastive levels of tone rather than three.

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Interestingly, female speakers in rural settings are the most likely to preserve contrasts that rely on tonal differences, whereas female speakers in urban settings are the most progressive towards a voicing contrast rather than a tonal one (Pearce 2013a; Pearce 2013b). These changes in women's speech could likely result in changes in men's speech in the next generation. The Kera context shows the importance of wide studies in different locations, with both male and female speakers, conducted over longer periods of time.

8 Conclusions and further research

In this chapter, we see that consonant production and vowel tone are phonologically linked in diverse ways in the continent of Africa. Many questions still exist. We start with questions regarding the more widespread nature of tone-lowering phenomena when compared to tone-raising phenomena. Descriptions of tone-lowering phenomena are found for Bantu languages (largely of the Mijikenda, Nguni and Shona subgroups), other languages of the non-Bantu Niger-Congo subfamilies, and Chadic languages, as well as occasionally elsewhere. Meanwhile, descriptions of tone-raising phenomena are limited mostly to Chadic languages and a relative few non-Bantu Niger-Congo languages. The extent of these phenomena and the role of language contact in establishing or maintaining them will be better understood as more tone analysis is completed for the languages of Africa.

Longitudinal studies are needed to show how diachronic changes in the relationship between consonants and tone are influenced by sociolinguistic and other factors. One question that continues to be of relevance is how depressor or raiser categorization is maintained from one generation to the next, particularly in light of the diversity of sounds that are treated as depressors or raisers in various languages.

Abbreviations

aspir	aspiration
C	consonant
CL	noun class
D	depressor consonant
F0	fundamental frequency
H	high tone
implos	implosive
L	low tone
M	mid tone
N	nasal
NC	nasal-consonant sequence
obj	object
R	raiser consonant
son	sonorant
TBU	tone-bearing unit
UR	underlying representation
vcd	voiced
vcless	voiceless
VOT	voice onset time

Acknowledgements

We are indebted to Larry Hyman and Jim Roberts for feedback that led to significant improvement of this chapter.

Mary Pearce, the second author of this chapter, passed away before seeing the final publication of this work. She will be remembered by many as someone who was lovely to interact with and passionate about linguistics, including this topic of consonant-tone interaction. On a personal note, I will miss her example of finding beauty in the everyday moments and sharing that with others.

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Chapter 7

Tone polarity

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Tonal polarity is a phenomenon where in defined contexts, a tone must be opposite to an adjacent tone. This chapter considers the description and analysis of a range of such cases. The chapter begins with languages with two-tone systems, going on to examine several languages with three-tone systems. Various approaches to polarity are considered, including rule-based and constraint-based analyses. The role or lack thereof of underlying representations is examined. Whether tone polarity should be considered a type of dissimilation is addressed.

1 Setting the stage: Hausa

‘Polarity’ can be used in both a superficial descriptive way and in a deeper analytic way. Descriptively, tone polarity is used to describe a situation where a tone is realised as opposite to an adjacent tone. This is illustrated here by the ‘stabiliser’ morpheme in Standard Hausa (Chadic; Nigeria and numerous other African countries; glottocode: haus1257, [Newman 1995](#)).

- (1) Hausa: polarity in the stabiliser

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- | | | |
|----|---------------|-----------------------------|
| a. | jàakíi nèe | ‘it’s a donkey’ |
| | riigáa cèe | ‘it’s a gown’ |
| | nàa-wá nèe | ‘it’s mine’ |
| | tàa-wá cèe | ‘it’s mine’ |
| b. | zòobèe née | ‘it’s a ring’ |
| | móotàa cée | ‘it’s a car’ |
| | húulúnàa née | ‘they’re caps’ |
| | hàrâm née | ‘it’s religiously unlawful’ |
| | mâi née | ‘it’s oil’ |
| | húulúnàa née | ‘they’re caps’ |
| | ná yáaròo née | ‘it’s the boy’s’ |
| | tá yáaròo cée | ‘it’s the boy’s’ |
| | náa-sà née | ‘it’s his’ |
| | táa-mù cée | ‘it’s ours’ |

The stabiliser is *nee* ‘masculine’ or *cee* ‘feminine’, depending on gender; its tone is the polar opposite of the final syllable of the preceding noun, L when the preceding noun ends in a H (1a), H when the preceding noun ends in a L (1b).¹

In certain cases where root tones alternate, we can see that the stabiliser is tonally appropriate to whichever root alternant is observed.

(2) Hausa: polarity in the stabiliser

- | | | | |
|----|-----------|--------------|------------------|
| a. | kèekè née | kèeké(e) nèe | ‘it’s a bicycle’ |
| b. | ʔílmù née | ʔílimí nèe | ‘it’s knowledge’ |

In addition to being descriptively polar, the Hausa stabiliser can be considered analytically polar. As observed by Newman 1995: p. 771, ‘there is no compelling synchronic reason to presume that the tone started underlying as some specified tone or other.’ Had there been reason to consider *nee/cee* to have a particular underlying tone, then we could account for the pattern in (1) and (2) by dissimilation: if H in the underlying form then the L-tone cases, *nèe* and *cèe*, would be derived by dissimilation of H after H; if L in the underlying form then the H-tone cases, *née* and *cée*, would be derived by dissimilation of L after L.

As even this fairly straightforward case illustrates, whether a pattern is truly or deeply ‘polar’ thus depends on the analysis. Indeed, the Hausa example has been the source of some controversy. Leben 1971 has argued for a rule in Hausa that raises a L on a final long vowel when it follows another L; see also Hyman

¹Examples with falling tones, such as *mâi née* ‘it’s oil’ show unsurprisingly that falling tones behave as though they are L-final.

& Schuh 1974 for discussion. If this raising rule holds synchronically in Hausa then the pattern in (1) and (2) could be analysed as dissimilatory rather than strictly polar. The validity of a raising analysis has been questioned, however, in Newman & Jaggar 1989, and if a synchronic rule of dissimilation cannot be motivated, then we may want to return to analytic polarity. As frequently noted in a general sense – and specifically by Hyman & Schuh 1974, Newman 1995 for the Hausa case – if a morpheme is considered truly polar then as alluded to above, one possible analysis is to consider it unspecified underlyingly with its surface tone assigned in polar fashion.

In any given case, therefore, whether a pattern is to be considered ‘polar’ or not depends on whether or not it is desirable to analyse the pattern as due to dissimilation. Cross-linguistically, whether polarity exists as a phenomenon independent of dissimilation depends on whether there are cases where a dissimilation analysis is non-optimal.

In the sections below, we begin by discussing several cases of polarity, illustrating various analytic approaches. The goal is not to argue for a particular approach, but to illustrate various ways in which polarity can be analysed. We begin with polarity in languages with two tones, subsequently moving on to languages with more than two tones. While a variety of languages and patterns will be discussed, there are numerous cases in the literature which will not be treated – cases from a variety of languages and language families: Bambara (Dwyer 1976), Etsako (Hyman & Schuh 1974, Elimelech 1976), Fur (Kutsch Lojenga & Waag 2004), Igbo (Welmers & Welmers 1968, Hyman & Schuh 1974, Schuh 1978), Kanakuru (Newman 1974, 1995), various Mabia languages including Dagbani, Lama, Mooré (Kenstowicz et al. 1988, Hyman 1993, Olawsky 1999), Ngizim (Schuh 1971, Schuh 1978). Cases outside of the African continent are not commented on here. Note that one difficulty in surveying cases of polarity arises due to cases treated as dissimilation; as discussed already but in more detail below, it is not always clear whether a given example is polar or not.

2 Lack of underlying specification: Hausa

If a pattern is analytically polar it must be distinguished from dissimilation in one way or another. One commonly adopted possibility is to distinguish polarising morphemes from other morphemes by marking the polarising set by a lack of underlying tone features. Consider again the case of Hausa discussed in §1. Noun roots, which do not alternate, would bear tones underlyingly; the stabiliser morpheme, which exhibits polarity, would be underlyingly unspecified for tone.

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In a rule-based framework, such an underlying representation can be combined with a structure-building rule. Following Yip 1980, 2002, Pulleyblank 1986, I assume hierarchical tone features throughout: the pitch range is split into two ‘registers’ with the higher register corresponding to a high tone ([+Upper (register)]) and the lower register to a low tone ([−Upper (register)]); in languages with multiple tones, these registers can be further divided into higher and lower upper and lower register tones ([±Raised]).² The rule is structure-building in the sense that it adds tone feature values but does not change any tones present in the input.

(3) Polarity rule

$V' \rightarrow [-\alpha\text{Upper}] / [\alpha\text{Upper}] \text{ ____}$

The prime in V' indicates that the vowel is unspecified for tone (Clements & Ford 1979). This exempts any vowel lexically specified as L or H.

Assuming for Hausa that the stabiliser is underlyingly unspecified for tone, this would allow the following derivations for examples such as [jàakíi née] ‘it’s a donkey’, [zóobèe née] ‘it’s a ring’, and [húulúnàa née] ‘they’re caps’.

(4) Derivations using a polarity rule: Hausa

<i>Underlying</i>	/jàakíi nee/	/zóobèe nee/	/húulúnàa nee/
<i>Polarity</i>	èè	ée	ée
<i>Surface</i>	[jàakíi née]	[zóobèe née]	[húulúnàa née]

A comparable effect can be achieved within Optimality Theory (Prince & Smolensky 1993, McCarthy & Prince 1993) by reinterpreting the polarity rule as a constraint operating on the tonal tier.

(5) POLARITY constraint

*[αUpper] [αUpper]

Assuming that all vowels require a tone on the surface (derivable by an independent constraint requiring all vowels to have a tone), the effect of the POLARITY constraint is restricted to unspecified vowels by ranking it below IDENT(tone), which requires that input-output segment pairs be identical in their tone.³

²See §9 for discussion of three-tone systems.

³See McCarthy & Prince 1995 for detailed discussion of faithfulness constraints like IDENT. Note that DEP(tone) would also be violated in cases involving a surface tone and no underlying tone, but this would not differentiate inserted-L and inserted-H and would therefore have no bearing on these cases.

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- (6) IDENT(tone): Correspondent segments have identical values for [Upper].
IDENT(tone) >> POLARITY (*[αUpper] [αUpper])

- (7) Hausa: polarising suffixes

- i. jàakíi nèe ‘it’s a donkey’

	/jàakíi nee/	IDENT(tone)	POLARITY
☺	a. [jàakíi nèe]	*	
	b. [jàakíi née]	*	*!

- ii. zóobèe née it’s a ring’

	/zóobèe nee/	IDENT(tone)	POLARITY
	a. [zóobèe nèe]	*	*!
☺	b. [zóobèe née]	*	

- iii. húulúnàa née ‘they’re caps’

	/húulúnàa nee/	IDENT(tone)	POLARITY
	a. [húulúnàa nèe]	*	**!
☺	b. [húulúnàa née]	*	*
	c. [hùulúnàa née]	**!	

One violation of IDENT(tone) is unavoidable since the stabiliser is toneless underlyingly but bears a tone on the surface; any change in underlying tones, as in (7iii.c), is disallowed.

An alternative to underlying lack of specification, but still an account of polarity based on a morpheme’s underlying representation, is to assume that a polarising morpheme underlyingly has both a L and a H, but that only one of the two is realised on the surface: the L surfaces adjacent to H and the H surfaces adjacent to L. Such a multi-tone account could be applied to the case just seen in Hausa, but is illustrated here by polarity in Laarim.

3 Polarity by including both L & H: Laarim

Laarim is a Nilo-Saharan/Surmic language spoken in South Sudan (glottocode: nari1240). [Stirtz 2012](#) demonstrates that polarity is exhibited on nominal case markers. I illustrate with nominative, ablative and genitive case suffixes here, the latter two exhibiting tone polarity. For additional cases, refer to [Stirtz 2012](#).

- (8) Laarim: three singular cases, invariant L vs. polarity

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<i>stable L</i>		<i>polar</i>		
	<i>nominative</i>	<i>ablative</i>	<i>genitive</i>	<i>gloss</i>
a.	ɲérér-ì	ɲérér-à	ɲérér-ò	‘fruit’
b.	bàlàŋ-ì	bàlàŋ-á	bàlàŋ-ó	‘salt’
c.	mìḍìl-ì	mìḍìl-à	mìḍìl-ò	‘shin’
d.	ḍḍlòw-ì	ḍḍlòw-á	ḍḍlòw-ó	‘shrew’ ⁴
e.	dòwóc-ì	dòwóc-à	dòwóc-ò	‘bowl’

As can be seen in (8), the tone of the nominative singular suffix is invariably L. The tones of the ablative singular and genitive singular, however, are polar: the suffix is L following a H or rising tone (8a,c,e) and H following a L (8b,d). Note that a rising tone conditions a polar L to its right, consistent with an analysis of a rising tone as being composed of a LH sequence.

The polar pattern cannot be attributed to a putative underlying H property. As shown in the verbal and adjectival forms in (9), invariant H, like invariant L, can contrast with a polar pattern.

(9) Laarim: invariant H vs. polarity

<i>stable H</i>		<i>polar</i>	
	<i>plural stative verb</i>	<i>singular adjective</i>	<i>gloss</i>
a.	máán-tí	máán-ì	‘yellow’
b.	gídán-é	gídán-ì	‘brown’
c.	tùùr-é	tùùr-í	‘short’
d.	lèhèḍ-é	lèhèḍ-í	‘sharp’

The tonal patterns of Laarim deserve much more attention than the brief discussion here; see Stirtz 2012. Crucial for our purposes, both stable invariant H and stable invariant L exist. This means that the polar morphemes cannot be considered either L or H without some additional distinction, either phonological or morphological. That is, polar tones must in some way be distinguished from the invariant level tones. The required distinction could be achieved using tonally underspecified lexical entries, as seen in §1 for Hausa or by morphological marking, as seen in §7 below for Margi. Here we illustrate an alternative multi-tone account as proposed in Stirtz 2012.

Like the proposal that polarity be achieved by tonal underspecification, the analysis in Stirtz 2012 is representational. Following a proposal made by Trommer 2005 for Kanuri, Stirtz proposes that the polar morphemes have a tonal sequence underlying, specifically, HL. Noting that falling tones never occur on a

⁴Stirtz 2012 points out that the initial syllable of a HL root like ḍḍlòw appears with a L in certain suffixed forms, noting that the conditions governing this root alternation require further investigation.

single mora, Stirtz proposes that a prohibition on sequences of H tones results in only the L of the polar HL sequence occurring when the root ends in H; since both tones cannot be realised on the suffix, the H is retained and the L is lost when the root ends in a L tone.

In a standard Optimality Theory analysis, these effects can result directly from the Polarity constraint given in (5) in conjunction with basic conditions governing falling contours. Specifically, since it is not possible for both H and L tones of the proposed underlying representation to survive (without creating an illicit falling tone), we need to assume that the prohibition on a falling tone (*HL) outranks the constraint that would otherwise require the preservation of underlying tones (MAXTONE).

(10) Constraints on tone sequences

- *HL Assign a violation to any HL sequence associated to a single mora.
- MAXTONE Assign a violation for any tone in the input that does not have a corresponding tone in the output.

(11) Laarim: polarising suffixes

i. ηérér-à ‘fruit-SG.ABL’

/ηérér-a HL/	POLARITY	*HL	MAXTONE
a. [ηérér-â]		*!	
b. [ηérér-á]	*!		*
☺ c. [ηérér-à]			*

ii. bàlàn-á ‘salt-SG.ABL’

/bàlàn-a HL/	POLARITY	*HL	MAXTONE
a. [bàlàn-â]		*!	
☺ b. [bàlàn-á]			*
c. [bàlàn-à]	*!		*

In its essence, the Laarim pattern exhibits a classic polarity pattern. The polar suffixes are L after a preceding H, H after a preceding L; polar suffixes must be distinguished in some way from both stable L-tone suffixes and stable H-tone suffixes and this is accomplished by including a tone sequence underlyingly. The proposal is abstract in that the proposed HL sequence never occurs on the surface. In the next section, we consider a more concrete alternative.

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4 Abstractness and morph sets: Laarim, continued

By definition, polarising morphemes have more than one surface form: were they to be consistently low, or consistently high, they would not be polarising. In addition, at least in some cases of polarity, the polarising morpheme cannot be intrinsically low or high since instances of such morphemes occur, and do not alternate. We have just seen this type of three-way distinction – low vs. high vs. polarising – in Laarim, and it is cases like these that arguably differentiate polarity from dissimilation.

A possible conclusion is that polarising morphemes must be abstract in some way. We have just seen two versions of abstractness. The first was to posit toneless morphemes underlyingly, even though tonelessness is never attested on the surface.⁵ The second was to posit a contour tone even though such a contour is never attested on the surface. These approaches to polarity share the property of positing an abstract representation for the polar morphemes, one which must be eliminated during the derivation.

An alternative ‘concrete’ approach is possible, one that does not require the postulation of abstract underlying representations. Since polarising morphemes involve more than one surface realisation, a concrete option is to encode surface forms directly in lexical representations, along the lines of Emergent Phonology (Archangeli & Pulleyblank 2022).⁶ While invariant morphemes have only one representation, polarising morphemes have more than one surface form:

- (12) Lexical morph sets, illustrated with Laarim
- | | | | |
|----|-------------|--------|----------------------------------|
| a. | invariant H | {H} | e.g., {é} ‘plural stative’ |
| b. | invariant L | {L} | e.g., {i} ‘nominative’ |
| c. | polarising | {L, H} | e.g., {à, á} ‘singular ablative’ |

The fundamental ‘concrete’ property of Emergent Phonology is the assumption that a speaker’s knowledge of language results from general cognitive properties, that there are no – or at least very few – language-specific properties. Human learners encounter language data and develop an analysis using cognitive properties such as memory, identification of similarity, ability to generalise,

⁵This is not to say that tonelessness is impossible in a ‘concrete’ analysis. Arguments in favour of surface tonelessness have been made, for example, in cases of an accentual nature, e.g. Japanese (Pierrehumbert & Beckman 1988). In the cases involving polarity discussed here, however, all tone-bearing units bear a tone on the surface.

⁶As pointed out by a reviewer, the appeal to morph sets relates conceptually to the proposals for phonologically conditioned allomorphy in Bonet et al. 2007, Mascaro 2007. The two kinds of approaches differ in their applicability to phonologically related morphs such as those here. See Archangeli & Pulleyblank 2022 for discussion.

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attention to frequency, and so on (Archangeli & Pulleyblank 2015, 2022). Two specific properties of Emergent Grammar are relevant here. First, constraints are not derived from some universal set but developed through exposure to data (Hayes & Wilson 2008). Hence conditions such as POLARITY would be derived through the identification of skewed frequency distributions of morphological forms. Second, the candidates considered in the assessment of possible words are based strictly on morphological possibilities that have either been observed or postulated on the basis of lexical redundancy. The candidate set is finite, not infinite as in OT, and directly reflects observed possibilities. Consider the application of these two principles to polarity in Laarim.

The condition POLARITY would result from the observation of skewing towards sequences that do not exhibit identical tones. Where possible, morphemes will avoid such sequences. The candidate sets result from the combination of observed forms. Roots in Laarim are not observed to alternate and would therefore be invariant; there would be a single tonal form for each root, exhibiting the observed surface tone pattern. Suffixes too would simply reflect what is observed. As encoded in (12), some morphemes never alternate and would therefore have a single form in the possible set of morphs, H or L as appropriate (12a, b). Polarising morphemes would have both H and L morphs since both are observed in combination with different roots. Since there is no pattern to this set of possibilities (morph sets may have only a H, only a L, or both H and L), there would be no lexical redundancies to augment these lexical options for suffixes.

Compare H and L roots with a polar suffix, and a H root with a stable H suffix.

(13) Laarim: morph sets

i. ɲérér-à ‘fruit-SG.ABL’

{ɲérér} _{FRUIT} -{à, á} _{SG.ABL}		POLARITY
a.	[ɲérér-á]	**!
☺ b.	[ɲérér-à]	*

ii. bàlàn-á ‘salt-SG.ABL’

{bàlàn} _{SALT} -{à, á} _{SG.ABL}		POLARITY
☺ a.	[bàlàn-á]	*
b.	[bàlàn-à]	**!

iii. gídán-é ‘brown-PL.STATIVE’

{gídán} _{BROWN} -{é} _{PL.STATIVE}		POLARITY
☺ a.	[gídán-é]	**

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I assume that LL and HH roots unavoidably violate POLARITY given their sequence of identical tones; roots are invariant so if a form violating POLARITY is observed there is no way to improve on it. In the singular ablative cases ([ɲérér-à], [bàlàn-á] (13i, ii)), the suffix has two possible forms hence the form with a different tone than the root is chosen. In the example [gídán-é] (13iii), there are violations of POLARITY but because both the root and the suffix have a single observed tonal form there is no way to repair the violation.⁷ Critically, the number of candidates in each case directly reflects the observed possibilities for each morpheme involved.

To conclude to this point, we have seen that positing a rule or constraint of polarity provides an account of tone polarisation, provided that we have some mechanism for identifying the morphemes with polar behaviour. Two types of approaches can be contrasted. Abstract approaches posit an underlying representation, typically underspecification or contour tones, that do not occur on the surface. These representations are repaired in such a way as to achieve polarity. A concrete approach is to actually include lexically the surface variants attested for a given morpheme, with a constraint on polarity choosing between the available options. We will consider both types of approaches below as additional cases of polarity are considered.

In the next section, a case is considered that in broad strokes is comparable to those already seen. Certain details, however, will be seen to demand attention.

5 Lexical distinctions between roots: Kɔnni

Mabia languages have figured prominently in the literature on tone polarity. Cases include Dagaare (Anttila & Bodomo 2000, Angsongna 2023), Dagbani (Hyman 1993, Olawsky 1999), Kɔnni (Cahill 2004), Lama and Mooré (Kenstowicz et al. 1988). I consider here the case of Kɔnni (Mabia; Ghana; glottocode: konn1242).

Like other Mabia languages, nouns exhibit suffixes specific to their membership in a number of classes. Cahill 2004 shows that Kɔnni has five such classes, each class distinguishing singular and plural suffixes. In most instances, class

⁷Crucial here is the concreteness of Emergent Phonology. Since the plural stative suffix never alternates, the learner never observes – and therefore never postulates – a non-H form for the suffix. As pointed out by a reviewer, in a classic OT analysis, some condition would be required to rule out a candidate like *[gídán-è] which would respect POLARITY. Presumably the impossibility of a L on the plural stative suffix could be accounted for by a combination of abstract underlying representation with appropriate faithfulness constraints, e.g., IDENT(tone).

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suffixes are high-toned; in noun class 1 plural, however, the suffix exhibits polar behaviour. The behaviour of both tonal types of suffixes can be seen in (14).⁸

(14) Kɔnni: Polarity 1, noun class 1 singular (non-polar) vs. plural (polar)

	<i>singular</i>	<i>plural</i>	<i>gloss</i>
a.	tǎ-ŋ	tàn-á	‘stone’
b.	sí-ŋ	sí-à	‘fish-sp’
c.	bìis-ín	bìis-á	‘breast’
d.	tíg-ín	tíg-è	‘house’
e.	zònzó-ŋ	zònzó-à	‘maggot’

As seen in (14), the plural suffix is H after a L and L after a H, hallmark polarity. In other classes and in the class 1 singular, the suffix is consistently H; see Cahill 2004, as well as the singular forms in (14).

Using the morph set approach of Emergent Phonology, the analysis of polarity in Kɔnni can be comparable to that just seen for Laarim.

(15) Kɔnni: morph sets

i. tàn-á ‘stone-PL’

{tàn} _{STONE} -{à, á} _{PL}		POLARITY
☺ a.	[tàn-á]	
b.	[tàn-à]	*!

ii. zònzó-à ‘maggot-PL’

{zònzó} _{MAGGOT} -{à, á} _{PL}		POLARITY
a.	[zònzó-á]	*!
☺ b.	[zònzó-à]	

The polarity pattern seen in (14) could equally well be treated using a lack of tonal specification in the input and a rule of polarity or a constraint set including polarity.

Interestingly, one class of cases complicates the polarity picture in Kɔnni. As seen in (16), the noun class 1 plural suffix bears a falling tone in a sub-class of nouns. Contrast the expected case (16a), where the polar suffix is L after H, with the falling tone case (16b), where the polar suffix is unexpectedly a falling tone after H.

⁸The a~e alternation is due to vowel harmony and is not treated here.

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- (16) Kɔnni: Polarity 2, noun class 1 sub-class
- a. móg-úŋ móg-à ‘river; rivers’
- b. ɖág-ín ɖág-â ‘shade; shades’

The special property of such cases is a root property: this is not a new suffix – it is exactly the plural marker that in other cases exhibits routine polar behaviour. Some sort of special marking must be associated with the root. As discussed in §4, there are two sorts of strategies that can be adopted in such an instance.

A ‘concrete’ approach observes that the suffix has a special form, a falling tone, giving three surface options for the the plural marker: {à, á, â}. In general, the dispreference for the falling tone can be attributed to a condition against contour tones:

- (17) Dispreference for contour tones
- NoCONTOUR Assign a violation to any contour tone.

In the general case, the NoCONTOUR condition rules out the falling tone suffix. Consider revised illustrations for ‘stone-PL’ and ‘maggot-PL’, seen above in (15).

- (18) Kɔnni: morph sets version 2
- i. tàn-á ‘stone-PL’

{tàn} _{STONE} -{à, á, â} _{PL}		POLARITY	NoCONTOUR
☺ a.	[tàn-á]		
b.	[tàn-à]	*!	
c.	[tàn-â]		*!

- ii. zònzó-à ‘maggot-PL’

{zònzó} _{MAGGOT} -{à, á, â} _{PL}		POLARITY	NoCONTOUR
a.	[zònzó-á]	*!	
☺ b.	[zònzó-à]		
c.	[zònzó-â]		*!

In the special sub-class of roots like ‘shade’, the special root property of selecting for a contour can be directly encoded. Unlike the regular case where contour tones are avoided, roots like ‘shade’ are preferentially followed by a contour. Following proposals for lexical conditioning such as Pater 2000, Archangeli & Pulleyblank 2022, this results from the condition in (19).

- (19) Roots selecting a falling suffix

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NOLEVEL $\ast\{\}_{\alpha}$ LEVEL; $\alpha = \{\text{shade}, \dots\}$; Domain = Word
Assign a violation to any root of the α class followed by a suffix with a level tone.

Membership in the class of α roots is completely arbitrary and must be learned; there is no phonological property identifying this class. Their surface property, as encoded, is simply that they cannot be followed by a level tone.

In cases like *tàn-á* ‘stone-PL’ and *zònzò-à* ‘maggot-PL’, this condition will have no effect since the roots are not members of this arbitrary class. For a root like ‘shade’, however, assignment to the arbitrary α class invokes $\ast\{\}_{\alpha}$ LEVEL. As illustrated in (20), $\ast\{\}_{\alpha}$ LEVEL must outrank NOCONTOUR; it must also outrank POLARITY under the assumption that a high tone followed by a falling tone is a POLARITY violation, as would be the case if a fall consists of a HL sequence. Nothing crucially depends on this assumption here, however.

The effect of $\ast\{\}_{\alpha}$ LEVEL is illustrated in (20). In comparing the two cases, keep in mind that the number of candidates is determined by the number of options in each morph set, hence a single candidate in (20i) and three candidates in (20ii) because of the number of observed suffixal possibilities.

(20) Kɔnni: morph sets version 3

i. *ɕág-ín* ‘shade’

$\{\text{ɕág}\}_{\text{SHADE}, \alpha} - \{\text{ín}\}_{\text{SG}}$	$\ast\{\}_{\alpha}$ LEVEL	POLARITY	NOCONTOUR
☺ a. $[\text{ɕág}_{\alpha} - \text{ín}]$	*		

ii. *ɕág-â* ‘shades’

$\{\text{ɕág}\}_{\text{SHADE}, \alpha} - \{\text{â}, \text{á}, \text{à}\}_{\text{PL}}$	$\ast\{\}_{\alpha}$ LEVEL	POLARITY	NOCONTOUR
a. $[\text{ɕág}_{\alpha} - \text{á}]$	*!	*	
b. $[\text{ɕág}_{\alpha} - \text{à}]$	*!		
☺ c. $[\text{ɕág}_{\alpha} - \text{â}]$		*	*

As seen by comparing the singular and plural forms, (20i) vs. (20ii), in order for $\ast\{\}_{\alpha}$ LEVEL to have an effect two conditions must be met: the root must be a member of the phonologically arbitrary class that triggers this effect, and the suffix must have a form with a contour tone. Just as a typical high-tone suffix cannot satisfy POLARITY when following a H, such a suffix cannot satisfy $\ast\{\}_{\alpha}$ LEVEL when following a member of the α class.

Cahill 2004 proposes a more abstract analysis, abstract in the sense that the special behaviour of this kind of root is attributed indirectly to a phonological property for which there is no phonetic evidence. Specifically, the phonetic H

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tones of [móŋ-úŋ] ‘river-SG’ and [ɖág-ín] ‘shade-SG’ are assumed to be phonologically different: /móŋ/ is proposed to have a single H tone while /ɖág/ has two H tones. This interacts with a particular definition of the ‘POLAR’ constraint:

(21) Definition of polarity in Cahill 2004

POLAR: in a Noun Class 1 plural, the last tone of the plural is opposite in value to the immediately preceding tone.

Constraints are ranked in such a way as to make it optimal for the second H to link to the suffix in [ɖág-â] ‘shade-PL’ with a L inserted to satisfy the requirement that ‘the last tone’ of the plural be opposite.

Cahill’s definition of POLAR is interesting in two respects. First, as just described, the constraint triggers contour formation in appropriate cases though polarity typically involves level tones. Second, the constraint encodes both phonological information (‘the last tone ... is opposite in value to the immediately preceding tone’) and morphological information (‘in a Noun Class 1 plural’). The reason for the morphological encoding is twofold. First, in terms of observational facts, polarity is specific to this morphological context. In the morph set approach, this is directly encoded by building H and L morphs into the morph set. In classic Optimality Theory, the candidate set for any morpheme could contain both faithful tones and polar opposite tones; whether a morpheme exhibits or does not exhibit polarity must therefore appeal to morphological conditions built into otherwise phonological constraints. Since such morphological content is built into the structure of a morph set in the Emergent approach, the condition itself (POLARITY (5)) does not need to make any reference to morphology.

One way in which the case of Kɔnni is interesting is that neither the morph-set analysis nor the analysis in Cahill 2004 invokes underspecified lexical entries. We consider this issue in more depth in the next section, looking at another Mabia language, Dagaare.

6 Separating polarity from underlying lack of specification: Dagaare

Dagaare (Mabia; Ghana; glottocode: sout2789; Bodomo 1997, Anttila & Bodomo 2022, Angsongna 2023), exhibits polarity in a large number of verbal and nominal suffixes. This is illustrated here with the nominal plural suffix *-ri/-rɪ*.⁹

⁹Dagaare has a system of noun classes involving morphological class marking combined with number (Bodomo 1997, Bodomo & Marfo 2006, Ali et al. 2021), as mentioned above for Kɔnni.

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(22) Dagaare: polarity in number suffixes (initial)

- a. pìi-rí ‘rock-PL’
 sùù-rí ‘knife-PL’
 dò-rí ‘warthog-PL’
 pè-rí ‘basket-PL’
- b. kóg-rì ‘mahogany-PL’
 dág-rì ‘box-PL’
 ʃúú-rì ‘moon/month-PL’
 wég-rì ‘log-PL’

As seen in (22), the suffix is H when the root is L (22a) and L when the root is H (22b). This core pattern could be treated straightforwardly by the kinds of mechanisms discussed so far.

The situation, however, is more complex. In addition to cases where suffixes polarise, there are roots that agree in tone with the same otherwise polarising suffixes. Consider the data in (23) which do not exhibit polarity.

(23) Dagaare: polarity in number suffixes (expanded)

- a. tòò-rì ‘tobacco-SG’
 gbàg-rì ‘savannah monitor lizard-PL’
 ɖùgl-ì ‘vulture-SG’
- b. bíi-rí ‘child-PL’
 bóó-rí ‘goat-PL’
 nág-rí ‘root-SG’
 pég-rí ‘shell-PL’

The situation for the L+L cases in (23a) is quite clearly exceptional. Anttila & Bodomo 2000: p. 121 note ‘the systematic absence of LL nominals’; Angsongna 2023 also considers this tone type marginal, with only the three nouns given in (23a) appearing in his database.

The H+H cases, however, form a robust class. Indeed, in Angsongna’s appendix of ‘tone classes in nouns’, the following distribution is observed:¹⁰

(24) Dagaare: numbers of root types; sample of 131 noun roots

Alternations in the vowel quality of *-ri/-rì*, advanced vs. retracted vowels, are determined by vowel harmony and are orthogonal to the discussion of tone. See Bodomo 1997, Ozburn et al. 2018, Angsongna & Akinbo 2022, Angsongna 2023.

¹⁰Not included here are an additional 7 nouns with rising contours on the root and 2 nouns with falling contours. Nominalisations are also excluded.

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[Root]+suffix	example	count
[L]+L	tòò-rì (23c)	3
[L]+H	sùù-rí (23a)	43
[H]+L	ʈúú-rì (23b)	26 ¹¹
[H]+H	bóó-rí (23d)	59

As can be seen in (24), the H+H class is in fact the most numerous.

In both Anttila & Bodomo's analysis and Angsongna's, a key observation is that the number suffix is H in most cases. This can be accounted for either by assuming that the suffix is underlyingly H (Anttila & Bodomo 2022, Angsongna 2023) or that it is underlyingly toneless and assigned a H (Anttila & Bodomo 2000). Setting aside the [L]+L cases as exceptional, either of these proposals in combination with POLARITY derives the observed results for the [L]+H and [H]+L cases. For the special, non-polarising [H]+H cases, the key to their behaviour is an additional fact.

When a nominal root appears in a noun-adjective compound, the root does not have a number suffix. In this type of construction, roots of the types seen in the polarising patterns are unchanged, but roots of the [H]+H type exhibit H~L alternation. Consider the data in (25), illustrating the three productive classes.

(25) Dagaare noun-adjective compounds

a.	pìi-rí	'rock-PL'	pì-bílé	'small rock'
	dò-rí	'warthog-PL'	dò-bílé	'small warthog'
b.	dág-rì	'box-PL'	dág- [↓] bílé	'small box'
	só-rì	'path-SG'	só- [↓] bílé	'small path'
c.	bíi-rí	'child-PL'	bì-fáá	'bad child'
	váál-á	'leaf-PL'	vàál-zíé	'red leaf'

The proposal in Anttila & Bodomo 2000, Angsongna 2023, following proposals by Kenstowicz et al. 1988 for Moore and Lama, and by Hyman 1993 for Dagbani, is that the difference between the [H] roots, (25b), and the alternating [H]~[L] roots, (25c), is that the former are underlyingly H and the latter are underlyingly toneless. Polarity holds of the underlyingly specified tones, (25a,b), while some other mechanism accounts for the HH sequence attested with an underlyingly toneless root. Anttila & Bodomo 2000 proposes an account based on stress while Angsongna 2023 proposed an account based on the alignment of root tones. I will not go into the details of either account since the details are essentially orthogonal to polarity per se.

¹¹ Angsongna 2023 gives a total of '27' for this class but lists 26.

The crucial observations for polarity are twofold. First, polarity, while robust, is restricted to a subpart of the lexicon. Second, using the device of unspecified underlying representations does not automatically result in polarity; here the roots that would be analysed as unspecified do not exhibit polarity, while suffixes may or may not depending on other factors. We pursue this latter point further in the next section, examining polarity in Margi.

7 Lexical specificity: Margi

In the preceding discussion, attempts have been made to trigger polarity through reference to properties, abstract or concrete, of the phonological and morphological structure. An additional option, however, is to mark the morphemes that exhibit polar effects in a completely arbitrary fashion. Invariant morphemes would be unmarked; polar morphemes would be signalled by a diacritic which we can refer to here simply as, ‘polar’. In the Hausa example of §1 and §2, for example, the stabiliser morpheme could simply bear the diacritic mark ‘polar’: /nee_{polar}/. As such, a rule comparable to (3), but without requiring reference to underspecification, would apply only to morphemes bearing the diacritic, *polar*; a constraint comparable to (5) would be similarly restricted.

While at first blush, this appears to be an undesirable approach, we will see in this section a set of polarity cases where it appears that some nonphonological means must indeed be invoked to identify polarising morphemes. That is, it is not enough to say that a morpheme is L, or H, or even underlyingly unspecified – in certain cases some apparently arbitrary means must be found to indicate that a specific morpheme is polarising. A clear example of this is found in Margi.

Margi, a Chadic language of Nigeria (glottocode: marg1265, Hoffmann 1963) has several polarising morphemes. Consider as a first example the data in (26) illustrating the present tense.¹²

(26) Margi present tense

	<i>verb stem</i>	<i>present</i>	
a.	sá	à-sá	‘err, go astray’
b.	wì	á-wì	‘run’
c.	vəl	á-vəl	‘jump, fly’

¹²In transcriptions below, the symbol ‘y’ is used for a palatal glide; ‘y’ represents a glottalised palatal glide; the symbol ‘j’ represents a voiced alveopalatal affricate; ‘ny’ represents a palatal nasal. Hoffmann uses ‘u’ to represent a ‘centralized (?) close back vowel without lip-rounding’ (Hoffmann 1963: p. 21); I use ‘u’ for this vowel here.

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As seen in (26), the prefix is low before a high stem, and high before a stem that is either low or rising. We can assume that the behaviour before a rising tone, as in (26c), is because the rise consists phonologically of a LH sequence.

The present tense prefix behaves in a polar fashion in additional related constructions, in some cases involving more than one morphological marker. For example, (27a,b) show the same prefix in what Hoffmann refers to as the “adverbial present” while (27c,d) illustrate the past tense where the prefix combines with a suffix.

(27) Margi tenses involving the present tense prefix

	<i>verb stem</i>	<i>present adverbial</i>	
a.	lù ~ lù ¹³	à-lù	‘go to; direction away’
b.	bà	á-bà	‘go out, across; out, over, across’
	<i>verb stem</i>	<i>past tense</i>	
c.	sá	à-sá-rì	‘err, go astray’
d.	wì	á-wì-rì	‘run’

Before turning to possible analyses of polar morphemes in Margi, it is important to establish a couple of additional properties. First, polarising morphemes need not be prefixes. For example, the present tense prefix seen in (27) above appeared before the conditioning stem while the pronominal enclitics seen in (28) below appear after the stem conditioning their tonal value.

(28) Margi: subject pronouns, part 1

- | | | | |
|----|--------------------------|-----------|--------------------|
| a. | Stem-final H; enclitic L | màrgyí gù | ‘you-SG are Margi’ |
| b. | Stem-final L; enclitic H | hègyì gá | ‘you-SG are Higi’ |

Second, not all morphemes within a particular semantic or syntactic class behave the same way. Of the eight subject pronouns, for example, six alternate and two do not.

(29) Margi: subject pronouns, part 2; ‘X am/are/is Margi’, ‘Y am/are/is Higi’

¹³This is a ‘changing’ verb which has a high or low tone depending on context. This is discussed further below.

7 Tone polarity

	<i>singular</i>		<i>dual</i>		<i>plural</i>
<i>1p</i>	màrgyí yú	<i>exclusive</i>			màrgyí 'yà
	hègyì yú				hègyì 'yá
		<i>inclusive</i>	màrgyí mà		màrgyí mër
			hègyì má		hègyì mér
<i>2p</i>	màrgyí gá				màrgyí nyì
	hègyì gá				hègyì nyí
<i>3p</i>	màrgyí já				màrgyí ndà
	hègyì já				hègyì ndá

As seen in (29), the first person singular and the third person singular have stable high tones while all other subject pronouns exhibit polar behaviour.

The third property relates to the frequent suggestion that polarity should be analysed by positing morphemes that are underlyingly unspecified for tone. The crucial observation is that there are numerous cases involving a different kind of L~H alternation, one that does not involve polarity.

Consider, for example, the object pronoun. For six of the eight pronouns, the tone is a stable L. Examples are given in (30) with the first person singular object pronoun -*dǎ* after L-final and H-final verbs.

(30) Margi: first person singular object pronoun; pronoun underlined

L-final verb skà-dǎ | 'wait for me!'

H-final verb lèbá-dǎ | 'lead me!'

Differing tonally are the second and third person singular object pronouns. These two pronouns are L in phrase-final position but appear with the same tone as the preceding vowel in non-final position. Examples are given in (31) with the third person singular object pronoun.

(31) Margi: third person singular object pronoun; pronoun underlined

L-final verb nàjà ndà nà-nyì mài | 'he/she did not call him/her/it'¹⁴

H verb à-nǎ-nyí-r já | 'he/she called him/her/it'

This kind of agreement behaviour is canonical motivation for a tonally unspecified morpheme – but this leads to a conundrum. If an underlying lack of tonal specification leads to agreement in object pronouns as in (31), why would

¹⁴The verb *nǎ* 'call' has a changing tone, L or H depending on the context.

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it lead to polarity in subject pronouns as in (29)? The problem, moreover, is not restricted to objects.

Consider derived verbs, formed by attaching a suffix to a verbal root. Hoffmann 1963: p. 114 gives a list of 11 common verbal suffixes. Of these 11, 6 are consistently H, and 5 alternate between L and H, agreeing with the preceding verb root. An example of the stable H type is given in (32) and an example of the tonally agreeing kind is given in (33).

(32) Margi: verbal suffixes, stable H

a.	ptsà	‘roast’	ptsà-bá	‘roast thoroughly’
	ndàl	‘throw’	ndàl-bá	‘throw out’
b.	tá	‘cook’	tá-bá	‘cook all’
	ngúli	‘roar’	ngúli-bá	‘surpass in roaring’
c.	fà	‘take (many)’	fà-bá	‘take out (many)’
	ɲà	‘call’	ɲà-bá	‘call out’

(33) Margi: verbal suffixes, alternating

a.	ghàl	‘grow old’	ghàl-nà	‘wear out (garment)’
	ndàl	‘throw, cast’	ndàl-nà	‘throw away’
b.	tá	‘cook’	tá-ná	‘cook and put aside’
	sá	‘get lost’	sá-ná	‘lose’
c.	fà	‘take (many)’	fà-nà	‘take away (many)’
	bèl	‘break’	bèl-nà	‘break (off)’

These examples, and many others like them, illustrate two types of alternation, both of which are assimilatory – not polarising. First, in contrast with *-bá* which is consistently H, *-na* is L after a L root and H after a H root. Second, these examples illustrate three kinds of roots: (i) some roots are consistently L (32a, 33a); (ii) some roots are consistently H (32b, 33b); (iii) some roots alternate between L and H, agreeing with the suffix (32c, 33c).¹⁵ Overall it seems that agreement is the general pattern and polarity occurs with a special class of designated morphemes. Even when a class of morphemes, such as subject pronouns, exhibits polarity, it may not be all morphemes in the class that polarise. This was seen for the subject pronouns in (29) where dual and plural subjects polarise, but where in singulars it is only the second person singular that polarises.

The important point to note is that the phonological representation by itself is not enough in Margi to identify polarising morphemes. Non-polarising morphemes may be stable L or stable H; they may alternate between L and H. This

¹⁵Not given here is a class of roots that alternates between a rising tone and a LH sequence.

suggests that being L, or being H, or even being toneless underlyingly is insufficient to identify polarity. At least in some cases, some sort of morphological feature is needed to trigger polarity. This point has potential relevance of a broad nature, to which we turn in the next section.

8 Polarity vs. dissimilation

An interesting analytic question arises as to whether polarity exists analytically or whether instances of apparent polarity are the result of dissimilation. The issue has been argued in both directions. For example, Newman 1995, Cahill 2004, Hantgan 2009 argue in favour of polarity while Pulleyblank 1983, Suzuki 1998, Anttila & Bodomo 2000 suggest that polarity is a type of dissimilation.

The issue warrants detailed discussion that goes beyond the scope of this chapter. In this section, a few basic considerations are presented, but no attempt is made to adequately resolve the issue.

One basic consideration is *arbitrariness*. Newman 1995: p. 771 states this as follows: ‘In the case of “true polarity,” the tone of some element is always assigned as opposite to that of a neighboring tone, but there is no compelling synchronic reason to presume that the tone started underlying as some specified tone or other.’ He cites the example of the Hausa stabiliser discussed here in §1. Hantgan 2009 presents a similar argument for Bangime, but see Heath & Hantgan 2018. A frequent consideration in such cases is the idea that tonal stability results from lexically specified tones while tonal polarity can result from a lack of underlying tonal specification.

A rather different consideration is one of theoretical *economy*. Kenstowicz et al. 1988, Suzuki 1998 suggest, for example, that independently required constraints governing dissimilation are adequate to account for polarity. As such, polarity can be seen as an ‘epiphenomenon’, not a distinct type of process formally.

Perhaps more importantly, arguments can be made based on *descriptive adequacy*. I will give two brief examples here, one favouring a dissimilation account, the second favouring polarity.

Consider again the case of Margi seen in the last section (§7). A somewhat intricate set of patterns bears on the question of whether observed polarity is truly ‘polarity’ or more correctly dissimilation. The issue surrounds the behaviour of L, H and alternating verbs in a tense like the present as compared with a tonally neutral tense.

(34) Margi: alternating verbs with a polar prefix

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a. L verb	wì	‘run’	á-wì	‘PRES-run’
b. H verb	sá	‘err, go astray’	à-sá	‘PRES-err, go astray’
c. alternating verb	sà	‘drink’	à-sá	‘PRES-drink’

Low verbs are consistently L, as seen with ‘run’ in (34a); as expected the polar prefix for the present is H with such a verb. High verbs are consistently H, as seen with ‘err, go astray’ in (34b); also as expected, the polar prefix for the present is L with this class of verb.

What is relevant for the issue of dissimilation vs. polarity is the third type. Hoffmann 1963 clearly shows that alternating verbs of the type illustrated by ‘drink’ (34c) are L in a neutral context; Pulleyblank 1983 interprets this as meaning that the alternating verbs are tonally unspecified underlyingly, receiving a default L in a neutral context. At issue then is the source of the verb’s high tone in the present tense. Why do we observe *à-sá* ‘PRES-drink’ rather than **á-sà*? While what Newman 1995 characterises as a ‘straightforward’ account assuming polarity would account for the tone of the prefix, it leaves unexplained the tone of the verb root in these alternating examples. Pulleyblank 1983 suggests that the root tone can be accounted for by interpreting surface polarity as dissimilation, proposing that the present tense prefix has an underlying floating H that links to the root when the root is underlyingly unspecified for tone (cf. Rolle et al. 2026 [this volume]). Such an analysis makes superficial polarity in effect analytic dissimilation.

A case of the opposite persuasion can be seen with Kɔnni (§5). Cahill 2004 advances various arguments against a dissimilation account of Kɔnni polarity, and therefore in favour of a formal polarity account. His first class of arguments is based on special cases which seem to point towards positing sequences of identical tones, in violation of the OCP.

Consider again the pair seen above in (16), reproduced here as (35).

- (35) Kɔnni: Polarity 2, repeated
- | | | | |
|----|----------|--------|-----------------|
| a. | mɔ́g-ɔ́ŋ | mɔ́g-à | ‘river; rivers’ |
| b. | ɕʒág-ín | ɕʒág-â | ‘shade; shades’ |

Cahill 2004 proposes uncontroversially that the singular forms involve a high tone suffix.¹⁶ The plural form *mɔ́g-à* ‘rivers’ would be the straightforward result of polarity applying to the appropriate morphologically marked class. At issue is how to encode the special behaviour of forms like *ɕʒág-â* with a final falling

¹⁶This would appear to involve an OCP violation itself, though one might argue that the HH sequence in *mɔ́g-ɔ́ŋ* was the result of a single multiply-linked H.

tone in the plural. Cahill's proposal is that the roots followed by a falling tone are the result of the root having two H tones underlyingly. While this analysis would be problematic for a dissimilation account since the HH sequence violates the OCP, the form with a falling contour does satisfy the polarity constraint that he proposes.

He interprets various examples with sequences of surface low tones as additional evidence against the OCP. Compare the two 'LLH' examples in (36).

(36) Kɔnni: L~H alternations

- a. b̀̀r̀̀m̀̀íń 'donkey' ń wó ↓b̀̀r̀̀m̀̀íń 'I lack donkey'
- b. d̀̀m̀̀p̀̀á 'bench' ń wó d̀̀m̀̀p̀̀á 'I lack bench'

The alternation in (36a) is due to a constraint against HLH sequences. Assuming that the 'LL' sequence of *b̀̀r̀̀m̀̀íń* is a single multiply-linked L, this (multiply-linked) L alternates with ↓H when *b̀̀r̀̀m̀̀íń* follows a H-final form: ń wó ↓b̀̀r̀̀m̀̀íń, *ń wó b̀̀̀r̀̀m̀̀íń. Positing a multiply-linked L is consistent with the OCP. How then do we account for the failure to alternate seen with an otherwise comparable LLH sequence as in (36b)? Cahill's proposal is that *d̀̀m̀̀p̀̀á* has two lexical L-tones, that it is formally LLH. As such, when *d̀̀m̀̀p̀̀á* occurs after a H in ń wó d̀̀m̀̀p̀̀á, there is no violation of the *HLH constraint and therefore no alternation.

Overall, Cahill 2004 argues that the OCP appears to play no role in Kɔnni either in HH or LL sequences. If correct, this means that an alternative explanation must be found for the cases of tone polarity, hence his 'Polar' constraint.

As stated above, the discussion here makes no attempt to resolve the issue of whether tonal polarity can be accounted for by relatively standard, sometimes morphologically conditioned, dissimilation constraints or whether specific polarity constraints must be envisioned – though see the conclusion (§10) for some additional discussion. Indeed, the POLARITY condition (5) used throughout this paper could without modification be considered an OCP condition. Resolution of this issue clearly depends on an in-depth examination of multiple tonal polarity cases, as well as cases where feature disagreement occurs with other features.

9 Expanding the range of patterns: three tones

In numerous cases, as seen above, polarity involves alternations along a two-way continuum of Low versus High. Polarity may exist however in languages with more differentiated tone systems. In this section, we will consider four such cases, each involving a three-way set of tone distinctions.¹⁷

¹⁷Throughout the discussion of three-tone systems, H is transcribed by an acute accent, á; M is transcribed with a macron or by the absence of diacritic, ā or a; L is transcribed with a grave

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9.1 Lower register mid: Mono

Mono is a Banda language spoken in the Democratic Republic of Congo (Atlantic-Congo; glottocode: mono1270; Olson 2005). The language has three tone levels (L, M, H), as illustrated in (37).

- (37) Mono: three-way tone contrast
- a. gbándà ‘net’
 - b. gbāndà ‘hunting shelter’
 - c. àwó ‘lung’
 - d. àwō ‘wife’
 - e. mārā ‘barren’
 - f. màrā ‘clan, fetish’

Olson 2005 notes polarity in the Mono preposition for ‘to’:¹⁸

- (38) Mono: polarity in the preposition for ‘to’
- a. before H
 - gà-lá ‘to-in’
 - gà-úfú ‘to-outside’
 - gà-pá ‘to-top’
 - gà-ndá ‘to-home’
 - gà-sá ‘to-place’
 - b. before M
 - gá-māngbá ‘behind’
 - gá-ũfũ ‘to-in.front.of’
 - gá-mā ‘to-1SG’
 - c. before L
 - gá-ndìrì ‘to the side of’
 - gá-sè-ndá ‘to-place-home’¹⁹
 - gá-tà ‘to-on:QUEST’
 - gá-zè ‘to-there’

An important reason for considering such a prefix as polar, both in Mono and in many other cases, is that the alternating morpheme must be distinguished from nonalternating morphemes. In Mono, there are prefixes that are consistently H and prefixes that are consistently L.

accent, à.

¹⁸See Olson 2005: p. 113 for comments on language consultants’ comments concerning polarity.

¹⁹The tone pattern on the adverbial [sè-ndá] involves a replacement of the lexical tones; see Olson 2005: p. 114.

(39) Mono: nonalternating prefixes

- a. fá- ‘at’
 fá-ndó ‘at-home’
 fá-kē ‘at-here’
 fá-tà ‘at-on:QUEST’
 b. à- ‘PLURAL’
 à-gbózū ‘PL-elder’
 à-bākòngō ‘PL-turtle’
 à-gbòlò ‘PL-child’

Given nonalternating prefixes like those in (39), we cannot simply assume that the polar prefix is L or H. If the prefix was H, we could posit a pattern for the polarity cases in (38) where H dissimilates to L before H but this would be a problem for fá- ‘at’ (e.g., fá-ndó, *fà-ndó). Similarly, if the prefix was L, we could posit a pattern where L dissimilates to H before M and L but this would be a problem for à- ‘PLURAL’ (e.g., à-bākòngō, *á-bākòngō; à-gbòlò, *á-gbòlò).

In order to develop an account of this tone polarity in Mono, it is important that M and L tones be classified together in opposition to H tones: we observe H before M and L versus L before H. Though a variety of tonal feature systems would be adequate for characterising these classes, for concreteness I will assume as discussed briefly in §2, the hierarchical tone features of Yip 1980, 2002, Pulleyblank 1986.

(40) Tone features

+Upper register	+Raised	H	High
	–Raised	M _H	Higher.mid
–Upper register	+Raised	M _L	Lower.mid
	–Raised	L	Low

In a language with four tone levels, all four tones are phonetically distinguished. In a language with three tones, however, there is an issue as to whether the mid tone is a ‘higher.mid’ (M_H) or a ‘lower.mid’ (M_L) since there is a single ‘mid’. Assuming that the M of Mono is a lower register tone (M_L) to account for its patterning with L, that is, [–Upper, +Raised], polarity can result from a rule or constraint affecting [Upper] – exactly as assumed throughout. The constraint given in (5) is repeated here for convenience.

(41) POLARITY (repeated from (5))

*[αUpper] [αUpper]

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I illustrate the effect of POLARITY by means of an Emergent Grammar analysis. Lexically, the three prefixes considered in Mono are given in (42).

(42) Mono prefixes

- a. ‘at’ {bá}
- b. ‘PLURAL’ {à}
- c. ‘to’ {gà, gá}

As seen in (38) and (39), and as encoded in (42), ‘at’ is only observed with a H tone, ‘PLURAL’ is only observed with a L tone, while ‘to’ is seen both as H and as L. The result is that ‘at’ and ‘PLURAL’ will obey or violate Polarity given the vagaries of lexical combinations; on the other hand, given that ‘to’ has both L and H forms, the result will always respect surface polarity. Nonalternating affixes are illustrated with ‘at’; L prefixes would be comparable though opposite in tone.

(43) Mono: invariant prefixes

- i. bá-tà ‘at-on:QUEST’

{bá} _{at} -{tà} _{on:QUEST}	POLARITY
☺ a. [bá-tà]	
- ii. bá-ndó ‘at-home’

{bá} _{at} -{ndó} _{home}	POLARITY
☺ a. [bá-ndó]	*

If a root is L, as in (43i), the Polarity condition is satisfied; if a root is H, as in (43ii), then the Polarity condition is violated but there is no alternative form that would repair the violation: the root is invariant H as is the prefix.

In contrast, if a prefix is polarising, then the Polarity condition ensures the correct surface form.

(44) Mono: polarising prefixes

- i. gá-tà ‘to-on:QUEST’

{gà, gá} _{to} -{tà} _{on:QUEST}	POLARITY
a. [gà-tà]	*!
☺ b. [gá-tà]	
- ii. gá-mā ‘to-1SG’

{gà, gá} _{to} -{mā} _{1SG}	POLARITY
a. [gà-mā]	*!
☺ b. [gá-mā]	
- iii. gà-ndó ‘to-home’

{gà, gá} _{to} -{ndó} _{home}	POLARITY
☺ a. [gà-ndó]	
b. [gá-ndó]	*!

7 Tone polarity

There are several points to take away from this Mono example. First, polarity may be attested in tone systems with more than two tone levels. Second, in Mono, L and M (M_L) tones function as a class as opposed to H tones. Third, just as in two-tone systems, some morphemes polarise and some morphemes do not and this distinction must be lexically encoded. Fourth, tone polarity is enforced by a rule or constraint. The same condition adopted for two-tone patterns, POLARITY – which prohibits sequences of like values for [Upper] – is adequate for a three-tone pattern as in Mono.

In the next section, another three-tone case is considered, one where mid tones form a class with high tones.

9.2 Higher register mid: Bangime

Bangime, a language isolate spoken in Mali (glottocode: bang1363; [Heath & Hantgan 2018](#)), also exhibits tonal polarity in a three-tone system. Setting aside contours, the three level tones are illustrated in (45) by ‘undetermined singular’ forms.

- (45) Bangime: three level tones
undetermined sg
- | | | |
|----|--------------------|----------------|
| a. | bùr ⁿ à | ‘stick’ |
| b. | ɖʒèndʒé | ‘crocodile’ |
| c. | dāndì | ‘chili pepper’ |
| d. | párí | ‘arrow’ |
| e. | jáámbè | ‘child’ |

Tone polarity in Bangime is observed in several forms, including particular nominal and verbal suffixes, a relative marker, and so on. Polarity interacts in interesting ways with other tonal processes; see [Heath & Hantgan 2018](#) for detailed discussion. Here we restrict attention to one example, that of the plural suffix *-ndɛ*.

- (46) Bangime: tone polarity with plural *-ndɛ*
plural
- | | | |
|----|-------------------------|----------------------|
| a. | bùr ⁿ à-ndé | ‘sticks’ |
| | dāndì-ndé | ‘chili peppers’ |
| b. | kūwō-ndè | ‘houses’ |
| | ɲāār ⁿ ā-ndè | ‘giant pouched rats’ |
| c. | párí-ndè | ‘arrows’ |
| | tùbààkú-ndè | ‘white people’ |

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As seen in (46), *-ndɛ* is high when following a low tone and low when following either mid or high. Depending on certain assumptions – to which we turn below – the same polarity constraint, * $[\alpha\text{Upper}] [\alpha\text{Upper}]$ (used through this chapter), can account for the alternation seen here. The one difference between Mono and Bangime is that mid tones group with high tones in Bangime where they group with low tones in Mono.

(47) Mid tone features

language	Mono	Bangime
H	[+Upper, +Raised]	[+Upper, +Raised]
M_H		[+Upper, –Raised]
M_L	[–Upper, +Raised]	
L	[–Upper, –Raised]	[–Upper, –Raised]

Illustrations of the Bangime alternations are given in (48).

(48) Bangime: polarising suffixes

i.	bùr ⁿ à-ndé ‘sticks’	$\{\text{bùr}^n\text{à}\}_{\text{stick}} - \{\text{ndè, ndé}\}_{\text{PL}}$	POLARITY
		a. [bùr ⁿ à-ndè]	*!
		☺ b. [bùr ⁿ à-ndé]	
ii.	kūwō-ndè ‘houses’	$\{\text{kūwō}\}_{\text{house}} - \{\text{ndè, ndé}\}_{\text{PL}}$	POLARITY
		☺ a. [kūwō-ndè]	
		b. [kūwō-ndé]	*!
iii.	párí-ndè ‘arrows’	$\{\text{párí}\}_{\text{arrow}} - \{\text{ndè, ndé}\}_{\text{PL}}$	POLARITY
		☺ a. [párí-ndè]	
		b. [párí-ndé]	*!

As can be seen in (48ii), as long as a mid tone is analysed as being [+Upper, –Raised] (that is, M_H), Polarity ensures that the low tone form of the suffix is chosen. Having a high tone on the suffix would result in a [+Upper][+Upper] sequence, in violation of the constraint. Hence while Bangime can be treated in a manner that is in essence comparable to Mono, the tonal effects are somewhat different due to differing classifications of mid tones with respect to high and low.

Before addressing two additional examples of polarity in three-tone systems, the next section will consider analytic complications raised by such cases.

9.3 Tonal feature interactions in three-tone systems

While the analyses of Mono and Bangime help to demonstrate the importance of three-tone systems in understanding polarity, certain problems arise when considering details of their analysis. As seen in §9.1 and §9.2, an analysis rooted in morph sets can straightforwardly account for both three-tone languages. Consider, however, an approach where polar morphemes are identified not by the postulation of morph sets but by analysing the polarising morphemes as underlyingly unspecified. This assumption, while common in the literature for two-tone systems, poses certain problems of detail in three-tone systems.

Consider first a rule-based approach. For Mono, we could assume that the polarising prefix is toneless in its base form, giving underlying representations such as /ga-tà/, /ga-mā/ and /ga-ndá/. A rule such as in (3), reproduced here as (49), could then be assumed to produce polarity.

- (49) Polarity rule (repeated from (3))

$$V' \rightarrow [-\alpha\text{Upper}] / [\alpha\text{Upper}] \text{ ___}$$

The problem of detail is that this rule does not provide a value for [Raised], the feature needed to add a mid tone to the system. Hence an additional rule is needed to supply appropriate values for this second feature. The issues of detail are twofold. First, no single value of [Raised] is adequate. The high tone on the prefix in [gá-tà] and [gá-mā] requires the value [+Raised] to be inserted (refer to (47)) while the low tone in [gà-ndá] requires the value [−Raised]. Hence the rules assigning [Raised] values to an unspecified vowel must be sensitive to the value of [Upper]. Second, the inserted value cannot be the result of a polarity constraint on [Raised]. When a H prefix is the result before a L tone [gá-tà], the two tones are indeed different both with respect to [Upper] and with respect to [Raised]; when the H prefix is before a M tone as in [gá-mā], however, both the H prefix and the M root are [+Raised]. Polarity in such a case is specifically with respect to [Upper], not [Raised]. If an unspecified vowel is assumed to be the underlying representation of a polarising affix, then the Polarity rule must assign the correct value for [Upper] while redundancy rules must assign [+Raised] to a [+Upper] vowel and [−Raised] to a [−Upper] vowel.

A comparable problem arises in a constraint-based approach if the underlying representation of a polarising morpheme is assumed to be tonally unspecified. While a Polarity constraint could correctly govern values of [Upper], feature cooccurrence constraints would be required to ensure the appropriate values for [Raised].

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This problem is not specific to cases where the mid tone groups with the low tone, as in Mono. In Bangime, where the mid tone groups with the high tone as seen in (48), we see an entirely comparable problem. Although both [Upper] and [Raised] are polar opposites when the root tone is H or L, the suffix after a M tone (M_H), [+Upper, –Raised], is [–Upper, –Raised], that is, opposite with respect to [Upper] and the same with respect to [Raised].

9.4 Non-peripheral alternations: Ogori

In Mono and Bangime, we see cases where tonal polarity involves H~L alternations. Ogori (Atlantic-Congo; glottocode: okoe1238; Chumbow 1982), another three-tone language, exhibits alternations between M and L. Consider the following examples of possessive pronouns, taken from Chumbow (1982: p. 93).²⁰

(50) Ogori possessive pronouns

	igilà ‘yam’	érú ‘farm’	ówó ‘mouth’	alternation
1p sg	miigilà	mèérú	mòówó	$m\bar{V} \sim m\check{V}$
2p sg	wiigilà	wèérú	wòówó	$w\bar{V} \sim w\check{V}$
3p sg	jüigilà	jèérú	jòówó	$j\bar{V} \sim j\check{V}$
1p pl	tiigilà	tèérú	tòówó	$t\bar{V} \sim t\check{V}$
2p pl	niigilà	nèérú	nòówó	$n\bar{V} \sim n\check{V}$
3p pl	biigilà	bèérú	bòówó	$b\bar{V} \sim b\check{V}$

As seen in (50), the tone of a possessive pronoun is M when the immediately following tone is L, but L when the immediately following tone is H. There are no examples in Chumbow (1982) of possessives before mid tones. To ensure that a ML sequence does not violate POLARITY (41), mid tones must be analysed as forming a class with high tones, as in Bangime.

To illustrate again using assumptions of Emergent Grammar, the examples in (51) show combinations of the first person possessive prefix with L-initial and H-initial nouns.²¹

(51) Ogori: polarising prefixes

²⁰Vowel place assimilation is observed in the possessive pronouns but abstracted away from here; see Chumbow (1982) for discussion. Consistent with the transcriptions in Chumbow (1982), M tones are represented by the absence of any diacritic on the mid-toned vowel.

²¹I continue to abstract away from the vowel quality alternations. Building them into the analysis would simply result in larger morph sets.

i.	miìgilà ‘my yam’	{mi, mi} _{1.SG.POSS} -{ìgilà} _{YAM}	POLARITY
		☺ a. [mi-ìgilà]	
		b. [mì-ìgilà]	*!

ii.	mòówó ‘my mouth’	{mi, mì} _{1.SG.POSS} -{ówó} _{MOUTH}	POLARITY
		a. [mo-ówó]	*!
		☺ b. [mò-ówó]	

The M prefix is chosen in *mi-ìgilà* to avoid a L+L sequence; the L prefix is chosen in *mò-ówó* to avoid a M_H+H sequence, which would be an illicit sequence of [+Upper] tones.

As in the case of Mono and Bangime, the three-tone pattern of Ogori raises problems of detail in a standard OT analysis. If we assume both (i) that the underlying representation is toneless, following Chumbow 1982, and (ii) that mid tones are [+Upper, –Raised], then two options are equally good for L-initial nouns. If the prefix has a M tone, *mi-ìgilà*, then POLARITY is satisfied; POLARITY is equally satisfied, however, if the prefix has a H tone, **mì-ìgilà*. Since a H is a well-formed tone, an additional constraint must be invoked in order for M, not H, to be the attested result. That is, if a H tone form is included in the candidate set, as it would be in standard OT, some additional constraint is needed to make the M+L sequence optimal.

Taken together, Mono, Bangime and Ogori point to several conclusions: (i) three-tone systems can exhibit polarity, like two-tone systems; (ii) the alternations in a three-tone system may be between the peripheral tones (Mono, Bangime), or may involve M (Ogori); (iii) the M tones themselves may function as a class with either the H tone (Bangime, Ogori) or the L tone (Mono)

9.5 Multiple patterns and multiple features: Yorùbá

The final three-tone case to be considered here is that of clitics in Yorùbá (Atlantic-Congo; Nigeria and other countries; glottocode: yoru1245).²² As discussed in Akinlabi & Liberman 2001, Yorùbá clitics show polarity involving both H and L tones. Consider first the tonal alternations seen with object clitics, illustrated here with first and second person singular clitics.²³

²² As in Ogori, the M tone in Yorùbá is transcribed by the absence of a diacritic on the mid-toned vowel.

²³ Second person plural clitics present additional patterns of interest; see Akinlabi & Liberman 2001.

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(52) Yorùbá object clitics

- | | | | |
|----|---------|---------|-----------------------------|
| a. | after H | ó kọ mí | ‘he/she/it taught me’ |
| | | ó kọ ẹ | ‘he/she/it taught you-SG’ |
| b. | after M | ó pa mí | ‘he/she/it killed me’ |
| | | ó pa ẹ | ‘he/she/it killed you-SG’ |
| c. | after L | ó kọ mí | ‘he/she/it divorced me’ |
| | | ó kọ ẹ | ‘he/she/it divorced you-SG’ |

As illustrated, ‘the clitic tone cannot be the same as the tone of the previous vowel’ (Akinlabi & Liberman 2001: p. 41). The clitic is H after M and L tones, but M after H.

The emphatic clitic is similarly different from the preceding tone, but alternates between L and M. The emphatic clitic is illustrated here following a verb, but as shown in Akinlabi & Liberman 2001, it may follow verbs, nouns, adjectives, as well as other clitics.

(53) Yorùbá emphatic clitic

- | | | | |
|----|---------|--------------------|--------------------------------|
| a. | after H | ó dé | ‘he/she/it arrived’ |
| | | ó dé ẹ | ‘he/she/it arrived (EMPHATIC)’ |
| b. | after M | ó lọ | ‘he/she/it went’ |
| | | ó lọ ọ | ‘he/she/it went (EMPHATIC)’ |
| c. | after L | ó sùn [...sù] | ‘he/she/it slept’ |
| | | ó sùn un [...sù ũ] | ‘he/she/it slept (EMPHATIC)’ |

The emphatic clitic is L after H and M tones, but M after L.

Having a pair of polarising morphemes with opposite values poses an interesting conundrum: one case behaves as expected, the second raises difficulties. Consider first the object clitics. As argued in Akinlabi 1985, Pulleyblank 1986, the mid tone in Yorùbá should be analysed as a ‘lower mid’ (M_L), that is, [–Upper, +Raised]. That is, it should be analysed as in Mono (47). With this feature representation in mind, consider the tonal possibilities for roots followed by object clitics, represented schematically in (54).

(54) Yorùbá H~M: schematic tone combinations for the object clitic

<i>root+{clitic}</i> T+{H, M _L }	POLARITY *[αUpper] [αUpper]
H+{H}	*!
☺ H+{M _L }	
☺ M _L +{H}	
M _L +{M _L }	*!
☺ L+{H}	
L+{M _L }	*!

If a M is [−Upper], then POLARITY is violated if a M follows either another M or a L; in such cases a H is preferred. H after a H, however, violates POLARITY while M after H satisfies the constraint. The result, as desired, is that a M is obtained on an object clitic after H but a H is obtained after both M and L.

The problem arises because of the L~M (emphatic) clitic in addition to the H~M (object) clitic. Consider a schematic representation of the emphatic pattern.

(55) Yorùbá L~M: schematic tone combinations for the emphatic clitic

<i>root+{clitic}</i> T+{L, M}	POLARITY *[αUpper] [αUpper]
?? H+{L}	
?? H+{M _L }	
?? M _L +{L}	*
?? M _L +{M _L }	*
?? L+{L}	*
?? L+{M _L }	*

The problem is that if a M is [−Upper], i.e., M_L, then neither M_L nor L violates POLARITY after a H, but both M_L and L violate POLARITY after M_L and L. The analysis so far fails to correctly determine the emphatic pattern in all cases. The problem arises because of the presence of two clitics in Yorùbá with opposite tone patterns, H~M vs. L~M.

The hierarchical nature of the tone features points to a hierarchical solution to this problem. Until this point, consideration of [Upper] alone was sufficient to invariably determine polarity effects. This was necessarily the case in two-tone systems since only [Upper] plays a role; it was also the case in Mono, Bangime and Ogori given appropriate characterisations of the M tone. In Yorùbá, however, we must factor [Raised] into consideration. Consider again the schematic representation of emphatic clitics but factoring in polarity with respect to both [Upper] and [Raised], as sketched in (56).

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- (56) POLARITY-UPPER & POLARITY-RAISED
POLARITY-UPPER * $[\alpha\text{Upper}]$ $[\alpha\text{Upper}]$
POLARITY-RAISED * $[\alpha\text{Raised}]$ $[\alpha\text{Raised}]$

POLARITY-UPPER >> POLARITY-RAISED

- (57) Yorùbá L~M_L: schematic tone combinations for the emphatic clitic
(revised)

<i>root+{clitic}</i> T+{L, M _L }		POLARITY-UPPER * $[\alpha\text{Upper}]$ $[\alpha\text{Upper}]$	POLARITY-RAISED * $[\alpha\text{Raised}]$ $[\alpha\text{Raised}]$
☺	H+{L}		
	H+{M _L }		*!
☺	M _L +{L}	*	
	M _L +{M _L }	*	*!
☺	L+{L}	*	*!
	L+{M _L }	*	

Invoking polarity with respect to both [Upper] and [Raised] will have no effect on the cases in Mono, Bangime and Ogori since the best option was determined with reference to the more highly ranked POLARITY-UPPER alone.²⁴ For Yorùbá, invoking reference to the sub-feature [Raised] when reference to [Upper] is insufficient resolves the issues in the interpretation of polarity. Independent of the particular solution proposed here to the Yorùbá problem, the general point is that in multi-tone systems it is important to pay attention to the details of tonal specification.

10 Conclusion, both specific and general

As seen in our discussion of Yorùbá, polarity results from a prohibition of sequences of identical tones. Just as H+H and L+L sequences are ruled out in two-tone languages with polarity, these sequences are also ruled out in three-tone languages with polarity. Where reference to the primary feature [Upper] is insufficient to establish the observed result, reference to the secondary feature [Raised] resolves the issue.

²⁴Additional considerations must be invoked in a standard OT analysis to rule out a possibility such as having a H-tone emphatic clitic after M or L tones (e.g., **ó lẹ ó* rather than *ó lẹ ó* ‘he/she/it went (EMPHATIC)’). A H emphatic candidate would satisfy both POLARITY-UPPER and POLARITY-RAISED.

7 Tone polarity

The revised POLARITY constraints bring us full circle, however, to the issue of whether polarity is best seen as a distinct phenomenon from dissimilation. Though not addressed as examples were considered, POLARITY is in effect an OCP constraint (5) or a pair of OCP constraints (56), appropriate in either instantiation for analysing surface polarity as formal dissimilation. Considering a pattern to be ‘polarity’ would then depend on other factors.

We might return to the question of what it means to be ‘polar’. In a two-tone L~H system, saying one tone is different than the other is equivalent to saying that the tone is the opposite of the other. In a system of three or more tones, however, it is less clear what it means to say that one tone is the opposite of another. One might take ‘opposite’ to mean ‘different in all tonal features’. This would make a H ([+Upper, +Raised]) the opposite of a L ([−Upper, −Raised]). If polarity implies opposition, then Mono and Bangime would exhibit polarity, since they exhibit alternations between L and H, while Ogori and Yorùbá would not, since the observed alternations are between a peripheral tone and M. Whether this potential distinction is of significance linguistically needs to be determined.

We might consider the distinction between polarity and dissimilation to depend not on the observed alternation or the driver for that alternation, but to depend on the underlying representation of the morpheme in question. A pattern would be considered dissimilation if there was a reason for assigning one tone as the ‘underlying’ tone, and considered polarity if there was no particular reason for considering one value as more basic than the other. As seen in §6, however, it is problematic to rely on an underlying lack of specification as an indicator of polarity. If the range of relevant cases includes both alternations between opposites and alternations between minimally different tones, then tonal ‘polarity’ might appear to be a subtype of tonal dissimilation (Kenstowicz et al. 1988, Bodomo 1997).

We have also seen that polarity is typically restricted to particular morphemes. This restriction may be encoded directly, as in Emergent Phonology, or it may be encoded diacritically. Diacritic encoding may involve a ‘phonological’ diacritic (lack of specification, multiple tones), a ‘morphological’ diacritic (particular features such as person and number), or a totally arbitrary diacritic. And returning to the discussion of dissimilation versus polarity in §8, we would be left with the need to do an in-depth analysis of cases like Kɔnni (Cahill 2004).

Acknowledgements

Thanks to Alex Angsongna as well as two reviewers for comments on an earlier draft. This chapter is dedicated to Mike Cahill, who should have written it.

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Chapter 8

Tone and music in Africa

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In African languages, linguistic tone can interact with musical structure in two primary ways: tone-melody mapping in vocal music and musical surrogate languages. In the former, melodies of sung texts tend to take into account tone contrasts, but with varying degrees of strictness depending on the language and musical style. In the latter, tone (and rhythm) of speech are transposed onto musical instruments (drums, xylophones, flutes, etc.), which can then be used to transmit messages. Since tone is the primary linguistic contrast in musical surrogate speech, its musical encoding typically follows a more rigid set of rules than in vocal music, where segmental information is maintained. This chapter provides a brief overview of both musical practices and also discusses how music can serve as a tool for tonal analysis.

1 Introduction

There has been long-standing linguistic and ethnomusicological interest in the relationship between tone and music in languages around the world. When pitch plays an important linguistic role, differentiating lexical items and even grammatical constructions, one naturally wonders: To what extent is it preserved in sung melodies? How does it shape, constrain, or otherwise interact with musical structure? Given the fact that the majority of Sub-Saharan African languages are tonal, Africa presents a natural testing ground for these questions.

Eminent African musicologist Kofi Agawu recognizes the tight connection between language and music in many African cultures, stating:

Like the spoken word, song lies at the heart of African musical expression and imagining. Without language, there would be no song; without song,

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African music would not exist. Language and music are thus tied, as if by an umbilical cord. No one who ignores its linguistic aspects can hope to reach a profound understanding of African music (Agawu 2016: 113)

I would likewise argue that anyone interested in tone in African languages would be remiss to ignore its important role in musical expression, which offers unique insights into speakers' metalinguistic awareness and conceptualization of this important phonological aspect.

This chapter offers a brief overview of the connections between tone and music in Africa. I focus on two main musical domains. In Section 2, I discuss vocal music, with the question of tonal textsetting—how to set the tones of lyrics to sung melodies. In Section 3, I turn to musical surrogate languages, traditions in which the prosodic structure of language (primarily tone and rhythm) is transposed onto musical instruments to transmit messages. Section 4 briefly highlights what these traditions can teach us about tone systems and speakers' awareness of them, while Section 5 concludes.

2 Tonal textsetting

Attention to the question of how to set linguistic tone to sung melodies dates back at least 100 years, when Nigerian composer Rev. J.J. Ransome-Kuti noted in the preface to his Yorùbá Anglican hymnbook, “No [hymn] tune . . . can possibly express the meaning of words in a ‘tonic’ language such as Yoruba, so well as one written specially for the words” (Ransome-Kuti 1923, cited in Carter-Ényì 2021: 11). In the intervening century, a small but growing body of work has examined this question of “tonal textsetting”—how the linguistic tones of a text relate to a musical melody. In the subsections that follow, I will review some of the fundamental literature that lays out the basic principles and methodologies (Section 2.1), then expand upon these principles to consider the roles of linguistic structure (Section 2.2) and genre (Section 2.3) in mitigating strictness of mapping, as well as any effect of interval size (Section 2.4).

2.1 Basic principles

Most studies of tonal textsetting focus on the transitions between sung syllables, that is, the directionality of musical and linguistic pitch: Is the sequence of notes rising, falling, or level? Is the sequence of tones rising, falling, or level? These two measures are then compared to assess the degree of tone-tune matching. If the directionality is exactly the same (e.g. falling and falling, level and level),

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then the textsetting is said to be “parallel”. If a falling sequence of notes is set to a rising sequence of tones, or vice versa, then this is said to be “opposing” or “contrary”. Finally, if a rising or falling sequence is set against a level one, then this is said to be “oblique”. Together, parallel and oblique textsetting form a category called “non-opposing”; see [Schellenberg \(2012\)](#) and [Ladd & Kirby \(2021\)](#) for good overviews and definitions of these measures.



Figure 1: Line from a Tommo So folk song, meaning ‘whoever is hungry for milk, come here and suckle’

We can illustrate these principles with a brief excerpt from a folk song in Tommo So, a Dogon language of Mali. In Figure 1, transitions are color-coded for easy comprehension. Green indicates parallel mappings. For instance, the first word, *nòm-báá* ‘here’ is a rising L-H tonal sequence sung on a rising melody; the falling tonal transition from the final H-toned syllable of *yèl-ée* ‘come’ to the initial L-toned syllable of *irùt-gíyè* ‘hungry for milk’ is sung on a falling melody; and the level H-H sequence of *gíyè* ‘hunger’ is sung on a level melody. Orange indicates oblique mappings. For instance, the L-L sequence on *irùt* is sung on a rising melody, and the fall from the final H tone of *gíyè* to the initial L of *ndè* ‘person’ is sung on a level melody. Notice that there are no contrary or opposing mappings here. This is, in fact, characteristic of Tommo So folk music; the data corpus analyzed in [McPherson & Ryan \(2018\)](#) shows a non-opposing rate of 96.3%. However, only 46.5% of transitions are parallel, which highlights the fact that the musical melody is not defined solely by linguistic tone. It can and does move according to aesthetic principles governing the musical style, as many authors have noted over the decades (including [Herzog 1934](#), [Fiagbedzi 1977](#), [Agawu 1984](#), [Pooley 2020](#), [Carter-Ényì 2021](#), among many others).

Likewise, the extent to which linguistic tone influences musical melody differs across African cultures. In a survey of the literature, [Schellenberg \(2012\)](#) found that even languages within the same family can differ drastically in their tonal textsetting, with Shona showing only 53% parallel and 67% non-opposing transitions ([Schellenberg 2009](#)), while Zulu shows 92% parallel and 97% non-opposing ([Rycroft 1979](#)). In other words, some musical cultures show strict adherence to tonal transitions (Zulu), some allow flexibility so long as contrary movement is avoided (Tommo So), and some pay little heed to the linguistic tone (Shona).

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The subsections below add nuance to the question of tonal textsetting by discussing the effects of linguistic structure (Section 2.2) and genre (Section 2.3) on the strictness of tonal textsetting, as well as how this relationship between tone and tone intersects with musical interval size (Section 2.4).

2.2 The effect of linguistic structure

While percentage measures like those cited above provide a preliminary gauge of tonal textsetting for a particular musical culture and language, composers and singers are sensitive to far more linguistic structure than a simple string of phonological tones. One element that modulates strictness is the word boundary: multiple studies have shown a tighter match between tone and tune within words than across word boundaries (see e.g. [Agawu 1984](#) on Ewe or [McPherson & Ryan 2018](#) on Tommo So).

Tone itself is multifaceted, so simply asking how “tone” influences musical melody can obscure many important details. A crucial first question is to ask whether musicians are drawing upon phonemic, underlying tones or whether they are instead sensitive to their surface realization, after postlexical and even phonetic processes apply. To illustrate, we can consider the case of Hausa vocal music. [Richards \(1972\)](#) reported 53.4% parallel mappings in the texts he analyzed—barely above chance. However, in a reanalysis of the same data, [Leben \(1983\)](#) argued that singers encoding not the underlying tones but rather their intonational realization; from this surface perspective, tone-tune correspondence becomes “near 100%” ([Leben 1983](#): 153). Similarly, [Carter-Ényì \(2021\)](#) notes that the Nigerian composer Laz Èkwùémé is sensitive to phrase-level intonational trends like declination in his textsetting of Igbo tone.

Even phonemic tone can be broken down further, as many African languages display both lexical and grammatical tone (cf. [Rolle & Bickmore 2026 \[this volume\]](#)). This distinction has received little attention in the literature on textsetting, but may prove to be a fruitful avenue for future study. [McPherson & Ryan \(2018\)](#), for instance, report that in Tommo So folk music, lexical tone is more strictly adhered to by singers than grammatical tone, which in Tommo So takes the form of replacive tonal overlays. Interestingly, deviations from the grammatical overlays tend to align with the underlying lexical tone, suggesting that singers remain conscious of these underlying contrasts even when they do not appear on the surface. For further discussion of how music can shed light on phonological awareness, see Section 4.

2.3 The effect of genre

In addition to the effects of linguistic structure, strictness of tonal textsetting may also be modulated by non-linguistic factors. One factor that has received ample attention in the literature is musical genre. Rycroft (1979) looked at the degree of correspondence across multiple genres of Nguni music (including Zulu, Xhosa, Ndebele, and Siswati) and found a continuum, with more speech-like genres such as war cries and praise poetry at the high end and modern and popular music at the low end. This spectrum from more speech-like genres to more strictly musical forms of expression is echoed in research from other parts of the world, such as Chinese (Chao 1965) and Thai (List 1961); see Schellenberg (2012) for more details.

However, as Carter-Ényì (2021) demonstrates, this continuum oversimplifies the reality of tonal textsetting, as art song (vocal music composed in the Western style, often accompanied by piano and meant to be sung in recital) is an unarguably musical form and yet it tends to show the strictest adherence to tone-tune matching. Many African composers, including Nigerian Laz Èkwùémé (Èkwùémé 1974, discussed in Carter-Ényì 2021) and Ghanaian Ephraim Amu (Agawu 1984), are acutely aware of linguistic tone in their respective languages and incorporate it into their compositional practices in systematic ways. This is true even in the composition of harmony, which is generally rare in indigenous (West) African music (Agawu 1984); while Western choral harmony tends to involve counterpoint, where different voices carry independent melodic lines that can move in different directions, African choral composers make greater use of harmonic parallelism, allowing all voice parts to adhere to the tonal contours of the language.

2.4 The role of interval size

A last point to consider is the extent to which musical interval size plays a role in tonal textsetting. Many studies do not take interval size into consideration, leaving this an unanswered question, but those that do show mixed results. Agawu (1984) explicitly states that in the Akan folk music he considers, “speech tones only influence melodic contour or direction, not the actual intervallic succession” (pg. 42). Carter-Ényì (2021) notes that based on interviews, Nigerian composers differ in whether the magnitude of a tone change (e.g. H to L vs. H to down-stepped H) is represented in the melodic interval.

McPherson & James (2021) discuss tonal textsetting in Seenku vocal music, showing a close correlation between tonal intervals and melodic intervals. Seenku has a denser tone system than most other languages studied for tonal

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textsetting, with four contrastive levels (as well as multiple contour tones). Given this tonal density, one can see how simple melodic direction (level, falling, rising) would be insufficient to indicate the full range of tone contrasts. Instead, larger tonal intervals (for example, an Extra-Low to a Super-High tone) are more likely to be set to non-adjacent notes of the scale (i.e. an interval size of 2 or greater) than a smaller tonal interval (e.g. Extra-Low to High). This effect is stronger for rising tonal sequences than falling tonal sequences.

It is not just dense tonal systems in which interval size plays a role, however. [McPherson & Ryan \(2018\)](#) in their study of Tommo So, a two-tone Dogon language, likewise found an effect of interval size. Rather than musical intervals representing tonal intervals of different sizes, instead it was found that the strictness of tonal textsetting was modulated by musical interval size: mismatches were avoided more ardently on larger musical intervals.

As more research takes into account interval size and not just directionality, we may begin to paint a clearer picture of how singers and composers craft their melodies to respect linguistic tone or to leave room for its interpretation.

3 Musical surrogate languages

In many African cultures, we find another musical practice with a tight connection to linguistic tone: musical surrogate languages. This category includes well-known cases of “talking drums”, which is itself an umbrella term for a number of linguistically and organologically different traditions like slit-log drums used to encode Lokele in Congo ([Carrington 1949](#)), a pair of barrel drums used to encode Akan in Ghana ([Nketia 1971](#), [McPherson & Obiri-Yeboah 2023](#)), and a double-headed tension drum used to encode Yorùbá ([Euba 1990](#), [Akinbo 2019](#), [Durojaye et al. 2021](#), [González & Oludare 2022](#)); but as we will see, musical surrogate speech is attested on a broad range of instruments beyond just drums.

Though seemingly instrumental, these traditions are inherently linguistic as they transfer linguistic content (phonological, syntactic, and/or semantic) onto the sounds of musical instruments in order to transmit messages ([McPherson 2018](#), [Winter & McPherson 2022](#)). [Stern \(1957\)](#) distinguishes between so-called “abridging” systems, which encode phonemic contrasts, and “lexical ideogram” systems, which encode concepts directly without any iconic representation of speech sounds. Some authors such as [Nketia \(1971\)](#) do not consider lexical ideogram systems (sometimes called “signal drumming”) to be true speech surrogates; however, the line between the two categories is blurry at best, and many traditions include elements of both ([James 2021](#)). Nevertheless, as this chapter

focuses on the role of tone in musical traditions, I will focus on “abridging”, that is to say, phonemic speech surrogates here.

I will draw on three case studies to illustrate a range of instruments and tone systems: the Sambla balafon (resonator xylophone), which encodes Seenku’s four-tone system (McPherson 2018, McPherson 2021); the Igbo oja (aerophone), which encodes Igbo’s two-tone plus downstep system (Carter-Ényì et al. 2021, Eze & McPherson 2025); and the Yorùbá bàtá (membrane drum ensemble), which encodes Yorùbá’s three-tone system in addition to limited segmental contrasts (Villepastour 2010, González & Oludare 2022).

3.1 Sambla balafon



Figure 2: A Sambla balafon in Toronsso, Burkina Faso.

We begin with the Sambla balafon, a wooden resonator xylophone used to encode the Mande language Seenku in southwestern Burkina Faso (McPherson 2018, McPherson 2021).¹ The instrument is a mainstay of Sambla musical culture and is played at all major community events, including weddings, funerals, and traditional religious festivals and rites (Strand 2009). A single instrument is played by three musicians simultaneously, and the player on the highest notes of the instrument (the most experienced player) spends much of his time engaged in

¹The term “Sambla” is an exonym for both the language and ethnicity, while Seenku is the endonym for the language. Following community practices, I will use Sambla to refer to people and culture.

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speech surrogacy, either “singing” proverb-rich lyrics through the notes of the instrument or communicating directly with the audience and other musicians. An image of the balafon is shown in Figure 2.

The foundation of this speech surrogacy is tone. Seenku has a complex tone system, with four contrastive levels (Extra-low <X>, Low <L>, High <H>, and Super-high <S>) and multiple contour tones (McPherson 2019b, McPherson 2020). These four tones map in a largely one-to-one manner to notes of the instrument’s pentatonic scale, considering that one note of the scale (a flat third in Western musical parlance) is reserved for largely spiritual purposes. The precise tone-note mappings change depending upon the musical mode of the song, but with few exceptions, the Super-high tone maps to the higher octave center of the mode (the tonic), and each lower tone works its way down the scale, with the Extra-low tone able to be played both on the second scale degree and the lower octave tonic. For more details, see McPherson (2018).

‘I will not buy a pig.’ (MD: November 24, 2014)

	S [8]							
	Sk [6]							
	T [5]							
	Bg [3]							
	J [♭3]							
➤	B [1]							
	Words	<i>mó</i>	<i>nǎ</i>		<i>bèè</i>		<i>sâ</i>	<i>ɲé</i>
	Duration	169	49	169	62	153	156	...

Figure 3: Example of Sambla balafon speech surrogacy meaning ‘I will not buy pigs’, from McPherson (2018: 272)

This system of surrogacy is illustrated in Figure 3, taken from McPherson (2018) (9b). The scale degrees are shown along the y-axis, with both abbreviations of traditional note names (e.g. B for *bâq-pǎ* ‘balafon mother’) as well as Western equivalences. The center of the mode is indicated with an arrow. The Seenku words are placed along the x-axis (together with the duration of each note in milliseconds). Squares of the grid are grayed out when that note is struck. Thus, we see that the H-toned word *mó* ‘1sg’ is played on the *səràkùà-kɛ̀-nòn*/Sk/6th, as is the H-toned negative particle *ɲé* at the end of the clause. L-toned words *bèè* ‘pigs’ and *sâ* ‘buy’ are both played on the *tərón-tərón*/T/5th. This example also illustrates how contour tones are encoded with the LS-toned future auxiliary *nǎ*: the contour tone is played as a flam, that is, a quick succession of two strikes, with each tone of the contour represented. Here, the initial L tone is played (also on

T/5th), followed 49ms later by the S tone on the higher octave tonic/*səràkùu*/8th. Astute readers will also note that long vowels are played with a flam, and thus *bɛɛ* ‘pigs’ receives two strikes of the same note.

As is true of many, though not all, musical surrogate languages, the Sambla balafon encodes only prosodic elements of Seenku: tone and rhythm (including vowel length and syllable shape). Segmental contrasts play no role in the explicit rules of surrogacy (though see [McPherson 2021](#) for subtle durational differences in performance). This, of course, results in a great deal of ambiguity; I will briefly discuss how surrogate languages, including the Sambla balafon, mitigate this ambiguity in Section 3.4 below.

We saw in Section 2.2 above that tonal textsetting in vocal music can be sensitive to different levels of tonal representation: lexical, grammatical, postlexical, etc. The same question must be asked of musical surrogate languages. In the case of the Sambla balafon, we find that all lexical and grammatical tone is encoded, but postlexical tonal rules do not apply. These principles can be seen in the example in Figure 3. The plural in Seenku is marked largely tonally through tone raising ([McPherson 2017](#)), and thus the plural of X-toned *bɛɛ* ‘pig’ is L-toned *bèè* ‘pigs’; we see this L tone represented in the balafon surrogate language. There is also a grammatical tone process I call argument-head tone sandhi ([McPherson 2019c](#)), in which the tone of an internal argument can influence the tone of its head. One environment in which this applies is between a direct object and an irrealis verb, such as in future constructions like the one in Figure 3. Thus, lexically X-toned *sǎ* ‘buy’ raises to L due to the preceding L-toned object *bèè* ‘pigs’. This is likewise reflected in the balafon surrogate language. However, a postlexical contour tone simplification process is not. The LS rising contour on *nǎ* is typically simplified in speech to either L or !S depending on the phonological environment. Between the H-toned subject *mó* and a following L tone, as in Figure 3, *nǎ* would be realized as [*!nǎ*], but in the surrogate language, the contour tone is realized faithfully. I will return to this point in Section 4.

3.2 Igbo oja

I now turn to a case of a wind instrument used for speech surrogacy: the Igbo oja from Nigeria. The oja is an end-blown wooden aerophone used by the Igbo ethnic group in southern Nigeria ([Carter-Ényì et al. 2021](#), [Eze & McPherson 2025](#)); an image is provided in Figure 4. It has three finger holes and produces a six-note diatonic scale. Like the Sambla balafon, tunes played on the oja are not just music but also encode phrases. Common uses include “singing” praises of chiefs

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Figure 4: Three sizes of oja (Photo courtesy Gerald Eze).

and important community figures and communicating with masked spirits in the masquerade arena (Eze & McPherson 2025).

It is interesting to compare the Sambla balafon and the Igbo oja, not just because the instruments are so different (a struck idiophone vs. an aerophone), but also because of differences in tonal structure, both musically and linguistically. The Sambla balafon uses a pentatonic scale to encode Seenku's four tonal primitives, while the Igbo oja uses a hexatonic scale to encode Igbo's two tones plus contrastive downstep. In other words, the oja has a larger musical space to encode a smaller tonal inventory. This has ramifications for how Igbo tone is encoded in speech surrogacy, namely that the oja has enough musical room to encode downstep and downdrift (cf. Snider 2026 [this volume]). Consider the example in Figure 5, which is representative of the tonal encoding principles found in both Carter-Ényì et al. (2021) as well as Eze & McPherson (2025), from which it was drawn.

Sequences of like tones are played on the same note, but unlike on the Sambla balafon, there is not a particular note of the scale that attracts these tones. The H tones on *ókú* 'fire' are played on G, while those on *àgbááná* 'burnt completely' are played on E. This is because of the encoding of downdrift, which reproduces



Figure 5: Example of Igbo oja speech surrogacy meaning ‘The fire that burnt the bamboo has burnt it completely’, from Eze & McPherson (2025)

Igbo’s terracing tone system. Notice that the fall from a H tone to a downstepped H, as between *ókú* ‘fire’ and *!gbára* ‘burnt (REL)’, is an interval of just one note of the scale (here, G to F#); a fall from H to L is an interval of two notes (here, F# to D); but when the melody ascends from L to H, it goes up just a single note, not the two that it fell (here, D to E). Thus, HLHL sequences will step downwards on the oja, mimicking the language’s downdrift.

I should note that the ratio of linguistic tones to notes on a scale is not an automatic determinant of how a surrogate language will encode tone (i.e. at an underlying vs. surface level). For instance, the Akan *seperewa* is a harp-like instrument with either 6 or 8 strings (Nketia 1994), which should leave ample space to encode Akan’s two tones; however, for speech surrogacy, only two strings are used and tone is encoded at a largely underlying level (McPherson & Obiri-Yeboah 2023). As with tonal textsetting discussed in Section 2, the relationship between tone and music in musical surrogate languages depends on linguistic, organological (that is, instrument-specific), and cultural factors.

3.3 Yorùbá bàtá

The last case study we will consider is the Yorùbá bàtá, an ensemble of membrane drums used to encode Yorùbá (Villepastour 2010). While perhaps the best known case of Yorùbá “talking drum” is the hourglass-shaped tension drum known as *dùndún* (Akinbo 2019, Durojaye et al. 2021, González & Oludare 2022), the bàtá drum ensemble has an equally long and rich history, but ties to traditional religion as well as a more acoustically opaque style of surrogate speech have curtailed its expansion in modern Nigerian cultural and musical spaces (Villepastour 2010). For an image of the drum ensemble, see Figure 6.

The drum ensemble consists of three drums. The main drum, which forms the basis of speech surrogacy, is known as the *iyáàlù bàtá*; it is a large wooden drum with skin membranes on each side, a large one called *ojú òjò* and a smaller one called *šášá*. The *ojú òjò* is played with the hand while the *šášá* is played with a leather strap (*bílálà*). In addition to the *iyáàlù bàtá*, the ensemble also

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Figure 6: Bata drum ensemble (By Mplanetech - Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=54943457T>)

includes the *omele abo* (a smaller membranophone with a similar construction as the *iyáàlù*) and a set of two or three small drums tied together (*omele akọ*, according to [Villepastour 2010](#), or the *omele méjì* for two drums or *omele méta* for three, according to [González & Oludare 2022](#)).

Unlike the balafon and the *ọja*, the *iyáàlù bata* is not a melodic instrument; it does not have a fixed scale, nor can the pitch be manipulated, as is possible on the *dùndún* talking drum (by changing the tension on the membrane). Instead of mapping Yorùbá's three phonemic tones to different pitches, then, *bata* speech surrogacy is instead grounded in timbral differences that arise from striking the drum in different ways. An open tone is produced when the hand strikes the *ojú òjò* membrane and bounces back, which allows for free vibration of the membrane and a low sustained sound; this open tone encodes Yorùbá's L tone. If the membrane is struck with the hand but the fingers are left on the membrane, this creates a muted tone with a short sustain time; this muted tone encodes Yorùbá's M tone. Finally, a slap mute produced with a rapid strike of the fingers coupled with muting the membrane with the palm produces a short high frequency burst of sound that encodes Yorùbá's H tone. Since each tonal category is mapped to a particular way of striking the drum, the *bata* provides a fairly faithful encoding of Yorùbá phonemic tones rather than a more surface-level encoding; phonetic effects like declination are simply not possible, given the acoustic output of the drums and the rules of encoding.

While the balafon and the *ọja* completely eschewed segmental contrasts, the Yorùbá *bata* makes a binary vowel quality distinction. Vowels belong to two classes, called soft and intense, which roughly correspond to high vs. non-high in terms of phonological features. The nomenclature of soft vs. intense represents

the intrinsic amplitude differences between high and non-high vowels (Villeepastour 2010). On the drum, this distinction is encoded through the smaller membrane, the *śáśá*. Intense vowels involve a strike on the *śáśá* with the leather strap, which adds high amplitude sound; on soft vowels, this strike is absent.

Villeepastour (2010) considers speech surrogacy to form a continuum, from systems that “encode” language to those that “mimic” language. She places the *bàtá* at the encoding end of the spectrum, noting that even Yorùbá speakers can have a difficult time understanding *bàtá* surrogacy; a talking drum like the *dùndún*, with its malleable pitch, more directly mimics the tonal contrasts. It is true that the different strokes produce quite different spectral characteristics and are thus perhaps best categorized as differences in timbre rather than pitch per se, but as González & Oludare (2022) point out, the acoustic properties of the different strokes do overlap with spoken acoustics. For instance, the three kinds of strokes do involve low, medium, and high frequency energy, which encode L, M, and H tones; see González & Oludare (2022) for further discussion.

3.4 Strategies for mitigating ambiguity

Collapsing nearly all linguistic structure to simply tone and rhythm (with few exceptions, like vowel encoding in Yorùbá) creates an enormous amount of ambiguity. While this can be seen as desirable in some contexts of use (for instance, spiritual contexts or courtly contexts where the general public should not understand, Falk 2003, Kaminski 2008), in other contexts, strategies must be employed to help disambiguate the message.

One strategy found in some Bantu slit-log drumming traditions, such as Lokele, is known as “enphrasing” (Carrington 1949, Stern 1957). Because Bantu languages generally have only a two-way tonal distinction (H vs. L, analyzable as privative H or privative L in some cases), many words end up identical when played on the drums. To disambiguate, words (generally nouns) are replaced with longer conventionalized phrases, which are tonally distinct. For instance, *likɔ̀ndɔ* ‘plantain’ and *lomata* ‘manioc’ are tonally identical (all L-toned), and so on the drum, *likɔ̀ndɔ* is rendered as *likɔ̀ndɔ libotúmbela* ‘plantain to be propped up’ (L-L-L H-L-H-L-L) and *lomata* as *lomata otíkala kóndo* ‘manioc remaining in fallow land’ (L-L-L L-H-L-L H-L). Here, message length and complexity compensate for the reduced set of contrasts, but crucially the longer replacements are themselves rote—they form a surrogate lexicon derived from but not identical to the spoken language (James 2021).

Rote phrases play an important role in reducing the hypothesis space for surrogate messages even without enphrasing. For example, in surrogate traditions

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like the Igbo oja or Yorùbá drums, messages tend to consist of established praises and proverbs, which are further disambiguated by their context of use. For instance, Eze & McPherson (2025) note that the Igbo phrases *ébélébé ègbúó* ‘the unthinkable has happened’ and *íféómá èméé* ‘a good thing has happened’ would be identical on the oja, but in the context of a burial, one would understand the message to mean the former rather than the latter. Once again, by “looking up” longer tone patterns in a mental lexicon of rote phrases, a listener can decipher surrogate messages even with ambiguity.

Finally, even without enphrasing or other rote phrases, context is crucial for understanding surrogate messages. This can be the environmental context (who the player is addressing, what kind of setting it is, what the community knows about that person), but it can also be linguistic context set up by the surrogate. This strategy is attested in the Sambla balafon tradition, where tonally identical phrases like ‘I will get water’ vs. ‘I will eat chicken’ can be preceded on the balafon with unambiguous phrases like ‘I am dying of thirst’ vs. ‘I am dying of hunger’ to allow the listener to properly interpret what follows (McPherson 2018).

Of course, even with all of these strategies, speech surrogacy involves a severely restricted set of contrasts, meaning it is a challenge for all but the most experienced listeners to decode the messages. Perception studies are largely lacking and would be an area ripe for future research.

4 Music as a tool for tonal analysis

So far in this chapter, we have seen how tone influences musical melodies and can form the foundation of musical surrogate speech. While these topics are interesting in their own right, they can also be unexpected tools for tonal analysis and raise interesting questions about the psycholinguistic reality of tonal categories.

Tonal analysis is especially aided by musical surrogate languages, where the relative pitches of spoken language get mapped to discrete notes or strikes on a musical instrument. Whether tonal encoding is postlexical, as with the Igbo oja, or at a lexical level, as with the Sambla balafon, both kinds of surrogate languages can help train a fieldworker’s ear. For instance, in Igbo, a pitch drop between two syllables could either be a H!H sequence or a HL sequence, with the former usually a smaller pitch interval than the second. However, the exact size of the interval will depend on speaker, speech style, and phrasal location, among other factors, making the distinction harder for a non-native speaker. Enter the

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oja surrogate language: with H!H intervals consistently played on adjacent notes of the scale and HL sequences consistently played on non-adjacent notes, the two kinds of falling sequences can be more concretely distinguished.

Surrogate languages can even prove useful when deciding how many phonemic tones a language has. This was the case in my work on Seenku, where I was still working to understand the spoken language's sound system while starting to work on the surrogate language. At the time, I was working under the assumption that Seenku had three phonemic tone levels, like many other languages in southwestern Burkina Faso. Under this analysis, I was having trouble understanding what I took to be downstep between certain M tones and unusual alternations in the prospective auxiliary verb *na* (left without tone marking here as an illustration of my uncertainty at the time). When the mystery phrases were played on the Sambla balafon, I found consistent evidence of four tone levels, and the auxiliary was exceptionlessly played with a Low to Superhigh rising tone; spoken alternations between L and !S were not encoded (as discussed in Section 3.1). What appeared to be a M-toned allomorph on the surface was instead L, a higher level than a preceding Extra-Low tone, but lower than other words I considered to be M (but which turned out, under a four-tone analysis, to be High). For more in-depth discussion and exemplification, see McPherson (2018) and McPherson (2019a).

While tonal textsetting is not as good a tool for phonemic analysis of tone, due to the much less strict relationship between tone and music, the ways in which different kinds of tone affect the melody can raise interesting questions about their psychological reality and mental representation. The most obvious difference is between underlying and postlexical tone, as discussed in Section 2.2, but differences between lexical and grammatical tone hold promise for future research. McPherson & Ryan (2018) found a “bleed through” effect of lexical tone on the textsetting of words with replacive grammatical overlays in Tommo So. For instance, comparing the textsetting of /H/ and /LH/ words with a grammatical L-toned overlay, /LH/ words were more likely to show mismatches favoring a rising musical melody than /H/ words. This suggests dual activation of both underlying and grammatical (surface) tone melodies, but it leaves open the question of whether this has any phonetic effect on spoken tone, i.e. whether we find any effects of incomplete neutralization of lexical tone contrasts with grammatical overlays. Further phonetic analysis could probe this question, demonstrating once again the symbiotic relationship that exists between work on tonal effects in music and the phonetic and phonological analysis of spoken tone systems.

For more on the role of music in phonological analysis, see McPherson (2019a).

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5 Conclusion

In this chapter, we have seen two general areas in which tone in African languages interacts with music: the accommodation of linguistic tone in vocal melodies (tonal textsetting) and the transposition of linguistic tone onto musical instruments for communication (musical surrogate languages). I have also suggested ways in which the study of music can bolster phonological analysis and provide ideas for future phonetic, phonological, and psycholinguistic studies. Another domain in which music could play a role in the study of African languages is language pedagogy, whether L1 or L2, but this remains little-explored. However, preliminary work on whistled languages suggests that musical surrogate languages could serve as an excellent tool in teaching about tone systems (Meyers & Casimiro 2008).²

As I hope to have shown, there is a great deal of diversity in both tonal textsetting as well as musical surrogate languages along multiple axes. Do vocal melodies adhere to tone at all, and if so, how much? Does this vary based on linguistic or cultural factors? Does a community have speech surrogacy, and if so, on what instrument? In what contexts is it used? Which linguistic parameters are encoded (beyond tone), and how? To what extent is ambiguity desirable or something to be mitigated through a variety of means? There is no single way in which tone plays a role in musical culture, and indeed, there may well be musical cultures in Africa in which it plays no discernible role at all.

Tone can have other influences on music beyond tonal textsetting and musical surrogate languages. For instance, Agawu (1984) notes that tone can influence broader musical organization in African music, such as the lack of strophic structure; if vocal melodies must adhere to tone (to some extent), then it becomes challenging to keep the same melody across verses with different lyrics, whose tones would likely differ. We also find unique musical traditions deeply rooted in tone that defy classification as either strictly tonal textsetting or a musical surrogate language. One such case is the Kirundi *inanga chuchotée*, or ‘whispered inanga’ (Fales 2013). The *inanga* is a zither-like string instrument that accompanies whispered singing. Fales describes how the combination of the plucked strings and the whispered voice creates an auditory illusion, whereby the f_0 of the strings decouples from the instrument and is perceived as the pitch of the voice, which closely follows the tone contours of Kirundi. This tradition has a foot in the realm of tonal textsetting, as the melodies are constrained by linguistic tone, but it also

²I would like to thank an anonymous reviewer for bringing up this important point for future research.

has a foot in the realm of musical surrogate languages, as the tones themselves are produced by a musical instrument.

The true extent of this diversity, and the varied ways in which tone and music interact in African communities, still remains to be uncovered. I would argue that a linguist’s perspective has much to add to this conversation, and likewise, linguistics and the study of tone in particular has much to learn from the inclusion of music. We would do well to turn our eye (and ear!) towards this form of expression, where pitch often takes center stage.

Abbreviations

H	High	S	Super-high
L	Low	X	Extra-low
REL	Relative	!	Downstep

Acknowledgements

Aspects of this work have been funded by the US National Science Foundation under grants BCS-1664335 and BCS-1943896. I would like to thank Kofi Agawu, Samuel Akinbo, Jonathan Barnes, Andre Batchelder-Schwab, Aaron Carter-Ényì, Mamadou Diabate, Mahuton Djessou, Gerald Eze, Lucas James, Bob Ladd, Theodore Levin, Michael Obiri-Yeboah, Ani Patel, Joe Pater, Kevin Ryan, and Yoad Winter for stimulating discussion and collaboration on language and music over the years, as well as numerous audiences and student research assistants who have provided many great ideas and insights. All errors are my own.

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Chapter 9

Tone in Cushitic languages

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Cushitic languages, by and large, exhibit “restricted” tone systems whose properties differ markedly from most other African tonal systems. Such systems manifest many but not all properties of prototypical stress accent systems, while their primary correlate of prominence and dimension of contrast relates to pitch. This state of affairs has long presented descriptive and analytical challenges in scholarship on these languages. This chapter focuses on exemplifying characteristics of Cushitic tone systems, among them privativity of contrast and the interplay between tone and word formation. It is shown that tonal distinctions are more robust in encoding grammatical relationships than in the lexicon and that the languages exhibit a preponderance of grammatical tone phenomena, including many instances of subtractive grammatical tone.

1 Introduction

The state of the art on the subject of Cushitic tone has long been [Appleyard \(1991\)](#), though the article covers only three languages in the Lowland East sub-branch of the family and explores only three phenomena (case marking, gender, and the jussive mood). Cushitic languages have prosodic systems that are at a “typological watershed” ([Hayward 1991: 118](#)) such that they are far from prototypical in comparison to most other African language groups when it comes to their use of tone. Somali, in particular, has long been pointed to as an example of just how non-prototypical a tone system can be ([Hyman 2019a,b](#); [Klingenberg 1949](#)). Some Cushitic languages are analyzed as having both tone and stress systems that are independent of one another ([Heine 1975](#); [Sasse 1979](#)), or a tonal system that “operate(s) in concert with” stress ([Appleyard 1991: 6](#); [Hayward 1978](#)).

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These facts have long complicated analyses and have sometimes had the unfortunate consequence of scholars setting aside matters of prominence altogether from their descriptions, or instead under-reporting on important details.

When matters of prominence are taken up, descriptions of most Cushitic languages report some contrastive use of tone. A major exception has long been the Highland East Cushitic sub-branch of the family (Tosco 2000: 111), though recent work on some languages in this group reports tone (Frosini 2008; Kawachi 2007) or “pitch accent” (Garkebo 2015).¹

Mous (2021) provides an update on what was learned in the 30 years since Appleyard’s article, offering not only a contemporary look at Cushitic tone, but also casting a broader net. Mous’s study focuses on a longstanding analytical matter – the treatment of most Cushitic languages’ prosodic systems as “tone accent”, “tonal accent”, “pitch accent”, and the like. At issue is that Cushitic prosodic systems exhibit many, but not all properties that Hyman (2006) attributes to prototypical “stress accent” systems. For example, Cushitic tonal distinctions tend to be limited in their distribution, often demarcating word or stem edges, and most make use of a privative High vs. non-High contrast. There tends to be a single culminative tonal prominence in a given domain, though there are oftentimes phonologically toneless words in a given language. Thus, prominence is often not obligatory, which is a hallmark of stress (accent) systems. In light of discussion in Hyman (2006, 2009), some recent works discussing Cushitic prosodic systems have begun to invoke the term “restricted tone system” (Green & Smith 2024, Mous 2021, Smith 2022), a term earlier suggested by Voorhoeve (1973), which captures the fact that languages like these exhibit many stress-like properties, yet their prosodic systems remain definitionally tonal. In summary, Mous (2021) argues that several factors conspire to yield the typologically unusual characteristics of Cushitic tone systems. Among them, i) the languages are almost exclusively suffixing, ii) there is a tendency toward (but often lack of) culminativity, and iii) tone primarily encodes grammatical, rather than lexical distinctions. These factors tend to bring about a single tonal prominence per domain, favoring the right edge.

Rather than rehash the issues so clearly explicated in Mous (2021), this chapter does not focus on matters of prosodic system typology. I will assume that,

¹Other notable exceptions include: Xamtaanga (Central) – there is culminative and obligatory stress, typically on the penultimate syllable (Darmon 2015); possibly Beja (Northern) – Vanhove (2017) cites Hudson (1973) in describing the language as having “pitch accent”, but Wedekind et al. (2007) refer to stress; Awngi (Central) – Hetzron (1976) claims there is demarcative penultimate stress, but this is not confirmed by Joswig (2010b) or Worku (1986); Lower Jubba Maay (Lowland East) – Paster (2007) discusses syllable-based prominence with no pitch correlates that is obligatory on all content words.

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except in the few notable instances mentioned above, Cushitic languages have “restricted” tone systems. This assumption is in the spirit of Hyman (2006, 2009), though I acknowledge that this designation leaves significant room for interpretation depending on a given language’s properties. For more perspectives on Cushitic prosodic typology, see Appleyard (1991), Green & Smith (2024), Hayward (1991), Hyman (1981), Kießling (2003), and Tosco (2000). The focus of this chapter is instead on exemplifying specific characteristics of Cushitic tone systems, with broad coverage from the entire family, to the extent that available data permit. It might appear that the discussion is skewed towards Eastern Cushitic, but this only because of the sheer number of languages in the sub-group and the state of available phonological descriptions for Cushitic overall.

Unfortunately, this chapter does not cover matters related to tonal phenomena across phrases or within larger intonational units. The reason for this is so little has yet been reported on these topics that it would render impossible any reasonable summary concerning their prevalence or their functions across the family, or even within a given sub-branch of the family. In my extensive review of the literature, declination in the form of downdrift and (automatic) downstep is mentioned explicitly only in works on Booran Oromo (Eastern, Oromoid: Owens 1980, Saho (Eastern, Saho-Afar: Tajebe 2015), Rendille (Eastern, Omo-Tana: Pillinger 1988, Pillinger 1989), Somali (Eastern, Omo-Tana: Green 2021, Le Gac 1997, 2003a, 2016), and Dhaasanac (Western: Tosco 2001). Intonation is mentioned more frequently, but most often only in passing to point out that while intonational phenomena appear to be present and may play a key role in a given language, they remain underexplored. Exceptions to this in which one or more intonational phenomenon is exemplified and discussed in some detail include: Awngi (Central: Joswig 2010a), Gorwaa (Southern: Harvey 2018), Burunge (Southern: Kießling 1994), Afar (Eastern, Saho-Afar: Kamil 2015), Saho (Eastern, Saho-Afar: Tajebe 2015), Booran Oromo (Eastern, Oromoid: Owens 1980), Boni (Eastern, Omo-Tana: Sasse 1979), Somali (Eastern, Omo-Tana: Green 2021, Le Gac 1997, 2001, 2002, 2003a,b, Nagano-Madsen et al. 2019), and Dhaasanac (Western: Tosco 2001).

With this limitation stated, the remainder of this chapter proceeds as follows: Section 2 presents an overview of Cushitic internal classification to situate the rest of the chapter. Section 3 covers basics of Cushitic tone: dimensions of contrast, tone bearing units, and matters of tone assignment. Section 4 discusses grammatical tone, while Section 5 turns to the close connection between tone and word formation. Section 6 discusses tonal alternations, Section 7 briefly summarizes perspectives on the historical development of Cushitic tone, and Section 8 offers summary remarks.

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2 Background on internal classification

Cushitic languages are spoken primarily in the Horn of Africa, mostly in Somalia, Ethiopia, Djibouti, and eastern Kenya. In addition, smaller pockets of Cushitic languages can be found further north in Sudan and Egypt, as well as further south in Tanzania. Several works discuss the internal classification of Cushitic or one of its sub-branches, notably Black (1974), Heine (1978), Sasse (1981), and Tosco (2000, 2020). The “traditional view” (Tosco 2020) of Cushitic internal classification is found in Tosco (2000), which is adapted here in Figure 1. Italicized nodes represent sub-groups containing one or more language. So-called “clusters” are indicated where relevant.

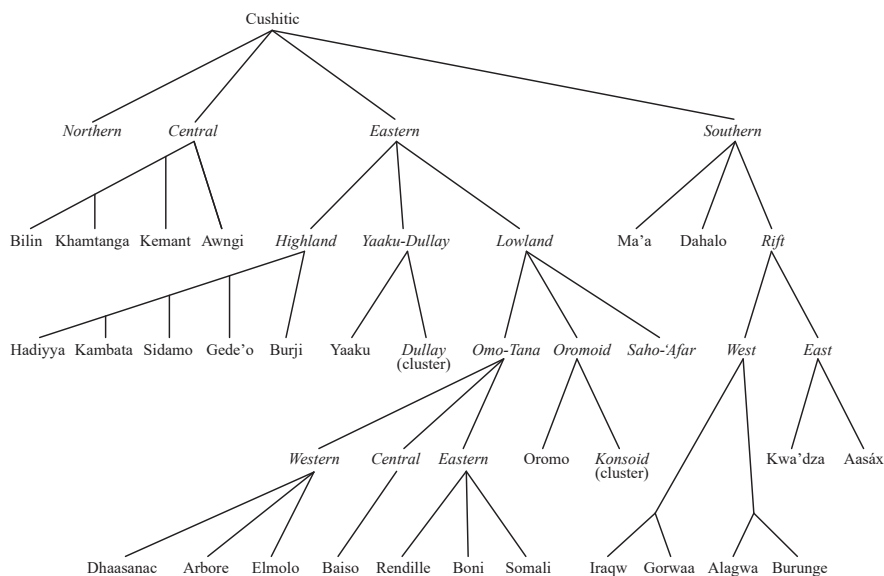


Figure 1: Cushitic internal classification, adapted from Tosco (2000)

This classification sub-divides Cushitic at the highest level between four geographically-oriented categories: Northern, Central, Eastern, and Southern. While the Northern and Central branches are quite small, containing only one and four languages, respectively, the Southern and Eastern branches are much larger. The Southern branch contains Dahalo and several other languages within the Rift group. Within Rift, the two East Rift languages are extinct. There is additionally the case of Ma’a (or Mbugu), which Mous (2003) describes as a mixed language that exhibits Bantu grammar but a Cushitic lexicon.

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Within the largest branch – Eastern Cushitic – one finds the Lowland subgroup, which contains some of the most extensively described Cushitic languages – Oromo, Afar, and Somali. The Highland East Cushitic languages fall within this branch as well, though they will not be discussed in this chapter, as all these languages have been traditionally analyzed as having stress accent systems (Tosco 2000: 111). Lastly, within Yaaku-Dullay, Yaaku is extinct, and the Dullay cluster will be represented by Gawwada.

The survey that I present below is based on data and generalizations that I have gathered from 29 languages across the Cushitic family. Table 1 provides a list of these languages alongside their classificatory branch within the Cushitic tree, their ISO 639-3 designators, and major data sources that I have consulted which contain information on their prosodic systems. Given the sheer size of the Eastern branch of the family, I have indicated its major internal sub-branches.

3 Dimensions of contrast

In most instances, descriptions of Cushitic tonal systems have been predicated upon a privative contrast, variously characterized as High vs. non-High, High vs. toneless, or High vs. Neutral. Some scholars have explicitly made an analytical claim of privativity, while it is implicit in other descriptions. Indeed, there are some descriptions that pose High vs. Low tone levels (e.g., Tosco 1997: 35 on Tunni), but ensuing discussion makes clear that these are *surface* tone levels, thus implying a privative contrast underlyingly.

An explicit argument for privativity is made in Hayward (1991) concerning Qafar (Afar) nouns. It is shown that (C)onsonant-final and (V)owel-final nouns bearing a High tone on their penult are lexically marked and require masculine grammatical gender agreement. Unmarked nouns are all C-final and bear a final High tone which is assigned by default to satisfy an obligatory High tone requirement. These nouns accordingly take feminine grammatical gender agreement; loanwords also pattern with this lexical class. Hayward & Megerssa (1996) likewise analyze Western Oromo based on a privative contrast.

Arguments for a privative tonal contrast are sometimes supported by the presence of items displaying largely static or immovable tones. For Boni, Sasse (1979) discusses two contrastive classes of High vs. Low (Neutral) stems for which the latter often has a final High tone, except that it is lost before pause (2a) or before another High tone (2b), but not before another toneless word (2c) or suffix (2d). The authors states that the tone of underlyingly High nouns is not affected by alternations in the same contexts, though unfortunately examples are not provided to illustrate this fact.

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Table 1: Languages and data sources

Branch	Language	Source	ISO 639-3
Northern	Beja	Hudson 1973, 1974; Vanhove 2017	bej
Central	Bilin	Palmer 1960; Lamberti & Tonelli 1996; Fallon 2001, 2004	byn
	Xamtanga	Darmon 2015	xan
	Awngi	Palmer 1959; Hetzron 1997; Joswig 2010b,a	awn
Eastern; Highland	Hadiyya	Garkebo 2015	hdy
	Kambaata	Treis 2008	ktb
	Sidamo	Kawachi 2007; Frosini 2008	sid
	Gede’o	Dawit 2019	drs
	Burji	Assefa 2015	bji
Eastern; Yaaku-Dullay	Yaaku	Heine 1975	muu
Eastern; Lowland	Gawwada	Tosco 2021	gwd
	Dhaasanac	Tosco 2001	dsh
	Arbore	Hayward 1984	arv
	Baiso	Hayward 1978	bsw
	Rendille	Sim 1981; Pillinger 1988, Pillinger 1989	rel
	Boni	Sasse 1979; Heine 1982; Mungai 1996	bob
	Somali	Saeed 1999; Green 2021	som
	Maay	Biber 1982; Saeed 1982 Paster 2007; Green & Smith 2024	ymm
	Tunni	Tosco 1997	tqq
	Oromo	Owens 1980, 1985; Banti 1988; Lloret-Romanyach 1988; Hayward & Megerssa 1996	orm
	Konso	Orkaydo 2013	kxc
	Dirayta	Abire 2006	gdl
	Saho	Welmers 1952; Tajebe 2015	ssy
	Afar	Hayward 1991; Kamil 2015	aar
Southern	Dahalo	Tosco 1991	dal
	Iraqw	Mous 1993	irk
	Gorwaa	Harvey 2018	gow
	Alagwa	Mous 2016	wbj
	Burunge	Kießling 1994	bds

Table 2: Low (neutral) stems in Boni (Sasse 1979: 31)

a.	boʔor	‘gourd’
b.	boʔor áado	‘take a gourd’
c.	boʔór aadid	‘he took a gourd’
d.	boʔórti	‘the gourd’

A similar argument is made by Smith (2022) for Kenyan Maay, where underlyingly High, masculine-agreeing nouns have a static High tone (except in a few instances, such as in the presence of an interrogative determiner), even on long vowels where it yields a rise. Feminine-agreeing nouns instead are treated as underlyingly toneless and receive a High tone whose location is predictable based on stem shape and context. Furthermore, their High tones are subject to a variety of alternations which sometimes depend on modifier shape and type. Table 3 shows that masculine nouns in Kenyan Maay maintain a High on their final vocalic mora whereas High tone in feminine nouns alternates between a stem and determiner.

Table 3: Noun + determiner in Kenyan Maay (Smith 2022)

	Masculine		Feminine	
a.	fərás	‘horse’	qahwá	‘coffee’
b.	fərás kə	‘the horse’	qahwá ðə	‘the coffee’
c.	fərás kii	‘the (REM) horse’	qáhwa ðii	‘the (REM) coffee’
d.	fərás kej	‘my horse’	qáhwa ðej	‘my coffee’
e.	fərás kəŋ	‘this horse’	qahwə ðéŋ	‘this coffee’

In Harar Oromo (Owens 1985), there is a privative High vs. non-High contrast centered upon the penultimate syllable of nouns; for underlyingly High penult nouns, the High is present in all instances. Final syllables tend also to be High, including in the underlyingly Low/non-High class, but these Highs fail to appear in many contexts.

For some languages, scholars have argued for an active Low tone primitive. In Awngi, for example, Joswig (2010b: 21–23) illustrates that roots may have raising and lowering effects on their suffixes and that the suffixes themselves may surface with a contour, even on a short vowel. He accounts for these outcomes by proposing that stems and suffixes not only have associated tones but also a

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floating High or Low tone beyond the stem that is responsible for the raising and lowering effects, respectively (cf. [Rolle et al. 2026 \[this volume\]](#)). For example, the verb *itt* ‘fall’ lowers an otherwise High suffix *-é*, yielding [ittè] ‘he falls’ – this is attributed to a floating Low belonging to the stem that docks onto the suffix lowering it. The verb *giŋ* ‘run’, on the other hand, introduces a Fall on an otherwise Low suffix *-à*, yielding [giŋâ] – this is again attributed to a floating tone, this time a High tone, that comes to be realized on the suffix. [Tajebe \(2015: 113–114\)](#) also argues for a floating Low tone in Saho that encodes accusative case marking; depending on the context, this yields a falling H-L contour on an accusative noun or instead initiates a downstep on the first syllable of a following verb (cf. [Snider 2026 \[this volume\]](#)). [Le Gac \(2003b\)](#) has argued for a “Low tone of focus” in Somali, though this might best be seen as an intonational boundary tone.

As should be clear from the preceding, it is typically the case that accent *cum* tone is assigned underlyingly or instead by rule. Many descriptions have invoked the mora as the tone/accent bearing unit, though in some languages, it is the syllable. The clearest illustration of a mora tone bearing unit (TBU) are in those languages for which tone can occupy one or the other mora of a long vowel. Examples of such languages include Saho, Dhaasanac, and Somali, among several others. In fact, grammatical gender is sometimes encoded by precisely such a distinction, as in *máar* ‘bullock’ vs. *maár* ‘heiffer’ in Rendille ([Pillinger 1988: 505](#)); see further discussion in Section 4. In some cases, this mora-based distinction may be obscured by tonal rules that serve to flatten contours (see [Banti 1988](#) for Somali) thus making it appear as though the syllable might be the relevant prominence bearer.

This can be compared to Central Somali ([Saeed 1982](#)), aka Baydhabo Maay, for which the TBU is undoubtedly the syllable. High tone is assigned to a final or penultimate syllable, in masculine- and feminine-agreeing nouns, regardless of word shape, i.e., the presence of a long vowel in one syllable vs. the other. A related argument for a syllable prominence bearer has been made for Boorana Oromo ([Banti 1988](#)) where falling tones appear on both short and long syllables.

There are some languages for which it has been explicitly argued that, in addition to vowels (or vocalic moras), consonants (or consonantal moras) function as TBUs. Notable cases include Rendille ([Pillinger 1988](#), [Pillinger 1989](#)) and Somali. While [Orwin \(1996\)](#) shows that consonantal moras must be counted in satisfying a partial prefixing reduplication template in Somali, [Green \(2022\)](#) further argues that consonantal moras count in Somali tone assignment, at least in short stems that have just a single vocalic mora.²

²[Hyman \(1981\)](#) accounts for these outcomes by treating such vowel-glides as vowel-

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Unlike many more prototypical tone systems, the locus of tonal contrast in Cushitic languages tends to be limited to a window, typically calculated from the right edge of a word or stem. The final or penultimate TBU is a common location of tonal contrast, particularly in words with two or more syllables. Monosyllabic words often differ – they may maintain a toned vs. toneless contrast or instead neutralize the contrast entirely, either in favor of all words having High tone or otherwise being toneless. The examples in Table 4 from Saho show nouns with a lexical tonal contrast on the penultimate vs. final syllable in disyllabic words, but the distinction is on the penultimate vs. final mora in monosyllabic CVVC words, and in monosyllabic CVC nouns, the dimension of contrast is toned vs. toneless.³

Table 4: Noun tone in Saho (Tajebe 2015)

a.	gíra	‘face’	girá	‘fire’
b.	álsa	‘moon’	alsá	‘month’
c.	gáana	‘container’	gaaná	‘season’
d.	bóol	‘hundred’	koón	‘five’
e.	ból	‘cliff’	gul	‘time’

In some languages, prominence assignment is predictable (almost entirely) by word shape. In Bayso (Hayward 1978), quantity sensitivity also plays a role. A disyllabic word with one CVV syllable will attract accent (i.e., tone) to that syllable, with a preference for the penult when there are two such syllables. The penult is also favored in longer words without a CVV syllable, as well as in disyllabic words with an initial CVC syllable. It is only in CVCV words that the location of prominence is unpredictable. A similar but more complex case is found in Dirayta (Abire 2006) wherein monosyllabic CVC words are predictably High toned while CVVC and CVCC words are Low (toneless). Tone location

vowel sequences and also points out that Klingenheben (1949) records words of this type as containing two syllables.

³A reviewer asks whether one might instead analyze the contrast in Saho as HL vs. LH, such monosyllabic forms in 4(e) might contain a floating tone at their left edge that simply cannot associate due to lack of space to do so. Unfortunately, the source does not offer us much to tease apart the merits of this analytical alternative, though one might look to CVVCV forms for further insight. That is, if (L)H and (H)L were the appropriate representations in 4(e), this might raise a question as to why, if tones were indeed assigned right-to-left, we find *gáana* ‘container’, and not **gaána*, given that the mora is clearly the language’s TBU, and rises are otherwise accommodated by the system, as seen in *koón* ‘five’.

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is also predictable in trisyllabic words, where it is always on the final syllable. Once again, it is in disyllabic words, regardless of the presence of CVV syllables, that High tone location is unpredictable. Dhaasanac (Tosco 2001) and Rendille (Pillinger 1989: 42) present other exceptional cases in which prominence may appear on the antepenultimate syllable in a small subset of the nominal lexicon.

Lexical tonal minimal pairs are not widespread, though examples can easily be found, particularly among nouns. In Afar, for example, one finds minimal pairs like *gári* ‘proximity’ vs. *garí* ‘girafe’ and *ála* ‘animal’ vs. *alá* ‘camel’ (Kamil 2015: 105). Analogous pairs are found in closely-related Saho like *kálo* ‘water body’ vs. *kaló* ‘land’ (Tajebe 2015: 47). These are not limited to the Saho-Afar sub-branch though, as one can equally find such pairs in Dhaasanac like *ʔár* ‘bull’ vs. *ʔar* ‘song’ and *kólmu* ‘he-goats’ vs. *kolmu* ‘water basins’ (Tosco 2001: 37–38), and elsewhere.

Despite not having a major role to play in the encoding of lexical contrasts, one finds a robust array of tonal contrasts that encode grammatical distinctions, among them case and number in nouns, and also a variety of inflectional distinctions in verbs, including tense, aspect, mood, and negation, as well as grammatical gender. These are discussed further in Section 4.

4 Grammatical tone

Grammatical tone (GT) refers to “tonological operations which are not part of the general phonological system and are only licensed by specific grammatical conditions” (Rolle 2018: 2) (cf. Rolle & Bickmore 2026 [this volume]). Such operations are diverse but generally involve processes in which tone is one of, or the only phonological exponent of some morphological operation. As a phenomenon, GT is widely attested across Cushitic though seldom referred to in such terms. GT is variously involved in word level inflectional or derivational operations, as well as in compound formation and in the marking of genitival relationships between juxtaposed nouns, among other operations. These processes may involve tone addition or subtraction, or may even be replacive. Subtractive grammatical tone seems less well-attested cross-linguistically (Rolle 2018: 56), but it is a common occurrence in Cushitic.

In the simplest instances, GT operations function in the derivation of one lexical category from another. For example, in Maccaa Oromo (Lloret-Romanyach 1988: 27), verb infinitives and related nouns differ only in that verbs have an all-High melody while nouns are Low-High: *lágúu* ‘to prevent’ vs. *làgúu* ‘abstention’. Such nouns likewise differ minimally from corresponding third person singular

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verbs in that the latter exhibits a (Low)-High-Low melody assigned by syllable from the right edge: *kùrrúufà* ‘snoring’ vs. *kùrrúufà* ‘he snores’. Similar distinctions reveal themselves in Saho nouns vs. imperative verbs, with the former being toneless and the latter exhibiting a High tone: cf. *qín* ‘sleeping’ vs. *qín* ‘(you.sg) get sleep.IPV’ (Tajebe 2015: 48). Such tonal distinctions are widespread across the family.

In addition, there are well-known correlations between prominence (i.e., High tone or “accent”) and grammatical gender (and, where relevant, biological gender) across much of Cushitic, with notable exceptions being in Northern and some Central Cushitic languages (Appleyard 1991). Elsewhere, grammatical gender and prominence location and/or the presence vs. absence of prominence often but not always align with one another. An illustration of correlations between prominence and gender can be seen by comparing the tonal behavior of Boni nouns and the allomorphs of the definite suffix that they select in Table 5; the stem-suffix boundary is indicated by a hyphen. Here, masculine nouns (a-c) have High tone on their penultimate vocalic mora in the “basic” (aka “unit reference” form), and this High remains in the definite, which takes the suffix *-i*. Feminine nouns (d-f) instead are toneless in isolation, but they receive a High tone on the first mora of the final stem syllable when the feminine definite *-ti* suffix is present.

Table 5: Nouns and definite forms in Boni (Sasse 1979)

	Basic	Definite	
a.	béer	béer-i	‘stomach, liver’
b.	galáan	galáan-i	‘river’
c.	áɲal	áɲal-i	‘honey-wine’
d.	moor	móor-ti	‘home’
e.	misraar	misráar-ti	‘axe’
f.	shimir	shimír-ti	‘bird’

How best to characterize these correlations is a matter of longstanding debate. Some scholars argue that gender is lexically specified (Appleyard 1991, Hyman 1981) and that tone is a “manifestation” of grammatical gender marking. Others like Hayward (1991: 125) contend that grammatical gender is instead “derivative” of lexical tone/prominence specification. Nilsson (2018) argues, for Somali in particular, that both tone and grammatical gender are entirely predictable by stem shape. Exceptions, however, would be nouns with identical stem shape but different tonal patterns which are strictly based on a biological gender distinction (e.g.,

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inan ‘boy’ vs. *inán* ‘girl’, in Somali), but these have elsewhere been accounted for in some instances by appealing to the presence of catalectic slot at the right edge of grammatically feminine stems (Godon 1998, Green & Lampitelli 2024, Lampitelli 2011, 2013, Le Gac 1997).

The noted prominence/gender correlations often apply to stems and to certain affixes that neutralize stem-level grammatical gender distinctions in favor of one particular value entailed by the affix itself. For example, in Somali, plural marking via the suffix *-yaál* entails feminine agreement, and the same operation via *-óyin* requires masculine agreement. Tone on these affixes follows the same final vs. penultimate pattern as in singular nouns. In other instances, no correlation exists. For examples, deverbal nouns in Somali formed by the suffix *-tín* unpredictably require either masculine or feminine agreement (e.g., *cabatín-ka* ‘the complaint’ vs. *baqdín-ta* ‘the act of being afraid’, respectively – examples from Green 2021: 136). Depending on one’s point of view, grammatical gender in Cushitic may be best seen as an epiphenomenon, rather than being distinctly tied to tone. As such, it may or may not rightfully fall under the heading of grammatical tone.

Also widespread are GT operations that encode other inflectional distinctions in various lexical categories. GT affects nouns in several ways, one of which is in the manifestation of case marking.⁴ To my knowledge, the only detailed treatment of case marking as GT is Green & Lampitelli (2024), which concerns Somali subject marking. Examples in Table 6 show that whereas Somali nouns typically have a High tone on their penultimate or final mora, subject (nominative) marking entails loss of High tone in all but a few exceptional instances (e-f), provided the noun is not marked for focus. Feminine nouns (c-d) additionally realize the suffix *-i*.

Case marking is a phrase- or clause-level phenomenon such that it is marked on just one element of a phrase or clause, often at the right edge. In (1), nominative marking in Rendille via High tone loss (and the suffix *-e* for the feminine noun) appears on the rightmost word of a complex noun phrase. Nominative marking likewise appears on a phrase-final relative clause verb in (2).

- (1) Rendille (Pillinger 1989: 131)
 [ínám ɪçóòw ínàmè] íntàné kíjírààn.
 boy and girl.NOM here be.at
 ‘The boy and girl are here.’ (cf. *ínám* ‘girl’)

⁴I take no stance on whether “case marking” is truly applicable to Cushitic languages – compelling critiques of case marking in Somali, which may be extensible more broadly in Cushitic, include Nilsson (2017) and Lampitelli (2019).

Table 6: Subject marking in Somali (Green & Lampitelli 2024)

	Basic		Subject-marked
a.	búug	‘book.UMRK’	buug ‘book.MRK’
b.	nácas	‘fool.UMRK’	nacas ‘fool.MRK’
c.	naág	‘woman.UMRK’	naagi ‘woman.MRK’
d.	galáb	‘afternoon.UMRK’	galabi ‘afternoon.MRK’
e.	dáwo	‘medicine.UMRK’	dáwo ‘medicine.MRK’
f.	beége	‘gauge.UMRK’	beége ‘gauge.MRK’

(2) Rendille (Pillinger 1989: 131)

[étí čèlé yìmi] á ùrdà.

man yesterday come.NOM PFM sleep

‘The man who came yesterday is sleeping.’ (cf. yìmi ‘come’)

Other languages for which subtractive subject/nominative marking of nouns is attested in at least some subset of the lexicon include Afar (Hayward 1991), Konso (Orkaydo 2013), Arbore (Hayward 1984: 149–150), and Maay (Saeed 1982: 9). Analogous operations entail a High-tone nominative suffix which consequently suppresses the stem’s High tone, at least in some nouns. This can be seen in both masculine (a-c) and feminine (d-e) nouns in the Booran dialect of Oromo in Table 7. Additive nominative marking, via the addition of H tone, has been reported for Boni (Mungai 1996), but with no details provided.

Table 7: Booran Oromo subject marking (Appleyard 1991: 15)

	Absolutive	Subject
a.	nám	namí ‘man’
b.	fárd	fardí ‘horse’
c.	nagáa	nageenní ‘peace’
d.	ráad	raaddí ‘heifer’
e.	íntal	intaltí ‘girl’

There are other instances where it is the marking of absolutive case that entails GT. One example is found in Declension 2 and 3 nouns in Rendille which involve the imposition of a right edge H tone (Pillinger 1989: 142). Absolutive marking via a floating Low tone is Saho was mentioned above. Other types of object marking are reported that involve tonal subtraction. These are particularly robust in

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Arbore where plural nouns and vowel-final feminine nouns functioning as direct and/or indirect objects lose their High tone (Hayward 1984: 158–159). Some nouns (non-feminine nouns ending in a vowel) in “predicative” position similarly lose their High tone in the language and also undergo an alternation in the quality of the stem-final vowel (Hayward 1984: 135–145).

Genitive (possessor, associative) relationships are typically encoded tonally, either with or without an additional affix, the form of which depends on declension class or stem shape. Genitive marking is tonally subtractive in languages like Rendille (Pillinger 1989: 130), Boni (Sasse 1979: 73), Afar (Hayward 1991: 127), and Arbore (Hayward 1984: 154–155). Examples (3) and (4) from Arbore show genitive marking at the phrase’s right edge via High tone loss, and the addition of a suffix, where appropriate based on the noun’s gender.

- (3) Arbore (Hayward 1984: 154)
 dórr-a kirmate-t
 feather-PL bird-GEN
 ‘(the) feathers of a bird’ (cf. kirmaté ‘bird’)
- (4) Arbore (Hayward 1984: 155)
 ñawoló naag
 panga boy.GEN
 ‘(the) boy’s panga’ (cf. náag ‘boy’)

The genitive is instead marked by a High tone on the possessor in Saho (Tajebe 2015: 115–118), which may or may not occur on a suffix. In the case of the latter, the genitive High suppresses stem tone. Genitive nouns are marked by a long High-toned vowel in Oromo which similarly suppresses stem tone (Banti 1988: 34). Elsewhere, such as in Tunni (Tosco 1997: 48) and Somali (Green 2021: 111), the Genitive is marked by a High tone at the possessor’s right edge. This can be seen in example (5) where tone on the noun ‘sheep’ is typically on the penult but appears on the final when in a genitive construction.

- (5) Tunni (Tosco 1997: 48)
 náf arán
 head sheep.GEN
 ‘a (head of) sheep’ (cf. áran ‘sheep (coll.)’)

Other case forms may be tonally encoded in some languages, such as the dative-ablative in Oromo, which is indicated by the combination of a suffix and

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loss of a preceding tone. So-called vocative marking is encoded by a tonal overlay in languages like Somali (Green 2021: 142) and Tunni (Tosco 1997: 48).

One particularly intriguing instance of tonal inflection is found in Iraqw wherein agreement between a noun and its modifying attributive adjective is encoded by prominence pattern, as well as by a gender-marking suffix on the head noun. Consider the following examples in (6) and (7) whose head nouns differ in gender but on their own otherwise display the same “accent 1” (Low pitch throughout). When modified, their gender comes to be encoded tonally in these “construct” forms, but on the modifier itself. That is, the “accent 1” pattern is realized after feminine nouns and thus appears on *tlèer* in the first example. An “accent 2” (Low-High) pattern is realized after masculine nouns and accordingly appears on *tléer*. The head noun’s prominence patterns remain tonally neutralized, but now, both realize “accent 2”.

- (6) Iraqw (Kießling 2003: 6)
 hhàrtá tlèer
 stick.GEN.F1 long.F
 ‘long stick’ (cf. *hhàrà* ‘stick’)
- (7) Iraqw (Kießling 2003: 6)
 hímúu tléer
 rope.GEN.M long.M
 ‘long rope’ (cf. *hìmà* ‘rope’)

In this instance, the simplest analysis would appear to be that agreement with the noun’s grammatical gender is dictating these outcomes, expounded by the adjective’s resulting pattern of prominence. The patterns on the nouns themselves would be a type of “overlay” associated with various “constructs”. Agreement patterns reminiscent of these are reported for other languages and attributed to phonologically conditioned concord (Dobrin 1998, Sande 2019) whereby some aspect of a head noun’s phonological form dictates patterns of agreement. This seems not to be applicable in the Iraqw case, however, as the key factor would appear to be gender, rather than some phonological characteristic of the noun, that is dictating the outcomes.

Before moving on to GT in verbs, it is important to note that plural formation, like many other instances of suffixation and related word formation processes discussed in Section 5 often has an effect on stem tone. In Somali, for example, nearly all instances of pluralization suppress stem tone in favor of High appearing on the suffix itself (e.g., *hilib* ‘meat’ vs. *hilib-ó* ‘meat’ or *sán* ‘nose’ vs. *san-án*

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‘noses’).⁵ This has sometimes argued to be due to a culminativity requirement that only one High tone appears on a word (Hyman 1981; Green & Morrison 2016) and reflects the widespread tonal “dominance” effects that appear across Cushitic.

Indeed, most instances of pluralization via suffixation in Dhaasanac similarly lead to stem tone loss (Tosco 2001: 73–75), just as they do in Boni. Plural formation in Dahalo has many patterns (Tosco 1991: 22–33), some of which simply involve stem tone loss, while others involve stem-final vowel mutation and the addition of stem prominence. In Yaaku, the plural *-ndi* leads to stem tone loss but the suffix itself does not bear a tone; a similar outcome obtains in Iraqw (Mous 1993: 21). In Yaaku (Heine 1975), however, the singulative suffix *-ní* instead compels the entire stem to be realized High: cf. *kit’ε* ‘bees’ and *kít’ε-ní* ‘bee’.

As seen just above, dominance effects typically reveal themselves in the interplay between tone and affixation, and sometimes in cliticization and other word building operations like compound formation. The presence of such effects and their associated alternations offers support to cyclic theories of the phonology-morphology interface, as discussed specifically for GT in Inkelas (2008) and Rolle (2018), more broadly in works like Inkelas & Zoll 2007, and specifically for Somali in Green & Lampitelli (2024). For additional examples, see Section 5.

Turning to verbs, GT operations are diverse. In the simplest instances, tone alone differentiates between inflectional categories, such as imperfective *saá* ‘he enters’ and imperative *sáa* ‘you(PL) enter!’ in Saho, where location of prominence on one mora vs. the other of a verb encodes the relevant distinction (Tajebe 2015: 47). Likewise, in Afar, tone location differentiates between the “incomplete assertive” as in *ábá=h* ‘It does (effectively)’ and a “converb with a value of simultaneity” as in *ába=h* ‘by doing’ (Kamil 2015: 107). In Dhaasanac, it is instead the presence vs. absence of tone that encodes some distinctions, like the imperative singular vs. “A-form” of the Simple Past: cf. ‘to open’ *fúr* vs. *fur*, respectively (Tosco 2001: 38). Elsewhere, inflectional suffixes may subcategorize for a particular prosodic configuration, thus affecting tone. In Boni, for example, all Habitual suffixes require a High to appear immediately before them on stems, which are otherwise toneless (Sasse 1979: 110–111). Compare, for example, 1SG Imperfect *ahan-a* to its corresponding Habitual *ahán-sii* and Habitual Negative *ahán-sine*.

The importance of tone in modal contrasts was established in Appleyard (1991) though it extends beyond those few languages discussed. In many languages, the jussive and related moods like the optative and subjunctive (depending on the

⁵The exception is so-called prosodic plurals where High appears on a word’s rightmost TBU (cf. *mádax* ‘head’ and *madáx* ‘heads’).

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language and description) are characterized by a particular tonal configuration. In Boni, the jussive is formed from the toneless subjunctive by the addition of a High tone of the “prefinal syllable of the subjunctive forms” (Sasse 1979: 103). The subjunctive is also toneless in Iraqw (Mous 1993: 162). The subjunctive in Booran Oromo is invariably marked by High tone on the first syllable of the verb (Owens 1980: 170). In Somali, so-called “irrealis” marking (Green 2021: 160) involves a unique inflectional paradigm characterized by High-toned suffixes that suppress stem tone; most other inflectional suffixes are toneless in the language. In Saho, the subjunctive requires a penultimate High tone, while High is final in the jussive (Tajebe 2015: 50).

In some instances, verbs that bear a High tone in some contexts are realized toneless in contexts that have variously been described as affirmative, declarative, indicative, or realis. Such behavior can be observed in Somali (Saeed 1999) and Dhaasanac (Tosco 2001: 41), as well as in Afar and Booran Oromo (Hayward 1991: 26–27).

5 *Tone in word formation*

Tone plays a major role in word formation owing to the fact that many Cushitic languages permit only a single, culminative prominence within some domain, typically the “word” (Arbore: Hayward 1984; Bilin: Fallon 2004) or prosodic word (Somali: Downing & Nilsson 2019, Green & Morrison 2016, Le Gac 2002), though sometimes a larger domain like the phrase (Afar: Parker & Hayward 1985) or “accent-unit” (Beja: Hudson 1974) have been proposed as key unit influencing tone assignment. Tone has sometimes been proposed as a diagnostic for wordhood, though not uncontroversially (on Somali, see Kaldhol & Stausland Johnsen 2025). Rendille is one language which permits multiple High tones per tonal domain, thus making its prosodic system appear “more tonal” than most other Cushitic languages. It is claimed elsewhere, such as in Arbore (Hayward 1984), that multiple High tones are permitted per word, though this arguably raises an analytical question about orthographic words vs. the prosodic units that compose them.

While in some Cushitic languages, like Somali, the appearance of toneless words is exceptional, in others, and notably across South Cushitic, the vast majority of words are toneless (Mous 2021). In this sub-group, however, tones are often contributed by lexically-toned affixes. This can be seen in the examples from Gorwaa (Harvey 2018: 121–123) in Table (8) where several High-toned suffixes derive extended verb stems from an otherwise toneless base (a-c), but if the verb stem

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happens to be toned, the presence of the suffix brings about loss of stem tone (d). This is most certainly due to a culminativity constraint on word prominence. Such outcomes are commonplace in Cushitic: in Booran Oromo, for example, a class of adjective-formation suffixes systematically requires lowering of the stem High tone (Owens 1980: 153): *nám* ‘person’ vs. *nam-ic* ‘certain person’.

Table 8: Gorwaa verb stems (Harvey 2018: 121-123)

a.	tlatu	‘debt’	tlatuús	‘be indebted’
b.	da’aye	‘fear’	da’ayuút	‘fear’
c.	niinaw	‘small/	niinaweés	‘lessen’
d.	/awaákw	‘white’	/awaakeés	‘whiten’

It is often the case that when more than one toned affix is present, affixal tones may also be suppressed, as only the morphologically highest, “dominant” prominence is realized (Appleyard 1991). Similar outcomes may obtain in the presence of some (but not all) determiners and even other modifiers, thus contributing to the complexity of defining the domain of prominence realization in some languages. This is discussed extensively for Somali in Green & Morrison (2016) and Downing & Nilsson (2019). Along similar lines, derivational and inflection affixes, as well as clitics, may contribute differently to the tonal morphophonology of a given language (Green & Morrison 2018, Tajebe 2015).

Once again using Somali as an example, some but not all derivational suffixes are tonal and suppress stem tone, while inflectional suffixes, except those of the verbal irrealis paradigm, are toneless; pluralization tends to pattern tonally with other instances of derivation. Green (2021: 147) illustrates that in the Somali verbal system, all derivational extensions to the verb stem suppress stem High tone and themselves surface with a High, exception for the reciprocal *-tan* and the weak causative *-i*. Inflectional suffixes on verbs that encode person, number, and grammatical gender do not affect tone. In the nominal system, Green (2021: 130–139) and Saeed (1999: 149–154) show that most derivational suffixes exhibit similar behavior, suppressing stem High in favor of hosting a High themselves. The only exceptions to this behavior are the deverbalizer *-niin* which only sometimes behaves as just described and otherwise does not affect stem tone, and a second deverbalizer *-n* which selects only weak causative verbs. When this suffix is present, High is suppressed on the stem and appears instead on the weak causative suffix itself.

Analogous outcomes arise in compound formation. For example, in Saho, Tajebe (2015: 133–134) describes compound formation as the creation of a single

phonological word, whereby two otherwise independent and toned words join to form a single word with a single High. Consider the phrase *sek-tí kabellá* ‘shoe of sheik’ which has come to be lexicalized as the compound *sektikabélla* ‘accessible’. In the former, each word bears its own prominence, but there is just a single prominence on the penult in the latter. Here and elsewhere in Saho, the location of the High tone, whether penultimate or final on the final element of the compound, is determined by grammatical gender. Not all analyses of Cushitic compounds assume that the domain of tone association is the prosodic word; for example, to account for the behavior of compounds in Somali, [Downing & Nilsson \(2019\)](#) propose that the domain of High tone assignment is the *Complex Word Group*.

6 Tonal alternations

The diverse array of tonal alternations seen in many other African language phyla is not observed in Cushitic. As shown below, alternations in Cushitic are generally uncomplicated and mostly involve tone spreading, contour avoidance, non-finality, and, in some instances, tone (accent) polarity (cf. [Pulleyblank 2026 \[this volume\]](#)).

Not discussed in this section are matters related to tone assignment nor alternations falling under the heading of grammatical tone. The former typically pertain to the assignment of High tone to a particular word position or TBU in privative systems like those discussed in [Section 3](#), or to the assignment of Low tone to otherwise tonally unspecified TBUs. For example, [Hayward & Megerssa \(1996\)](#) propose a *Default High Assignment* rule in Oromo whereby High tone is assigned word-finally to lexically unaccented nouns. [Tosco \(1991: 14\)](#), on the other hand, proposes an *Unmarked Low Rule* for Dahalo that assigns Low tone to any non-High TBU. Similar rules can be found that assign a Low tone to the second mora of a long vowel whose first mora is already associated with High, thus creating a Fall. In other instances, when “accent” analyses have been proposed, rules may formalize the assignment of tone to a lexical accent. Operations involving grammatical tones, which are phonological exponents of certain morphological or morphosyntactic configurations, are discussed above in [Section 4](#).

The tonal alternations that I have most often encountered in my survey of the Cushitic literature involve tone spreading (cf. [Smolders 2026 \[this volume\]](#)). These generally are realized in either anticipatory or perservative spreading of High tone by a single TBU, though the conditions leading to spreading and its domain differ between languages. [Hayward & Megerssa \(1996: 33\)](#) propose a

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Rightward High Spread rule for several Oromo dialects the result of which is to spread High from the penult of some nouns onto the final. Analogous spreading is attested in Arbore nouns when they appear in the predicative form (Hayward 1984: 225). Also in Arbore, High will spread from the penult of a verb to the final, if a High suffix is attached, thus creating an all-H verb (Hayward 1984: 258). In Yaaku (Heine 1975: 38), High tone will spread onto a toneless suffix, but in others instances, a H-toned suffix like the singulative *-ni* spreads its High leftward onto a toneless stem. Anticipatory High spreading is also reported in Tunni (Tosco 1997: 36).

High Spreading in Rendille (Pillinger 1989: 89) is unique not only in that it will spread to the first mora of a long vowel creating a Fall (e.g., *mín* ‘house’ + *kààya* ‘my’ → *mínkáàya* ‘my house’), but if High Spreading displaces a Low tone, the resulting floating Low yields a downstep on the next High tone (e.g., *yábàr* + *-í* → *yábàr’í* ‘the rope of...’). Kießling (2003: 7) reports a high spreading rule in Iraqw the motivation for which is said to be one way to avoid of two instances of rising “accent 2” within the same phonological word: /waqát-uu-í/ → *waqatuuwí* or *waqátúuwí*, but **waqátuuwí*. Spreading is also implicated in various decontouring operations, including in Somali (Banti 1988), Kenyan Maay (Smith 2022), Arbore (Hayward 1984), and Rendille (Pillinger 1989). Contour tones are instead created by the docking of floating High tone case markers onto word-final Low tones in Awngi (Joswig 2010b: 22–23).

Though there is a widespread tendency for tonal prominence to be attracted rightward onto affixes in the suffix-heavy Cushitic languages, High sometimes fails to be realized on word-final affixes. This outcome can perhaps be seen as the result of non-finality constraints whereby (typically) some languages disprefer High tone on a final open syllable. In such instances, High may appear on the stem instead of on an otherwise toned affix. Such “accent shift” alternations are particularly well-known to affect Somali nouns, which appear with a penultimate High tone in final position but otherwise with a final High tone when followed by a modifier (Green 2021: 109; Saeed 1999: 47). Similar alternations have been described as *prepausal high tone shift* in Tunni (Tosco 1997: 39) and as the *tonal regression rule* in Dahalo (Tosco 1991: 14). In Tunni, for example, various vowel-final nouns are underlyingly prominent on their final syllable, but when they appear in isolation, prominence appears instead on the penult: cf. /dooró/ → *dóoro* ‘chicken’.

Instances of tone polarity, or accentual “see-saw” (Tosco 2001: 41–42), are reported for some determiners and the negative declarative marker in Dhaasanac. For example, when modifying a singular noun bearing prominence, the 1SG possessive is toneless (*gíl cu* ‘my hand’). But, the corresponding toneless plural is

instead modified by a possessive bearing an “accent” (*gil cú* ‘my hands’). Similar polar alternations affect the locative and ablative case suffixes in Awngi (Joswig 2010b).

7 Historical perspectives

It is generally assumed, following work on Proto-Cushitic by Heine (1978) and Ehret (1987, 1995), that Proto-Cushitic was a tonal language. Indeed, Heine (1978) assumes that Proto-Cushitic was tonal but also “accentual”, and Appleyard (1991) posits that the limited role for “tone/accent” in some contemporary Cushitic languages may, in fact, stem from a more robust, productive proto system with more substantial tonal oppositions. This is a view also implicit in Lamberti (1986), but not all scholars necessarily agree.

As suggested by Kießling (2003), there are well established tonal oppositions in Eastern Cushitic languages which several scholars have variously attributed to what Kießling calls “terminal erosion”, or the loss of segmental material from the right edge of some nouns. Kießling provides the famous example of **ínan* ‘boy’ vs. **ínám-et* ‘girl’ which in contemporary Somali is realized *ínan* ‘boy’ vs. *ínán* ‘girl’. Here, an inflectional suffix once marking feminine grammatical gender is lost from nouns, rendering a once predictable penultimate stress pattern no longer predictable and, thus, contrastive. Accordingly, Kießling suggests that such a development and others like it, which are widely reported in the literature, including back to Proto-Sam (Heine 1978), could be seen as tonogenetic in nature. The implication, thus, is that the prosodic opposition (whose primary correlate is pitch), was not inherited from Proto-Cushitic, but rather emerged diachronically as the result of other developments in these languages.⁶ To this, Kießling (2003) adds that tonogenesis is also independently underway in Southern Cushitic, as Iraqwoid languages have undergone tonogenesis, albeit from a less distant tonal ancestor, Proto West Rift. It is also well known that the Highland East Cushitic languages exhibit stress-accent systems, with the possible exception of Sidaamo (cf. Frosini 2008, Kawachi 2007). These facts, Kießling suggests, may call into question the verity of the aforementioned claims about tone in these languages’ shared ancestor, Proto Cushitic.

⁶According to Appleyard (1991), this development in Somali was not due to a simple loss of the feminine suffix. Rather, his analysis is that the feminine suffix itself once was the bearer of prominence – the segmental material of the suffix was lost only after “accent shift” brought prominence onto the stem. It is important to note that not all languages lost segmental material from their right edge, and not all experienced the same tonal adjustments; see Green & Smith (2024) for further discussion on this topic.

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8 Summary remarks

It has been shown in this chapter that most Cushitic languages have prosodic systems which can be considered “restricted tone” systems. These are non-prototypical compared to most other African tonal systems and often exhibit properties that are more often encountered in stress accent systems. I have provided an overview illustrating examples of many of these characteristics from languages across the family, including many instances of privative High vs. non-High contrast, culminative but not always obligatory prominence in some domain, a major role for tonal prominence in word formation, and evidence for operations involving grammatical tone. It is remarkable just how few tonal processes (both in type and frequency) have been observed in Cushitic languages, and as a reviewer has suggested, this may prove to be a consequence of the relative simplicity of these tonal systems overall – with just a privative H vs. Ø contrast in most instances, choices for alternation are to i) add a High tone (at the possible consequence of removing another), ii) remove a High tone, or iii) move an existing High tone elsewhere within some domain.

While a number of careful descriptions of Cushitic prosodic systems have been made available in recent years, the study of prominence in these languages, be they rightfully tone- or stress-based, still lags behind research on other aspects of these languages’ grammar, and their morphology and syntax in particular.

Abbreviations

C	consonant
coll.	collective
F	feminine
GEN	genitive
GT	grammatical tone
H	High tone
IPV	imperative
L	Low tone
M	masculine
MRK	subject-marked
NOM	nominative
PFM	predicate focus marker
PL	plural
REM	remote definite determine
SG	singular
TBU	tone bearing unit
UMRK	non-subject-marked
V	vowel

Acknowledgements

TBD

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Chapter 10

Tone in Chadic languages

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The Chadic family of languages is rather exceptional among African languages with respect to the behavior of its tonal systems. Constituting a branch of the Afroasiatic phylum, Chadic languages are often characterized as “minimally tonal” because of the low functional load of tone in making lexical distinctions. The present article surveys all four branches of the family in order to understand the peculiar aspects of its tonal systems. As background, I begin by presenting the phonological characteristics of Chadic languages that have bearing on tone and its behavior. Next follows a consideration of two recurring tone phenomena in languages of the family, namely the use of tone to make grammatical distinctions and the effect of consonant quality on the realization of tone. It is also shown that other common tone phenomena that figure prominently in other African languages play only a marginal or nonexistent role in Chadic languages. In fact, there are even tone phenomena in Chadic languages that do not easily fit within the usual frameworks for tonal behavior, specifically tone reversal and the role of metrical structure in determining tone. So, given the exceptional behavior of tone in Chadic languages, how did this situation come about? Chadic languages may not have always been tonal, but I doubt that external factors from language contact are principally responsible for tonogenesis. It is much more likely that tonogenesis is due to factors internal to the language(s), such as the influence of consonants and metrical structure.

Russell Schuh, in his massive compendium of data on Chadic languages, states flatly (2017: 89) that “all Chadic languages are tone languages.” This statement cannot be taken without some reserve, however. Schuh himself goes on to say (p. 89) that it is unlikely that the parent language, proto-Afroasiatic, was a tone language. Semitic, Berber, Egyptian, and Cushitic – Chadic’s sister families within the Afroasiatic phylum – are nontonal or only marginally tonal. So how is it that all Chadic languages of today should be tonal? In response, Schuh

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repeats a common assumption (p. 89), namely, that it must be due to the contact of Chadic languages with neighboring Benue-Congo languages, which are manifestly tonal. This hypothesis may be plausible for the languages spoken in Nigeria (the West Chadic branch), but there are three other branches of the Chadic family which are not in direct contact with Benue-Congo languages at all, nor with many other Niger-Congo languages. Later in this article I present at least three other reasons why Chadic languages might have become tone languages. The article begins, however, by surveying some general characteristics of Chadic languages that have bearing on tone (Section 1), and the role of lexical tone in Chadic languages (Section 2). Then it considers the various sorts of tone phenomena that have been attested in some of the 200 languages that comprise the Chadic family (Section 3–Section 5). Section 6 then presents some phenomena from Chadic languages that are not prototypically tonal. Finally, a return to the question of tonogenesis (Section 7) is followed by a synthesis of what can be said about tone in the Chadic language family (Section 8).

1 General characteristics of Chadic languages

The Chadic language family is one of the principal divisions of the Afroasiatic phylum, and comprises about 200 languages spoken in the three countries of Nigeria, Cameroon, and Chad. Four branches of the family are generally recognized: West Chadic (found exclusively in Nigeria), Central Chadic (in Nigeria and Cameroon, with a small spillover into Chad), East Chadic (spoken exclusively in Chad), and the Masa branch (in Chad along the Cameroon border, with some representation in the latter country as well). Hausa is the best-known Chadic language (from the West branch), but because of its long and widespread use as a language of wider communication, some specialists question how well it represents Chadic characteristics.

It is well known that the semantic load of a lexical root in Afroasiatic languages rests on its consonants, and not on its vowels. The Semitic languages with their trilateral roots are best known, but a similar dependency on consonants can be adduced for Chadic languages as well. For one thing, Chadic languages often have large consonant inventories. Schuh (2017: 22) shows a table with more than 50 consonants, indicating that most of them can be reconstructed for Proto-Chadic, and that each Chadic language has a large subset of them.

For Chadic verbs, a lexical root will be made up of a series of 1–6 consonants, and only one vowel (Frajzyngier 1983: 125, Roberts 2002). There seems to be a small set of verbs in languages across all branches which are composed of

one underlying consonant, and nothing more, i.e. no underlying vowel at all (Tourneux & Jungraithmayr 1990). The structure of Chadic noun roots is not as highly constrained as that of verb roots, but the same tendencies prevail.

If consonants predominate, then, it follows that vowels play a secondary role in Chadic languages. Some languages may have as many as eight or nine surface vowels, but the number of underlying vowels is much fewer. A common system, especially in the Western, Eastern, and Masa branches, is the classic five-vowel set /a e i o u/. An extreme system is found in most languages of the Central branch, wherein the only underlying vowel is /a/, despite a more complete array of surface vowels. Although the minimal system is commonplace in the Central Chadic languages (for example Moloko, see Bow 1999), there are languages in each of the other three branches which also have the minimal vowel system: for example, Miya (Western Chadic, see Schuh 1998, 2017: 51), Somrai (Eastern Chadic, see Roberts 2007, 2012, 2025a), and Hede or Zimé of Pala (Masa branch, see Roberts 2007, 2025a). Regardless of the underlying vowel system, though, all Chadic vowel systems display two phonological levels, wherein the high vowels (i, ɨ, u) are more marginal, even nonphonemic, while the nonhigh vowels (e, a, o) have a more solid lexical value (Roberts 2025b). Graphically, a simple Chadic vowel system will take on the rectangular shape shown in Table 1, even though not all vowels (especially the high vowels) may be recognized as underlying.

Table 1: Simple Chadic vowel system

High	i	ɨ	u
Nonhigh	e	a	o

Chadic languages make substantial use of vowel epenthesis in order to break up clusters in their consonant-heavy roots. The vowels regularly used for this purpose are the high vowels, usually [i] or [ɨ], a fact which follows if they are not in fact underlying vowels.

There is another common morphophonological phenomenon that concerns vowels in many Chadic languages, that of apophony or ablaut, i.e. inflectional changes to a root-internal vowel. In many cases, this involves the regular alternation of a full-fledged low vowel with one of the marginal high vowels. It is possible to analyze this change as the deletion of the low vowel, with the epenthesis of a default high vowel to provide the needed transition between the consonants which have come into contact. See the examples of (1):¹

¹Transcriptions in this article follow IPA conventions, with an exception common to African-

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- (1) a. Ron (Western branch)
 duk (perfective) *dwaak* (imperfective) ‘beat’
- b. Mukulu (Eastern branch)
 goome (subjunctive) *guume* (completive) ‘swallow’

The preceding comments about the reduced vowel systems of Chadic languages lead us to consider another feature which must be reckoned with, namely the functioning of the suprasegmental prosodies of labialization (LAB) and palatalization (PAL). These suprasegmental features normally affect complete morphemes or words, which justifies their suprasegmental status as prosodies. Also, LAB and PAL are distinctive underlying elements of the phonologies of these languages, alongside the phonemic consonants and vowels. Languages which have been analyzed as having the single underlying vowel /a/ always utilize LAB and PAL to flesh out their surface vowel inventories: when the vowel /a/ falls within the domain of LAB, it will be rounded and realized as [o], and when it falls within the domain of PAL, the surface vowel will be a fronted [e]. In similar fashion, an epenthetic [i] which falls within the domain of LAB will be realized as [u], and within the domain of PAL it will surface as [i]. Some languages even allow both prosodies to occur on the same vowel, producing the front rounded vowels [ø] and [y]. The prosodies are also responsible to some degree for augmenting the consonant inventories of some Chadic languages, for the prosodies may affect certain consonants as well as the vowels that fall within their domain. Schuh’s table of consonants (2017: 22) includes some palatalized consonants and labialized consonants, which may be attributable to the functioning of PAL and LAB, if in fact those complex consonants are not to be considered as underlying.

Table 2 presents a generalized chart of Chadic consonants, similar to that shown in Schuh (2017: 22), but arranged in a way that will be shown to be relevant to the tonal phenomena presented in Section 4. As Schuh comments, no Chadic language would have as many as appear in the table, but most languages would use a large subset of them.

Syllable structure in most Chadic languages is quite simple. Because of the paucity of underlying vowels, epenthetic vowels may be called upon to fill the nucleus of some syllables, and so it is necessary to specify that we are speaking of the structure of surface syllables. In addition to CV syllables, most languages allow for the integration of a second consonant into the coda of the syllable, producing CVC syllables as well. These two surface syllable types are by far the most common in Chadic. The clustering of consonants within the syllable is rare, for epenthesis is used liberally to break up consonant clusters.

ists: [y] represents the palatal glide, and [j] the palatal plosive/affricate.

Table 2: General table of consonants in Chadic languages.

Voiceless obstruents							
Plosives	p	t	ts	c / tʃ	k k ^w k ^y		
Fricatives	f	ɬ	s	ʃ	x x ^w x ^y	h	
Glottalized			ts'		k' k' ^w k' ^y	[ʔ]	
Voiced obstruents							
Plosives	b	d	dz	ɟ / dʒ	g g ^w g ^y		
Prenasalized	mb	nd	ndz	ɲɟ / ɲdʒ	ŋg		
Fricatives	v	ɓ	z	ʒ	ɣ ɣ ^w ɣ ^y	ɦ	
Sonorants							
Glottalized	ɓ	ɗ		f			
Nasals	m	n		ɲ	ŋ		
Liquids		l	r ɾ				
Glides	w			y	(w)		

The templatic morphology found in Semitic languages (see [McCarthy 1981](#)) also plays a role in a number of Chadic languages, establishing the consonantal structure of lexical roots (see [Roberts 2002](#) for Migaama, [Prickett 2012](#) for Mubi). Most Chadic languages allow the initial syllable of some lexemes to be onsetless, that is of shape V and VC, alongside the CV and CVC syllable types mentioned above. In those cases the initial consonant slot in the template of these lexemes is empty, but is filled by an optional phonetic [ʔ]. A number of Chadic lexicons include the [ʔ] in the transcription of such items, even though it is not phonemic.

One additional complexity must be mentioned, namely the existence of long vowels, for there are also surface syllables of type CVV and even CVVC in some languages. Long vowels are mostly found in West Chadic languages and in the Guéra group² of East Chadic, as well as in a smattering of other languages, but length is certainly not a universal feature of Chadic vowel systems. In some cases, the long vowel arose historically when a consonant dropped out between two short vowels. If the consonant did not simply disappear, it sometimes left an empty slot in the structure, which was filled by a nearby vowel, resulting in compensatory lengthening. Such explanations for today’s long vowels are not at

²Newman ([1977](#)) uses abstract labels A and B to designate the two subdivisions of Eastern Chadic; I prefer the geographic labels Chari-Logone and Guéra, respectively, to refer to these two subgroups.

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all surprising for Chadic languages, because of their predilection for consonants in basic lexeme structure.

2 Lexical tone

First of all, it must be noted that throughout all four branches of the Chadic family, the functional load of lexical tone is very low. Essentially all Chadic languages that have a practical orthography have been written successfully with no indication of tone. Hausa is the best-known example, of course. But as linguistically-informed orthographies for other Chadic languages have been developed in recent years, the conclusion is always the same: there is no need to mark lexical tone in general, for no significant confusion results from this omission. As might be expected, minimal pairs for tone are often exceedingly rare. African languages whose lexical roots are very short, having only one or two syllables, may rely heavily on tone (in addition to the segmental material) to make lexical distinctions between words. But Chadic languages often involve three, four, or possibly more consonants in their lexical roots, which may extend beyond two syllables; in such cases, the dependence on tone for distinguishing between lexemes becomes much less. A survey made by Schuh (2017: 90) of five Western and Central Chadic languages showed that, on the average, the word pairs distinguished only by pitch was only about 2% of the whole lexicon.

Most Chadic languages have only two contrastive levels, High (H) and Low (L), and this of course would be the minimum for a language to be considered tonal. There are languages in all the main branches of Chadic that have two-tone systems: Hausa, Bole, Goemai (West Chadic); Gude, Podoko, Margi (Central Chadic); Lele, Bidiya, Mukulu (East Chadic). In languages which allow for heavy syllables, H and L may combine to form falling and rising contours, with the falling tone by far the most frequent. Some languages prohibit a rising tone altogether, allowing only falls (e.g. Hausa, West branch; Mukulu, East branch).

Despite the absence of minimal pairs, nouns and verbs in most languages can be sorted into groups which display a small number of distinctive tone patterns. Witness first the data from Kirya (Western branch), taken from Blench & Ndam-sai (2009). With two underlying tones, four tone patterns are attested in two-syllable nouns, displayed in Table 3.

A similar display can be made for Mawa (Eastern branch) in Table 4, where exactly four tone patterns are found in monomorphemic nouns, regardless of lexeme shape (Roberts 2013; M = mid tone).

Throughout the Chadic family, the lexical tone patterns among verbs are much more restricted than for nouns. In fact, there are even languages in which verbs

Table 3: Tone patterns in Kirya nouns

	LL	LH	HH	HL
CV.CV	<i>pàlà</i> 'hand'	<i>fàfà</i> 'bush'	<i>péíí</i> 'stone'	<i>páàà</i> 'rock'
CVC.CV	<i>gàrmà</i> 'large hoe'	<i>tàrpá</i> 'rib'	<i>húlfá</i> 'relation'	<i>málkà</i> 'reward'
CV.CVC ^a	<i>tàyàl</i> 'chaff'	<i>xùláy</i> 'chameleon'	<i>mbáláy</i> 'tamarind'	<i>fímál</i> 'crossbar'

^aThis structure is not common in Kirya; most words end in a vowel.

Table 4: Tone patterns in Mawa nouns

	LL	LM	MM	MH
CV.CV	<i>pàlà</i> 'giraffe'	<i>kùlū</i> 'knot'	<i>kālī</i> 'jaw'	<i>kāsú</i> 'leper'
CVV.CV	<i>jòòjò</i> 'hump'	<i>gòòlō</i> 'sorrel'	–	<i>māālā</i> 'animal'
CVC.CV	<i>gàrpà</i> 'blacksmith'	<i>pùrkā</i> 'bark'	<i>kōllē</i> 'pot'	<i>īndí</i> 'scorpion'
CV.CVC	<i>sògòn</i> 'dream'	–	–	<i>jībín</i> 'enemy'
CVV.CVC	–	–	<i>lōōcōm</i> 'flea'	–
CVC.CVC	<i>tìndìr</i> 'ant'	<i>wàlwāc</i> 'jackal'	<i>bōngōm</i> 'baboon'	<i>mūrgám</i> 'red millet'

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have no lexical tone at all; surface tone of verbs is determined in part or in whole by the grammatical tone associated with tense-aspect marking (e.g. Musey from the Masa branch; see Section 3) and/or by the effect of consonants (e.g. Kera from the Eastern branch; see Section 4). Schuh (1971: 34–36) indicates that the tone of a Ngizim verb (Western branch) is predictable from the quality of the vowel in its initial syllable. In Karekare (Western branch), although there are no lexical tone restrictions on nouns, verb tone is predictable: polysyllabic verbs all take a LH pattern, and monosyllabic verbs take H tone. Nonetheless, a number of languages do make lexical distinctions of tone in verbs, e.g. Miya (Western branch, Schuh 1998: 40, 84), Margi (Central branch, Schuh 2017: 94), Mukulu (Eastern branch, Jungraithmayr 1990: 20).

There is also a number of languages throughout the Chadic family which have three distinctive levels, although they are fewer in number. These include Gwandara, Angas, Ron-Fyer (for the Western branch); Daba, Muyang, Zulgo (for the Central branch); and Kera, Lele, Somrai (for the Eastern branch). Interestingly, all the languages of the small Masa branch have three tones today, but they are also characterized by consonant-tone interaction (see Section 4), which may account for the emergence of the third level. Geographically close to them are the East Chadic languages of the Chari-Logone group, and almost all of these have three-tone systems too. In the Guéra group of East Chadic, though, two-tone systems are the rule.

It is most likely that the three-tone systems of today developed from simpler two-tone systems, by at least two paths. The first path is evidenced in the Masa branch, whose languages are all affected by the phenomenon of consonant-tone interaction (see Section 4 below), in which consonant quality is actually able to raise or lower the pitch of its syllable. When the lowering or raising does not effect an absolute switch from H to L or L to H, an intermediate level is created, which through time becomes phonologized into a Mid tone. Musey is one of the languages of this group, and those who have analyzed the language (Duncanson 1972, Shryock 1995, Soulokadi 2024) all recognize three underlying levels. The three tones further combine into a number of tone patterns on lexemes which, although robust, have certain restrictions due to the effect of the initial consonant (Soulokadi 2021). Aaron Shryock (p.c.) reports that both he, and Duncanson before him, had the feeling that the system really has only two underlying levels, but that the analytical means of showing that have eluded us up to the present. As a result, three underlying tone levels must be recognized for the language. From the Western branch, Schuh (2017) cites Zaar as a language which has clearly evolved from a two-tone to a three-tone system because of consonant-induced modifications.

There is another means of splitting two tones into three, via the interplay of grammatical tone (see Section 3 below) with lexical tone. In the Mawa data of Table 4, it can be shown that the phonetic H tone has a very restricted distribution (Roberts 2013). The two tones that contrast robustly are those labeled L and M. The H tone arises predictably in one of two ways. First, in order to “repair” the illicit sequence *ML, the H becomes the allophone of L tone after a M, as shown in the rule of (1):

(2) $L \rightarrow H / M _$

The other (independent) occurrence of H is in grammatical morphemes, such as the irrealis auxiliary *kár*. As a result, a revised understanding of the system recognizes two underlying tones in contrast, henceforth labeled H (formerly M) and L. In addition, there is a super-High tone (formerly simply H) which occurs in restricted contexts and is marginal. In time, the extra level may phonologize to the extent that the language ends up with a robust three-tone system.

Given the low functional load of tone in Chadic languages in general, I conclude that the complexity of tone systems in Chadic languages is limited to 2-tone and 3-tone systems. The seeming claim for as many as five distinctive tone levels in Mushere (West branch) in Jungraithmayr & Diyakal (2013) is surely exaggerated. A close examination of the data in that article shows that the language has no more than three underlying levels; the authors themselves admit (p. 299) that only three levels are basic. The additional two levels, while perhaps distinct phonetically, arise predictably in restricted contexts which are heavily dependent on grammatical factors, e.g. in the locative forms of nouns, or in the genitive construction.

If Chadic languages do not depend heavily on tone, one may well ask: Is tone actually predictable in some languages? The answer seems to be a qualified “yes,” although tone usually retains a distinctive character in some corner of such languages, most often within the nominal system. In Malgwa (Central branch), Löhr (2002: 78) reports that most words have two syllables, and that they almost all take the tone pattern HL; other patterns are quite rare. Further, when affixes are added, the H tone shifts in a predictable fashion to the penultimate syllable. As to verbal systems, Newman (1972) shows that Bolanci verbs (Western branch) will always have High tone on the second syllable; the tone of the first syllable will also be H if it is light (of shape CV), but will be L if it is heavy (of shape CVV or CVC). For Lagwan (Central branch), Ruff (2007: 2) states that “tone on the overwhelming majority of verbs is predictable...it is determined by default rules, dependent on syllable structure.” A final example is Migaama verbs (Eastern branch). The tone pattern of “long” verbs (with three, four, or five moras)

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is always predictable. These verbs all contain a heavy syllable, and H tone will always fall on it; before and after the H, tone is L (Roberts 2005). In “short” verbs (with one or two moras), the tone patterns L, H, and HL are still contrastive, however.

A final question often asked about tone languages concerns the unit that bears the tone. Since tone is a suprasegmental feature, it is of necessity associated with a larger unit than the vowel, even though traditional transcriptions mark tone on vowels. In most languages the individual tone unit is borne either by the (surface) syllable or the mora. Chadic languages are not uniform in their handling of this question. Rising and falling tones, which are actually LH and HL sequences, are only allowed in Chadic on heavy syllables, that is, syllables composed of two moras. For languages that allow these contours, the tone-bearing unit must be the mora, so that each element of the contour is associated with its own mora. Languages like this would be Ngamo (Western branch, Schuh 2017: 133) or Mukulu (Eastern branch, Jungraithmayr 1990: 20, Roberts 2021). In other languages which do not allow contours, the tone-bearing unit (TBU) is probably simply the syllable, and this may be a factor that inhibits the development of contours. The case of Mawa (Eastern branch, Roberts 2013) is instructive in this regard, for it allows syllables of shapes CV, CVV, CVC, and CVVC. Yet, regardless of complexity, each syllable may take only one level tone, as seen in (3), and no contours are allowed:

- (3) L tone: *bòo* ‘person’, *pìs* ‘horse’, *tùuk* ‘skin’
 M tone: *dāa* ‘compound’, *būt* ‘mouth’, *dūus* ‘knife’

3 Grammatical tone

If lexical tone does not enjoy a place of privilege in the analysis of Chadic languages, it must be admitted that distinctive tone is deployed again and again in Chadic languages to effect grammatical distinctions. The most recurring cases of grammatical tone throughout Chadic affect the verbs, specifically in the tense-aspect-mood (TAM) system. In a number of languages, the verb has no lexical tone of its own at all, and its surface tone is completely determined by the tonal marking of TAM.

In Daffo (Western branch) the perfective aspect is differentiated from the imperfective aspect by tone (Jungraithmayr 2009), as shown in the examples of (4):

- (4) a. *à mōt* ‘he died’ (perfective) *à mwāāt* ‘he dies’ (imperfective)
 b. *à cūh* ‘he ate’ (perfective) *à cwāáh* ‘he eats’ (imperfective)

In the forms of (4), there is an internal vowel difference as well as the tonal difference between Mid tone (perfective) and Mid-High tone (imperfective). According to Junggraithmayr (2009: 211), the distinction between the aspects in Chadic was originally carried by the vowels alone, but is eventually taken over by tone in the diachronic development, while at the same time the vowels lose their distinctive character. In that case Daffo must display an intermediate stage in this evolution, since the aspectual distinction is still manifested by both vowels and tone.

An influential paper by Newman (1975), drawing on data from Central and Western Chadic languages, proposes that a lexical High-Low tone contrast was a principal factor in establishing Proto-Chadic verb classes. In a footnote, however, he says that “whether Proto-Chadic verbs manifested distinctive tone is far from clear.” An examination of the means of marking TAM distinctions throughout Chadic today, however, reveals that grammatical tone plays a role at least as important as any lexical tone distinction in the verbal system.

In all languages of the Masa branch, the distinctions of TAM in the verb are effected by a change of tone. In Musey, it can even be said that the verb has no inherent tone of its own. Tone alone signals the difference between the perfective and imperfective forms of the verb (Shryock 1995, Soulokadi 2024), but is complicated by the tonal interaction triggered by the initial consonant (see the following Section 4). For verbs that begin with a depressor consonant, the perfective form takes High tone on the first syllable, and the imperfective takes Mid tone. For verbs that begin with all other consonants, the perfective form has Mid tone, and the imperfective has High tone (on the first syllable). As a result, it can be said that all verb roots in this language are toneless, and the surface tone on a verb is the result of the interplay of the grammatical tone of the aspect (perfective or imperfective) with the verb’s initial consonant (depressor or non-depressor). In the irrealis mood (used in imperative, conditional, etc.) another grammatical tone comes into play; it is a Low tone which overrides all other factors. But verb tone is everywhere predictable in this language, according to the grammatical marking. Examples are shown in Table 5.

Gravina (2010: 225) states that in Mbuko (Central branch) “verb tone is based on grammatical function,” leaving the verb roots without lexical tone. In Muyang (Smith 2010), also from the Central branch, grammatical tone determines the surface tone of the complete verbal complex, which includes the subject person markers, the verb root, and a few other affixes. The grammatical tone will be lowered by depressor consonants, and in some cases will be modified by the structure of the root. When each of the relevant factors is taken into account, the tone on the verbal complex is by and large completely predictable, despite the exceptions

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Table 5: Grammatical tone in the Musey verb

	Infinitive	Imperfective	Perfective	Irrealis	
Initial depressor	<i>zàl-là</i>	<i>zāl</i>	<i>zál</i>	<i>zàl</i>	‘boil’
	<i>vàràk-kà</i>	<i>vārāk</i>	<i>vārāk</i>	<i>vàràk</i>	‘replace’
Initial non-depressor	<i>kāl-lá</i>	<i>kál</i>	<i>kāl</i>	<i>kàl</i>	‘come’
	<i>tūrbūt-tá</i>	<i>tūrbūt</i>	<i>tūrbūt</i>	<i>tūrbūt</i>	‘crumple’

that Smith notes. The forms in Table 6, taken from [Smith \(2010: 125\)](#), show the distinction between imperfective aspect vs. perfective aspect, manifested by the H vs. M tone contrast on the subject marker prefix. An initial depressor consonant lowers the tone on the verb root, otherwise High, to Low. In the subjunctive (together with all other irrealis forms), the tone is L everywhere. (All forms are in the 3rd person singular.)

Table 6: Predictable tone in Muyang verbs

	Imperfective	Perfective	Subjunctive	
Initial non-depressor	<i>á-sák</i>	<i>ā-sák</i>	<i>mà-sàk</i>	‘sift’
Initial depressor	<i>á-bàz</i>	<i>ā-bàz</i>	<i>mà-bàz</i>	‘harvest’

In Mawa (Eastern branch), the lexical tone of the verb, Mid or Low, is distinctive only in one form, the nonpast; consonant quality has no effect on tone. In each of the other forms of the verb, a suffix is added with a grammatical tone that blankets the whole word, replacing the lexical tone. Table 7 shows one verb with lexical Low tone in the nonpast, and another verb with lexical Mid tone. Elsewhere the tone of all verbs is the same, because each verbal form is characterized by a neutralizing grammatical tone ([Roberts 2013](#)).

Grammatical tone usually signals inflectional distinctions. Apart from TAM in the verbal system, it is sometimes found in the nominal system for marking categories such as gender or number.

Newman (1990), in his compilation of plural marking throughout the Chadic family, does not mention tone as a means of marking it, so it is clear that grammatical tone is only marginally used for this purpose. Nonetheless, a few exam-

Table 7: Grammatical tone of verbal forms in Mawa

	Low-toned verb	Tone	Mid-toned verb
nonpast	<i>dìpì</i> ‘pour’ (Low)		<i>dēmī</i> ‘fold’ (Mid)
past	<i>dīp-é</i>	MH	<i>dēm-é</i>
infinitive	<i>dīp-īŋ</i>	LM	<i>dēm-ēŋ</i>
imperative 2SG	<i>dīp</i>	M	<i>dēm</i>
imperative 2PL	<i>dīp-īŋ</i>	M	<i>dēm-ēŋ</i>
participle	<i>dīp-nō</i>	HM	<i>dēm-nō</i>

ples can be found, as shown in (5) for Hausa (Western branch) and Dangaleat (Eastern branch, [Yakoub 2000](#)). It should be emphasized, though, that these seem to be isolated cases, and that tone is not the usual way of marking plurality in these languages.

- (5) a. Hausa
 màatàa (singular) *máatáa* (plural) ‘woman, wife’
 b. Dangaleat
 ásé (singular) *àsé* (plural) ‘leg’

Gender is also marked by tone in a limited way in several languages of the Guéra in Chad (Eastern branch). The examples of (6) from Dangaleat ([Yakoub 2000](#)) are illustrative, but again, these examples do not represent a widespread system of gender marking.

- (6) *támɡá* (masc.) ‘ram’ *tàmɡà* (fem.) ‘ewe’
 kókírà (masc.) ‘cock, rooster’ *kòkírà* (fem.) ‘hen’

A final example shows grammatical tone used in derivational morphology. In Hausa, locative nominals are derived from verbs by the prefix *má-* and suffix *-áa*; the whole derived noun takes a grammatical high tone throughout, regardless of the lexical tone of the root, as seen in the examples of (7):

- (7) *fitá* ‘go out’ → *má-fít-áa* ‘exit’
 kárántáa ‘read’ → *má-káránt-áa* ‘school’

4 Consonant-tone interaction

The Chadic family provides some of the best exemplars of the phenomenon called consonant-tone interaction (cf. [Beavon-Ham & Pearce 2026 \[this volume\]](#)).

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Acoustically, it is found that voiced obstruents have a phonetic lowering effect on fundamental frequency (F_0), i.e. pitch (Hombert 1978; see also **chapter... in this volume**). A number of languages have incorporated this phonetic effect to varying degrees into their phonology. In some cases, the effect has been phonologized to the point of creating tonal contrasts (see Section 2). Languages which display consonant-tone interaction divide the consonant inventory into two or three categories. Voiced obstruents, mentioned above, constitute one group which is often called “depressors” because of their lowering effect on tone. All other consonants may be neutral or non-affecting. However – and this is the case in several Chadic languages – the remaining consonants divide again into two groups. Voiceless obstruents do have the opposite phonetic effect to the depressors, in that they raise pitch, and they may constitute a separate class of “raiser” consonants. In that case, the only neutral consonants, as far as their effect on tone, would be the sonorants. Table 2 (in Section 1) displays Chadic consonants according to this three-way distinction. Most frequently, it is the onset consonant that exercises the influence on tone, and usually it is the initial consonant of the word. There are languages from all four branches of Chadic which manifest consonant-tone interaction.

A simple example of this effect comes from Ngizim, a two-tone language of the Western branch. Schuh (2002: 8) indicates that rising tones in this language are always caused by a depressor consonant in the onset of a High-toned syllable; the rise never comes from an underlying LH sequence. Witness the contrast in (8). In parallel fashion, Falling tones in this language which are not HL sequences are produced when a High-toned syllable precedes a syllable beginning with a depressor. Example (9) compares the High tone before the non-depressor [n] with a Falling tone before the depressor consonant [j]. Significantly, the syllables with the contour tones are all heavy, i.e. containing two moras; in this language the TBU is the mora.

- | | |
|---|---|
| (8) a. Initial non-depressor
<i>páu</i> ‘pour’
<i>táu</i> ‘eat’ | b. Initial depressor
<i>báu</i> ‘get’
<i>dáu</i> ‘remove’ |
| (9) a. Before a non-depressor
<i>ákurnà</i> ‘gruel’ | b. Before a depressor
<i>wúrjì</i> ‘scorpion’ |

A more complex example is from Zaar (Western branch), presented by Schuh (2017: 103ff) with data taken from Schneeberg (1974) and Caron (2005). Two groups of consonants are recognized: the depressors (the voiced obstruents b, d, dz, j, g, v, z, ɣ) on the one hand, and all other consonants on the other. Today

there are three distinctive tone levels in the language, but there are severe restrictions on the tone patterns that occur because of the effect of the depressors. In nouns, the Low tone only occurs on syllables which start with a depressor; otherwise the tone of those syllables will be Mid or High. Examples (10–11) shows how two tone patterns are realized on nouns, depending on the quality of the onset of each syllable. In (10) each syllable begins with a non-depressor, showing the unaffected shape of the two tone patterns, HM and MH; in (11), the onset of each syllable is a depressor consonant, and each of the two preceding patterns is shifted as a whole one level down, to ML and LM, respectively. (If only one of the two syllable onsets is a depressor, the resulting tonal patterns are mixed, and there is some interdialectal variation; see Schuh 2017: 105.)

- (10) Non-depressors in onset

a. Falling tone pattern: HM
nyérām ‘monkey’

b. Rising tone pattern: MH
kārí ‘vervet monkey’
- (11) Depressors in onset

a. Falling tone pattern: ML
būbzàṅ ‘beard’

b. Rising tone pattern: LM
gàbzā ‘ladder’

From the Central branch of Chadic come the well-known examples of Lamang (Wolff 1987: 200), a two-tone language. Syllables beginning with a depressor consonant (voiced obstruent) are systematically Low, and those beginning with non-depressors are High. The examples of Table 8 show that the effect extends to both nouns and verbs³.

Table 8: The effect of depressor consonants in Lamang nouns and verbs

	Noun		Verb	
Initial non-depressor	tlá	‘cow’	tsxúrá	‘sit’
Initial depressor	ghwà	‘mountain’	gùrvà	‘dance’

The effect of consonants on tone is of particular interest in Musey (Masa branch), in that the effect of depressor consonants is passive; that is, these consonants do not actually lower the tone of their syllable. Nonetheless, words which begin with a Low tone always have an initial depressor consonant. On the other

³For conjugated verbal forms, grammatical tone in the TAM system overrides the influence of the initial consonant.

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hand, all other consonants (voiceless obstruents and sonorants) are active raisers. When they occur in initial position of a word which has an underlying Low tone, the first syllable of such words is raised to Mid, before returning to the underlying Low in the following syllable(s). Observe the patterns in Table 9, from Soulokadi 2024. (Note that the raising only affects Low tone; Mid and High are undisturbed.)

Table 9: The effect of raiser consonants in Musey nouns

Surface tone	Initial depressor consonant	Surface tone	Initial raiser consonant
Mid	<i>dēmē</i> ‘sieve’	Mid	<i>sīmī</i> ‘tear’
Low	<i>gàrì</i> ‘guinea fowl’	Mid-Low	<i>sēnē</i> ‘field’

A final example comes from Kera, in the Eastern branch (Pearce 1998). Consonants break into three groups, roughly along the distinctions indicated in Table 2 (Section 1): voiceless obstruents corresponding to raisers, voiced obstruents corresponding to depressors, and sonorants being neutral. The behaviors of the tone of nouns and verbs are governed by different principles. Kera is another Chadic language in which verbs are underlyingly toneless. A verb may have one of three tone patterns which are determined entirely by the nature of its consonants. If it contains no depressors, its tone will be realized with a default High tone. If the verb has a single depressor in initial position, the tone will be Low throughout. And if the verb contains two depressors, the tone pattern will be Low-High, as though the second depressor exercises a polarizing effect on the second syllable. Witness the examples in (12):

- (12) a. No depressor: *mírkí* ‘to greet’
b. Initial depressor: *gùni* ‘to awaken’
c. Two depressors: *zàldé* ‘to hit’

Nouns in Kera, on the other hand, have four distinctive tone patterns, which are further modified by the consonant type of the initial consonant. Here the effect of raiser consonants comes into play. Nouns which begin with a depressor or neutral consonant will have one of the following tone patterns: Low, High, Low-High, High-Low. But nouns which begin with a raiser consonant will systematically raise the Lows of these patterns to a Mid. Table 10 shows the resulting patterns.

Table 10: The effect of raiser consonants in Kera nouns

	Initial C depressor	Initial C neutral		Initial C raiser
L	<i>dòrgòdī</i> ‘granary’	<i>màrlà</i> ‘vulture’	M	<i>sāāmā</i> ‘rope’
H	<i>bályákí</i> ‘bird (sp.)’	<i>wálká</i> ‘fear’	H	<i>párkí</i> ‘mountain’
LH	<i>gùùgúr</i> ‘chicken’	<i>mòòmó</i> ‘grandfather’	MH	<i>tālāṅká</i> ‘hole’
HL	<i>zógòy</i> ‘exorcism’	<i>lógòm</i> ‘humidity’	HM	<i>kítír</i> ‘moon’

As a result, it is shown that depressor consonants are active in the verbal system of Kera, while raiser consonants are active in the nominal system. The remaining neutral consonants have no effect.

The phenomenon of consonant-tone interaction recurs with regularity in the description of Chadic languages, and often combines with other tonal phenomena, so that surface tone may be the result of a complex of factors. For example, verb tone in Musey (Section 3) results from the interplay of grammatical tone with the influence of word-initial consonants. In Bole, the spreading of High tone (cf. Section 5.2) will be blocked if it encounters a depressor consonant. The discussion in the following sections will continue to demonstrate consonant quality playing a role in other tone phenomena too.

5 Other tone phenomena

There are other phenomena that have been cited in the Chadic literature which speak of tonal processes similar to those in other African languages. Some, such as tonal polarity, are widespread throughout the Chadic family, but others, such as downstep, seem to be exceptional for a Chadic language.

5.1 Tonal polarity

Polar tone characterizes specific grammatical morphemes, and examples of it have been adduced from all branches of the Chadic family (cf. Pulleyblank 2026 [this volume]). It is interesting that this phenomenon, while it occurs widely through African languages, does not fit easily within current theories of tone, because it causes opposite effects depending on context. Insofar as its function is to contrast or highlight a morpheme with respect to its neighbor, tonal polarity seems as much a characteristic of a stress system as that of a purely tonal system.

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Our first example comes from the copula *nee* of Hausa (Western branch; Schuh 2017: 132). This morpheme always takes the opposite tone from the tone that precedes it, regardless of the lexemes in the construction, as demonstrated by the examples of (13):

- (13) a. *dóókìì néé* ‘it’s a horse’
 b. *bàà dóókìì bá nèè* ‘it’s not a horse’

Ruff (2007: 4) demonstrates that the 1st person singular subject pronoun *wu* in Lagwan (Central branch) exhibits polar behavior with respect to the initial tone of the verb which follows. The relevant examples are shown here in (14):

- (14) a. *wù bân* ‘I bathe’
 b. *wú tkàm* ‘I measure’

The final examples come from the definite markers of Mawa (Eastern branch; Roberts 2013). There are three, differentiated by gender and number, but they all demonstrate the same polar behavior with respect to tone: *di* (masculine), *ti* (feminine), and *kɛ* (plural). If the syllable that precedes is L, the tone of the marker will be M. If the syllable that precedes is M, the marker will take H tone, which is the allotone of L after M according to the rule of (2) above. It follows, then, that after a syllable with H tone, the marker will again take M tone, and the polarity of M vs. underlying L is respected in each case; witness the examples of (15):

- (15) a. *pìs dī* ‘(the) horse’
 b. *kūy dí* ‘(the) dog’
 c. *āmí dī* ‘(the) water’

5.2 Tone spreading

Chadic languages exhibit a limited amount of tone spreading, but the nature of the spreading is often different from that of other African languages. According to the typology of tone systems proposed by Kutsch Lojenga (2014), Chadic languages are generally of the “stable” type, i.e. tones are not changed by their tonal environment, and do not often spread beyond their point of origin (cf. Smolders 2026 [this volume]). Nonetheless, a couple of examples of tone spreading are noted here, along with their peculiarities.

In Bole (Western branch), H tone spreads forward one syllable in certain grammatical constructions, but is sensitive to consonant types (Schuh 2017: 128). Spreading takes place when the onset consonant of the following syllable is a

voiceless obstruent, an implosive, or a nasal, but the spreading is blocked when the following consonant is a voiced obstruent, as shown in (16).

- (16) a. *ló tēmshí* → *ló tēmshí* ‘meat of sheep’ (spreading takes place)
 b. *ló bìdò* → *ló bìdò* ‘meat of monkey’ (no spreading)

The remarkable behavior to note in (16a) is that the Low which is dislodged by spreading is simply lost; it does not manifest tonal stability according to the definition of Goldsmith (1976)⁴, wherein the dislodged tone would remain as a floating tone and might trigger downstep of the following High tone.

A common reason for spreading is to provide toneless morphemes with a surface tone, and Chadic languages do exhibit this behavior. In Musey (Masa branch), L and M will spread forward onto toneless clitics. In the same context, however, the H tone does not spread. Instead, the toneless morpheme receives a default Mid tone (Soulokadi & Roberts 2022). This behavior can be observed in (17) for the toneless enclitic =*na*, which marks the masculine gender and definiteness of the preceding noun:

- (17) a. Low: *bùù=na* → *bùù=nà* ‘day’ (spreading)
 b. Mid: *wīī=na* → *wīī=nā* ‘pregnancy’ (spreading)
 c. High: *kée=na* → *kée=nā* ‘belt’ (no spreading)

This behavior contrasts with many other languages in which the H tone is the one tone which is most susceptible to spreading. In Musey the H tone is the one tone which does not spread.⁵

5.3 Floating tones and floating tone morphemes

A floating tone can be one of four types: (a) a tone which forms part of the tone pattern of a lexeme, but is not associated with a TBU for whatever reason; (b) a tone which has been dislodged from its association to a TBU by the spreading of another tone or the loss of its segmental support; (c) a tone which constitutes a grammatical morpheme in itself, and is affixed onto a lexeme; and (d) a tone which constitutes a morpheme in itself, but is syntactically separate, having the

⁴Frajzyngier (2003: 84, 87) notes two other cases of a lack of tonal stability, although spreading is not involved. In Mina (Central branch) and Lele (Eastern branch), when certain vowels are deleted, the tone they bear is deleted too.

⁵If the function of High tone is to highlight one syllable with respect to its neighbors (see Section 6.2), it should be noted that this function would be compromised if the High were allowed to spread over several syllables.

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status of a clitic in the grammatical structure, rather than that of an affix (cf. [Rolle et al. 2026 \[this volume\]](#)).

According to this taxonomy, it is difficult to find examples of (a) and (b) within the Chadic family. Floating tones of type (a) are usually posited as a construct of the analyst, and the data for which their existence has been claimed may be open to alternative analyses.⁶ The dislodging of a tone which characterizes type (b) has been claimed in a limited number of languages, mostly of the Western branch (e.g. Kanakuru, cf. [Newman 1974](#)). However, such cases seem not to be widespread throughout the family, and in fact some of the floating tones in question may better be treated as grammatical tones of type (c) wherein no dislodging need be recognized.

Type (d) in the taxonomy would include boundary tones and particles with phrase-level or clause-level function. Here also it is difficult to find good examples in the literature on Chadic languages, and potential examples of this type of floating tone might also be susceptible to other analyses.

Floating tones of type (c), on the other hand, are largely indistinguishable from grammatical tone markers, and these are the floating tones most commonly found in Chadic languages. For example, the irrealis Low tone in Musey (see Section 3) is a floating tone which should be considered a superfix attaching to the toneless lexical verb root. [Schuh \(2017: 115–116\)](#) says that the aorist in Zaar (Western branch) is a floating High tone which attaches to the subject pronoun; this also is a floating tone of type (c). Another example from the Western branch is Ngamo ([Schuh 2017: 142](#)), where the genitive morpheme is a floating High tone which attaches as a prefix to the second noun in the associative construction, as in (18):

(18) *tìlì* ‘heart’ + *kòorò* ‘donkey’ → *tìlì kóorò* ‘heart of donkey’

Schuh adds that there is no evidence that this morpheme ever had any segmental part that has been lost, so that this could not be a floating tone of type (b) or (d).⁷

⁶A potential example would be the floating L tone posited in the tone pattern H(L) for one group of verbs in Miya ([Schuh 1998: 55](#)). Because of the complex set of tone rules that Schuh proposes for the language, and because of some exceptional behavior of the floating tone (indicated on pp. 58, 61), it is difficult to evaluate his claim.

⁷Contrasting with this situation is the genitive morpheme in a number of Niger-Congo languages, which was once a distinct particle containing both segments and tones, but whose segmental content has fallen away, leaving us with a floating tone clitic.

5.4 Downdrift and downstep

Downdrift (i.e., automatic downstep) and downstep (i.e., non-automatic downstep) have been reported for certain languages of the Western and Central branches, but these phenomena do not seem to be typical of Chadic languages in general (cf. [Snider 2026 \[this volume\]](#)). A slight automatic lowering of High tones throughout the phrase is sometimes attested, perhaps more than what would be expected in declination, but it is not usually perceived, and does not take on any phonological significance. Schuh (2002) speaks of High-Low-High sequences in Bade-Ngizim, for example, where the second High tone is phonetically lower than the first (i.e. downdrift). He also finds in these languages that the suppression of a Low tone between two Highs can trigger downstep, adding (2002: 12) that although this is a natural source for downstep in many languages, Chadic languages do not commonly manifest it. He also allows (2002: 13) that the auditory impression of downstep might actually be the realization of decreased intensity, rather than a true tonal downstepping of the pitch.

Schuh's grammar of Miya (Western branch) uses the term downstep (1998: 37) to describe a lowering in pitch between successive High tones within the phrase, and even within words. This lowering might be viewed as a strategy to repair a violation of the Obligatory Contour Principle, which prohibits adjacent identical elements such as the successive High tones here. However, it is not clear that this predictable phenomenon creates additional phonologically significant levels of tone. Schuh's example shows six downsteps in a phrase of five phonological words, and he indicates that the downstepping tends to flatten out after the first few instances. The remarkable number of downsteps that are claimed to be possible within a short space makes one wonder if they actually represent a type of declination.

Podoko (Central branch) also demonstrates the phenomenon of downstep ([Anderson & Swackhamer 1981](#)), but downstep of a different nature from that found in most African languages. In Podoko, it is the Low tone (not High tone) which is downstepped after a High, and the downstep is triggered by depressor consonants rather than by a tonal configuration. Again the nature of the consonants is the factor that controls the occurrence of downstep, rather than simply the interaction of tones among themselves. The examples of (19), from [Anderson & Swackhamer \(1981: 141\)](#), show two Low-toned nouns and how they are realized after the High tone of the associative construction, which falls on the first syllable of the second noun. In (19a) the noun has no depressor consonants, and the second syllable has Low tone at the normal level. In (19b), however, the second syllable begins with a depressor consonant, and its Low is downstepped to an

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even lower level. (The down arrow (↓) indicates the position of the downstep.)

- (19) a. *hwàdɛ́ tákà* ‘inside of thorn’
b. *hwàdɛ́ ná↓và* ‘inside of body’

Downstep has also been reported sporadically in languages of the Western branch (Kanakuru, P. Newman 1974) and in the Central branch (Ga’anda, R. Newman 1971; Lagwan, Ruff 2007). I am not aware of any claims of downstep among languages of the Eastern or Masa branches, however. To the extent that down-drifting (especially of High tone) is a universal tendency (Schuh 2017: 118), the phenomenon can be observed to a limited extent in a number of Chadic languages. But downstep is simply not a phenomenon that characterizes Chadic tone systems in general. A number of descriptions of Chadic languages contain statements similar to that of Richard Gravina (2001: 7) for the language of the Central branch that he has studied extensively: “There is no downstep or down-drift in Mbuko.”

6 Atypical behavior of tone in Chadic languages

6.1 Tone reversal

Jacques Fédry reports (1974, 1977) on a very curious situation among the dialects of Dangaléat (Eastern branch) which does not find a satisfying tonal explanation. The tone patterns of words in the eastern dialect of that language are systematically flipped for the same words in the western dialect, as seen in Table 11.

Table 11: Reversed tone patterns in Dangaleat dialects

Eastern dialect		Western dialect		
HH	<i>pátó</i>	LL	<i>pàtò</i>	‘sun’
MM	<i>gūrō</i>	LL	<i>gùrò</i>	‘caterpillar’
LM	<i>dùwō</i>	HL	<i>díwò</i>	‘fly’
LL	<i>gàrpò</i>	HL	<i>gárpò</i>	‘blacksmith’

An added peculiarity of the eastern dialect is that a third Mid tone has developed, although the western dialect has only two tones, High and Low. In the correspondences of Table 11, the Mid tone in the eastern dialect has the same value as the High tone. Peust (2017: 206) indicates that this allows the three-tone

system of the east to be considered as having an underlying two-tone system like the western dialect. It is also interesting to note that depressor consonants also play a role in the realization of the surface tones. Peust then attempts to explain the synchronic inversion by positing a diachronic tone shift which happened morpheme-internally in the western dialect, similar to the diachronic tone shift reported on by Schuh (2017: 133ff) for Ngamo (Western branch). Whether or not this explanation turns out to be an accurate account of the Dangaleat data, there are other documented cases of tone reversal in Chadic languages, and it remains to be seen whether a diachronic tone shift is plausible to explain each one.

Fédry (1977) shows similar cases of tone reversal in comparing data from Dangaleat with two of its neighbors, Bidiya and Migaama, all languages from Eastern branch. Elsewhere, tone reversal has been reported in the Zimé cluster in the Masa branch (Wolff 1985), for Kotoko varieties in the Central branch (Tourneux 2007) and for Geji dialects in the Western branch (Caron 2013).

One more example of tone reversal patterns is presented in this section, taken from two Kwang dialects (Jungraithmayr & Peust 2023: 34–37), and displayed in Table 12.

Table 12: Reversed tone patterns in Kwang dialects

Mobu dialect		Ngam dialect		
H	<i>kúwáár</i>	LM	<i>kòwāār</i>	‘blood’
M	<i>cīn</i>	H	<i>cín</i>	‘thirst’
L	<i>kùlù</i>	HM	<i>kúlū</i>	‘house’
L	<i>sàbáy</i>	MH	<i>sūpáy</i>	‘three’
HL	<i>gódò</i>	LH	<i>gòtó</i>	‘rat’
LH	<i>sèní</i>	HM	<i>séénī</i>	‘brother’

Reversals of tone such as these do not arise through normal tonal processes, but it is interesting that several examples of it have been documented among Chadic languages. In another search for an explanation of the reversal phenomenon, Wolff (1985) suggests that the difference is due to consonant-tone interaction, and also to the effect of accent. Although consonant-tone interaction may elucidate certain items in Dangaleat, it is not clear all cases of tone inversion are due to the effects of consonants. Normally the effect of a consonant on tone is limited to the syllable that contains the consonant, and would not cause a wholesale perturbation of the tone pattern of the word.

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The effect of accent provides another explanation, if it is assumed that Chadic languages were originally accentual, and not tonal. This view is espoused in part by both [Fédry \(1977\)](#) and [Tourneux \(2007\)](#), but they both go on to opine that sociolinguistic factors may also be involved: tone reversal, however it came about, allowed each of the dialects to achieve a degree of distinctiveness without making great changes to the grammar. Insofar as there is something useful about the accentual explanation for tone reversal (see also [Section 7](#) below), it should be noted that two-tone systems are by nature polar. One of the two tones can be used to highlight certain syllables, and the neighboring syllables bearing the other tone will be downplayed. If one speech variety chooses High tone to do the highlighting, the neighboring variety might use Low tone for the same purpose. In that way the same syllables would be highlighted in both varieties, but the means of doing so is simply switched.

6.2 Metrical structure

The structural organization of words into syllables and feet is a foremost consideration for the placement of stress in stress/accent systems, but is not normally relevant to tone languages. Nonetheless, there are recurring references in the literature to the intimate relationship between tone and metrical structure in a number of Chadic languages.

[Newman \(1972\)](#), in an early article, drew attention to the relevance of metrical structure (in specific, syllable weight) to phenomena in Chadic languages, including tone. In a later article ([Newman 1981](#)), he focuses on the relevance of syllable weight in determining tone in Bole and Hausa (Western branch). The concluding remarks of the article (1981: 73) imply that syllable weight is not a “normal” factor for characterizing tone languages, despite the fact that he has demonstrated that it is relevant in Chadic.

Along the same lines, [Barreteau \(1983: 309–311\)](#) speaks of accent rather than tone in describing the verbs of Mofu-Gudur (Central branch). The accent is realized on a higher pitch (“High” tone), and there is never more than one High within a verb, similar to an accentual system which does not allow two primary accents within a word. [Roberts \(1999, 2021\)](#) describes the metrical structure of Mukulu verbs (Eastern branch) in a similar way, in which “accent” (realized as High pitch/tone) must fall on either the first, the second, or the third mora of the verb. Each verb must have at least one accent within it; some verbs have two accents, but in that case they are always separated by an unaccented mora. Also, the second of the two accents is often realized on a higher pitch than the first, as though there were a distinction of primary vs. secondary accent.

A more recent argument for including metrical structure in Chadic tone descriptions comes from [Pearce's \(2006, 2013\)](#) work on Kera (Eastern branch). To account for tone in that language, Pearce is led to establish the metrical foot and iambicity as key factors. Kera words and phrases are organized into feet, and the foot as well as the syllable is the unit that bears tone. Table 13 shows the tone patterns on a sample of Kera nouns with three syllables, taken from [Pearce \(2013: 141\)](#). The words are organized into metric feet (shown within parentheses), according to the language's principles of foot construction. Whenever a foot is formed, it takes only one tone for the whole foot.

Table 13: Tone patterns on three-syllable Kera nouns

Tone pattern	One foot	Two feet
LL	(<i>dìmìì</i>) <i>mì</i> 'clothes'	(<i>bòm</i>)(<i>bòròṅ</i>) 'carp'
LH	(<i>gàdàa</i>) <i>mó</i> 'horse'	(<i>dàk</i>)(<i>táláw</i>) 'bird (sp.)'
MH	(<i>tīlīṅ</i>) <i>kí</i> 'hole'	(<i>tāa</i>)(<i>mókáa</i>) 'sheep'
HM	(<i>kúbúr</i>) <i>sī</i> 'burning coal'	(<i>sáa</i>)(<i>tārāw</i>) 'cat'

Of course, the foot, and specifically the iambic foot, was originally recognized as relevant to stress/accent systems, where strong-weak relationships characterize the alternating patterns of prominence accorded to syllables in sequence. However, such units and concepts are not to be found in accounts of tone systems. In fact, [Hyman \(2009: 215\)](#) indicates that the relevance of metrical structure is a characteristic of stress systems, not of tone systems, even though he reckons that tone systems may allow some metrical considerations.

6.3 Anomalies arising through affixation

In Mawa (Eastern branch) there is another perplexing tone change which is triggered by the addition of the oblique suffix *-o* onto nouns ([Roberts 2013](#)). The changes are completely regular, but once again, they defy description within a tonal framework. To understand the tone patterns in this case, a distinction must be made between the result of suffixation for nouns ending with a closed syllable (i.e. with a consonant) as compared to nouns ending with an open syllable (i.e. with a vowel). So once again, the overall structure of the word comes into play.

For closed-structure nouns, the oblique form takes one of two tone patterns: M or LM. If the tone pattern of the base noun is L throughout, its oblique form will take the pattern M throughout. If the tone pattern of the base noun has any

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other pattern, the corresponding pattern in the oblique form will be LM. (Refer to Table 4 to see the range of tone patterns for the base forms.) These patterns are presented in Table 14.

Table 14: Tone of oblique form of Mawa nouns ending in a consonant

	Base form		Oblique form	
L	<i>pìs</i>	‘horse’	<i>pīsō</i>	M
	<i>gàwàr</i>	‘plant (sp.)’	<i>gāwārō</i>	
	<i>kòndòlòm</i>	‘recipient’	<i>kōndōlōmō</i>	
M	<i>kūy</i>	‘dog’	<i>kùyō</i>	LM
MH	<i>guātál</i>	‘calf of leg’	<i>guātālō</i>	
LM	<i>dìbilīṇ</i>	‘worm’	<i>dìbilīṇō</i>	

For nouns that end in a vowel, an epenthetic *-r-* is inserted before the suffix *-o*, in order to avoid the contact between two vowels at the morpheme boundary. Again, the tone of the oblique form will take one of two patterns: MH or LMH. If the tone of the base noun is simple (all L or all M), the oblique form will have the pattern MH. But if the base tone pattern is complex (LM or MH), the oblique form will have the pattern LMH. See Table 15.

Table 15: Tone of oblique form of Mawa nouns ending in a vowel

	Base form		Oblique form	
L	<i>àcò</i>	‘tamarind’	<i>ācōró</i>	MH
	<i>kuàdàmà</i>	‘desert’	<i>kuādāmāró</i>	
M	<i>buaā</i>	‘grandmother’	<i>buaāró</i>	
	<i>kūlgūtū</i>	‘turtledove’	<i>kūlgūtūrú</i>	
LM	<i>gārā</i>	‘basket’	<i>gārāró</i>	LMH
	<i>mòdògō</i>	‘yeast’	<i>mòdògōró</i>	
MH	<i>jēté</i>	‘big basket’	<i>jètēró</i>	
	<i>lōkōmbó</i>	‘camel’	<i>lòkòmbōró</i>	

Roberts (2013) attempts a nontonal explanation of these patterns, suggesting that the principles common to stress systems may play a role here. He proposes that the Mid tone in these words represents the presence of a type of “accent,”

and that the suffixation of *-o* introduces an additional “accent” onto the word. The resulting interaction of the “accents,” along with syllable structure considerations, triggers readjustments of the tonal shape of the word. When the noun root ends in a consonant, the Mid tone is directly realized on the final syllable of the suffixed form; but when the root ends in a vowel, the Mid tone is realized on the penultimate syllable of the oblique.

In the patterns of these changes, the structure of the base form plays a critical role in determining the surface tone of the oblique form, in addition to the original tonal shape of the noun. Structural considerations are also relevant to the analysis of tone in other Chadic languages. The case of Bolanci has already been mentioned, in which the weight of the initial syllable of a word determines its tone; similarly, Pearce’s (2006) analysis of Kera tone relies critically on metrical structure, in her case foot structure.

7 Tonogenesis

Earlier in this article, I mention evidence that certain Chadic languages have evolved diachronically from one type of tonal system to another, e.g. from one with 2 tones to one with 3, through the effect of consonant-tone interaction, or else the operation of grammatical tone. But that is altogether a different type of development from the question of tonogenesis, that is, the emergence of distinctive tonal contrasts in a language that previously had none.

This article began with a mention of the hypothesis of language contact having been the catalyst for tonogenesis in the Chadic family. Today there is contact between Chadic languages and the tonal languages of the Niger-Congo phylum in Nigeria, certainly. However, as one moves eastward within the Chadic zone to where the Central, Eastern, and Masa branches of the family are spoken, there are far fewer tonal languages in contact with them. In Cameroon and Chad, the principal contact languages are Fulfulde and Arabic, neither of which is tonal!⁸ There is also some contact with tonal languages of the Adamawa group, and the Nilo-Saharan Sara group, but of a much more marginal nature. Of course, it is possible that the dynamics of population settlements, movements, and interactions was much different centuries ago, when the Chadic languages became tonal. But

⁸See also Hyman & Shuh (1974: 83) who say: “It is entirely possible that the Chadic languages owe their tonal nature to long contact with Niger-Congo languages. However, the examples of tonal processes we shall cite have evolved independently in most cases in Chadic languages having no present contact with Niger-Congo languages... [and] cannot therefore be directly attributed to contact phenomena.”

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to suppose that Proto-Chadic became tonal at one point in far-distant history, under very different societal conditions from those observed today, seems at best speculative, given the paucity of concrete information available on these matters. Current scholarship (Dimmendaal 2019, Ehret 2000: 292) indicates that the proto-Chadic peoples came from northeast Africa, rendering doubtful the hypothesis of its tone coming from contact with Niger-Congo languages. If tone in Chadic extends far back in time, a more plausible contact scenario for tonogenesis would be the influence of tonal Nilo-Saharan languages of northeast Africa. And these considerations leave completely to the side the mechanism(s) by which language contact might have transferred tonal distinctiveness to Chadic languages, whether at the stage of Proto-Chadic, or more recently.

In any case, Frajzyngier & Shay state (2012: 236) that “we will not have a clear picture of which characteristics have resulted from language contact and which are products of independent language-internal change until we have reconstructed the phonological, morphological, and syntactic characteristics of proto-Chadic, a task that has yet to be accomplished.” In that vein, I prefer to explore reasons that are internal to Chadic languages themselves in order to explain why they should be tonal today, rather than influences due to language contact. Wolff (1987) also attributes tonogenesis in Chadic to language-internal factors. Thus I return to several issues that were examined in earlier sections of the article: (a) consonant-tone interaction; (b) metrical structure and accentual prominence; and (c) grammatical tone.

Yip’s classic work on tone devotes several pages to tonogenesis (Yip 2002: 35–38), but the only trigger for it that she envisages is consonant quality (voicing, aspiration, etc.). Along the same line, Wolff (1987) lays particular emphasis on the well-documented synchronic cases of consonant-tone interaction (see Section 4 of this paper) to explain tonogenesis in Chadic. That very factor, responsible in some Chadic languages for having split two tones into three, might well have caused pitch differences to become phonologically distinctive in the first instance.

While consonant-tone interaction is surely one mechanism by which Chadic languages might have acquired tone, I believe that it is not the only one. There are other language-internal factors that surely have helped to trigger tonogenesis in a number of Chadic languages. When addressing the tonal anomalies of Chadic languages in Section 6, I drew upon similarities between those phenomena and characteristics common to stress/accent systems. I continue, then, by considering issues related to metrical prominence.

Newman opines, very relevantly (2006: 193): “Whether tone can be reconstructed all the way back to PC [Proto-Chadic] or whether PC was characterized

by stress or pitch-accent is an open question.” This statement evokes another phenomenon that must be taken into account in this context, that of stress. Most Afroasiatic languages today (Semitic, Berber, Cushitic) are stress languages, so it is reasonable to believe that Proto-Chadic was a stress language too, as Schuh (2017: 89) also intimates (cf. Green 2026 [this volume]). The most relevant characteristic of stress is syntagmatic prominence, that is, the prosodic highlighting of one unit (syllable) to the detriment of its neighbors in the string of speech. Since the most common manifestation of stress prominence is contrastive pitch, it is a small step to move from contrastive stress to contrastive tone, even though such a tone system would be very restricted initially. For example, if the stressed syllable of a word is systematically realized with higher pitch than the other syllables, a simple tone system is soon born which contrasts High and Low. And voilà, tonogenesis! Kingston (2011) also traces the transition from stress to distinctive tone in certain Scandinavian languages, showing how the pitch peak of F_0 associated with a stressed syllable was reanalyzed as a lexical H tone.

Wolff (1988: 166) also sees the synchronic tones in Chadic as reflexes of a previous accent system. He points out that the accentual hypothesis could also explain another phenomenon, i.e. the ablaut observed in certain inflectional forms. The placement of the accent, and its shift, might be responsible both for changes in pitch/tone and corresponding differences in vowel quality in languages like Mubi (Eastern branch), the case that Wolff cites.

The previous sections have shown that phonological structure is often relevant in elucidating the nature of tone in Chadic languages. In stress accent languages, syllable weight and metrical structure are key elements that determine the placement of stress. If Chadic languages were originally stress languages rather than tone languages, it is not surprising that structural factors continue to play a role in their tone systems today. It also would explain why Chadic languages give the impression of not being fully tonal.

Hyman (2006, 2009) lays special emphasis on two word-level characteristics that stress accent systems all share. The first, obligatoriness, means that each lexical word must have at least one unit marked for metrical prominence. Culminativity is the other characteristic, meaning that each lexical word has at most one unit marked for this prominence. If “accents” are proposed as an alternative to tones, many Chadic languages will meet the culminativity criterion. For example, if High tone/pitch is intended to correspond to the accent, then no lexical word will have more than one High. But many of these same languages fall short with respect to the obligatoriness criterion. They have words with all Low tone, i.e. without any High. As a result, with High tone/pitch as the realizer of the accent,

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these languages would not qualify as having a prototypical accentual system. Another drawback to adopting an accentual analysis for a Chadic language is that the tones are always realized as pitch, and not by any other suprasegmental, i.e. intensity or length.

The final internal factor that may have contributed to the development of tone is one that recurs in the description of many Chadic languages, namely grammatical tone (cf. [Rolle & Bickmore 2026 \[this volume\]](#)). As [Section 4](#) points out, this phenomenon is commonly employed for making distinctions of TAM in the verb. It is possible that these nuances were originally distinguished by the expressive use of pitch (e.g. for deictic distinctions), which eventually became regularized into a system of distinctive grammatical marking. It is not likely that lexical distinctions of tone had their origins in grammatical marking, nor that pitch differences for grammatical marking were responsible in themselves for tonogenesis. Still, pitch differences for grammatical distinctions may have played a role, together with other mechanisms such as the influence of consonant quality or metrical structure, in the transition from a nontonal to a tonal system.

The Chadic literature on tone frequently uses the controversial term “pitch-accent,” a term which is intended to draw attention to the restrictive and peculiar nature of tonal behavior in many Chadic languages. [Hyman \(2009\)](#) has appropriately criticized this term for its lack of precise definition, and for the abusive use of the term to hide poorly-understood suprasegmental phenomena. Nonetheless, the present survey of Chadic languages demonstrates that the atypical behavior of tone in some of these languages falls somewhere between a tone system and a stress/accent system. To simply describe them as having a restricted tone system does not seem to do justice to the peculiarities. It is probably inappropriate to see tone and stress/accent as a polar, “either-or” opposition for suprasegmental systems. Rather, our understanding of such systems must allow languages to move gradually on a continuum between a prototypical tone system and a prototypical stress system. This has been shown to have happened diachronically in the languages of the world, and such changes have apparently occurred without an abrupt shift. Alternatively, each language may fashion its suprasegmental system by exploiting a variety of features, some of which are properly tonal and some of which are properly accentual. In that case, no language is ever shut up to being purely tonal or purely stress-based, and the terms “tone language” and “stress language” could only be used to indicate a predilection for the use of one type of feature rather than the other in the suprasegmental system of the language.

One question related to tonogenesis has not been answered, however. Are all Chadic languages tonal today because the Proto language adopted distinctive tone at one point in history, through the internal pressure and mechanisms

suggested here? Or, have subgroups of Chadic languages become tonal independently of one another? That question is left for further research.

8 Concluding remarks

Here is what can be said about the tonal systems of Chadic languages, based on the present survey. If the premise is accepted that Chadic languages are all tonal in some sense, it is clear that the functional load of lexical distinctions of tone is very low. Nouns are more likely to make full use of the tonal distinctions that the language affords. Verbs, on the other hand, use tone in a more restrictive way, often reserving tone for making grammatical distinctions of TAM; in some languages the verb root has no lexical tone of its own. Another factor that influences tone in many Chadic languages is the effect of consonants, certain of them lowering and others raising the tone, according to the rules of the language. This factor may interact with lexical tone, and in some cases creates an additional phonological level. However, no Chadic language need recognize more than 3 distinctive tone levels. A survey of other tonal phenomena (tone spreading, downdrift and downstep, floating tones) shows that, while some examples of each may be found among Chadic languages, none of them is general throughout the family, as are the phenomena of consonant-tone interaction and grammatical marking by means of tone.

It was also found that phonological structure and properties common to stress-accent systems play a significant role in Chadic tone. While I do not use the term pitch-accent to characterize Chadic suprasegmental systems, I recognize that Chadic languages often integrate nontonal features together with the tonal features that were cited earlier. These nontonal features, characteristic rather of stress-accent systems, are realized by tone nonetheless, and include: metrical prominence marked by tone on individual units such as the syllable or mora; culminative use of one of the tones; tonal polarity; and the determination of tone by factors of phonological structure.

Finally, the consideration of Chadic tone led to an examination of the mechanisms of tonogenesis, since it is not clear that Chadic languages have always been tonal throughout history. I propose that the factors internal to Chadic languages provide sufficient pressure to transform these languages from nontonal to tonal. Consonant quality and prominence within metrical structures would be the principal triggers, perhaps working together with grammatical distinctions achieved through pitch. These are more likely explanations for tonogenesis than the claim that distinctive tonality has arisen through borrowing from neighboring languages.

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Chapter 11

Tone in Nilo-Saharan

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The majority of Nilo-Saharan languages, bridging an expanse from Mali to Ethiopia, are tonal. A comprehensive summary of how tone works across Nilo-Saharan is difficult, because many of its tone systems remain underexplored, because the languages can differ with regards to almost any element of the tone system, and because Nilo-Saharan’s status as a phylum is still subject to debate. Therefore, Nilo-Saharan is framed as a group of languages exhibiting substantial variation, and this chapter examines the types of tone systems found across these languages, including inventory, processes, and tone’s role in lexicon and morphology.

1 Introduction

One of the most consistently debated major African language phyla since it was postulated (Greenberg 1963), “Greenberg’s wastebasket” (Campbell & Mixco 2007) of Nilo-Saharan languages stretches from western Mali and Mauritania to Ethiopia, Kenya, Tanzania, and Uganda, north of the majority of Niger-Congo and south of the majority of Afro-Asiatic. Genealogical debate spans the range of the family: Some question whether the hypothesized (sub)families of Songhay, Koman/B’aga/Gumuz, and Kuliak are part of Nilo-Saharan at all (Sands 2009, Dimmendaal et al. 2019, Dimmendaal 2019); other hypotheses suggest that isolates are actually part of the phylum (Ehret 2001); still others toss languages back and forth between Nilo-Saharan and Kordofanian (or Niger-Kordofanian) (Bender 1996, Ehret 2001). To a large extent, this chapter does not touch on genealogical detail, though it is important to keep in mind that the makeup of the phylum is still debated. While some sub-families are clearly related, consensus may

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change as in-depth research continues. A hypothesized family tree can be seen in Figure 1. Further subdivisions of these families are still contested—for example, Eastern Sudanic has had numerous postulated subdivisions (Bender 2000, Rilly 2009, Starostin 2015), and Blench (2012) postulates a Songhay-Saharan branch of Nilo-Saharan, instead of considering Saharan a subfamily of Northeastern Nilo-Saharan. Some (e.g., Güldemann 2018) even consider the existence of a single ‘Nilo-Saharan’ phylum to be as doubtful as that of Khoisan—the latter of which is largely accepted today as consisting of unrelated groups of languages that happened to share click consonants. Though I do sometimes refer to the languages discussed here as *Nilo-Saharan*, grouping them under one name, this is meant simply to refer to the languages, essentially for convenience’s sake, not to comment on the genealogical perspective.

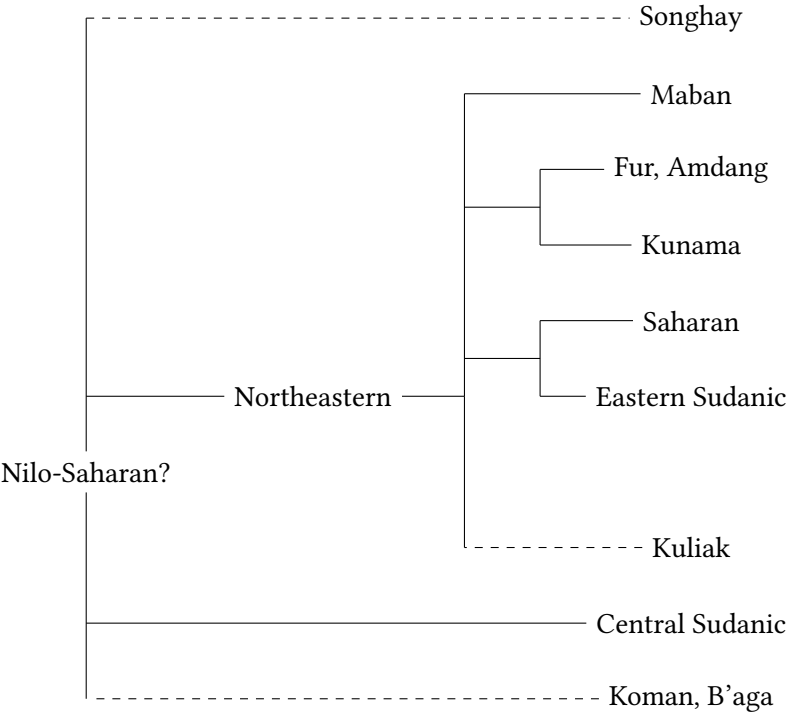


Figure 1: Subclassification of Nilo-Saharan (Dimmendaal 2019). Dotted lines represent particularly highly-debated subfamilies.

Despite the major question of whether the phylum exists, the languages commonly referred to as *Nilo-Saharan* are rich when it comes to tone and tonal phenomena. Of the roughly 200 languages that might be considered part of the phy-

lum (Eberhard et al. 2024),¹ the majority are tonal, with only a few described non-tonal languages at present, such as the Songhay languages Koyra Chiini, Djenné Chiini, and Koroboro Senni (Heath 1999).² Walters (2016) suggests that Dazaga, a Saharan language spoken in Chad and Niger, may have a pitch-accent system (citing the van der Hulst 2011 and Hyman 2006, 2009 properties of pitch-accent languages), contra Alidou (1988) and Amani (1986). Other languages sometimes cited as non-tonal, such as some languages of Nubian—a subfamily of Eastern Sudanic—have since been proven tonal (e.g., Bell 1968, Jakobi & Dimmendaal 2022, see also Almkvist 1911).

This latter point is worth looking at further in the context of this volume. Nubian is not the only language family previously thought to be non-tonal, or for which tone was acknowledged but discounted or set aside. This has also been seen in the languages Dinka and Nuer, which are Nilotic languages (a subfamily of Eastern Sudanic). Tone is vitally important to these languages, and while tone was acknowledged as present in Dinka and Nuer in the first half of the 20th century (Crazzolara 1933, Tucker 1939, Nebel 1948), it was not until Andersen (1987) that tone in Dinka was seriously approached, and not until the late 2010s that Nuer tone was described (Gjersøe 2016, 2017, 2020, Reid 2020, Monich 2020).

A similar situation occurred in Majang, a Surmic language also from the Eastern Sudanic branch. While earlier sources on the language exist (Bender 1983, Unseth 1988, 1989, 1992, 2007, Moges 2007, Getachew 2014), the first in-depth examination of tone in the language was Joswig (2010, 2015). This is true despite the fact that, as is true for Dinka and Nuer, tone is key not only for Majang's lexicon but also for its grammar. There are many other similar cases, in part because the downplaying the importance of tone is not isolated to Nilo-Saharan—or, for that matter, to African languages (cf. e.g., McPherson 2019). But grammatical tone is common across Nilo-Saharan; this is true even for languages that also use affixation or consonant and vowel shifts—i.e., languages with extensive segmental morphology (cf. Rolle & Bickmore 2026 [this volume]). There are also a number of languages, such as some of the Western Nilotic languages, in which segmental morphology is highly reduced, and in which tone takes on an especially high functional load. As such, ignoring tone in descriptive work on Nilo-Saharan exemplifies how vulnerable languages are to the people describing them. In other

¹As always, what exactly constitutes a language versus a dialect may vary depending on the source. For some more comments on this with regards to Nilotic, a subfamily of Nilo-Saharan, see Blum (2023).

²Since Songhay is a contested family, it may be the case that all of Nilo-Saharan is tonal—proving this comprehensively, however, may be near impossible.

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words: discounting tone in these languages means discounting much of a given language's grammar.

As is true in many other groups of languages, it is also necessary to tread carefully to avoid confusing different families, subfamilies, and languages. For example, *Maban* is a subfamily of Northeastern Nilo-Saharan, but *Mabaan* is a language belonging to the Burun branch of Western Nilotic, itself a sub-subfamily of the Eastern Sudanic branch of Nilo-Saharan (Andersen 1992). There is also *Bor*, a dialect/dialect cluster of the Dinka language, as well as *Belanda Bor/Boor*, which belongs to the Luo branch of Western Nilotic; the Luo branch in question is sometimes spelled *Lwo* or *Lwoo*, which is not to be confused with its constituent languages *Luwo* and *Dholuo*, the latter of which is also often referred to as *Luo*. Being as specific as possible is key. These complexities are not limited to any one subfamily, either: there are also terms that sound similar, perhaps like they should be synonymous, but they are not—like *Nubian languages*, which is genealogical, and *languages of the Nuba mountains*, which is geographical, referring to languages of any family that are spoken in the Nuba mountains of Sudan and the surrounding area. Though some languages spoken in the Nuba mountains are Nubian, the two terms are not interchangeable.

As Kouneli and Nie (2021) note, despite detailed work on the tone systems of Nilo-Saharan languages, many languages' tone systems, as well as the variation across lects of those languages, remain undocumented. And yet, as will be discussed below, what is known about these languages suggests that they are rich with examples of interesting tonal phenomena, with much to contribute to the studies of the phonology-morphology and phonology-syntax interfaces—indeed, there are even unusual cases that have the potential to disprove implied or argued tonal universals. There is a danger here, too, of making assumptions about any given language, and I would caution researchers studying these languages against assumptions when beginning investigation. Nothing should be assumed: not the tonal inventory, nor the tonal phenomena, nor how tone interacts with other areas of the phonology or the grammar at large.

Finally, though some of the described phenomena may only be exemplified thoroughly by one language or in one example, this is not to suggest that the phenomenon in question occurs only in those examples. For some topics, a range of differing examples are presented; in others, a particularly unusual case is shown, or a case that makes a particular typological point is displayed. This is, in part, because the one of the larger purposes of this chapter is to argue that these languages deserve focus and work, and can and should be treated much in the same way that other phyla have been examined, with regards to tonal description and language use, as well as to possible theoretical discoveries.

2 Tonal inventory

Nilo-Saharan languages usually have two- or three-level tone systems, consisting either of H(igh) and L(ow) levels, or H(igh), M(id), and L(ow) levels. These *levels* are sometimes referred to as *registers*.³ Dimmendaal et al. (2019) also note that the division between a two-level and three-level tone system can be somewhat blurry, with some varieties of a language having a two-level system with downstep (cf. Snider 2026 [this volume]), and others a three-level system. There are exceptions to the High/(Mid)/Low system: the Koman language Opo is attested to have four levels, adding an Extra-high (X) level in addition to the aforementioned three (Smolders 2019), and the Central Sudanic language Yulu, which belongs to the Bongo-Bagirmi subfamily, has an Extra-low tone in addition to High, Mid, and Low (Boyeldieu 2009). The Western dialects of the Moru-Madi language Lugbara—also a member of the Central Sudanic family—have also been argued to have four tone levels. In this case, the fourth level in question is Extra-high (Andersen 1986). In the case of both Yulu and Lugbara, the case has been made that the extra-low and extra-high tones result from the phonemicization of segmental effects—of depressor consonants and +ATR vowels, respectively (cf. Beavon-Ham & Pearce 2026 [this volume]).

Level tones can combine further into contour tones, though how exactly this manifests depends in part on syllable structure, word structure, and the tone-bearing unit (TBU) in a given language. Some languages, like Nara, an East Sudanic language (Dawd & Hayward 2002), have a tone-bearing unit of the mora. Examples of this in the Higir dialect of Nara can be seen in (1). In (1a–d), one sees that monomoraic syllables carry either High or Low tones. Bimoraic and trimoraic syllables, on the other hand, carry more than one tone, as can be seen in (1e–g), and can be interpreted as consecutive sequences of Low and High tones. This includes both contour tones (1e, f, h), as well as level tones (1g). Note that /n/ and /m/, as well as other sonorant consonants and post-vocalic glides, are tone-bearing in addition to vowels, as can be seen in (1e,f,h).⁴ The one-to-one correspondence between moras and tones in Nara means that contour tones appear only on long vowels, which matches a cross-linguistic tendency towards the avoidance of contour tones on short vowels (Zhang 2001).

³I intentionally do not use this term in this chapter. The term can also be used to describe the redundant combination of tone/pitch, phonation/voice quality, vowel quality, and vowel length in Southeast Asian tone languages (Brunelle & Kirby 2016). Because some Nilo-Saharan languages also have a phonation contrast, albeit not a redundant one, the term *register* as a synonym of *level* is avoided.

⁴Tone-bearing consonants are seen across a number of languages; see Kutsch Lojenga (1989) for an interesting interaction between tone and voicing in the Lendu language.

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- (1) a. nì.hì
 'tusk'
- b. tà.wó
 'form'
- c. tú.sá
 'ears'
- d. kú.tì
 'tree bark'
- e. sùm
 'grass'
- f. súm
 'poison'
- g. gòò
 'frog'
- h. còón
 'four'

Nara examples from Dawd & Hayward 2002.

Many languages have the syllable as the tone-bearing unit, such as Gaahmg, an Eastern Sudanic language spoken in Sudan, near the Ethiopian border. Gaahmg has three tones: High, Mid, and Low (Stirtz 2011). As before, contour tones are analyzed as sequences of level tones, and all of the levels can combine, making a total of nine possible tone categories on monosyllables: besides L, M, H, there are HL, HM, ML, LH, LM, and MH. In this case, the distribution of level and contour tones is directly related to syllable weight, as can be seen in Table 1. One sees that in Gaahmg, only the level tones are found on light syllables. Contour tones, on the other hand, appear on both mid-weight and heavy syllables. Note that the examples presented in Table 1 do not represent the full range of syllable structures in the language, and that the tonal distribution across polysyllabic words differs slightly; see Stirtz 2011:43–48 for further detail.

The Shilluk language, which is Nilotic (Eastern Sudanic) represents an extreme end of the spectrum—in which H, M, and L combine with a tone-bearing unit of the syllable, to form an additional five contour tones (Remijsen & Ayoker 2014). Thus, Shilluk has eight tone categories: High, Mid, Low, Rise, Early High Fall, Late High Fall, High Fall to Mid, and Early Low Fall, as can be seen in (2). In the previous case of Gaahmg, the tone categories were distributed unevenly across syllable weights. In contrast, in Shilluk, all contours appear on short vowels.

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Table 1: Tonal categories across syllable weights in Gaahmg. Examples from Stirtz 2011.

	Light		Mid		Heavy
H	ḍínì ‘world’		cíl ‘spine’		póór ‘boat’
M	kāśá ‘boy’		mēl ‘tree type’		bāāl ‘cave’
L	fàrì <i>hill name</i>		ḍàl ‘pot’		dèèl ‘collar bone’
HL	-		máà ‘house’		sèèn ‘ruler’
HM	-		múū ‘mosquito’		ḡí-ḍ ‘tooth’
ML	-		ṭī ‘cassava’		bēèl ‘metal’
LH	-		ḍǎlgǎ ‘drum’		ḍóór ‘fence’
LM	-		gǎ n ^a ‘responsibility’		ḍēēr ‘leach’
MH	-		mīí ‘chicken’		būúl ‘bread’
LHL	-		-		ḍùúl ‘instrument’

^aThe tone diacritics here are meant to represent a Low-Mid contour on /ɔ/.

Additionally, unlike previous examples, the tones cannot all be distinguished via representation as a sequence of level tones, as explained further below. Note that in Shilluk, diacritics on the first vowel symbol apply to the entire syllable.

- (2) a. njēen á-lēṅ pāac *Late High Fall*
money.PL PST-take.FUG village.SG
‘Somebody went to the village to take money.’
- b. njēen á-lēṅ kì kēṅ *Low*
money.PL PST-take.2SG PREP place.SG;DEM
‘You have taken money in this place.’
- c. bùul á-lēṅ kì kēṅ *Early Low Fall*
drum.SG PST-beat.2SG PREP place.SG;DEM
‘You have beaten the drum in this place.’
- d. bùul á-lēṅ kì kēṅ *Early High Fall*
drum.SG PST-beat PREP place.SG;DEM
‘Somebody has beaten the drum in this place.’
- e. mǎḷḷt á-lēṅ njēen *Mid*
friend.SG PST-take.APPL money.PL
‘Somebody used a friend to take money.’

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- f. m̩ʌʌṭ á-lěŋ njēep *Rise*
 friend.SG PST-take.APPL.2SG money.PL
 ‘You used a friend to take money.’
- g. l̩ʊṭ á-lěŋ bùul *High*
 stick.SG PST-take.APPL;2SG drum.SG
 ‘You have used a stick to beat the drum.’
- h. l̩ʊṭ á-lěŋ bùul *High Fall to Mid*
 stick.SG PST-take.2SG PREP place.SG;DEM
 ‘Somebody used a stick to beat the drum.’
- Shilluk examples from Remijsen & Ayoker 2014:441–2.*

The existence of four phonemic fall tones on short vowels is highly unusual, as is the ways in which they differ. Shilluk is a language with a particularly rare sound system—along with the complex tone system, it has a three-level vowel length contrast, distinguishing between short, long, and overlong vowels (Remijsen et al. 2019). But in related languages with similarly-complex sound systems, such as Nuer, Thok Reel, and Dinka⁵ (Andersen 1987, Reid 2010, Remijsen & Ladd 2008), it is possible to conceptualize contour tones as adjacent level elements within the same tone-bearing unit—HL or LH on a single syllable, similar to the examples from Gaahmg and Nara. This is consistent with tonal analysis across many African tone languages. But this is not the case for the Early High Fall and Late High Fall in Shilluk, because both have as their initial target the High level tone, and as their final target the Low level tone. That is, they could both be represented as *HL*; from that analytical perspective, they are indistinguishable. So how are they differentiated?

Instead, timing is inseparable from the tonal inventory in Shilluk: essentially, the tones are differentiated by the time point in the vowel at which the fundamental frequency (*f*₀) begins to fall (‘contrastive alignment’) (Remijsen & Ayoker 2014). The Early High Fall has an earlier onset than the Late High Fall; the Early High Fall and Late High Fall appear across all three vowel lengths, and cannot be explained by the association of tones to different timing units within the syllable (Remijsen & Ayoker 2014). This is because the timing is similar across vowel lengths; the onset of the *f*₀ drop begins at a mean of 13 milliseconds (ms) into the vowel for the Early High Fall and at 77ms for the Late High Fall (Remijsen & Ayoker 2014). Such a difference in timing was previously hypothesized as nonexistent in language (Odden 1995); contrastive alignment is an excellent example of why careful work on tone can reveal previously unattested phenomena.

⁵Dinka also has a related case of contrastive alignment in tone (Remijsen 2013), but the early-aligned fall has an allophonic relationship with the Low tone.

3 Lexical and grammatical tone

While Nilo-Saharan languages commonly have a high functional load of tone, tone’s distribution between lexical and grammatical varies across languages (though grammatical tone is common). For example, the Fur language (Fur, spoken in Sudan and Chad) has what Tucker and Bryan (1956), Jakobi (1990), and Kutsch Lojenga and Waag (2004) all agree is a relatively high lexical function of tone. On the other hand, in the Maa language (Eastern Nilotic), tone is key to the verbal morphology, and possibly not at all involved in the verb system (Rasmussen 2002). While tone certainly can be involved in lexical differentiation in the language, it is rarely the difference in a minimal pair or set (Tucker & Mpaayei 1955). In many other languages, tone plays a key role both lexically and grammatically; consider Table 2, in which the same four tones can differentiate words both lexically and grammatically in Dinka (Western Nilotic). A comprehensive description of all of the manners in which these languages use tone—lexically, grammatically, or both—would be impossible. As Billington (2017) notes, tone’s role in the grammar of Nilo-Saharan languages is under-described. What is known, though is that tone can play a role, across these languages, in a diverse variety of functions, distinguishing, among other elements, nouns and verbs, inflections and derivations, tense, aspect, and mood, person, and number.

Table 2: Lexical and grammatical contrasts in Bor South Dinka.
Note that in Dinka, as in Shilluk, diacritics on the first vowel symbol apply to the entire syllable.

Tone	Form	Gloss	Contrast type
Low (L)	c̣iir	thorny_tree.SG	Lexical
High (H)	c̣iir	pour:BEN;NMZ	Lexical
Fall (HL)	c̣iir	big_river.PL	Lexical
Rise (LH)	c̣iir	star.PL	Lexical
Low (L)	ɲòooṭ	point:3SG	Grammatical
High (H)	ɲóooṭ	point:FUG	Grammatical
Fall (HL)	ɲôooṭ	point:FUG;3SG	Grammatical
Rise (LH)	ɲöooṭ	point:NF	Grammatical

Tone can be the sole exponent of case-marking, as can be seen in Table 3. This is true in Fur, in which tone marks the locative case (Kutsch Lojenga & Waag 2004), and in Dinka, in which tone marks the genitive (Andersen 2002). Exponentence of tone in case-marking is not necessarily predictable; that is, while a shift

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from Low tone to High tone might reliably mark a certain case in one language, it should not be assumed that a single pattern covers all case-marking in another language—one can see in the Dinka examples that a shift from Low tone to High tone, as well as a shift from High tone to Low tone, can mark the genitive. Similarly, while the tone patterns that nouns exhibit in the absolutive case are not predictable in Datooga (Southern Nilotic), the number of possible patterns is constrained; the two items seen in Table 3 are representative of two different absolutive tone patterns (Kießling 2007). In other languages, tone can be the sole exponent of some cases but not others—this can be seen in Tugen (Kalenjin, Southern Nilotic), which differentiates nominative and absolutive cases via tone, but benefactive and genitive via affixation (Jerono 2013). And in Majang, non-automatic downstep—the lowering of the f0 ceiling for High tones based on a floating Low tone—can be the marker of case (Joswig 2019).

Table 3: Tone as the sole exponent of case-marking.

Language	Base	Gloss	Case-marked	Gloss
Fur	tòŋ	house	tóŋ	house:LOC
Fur	tindíl	rubbish_heap	tíndíl	rubbish_heap:LOC
Dinka	ràaan	person	ráaan	person:GEN
Dinka	bàŋ	chief.sg	báŋ	chief.sg:GEN
Datooga	qàarèemángà	youths:NOM	qàarèemángà	youths:ABS
Datooga	jàawúudà	cat:NOM	jàawùudà	cat:ABS
Tugen	láàkwéé	child:NOM	láàkwéé	child:ABS
Tugen	kípchóríúréé	squirrel:NOM	kípchóríúréé	squirrel:ABS
Majang	wár	dog.sg:NOM	wár ^L	dog.sg:ABS
Majang	tân	cow.sg:ERG	tân ^L	cow.sg:ABS

Broadening the lens past case-marking in nouns, sub-groups of lexical items can often be categorized based on their tone patterns alone. In addition to the Datooga absolutes, this can be seen in a language such as Dinka, in which the major divisions of transitive verb class are differentiated by the tone of the infinitive (Andersen 1993). This is also true for the intransitive verbs to some extent (Andersen 2020, Blum 2021). A similar pattern can be seen in Shilluk, in which transitive verbs can be separated into seven classes based on the tones of their past-tense, second-singular past-tense, imperfective, and past-tense centrifugal forms (Remijnsen et al. 2016). In the Barka dialect of Kunama (spoken in western

Eritrea), a floating High tone marks a lexical class of nouns (‘Class II’); this independent, purely tonal morpheme adds no semantic value (Connell et al. 2000). And in Kanuri, a Saharan language of Cameroon, Chad, Niger, and Nigeria, verbal nouns, derived nouns, and a subset of adverbs may all be distinguished by tone pattern (Hutchison 1981). Some of these are seen in Table 4; these derived nouns display a persistent pattern in which the final syllable is High-toned, and all preceding syllables are Low-toned, regardless of the tone of the underived noun (Hutchison 1981).

Table 4: Tonal patterns in morphologically-derived nouns in Kanuri (Hutchison 1981):28.

Underived	Gloss	Derived	Gloss
fúfù	‘lungs’	fùfú	‘permanant cough’
shímàlò	‘tears’	shìmàló	‘cattle disease with tearing’
fèrò	‘girl’	fèró	‘girlish’
cî	‘mouth’	cìkùndùlí	‘moustache’
kùndùlí	‘hair’	”	”
kànúrí	‘Kanuri people’	kànùrí	‘Kanuri language’

Tone can also be the exponent of number distinction, as can be seen in Table 5, for many languages in the Eastern subdivision of the Central Sudanic family (Dimmendaal et al. 2019), such as Ngiti, spoken in the Democratic Republic of Congo (Kutsch Lojenga 1994 as cited in Dimmendaal et al. 2019). This can also be found in Beria (also called *Zaghawa*), a Saharan language spoken in Chad and Sudan (Jakobi & Crass 2004, Gayler et al. 2023), as well as in the Dinka-Nuer family (Western Nilotic, Eastern Sudanic) of South Sudan and Ethiopia. The contexts in which tone is the sole exponent of number marking, however, vary across languages. For example, in many Central Sudanic languages, tone is the sole marker of singularity vs. plurality for pronouns and some human nouns (Kutsch Lojenga 1994, Dimmendaal et al. 2019), whereas in Dinka-Nuer, tone is one of numerous exponents of noun number (Ladd et al. 2009, Andersen 2014, Bond et al. 2020, Ladd & Blum 2021a, Nyang 2021). In Beria, noun number can be distinguished solely by tone (Jakobi & Crass 2004). Possessive pronouns can be distinguished for number as well, though there is syncretism between the 1PL inclusive (with a singular possessum) and the 1PL exclusive with both singular and plural possessa (Gayler et al. 2023).

Along with noun morphology, tone is also responsible for extensive verbal morphology across these languages. As mentioned above, tone can be a defining

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Table 5: Tone as the sole exponent of number. Beria examples from Gayler et al. 2023, Ngiti examples from Kutsch Lojenga 1994 as cited in Dimmendaal et al. 2019. Dinka examples from Ladd & Blum 2021b. Nuer examples from Bond et al. 2020.

Language	SG	Gloss	PL	Gloss
Beria	bōr	child.SG	bō`r	child.PL
"	kō`	razorblade.SG	kō´	razorblade.PL
"	dèrì	feather.SG	dèrì`	feather.PL
"	tēlē	girl/daughter.SG	tēlē`	girl/daughter.PL
"	ōrò	throat.SG	ōrò`	throat.PL
"	ēgī	1SG.PRO (SG possessum)	ēgì	1SG.PRO (PL possessum)
"	kōlō	2SG.PRO (SG poss.)	kōlò	2SG.PRO (PL poss.)
"	kōgò	3SG.PRO (SG poss.)	kōgò´	3SG.PRO (PL poss.)
"	kārō	1PL.EXCL.PRO (SG poss.)	kārò	1PL.EXCL.PRO (PL poss.)
"	kārò	1PL.INCL.PRO (SG poss.)	kārò	1PL.INCL.PRO (PL poss.)
"	kōlū	2SG.PRO (SG poss.)	kōlù	2SG.PRO (PL poss.)
"	kōgù	3SG.PRO (SG poss.)	kōgù´	3SG.PRO (PL poss.)
Ngiti	īmā	1SG	imă	1PL
"	īŋī	2SG	inĭ	2PL
"	àbādī	3SG	àbādī	3PL
Dinka	ròok	kidney.SG	ròok	kidney.PL
"	tùuŋ	horn.SG	tùuŋ	horn.PL
"	tòon	pot.SG	tòon	pot.PL
Nuer	lēj	beast.SG	lēj	beast.PL

feature of verb sub-classes, but can equally be an exponent of verbal morphology itself—and while it can appear in combination with other phonological elements as morphological exponence, it can also function by itself in this manner. The roles tone can play in verbal morphology are not constrained, even within a single language; for example, in Opo (Koman, South Sudan and Ethiopia), a tonal contrast can be the difference between various types of derived verb forms, including transitivity and intransitivity (3a,b), singular versus plural subjects (3c,d), distributed and non-distributed objects (3e,f), singular and pluractional (3g,h), among others (Smolders 2024).

- (3) a. b̀à
‘lay (vi. SG)’

- b. ɓǎ
'lay, put (vt. SG)'
- c. k^hālā
'be small (SG)'
- d. k^hálá
'be small (PL)'
- e. ɟùk'à
'grind (grain)'
- f. ɟúk'ǎ
'grind (various kinds of grain)'
- g. sāk'ā
'carry'
- h. sák'á
'carry (PLAC)'

Opo examples from Smolders 2024.

As can be seen in the examples in Table 6, tone's role in verbal morphology does not need to be class-marking, or class-defining, as it has been in some examples—or even predictable. In Fur, though tone is clearly key to verbal inflectional morphology, as can be seen in comparing forms like ɗɗɗɗ and ɗɗɗɗ, the tone patterns themselves are not predictable (Kutsch Lojenga & Waag 2004). In Beria, tone can be the difference between a singular and plural agent; in Gaahmg, tonal marking is responsible for subject-person inflection, for distinguishing non-past and past tense, and marking antipassives, causatives, and verbal nouns (Stirtz 2011). In Lendu (Central Sudanic, Democratic Republic of the Congo), tone is the difference between present and present continuous (Kutsch Lojenga 1989); and in Kanuri, the difference between present and past (Hutchison 1981).

Many of the examples in Table 6, including as the *těngī/těngí* ('I put it down'/'I have put it down') contrast, bridge the divide between tonal inflection of a verbal root, and tense-aspect-mood (TAM) marking. Whereas TAM is incorporated into verbal morphology in Table 6, tone can also mark TAM in the non-verbal or auxiliary settings. For example, in addition to tone's role in its verbal morphology as shown above, Opo also exhibits tonal alternation expressing TAM, which appear as proclitics on the verb, as seen in (4). The TAM markers can also be solely tonal. In Lendu, imperfective aspect is expressed via a floating High tone (Dimmendaal et al. 2019). In Ma'di, a floating Low tone expresses present and future–non-past tenses (Blackings & Fabb 2003: 155, 157, as cited in Dimmendaal et al. 2019).

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Table 6: Tone as an exponent of verbal morphology. Examples from [Dimmendaal 2019](#), [Dimmendaal et al. 2019](#), [Hutchison 1981](#), [Kutsch Lojenga & Waag 2004](#), [Connell et al. 2000](#), [Stirtz 2011](#), [Noonan 1992](#).

Language	SG	Gloss	PL	Gloss
Fur	kùrtò	dig.3SG;PST	kúrtó	dig.1PL;PST
Fur	óóŋó	dig.1SG;PST	òòŋò	dig.3SG;PST
Beria	gèrì	‘s/he looks for it’	gèrí	‘they look for it’
Beria	těngī	‘I put it down’	těngí	‘I have put it down’
Lango	ànêní	‘I see you’	ànéní	‘(that) I see you’
Gaahmg	kóm-só	chop-COMP.3SG	kóm-sò	chop-COMP.3PL
Lendu	bṛ	‘jump’	bř	‘jump continuously’
Kanuri	díwì	‘you do it’	díwí	‘you did it’
Kunama	álâ	‘Leave me (pl.)!’	álá	‘Did you (sg./pl.)/they leave me?’

- (4) a. ār=ǰō
3SG.M:PRF=come.SG
‘He came.’
b. ár=ǰō
3SG.M:JUSS=come.SG
‘Let him come!’
c. àr=à-ǰō
3SG.M=IMPV-come.SG
‘He comes/is coming.’
d. ār=á-ǰō
3SG.M=FUT-come.SG
‘He will come.’
Opo examples from Smolders 2024.

Tone can also play a role as a function morpheme. For example, Gumuz (a B’aga language of both Ethiopia and Sudan) has an associative morpheme consisting solely of a High tone, appearing between nouns in noun-noun constructions ([Dimmendaal 2019](#)). This can be seen in (5a) in Northern Gumuz. Similarly, the Uduk (Koman, Sudan) language’s associative construction for Class II nouns involves a tone change from Low to Mid on the class proclitic /à=/ of the second noun, as seen in (5b). (The final surface realization also involves a transfer of the

second noun’s proclitic onto the first noun as a postclitic; see Killian (2015) for more details.)

- (5) a. b̥àtʃ’à (H) χòsà [b̥àtʃ’áχòsà] Northern Gumuz
 meat ASC bovine
 ‘beef’
 b. à=ōm à=ɲwá [à=ōm=ā ɲwá] Uduk
 II=egg II=chicken
 ‘chicken egg’

Examples from Dimmendaal (2019) and Killian (2015).

The Bor South dialect of Dinka displays a case similar to that of Northern Gumuz, in which a floating tone—in this case, a Low tone—appears in modificational phrases, directly after the head noun and before the modifier; this includes noun-noun constructions, as all lexical categories may be used as postnominal modifiers in Dinka (Blum 2021). Similar also to the Uduk case, in which tone is part of associative constructions for a subclass of nouns, the Dinka floating Low tone has a phonological effect on all following nouns. Instead, an effect appears only in two cases: first, on High-toned short vowels, in which case the vowel in question is realized as a Rise tone, and second, on the prefix /a-/, which is underlyingly toneless (Remijsen & Ladd 2008) but which surfaces as a High tone by default; in the presence of a floating Low tone, the toneless syllable is realized as a Low tone (cf. Rolle et al. 2026 [this volume]). These can be seen in (6), both of which are noun-noun constructions, and which use the same head noun and modifier for the purposes of comparison.

- (6) a. b̥ɔ́ɔ̃ (L) b̥ɔ́ɔ̃ [b̥ɔ́ɔ̃ b̥ɔ́ɔ̃]
 chief.SG MOD chief.SG
 ‘peaceful chief’ (lit. ‘chieflike chief’)
 b. ab̥ɔ́ɔ̃ (L) ab̥ɔ́ɔ̃ [áb̥ɔ́ɔ̃ àb̥ɔ́ɔ̃]
 orphan.SG MOD orphan.SG
 ‘orphaned orphan’

Whereas the previous examples in this section have focused on the role of tone on the level of the word or phrase, tone can also function on the scale of domains larger than these units, indicating boundaries and marking domain. As has been shown in Table 3, downstep can be an exponent of case-marking in Ma-jang. However, downstep can also be a grammatical indicator of word boundary

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in Majang. Indeed, Joswig (2019) considers these two types of downstep in Majang as separate, noting that when High tones are adjacent across a morpheme boundary, the initial syllable of the second morpheme has a lowered f₀ ceiling. This is termed *morpheme downstep* (Joswig 2019:90). However, the same lowered f₀ ceiling does not apply across word boundaries; adjacent High tones across a word boundary do not trigger downstep, unless a floating Low tone is present (Joswig 2019). An example of this is presented in (7). The key difference is found in comparing the downstep in (7b) and (7d). In (7d), *kó* is not downstepped relative to the preceding High tone on *rí* (though *rí* is downstepped relative to the High tone on *gá* due to the intervening Low tone on *rà*). However, *kó* is downstepped in (7b), due to the floating Low tone on the preceding syllable *-i*. As such, the boundary between *rí* and *kó* can be analyzed as a word boundary, not a morpheme boundary—Joswig’s analysis runs contrary to Bender (1983) and Unseth (1989), who analyzed *kó* as a clitic. Though the morpheme downstep is also shown to have a number of complex exceptional cases (Joswig 2019:92), it is also an exemplar of the way in which careful tonal work is key to morphosyntactic analysis. Moreover, instances of tonal grammar such as this type of downstep sit very much at the phonology-syntax interface; grammatically-sensitive downstep is also seen in, among other cases, Maa (Rasmussen 2002).

- (7) a. dé:gár-i:^L [dé:gári:]
 sleep-1PL.DJ
 ‘we sleep’
 b. dé:gár-i:^L kó [dé:gári: ^Lkó]
 sleep-1PL.DJ RECPST
 ‘we just slept’
 c. dé:gár-àrí [dé:gàrà^Lrí]
 sleep-2PL.DJ
 ‘you (PL) sleep’
 d. dé:gár-àrí kó [dé:gàrà^Lrí kó]
 sleep-1PL.DJ RECPST
 ‘you (PL) just slept’

These are certainly not all of the roles that tone can play in Nilo-Saharan languages; instead, this is meant to show the range of areas in which tone can be an exponent of lexicon and of grammar. Additionally, and importantly, tone can play all of these roles in a language. That is, tone can express case marking, number marking, and other instances of inflectional and/or derivational processes, in

a single language—not to mention that a language in which tone plays these roles can also have purely phonological tonal phenomena that interact little, if at all, with the rest of the grammar.

4 Tonal processes, theory, and the role of description

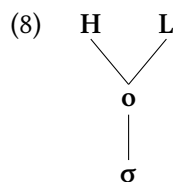
Separating ‘purely’ phonological phenomena from phenomena that lean more towards the phonology-morphosyntax interface is difficult, as can be seen from the role that downstep plays in Majang; a phenomenon prompted at its root by the adjacency of two High tones can be (and often is) governed by syntactic domain or morphosyntax. Tonal processes that might be considered more strictly phonological—especially more common processes such as those hypothesized as universals by Hyman and Schuh (1974), and later reaffirmed by Hyman (2007)—can also be conditioned by various grammatical factors. That said, processes like downdrift and downstep, spreading and shifting (cf. Smolders 2026 [this volume]), absorption and simplification, dissimilation and polarization, can commonly be found across Nilo-Saharan, though their nuances differ language to language, as is often the case.

However, the tonal processes in these languages can also be sources of both descriptive novelty and theoretical innovation. Already, tonal properties of some of these languages—in particular the Western Nilotic languages, some of which have highly unusual suprasegmental systems—have proven wrong assumptions about what can be contrastive. And as Kouneli and Nie (2021) point out, some of the tonal properties of Nilo-Saharan might differ from those of more widely-described phyla. As description of the tone systems of these languages continues and expands, assumptions or generalizations that are made about African tone languages might change.

One good example of this is the Obligatory Contour Principle (Leben 1973, Goldsmith 1976). Many Bantu languages avoid adjacent High tones, often realizing the second of two as a Low tone (i.e., H-H → H-L); in the Bantu context, this is also known as Meeussen’s Rule (Meeussen 1963, Goldsmith 1984). Nuer (Western Nilotic, South Sudan and Ethiopia), on the other hand, bans adjacent Low tones (Gjersøe 2020); while this is still a manifestation of the OCP in its most general definition (McCarthy 1986), it is a typologically unusual realization. The Nuer phenomenon (‘OCP-L’) has a few repairs, namely the epenthesis of a floating High tone (Gjersøe 2020:91, example 69). The floating High tone in turn serves as a diagnostic for underlying tonal specification, and leads to a conclusion that (lexical) HL tones (i.e., Fall tones) are a unitary contour tone with the structure

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seen in (8) (Gjersøe 2020:16, example 16). This is a structure seen far more often in East Asian tone (Yip 1989), and is unusual in African tone.



Tone can also interact with other elements of a language’s phonology, and this is another area in which Nuer exhibits unusual behavior. For example, in Nuer, the High (H) and Fall (HL) tone categories share an allotonic relationship based on phonation. Nuer has a voice quality contrast between modal and breathy, and though an increased f_0 associated with breathy voice is certainly not unexpected—this correlation is also seen in the closely-related language Dinka (Remijsen et al. 2025)—Nuer has taken this a step further; the High and Fall tones are in complementary distribution, associated with breathy voice and modal voice, respectively (Monich 2023). This bears some resemblance to the South-east Asian ‘register’ tone languages, in which f_0 and phonation, among other phonological elements, are used in combination to create contrast categories (Brunelle & Kirby 2016). This is once again, in some sense, non-African—or, at least, not what those who study, describe, and analyze African tone languages have come to expect. But it may be the case that there are more languages in the Nilo-Saharan grouping—within or outwith Nilotic—that exhibit this kind of typological behavior; with further description of the tone systems of these languages, both descriptive and typological norms can become more nuanced and better understood.

Another source of both descriptive and theoretical novelty is tone polarity (cf. Pulleyblank 2026 [this volume]). At its core, tone polarity is a process in which adjacent tones oppose one another; a canonical example is that a High-toned stem takes a Low-toned suffix, and a Low-toned stem takes a High-toned suffix (Yip 2002). Polarity is not uncommon in African tone languages, and has been analyzed as the result of polar morphemes in, for example, Hausa and Kɔnni (Newman 1995, Cahill 2004), and as dissimilation in Mooré (Kenstowicz et al. 1988) (none of which are Nilo-Saharan). The Nilotic languages Kipsigis and Dinka, however, exhibit polarity or polarity-like phenomena that broaden the understanding of the nature of tone polarity from both theoretical and diachronic perspectives.

In Kipsigis (Southern Nilotic, Kenya), the case inflection of noun modifiers exhibits paradigmatic tone polarity, applying between obliques and nominatives

Table 7: Polarity in Kipsigis, examples from [Kouneli & Nie 2021:e131,e133](#).

Lexical category	Oblique	Nominative	Gloss
Adjective	jà	já	‘bad.SG’
Adjective	númɲùm	ɲùmɲúm	‘easy.PL’
Adjective	tʃéptʃép-è:n	tʃèptʃèp-é:n	‘swift.PL’
Adjective	míntíli:l-è:n	mìntíli:l-é:n	‘sour.PL’
Demonstrative	ná:n	nà:n	‘MED.SG’
Demonstrative	tʃá:n	tʃà:n	‘MED.PL’
Demonstrative	nì/nì	ní/ní	‘PROX.SG’
Possessive	ɲwà:n	ɲwá:n	‘their.SG’
Possessive	tʃwà:k	tʃwá:k	‘their.PL’
Possessive	kù:k	kú:k	‘your.PL’

across an entire word, and across sequences of tones ([Kouneli & Nie 2021](#)). This can be seen in Table 7; one sees that for the adjective ‘sour,’ for example, the oblique HLHL tone pattern becomes LHLH in the nominative. Kipsigis represents (to the best of the authors’ knowledge, as well as mine) the first documented case of fully paradigmatic tone polarity, allowing innovation to be made in how across-the-board phonological processes are analyzed, as well as how phonological processes such as tonal polarity relate not only to phonological theory but also to the phonology-morphology interface ([Kouneli & Nie 2021](#)).

In Dinka, on the other hand, a seeming case of tone polarity can actually be traced back to contour simplification. The result is a different type of perspective: that the mechanisms underlying “polarity” are the vestiges of a historical contour process (LH → L / __ H) followed by a merger of tone categories. Hyman and Schuh (1974) postulate exactly this process as a potential root of polarity; in comparing closely-related dialects of the language, Dinka shows that it is the case. An example can be seen in (9). In a synchronic sense, identical surface realizations across dialects in (9a,c) (i.e., [rɛ̃ɛc dɔ̃ɔŋ]) are due to different underlying mechanisms in their individual dialects—seeming polarity in Twic Mayardit (9a,b) and contour simplification—specifically, High Absorption—in Bor South (9c,d). But whereas the Twic Mayardit dialect has only three tone categories—Low, High, and Fall (HL)—the Bor South dialect, which is more conservative ([Remijsen 2010](#), [Blum 2023](#)) has four tones: Low, High, Fall (HL), and Rise (LH), and exhibits High Absorption for every case in which Twic would have the “polarity” process. The

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High tone on ‘fish’ in Twic is the result of a merger of two historical tone categories, Rise and High. These categories did not collapse in Bor South, but did in Twic, leaving only a High tone in the latter variety. In other words, at an earlier stage of the language, ‘fish’ would have been Rise-toned across all varieties.

- (9) a. r̥ɛ̃ɛc d̥ɔ̃ɔŋ [r̥ɛ̃ɛc d̥ɔ̃ɔŋ] [L-H] *Twic Mayardit dialect*
 fish.SG remain
 ‘leftover fish’
- b. r̥ɛ̃ɛc d̥iit [r̥ɛ̃ɛc d̥iit] [H-L] *Twic Mayardit dialect*
 fish.SG big
 ‘big fish’
- c. r̥ɛ̃ɛc d̥ɔ̃ɔŋ [r̥ɛ̃ɛc d̥ɔ̃ɔŋ] [L-H] *Bor South dialect*
 fish.SG remain
 ‘leftover fish’
- d. r̥ɛ̃ɛc d̥iit [r̥ɛ̃ɛc d̥iit] [LH-L] *Bor South dialect*
 fish.SG big
 ‘big fish’

Unlike Kipsigis, the Dinka case is not paradigmatic tone polarity; nor is it meant to suggest a universal underlying mechanism for polarity itself. However, in showing these data I make the point that even a single case of this kind of underlying mechanism adds to what is known about the possibilities of tonal languages. Nilotic has been a focus of African tonal description and analysis in recent years, and these somewhat unusual cases push the boundaries of what is known about tone in African languages, as well as their typology. But perhaps these are not so unusual: any language can contribute significantly to a fuller understanding of trends and patterns across African tone languages, some of which have the potential to be notably different in the Nilo-Saharan grouping than in others.

Abbreviations

1	first person	INCL	inclusive
2	second person	JUSS	jussive
3	third person	LOC	locative
ABS	absolutive	M	masculine
APPL	applicative	MED	medial demonstrative
ASC	associative	MOD	modification
BEN	benefactive	NMZ	nominalization
COMP	completive	NOM	nominative
DEM	demonstrative	NF	nonfinite (infinitive)
DJ	disjoint marking	PL	plural
ERG	ergative	PLAC	pluractional
EXCL	exclusive	PRO	pronoun
FUG	centrifugal	SG	singular
FUT	future	PST	past
IMPV	imperfective	REC	recent

Acknowledgements

Thank you to two anonymous reviewers for attentive reviews and comments, and to Laura McPherson for helpful conversation. This material is based upon work supported by the National Science Foundation (NSF SMA-2313787). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author and do not necessarily reflect the views of the National Science Foundation.

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Chapter 12

Tone systems in the Bantoid languages

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The Bantoid languages are a body of some 150–200 languages positioned geographically between Nigeria and Cameroun. They do not form a genetic group, but all are in some way related to Bantu. The most well-known branches are Dakoid, Mambiloid, Tivoid, Beboïd, Grassfields, and Ekoid. Bendi, formerly Cross River, is probably Bantoid, while Jarawan may be Narrow Bantu. This chapter describes the tone systems recorded for the different branches of Bantoid. Mambiloid and Grassfields are only touched on briefly, since they are discussed in more detail in other chapters. Tone systems range from ultra-complex, in languages with strong erosion of morphology, to reduced, where morphology has been either retained or rebuilt. The paper refers briefly to the importance of speech surrogacy, i.e. “talking” musical instruments, and the challenges of developing orthographies for complex tone languages.

1 Introduction

The Bantoid languages are a body of some 150–200 languages positioned geographically between Nigeria and Cameroun and between Benue-Congo and Bantu in terms of their position within Niger-Congo (cf. [Blench 2026 \[this volume\]](#)). Often referred to as Bantu, for example in the term “Ekoid Bantu”, their classificatory position remains uncertain both in relation to Narrow Bantu and to Benue-Congo (cf. [Grimm 2026 \[this volume\]](#)). “Bantoid” does not represent a genetic group in the sense that there are a series of undisputed lexical or morphological isoglosses which argue for its coherence. The literature on many Bantoid subgroups is sparse, to say the least, and many important sources are unpublished. This is in part a reflection of accessibility, since the poor roads are often

Roger Blench. 2026. Tone systems in the Bantoid languages. In Keith L. Snider & Virginia Beavon-Ham (eds.), *African tone phenomena*, 359–395. Berlin: Language Science Press. DOI: ??

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cut in the wet season, discouraging extended fieldwork. In addition, civil insecurity in several areas where Bantoid languages are spoken has deterred research on the ground. Overviews of Bantoid include Johnston (1919, 1922), Hedinger (1989), Watters (1989), Piron (1998), Marten (2020), Blench (2022, 2024). References to individual subgroups are given in the appropriate sections below.

The main theme of the paper is the description of the tonal systems of Bantoid languages. It begins with a broad overview of Bantoid and then discusses individual branches in more detail. Mambiloid and Grassfields are only referred to briefly as they are described in greater detail in other chapters (cf. Connell 2026 [this volume] and Hamm et al. 2026 [this volume]). It should be emphasised that the data is very weak for some branches. There may be no reviewed publication dealing with tone, or accounts that clearly contain lacunae. There are no reconstructions of tone for the proto-language of any Bantoid branch except Grassfields, although the internal diversity of Bantoid is such that this would have few implications for the broader characterisation of tonal systems.

Table 1 presents a summary of the language subgroups described that “stand between” Eastern Benue-Congo and Narrow Bantu. It lists the country where they occur, their location broadly defined, and some representative languages are cited. Bantoid languages are very numerous (>200) and also highly diverse morphologically. A more complete list of all known languages is given in the *Ethnologue*¹ and *Glottolog*².

Bantoid is a member of Benue-Congo, a large and complex group of languages, whose exact membership remains disputed. Originating with Westermann’s (1927) *Benue-Cross-Fluss*, it took shape in Greenberg (1963), Williamson (1971, 1989) and De Wolff (1971). With this in mind, Figure 1 (page 362) provides a revised subclassification of Benue-Congo languages, including recent research on the status of South Jukunoid³.

Bendi, previously considered part of Cross River, has been shifted to Bantoid, a change of affiliation proposed by Blench (2001). In the light of this, Figure 2 (page 363) presents a tentative tree of South Bantoid.

¹<http://www.ethnologue.com>

²<https://glottolog.org>

³I hesitate to include a classification which has not been justified in print, but older versions can be found online

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Table 1: Major subgroups of Bantoid

Group	Country ^a	Location	Representative languages
Dakoid	N	Around Ganye	Daka, Taram, Gãã (=Tiba)
Mambiloid	N/C	Around Gembu	Mambila, Kwanja, Vute, Ndoro
Tikar	C	NE of Foumban	Three dialects
Bendi	N	Around Ogoja	Bokyi, Bekwara, Alege
Tivoid	N/C	Around Obudu	Tiv, Iyive, Ugarə
Buru	N	Buru	Buru
Furu	N/C	Furu Awa	Furu
East Beboid	C	West of Nkambe	Noone ^b , Ncane
Yemne-Kimbi	C	North of Grassfields	Mundabli
Nyang	C	West of Mamfe	Kenyang
Ekoid	N/C	Mamfe & westwards	Ejagham, Etung
Mbe	N	Ogoja	Mbe
Ambele	C	Grassfields	Ambele
Menchum	C	Grassfields	Menchum
Grassfields	C		
Ndemli	C	Nkam, Littoral region	Ndemli
Ring	C	Grassfields	
Centre	C	Grassfields	Babanki, Kom, Mmen, Oku
East	C	Grassfields	Lamnso?
South	C	Grassfields	Bamunka
West	C	Grassfields	Aghem, Isu
Momo	C	Grassfields	Moghamo
Southwest	C	Grassfields	Manta
Eastern	C	Grassfields	
Bamileke	C	Grassfields	Bamileke, Ngiemboon, Ngomba
Ngemba	C	Grassfields	Bafut, Mankon, Ngemba
Nkambe	C	Grassfields	Limum, Mfumte, Yamba
Jarawan	N/C	East-Central Nigeria	Jar, Mbula-Bwazza, Kulung
Bantu A	C	Southern Cameroon	Akɔɔse

^aN=Nigeria, C=Cameroon

^bThis language name is spelt in various ways (Noni, Nooni) in bibliographic references and even within the Noone community.

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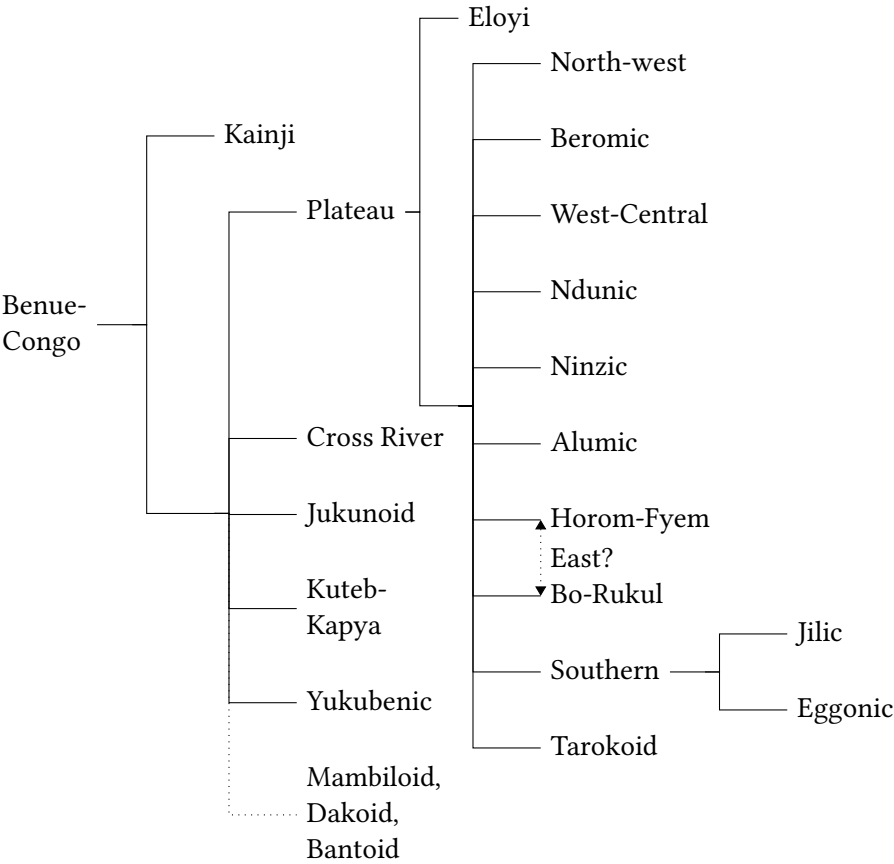


Figure 1: Revised subclassification of Benue-Congo languages

2 Overview of Bantoid subgroups

2.1 Typological features of Bantoid

The descriptive data required to characterise Bantoid languages in ways which would satisfy typologists is not available for many branches. One author who has attempted to summarise the typological differences between Bantu and Bantoid is Hyman (2017). He summarises these in the formula *from syntheticity to analyticity* and discusses the way in “which [Bantoid] languages compensate for the loss of valence-adding extensions, e.g. the applicative, which has multiple functions in Common Bantu” (Hyman 2017: 69). He identifies periphrasis, unmarked double objects, adpositions and nominal constructions as strategies for dealing

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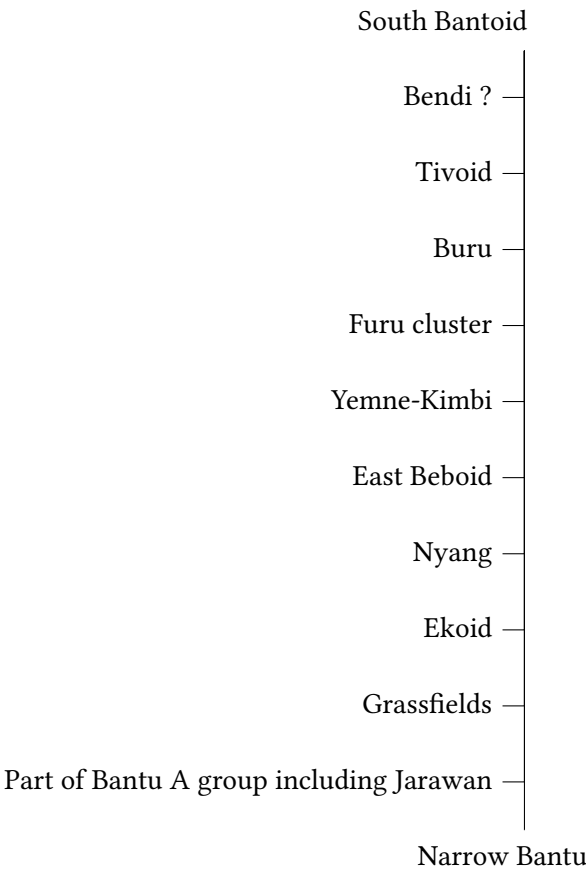


Figure 2: Genetic tree of South Bantoid languages

with the loss of verbal extensions. Table 2, adapted from Table 3 in Hyman (2017: 74), summarises the sort of contrasts which can be expected.

This view of Bantoid is heavily influenced by Grassfields and would not apply to Dakoid, or indeed cover the internal diversity of Mambiloid. Nonetheless, it provides a useful division between the complexities of a significant body of languages and the very different features of Bantu. Bantoid tone systems are the outcome of the erosion of Bantu-like morphology, with corresponding consequences in other areas of morphosyntax. The shortening of words almost inevitably leads to more complex tones, through the avoidance of homophones.

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Table 2: Canonical Bantu compared with Bantoid (Hyman 2017: 74)

Feature	Canonical Bantu	Bantoid
phonology	minimum word = 2 syllables	maximum stem =mostly 2~3 syllables
morphology	highly synthetic, agglutinative	less so, gradual move towards analyticity
verb extensions	many, mostly marking valence	few, mostly marking aspect
unmarked objects	multiple	at most two, ultimate limitation to one per verb
object marking	head marking on verb	various prepositions and/or serial verbs
ditransitive verbs	a few (* <i>pá</i> ‘give’)	few or none

2.2 Issues in tone languages in the Bantoid area

2.2.1 The changing nature of tonal description

The recognition that African languages are tonal goes back more than a century. Banfield (1914, 1916) published a dictionary of Nupe which marked all three tone levels and glide tones, completely and accurately. Nonetheless, the difficulties of hearing and marking tone have meant that various types of publications continue to omit tone-marking up to the present. In addition, if the focus is on orthography, as it relates to bible translation, tonal diacritics are seen as potentially deterring readers and are either severely reduced or omitted.

Readers should be aware that older descriptions of tone tend to under-discriminate tonal distinctions. For example, earlier analyses often underestimate the number of tone heights. A failure to understand the operation of downstep allowed languages with two levels plus downstep to be assigned three tones (cf. Snider 2026 [this volume]). Most importantly, the different types of glide tones were simplified to rising (ˆ) and falling (˘). However, many languages distinguish between multiple contour tones, for example Low-High (ˆ˘) and Mid-High (˘ˆ) and not marking these would fail to make key distinctions. A lesson must be drawn from this, that some descriptions of tone are simplified for non-linguistic reasons, and that modern descriptions are likely to be more

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nuanced than older ones. An additional problem with interpreting descriptions of tone is the use of outmoded theoretical frameworks, for example, Government & Binding. Mbuagbaw (1996) uses this to interpret downstep in Denya. I have omitted redescribing such materials since I remain uncertain as to exactly the nature of the phenomena being analysed.

Tone has at least two major roles in this area, lexical and grammatical. Typically, in languages with large numbers of very short words, tone develops and becomes more complex in order to distinguish short items. Mambiloid is a good example of this, where languages like Vute have longer words and noun morphology (Thwing 1987), whereas Gembu Mambila has four tones plus complex contour tones, and has lost almost morphology (Connell 2001). Grammatical tone is also very common and is typical of many Grassfields languages but also Tikar and some Tivoid languages, such as Iyive (cf. Rolle & Bickmore 2026 [this volume]). Ideophones usually have highly distinctive tonal patterns and these are featured in some descriptions.

2.2.2 Tone in musical performance

Africa is well-known externally for its “talking drums” (Carrington 1949, Dugast 1955, 1960). This is a popular reflection of the capacity of instrumental performers to mimic the contours of speech tones and thus to communicate utterances to an audience, also known as “speech surrogacy” (for reviews see papers in Sebeok & Umiker-Sebeok (1976) and Blench (1985, 2010). The use of instruments in this way is covered in more detail in McPherson (2026 [this volume]). These practices are often aligned with “whistle-speech”, where performers communicate using whistling, most famously the *silbo* in the Canary Islands, though also recorded in other parts of the world (e.g. Turkey) (Meyer 2015). However, these systems are equally used in non-tonal languages, reflecting second formants, and so not exactly comparable. Whistle-speech is not unknown in this region, and found in mountain areas, for example among the Chamba Daka in Eastern Nigeria. In northern Cameroun it is used by initiates in circumcision cycles, when the initiates are forbidden to speak to one another.

However, the vast body of speech surrogacy in Africa is instrumental, and almost any instrument can be adapted to “talk” in this way. In the Bantoid area, flutes, whistles, xylophones and slit-gongs are the most common instruments used. There are a wide variety of systems, which vary from a few coded phrases which have to be known to the audience in advance, to complex imitations of contours, most typical of flutes, where hearers can decode previously unfamiliar phrases.

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2.2.3 Tone in orthography

As previously mentioned, the goal of language description is frequently the development of a working orthography. Highly tonal languages thus need to mark tone. Globally speaking, there are five categories of orthographic tone-marking;

- a) Diacritics, most often developed from European accent marks, and common in Africa (but also in Chinese)
- b) Superscript numerals, where tone levels are assigned numbers and follow either the word or syllable (problematic with contour tones)
- c) Arbitrary punctuation conventions, for example, placing a single arbitrary diacritic before a word, where for example singular and plural are distinguished only by tone
- d) Use of ideograms, which contain segmental and tonal information (typically East Asia)
- e) Marking word level tone with final consonants unused in the orthography (most common in SE Asia, where word tone is generally static and lexemes are uniformly vowel-final, for example Hmongic)

Ideograms are most common in East Asia, but there is one example in the Bantoid area, namely Bamun, an Eastern Grassfields language which has an ideographic script, developed by Sultan Njoya in 1896, where the symbols stand for entire words (Dugast & Jeffreys 1950, Orosz 2015). Unused consonants as finals has been trialled in Africa, but this is not definitively in use anywhere. African languages more commonly have mobile tones, that is they can change within specific morphosyntactic contexts and thus a fixed equivalence is not easy to define for orthographic purposes.

3 Tonal systems by subgroup

3.1 Overview

Bantu is reconstructed as having two tones, without any complications such as glides and downstep (Guthrie 1967–1971, Marlo & Odden 2019). While this is characteristic of numerous Bantu languages in the central area, many languages, especially in Zone A, have a more elaborate system, which resembles more closely the three and four-tone systems which are common in Bantoid. Table 3 compiles

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a sample of descriptions of tone systems for Bantoid and Bantu Zone A, only listing languages with three tones or more. It is not always possible to be certain of the phonemic status of these contrasts, as some authors present the “tone system” without comment, but it can be assumed that the description is intended to be phonemic in most cases.⁴ An asterisk * marks descriptions which are probably phonetic.

In principle, these tones apply to single vowels and where differing tones occur on VV sequences, contour tones are heard. However, some authors show VV sequences to clarify contour tones, even where vowels are short (e.g. Eyoh 2010). There is clear relationship between the year of a description and the complexity of a tone system. Older accounts broadly show simpler tone systems; as researchers are more attuned to these systems, greater richness has emerged.

A characteristic feature of Bantoid is the widespread presence of floating tones (cf. Rolle et al. 2026 [this volume]). A floating tone is a morpheme or an element that contains neither consonants nor vowels, but only tone and which cannot be pronounced by itself. They arise from the deletion of tone-bearing segments, usually vowels, where the tone remains and interacts with those following it. Floating tones are common across Niger-Congo and a substantial literature describes their function, for example in Ga (Paster 2003). They are also widespread in some Bantoid branches, including Grassfields (e.g. Clark 1993).

Tone reconstruction is a more complex task than vowels and consonants in part due to the differing descriptions of tone in diverse authors and a failure to distinguish clearly surface and underlying tone. Some branches, such as Jarawan, lack reliable synchronic descriptions of tone. However, if the evidence from North Bantoid is relevant, then early Bantoid probably had either two or three level tones, as well as rising and falling. However, in many branches of Bantoid, the downstep derives from an underlying L tone between two surface Highs, the L tone persisting after a tone bearing unit had been lost. Surface contour tones are common but only further work will determine if they are an underlying feature of Bantoid.

3.2 Dakoid

Dakoid consists of three languages, Sama Mum [=Chamba Daka], Gãã [=Tiba] and the little-known Dō language (Blench 2012). Both Sama Mum and Gãã have

⁴In Table 3, the notation of tones follows the source material and some authors are more precise than others in marking contour tones. In languages with just two tones, Rising is the equivalent of LH and Falling HL. However, where there are more than two tones, the exact profile of contour tones is unclear in many sources.

Table 3: Tone systems in Bantoid and Bantu Zone A

Bantoid Group	Language	Tones ^a	Reference
Dakoid	Sama Mum	H,M,L,R,F	Boyd & Sa'ad 2010
	Mambila of Gembu	Four level tones, numerous glides	Perrin 1974, Connell 2016; Kwanja (Weber & Weber 2008); Wawa (Martin 2012)
Tikar	Tikar	H,L,R,F	Stanley 1991
Bendi	Bekwara	H,M,L, four glide tones*	Stanford 1967
Tivoid	Ugarə	H,L,U,D	Cassetta & Cassetta 1994
Tivoid [?]	Esimbi	H,M,L,R,F	Koenig et al. 2015
	Buru	H,M,L	Koops (ms.)
Nyang	Denya	H,L,D,R,F	Mbuagbaw 1996
Beboïd	Noone	H,M,L,HL,HM,H,M,L,F,MR,LH*	Lux & Lux 1996
Furu	Bikyak	H,M,L, (?VL), F*	Kießling (p.c.)
Ekoid	General	H,M,L,R,F	Crabb 1969
Mbe	Mbe	H,L	Pohlig 2006
Ambele	Ambele	H,M,L,R,F	Nganganu 2001
Western Momo	Manta	H,M,L	SIL Wordlists
Menchum	Modele	H,L,F	Boum 1981
Ring	Isu	H,M,L, four glide tones*	Anderson 2014
Momo	Ngwo	H,M,L,R,F	Eyoh 2010
Eastern Grassfields	Fe'fe'	H,M,RL,L,R,F	Hyman 1972
	Ngomba	H,D,L	Satre 1997
Bamileke	Bamileke	H,L or H,M,L	Tamanji 2009, Mfonyam 1990
Ngemba	Bafut	H,M,L, three glide tones	Hamm et al. 2013
Nun	Chufie'	H,M,L,R,F	Eyoh & Hedinger 2008
Nkambe	Mfumte	H,L,R,F	Hedinger 1987
Bantu Zone A	P-Manenguba	H,L,R,F	Van de Velde & Idiatov 2022
Jarawan	Mbula	H,L,R,F	

^aH = High, M = Mid, L = Low, R = Rising, F = Falling, U = Upstep, D = Downstep, VL = Very Low. Sequences of two letters describe combination tones, thus LF is Low falling.

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three level tones and contour tones. The inventory for Sama Mum is shown in Table 4.

Table 4: Sama Mum tones (Boyd & Sa’ad 2010)

Level	Diacritic	Contour	Diacritic
high	´	high-low falling	^
mid	-	high-mid falling	⊠
low	`	low-mid rising	⊠
		low-high rising	˘

Sama Mum has characteristic doubled vowels and thus multiple sets which are segmentally identical but distinguished by changing tone levels. Thus;

- (1) a. sáá ‘ooze’
- b. sǎā ‘oribi’
- c. sàà ‘co-wife’
- d. sǎā ‘bird, net’
- e. sàà ‘to walk about’

Boyd & Sa’ad (2010) also describe downstep for Sama Mum, which has the effect of lowering the tone heights of following words. This is connected with the presence of contour tones. For example, the word *wōók* ‘water’ has a downstep, such that in a sentence such as *wōók dǎóri* ‘water fell’, any subsequent words are automatically lowered, in contrast to words where a downstep is not present. Sama Mum has tone-bearing nasal consonants and in particular, the imperative marker is a suffixed *-n̩*. It can agree both segmentally and tonally with the verb to which it is attached. For example, *sāt(-)n̩* ‘speak!’, can be realised either as *sān̩* or even *sān̩̩*.

3.3 Mambiloid

Tone is one of the most difficult aspects of Mambiloid languages, since apart from the level tones, almost all languages also have complex glide tones, on both single V segments and VV sequences. They are described in greater detail in Hamm et al. (2026 [this volume]). Very often transcriptions have changed in the course of research. For example, Martin (p.c.) reports initially transcribing Wawa [www] with three tones, only to revise her notation to incorporate a fourth tone. Kwanja, similarly, is now recognised as having four tone levels, although in the

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orthography they are only marked by diacritics, acute for both tones 1 (high) and 2 (upper mid) and grave for tone 4 (low). Tone 3 (lower mid) is unmarked (Weber & Weber 2008). Table 5 presents evidence for the four-way tone height contrast in the Njanja dialect of Kwanja [knp];

Table 5: Four-way tone height contrast in Njanja (Robson 2010)

Level	Njanja	Gloss	Njanja	Gloss
Low	mgbà	‘chief’	sìŋ	‘chimpanzee’
Mid	njē	‘mother’	fāŋ	‘rope’
High	mbá	‘tribute’	tíŋ	‘vagabond’
Super-High	ŋgǔ	‘warthog’	sǐŋ	‘chimpanzees’

Mambila lects are divided into two clusters, some with four tones, others three (Connell 2001).

3.4 Tikar

Tikar [tik] is a cover term for three related dialects spoken in the Cameroun Grassfields, Tikari, Tige and Tumu. The principal grammatical description is Stanley (1991). Tikar has three contrastive phonetic tone heights;

- (2) a. sòn ‘slave’
- b. sōn ‘axe’
- c. sòn ‘raffia fibres’

In Tikar orthography, mid-tone is unmarked, but I use a macron here to avoid the assumption that tone marking has been omitted.
and four contour tones;

High-Low (ˆ), Mid-Low (⌘), Low-High (ˇ) and mid-High (⌘)

as in Table 6.

Stanley (1991: 18) notes that the only nouns of classes 1 and 6 can take a mid-tone, and the others take a low tone by default. She argues that the underlying tonal system could in principle be reduced to a two-way contrast, low and non-low, in which case ‘slave’ and ‘axe’ above would be merged, analytically. This affects the determiner system as the associative toneme is suffixed to the tone of the determiner. This occurs when;

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Table 6: Tikar contour tones

Tikar	Gloss
ṁḃlî	‘Vute people’
ḃliḂ	‘Vute language’
glwǎ	‘frog’
mḃaḂ	‘water’

- a) The determiner is from classes other than 1 and 6
- b) The determiner is low tone
- c) The determiner is from classes 3 and 5

For class 1 it has no effect;

- (3) ṁjî ḡàṁ
cl.1 cl.1
stalk okra
‘okra stalk’

For classes 3,5 the high tone determiner combines with the tone of the head noun;

- (4) kòn + ‘ + zǎṁ → kǒṁ zǎṁ
pot DET yam cl.3 cl.5
‘the pot of yams’

resulting in a rising tone in this example.
The high tone determiner raises the prefix of the plural for classes 2,4,6.

- (5) ṁsòn + ‘ mē-tǎ’ → ṁsòn mē-tǎ’
village Bamoun village Bamoun
cl.5 cl.2
‘the Bamoun village’

In this case, the low tone mē is raised to mē̄.

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3.5 Bendi

The Bendi languages are located in the northern part of Cross River State in south-eastern Nigeria and adjacent parts of Cameroon. The most well-known is Bokyi [bky], which has a large number of speakers which still lacks a modern phonological description. However, there is a comprehensive grammar of Bekwara [bkv] which has a three-level tone system, with four distinctive glide tones (Stanford 1967: 32). These are;

High-Low (^), Mid-Low (⊠), Low-High (˘) and Low-Mid (⊞)

Bekwara shows vowel length contrast, and tones can be spread across doubled vowels, giving the impression of contour tones, as in (6), which strongly resembles the Sama Mum system (Section 3.2) (cf. Smolders 2026 [this volume]).

- (6) a. fāā ‘to roast’
b. fāà ‘to teach’
c. fàà ‘to grind’

Monosyllabic verbs fall into tone classes according to their height, as in the following example.

- (7) High bé ‘come’
Mid jī ‘eat’
Low jì ‘steal’

However, the high tone class only has a single member, and so could effectively be reduced to low/non-low. Disyllabic verbs are similarly restricted, including only three patterns and no high tones (Table 7).

Table 7: Tone patterns in Bekwara disyllabic verbs

Tones	Bekwara	Gloss	Bekwara	Gloss
Mid-Mid	fāā	‘roast’	tībī	‘dig’
Mid-Low	fāà	‘teach’	tībì	‘be thick’
Low-Low	fàà	‘grind’	tìbì	‘weave’

Bekwara has a large number of what would now be called ideophones (Type 2 adverb in Stanford’s terminology, Stanford 1967: 253ff.). Restrictions on tone in ideophones are quite different. Almost all ideophones are reduplicated, with

both segments and tone copied exactly. The great majority of ideophones have “flat” tonal profiles, i.e. each syllable is at the same tone height, although there are some exceptions (Table 8). Ideophones do not permit contour tones and although Bekwara allows all three tone heights, no high tones occur in the examples with internal tone change.

Table 8: Bekwara ideophones

Bekwara	Gloss
gógó	‘describes illness’
gbūrūgbūrū	‘long ago’
kpàŋkpàŋ	‘smoothly I’
tèrètèrè	‘smoothly II’
byèrèbyèrè	‘long ago’

Further work on Bendi languages remains a high priority.

3.6 Tivoid

Tivoid is a group of nineteen languages spoken principally in the Cameroun Grassfields, but with a significant expansion of the Tiv language into southeast Nigeria, especially Benue State. Civil insecurity in the homeland of smaller languages such as Iyive and Iceve has forced many speakers from Cameroun to become refugee communities in Nigeria. The affiliation of Esimbi [ags] to Tivoid remains controversial. Examples of languages with descriptions of the tonal system are Ugare [iyo] (Cassetta & Cassetta 1994), Esimbi [ags] (Koenig et al. 2015), Iyive [uiv] (Foster 2012) and Iceve [bec] (Cox 2023).

There are five contrastive tones in Esimbi: three level tones (H,M,L) and two contour tones. Rising tone is low-mid (̎), and the Falling tone is mid-low (̏). Tone plays an important role in Esimbi morphosyntax. Coleman et al. (2004: 28) say for Esimbi;

Tense/aspect is usually indicated morphologically through verbal affixation and tonal distinctions, and also syntactically through the use of auxiliary verbs. Verbs are inflected for tense, aspect and mood, and also for agreement with the subject in person and number or noun class. Most affixation is done with prefixes, though there are a few cases of verbal suffixes. Tonal

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distinctions are primarily on the verb root. A number of tense/aspect categories exist in the language: Imperfective, Stative Imperfective, Perfective, Perfect, Progressive (Simple and Reduplicated), Past Habitual, and Future.

As in this example;

- (8) a. ùhìndī ‘he/she ran’ Perfective
b. ùhìndí ‘he/she runs’ Imperfective

Suzanne Foster (2012) has described the phonology of Iyive. Iyive recognises two tone heights and Falling and Rising contour tones (Table 9).

Table 9: Iyive contrastive tones

Height	Example	Gloss
H	/ná/	‘to give’
L	/dù/	‘to remove’
HL	/wêk/	‘to breathe’
LH	/wǎ/	‘to grind’

Although a noun-class language with alternating prefixes for some classes, many nouns are distinguished only by tone (Table 10).

Table 10: Iyive tonal distinctions in nouns

Iyive	Gloss	Iyive	Gloss
/ífú/	‘fish sg.’	/ífú/	‘fish pl.’
/ihó/	‘knife’	/ihó/	‘day’
/úkù/	‘clothes’	/úkú/	‘death’
/ítǎ/	‘hair’	/ítǎ/	‘name’

Changes on the tone in pronouns mark tense/aspect distinctions;

- (9) à sòl á sòl
he looked he will look
(10) è dzènd é dzènd
he walked he will walk

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as in Plateau languages, this almost certainly arises from the assimilation of an auxiliary, although this hypothesis is not given in the source.

Changing tone on both the pronoun and the verb stem distinguishes between 2nd and 3rd person in the present tense;

- (11) a. ú dzèndí ‘you are walking’
b. ù dzèndì ‘he is walking’

A question always ends with a falling tone which argues that the question marker is a final HL clitic. Compare;

- (12) a. ù nóndón ‘she is cooking’
b. ù nóndónô ‘is she cooking?’

There is also vowel insertion to give a TBU for this morpheme to surface, evidenced in the first example.

or;

- (13) a. ù jàtè ‘she is farming’
b. ù jàtê ‘is she farming?’

The question marker might be replacive since it replaces the low tone of the second question.

Cox (2023: 41 ff.) describes the tone system of Iceve-Maci [bec] which has two contrastive tone levels, High and Low. A word-final low tone is normally realised with low falling pitch, but the pitch is low when there is a following floating High tone. In Table 11 pairs of words with identical segments compare those with floating tones and those without.

Downstep is well-attested in Iceve-Maci. Automatic downstep (where the low tone is overt) is particularly noticeable in HLH tone patterns, which are realised as high-mid-low. For example;

- (14) /táⁿb-ŲÍŲ/ [tambili] - - - ‘listen’
/í-fíⁿga-/ [Ifɪŋga] - - - ‘nose’
/ò-nóⁿbòsós/ [ɔnɔmbɔsɔs] - - - ‘man’

Ugare is analysed as having two level tones plus downstep (Cassetta & Cassetta 1994). Downstep is triggered by a sequence of two low tones, which occurs both within a word and across word boundaries. As a result, the sequence high-low-high always comes out as high-low-downstepped high (Table 12).

However, Ugare also has upstep ($\uparrow$), a consequence of floating tones. Table 13 shows an example of how a (high) floating tone triggers upstep when marking the perfect in Ugare.

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Table 11: Iceve floating tones in nouns (2 = Low, 3 = Mid, 4 = High)

Surface	Tonal realisation	Gloss
ì-vú	[23]	‘goat’
í-vú-’	[44]	‘goats’
ì-kpà	[2+ Falling]	‘bag’
ì-kpá	[23]	‘tree sp.’
í-ká ⁿ bì-’	[442]	‘grasshopper sp.’
í-ká ⁿ bí-’	[444]	‘crab’
ì-kòwú	[223]	‘chicken’
ì-kówú	[233]	‘spoon’

Table 12: Downstep in Ugare (Cassetta & Cassetta 1994)

Utterance	Tone Pattern	Gloss
(v)ínùnú	312	‘birds’
úkìhí	312	‘worm’
úfò úsóhɔ	31 222	‘animal’s hair’

3.7 Buru

The Buru [bqw] language is spoken in a single village east of Baissa, below the Mambila escarpment in Sardauna LGA, Taraba State, Nigeria. The only data on Buru is a manuscript wordlist collected by Robert Koops in the 1970s. As transcribed, Buru has three level tones and a falling tone, but there is no tonological analysis.

Table 13: Upstep in Ugare (Cassetta & Cassetta 1994)

Tense	Ugare	Tone pattern	Translation
Past	ísóhó íkúndú	- - - - -	‘The animals jumped.’
Perfect	ísóhó ↑íkúndú	- - - - -	‘The animals have jumped.’

3.8 Nyang [=Mamfe]

Nyang [=Mamfe] consists of three languages, Kenyang [ken], Denya [anv] and Kendem [kvm], spoken in Manyu & Kitwii divisions of Southwest Region in Cameroun (Tyhurst 1983). There are phonological descriptions of both Denya (Mbuagbaw 1996) and Kenyang (Mbuagbaw 2000). Ojong (2005) is a reconstruction of Proto-Nyang which proposes a two-tone height system for the proto-language. Denya has two level tones, High [ˈ] and Low [ˌ], as well as the contour tones, Falling [ˈˀ] and Rising [ˌˀ] as well as downstepped High [ˈ↓H] and down-drift (Mbuagbaw 1996). Kenyang is less well-described but presentations of other aspects argue for a similar system.

Denya verbs and shorter have four tone patterns, H, L, HL, and LH. Nouns and verbs can be monosyllabic and disyllabic (Table 14).

Table 14: Tone patterns on Denya verbs

Tone	Example	Gloss	Example	Gloss
H	tí	‘to escape’	ɲméré	‘to press’
	tú	‘to collide’	gáré	‘to tell’
	wé	‘to fasten’	pélé	‘to watch’
L	tì	‘to erase’		
	kè	‘to sink’		
	kpè	‘to enter’		
LH			dèmbyó	‘to doubt’
			gèngá	‘to intervene’
HL			ɖǣntè	‘to intervene’

Trisyllabic nouns are common with a correspondingly wider range of tone patterns. Mbuagbaw (1996) explores the rules of downstep in the framework of Government and Binding Theory, but his argument is not reproduced here.

3.9 Beboid [= former East Beboid]

Hombert (1980) first identified a Beboid [=Misaje] subgroup of Bantoid languages spoken principally in the North West Region of Cameroun, although two languages are also spoken over the border in Nigeria. Hamm et al. (2002) followed Hombert in splitting them into two major divisions, Eastern and Western. However, research affiliated to a project directed by Jeff Good has established that

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Beboid is not a unity and that the languages in Western Beboïd constitute a distinct branch of Bantoid, for which the name Yemne-Kimbi has been proposed (Section 3.10). Descriptions of tonal systems exist for several languages, including Noone [nhu] (Hyman 1981), a dissertation by Richards (1991) concerning the phonology of three Eastern Beboïd languages (Noni, Ncane, and Nsari), Cox (2005) a phonology of Kemezung [dmo], Boutwell (2014) a sketch grammar of Mungong [ncr], Tabah (2020) a grammar sketch of Naami [asj], Tschonghongeï (2018) a grammar of Mbuk [bpc] and Boutwell (2020) a full grammar of Ncane [ncr].

Noone has three contrastive tones, L,M,H can be demonstrated by a series of minimal triplets, for example;

- (15) a. byé ‘fishes’
b. byē ‘goats’
c. byè ‘fish’

Any VV sequence can have level tones, but up to five contrastive possibilities, where the two vowels have different tones. Of six potential contrasts, only Mid-High is excluded.

Cox (2005) analyses Kemezung with three level tones and four contour tones (Table 15).

Table 15: Kemezung tones (Cox 2005)

Level	Diacritic	Contour	Diacritic
high	ˈ	high-low falling	ˆ
mid	-	mid-low falling	⌘
low	ˋ	low-mid rising	⌘
		mid-high rising	⌘

Before a pause, low tone becomes extra low, which could be treated as a type of automatic downstep.

Ncane tones have been described in some detail in Boutwell (2014, 2020). Apart from three level tones there are five contour tones and both super-high and super-low (Table 16).

Ncane is a good example of how more in-depth research results in an expansion in the complexity of tonal systems. The preliminary description of Ncane tonology in Boutwell & Boutwell (2008) has fewer contour tones and did not

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Table 16: Ncane tones (Boutwell 2020)

		Tone	Notation	Diacritic
Level		High	H	ˈ
		Mid	M	-
		Low	L	ˋ
		Super-high	SH	ˊ
		Super-low	SL	ˋˊ
Contour	Falling	High-Mid	HM	ˆ
		High-Low	HL	˘
		Mid-Low	ML	ˋ˘
	Rising	Mid-High	MH	ˊ˘
		Low-High	LH	ˋˊ

Table 17: Ncane tone melodies

Tone melody	Example	Gloss	Class
H	Ø-lú	‘marriage’	5
M	Ø-ñ sū	‘pepper’	1
L	Ø-ŋ kù	‘box’	1
HM	Ø-mŋ	‘razor’	5
HL	ŋ f-ɸ	‘stirring stick’	3
ML	Ø-mû	‘person’	1
MH	Ø-ŋ gū́lú	‘vulture’	1

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recognise super-high and super-low. Table 17 shows examples of the principal Ncane tone melodies;

The occurrence of Super high (SH) and super low (SL) tones is usually attributable to specific tonal or grammatical contexts and they are not phonemic. The source of a SH tone can be the result of a grammatical H realized on a word with a lexical H, as illustrated in the examples below. The example demonstrates a grammatical H (indicating Habitual) superimposed on the lexical H of the verb *tó* ‘come’, resulting in a SH.

- (16) *ɲɔ̀ɲi tɔ́* *wù kà:* *ɲgɔ́*
 L H=H L ML HM
 Nji HAB=come 3SG catch termite
 ‘Nji always catches termites.’

The Super-Low (SL) tone has multiple origins. In the case of the ML noun, the contour is realized as a LSL fall. In phrases with three or more consecutive Lows, the phrase-final L is realized as a SL, as below.

- (17) a. *Ø-kùŋ* *wù* *bǐ*
 L - L L LM
 c1-python c1AM c9.goat
 b. *Ø-kfũ* *wù* *Ø-kà(ǎ)*
 L - ML L L - ML
 c1-python c1AM c1-rattle

3.10 Yemne-Kimbi

The Yemne-Kimbi languages [formerly Western Beoid] consist of some eight languages (dialect divisions are unresolved) spoken in the Cameroun Grassfields, geographically intertwined with the Beoid languages (Section 3.9). The overall situation is described in Good et al. (2011) and there are full-length grammars of Mungbam (Lovegren 2013) and Mundabli (Voll 2017) and a grammar sketch of Bu in Yonga (2012).

Lovegren (2013) describes Mungbam tones in a rather unconventional style, so I have extracted what I take to be the tonal system and represented it in a more standard fashion (Table 18).

Mungbam has what Lovegren (2013: 83) describes as “possessive lengthening”, where the stem vowel of a possessed noun is lengthened with respect to the unpossessed form. The reduplicated vowel is either higher or lower, depending

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Table 18: Mungbam tones

Prosody	Symbol	Tone profile
high	´	H
mid	-	M
low	`	L
low-mid	⊠	LM
mid-low	⊠	ML
mid-high	⊠	MH
high-low	^	HL
high-mid	⊠	HM
superhigh	"	SH
superhigh-mid	"-	SM

Table 19: Possessive lengthening in Mungbam nouns

Tone pattern	Non-Possessed	Gloss	Possessed	Gloss
L ⁻ → LM ⁺	bə̀-mfi	‘cl2-slaves’	bə̀-mfiĩ mǎ	‘my slaves’
M ⁻ → M [○]	fi-nsā	‘cl19-needle’	fi-nsā mǎ	‘my needle’
M ⁻ → M ⁺	ú-wō	‘cl3-moon’	ú-wōó mǎ	‘my moon’
H ⁻ → H ⁺	bə́-ndjón̩	‘cl2-stories’	bə́-ndjón̩ mǎ	‘my stories’
S → S	ú-gbǎ	‘cl3-rope’	ú-gbǎ mǎ	‘my rope’
HM → SM	ká-tsɔ̃⊠	‘cl12-rooster’	ká-tsɔ̃ mǎ	‘my rooster’

on the tone pattern of the original. Table 19 illustrates the operation of this system with different nouns;

The notation reproduces Lovegren (2013: 84) exactly. Nouns with an M[○] or S [=Superhigh] stem tone do not undergo extension.

The tone system of Mundabli is characterised by a similar level of complexity. There are four level tones and several contour tones (Voll 2017: 15). Table 20 shows Mundabli tone heights and contour tones.

Mid, high and superhigh tones are phonetically level in both final and non-final position, but the low tone is also level in non-final position but phonetically falling before a pause. As this alternation is completely predictable, low tone is always transcribed with a grave accent. Contour tones may occur in non-final position or before a pause, but their pronunciation is generally the same in any

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Table 20: Mundabli tones

Prosody	Symbol	Example	Gloss	Note
high	H	kú	‘home.LOC’	
mid	M	kū	‘mole rat’	
low	L	kù	‘rope’	
superhigh	SH	kǔ	‘belly’	
low-high	LH	kǔ	‘to grab’	(clause-final)
mid-low	ML	kuǎ	‘to clap’	(clause-final)
high-mid	HM	kǔ	‘wisdom’	
high-low	HL	gân	‘gown’	

position.

Monosyllabic noun stems, which account for the great majority of nouns, may bear any of the eight level and contour tones. Nouns usually retain the same tone in all contexts and they are generally not influenced by adjacent elements or prosodic boundaries. The respective singular and plural forms of nouns belonging to genders other than 9/10, including their tones, are usually identical. A few exceptional nouns which belong to genders other than Gender 9/10 show tone-raising from mid to high in singular and plural forms as well as segmental change (Table 21).

Table 21: Mundabli irregular nominal tone plurals

Singular	Gloss	Plural	Gloss
wān	‘child (cl1)’	ɲwóm	‘children (cl2)’
fō	‘head (cl3)’	fā	‘heads (cl7a)’
ɲū	‘farm (cl3)’	ɲwé	‘farms (cl7a)’
tʃyē	‘stone (cl3)’	té	‘stones (cl7a)’

Nouns in Gender 9/10 (Voll 2017: Table 3.13) retain identical segments but show tone-raising in the plural as shown in Table 22.

The class pairing has no obvious semantic unity.

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Table 22: Mundabli tone plurals in Gender 9/10

tone Cl. 9 sg.	tone Cl. 10. pl.	Example	Example	Gloss
L	ML	kù	kuꞤ	‘rope/s’
LH	S	tsǎ	tsǎ̃	‘baboon/s’
LH	ML	tsǎŋ	tsaꞤ	‘palm kernel/s’

3.11 **Furu**

Furu is the name given to a cluster of three languages, Busuu, Bishuo and Bikyak, spoken in a remote area of northwest Cameroun in Furu-Awa division (Breton 1993). By the time linguists were able to reach the area, these languages were virtually moribund. The only one for which modern information is available is Bikyak, following an investigation by Roland Kiessling in 2007 (p.c.). They are probably an unresolved branch of Bantoid, although this is yet to be demonstrated. There are three phonetic tone-heights although a minimal triplet to exemplify this has not been recorded. The following pair illustrates a simple two-way contrast.

- (18) a. nám ‘cook porridge’
- b. nàm ‘lie down’

3.12 **Ekoid-Mbe**

The Ekoid-Mbe languages are spoken principally in SE Nigeria and in adjacent regions of Cameroun. They consist of around nineteen closely-related lects, the majority of which are spoken in Nigeria (Crabb 1969). Mbe [Mbè] is a single language spoken in seven villages in Ogoja LGA, Cross River State, Nigeria. The area and the people are known to the administration as Mbube, a designation which includes the Mbe and the Upper Mbe, who speak a different language.

Watters (1978) is a comparative phonology and classification of Ekoid which presents an internal classification shown in Figure 3 (page 384) with some updating.

Both Mbe and Ekoid have High, Low, Rising and Falling tones plus Downstep. Mbe phonology has been described by Bamgboṣe (1967) and more recently in an unpublished phonology and orthography statement (Pohlig 2006). In regards to tone, Pohlig (2006) focuses on the associative system which links two nouns. The associative morpheme is an underspecified high tone vowel. He states that their

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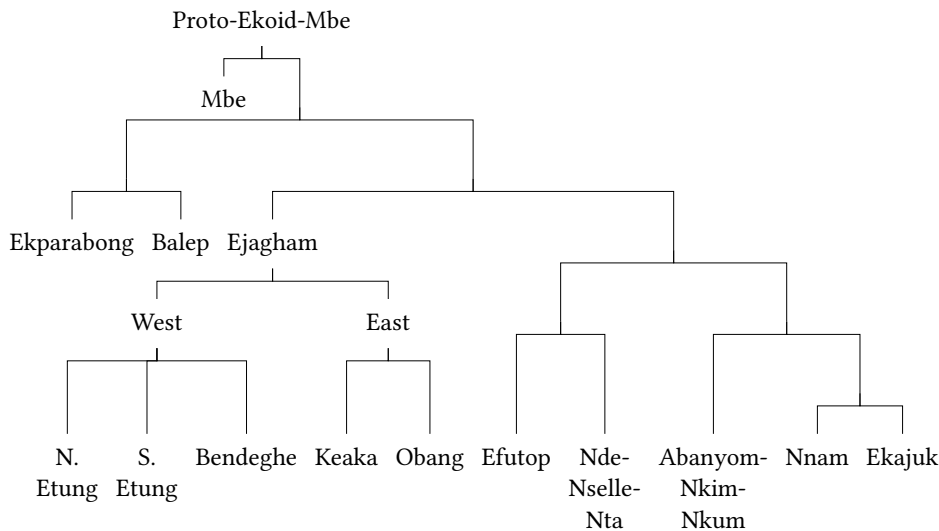


Figure 3: Genetic classification of Ekoid-Mbe languages

allomorphs reflect the noun classes in the phrase but does not actually mark the noun classes. Table 23 shows the tonal outcomes with the associative particle represented as á. The symbol (ˊ) signals a downstepped high tone following. The proposed orthography signals the downstep with a hyphen.

For Nkem, Sibomana (1986) provides a rather brief summary of the system as follows;

Two level tonemes – High (ˊ) and Low (ˋ) with contour tones analysed as combinations of High and Low. High tones can be lowered by Downdrift or Downstep.

In the case of Etung, Edmondson & Bendor-Samuel (1966) argue that by positing word-level tone, it is possible to account for the rising and falling tones, leaving only a three-way contrast. As so often, this is largely a matter of diachronic versus synchronic accounts; the five-way contrast exists on the surface but can be shown to derive from VV sequences that evolve through segment deletion.

Kleiner & Kleiner (1976: 28) have described the tonal system of Ekajuk. It has two contrasting tone levels, high and low, as well as downstep, lowering high tones but not low. In addition there is automatic downstep lowering all high tones after low. A predictable downstep and upstep are found as juncture features. They argue that tone is a feature of the phonological word although each segment

Table 23: Tonal outcomes of the Mbe associative morpheme

Underlying form	Phonetic form	Gloss	Proposed orthography
/ètʃi/ handle	á kēs ^w òm/ ASS matchet	‘matchet handle’	èchi-kèsuòm
/èf ^w ò/ driver	á Ñpób/ ASS air	‘a fan’	èfuò-ñpòb
/èji/ place	á Ñmù/ ASS snake	‘place of the snake’	èji-ñmu
/wùkàb/ heaper	á Ñm ^w àl/ ASS farm	‘maker of yam heaps’	wùkàb-ñmuàl
/Ñk ^w è/ soup	á Ñm ^w àn/ ASS child	‘the child’s soup’	ñkuè-ñmuàn
/Ñwêr/ book	á òtʃúwè/ ASS God	‘the Bible’	ñwêr-Ochuwè
/wùñén/ owner	á ètò/ ASS compound	‘the owner of the compound’	wùnyen-ètò
/beñàm/ meat	á Ñm ^y òr/ ASS body	‘flesh’	bènyàm-ñmyòr

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bears a tone. Sequences of tone occur over the length of a word irrespective of the number of syllables. Table 24 shows examples of tone patterns spread over two syllables but occurring compressed in a single syllable.

Table 24: Tone as a suprasegmental in Ekajuk

One syllable	Gloss	Two syllables	Gloss
bâm	‘close sg.’	bámèn	‘close pl.’
‘ǰwêr	‘book’	‘ǰwérà	‘a book?’
‘édínôṅ	‘bed’	nó:ṅò	‘to lie down’
râṅṅ kámé	‘break the wall’	râṅṅè	‘break’

Ekoid remains poorly covered in modern linguistic descriptions.

3.13 Grassfields

The Grassfields languages, situated around the Ring Road in West, Northwest and Southwest Provinces of Cameroun consist of some seventy-one languages. Figure 4 shows a tentative classification of Grassfields.

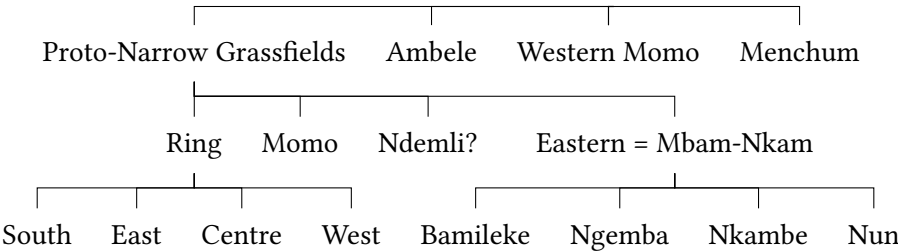


Figure 4: Classification of Grassfields languages

They are characterised by great diversity of tonal and morphological systems, covered in more detail in Hamm et al. (2026 [this volume]).

3.14 Jarawan Bantu

Jarawan Bantu languages are spoken today in eastern and central Nigeria, although there is evidence for populations in Cameroun in the first part of the twentieth century. There are some fourteen languages in total. Strümpell (1910) has a wordlist of the Jarawan Bantu language Mboa, formerly spoken on the

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Cameroun/CAR border near Meiganga. Strümpell (1922–1923) is the only record of Nagumi, formerly based around Natsari, SE of Garoua in northern Cameroun.

Published descriptions of Jarawan Bantu phonology are limited, although Green (2020; 2021) has published on Mbat and Van de Velde & Idiatov (2022) have circulated a powerpoint on Mbula. However, work is under way in several languages, including Mbat, Kulung and Mbula. The author, together with Timothy Terna, are working on Gura [bma] and Labir [=Jaku] [jku], and the notes below reflect these languages. Gura has two level tones plus a single falling tone. Occasionally a phonetic mid-tone was recorded, but these may be downstep. Some examples of tonal contrast;

- (19) a. bàrà ‘to count’
 b. bàrá ‘to see’
- (20) a. díírì ‘to mix’
 b. diìrì ‘to flow’

Labir has three level tones, plus rising and falling tones. Tones are marked as follows;

- (21) High ´
 Rising ˘
 Mid -
 Falling ^
 Low ˘

Below are examples of minimal triplets;

- (22) a. bál ‘to count’
 b. bāl ‘liver’
 c. bàl ‘large river’

Glide tones are very rare and do not occur in nouns. Rising tones are restricted to numerals;

- (23) a. lǔm ‘ten’
 b. nǎn ‘eight’

and falling tones, to lexemes other than nouns;

- (24) a. māmkin ‘someone (female)’
 b. nâpín ‘yesterday’
 c. fákâm ‘to shift’

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Labir, as is common in this area, combines an underlying aspectual system with tense marking. The perfective is marked by a suffix on the verb root -(a)ŋ as in Table 25. The first person has a zero pronoun. The tone of the verb stem changes in the third person singular.

Table 25: Labir subject pronouns with perfective of *wāl* ‘to hit’

Root	Gloss	Example	Gloss
I	‘∅’	-wàlán	‘I hit’
you sg,	‘wú’	wú wàlán	‘you.sg hit’
he/she/it	‘yí’	yí wàlán	‘he; she; it hit’
we	‘sù’	sù wàlán	‘we hit’
you pl.	‘wùŋ’	wùŋ wàlán	‘you pl. hit’
they	‘yà’	yà wàlán	‘they hit’

Green & Haliru (2025) in a so far unpublished powerpoint have argued there is evidence for Consonant-tone interaction (CTI) in Kulung (cf. Beavon-Ham & Pearce 2026 [this volume]). Their hypothesis is that this is an innovation which may be due to contact with Adamawa languages. Their summary findings are that;

- Sonorants are transparent to Consonant-tone interactions – H vs. L contrast maintained
- Voiceless obstruents and implosives are “raisers”
- Voiced obstruents are depressors – always and only permit [L]
- Both voiceless and voiced “prefixed” nasals yield only register lowering – [M] vs. [L]

They argue that there is no evidence for CTI in other Jarawan languages. Van de Velde & Idiatov (2022) have posted a powerpoint on Mbula [mbu] in which they posit a two tone system with the following features;

High and Low tones, plus rising and falling. LH rising tones tend to be realised as M. It is possible to have three tones on one syllable (☒).

This is the first report of a double contour tone in Jarawan Bantu and further detail will be of great value.

4 Conclusions

The Bantoid languages are a nexus of families on the Nigeria/Cameroon borderland, all of which show features which link them to Bantu. This chapter reviews the tonal systems in its diverse branches, although Mambiloid and Grassfields are covered separately. The linguistic interest of Bantoid is its extraordinary morphological and phonological diversity arising from elided syllables, especially nouns with alternating prefixes, which can be linked to the abundance of contour and floating tones. Tones most typically occur on vowels, but can also be assigned to nasals and liquids.

Benue-Congo is similarly diverse, ranging from the reduced systems in some Kainji to the multiple tone heights in Jukunoid. This suggests strongly that early Bantoid had a rich system of noun-classes and verbal extensions, but it is less easy to make proposals for its tone systems. These typically erode and are rebuilt, sometimes into systems of great complexity. Bantoid tonal systems show great variability even within single branches. Moreover, it is very noticeable that more recent descriptions show greater nuance in describing tones, even within the publications of a single author (e.g. the example of Ncane in the publications of Richard Boutwell). The concept of speech surrogacy, the use of musical instruments to “talk”, i.e. to make decipherable utterances by imitating tonal contours is widespread in almost all Bantoid branches. Tracing the pathways by which this occurred has hardly begun, and remains a challenge for future linguists.

Abbreviations

Three-letter ISO codes in square brackets next to individual language names are given throughout. These are taken from Ethnologue.

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Chapter 13

Lexical and non-lexical tone in Mambiloid

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Mambiloid languages have systems of three or four register tones. As is usual in tone systems in African languages, tone functions to establish lexical contrasts and to encode grammatical functions. Rarely found or reported, in some Mambiloid languages tone also serves a deictic function, used, for example, to indicate directionality (ventive vs itive). Tone also plays a role in ‘expressive language’, particularly in shaping ideophones. This chapter presents data from five Mambiloid languages, drawn from both the available literature as well as unpublished work, to illustrate the range of roles, both lexical and non-lexical, that tone fills in Mambiloid. Before discussing tone in these five languages, each in turn, the chapter begins with a brief introduction to the Mambiloid languages, and the extent to which information is available about them. The chapter concludes with a summary discussion of commonalities in tone systems across the five languages, and some speculation as to how the different tone systems came to be as they are.

1 The Mambiloid languages

The Mambiloid languages comprise a sub-group within Bantoid, in some classifications delineated as North Bantoid. They are situated in the Nigeria-Cameroon borderland, on both sides of the frontier, north of and adjacent to the Grassfields in Cameroon. In Cameroon they are found mainly in Adamawa Region, to a limited extent in Northwest Region, and extending into the Centre Region. In Nigeria they are found, with one or two exceptions, on the Mambila Plateau in Taraba State. The area in which they are found is geographically diverse but can broadly be described as comprising three elevations: the Mambila Plateau,

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a dissected plateau largely devoid of trees, and having a mean elevation of c. 1600m; the Adamawa Plateau, at lower elevation (c. 1200m); and lower still, the Tikar Plain (c. 600m). Steep escarpments separate the three levels. The escarpment that marks the division between the Mambila Plateau on one hand and the Adamawa Plateau and Tikar Plain on the other hand also defines the border between Nigeria and Cameroon through this region. On the plateaus the climate is relatively arid, while on the plain it is more humid, and the rainy season is longer.

Of the various ethnolinguistic groups that make up Mambiloid (see [Connell 2001](#) for details), the Mambila is the largest, numbering over 100,000 speakers, and with the bulk of their population inhabiting the Mambila Plateau. There are some 25 Mambila (ISO 639-3 [mzk]/[mcu]; Glottocode: mamb1312) lects.¹ These are grouped into two dialect clusters, with members of each cluster being found on both sides of the Nigeria-Cameroon border. Ndoro (aka Ndoola; ISO 639-3 [ndr]; Glottocode: ndoo1241) is the next largest group with a total population of approximately 60,000 to 65,000. The Ndoro are scattered in several disparate villages, all but one being in Nigeria; of these, some are situated off the Mambila Plateau, to the north or northeast. Other, much smaller, ethnolinguistic groups are also present on the Mambila Plateau: Tep (Glottocode: tepp1235), Mbongno (ISO 639-3 [bgu]; Glottocode: mbon1254), Mvanip (ISO 639-3 [mcj]), Ndunda (ISO 639-3 [nuh]), and Somyev (ISO 639-3 [kgt]; Glottocode: somy1238). Of these, Mbongno is the largest, with perhaps 1,000 speakers; Tep has approximately 300-500, Ndunda 300, and Mvanip approximately 100 speakers, though these figures may be out-dated. At last report Somyev had just two remaining speakers and may now be extinct. On the Cameroon side of the border, in addition to several Mambila varieties, Kwanja (ISO 639-3 [knp]; Glottocode: kwan1276) has two lects (Sundani and Ndung, not including the near-extinct Njanga). Kwanja has a total population of approximately 10,000 speakers. Wawa (ISO 639-3 [www]; Glottocode: wawa1246) has approximately 3,000 speakers divided among four principal lects, Oumyari, Gandoua, Mbenguedje and Ndi (glossonyms are drawn from the principal village in which each is spoken). For Vute (ISO 639-3 [vut]; Glottocode: vute1244) [Thwing \(1987a\)](#) and [Thwing & Watters \(1987\)](#) refer to the ‘Central and Eastern dialects’, which lumps together at least some of the eight varieties mentioned in [Glottolog \(2026\)](#), and conceivably includes others not mentioned. Vute has a total population of approximately 21,000 speakers. Finally, in the north-eastern extreme of the Mambiloid area is Nizaa (ISO 639-3 [sgi]; Glottocode: suga1248), spoken in and around Galim by approximately 10,000 people.

¹Both the ISO code (from ethnologue.org) and the Glottocode (glottolog.org), where available, are given at first mention of a language.

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Apart from Wawa and Nizaa, which are located entirely on the Adamawa Plateau, dialects of the Cameroon-based Mambiloid languages are distributed across both the Adamawa Plateau and the Tikar Plain.

2 Tone in Mambiloid

Tonally, the Mambiloid languages are of interest for their relatively complex inventories and in that tone is found to function both lexically and grammatically, and in at least some languages deictically and in expressive language. As in other African languages, in the Mambiloid languages have what are referred to as ‘register’ tone systems; tones are realized with a relatively steady-state pitch, of relative contrastive height. These may be contrasted with languages where pitch movement and direction of movement are important, e.g. those found primarily in Asia; these are described as ‘contour’ tone languages.² This is not to suggest ‘contours’ are not present, i.e. pitch is not dynamic, in Mambiloid or other African languages. Pitch contours are in fact relatively common, with the key difference being that pitch contours in most African languages, including the Mambiloid languages, can be readily analyzed as combinations of different tones of different levels: e.g. a falling tone as a combination of High + Low; they are thus best seen as tone clusters, or “compound” tones, the term adopted here.

Lexically, Mambiloid languages all appear to have at least three tonal registers, High (H), Mid (M), and Low (L); several have four, splitting the Mid register into Upper and Lower Mid. Languages with three contrastive tone registers include Vute, Nizaa and possibly some Wawa lects, along with East Mambila lects. Those with systems of four contrastive register tones include all West Mambila lects; the Kwanja lects Sundani and Ndung, Njanga, and Oumyari Wawa. The basic tone inventory of others, including Mbongno, Mvanip, Ndunda, Tep and Ndoro, is not entirely clear, but it appears each has at least three contrastive registers. This is interesting given that, according to Maddieson’s (1978) survey, more than half of all tone languages have a basic two-way contrast of H, L; approximately 30% have three level tones, while fewer than 4% have systems of four or more level tones. Seen from this perspective, the Mambiloid languages, and especially those with four tone registers, are of particular interest. As mentioned, level tones can combine to form compound tones. Restrictions exist on possible combinations; for example, in Ba Mambila (Ju Ba, discussed in detail below) at least eight compound tones are found, giving a total inventory of at least 12 surface tone melodies or patterns.

²This basic distinction, still useful, goes back to Pike (1948).

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There is less information available on non-lexical functions of tone in Mambiloid, since for many languages only wordlists are available. However, as is common throughout Niger-Congo, tone is known to serve grammatical functions in all Mambiloid languages for which relevant information is available. These have been described in most detail for Ju Ba and Sundani Kwanja. [Martin \(2012\)](#) discusses grammatical functions of tone in Oumyari Wawa, and some information can be gleaned from [Thwing \(1987a\)](#) and [Guarisma \(1978\)](#) for Vute, and from [Endresen \(1992\)](#) and Theil (ms) for Nizaa. Deictic functioning of tone, to my knowledge, has not been reported other than in Ju Ba, though in Sundani it factors into this function. Information on tone used in “expressive language”, particularly ideophones, is sketchy but has been commented on by [Martin \(2012\)](#) for Oumyari Wawa.

The following sections of this chapter look at tone in five Mambiloid languages: Ba Mambila (Section 3), Sundani Kwanja (Section 4) and Oumyari Wawa (Section 5) each of which has been described as having four contrastive registers. In Section 6 and Section 7 Vute and Nizaa, both of which have three contrastive registers, are examined. For each, the detail and depth of description and discussion is limited by the material available. The different authors use different conventions for marking tone, whether in orthographic rendition or transcription. After describing the conventions use by each at the outset of each section, a common system is adopted to facilitate comparison and discussion. Section 8 summarizes the preceding sections, drawing attention to commonalities found across the different languages discussed in those sections and offering speculation as to how the different tone systems came to be as they are. The tone systems of these languages are complex and operate in multiple ways, such that it is not feasible to give more than a partial account of tone in any one of them in a chapter such as this.

3 Tone in Mambila

Mambila is the largest of the Mambiloid languages with respect to both number of speakers and lectal variation. Broadly speaking, it can be divided into two dialect clusters, named East and West Mambila, roughly following their geographical distribution. Collectively there are some 25 varieties. There are several distinguishing diagnostic features of cluster membership, including their tonal systems. West Mambila lects contrast four tones at the lexical level whereas, as far as has been determined, those comprising East Mambila have three contrastive tone levels. Studies of tone in Mambila have been carried out by Connell (e.g.

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2003, 2017) and Perrin (1974, 1991). Both authors have focussed on Ju Ba, a variety of West Mambila; Connell's work has been concerned with Ju Ba as spoken in the village of Somié, Perrin's the village of Sonkolong and to a lesser extent Atta. The tonal differences across these villages are minor and are only referred here to where an interesting point is to be made. Perrin (1974) also describes tone in Tungba, the variety spoken at Gembu, on the Mambila Plateau. Tungba, like Ju Ba, is a West Mambila lect. To date there have been no detailed studies of tone in any East Mambila lect; here the focus is on Ju Ba.

3.1 Tone in Ba Mambila

Ju Ba has four contrastive register tones, High (H), Upper Mid (UM), Lower Mid (LM) and Low (L). In Perrin's work these are numbered 1 to 4, High to Low; in her transcriptions, superscript numbers follow the syllable, thus: /gu²nu²/ 'squat'; /gu³nu¹/ 'become cloudy'; gu⁴nu⁴ 'bale of straw'. In Connell's transcriptions tone marking follows standard Africanist/IPA conventions: acute accent represents H; macron indicates M (in this case UM); LM is unmarked as it is the least frequently occurring and leaving it unmarked avoids introducing an uncommonly used diacritic; and the grave accent marks L; so, /gūnū/ 'squat'; /gunú/ 'become cloudy'; gùnù 'bale of straw'. In the orthography developed by Perrin (1987/2005) only tones 1 and 4 are marked, with acute and grave accents respectively: *gunú* 'become cloudy'; *gùnù* 'bale of straw'. Tones UM and LM (tones 2 and 3) are unmarked, e.g. *galé* (UM.H or 2.1) 'tremble, shiver' vs *galé* (LM.H or 3.1) 'be spread'. The main rationale behind this system is that it is these two mid tones that are most subject to alternation at the grammatical level. Vowel length is not contrastive in Ju Ba. In the orthography vowel symbols are doubled to accommodate compound (contour) tones, where one element of the compound tone is either UM or LM: *ndóó* /ndóó/ 'eggplant'; *ndóó* /ndóó/ 'frighten'. As this last example shows, the analysis of two mid tones leaves problem in tone marking in the case of compound tones: while there is a standard diacritic for MH (and ML), used here for UMH as in 'eggplant', there is no corresponding diacritic available for LMH (or LML). In transcription, whether phonemic or phonetic, one does not have recourse to doubling the vowel symbol as is done orthographically and the acute or grave accent marking the H or L component comes after the vowel symbol; see also the examples in Table 2. In what follows, we use Perrin's numbering, 1–4, High to Low, to facilitate the more general discussion across Mambiloid varieties, though H, UM, LM, L are also used.

Every syllable bears at least one of the four register tones. Compound tones, i.e. contour tones which can readily be analyzed as combinations of level tones,

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are not uncommon, though possible combinations are restricted, as is their distribution, as discussed below. The different levels on which tone functions in Ju Ba, viz. lexical, grammatical, and deictic, as well as their role in expressive language, are discussed in turn.

3.2 Lexical tone in Ba Mambila

There is sufficient lexical contrast based solely on tone that it is not problematic to find minimal pairs or triplets; minimal quadruplets are less common but do exist, as shown in Table 1. In the table each word is given first in orthographic form, then in phonemic transcription. Tone marking is given following the conventions mentioned above, together with the IPA ‘tone letter’.

Table 1: Examples of lexical tonal contrasts, level tones

<i>Orth</i>	<i>Phon</i>	<i>IPA</i>	<i>Tone</i>	<i>Gloss</i>
báj	báj	ⓘ	1	trench defense around village
baɲ	bāɲ	ⓘ	2	a type of wild cat (genet sp)
baɲ	baɲ	ⓘ	3	begin
bam	bam	ⓘ	3	traditional week of 10 days
bàɲ	bàɲ	ⓘ	4	wound (n.)
gó	gó	ⓘ	1	guardian
go	gō	ⓘ	2	habit, custom
go	go	ⓘ	3	sell
gò	gò	ⓘ	4	journey (n.)
kér	kér	ⓘ	1	sparrowhawk
ker	kōr	ⓘ	2	new field
ker	kər	ⓘ	3	cut, dig
kèr	kər	ⓘ	4	cup used for palm wine
jéré	džéré	ⓘⓘ	1	dirtiness
jere	džeré	ⓘⓘ	2	pity (n.)
jere	džeré	ⓘⓘ	2.1	make bad
jere	džeré	ⓘⓘ	3.1	go up (e.g. smoke)
jèrè	džèrè	ⓘⓘ	4.4	flea (sp.)

These examples reflect the fact that the four-way contrast is restricted to nouns, and occurs only on the initial syllable. Most words in the language are monosyllabic, though disyllabic words are not uncommon; trisyllabic words are

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rare. Nouns bearing tone 3 are rare and many of those that do exist have been shown to be relatively recent innovations (Connell 2016, Connell & Ladd 2020). With monosyllabic verbs only tones 2 and 3 are found, and with disyllabic verbs, tone patterns are restricted to 2.2, 2.1, 3.3, 3.1.

Tonal combinations or sequences of tones on one syllable are limited to 21, 31, 41, 14, 24, 34, 212, and 214 and 314. These patterns appear to define the phonological word, along with others introduced through the addition or influence of grammatical tones, whether a word is of one, two, or three syllables. The tritonal compounds are rare; two, 214 and 314, are attested on monosyllabic or disyllabic nouns; the third, 212, on the past tense marker *naâ*, /na̠˥˥/. Compound tones are invariably on the final syllable of the word; that is, they occur on monosyllabic words, though on disyllabic words they occur only rarely (unless a result of grammatical functioning, discussed below) and when they do, the contour is always on the final syllable. Examples of compound tones in contrast with level tones are given in Table 2.

Evidence that these surface contours on monosyllabic words are indeed combinations of the four register tones, i.e. are compound tones or tonal clusters, comes in different forms. First, within Ju Ba, sub-lectal variation shows the second tone of the tonal compound disappearing, as in (1). This loss affects mainly 21 compounds (i.e. 21 > 2), commonly with nouns where disyllabic forms are also affected, though less frequently with verbs.

- (1) *Sonkolong Somié*
- | | | |
|--------------|--------|---------|
| boór ‘dog’ | /bo/ | /bōr/ |
| ceér ‘road’ | /tʃɛr/ | /tʃē̃r/ |
| temé ‘liver’ | /tēmé/ | /tēmē̃/ |

Second, as described in Section 3.4 below, in some instances the contour comes about through the application of grammatical tone, morphemes comprised only of a tone (cf. Rolle & Bickmore 2026 [this volume]).

3.3 Non-lexical tone in Ba Mambila

Tone in Mambila serves grammatical functions affecting both nominal and verbal constructions, marking for example genitive or associative forms, negation, imperative, and past tense, and deictic functions such as motion toward (ventive) or away from (itive) the deictic centre.

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Table 2: Examples of Ju Ba lexical tonal contrasts, compound tones

<i>Orth</i>	<i>Phon</i>	<i>IPA</i>	<i>Tone</i>	<i>Gloss</i>
mân	mân	ⓘⓘ	14	thus
maáb	maⓘb	ⓘⓘ	21	be quiet
maàn	mân	ⓘⓘ	24	small
maán	ma´ n	ⓘⓘ	31	learn
béh	bó	ⓘ	1	1PL
beh	bə	ⓘ	3	ASP.HAB
bèh	bə	ⓘ	4	place
bêh	bê	ⓘⓘ	14	1SG + 2SG
beéh	bəⓘ	ⓘⓘ	21	to spoil, ruin
beéh	bə´	ⓘⓘ	31	to braid
beèh	bê	ⓘⓘ	24	1PL.POSS
lám	lam	ⓘ	1	row
là	lam	ⓘ	4	tree (sp)
laám	lam	ⓘⓘ	21	lose consistency
lan	lan	ⓘ	3	today
làn	lan	ⓘ	4	society
laán	lan	ⓘⓘ	31	invite to work
baɲ	bāɲ	ⓘ	2	a type of wild cat (genet sp)
bam	bam	ⓘ	3	day traditional in the 10-day week
baàm	ba` m	ⓘⓘ	34	day traditional in the 10-day week

3.3.1 Grammatical functions of tone in Ba Mambila

As shown in the preceding section, tone serves to distinguish lexical items in Ju Ba. In this section first grammatical and then deictic functions of tone in the language are described. The grammatical functions of tone are most readily discussed as operating in nominal constructions and as pertaining to the verb phrase.

3.3.2 Grammatical tone in nominal constructions

The only clear case of grammatical tone in nominal constructions is found in what are generally referred to as associative constructions (i.e. including among them attributives and possessives) are encoded in Ju Ba through a tone 4 suffix: e.g. final tone 1 > 14; 2 > 24; 3 > 34; 21 > 214; 31 > 314. This may be described as

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a tonal morpheme which has developed from the loss of its formerly associated segmental material. This is confirmed by the existence of compounds in which the segmental suffix /-è/ is preserved, fossilized, as well as through comparison with other Mambila lects; e.g. Ju Tungba (spoken at Gembu) and Ju Mvure (spoken at Warwar) in the West Mambila cluster, as well as certain East Mambila lects and other Mambiloid languages (see below, as well as [Connell \(In preparation\)](#)). Indeed, [Perrin \(1991\)](#) reports the segmental morpheme to still be in use by older Ba speakers in Sonkolong and Atta at that time with CVC nouns, though for nouns ending in a vowel, the vowel of the suffix is lost with only the tone remaining. Examples of the associative tonal morpheme are given in 2, with tones indicated by number.

(2) Tone 4 as a marker of the associative construction

a. feh 2, 'head'

wò sié julé feèh

4 31 2.1 2-4

2SG comb hair head\ASSC

'You comb (your) hair of head.'

b. jùkum 4.2 'owl (sp)'

hueh jùkuùm

2 4.2-4

voice owl\ASSC

'(The) voice of (the) owl.'

The compound in 3 shows the fossilized segmental suffix.

(3) bècénè 'together'

bè cén -è

place one ASSC

The ASSC marker also appears as an inherent part of possessive pronouns in Ju Ba, where again comparative evidence, here from Ju Mvure, reveals the tonal morpheme to be the remnant of the suffix /-è/.³

3.3.3 Grammatical tone functioning in verbal constructions

Grammatical functions of tone in the verb phrase are more numerous than for the noun phrase. Inherent tone patterns change, and in some cases underlying lexical

³The final è of 'Mvurè' is itself the associative marker; Jū Mvurè, 'language of Mvur'.

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Table 3: Possessive pronouns

	1SG	2SG	3SG	1PL	2PL	3PL
Jū Bā	mò	yeè	seèn	beèh	biì	bɔɔn
Jū Mvurè	mà	tʃĩ	hēn.è	bī.è	bēl.è	bōn.è
Tone	4	2+4	2+4	2+4	24	2+4

contrasts are neutralized through use of grammatical tone. First, both imperative and negative formation involve changes to the inherent tonal pattern of verbs. As noted above, verbs bear only a restricted number of tone patterns, viz, 2, 21, 3, and 31. In the imperative form, 2, 21 > 1 while 3, 31 > 4, thus having the effect of neutralizing the 2 – 21 contrast and the 33 – 31 contrast. That is, verbs may be grouped into one of two classes, upper tier (or Class 1, bearing 2, 21) or lower tier (or Class 2, bearing 3, 31), according to the grammatical behaviour of tone. The examples in 4 through 11 show this alternation between indicative, imperative, and negative forms for both monosyllabic and disyllabic words. Affected verbs are given in bold. In the imperative the 2SG subject is omitted (it may be retained in the plural); the negative form, as well as undergoing the tone change, also involves the addition of the negative marker *ngwéh* and inversion of the subject and object.

3.3.3.1 Mood and polarity

- (4) Imperative and negative formation; tone 2 > tone 1
- a. Indicative
à **ke** nyam
4 2 3
3SG **grill** meat
‘S/he grills meat.’
 - b. Imperative
ké nyam
1 3
grill\IMP meat
‘Grill meat!’
 - c. Negative

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à nyam ké ηgwéh
 4 3 1 1
 3SG meat grill\neg NEG
 ‘S/he doesn’t grill the meat.’

(5) Imperative and negative formation; tone 21 > tone 1

a. Indicative

à kééh njágá
 4 21 1.1
 3SG break stick
 ‘He breaks (the) stick.’

b. Imperative

ké njágá
 1 1.1
 break\IMP stick
 ‘Break (the) stick !’

c. Negative

à njágá kéh ηgwéh
 4 1.1 1 1
 3SG stick break\neg NEG
 ‘S/he doesn’t break (the) stick.’

(6) Imperative and negative formation; tone 2.2 > tone 1.1

a. Indicative

à yila ké gwò
 4 2.2 1 4
 3SG enter PREP house
 ‘S/he enters (into) the house.’

b. Imperative

yílá ké gwò
 1.1 1 L4
 enter\IMP PREP house
 ‘Enter the house!’

c. Negative

à ké gwò yílá ηgwéh
 4 1 4 1.1 1
 3SG PREP house enter\NEG NEG
 ‘S/he doesn’t enter the house.’

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(7) Imperative and negative formation; tone 2.1 > tone 1.1

a. Indicative

à yílá mì
4 2.1 4
3SG call 1SG.OBJ
'S/he calls me.'

b. Imperative

yílá mì
1.1 4
call\IMP 1SG.OBJ
'Call me!'

c. Negative

à mì yílá ngwéh
4 L 1.1 1
3SG 1SG.OBJ call\NEG NEG
'S/he doesn't call me.'

(8) Imperative and negative formation; tone 3 > tone 4

a. Indicative

à le vún
4 3 1
3SG forge hoe
'He forges (a) hoe.'

b. Imperative

lè vún
4 1
forge\IMP hoe
'Forge (a) hoe !'

c. Negative

à vún lè ngwéh
4 1 4 1
3SG hoe forge\NEG NEG
'He doesn't forge (a) hoe.'

(9) Imperative and negative formation; tone 31 > tone 4

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- a. à **leér** vún
 4 3 1
 3SG **hide** hoe
 ‘S/he hides (the) hoe.’
- b. Imperative
lèr vún
 4 1
hide\IMP hoe
 ‘Hide (the) hoe !’
- c. Negative
 à vún **lèr** ɲgwéh
 4 1 4 1
 3SG hoe **hide\NEG** NEG
 ‘S/he doesn’t hide (the) hoe.’

(10) Imperative and negative formation; tone 3.3 > tone 4.4

- a. Indicative
 à **lobo** jùàgà
 4 3.3 4
 3SG **beat** drum
 ‘He beats (the) drum.’
- b. Imperative
lòbò jùàgà
 4.4 4
beat\IMP drum
 ‘Beat (the) drum.’
- c. Negative
 à jùàgà **lòbò** ɲgwéh
 4 4 4.4 1
 3SG drum **beat\NEG** NEG
 ‘He doesn’t beat the drum.’

(11) Imperative and negative formation; tone 3.1 > tone 4.4

- a. Indicative
 à **loró** bò gèh
 4 3.1 4 4
 3SG **remove** PL seed
 ‘She removes the seeds (and pulp).’

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b. Imperative

lòrò b̀ò gèh
 4.4 4 4
remove\IMP PL seed
 ‘Remove the seeds (and pulp) !’

c. Negative

à b̀ò gèh **lòrò** ɲgwéh
 4 4 4 4.4 1
 3SG PL seed **remove\NEG** NEG
 ‘She doesn’t remove the seeds (and pulp).’

The negative imperative maintains the tone alternation of the imperative and the inverted word order of the negative, but makes use of a specific negative marker, which occurs phrase initially rather than finally.

(12) Negative imperative

té ké gwò yílá
 1 1 4 1.1
 NEG.IPM PREP house **enter\NEG\IMP**
 ‘Don’t enter the house.’

3.3.3.2 Past tense

In past tense formation, verbs with inherent level tone, whether tone 2 or 3, form a contour with the addition of a polar tone, i.e. one or other of the two extreme tones: 2 > 2.4; 3 > 3.1. This is illustrated in 13, which may be compared with 6 and 10 respectively. The alternation does not affect existing compound tones; i.e. 2.1 > 2.1; 3.1 > 3.1.

(13) Tonal marking of past tense

a. Tone 2.2 > tone 2.4 (yila > yilà)

à la **yilà** naâ gwò
 4 3 2.4 212 4
 3SG PST1 **enter\PST** PST house
 ‘S/he entered the house (earlier today).’

b. Tone 3.3 > tone 3.1 (lobo > lobó)

à la **lobó** naâ jùàgà
 4 3 3.1 212 4
 3SG PST1 **beat\PST** PST drum
 ‘He beat (the) drum (earlier today).’

13 *Lexical and non-lexical tone in Mambiloid*3.3.3.3 **Nominal derivation**

Tone alternations may also signal the derivation of nouns from a verbal base though this appears no longer to be a productive process. It is seen most regularly with Class 2 verbs and parallels the encoding of imperative and negative forms, discussed above. That is, for verbs of tone 3 and 31, whether monosyllabic or disyllabic, the derived noun carries tone 4. Examples are given in Table 4.

Table 4: Derivation of nouns from Class 2 verbs

<i>Verb</i>	<i>Noun</i>	<i>Tone change</i>
dua ‘pray’	dùà ‘prayer’	3 > 4
gbaj ‘enclose’	gbàj ‘den, lair’	3 > 4
mena ‘swallow’	mènà ‘neck’	33 > 44
leér ‘hide, lose’	lèr ‘mystery’	31 > 4
diélé ‘fall’	dìèb ‘a fall’	31 > 44
laré ‘extend in all directions’	làrè ‘crossroads’	31 > 44

On the other hand, while showing tonal alternations, nouns derived from Class 1 verbs are not consistent in this marking; where it might be expected that the nominal forms would bear tone 1, this is not always the case. Some examples:

Table 5: Derivation of nouns from Class 1 verbs

<i>Verb</i>	<i>Noun</i>	<i>Tone change</i>
kaga ‘bar, encircle’	kágá ‘barrier, fence’	22 > 11
sugó ‘suspend, hang’	súgó ‘hook’	21 > 11
yueé ‘cry’	yúé ‘tears’	21 > 1
megé ‘make a ball’	mege ‘ball’	21 > 22
mvuú ‘reduce to a powder’	mvuu ‘something small, microbe’	21 > 2

While these all remain in the upper tier, not all alternate to tone 1. And, there is at least one instance in which a lower tier noun is derived from a Class 1 verb:

- (14) debbé 21 ‘tell’ > dèbbè 44 ‘(true) story’

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3.3.3.4 Deixis marked by tone alternation

The encoding of directional deixis, ventive (VNT) or itive (ITV), through tone has been reported only infrequently (see Andersen 1992 on Agar; Reh 1996 on Anywa⁴), so its presence in Ju Ba is of particular interest. Verbs of motion in Ju Ba are all Class 1 verbs, i.e. bearing either tone 2 or 21, and these tones are maintained with motion away from the speaker (or deictic centre). When motion is towards the speaker (or deictic centre), tone 4 is suffixed, thus giving 2.4, 21.4. Interestingly, this directionality (i.e. ventive) is also encoded on the preposition, not through a change of tone but of vowel quality:

(15) Motion away (a) vs motion toward (b)

- a. à nde ké gwà
 4 2 1 4
 3SG go prep house
 ‘S/he goes to (the) house.’
- b. à ndeè ká gwà
 4 24 1 4
 3SG go\VNT prep.VNT house
 ‘S/he comes to (the) house.’

However, if an auxiliary verb is present, or if the phrase is in the negative, the tone change is born by the AUX or NEG marker; tone 4 is suffixed with motion toward the deictic centre. The examples in (16), with the verb *gi* (tone 2) ‘finish’ functioning as AUX, and in 10 the NEG marker *ngwé* h illustrate this.

(16) Directionality indicated on the AUX; motion away (a) vs toward (b)

- a. à nde gi ké gwà
 4 2 2 1 4
 3SG go AUX.finish prep house
 ‘S/he has already gone to (the) house.’
- b. à nde giì ká gwà
 4 2 24 1 4
 3SG go AUX.finish\ VNT prep.VNT house
 ‘S/he has already come to (the) house.’

(17) Directionality indicated on NEG; motion away (a) vs toward (b)

⁴My thanks to Philippe Bourdin for bringing these to my attention.

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- a. à ké gwà ndé ngwéh
 4 1 4 1 14
 3SG prep house go\ NEG NEG
 ‘S/he doesn’t go to (the) house.’
- b. à ká gwà ndé ngwéèh
 4 1 4 1 14
 3SG prep.VNT house go\ NEG NEG\ITV
 ‘S/he doesn’t come to (the) house.’

Three further examples are available in our data which deviate from the the general pattern, though nonetheless can be considered tonal encoding of directional deixis. These are shown in 11. The verb *njií* indicates action in a particular (but unspecified) direction, while *njiì* is used to indicated action toward the speaker; in this respect it follows the general pattern. However, *ɲueèh* ‘field’ is neutral with respect to direction, though *ɲueh* appears to be itive. Lastly, *toón*, too, is neutral but *tān* is itive but shows the vowel alternation associated with the ventive preposition see in the examples above.

(18) Exceptions to the general pattern

- | | |
|---|---|
| a. <i>njií</i> 21 v. ‘directional action’ | <i>njiì</i> 24 (v) ‘action directed toward’ |
| b. <i>ɲueèh</i> 24 ‘field’ | <i>ɲueh</i> 3 ‘to the field’ |
| c. <i>toón</i> 21 ‘market’ | <i>tān</i> 2 ‘to market’ |

The examples above pertain to verbs of motion and related elements. Tonal marking of spatial deixis in Ju Ba however is not restricted to verbs. Nouns too may undergo tonal modification, as the following examples illustrate. The village name of Somié in Ju Ba is *Ndeba* /*ndeba*/ (tone 3). (19)a describes someone moving away from the speaker; (19)b can mean either someone is coming from Somié (e.g. arriving) or comes from Somié in the sense of being an indigene of the village. Note in this example the tone alternation appears to be replacive rather than one of suffixation. It is not clear whether this difference is a general characteristic of nouns as opposed to verbal elements.

(19) Directional deixis marked on the noun *Ndeba*

- a. à nde ké ndeba
 4 2 1 3.3
 3SG go prep Ndeba
 ‘S/he goes to Ndeba (Somié).’

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- b. à nde ké ndèbà
4 2 1 4.4
3sg go prep Ndeba
'S/he comes from Ndeba (Somié).'

Spatial deixis with respect to location may also be marked through tone alternations, sometimes accompanied by segmental modification, as the examples in Table 6 show.

Table 6: Locatives encoded through tone alternations

<i>Base form</i>	<i>Derived locative</i>
feéh 21 'savanna'	fei 2 'in the savanna'
lɔ́ 21 'village'	lɔ̌ 2 'in the village'
hobó 21 'forest'	homo 22 'in the forest'
cíbí 21 'night'	cíbi 11 'in the night'
gwà 4 'house' (to the...)	gwò 4 'in the house'

3.3.3.5 Other tone changes related to verbs in Ju Ba

Finally, we note the existence of deverbal modifiers derived through tone changes; as there are few transparent examples it is not possible to state with confidence the process involved; however, the examples that are available show Class 1 verbs leading to adjectival forms also of tone 1 or 2 and the examples of Class 2 verbs give adjectives of tone 4.

Table 7: Examples of deverbal modifiers

<i>Verb</i>	<i>Adjective</i>
kóór 21 'be vigilant'	kór 1 'vigilant, attentive'
soú 21 'forbid'	sóu 1 'sacred'
veé 21 'spoil, go bad'	ve 2 'bad, ugly'
mbierré 31 'do something valueless'	mbièrrè 44 'worthless'
naŋ 3 'rise up, raise'	nàŋ 4 'tall'

3.3.3.6 **Tone in expressive language: Ideophones**

Ideophones, “marked words that depict sensory imagery” (Dingemanse 2012: 654). As “words” ideophones may be considered part of the lexicon and thus their tones, lexical. Ideophones are also a form of expressive language, much of which is non-lexical in nature; e.g. manipulation of f0 to indicate different emotions. Just as these f0 variations are not considered grammatical in the way using f0 to distinguish sentence types is, the use of tone in ideophones may be looked at as non-lexical. Ideophones are known to be phonetically or phonologically deviant, a fact often cited as one of their defining characteristics (Childs 1994, Childs 2003). This deviance, or distinctness, is illustrated segmentally in Ju Ba, where 70% of all documented lexical items beginning with a labial velar are ideophones. It is also true of ideophones tonally. In the regular lexicon of Ju Ba, tones occur in a frequency hierarchy of tone 2 > 4 > 1 > 3, however in ideophones the frequency of occurrence is: tone 1 (64%) > tone 4 (20%) > tone 2 (13%) > tone 3 (4%); this count includes the initial tone of ideophones bearing a compound tone. Interestingly, among these ideophones, tone 14 compounds are most frequent, i.e. again utilizing the extremities of the tone system and serving to emphasize the distinctness of ideophones.

While the study of ideophones in Ju Ba is as yet in its early stages, it seems apparent that with respect to f0 utilization their depictive nature broadly fits the ethological view of f0 first discussed in Ohala (1984). The examples in Table 8 illustrate this. The colour ‘red’ bèlè is modified as ‘deep red’, or ‘red close to black’ with an ideophone bearing tone 4, while the ideophone for ‘clear red’ bears tone 1. Likewise, ideophones for ‘white’ wulé which may also have the sense of ‘transparent’ bear tone 1. Similarly, ideophones depicting ‘bigness’ (literal or metaphorical) carry tone 4, shown in Table 9.

Table 8: Ideophones pertaining to intensity of colour

hùn 4	bèlè hùn	‘deep red’
jùm 4	bèlè jùm	‘red close to black’
cúéí 111	Njer né cúéí.	‘The wound is clear red.’
fér 1	wulésé fér	‘very white’
pób 1		‘very white’

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Table 9: Ideophones referring to ‘bigness’

kèkèlàg 4.4.4		big
kpàm 4		something wide open
kpèlàŋ 4.4	Wò mé feh tà kpèlàŋ	You and your big head.
ŋàb 4	nùàr ñàb	big person, i.e. important
yùòm 44		plenty

3.4 Summary

In the preceding paragraphs I have presented the basic characteristics of the tone system of Ba Mambila, its inventory and functions at both lexical and non-lexical levels. Its inventory is complex, with as many as 13 surface contrasts based on four underlying register tones. Broadly speaking, two subsystems have been identified and discussed, one applying to nouns, where all four register tones are contrastive on the initial syllable, and one pertaining to verbs, with only tones 2 and 3 contrastive on the initial syllable. The grammatical functioning of tone reveals verbs belong to one of two classes, Class 1 or Class 2, with tone modifications encoding different grammatical functions; these typically involve alternating to one or other of the ‘extreme’ tones, tones 1 or 4, including, in some cases, in a polar fashion. Modification of this nature also applies to deictic and derivational functions of tone. The two extreme tones are also the most frequently occurring in ideophones, unlike in the main lexicon.

4 Tone in Kwanja

Kwanja has two primary dialects, Sundani and Ndung. Two other lects, Njanga and Bung, both now extinct or effectively so, are also often considered varieties of Kwanja (see Connell 2012 and Connell 2017 for discussion of their affiliation). The most detailed examination of tone in Kwanja is unpublished work by Joan Weber ‘Tone in Kwanja Verbs’ (Weber 2004) which describes the lexical and grammatical functions of tone in the Sundani verb phrase; the discussion here relies primarily on this work and an unpublished draft dictionary (Weber & Weber 2011).

4.1 Lexical tone in Kwanja

Like neighbouring Ju Ba, Kwanja has four lexical tones, numbered, in Weber’s work, 1–4, High to Low. The Kwanja orthography uses only two diacritics to mark

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tone: tones 1 and 2, High and Upper Mid, are both marked with an acute accent; the Lower Mid tone, tone 3, is unmarked and tone 4, Low, is marked with a grave accent. Sundani primarily has a disyllabic word structure while Ndung, like Ju Ba, is mainly monosyllabic. Also like Ju Ba, there are distributional restrictions between tones occurring on nouns or on verbs. All four tones are found on nouns but tone 3 is relatively infrequent. There are four tone patterns found with verbs: 2, 23, 3, 32; i.e. the initial syllable is either tone 2 or tone 3. This suggests verbs are grouped into two classes, which is supported by the grammatical behaviour of tone, discussed below. The data in Table 10 show a series of near minimal sets illustrating the four contrastive tone levels in Sundani Kwanja.

Table 10: Contrastive lexical tones in Sundani: Words are given first in orthographic transcription, followed by the IPA tone letter, the tone number, and gloss

<i>Orth</i>	<i>IPA</i>	<i>Tone</i>	<i>Gloss</i>
dájŋ	˥	1	earthen bench, indoor
dájŋ	˨˩˨	2	begin, start (v)
dan	˨˩˨	3	stretch, expand (v)
dàn	˨˩˨	4	earthen shelf in a fireplace
dúŋ	˥	2	moan, groan (v)
bájŋ	˥	2	place, plot (n)
bəŋ	˨˩˨	3	open (mouth, v)
bòŋ	˨˩˨	4	family, clan
duŋ	˨˩˨	3	create (v)
dùŋ	˨˩˨	4	plot of land, concession
gbóó	˥	1	chair, stool
gbóó	˨˩˨	2	conclude an event (v)

The four register tones of Sundani may combine to form contours. Of the 12 possible bitonal combinations, all but three, 12, 13, and 21 are attested; 43 is found in just one example in the available data, which appears to be bimorphemic. Disyllabic words predominate in Sundani and it is on these where contours are principally found; the cognate forms in Ndung are typically monosyllabic and maintain the same contour, as in Sundani gəni vs Ndung gən ‘egg’; Sundani kfwəri vs Ndung kfwə ‘bone’. Table 11 gives examples of bitonal melodies in Sundani. All the examples included in the table have cognate forms in Ndung which are monosyllabic. It is interesting to note however that a great majority of words with pattern 31 are also disyllabic in Ndung.

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Table 11: Bitonal melodies in Sundani: Words are given first in orthographic transcription, followed by the IPA tone letter, the tone number, and gloss

<i>Orth</i>	<i>IPA</i>	<i>Tone</i>	<i>Gloss</i>
cùùn	⚡⚡	14	five
cùrú	⚡⚡	41	piece (of sth)
curú	⚡⚡	32	waterfall
cùú	⚡⚡	42	leaf
ndùŋ	⚡⚡	24	heart, seat of emotions
fáj	⚡	2	hurry (v)
faŋ	⚡	3	breathe with difficulty
faŋgá	⚡⚡	31	mound
fǎŋ	⚡⚡	42	quiver (n)
gvwéré	⚡	22	shell
gvwəri	⚡⚡	32	ember
gvwàní	⚡⚡	42	announcement
kfwəri	⚡⚡	14	bone
kfwírá	⚡	22	mahogany (tree)
kfwíri	⚡⚡	23	cover (v)
kfwìrá	⚡⚡	42	toad
ŋgéri	⚡	11	mouse (sp)
njəri	⚡⚡	14	bad odour
ŋgéra	⚡	22	Tikar Plain
njéri	⚡⚡	23	pour a liquid forcefully
ŋgəri	⚡⚡	24	hawk (sp)
njwanì	⚡⚡	34	saltiness
njèrí	⚡⚡	42	chicken, pullet
njèrì	⚡	44	monkey (sp)
moní	⚡⚡	32	young (n. animal, plant)
monì	⚡⚡	34	child

4.2 Non-lexical tone in Kwanja

4.2.1 Tone in Kwanja verbs

As noted, Sundani Kwanja verbs may be grouped into two classes, Class 1 and Class 2, according to whether the initial tone of the verb is tone 2 or tone 3, i.e. upper tier or lower tier, similar to what was described for Ju Ba. This bipartite grouping is supported by the grammatical behaviour of tone, which marks different aspects (perfective, imperfective, progressive), and moods (imperative, subjunctive).

4.2.2 Temporal reference encoded on verbs

Six different tenses are distinguished in Sundani: the present is equivalent to the imperfective (considered by Weber as the unmarked); three past (P1, P2, P3) and two future (F1, F2) tenses, expressing different degrees of remoteness, are marked with both particles and tone alternations. P1 and F1 are marked with the particle *tì* and *tì* (tone 4 and tone 3) respectively; P2 and P3 are encoded with the particle *nyàá* (tone 4) and *nyaa* (tone 3).

A primary distinction between non-past and past (or, imperfective vs perfective) exists and is encoded tonally, as the examples in 20 and 21 demonstrate (Weber's examples below are adapted from Connell 2026). The Class 1 verb *jí* 'go' is used, indicated in bold. It bears tone 2 in the non-past forms (20) and tone 1 in the past forms (21). The first row of the examples is in the Kwanja orthography; tone numbers are given in the second row.

(20) Past vs non-past marked tonally; non-past

- a. *wu jí ɲvwàrkà*
3 2 4.4
3SG go field
'You go to the field.'
- b. *tì wu jí ɲvwàrkà*
3 3 2 4.4
F1 3SG go field
'He will go to the field (soon).'
- c. *sánj wu jí ɲvwàrkà*
1 3 2 4.4
F2 1SG go field
'He will go to the field (distant /indefinite future).'

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(21) Past vs non-past marked tonally; past

- a. *tì wu jí* *ɲvwàrkà*
 4 3 1 4.4
 P1 3SG go\PST field
 ‘He went to the field (earlier today).’
- b. *nyàa wu jí* *ɲvwàrkà*
 42 3 1 4.4
 P2 3SG go\PST field
 ‘He went to the field (recently).’
- c. *nyaa wu jí* *ɲvwàrkà*
 3 3 1 4.4
 P3 3SG go field
 ‘He went to the field (long ago).’

An equivalent set of sentential examples for Class 2 verbs is not available, however the following two sentences illustrate the different tonal behaviour of the two classes under the two different temporal conditions, imperfective (non-past habitual) and perfective (past). The sentences each contain the verbs *jí* (Class 1) and *ju* (Class 2), indicated in bold; in the non-past example 22a they are realized with their basic tones, 2 and 3, respectively; in 22b they show the modified tone for the past, with tone 2 > tone 1, and tone 3 > tone 4.

- (22) a. *tì bí ju* *ɲvwàrkà bí jí* *gùmbà*
 4 1 3 4.4 1 2 4.4
 P1 1PL arrive. field 1PL eat.IPFV corn
 ‘We arrive at the field (and) we eat corn (habitually until now).’
- b. *tì bí jù* *ɲvwàrkà bí jí* *gùmbà*
 4 1 4 4.4 1 1 4.4
 P1 1PL arrive.PFV field 1PL eat.PFV corn
 ‘We arrived at the field (and) we ate corn (once, earlier the same day, today).’

Whereas imperfective and perfective are encoded tonally, as shown, the progressive aspect involves use of a copula *má* and the infinitive form of the verb, indicated with the addition of the suffix *-ni*:

PROG: COP + VERB-INF

The tone of Class 1 verbs remains unchanged if tone 2, and the suffix added; if tone 23, the tone 3 spreads to or is absorbed by the infinitive suffix.

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(23) Progressive

- a. jí 'go' má jí-ni
2 1 2-3
- b. béémi 'lie down' má béémí-ni
223 1 222-3

With Class 2 verbs, whether tone 3 or 32, the tone becomes 42 and the suffix is added (i.e. 42-3).

- (24) a. dwa 'plant' má dwǎ-ni
3 1 42-3
- b. dǎŋdi 'teach' má dǎŋdí-ni
32 1 42-3

4.2.3 Mood

Both imperative and subjunctive moods are present in Sundani and encoded via tone alternations. Examples of imperative are given.

Imperatives are encoded through tone lowering; Class 1 verbs, whether tone 2 or 23, take tone 34 whereas Class 2 verbs take tone 4 or 434, depending on whether the base form of the verb is tone 3 or 32. The examples in 25, 26, and 27 illustrate this.

(25) Class 1

- a. 'greet' bíi biì
2 > 34 (IMP)
- b. 'accept' bámdì bǎmdì
23 > 34 (IMP)

(26) Class 2

- a. 'plant' dwa dwà.IMP
3 > 4
- b. 'teach' dǎŋdí dǎŋdì.IMP
32 > 434

Imperatives in the plural are marked with a plural suffix -ni and take the same tone pattern, spread on to this suffix:

- (27) a. 'steal' cǒŋ cǒŋ-nì.IMP
2 > 3-4
- b. 'refuse' keé kèe-nì.IMP
32 > 43-4

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4.2.4 Directional deixis

Sundani marks direction toward the deictic centre (ventive) with a verbal suffix -bá (which may be reduced to -á). Whether the verb to which it affixes is of Class 1 or Class 2, tonally it follows the pattern of Class 1 verbs, viz. bearing tone 2 in non-past and tone 1 in past reference. Etymologically -bá is presumed to derive from *ba* ‘come’, commonly found in Benue-Congo languages, including in some Mambiloid languages. There is no corresponding itive marker; direction away is marked by the absence of -bá.

4.3 Summary

In this section I have given a brief description of tone functioning in Sundani Kwanja. There are four register tones, with the four-way lexical contrast being found only on nouns, though tone 3, the lower mid tone, occurs only infrequently on nouns. Verbs bear one of four possible tone patterns, 2, 23, 3, 32 and are grouped into two classes according to their initial tone and the tone patterns filling different grammatical functions: tense, aspect, and mood. Directional deixis is partially encoded tonally, with the suffix -bá following the tone patterns of Class 1 verbs.

5 Tone in Oumyari Wawa

Oumyari Wawa, spoken in the village of Oumyari and neighbouring hamlets, is one of four named Wawa lects; the others are spoken at Mbenguedje, Gandoua, and Ndi, all in Adamawa Region. Two PhD dissertations, [Griffith \(2013\)](#) and [Martin \(2012\)](#) have focussed on Oumyari. For the other three lects, extended wordlists exist together with, in some cases, specialized lists for flora and fauna, such as [Todd \(2011\)](#); little or no information on their grammars is available. In addition to these four lects, there are other small settlements of Wawa speakers in the region, for instance at Hore Taram Torbi, though these have not been investigated to ascertain their status, i.e. whether the variety of Wawa spoken can be considered a distinct or aligned with one of the others. Martin’s dissertation is most relevant for our present purposes and unless otherwise noted the data presented here on Oumyari tone are drawn from this source. Overall, the system is described as discrete-level (in Welmer’s [1973a](#) terms), rather than terracing, i.e. downstep, whether automatic or non-automatic, is not present, and as a generalization, tones do not overlap (cf. [Snider 2026 \[this volume\]](#)).

5.1 Lexical tone in Oumyari Wawa

Oumyari Wawa contrasts lexical tones in four registers, High, Mid (or Upper Mid), Lower Mid, and Low; these are numbered 1 to 4, High to Low, and in Martin’s work are marked thus: H = ´ ; Upper Mid = ¯ Lower Mid = ˘ and L = ˙ . Here I continue the convention used above, leaving tone 3, LM, unmarked. Martin offers the following near-minimal sets (among other examples) to illustrate the four contrastive registers. Tones 1, 2, 4 are common while tone 3 is found only rarely on nouns. Tones 1, 2, 3 are found on verbs; tone 2 however is s relatively rare, while tone 4 does not occur on verbs except on modified forms, as described below. Table 12 gives examples of lexical contrasts with Oumyari tones.

Table 12: Examples of lexical tonal contrasts, level tones, given on three different vowels

Word	IPA	Tone	Gloss
bérá	☒	1	call.PRF
nē	☒	2	log.SG
dembā	☒	3	termite hill
bè	☒	4	how
bárá	☒	1	rock
gbāārā	☒	2	rib
gbarā	☒	3	grain
bàgbà	☒	4	fog
bórá	☒	1	wedding
gōṅgà	☒	2	lance
gombá	☒	3	hunt
gòr	☒	4	time

The four register tones can combine to form compound tones, surface contours, illustrated in Table 13. Nine of the 12 possible bitonal combinations are attested on monosyllabic nouns. Those not attested are 13, 21, 31, while the most frequently occurring contours are, 12, 14, 24 and 32. A majority of Oumyari lexical items are bisyllabic, but rarely longer than this.

5.2 Non-lexical tone in Oumyari Wawa

Tone in Oumyari Wawa serves grammatical functions in both the noun phrase and the verb phrase. In the noun phrase, associative constructions are marked

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Table 13: Examples of lexical tonal contrasts with contour tones

Word	IPA	Tone	Gloss
dʒéː r	ꞤꞤ	12	goat
dôr	ꞤꞤ	14	rat
nɛꞤnɛꞤ	ꞤꞤ	21	crow
bɛŋ	ꞤꞤ	24	nut
kɔ̃ː g	ꞤꞤ	32	back
ŋǣ̃ː r	ꞤꞤ	34	medicine
tũr	ꞤꞤ	41	forest
dʒǣ̃ː m	ꞤꞤ	42	smoke
bà r	ꞤꞤ	43	wound

both segmentally and tonally; verbs are subject to different tone alternations in different grammatical circumstances.

5.2.1 Grammatical functions of tone in nominal constructions

Martin (2012) distinguishes between possessive and associative constructions. In the possessive the head (possessor) normally precedes the possessed, whereas in the associative, this order is reversed: the modifier normally precedes the head. The basic marker of the possessive is the suffix –ǎ, which attaches to the modifier, as in the examples in 28. In noun stems ending with consonant, this suffix is affixed (28a, mũn); if the noun stem ends in /ə/, the suffix /-ǎ/ replaces /ə/ (28b, nòbǎ). However, if the noun ends in a vowel other than /ə/, only the low tone of the suffix is added to the noun (28c, nùgwǣ).

(28) Possessive construction in Oumyari

- a. mũnǎ gùr
mũn-ǎ gùr
child-POSS foot
the child’s foot
- b. nòbǎ gùrbǎ
nòr-bǎ-ǎ gùr-bǎ
people.PL.POSS foot.PL
the people’s feet

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- c. nùgwê gùr
 nùgwê-ò gùr
 woman.POSS foot
 the woman's foot

The associative, in addition to reversing the word order relative to the possessive, takes a different suffix, /-oi/, said by Martin often to be reduced to /-i/. Otherwise its realization follows the same conditioning described for the possessive.

(29) Associative construction in Oumyari

- a. gùr mūnòì
 foot child-ASSC
 gùr mūn-òì
 '(the) foot of the child'
- b. gùrbà nòbòì
 gùr-bà nòr-bà-òì
 foot.PL person.PL.ASSC
 '(the) feet of the people'
- c. gùr nùgwêòì
 gùr nùgwê-òì
 foot woman.ASSC
 '(the) foot of the woman'

5.2.2 Behaviour of tone in verbal constructions

Verbs in Wawa are said to fall into three classes depending on whether the base verb bears tone 1, 2 or 3 (i.e. giving Classes 1, 2, and 3, respectively; as mentioned, there are no verbs bearing tone 4). Martin (2012: 110) describes verbs as having two stems, one being the base form of the verb and the second a tonally modified form. If monosyllabic the latter can bear only tone 2 (Classes 1, 2) or 4 (Class 3), or if disyllabic tone patterns 1-4 (Class 1), 2-4 (Class 2), or 4-4 (Class 3). These are shown in the examples below; a commonality between Classes 1 and 3 may be noted, that in both the base tone undergoes lowering, whereas Class 2 verbs do not undergo this change. The tonal behaviour of verbs differs according to whether the verb is monosyllabic or disyllabic but is regular within classes; specifically, the second syllable of disyllabic verbs is toneless. In the following examples, infinitive (INF) and imperfective (IMPV) are included together as IMPV

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is based on the INF form preceded by the copula (COP) (similar to what was described earlier for Kwanja). ‘CONS’ denotes the ‘consecutive’ aspect, sometimes elsewhere referred to as the narrative tense. Allomorphic variation is explained in [Martin \(2012\)](#).

5.2.2.7 Class 1 verbs

In monosyllabic verbs, tone 1 lowers to tone 2 in certain aspects, mood, and polarity:

Table 14: Monosyllabic Class 1 verbs

Root	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
jáb-	jām-nè	jāb	jāb-ká	jāb-a	jāb-níi	be ill
nám-	nām-nè	nām	nām-ká	nām-a	nām-níi	bite
ér-	ēn-nè	ēr	ēr-ká	ēēr-a	ēr-níi	go

In disyllabic Class 1 verbs lowering does not occur, though the second syllable now bears tone 4:

Table 15: Disyllabic Class 1 verbs

Root	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
wéɲbə-	wéɲbè-n	wéɲbè	wéɲbè-ká	wéɲbè-ra	wéɲbè-níi	chat
bárə-	bárè-n	bárkè	bárè-ká	bât-ta	bên-níi	ask
túndə-	túndè-n	túndè	túndè-ká	tûn-ta	túndè-níi	peel

5.2.2.8 Class 2 verbs

Unlike Class 1, inflected monosyllabic Class 2 verbs do not undergo any change in the tone of the root: tone 2 remains tone 2:

However like Class 1, disyllabic verbs in Class 2 have tone 4 on the second syllable:

5.2.2.9 Class 3 verbs

Monosyllabic Class 3 verbs undergo a tone change similar to those of Class 1, with the tone lowering one register, 3 to 4:

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Table 16: Monosyllabic Class 1 verbs

Stem	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
tāā-	tāā-nè	tā	tāā-ká	tāā-ra	tāā-níi	say
jūg-	jūṅ-nè	jūg	jūk-ká	jūg-a	jūṅ-níi	run
kām-	kām-nè	kām	kām-é	kām-a	kām-níi	wring

Table 17: Disyllabic Class 2 verbs

Stem	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
mbāndə-	mbāndə-n	mbāndə	mbāndə-ká	mbānta	mbāndə-níi	climb
kōmbə-	kōmbə-n	kōmbə	kōmbə-ká	kōmbə-ra	kōmbə-níi	take out
kāṅgə-	kāṅgə-n	kāṅgə	kāṅgə-ká	kāṅgə-ra	kāṅgə-níi	cook

Unlike Classes 1 and 2 however, the tone of disyllabic Class 3 verbs changes in a manner corresponding to that of monosyllabic verbs, lowering again from tone 3 to 4 on the first syllable as well as the second syllable bearing tone 4.

These data suggest an analysis of two rather than three verb classes: verbs of both Class 1 and 2 remain high-toned (H or UM) after undergoing grammatical modification, as opposed to the Class 3 verbs which are low-toned, LM or L.

Table 18: Monosyllabic Class 3 verbs

Stem	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
ndug-	ndūṅ-nè	ndùg	ndùk-ká	ndùg-a	ndùg-níi	touch
sēm-	sèmnè	sèm	sèmká	sèma	sèmníi	to mix
so-	sò-n	sò	sò-ká	sò-ra	sò-níi	sprout

Table 19: Disyllabic Class 3 verbs

Stem	INF/IMPV	PFV.FOC	CONS	IMP	NEG	Gloss
akə-	àkə-n	àkə	àkə-ká	àkə-ra	àkə-níi	sell
bòrə-	bòrə-n	bòrə	bòrə-ká	bòt-ta	bòrə-níi	skin
bùmnə-	bùmnə-n	bùmnə	bùmnə-ká	bùmnə-ra	bùmnə-níi	carry

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Perfective aspect is marked with a suffix /-rǎ/ and allophones conditioned by the last segment of the stem. The segmental material however is described as optional and typically lost among younger speakers, leaving just the H as the marker of PFV. The following examples are given by Martin:

Verb root	Fully inflected PFV	Simplified inflection
ér- ‘go, leave’	ээр-ǎ	ēé
bā- ‘come’	bā-rǎ	baǎ
nā- ‘give’	nā-rǎ	naǎ

5.3 Sentence type

Oumyari Wawa marks questions through use of different strategies, some of which involve the use of tone. One means of distinguishing polar questions from the corresponding declarative statement (the word order for both is the same) is through use of a question marker -à, attached to the final element of the phrase; or in the case of this element ending in a vowel other than /ə/, the vowel is lengthened and only the L tone is added, similar to the marking of associatives described above. An alternative means of encoding this distinction is through use of a high boundary tone (H%), raising the pitch higher than that of a lexical H. The factors governing use of one means of marking polar Qs rather than the other are yet to be explored, but it may be noted that both, the /-à/ or simply L, or the H% are included in the constellation of characteristics available in what [Rialland \(2009\)](#) terms “lax prosody”.

Content questions are marked through use of a Q-word, however either of the two strategies just described for polar questions may be used in addition to the Q-word; pragmatically this appears to add or ask for greater specificity, though again this is an area requiring future research.

5.4 Tone in expressive language

The tones of ideophones in Oumyari do not differ from those of the main lexicon with regard to inventory (H UM, LM, L are all present) or possible melodies, combinations of these four. Where Oumyari ideophones do stand out tonally from the main lexicon is, like in Ju Ba, in exploiting the extremes: the greater prevalence of H, L, and HL compared to either UM or LM or melodies including these two tones. Ideophones may also feature more complex tone patterns, e.g. HLH occurring on a single syllable, than those found in the main lexicon.

5.5 Summary

This section has given a basic description of the tone systems of Oumyari Wawa, based largely on the work of [Martin \(2012\)](#). Like both Ju Ba and Sundani Kwanja, Oumyari has four register tones which combine to form multiple surface contours. Also like Ju Ba and Sundani, there is an imbalance in the distribution of tones: all four occur with nouns with the frequency of tone 3, LM, being less than that of the others. Not all tones occur lexically with verbs; L is here found only through grammatical functioning. Verbs are also classified according to the behaviour of tones in different grammatical contexts; [Martin \(2012\)](#) proposes three tone classes, however an analysis of two classes grouping High or upper tier tones (tones 1 and 2) against lower tier tone 3 is also plausible.

6 Tone in Vute

The tone system of Vute has yet to be described in detail. The information that is available is found in three main sources, [Guarisma \(1978\)](#), [Thwing \(1987a\)](#) and [Thwing & Watters \(1987\)](#), supplemented by a draft lexicon ([Thwing & Thwing 1981](#)).

6.1 Lexical tone in Vute

Both Guarisma and Thwing describe Vute as having three contrastive tone levels, High, Mid, and Low, though each uses different tone marking conventions. Here I continue the practice used above, taking into account that there is just one M: H is marked with acute, M with macron and L with grave accents. All six possible two-tone combinations occur on monosyllabic or disyllabic nouns, giving a total of nine tone patterns on nouns. Examples are found in [Table 20](#). One three-tone combination, HLH, is also attested. Verbs, on the other hand, utilize two registers, H, L in realizing three surface tones, H, L and HL.

Every syllable bears at least one tone, but tone functions at the word level, as shown by the fact that tones of monosyllabic stems spread over the whole word when inflected (cf. [Smolders 2026 \[this volume\]](#)), as the following associative forms show:

- (30) mî ‘good’ > mínñē ‘of good’
fě ‘sauce’ > fěhí ‘of the sauce’

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Table 20: Contrastive surface tone melodies in Vute

<i>Word</i>	<i>IPA</i>	<i>Tone</i>	<i>Gloss</i>
ŋgár	Ꞥ	H	hand
ŋgār	Ꞥ	M	medicine
ŋgàr	Ꞥ	L	castanet
mí	ꞤꞤ	HM	good
bîŋ	ꞤꞤ	HL	round
yò	ꞤꞤ	LM	tree
tí m	ꞤꞤ	MH	antelope (sp)
náám	ꞤꞤ	HH	knife, type of
nāām	ꞤꞤ	HM	mosquito
nààŋ	ꞤꞤ	HL	embarrassment
ôû	ꞤꞤ	HL	law
ôĩ	ꞤꞤ	MH	hatred
gūūm	ꞤꞤ	MM	fruit
gāām	ꞤꞤ	ML	horn
nàám	ꞤꞤ	LH	abundance
ôòòm	ꞤꞤ	LL	wrap

6.2 Non-lexical tone in Vute

6.2.1 Tone characteristics of nominal constructions

Nouns ending in H may be grouped into two classes, those which cause a lowering of a following H and those which do not. [Thwing \(1987a\)](#) refers to this as “downstep” (discussed below). Nouns that trigger the lowering have Proto-Bantu cognates which terminate in L and so may be considered to have retained the L as floating following the loss of the final vowel (cf. [Rolle et al. 2026 \[this volume\]](#)).

	<i>Vute</i>	<i>PB</i>	<i>gloss</i>
nyín	/nínꞤ/	*yínò	tooth
bír	/bírꞤ/	*bídà	oil palm
kúrúm	/kúrúmꞤ/	*kútà	oil

6.2.1.10 Associative constructions

Associative constructions in Vute follow the order HEAD + MOD. It is marked by a L which, similar to the class of H tone nouns described above, has the effect, not of creating a contour tone but of lowering a following H. With the associative construction, it is true of all H toned nouns not just those of that class. Thwing refers again to this as “downstep”, but also notes there is no “downdrift”: “[s]ucceeding high tones after low tones return to the level of preceding high tones” (1987a: 25); thus, in the absence of a new, lower ceiling being set for Hs, this may be considered a local phenomenon, one of tone coarticulation.

(31) Associative construction in Vute

- a. gòré á
monkey DEM
‘It’s a monkey.’
- b. sí gòréé □ ʼá
body monkey ASSC DEM
‘It’s the body of a monkey.’
- c. yàáj á
mother DEM
‘It’s mother.’
- d. sí yàáj □ ʼá
body mother ASSC DEM
‘It’s the body of mother.’

Like Thwing, [Guarisma \(1978\)](#) refers to this as downstep. Her examples show the effect persisting across the word (i.e. two syllables) though not beyond.

- (32) a. ǂár ʼyémi
H ʼH.H
spot leopard
‘spots (of) leopard’

The floating L marking ASSC is likely the remnant of a segmental suffix, –è, as noted for Ju Ba (3.3.2), which is still present in possessive pronouns. Similarly, as shown in Table 21, in Vute it is still present in the possessive plurals, all CVC, while assimilating to the final vowel in the singular forms, all CV.

⁵Thwing does not comment on the H here as opposed to the expected L.

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Table 21: Vute pronouns with associative marker as an inherent part of the possessive form

	1SG	2SG	3SG	1PL	2PL	3PL
SUBJ	mĩ	wũ	ŋgé	nĩm	ɲí	ŋgáb
POSS	mōò	wōò	ŋgéé ⁵	nĩmè	ɲínè	ŋgábè

6.2.2 Tone in verbal constructions

The data on tone in Vute verbal constructions are scanty, nowhere being presented in detail. Thwing (1987a) offers a summary statement, that the perfective consists of a floating high tone that has the effect of raising all low stem tones to mid and that the narrative is marked with a L suffix, changing H verb stems to HL.

Thwing & Watters (1987), in introducing their discussion of focus in Vute, give slightly more detailed discussion, a formulaic account of tense and aspect marking which indicate that tone plays some role in marking both tense and aspect. Vute has two future tenses, the second of which, referred to as indefinite, is encoded with a preverbal marker kwâ and a floating H.

- (33) Tense and aspect marking in Vute
- a. past: tí´ (hesternal, P1); yí´ (remote, P2)
 - b. hodiernal: Ø (HOD)
 - c. future: ɓa + VERB + ní (definite, more certain, immediate; F1); kwâ + □ (indefinite; F2)

The perfective is encoded with a preverbal floating H and a post verbal particle tí.

- (34) Tense and aspect marking in Vute; perfective and imperfective
- a. perfective: □ + VERB + tí
 - b. imperfective: á + VERB + ní

The PFV floating H leaves H-toned verbs unaffected but raises any L-toned verbs to M.

Additionally the narrative, often referred to as a tense, less frequently as an aspect, is marked by a L tone suffix which results in HL on H-toned verbs.

Lastly, with some verbs a rising tone pattern, LH, serves to derive nouns from verbs, though this is more commonly done though use of segmental suffixes. Examples in (35).

- | | | |
|------|-----------------------------------|-------------------|
| (35) | Nominal derivation marked tonally | |
| | děě-nì ‘to cultivate’ | děě ‘cultivation’ |
| | càm-nì ‘to deny’ | càám ‘denial’ |
| | túk-nì ‘to raise animals’ | túũ ‘raising’ |

6.3 Summary

Although detailed information is not available on the functions of tone in Vute, it is clear that both lexical and non-lexical functions are encoded via tone. Lexically, multiple surface contrasts are based on three contrastive registers, H, M, L on nouns, while a reduced number of contrast are found with verbs. Grammatical functions achieved with tone include marking the associative construction, as well as differentiating some tenses and marking at least the perfective aspect.

7 Tone in Nizaa

Nizaa is arguably the most divergent of Mambiloid languages from a historical-comparative perspective. Information on tone in Nizaa comes from the work of Rolf Theil (Endresen 1992; Theil n.d.). The former source gives some detail of tone with nouns, but neither much information about grammatical or non-lexical tone. Theil (ms) is a detailed diachronic study based on internal reconstruction and thus provides some discussion of synchronic tone. It is apparent from these sources that the tone system of Nizaa, is rather complex, and not unlike the tone systems found in the other Mambiloid languages discussed above.

7.1 Lexical tone in Nizaa

Nizaa has three contrastive lexical tones: H, M, L. These occur singly and in combination, giving surface melodies H, HM, HL, M, MH, ML, MHL, L, LHM, and LHL. Of these, only eight occur pre-pausally. The H of MH is a floating tone (i.e. M□) and does not surface pre-pausally, though in other contexts it does; see Table 23 below. Table 22 shows the nine tone melodies occurring in Nizaa on indefinite, monosyllabic nouns in prepausal position. Other tone melodies are found in non-prepausal environments, determined in part by the following tone, and whether the noun is definite or indefinite.

The three-way lexical contrast is found only with nouns, and at that only on the initial syllable of the noun. With a single exception, compound tones are found only on the final syllable, whether noun or verb. The exception, described as occurring on non-final syllables of polysyllabic nouns, is LH; examples of this

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Table 22: Lexical tone melodies in Nizaa on monosyllabic nouns in prepausal position

Word	IPA	Tone	Gloss
cún	☐	H	tree
ṇūn	☐	M	child
ḏùù	☐	L	rabbit
dīw	☐☐	LH	face
bó ^ː w	☐☐	HM	dog
ʃ ^w áà	☐☐	HL	saw
cā ^ː ā	☐☐	ML	guinea fowl
ḡṣṣ	☐☐☐	LHM	egg
ḡūū	☐	M☐	head

is rare and those available all appear to be morphologically complex; i.e. the LH occurs on the final syllable of the stem. Endresen (1992: 49) considers these four tones, H, M, L, and LH to be “les bases” and says that “les [syllabes] pénultimes peuvent généralement porter une seule base. La dernière syllabe peut porter une base ou une base plus une coda”; “coda” here refers to any of the tones added to create the compound tones.

As found elsewhere in Mambiloid, with verbs only a two-way tone contrast exists, giving two verb classes based on the behaviour of their ton. All monosyllabic verbs belong to one or other of these two classes. The contrast is described as H vs non-H: verbs bearing a H tone on the unmodified stem retain this tone; verbs bearing a Mid tone on the unmodified stem alternate with L in specifiable contexts.

7.2 Non-lexical tone in Nizaa

In the absence of more data, little can be said concerning grammatical use of tone in Nizaa. In nominal constructions, nouns undergo tonal modifications depending on whether the noun is indefinite or definite. In either case, the floating H referred to above surfaces. Otherwise the number of patterns attested is reduced relative to the prepausal patterns and, most notably, the definite form is marked with the addition of a suffixed L; examples are from Theil (ms):

For verbal constructions, the two verb classes referred to above are based primarily on alternations across PVF, IMPV and PTCP forms; whereas H-tone verbs remain unchanged tonally across the three, M-verbs become Low-toned when

13 *Lexical and non-lexical tone in Mambiloid*

Table 23: Tonal marking of the INDF – DEF contrast in Nizaa nouns

INDF form	IPA	Lexical Tone + L > DEF	IPA	DEF form	Gloss
cún	☐	H+L > HL	☐☐	cûn	tree
ɲûn	☐	M+L > ML	☐☐	ɲûn	child
dùù	☐	L+L > L	☐	dùù	rabbit
dīw	☐☐	LH+L > LHL	☐☐☐	dī☐w	face
bóː w	☐☐	HM+L > HL	☐☐	bôw	dog
ʃˈáà	☐☐	HL+L > HL	☐☐	ʃˈáà	saw
būū	☐☐	MH☐+L > MHL	☐☐☐	būū	head
cā̀lā̀	☐☐	ML+L > ML	☐☐	cā̀lā̀	guinea fowl
bḱḱ	☐☐☐	LHM+L > LHL	☐☐☐	bḱḱ	egg

derived as participle. Endresen gives as examples representative of each class the verbs lóm ‘bite’ and gur ‘buy’:

- (36) ‘bite’ ‘buy’
 PFV lóm gūr
 IMPV lówꞑí gūrcí
 PCTP lówꞑwâ gùrwâ

Kjelsvik (2002) mentions that imperatives are encoded with a L-tone suffixed to the verb root and that subjunctives are marked with the suffix –wâ as well as a L replacing the lexical tone of the root. A lone example is given of each. Kjelsvik hints at the existence of other tonal morphemes, though these are not identified.

- (37) verb root IMP SBJV
 píŋ pîŋ pîŋ–wâ
 ‘do/make’ ‘do!’ ‘want to do’

The three-way lexical tone contrast in Nizaa is considered by Endresen to be hierarchical: a basic H vs non-H bifurcation with non-H divided into M and L; this allows him to account for the two-way contrast found only with verbs, and the alternations between the aspectual and participial forms shown in (36) along with the added perspective that it reflects the fact that the H-toned verbs correspond regularly with the H-toned verbs of Proto-Bantu.

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8 Other facets of tone in Mambiloid

The focus of this chapter has been on the uses of tone, both lexical and non-lexical, in Mambiloid languages. Non-lexical functions have been understood to include grammatical and deictic functions, as well as tone as used in expressive language. Only brief mention has been made of other aspects of tone, for example the possible presence of downstep or a role for tone in marking information structure. Here I look briefly at these.

Several experimental studies (e.g. [Connell 2002](#); [Connell 2003](#); [Connell 2017](#)) have confirmed the absence of a role for f₀ in Ju Ba beyond those discussed in Section 3. While Ju Ba does show lowering of Hs following L, such affected Hs normally reach their original height within a span of two syllables. f₀ is not used to differentiate sentence types, e.g. polar Qs from corresponding declaratives. Ju Ba has, essentially, a discrete-level tone system ([Welmers 1965](#); [Welmers 1973b](#)), with each tone tending to stay in its own tonal space ([Connell 2009](#)). Similar experimental work has yet to be carried out on other Mambiloid languages. Impressionistic study however suggests a similar situation exists in Sundani [Martin \(2012\)](#) says the same for Oumyari Wawa. To date no research has been done to investigate the use of f₀ in expressive language such as variations that might be associated with different attitudes and emotions.

The two three-tone languages examined here, Vute and Nizaa, have both been described as having “downstep” ([Guarisma 1978](#); [Thwing 1987b](#) for Vute; [Endresen 1992](#) for Nizaa). This is of potential interest in that languages with three tones and downstep are rare. But for neither language do the authors show that the lowering is more than a local coarticulatory effect (see [Hyman & Schuh 1974](#), for discussion of tone coarticulation from a phonological perspective and [Gandour et al. 1994](#), or [Xu 1994](#), on the phonetics of tone coarticulation). Further, there is no mention of terracing and in neither case is there mention of the lowering being phonemic. On the other hand, Thwing confirms the absence of “down-drift”, i.e. automatic downstep, and describes the lowering as being corrected within two or three syllables. This in accord with what has been noted for Ju Ba ([Connell 2002](#); [Connell 2011](#); [Connell 2017](#)). At the same time, it is of interest to note that an effect that at least has occasionally been referred to as “upstep” may be present in Vute; as noted above, Thwing describes the PFV aspect involving a preverbal floating H raising L-toned verbs to M. By Thwing’s description however, this raising is limited in domain to the verb only.

9 Summary and conclusions

Lexical and non-lexical tone as found in the Mambiloid languages have been examined in this chapter. It was acknowledged at the outset and must be reiterated here that this is no more than a partial view, as relevant data for most languages of the group are not available.

One of the interesting characteristics shared across the languages discussed in this chapter is that the multiple lexical tone contrasts – whether based on three or four tone registers – occur only on the initial syllable of nouns, and essentially just a two-way contrast is found verbs. Similarly, in each language the three or four tone registers combine to create multiple compound or surface contour tones. With lexical tone, surface contours are typically found only on monosyllabic words, or the same tone pattern borne across the two syllables of disyllabic words. On the other hand contours also result from the grammatical use of tone and here they are generally found on the final syllable of the word and this has been shown to be due to the loss of the segmental portion of a suffix, e.g. in associative constructions: in Ju Ba though fossilized traces remain, the loss is complete; in Oumyari and Vute it appears to be a process in progress, with the segmental suffix still present with consonant final stems.

Verbs in each of the languages examined can be grouped into two classes: Class 1, a High tone or Upper Tier class, and Class 2, a Low tone or Lower Tier class. This follows from the two-way lexical tone contrast and is supported by the grammatical functions of tone present in the language, though the details differ from language to language. Tone modifications to verb stems encoding grammatical functions regularly adhere to the tier or class of the verb while in some cases a polar tone is used.

For both Vute and Nizaa, it was noted that verbs of the High tone class correspond regularly with the High tone verbs of Proto-Bantu; this is also true of Ju Ba and appears to be true of both Kwanja and Oumyari Wawa. Both [Thwing \(1987a\)](#) and [Theil \(n.d.\)](#), for Vute and Nizaa respectively, interpret this as evidence that these languages are descended from Proto Bantu.⁶ However [Connell \(2018\)](#) suggests the multiple contrast found only on the first syllable of nouns is a result of the loss of former nominal class marking prefixes; Theil (ms) suggests the same for the three-way contrast on Nizaa nouns. And just as the loss of nominal prefixes may have led to this result, the erosion of segmental material in suffixes appears to have led to increased or novel use of tone to serve grammatical roles.

⁶[Thwing \(1987a\)](#) is a study in the relations between the noun phrase in Vute and Proto Bantu and she presents other data, to support this notion beyond tonal.

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In any event, an adequate understanding of tone in Mambiloid, its lexical and non-lexical functions, depends on understanding its history.

An adequate understanding of tone in Mambiloid also depends on the availability of more complete data from both the languages discussed here and especially those Mambiloid languages only mentioned in passing in Section 1. As pointed out in Section 2, all Mambiloid languages appear to have at least a three-way tone contrast lexically. This seems true of Ndooro and Somyev; Tep, Mvanip, Ndunda, and Mbongno have at least and possibly four, yet nothing is known of the non-lexical functions of tone in these languages.

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Chapter 14

Tone in Grassfields Bantu languages

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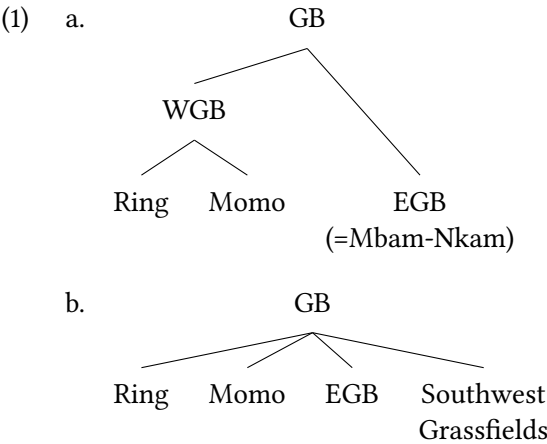
CNRS-LLACAN, Villejuif

The Grassfields Bantu languages of Cameroon are well known for their complex tone systems. Analysing the underlying tone of morphemes is thus crucial for accounting for surface tone patterns. Tone carries a heavy functional load both lexically and grammatically. We will describe lexical tone on noun and verb roots, as well as tone on their affixes. Tone in the noun phrase will then be examined (e.g. in noun class agreement and associative constructions), including the presence and influence of tonal morphemes. The grammars rely heavily on grammatical tone particularly in the verb phrase; we will demonstrate how tone contributes to the marking of tense, aspect, modality and mood. Finally, we will discuss some clause-level and sentence-level tonal phenomena.

1 Introduction

This chapter attempts an overview of the intricate systems of tone in Grassfields Bantu (GB) languages, known to “have some of the most complex tone systems of any group of languages in Africa” (Watters 2003: 236). We first present the geographical location and classification of the languages before turning to a general introduction of the tonal systems involved.

of the subclassifications adapted from Hyman (2018: 174) of “Narrow GB, i.e., ignoring Ndemli (cf. Stallcup 1980, Watters & Leroy 1989, Piron 1995, Watters 2003)” are given in (1):



As noted, “the older subclassification in (1a) recognizes a binary split between Western vs. Eastern Grassfields Bantu (WGB, EGB), while (1b) presents all of the subbranches as coordinate” (Hyman 2018: 174). Although with reservation due to lack of sufficient evidence/data, Blench (in preparation) proposes the following classification that recognizes “Narrow” and “Wide” (or Peripheral) GB: “The languages outside Narrow GB are treated as coordinate and not forming a genetic group in themselves as implied by Piron (1995)”.

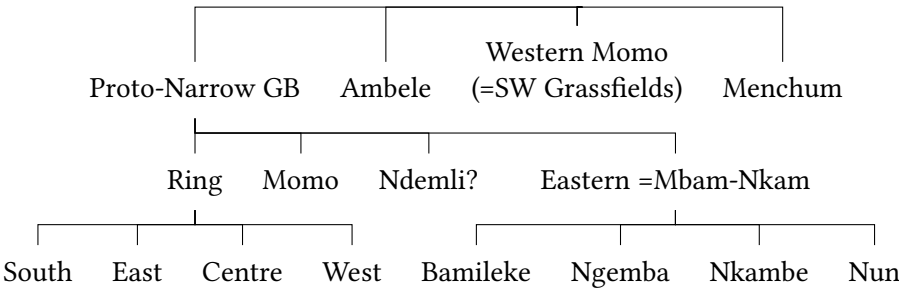


Figure 2: GB family (Blench in preparation)

The map in Figure 3 displays major subgroups of GB languages.

GB languages are generally assumed to have had two historical tone levels, high and low (Watters 2003: 236). Synchronically, some of the languages have

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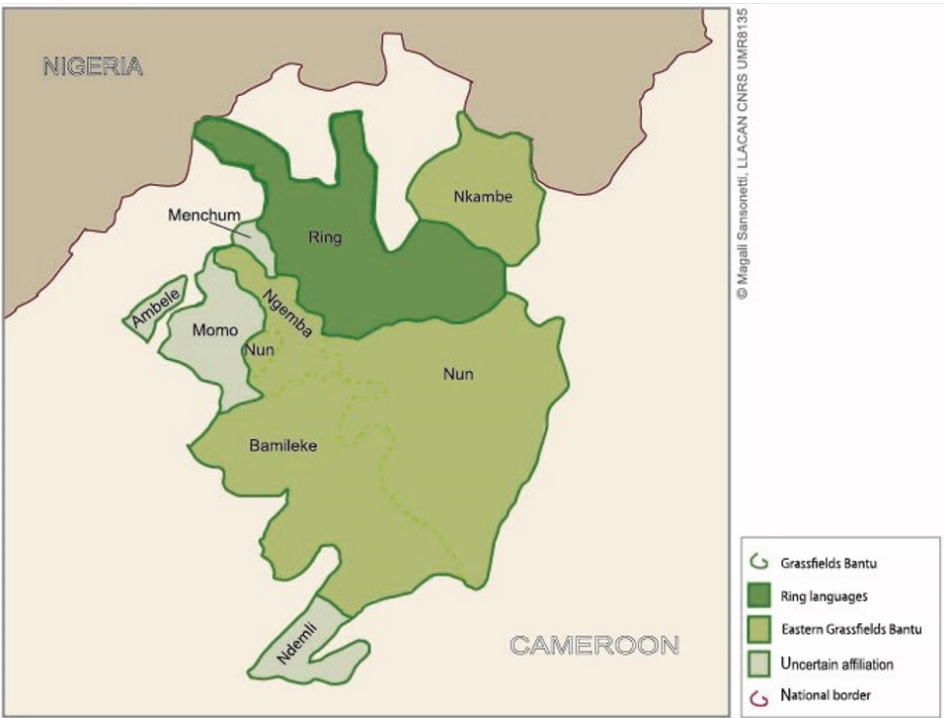


Figure 3: GB subgroups

at least four levels: high [H], mid [M], non-falling low [L°] and low that falls [L]. There is a divide between GB languages with /M/ tone and those with down-stepped H tone [$\downarrow$ H], whose distribution is areal (Hyman 1987: 217). Some of the languages, e.g., Bafut (Ngemba), Mfumte (Nkambe), Fe'fe' (Bamileke), Kom (Central Ring), Lamnso' (East Ring), Ngamambo (Momo), have [M] tone. On the other hand, others have [$\downarrow$ H], e.g., Yemba (Bamileke), Aghem (West Ring), Babungo (South Ring). An exceptional case is Babanki (Central Ring) with both [M] tone, that may only occur before a /H/ tone and never following pause, and [$\downarrow$ H] (Hyman 1979c, Akumbu 2019). Surface contour tones are common across GB, but they result from a combination of some of the level tones.

Floating tones are common in the area (Hyman & Tadadjeu 1976) due to a tremendous loss of syllables whose tones persist (cf. Rolle et al. 2026 [this volume]). Syllable loss has affected disyllabic roots that lost their final vowel or entire final syllable (except some Ngemba languages that have maintained a final vowel or syllable (Leroy 1979)), or nouns that lost their segmental class prefixes (cf. Hyman & Tadadjeu 1976: 69). For this reason, some roots have a prefix tone

and a final root tone that are not associated with a TBU (tone-bearing unit) underlyingly, but which influence neighbouring tones in various ways in individual languages (Elias et al. 1984).² It is important to note that the more edges of words have been eroded segmentally from proto-forms, the greater the likelihood of floating tones being present.

The TBU in GB languages is typically the syllable rather than the mora (see, for example, Robinson 2022: 9). Tones typically attach from left to right (see, for example, Anderson 1983 and Hamm & Majenpoh 2024: 4).

In the remainder of this chapter, we describe lexical tone on noun and verb roots, as well as tone on their affixes in Section 2, then examine tone in the noun phrase in Section 3 and verb phrases in Section 4, before discussing some clause-level and sentence-level tonal phenomena in Section 5.

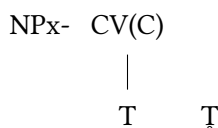
2 Lexical tone

2.1 Tone on nouns

2.1.1 Tone on noun roots

Most analyses of nominal tone in GB languages are based on the assumption that noun roots can (at least historically) be divided into four underlying tone categories: /L/, /LH/, /HL/ and /H/ (see, for example, Hyman & Tadadjeu 1976: 66). GB noun roots tend to be monosyllabic, but they have typically retained the four possible tone categories reconstructed for Proto-Bantu (Watters 2003: 236). Elias et al. (1984) reconstruct Proto-EGB noun roots as CVC with a final floating tone. This is presented below in (2) from a broader GB perspective, based on reconstruction of GB roots (Hyman 1979b) and noun prefixes (Hyman 1980b).

- (2) Structure of the typical Proto-GB noun (Hyman 1979b, 1980b)



The final floating tone forms part of the stem and has developed from the loss of an earlier vowel to which the tone was associated.

²Complicating the matter further, Chumbow & Nguendjio (1991) have argued in Bangwa (Nda'nda') (EGB, Bamileke) that lexical floating tones from proto-GB have been lost, but grammatical floating tones persist.

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These four tone categories for noun roots are illustrated in Table 1, using data from Chopechop (EGB, Nun). All four nouns are from the same noun class, which has the class 6 *mə̀-* prefix. The surface tone is for the noun pronounced in isolation.

Table 1: Nominal tone in Chopechop (Robinson 2023: 4)

Underlying tone of noun root	Example noun (prefix + root)
/L/	[mə̀-pùŋ] ‘stomachs’ ^a
/LH/	[mə̀-shǎ] ‘magic’
/HL/	[mə̀-fǝ] ‘fat’
/H/	[mə̀-lót] ‘hips’

^aThe final low tone in this word falls.

The tone analysis in Table 1 is fairly straightforward, but GB data is rarely this straightforward when nouns in isolation are examined. For example, in Ngienboon (EGB, Bamileke), /LH/ noun roots surface as non-falling low because the /H/ of the root floats following the noun, preventing the root’s low tone from falling. /HL/ noun roots surface as [[↓]H]. /H/ noun roots surface as rising due to the spreading of low tone from the prefix to the root. The only surface tone pattern that matches the Chopechop data in Table 1 is for /L/ roots; like Chopechop, these surface as low that falls (Anderson 2001).

There is even further complexity in the GB area. In some languages, more than four contrastive surface tone patterns are evident for nouns in isolation, whereas in others, fewer than four surface tone patterns are found. In these cases, it is only when nominal tone is analysed in context – in various tonal environments – that the four underlying tone categories for noun roots can be identified.

For example, in Mmen (Central Ring), multiple different surface tone patterns are evident when nouns are considered only in isolation. Möller (2014) describes eight different surface tone patterns for class 7 noun roots and five different surface tone patterns for class 9 noun roots. However, in the final analysis, Möller concludes (2014: 64) that “the underlying tones are closely linked to, and partly identical with, the reconstruction of tone sequences for Proto-GB” (i.e. the four-way underlying tone contrast for noun roots).

On the other hand, in the West Ring languages Aghem and Bu (Laimbue), there are two patterns of neutralization. *LH, *HL and *H merge in isolation if the noun prefix is /H-/. However, both *LH and *HL have a floating L after them. When

the prefix is /L-/ , it is *LH and *H that merge as L°. These patterns are observable when the noun roots are analysed within the context of associative constructions, and thus there is clear evidence for the four-way underlying tone contrast typical of Proto-GB (Kießling 2020).

More on nominal tone for EGB is found in Hyman & Tadadjeu (1976). For Ring languages, see Hyman (1980a). For Momo languages, see Asongwed & Hyman (1976), Hyman (1986), Hombert (1976), Stallcup (1978), and Spreda (1986).

2.1.2 Tone on noun class prefixes

Whereas noun class prefixes (NPx) in Proto-Bantu have been shown to unambiguously have low tone (Nurse & Philippson 2003: 8), the situation in GB languages is more complex, at least synchronically.

For some GB languages, like the EGB subgroup, NPx are still clearly /L/ (Hyman & Tadadjeu 1976: 66–67). This /L/ NPx often cooccurs with segments (CV̇-, V̇-), or it can be a floating tonal NPx as in Chufie' (Hamm & Majenpoh 2024: 7). However, in other GB languages, such as the Ring and Momo subgroups, the analysis is less clear. Many have argued for a /H/ NPx, but more recently, other theories have been proposed. The issue relates to the unity of the GB area which has a lot of lexical similarity but some surprising tonal variety.

One of the isoglosses that set apart Momo and Ring (formerly “WGB”) from EGB is that “NPx all carry a /L/ tone” in EGB while “most NPx carry a /H/ tone” in WGB languages (Stallcup 1980: 55). Since several languages in the Ring and Momo subgroups exhibit [H] tone on NPx on nouns in isolation and in various grammatical contexts, Hyman (1980b) reconstructed *H tone for Proto-WGB NPx. However, when these nouns occupy the N2 slot in an associative construction (‘N1 of N2’), their prefixes consistently have [L] tone. Akumbu (2024) speculates that the [L] tone of the prefix in this context is a direct reflex of the historical /L/ tone prefix, and that the [H] prefix observed in the citation form is a result of the influence of /H/ concord tones.

However, alternative analyses exist to account for the fact that NPx in Ring and Momo languages are not necessarily [L] in the citation form. For example, in the Ring language Mmen, prefixes always have [M] tone when nouns are cited in isolation. The prefix is analysed as /H/, but this /H/ is then “lowered in initial position because it is preceded by a low boundary tone found at the beginning of phonological phrases” (Möller 2014: 23).

/H/ to [M] lowering is also found in Kom: As Hyman (2005: 318) explains, the Kom “prefixed /H/ is realized mid on the surface—but not before it has been able

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to spread a /H/ feature onto a following /L/ tone stem”, e.g., /fĩ-kà?/ ‘tree’ → fĩ-kâ? → fĩ-kâ? [M-HL].

Ngamambo and Oshie (Momo) exhibit several /H/ NPx as well. In Ngamambo, Asongwed & Hyman (1976) postulate /H/ tones on both V and CV NPx with the /H/ on V NPx lowering to [M].

- (3) Ngamambo (Asongwed & Hyman 1976: 27, 30)

[ũ-kân]	‘message (cl.3)’
[ā-tsám]	‘home (cl.7)’
[má-nyíp]	‘water (cl.6a)’

Yet another analysis is that for NPx with [M] in the citation form, the prefix historically was /L/, but the /L/ was raised due to a /H/ pre-prefix (i.e. augment). This analysis has been proposed for Ngie (Momo) (Hombert 1976: 15).

It is also worth briefly considering the evidence for underlyingly toneless noun class affixes in the synchronic analysis of certain GB languages which are from various branches of GB, as highlighted by Neba (2024). He cites languages such as Bafut and Obang (Menchum), where toneless noun class prefixes have been proposed. In Wushi (South Ring), six noun classes are marked with suffixes rather than prefixes, and the synchronic analysis suggests that these suffixes are inherently toneless (Robinson 2022: 16). It is unclear how toneless noun class suffixes are related to the /H/ tone class 10 suffix in neighbouring Kenswei Nsei (South Ring), Babanki, Kom, Bum, Oku (Central Ring) and Lamnso’ (Hyman 1980a: 248).

With all of this variation in noun class affixes above, we note that for some languages, the facts are quite simple. It is reported for Mfumte (Nkambe, EGB) that there is no functional noun class system. Often noun plurality is marked by a /H/ noun suffix (McLean 2014). This plurality strategy is shown in (4).

- (4) Mfumte (McLean 2014: 14)

a.	nwà’	níní
	book	this
	‘this	book’
b.	nwǎ’	níní
	book.PL	this
	‘these	books’

The precise behaviour of noun class affixes continues to be debated among scholars and is important to both synchronic and diachronic tone studies of GB.

2.2 Tone on verbs

2.2.1 Tone on verb roots

Unlike lexical nominal tone, lexical verb tone is fairly straightforward in GB languages. Verb roots fall into two underlying tone patterns, /H/ and /L/ (Watters 2003: 236).

In order to separate verb roots into these two tone categories, a suitable ‘frame’ must be found where the tonal environment is conducive to revealing the underlying tone contrast of the verbs. For Wushi, one such frame involves citing the verb in the perfect aspect, as illustrated in (5).

(5) Wushi (Robinson 2021: 18, 85)

a. /H/ verb root

[í nâ' nǎ]

/í nâ' ná/

3SG PRF cook

‘He has cooked.’

b. /L/ verb root

[í nâ' jà]

/í nâ' jà/

3SG PRF catch

‘He has caught.’

The surface tone on the verb is rising in (5a) but low in (5b). This surface tone difference is evidence for the underlying contrast between the /H/ verb in (a) and the /L/ verb in (b).

If a suitable frame is not used, the underlying contrast between /H/ and /L/ verbs may be neutralized. This is illustrated in example (6), where the verbs are cited in the perfective aspect.

(6) Wushi (Robinson 2021: 41, 85)

a. /H/ verb root

[mè nǎ]

/mè LH-ná/

1SG PFV-cook

‘I cooked.’

b. /L/ verb root

[mè jǎ]

/mè LH-jà/

1SG PFV-catch

‘I caught.’

The surface tone transcriptions for (6a) and (6b) are identical, with the same rising surface tone on both verbs, showing that the underlying tone contrast between /H/ and /L/ verbs is neutralized in this context.

2.2.2 Tone on verbal affixes

Verbal derivational suffixes in GB are usually referred to in the literature as verb(al) extensions (as in Narrow Bantu). In comparison to Narrow Bantu, GB languages tend to have fewer verbal extensions (Hyman 2018: 193).

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GB verbal extensions are typically analysed as underlyingly toneless as they typically are realized with the same tone as the verb root (cf. Nurse & Philippson 2003: 8 for Narrow Bantu). Languages where this analysis has been proposed include Lamnso' (Grebe 1984: 36), Babanki (Akumbu et al. 2020: 3), Wushi (Robinson 2021: 21–22), Awing (Ngemba) (Fominyam 2021), Mankon (Ngemba) (Leroy 2007), Supapya (Nun) (Hamm, fieldwork), Ngomba (Bamileke) (Satre 2002), Medumba (Nun)³ (Nganmou 1991) and Meta (Momo) (Spreda 1986). The surface tone of the verbal extension is identical to the underlying tone of the verb root.

In contrast to verbal extensions, many GB languages have been analysed to have verbal inflectional suffixes which do have underlying tone. These are usually related to imperfective aspect (progressive and/or habitual) or perfective aspect (anterior, perfect, completive). The more conservative languages retain full segmental affixes (with tone) and the ones that have undergone significant segmental erosion often just have floating tones.

Note in Table 2 that some Ring languages such as Isu (West Ring) and Bamunka (South Ring), and some EGB languages such as Chufie' and Nda'nda' (Bamileke), have reduced their imperfective marking suffixes to the point of only grammatical tone (cf. Rolle & Bickmore 2026 [this volume]).⁴

A handful of different GB languages have verbal suffixes to mark one of two distinct aspects: perfect and perfective (those that mark perfect with a verbal suffix do not also mark perfective as a verbal suffix). When comparing them together, all but one include a [H] tone. As Table 3 shows, there is often more segmental erosion of perfective marking than perfect marking.

Some GB languages have a perfect/perfective distinction while others do not. It is not known currently if the various forms shown in Table 3 are related, but often a perfective GB form can be translated into English with a simple past ('he came') or a perfect ('he has come'). Further research on individual grammars is necessary.

Infinitive forms of verbs (Table 4) often include prefixes and sometimes are documented with a suffix as well.

Finally, homorganic syllabic nasal prefixes also have underlying tone in the GB languages. These are fairly common, particularly in the EGB subgroup, and analysing their underlying tone is key to disambiguating their grammatical functions.

³Medumba is considered Bamileke by some researchers and Nun by others. Piron (1995) and Elias et al. (1984) report that the distinction between Nun and Bamileke is not clear cut.

⁴No attempt is made here to reconcile the differing tones in the tables of verbal suffixes. Verbal TAM marking in GB is highly complex and many variables complicate the picture. The table only shows a small sample of how verbal suffixes contribute to imperfective marking.

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Table 2: Suffixes involved in marking of imperfective-related aspects.

Language	Form	Function	Source
Aghem (West Ring)	-à	incompletive	Anderson 1979
Isu (West Ring)	-L	imperfective	Kießling 2011: 76
Mmen (Central Ring)	-ǎ	imperfective	Möller 2014
Kom (Central Ring)	-ā	incompletive	Shultz 1997
Babanki (Central Ring)	-ǎ	progressive	Akumbu et al. 2020
Bamunka (South Ring)	-H	imperfective	Ingle, pers. comm.
Mundani (Momo)	-á	imperfective	Parker 1991
Meta (Momo)	-ə	imperfective and habitual	Spreda 1990
Chopechop (EGB, Nun)	-ghà	progressive	Robinson 2024
Chrambo (EGB, Nun)	-nú	progressive	Wright 2011 ^a
Chufie' (EGB, Nun)	-LH	imperfective	Hamm & Majenpoh 2024
Nda'nda' (EGB, Bamileke)	-H	imperfective	Chumbow & Nguendjio 1991
Ngwe (EGB, Bamileke)	-ā	imperfective	Nkemnji 1995
Ngiemboon (EGB, Bamileke)	-Ṽ	imperfective	Anderson 1983
Mankon (EGB, Ngemba)	-á/-í ^{''}	imperfective	Leroy 2007
Awing (EGB, Ngemba)	-ǎ	imperfective	Fominyam 2021

^aWright transcribes *nú* as a separate word, but notes that it could be a suffix (2011: 10).

Table 3: Perfect and perfective suffixes.

Language	Form	Function	Source
Ngombale (EGB, Bamileke)	-H	perfective	Voutsas 2003
Nda'nda' (EGB, Bamileke)	-H	perfective	Chumbow & Nguendjio 1991
Ngiemboon (EGB, Bamileke)	-H	perfect	Sonkoue Meli Epse Kamdem 2024
Bafut (EGB, Ngemba)	-mâ	perfect	Mfonyam 1989
Supapya (EGB, Nun)	-mǎ	perfect	Nashipu 2005
Mungaka (EGB, Nun)	-mā	perfect	Miehe 1992
Chufie' (EGB, Nun)	-H	perfective	Hamm & Majenpoh 2024

Table 4: Infinitive forms in some GB languages.

Language	prefix	suffix	source
Meta (Momo)	ĩ-	∅	Spreda 1990
Medumba (EGB, Nun)	nə̀-	-á/-H	Nganmou 1991
Bafut (EGB, Ngemba)	á-	∅	Tamanji 2009
Limbum (EGB, Nkambe)	à	∅	Becker et al. 2019 ^a
Supapya (EGB, Nun)	mà-	-á	Nashipu 2005
Babanki (Central Ring)	á L-	∅	Akumbu et al. 2020
Wushi (South Ring)	mə̀-	∅	Robinson 2021
Isu (West Ring)	í-	∅	Kießling 2011

^aThe authors have written the infinitive marker as a separate word. The assumption is this precedes the verb, but it is not clear from the source.

3 Tone in the noun phrase

Grammatical tones and lexical tones interact in the noun phrase as noun class concord appears on modifiers in the noun phrase. Some GB languages have reduced their noun class systems significantly, so it can be challenging to determine class membership in relation to reconstructed systems. In the Mfumte lects, the noun class system has been lost and the plurality in nouns is marked by tone (McLean 2014 and personal communication).

3.1 Tone in noun class concord

In most GB languages, concord appears as a prefix on the modifier, which puts it between the right edge of the noun and the left edge of the modifier. Hyman (1980b) reconstructs GB noun class affixes and concord in Table 5. Notice that for Proto-Momo & Ring⁵, the tone of the noun prefix is always identical to the tone of the concord⁶, but this is not the case for Proto-Eastern GB.

The tonal portion of noun class concord in PEGB for classes 1 and 9 is /L/ and the remaining classes have /H/. Ring and Momo languages add class 6a to the group which has /L/ tone concord. For Menchum, Boum notes that the tone of

⁵Hyman called these Proto-Western GB, but subsequently this group was shown not to be a coherent unit, hence we use Watters’ name “Momo & Ring” (Watters 2003).

⁶Hyman (1980b: 182) does not allocate a tone for the Proto-Momo & Ring class 6a, but Watters (2003: 240) concludes that it is *mə̀-*.

Table 5: Noun class formatives in Proto-Eastern GB and Proto-Momo & Ring (adapted from Hyman 1980b: 182).

Class	Proto-Eastern GB		Proto-Momo & Ring	
	NPx	Concord	NPx	Concord
1	Ñ-	ù-	ù(N)-	ù-
1a	∅	(=1)	–	–
2	bə-	bá-	bá-	bá-
3	Ñ-	ú-	ú-	ú-
3a	ì-	(=3)	–	–
4	–	–	í-	í-
5	lí-	lí-	í-	í-
6	(=6a)	(=6a)	á-	gá-
6a	mə-	má-	mə-	mə-
7	à-	í-	kí-	kí-
8	bì-	bí-	bí-	bí-
9	Ñ-	ì-	ì(N)-	ì-
10	Ñ-	í-	í(N)-	Cí-
13	–	–	tí-	tí-
19	fə-	fá-	fí-	fí-

the associative marker in classes 1 and 9 is /L/, classes 6a and 18 /HL/ and the rest have /H/ (1980: 76).

Individual languages differ in their behaviour when non-identical lexical and grammatical tones are adjacent in the noun phrase. Contour tone formations ([HL] and [LH]) and [↓H] are common in Chufie’ (Hamm et al. 2013). Others create surface [M], as in Ngamambo (Asongwed & Hyman 1976). Ring languages often have a particularly challenging variety of phonetic realizations of underlying tone as Hyman comments (1980a: 225).

The following sections give further details on what kinds of constructions can display these behaviours.

3.2 Tone in the associative construction

All GB languages have an associative construction which links two nouns together encoding a genitive function (Noun₁ of Noun₂). The tone of the associative marker (AM) depends on the noun class of Noun₁ (Voorhoeve 1971, Hyman

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& Tadadjeu 1976). The form of the AM can vary from a full CV morpheme as seen in all noun classes of Lidzonka (Nkambe) (Voorhoeve 1980: 67, 69), reduced to V as in Yemba (Harro & Haynes 1991: 37), or further reduced to just tonal morphemes, seen in Supapya (Nashipu 2005), Bamun (Nun) (Nchare 2012) and many others.⁷ The floating tonal AM is a common occurrence in GB languages that have undergone much erosion from fuller proto-forms. One finds tonal AM in subgroups such as Eastern Bamileke and Nun (Hyman & Tadadjeu 1976: 75) and the Menchum lects (Boum 1980: 76) where they typically occur for every noun class. However, floating tonal AMs are by no means confined to those subgroups. They can be found in at least one class⁸ in the other subgroups of GB: Bamunka (Ingle, personal communication), Ngie (Hombert 1976), Ngamambo (Asongwed & Hyman 1976), Bafut (Tamanji 2009) and Mfumte (McLean, personal communication).⁹ Thus, it is common to find a downstepped high [$\downarrow$ H] on Noun₂ when it is preceded by a /H/ noun from classes 1 or 9 that have a /L/ AM as shown in (7) (cf. Snider 2026 [this volume]).

- (7) Yemba-Dschang (Hyman & Tadadjeu 1976: 98)
- | | | |
|-----------------|----|--------|
| [ɲká | ↓ | sáɲ] |
| /ɲká-ḥ | L | sáɲ-ḥ/ |
| monkey | AM | bird |
| ‘bird’s monkey’ | | |

By a coalescence rule, the lexical floating tones after the nouns merge with the identical docked ones (the /H/ tone of *ɲká* merges with the following /ḥ/, and the same happens with *sáɲ* and its following /ḥ/). Then the associative marker causes downstep on Noun₂ by virtue of its /L/ being between two /H/ tones.

Recall that there has often been syllable loss at both the left edge and the right edge of nouns in much of EGB, while the tones often persist. When these facts combine with a floating AM, sometimes the result is three floating tones in sequence between Noun₁ and Noun₂: a floating lexical suffix on Noun₁, followed by the grammatical floating AM, and then a floating prefix to Noun₂. Hyman & Tadadjeu (1976: 76–82) report that Voorhoeve’s (1971) Medumba data has shown convincingly that two successive downsteps (also termed “doubled downstep”)

⁷McLean reports there is no segmental AM in any of the Mfumte lects (2014 and personal communication). There is no functional noun class system in these lects either. Plural nouns show evidence of ḥ as a suffix to Noun₁ in the associative construction, among other strategies.

⁸It is not rare in some GB languages to have a mix of AM forms across different classes in the same language: CV, V and tonal AM in one or more classes.

⁹It has been asserted that some classes in Lamnso’ have no AM (Hedinger 2015), however careful investigation may yet reveal a tonal AM.

can occur in such cases. In Medumba, this occurs when the three floating tones /L̥H̥L̥/ are found between attached /H/ tones, as seen in the following example.

- (8) Medumba (Voorhoeve 1971: 50–51 cited in Hyman & Tadadjeu 1976: 76–77)
 Proto-EGB: à-júmà á mù-ánà
 Medumba UF: /L̥-jú-L̥ H̥ L̥-mén-L̥/
 Gloss: thing AM child
 Medumba SF: [ʒú↓ú ↓mén] or [ʒú ↓↓mén]
 Translation: ‘child’s thing’

Hyman & Tadadjeu point out that grounding of floating tonal associative markers has the following tendencies: /L̥/ will attach leftward (to the end of Noun₁); /H̥/ will attach leftward if Noun₁ ends with /L̥/, otherwise it attaches rightward (to the front of Noun₂) (1976: 78). Due to lack of space, we refer the reader to Hyman & Tadadjeu (1976: 62) for more discussion.

3.3 Tone in locative morphemes

There is evidence for locative tonal morphemes in at least three branches of GB: Bamileke, Nun and South Ring. It is unclear if similar phenomena appear in other parts of GB, but it is likely underreported, since the languages that attest it also have segmental prepositions.

It was first documented by Voorhoeve (1974) for Medumba. Interestingly, where it is attested in GB, it is always analysable as a H̥.¹⁰ In the following example from Medumba, two utterances are contrasted, where the only difference is the locative in (9b), which entails two changes: a change of noun class for the possessive, and the tonal changes that begin after the verb.

- (9) a. Medumba (Voorhoeve 1974: 207, 215)¹¹
 [ā bó ā lá? sám]
 /à bó à L̥-lá? sám/
 3SG be ? compound my
 ‘It is my compound.’
 b. [ā bó à là? àm]
 /à bó à H̥ L̥-lá? àm/
 3SG be ? LOC compound my
 ‘It is in my compound.’

¹⁰This is likely reconstructed as *á since it appears as such in various parts of GB (L. Hyman, personal communication).

¹¹Voorhoeve’s tonal notation has been updated to standard usage: [á] = [H], [ā] = [M], [à] = [L], [â] = extra low

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In Chufie', the H locative attaches as a prefix to the noun to form a prepositional phrase. When the noun starts with a continuant, often the continuant is lengthened and becomes syllabic, hosting the /H/ tone.

- (10) Chufie' (Hamm, personal fieldwork)
 [pó^u tʃwó n-fà? rímátāⁱ]
 /pó^u tʃwó N-fà? H-mètāⁱ/
 3PL PROG CNS-work LOC-compound
 'They are working at the market.'

When the noun in such a prepositional phrase begins with a voiceless sibilant or fricative, an epenthetic schwa is inserted to host the locative tone.

- (11) Chufie' (Hamm, personal fieldwork)
 [pó^u tʃwó n-fà? á-səkúù]
 /pó^u tʃwó N-fà? H-səkúù/
 3PL PROG CNS-work LOC-school
 'They are working at school.'

In Bamunka, the locative H tone attaches to the end of the preceding word of the prepositional phrase.¹²

- (12) Bamunka (Ingle 2024: 9)
 [ɔ́ ↓nó kə̀ tùú ↓vyíí]
 /ɔ́ nó kə̀ (L)-tùù-(H) (L)-vyíí/
 3SG REAL P1 PFV-delay-LOC C5-market
 'He stayed in the market.'

Likely more work on locatives in other GB languages would uncover further variation.

4 Tone in the verb phrase

Since verbal parameters such as tense, aspect, mood and modality are often marked with separate grammatical markers within the verb phrase, rather than morphologically on the verb, these categories are discussed in this section.

¹²The tonal locative preposition is distinct from an associative marker.

4.1 Tone in tense marking

Tone often plays a significant role in the realization of many verbal categories in GB languages, and tense is no exception. This seems to be particularly true for past tense distinctions. For example, in Choepchop, the P3 (distant past) and P2 (near past) tenses are distinguished by tone alone. P3 *nə* has /L/ tone but P2 *ná* has /H/ (Robinson 2024: 3–4). Similarly, in another Nun language, Chrambo, the hesternal P2 (*nì*) and the hodiernal P1 (*nî*) are distinguished by tone alone (Wright 2011: 12).

In the West Ring language Isu (Kießling 2017: 245), the inferential remote past (P3) and evidential hesternal past (P2) are both marked segmentally with *kə*, and distinguished by tone: P3 is /H/ (with a following /L/), whereas P2 is /L/ (with a preceding /L/).

The situation becomes even more complex when tenses are marked solely by floating tonal morphemes, i.e. morphemes without any segments. For example, in Chufie', both P3 and P2 are marked solely by tone: P3 is marked with /L/, whereas P2 is marked with /H/ (Hamm & Majenpoh 2024: 32).

However, the segmental erosion of the tense marker is not always this complete. For example, in Isu, the P3 marker /*kə́*/ can undergo contraction morphosyntactically: in certain discourse contexts¹³ the segments may drop out, leaving only a /HL/ tone sequence.

- (13) Isu P3 contraction: /*kə́*/ followed by H tone verb *fyí* 'go out' (Kießling 2017: 251)
 [má kə́ ↓fyí] or [má ↓fyí] 'I went out.'

It is worth pointing out that it is not unusual to analyse the underlying tone of tense (and aspect) markers in GB languages as a tone sequence rather than as a single /H/ or /L/ unit. There are historical reasons accounting for these tone sequences. For example, in Chufie', three of the four segmental tense markers have underlying /LH/ tone, seen in (14).

- (14) Chufie' (adapted from Hamm & Majenpoh 2024: 33)
- | | |
|-------|-------------------|
| /pí/ | 'P1 tense marker' |
| /yũ/ | 'F1 tense marker' |
| /lʷǔ/ | 'F2 tense marker' |

¹³Kießling notes that "reduction indicates consecutivization in a string of narrated events... whereas retention in its full form entails a loss of a sense of sequentiality, evoking the impression of a report of isolated events" (2017: 252).

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It is likely that these tense markers were historically lexically /L/ verbs followed by a perfective /H/ tone, which accounts for the /LH/ sequence on the tense markers synchronically (Hamm & Majenpoh 2024: 34).

4.2 Tone in aspect marking

As with tense marking, tone has a significant functional load in aspect marking across the GB area.

Often perfective is the least marked aspect in GB, however, the perfective aspect may also be marked by tone. The data in Table 6 show how various subgroups in GB mark perfective aspect. The data show only non-focused marking, when using the P0 (unmarked tense).

Table 6: Perfective marking in some GB languages

Language	Form	Source
Mundani (Momo)	∅	Parker 1991
Nda’nda’ (EGB, Bamileke)	-H	Chumbow & Nguendjio 1991
Bafut (Ngemba)	L-	Tamanji 2009
Chufie’ (EGB, Nun)	-H	Hamm & Majenpoh 2024
Chopechop (EGB, Nun)	-H	Robinson 2024
Babanki (Central Ring)	L-	Akumbu et al. 2020 ^a
Lamnso’ (East Ring)	∅	Hedinger 2015
Wushi (South Ring)	LH-	Robinson 2021
Vengo (South Ring)	L-	Schaub 1985
Aghem (West Ring)	∅	Hyman 1979a

^aBabanki data is cited in the conjoint form. The disjoint is á-verb-L lí.

The data in Table 6 show a definite pattern: if perfective is marked pre-verbally, it is /L/ (or /LH/ in the case of Wushi), and if it is marked post-verbally, it is /H/. However, more research would be necessary to find out if this generalization based on a limited sample of languages holds for the family as a whole.

We were not able to access data on the perfective for Nkambe languages, which is why they are missing from Table 6.

Similarly, the imperfective aspect may be marked solely by tone in some GB languages (see Table 2). For example, in Babanki, the imperfective is marked with a floating /H/ tone following the verb (Akumbu 2015: 14). In Chufie’, it is marked with a /LH/ tone sequence following the verb (Hamm & Majenpoh 2024: 43).

Watters posits that imperfective aspect in GB is marked with a verbal suffix *-a*, which is related to the Bantu pre-final *-a(n)g* (2003: 245). In Ngiemboon, like a number of other Bamileke languages, the marking of imperfective aspect is by means of a prefix and a suffix. The prefix is a homorganic nasal and cooccurs with the *-a* suffix. These combine with realis/irrealis polar tones both before and after the verb in very complex patterns (Anderson 1983: 58–59) (cf. Pulleyblank 2026 [this volume]).

In various grammars of GB, imperfective aspect is sometimes not clearly separated from subtypes such as progressive, habitual, incomplete, etc. As more research is done, it will be possible to give a better characterization of the tones involved in imperfective marking and how it combines with other means for marking these subtypes.

4.3 Tone in modality marking

Grammatical marking in the GB verb phrase is complex, and in addition to the analysis of tense and aspect, there is often analysis of modality in terms of the realis/irrealis distinction (Watters 2003: 246). Clauses can be divided into realis (referring to events that have taken place, or are taking place) and irrealis (referring to events that have not (yet) happened).

The marking of realis/irrealis distinctions in GB languages typically involves floating tonal morphemes. For example, in Wushi, the realis modality is unmarked, but the irrealis modality is marked with a /H/ tone before the subject of the clause.

(15) Wushi (adapted from Robinson 2021: 45)

- | | |
|--------------|----------------------|
| a. [mè nǎ] | b. [mǎ ↓ ná] |
| /mè LH-ná/ | /H/ mè ná/ |
| 1SG PFV-cook | IRR 1SG cook |
| ‘I cooked.’ | ‘I will be cooking.’ |

All clauses in past tenses are therefore unmarked for modality, whereas in future tenses, the irrealis tonal morpheme is always evident as a /H/ replacive tone on the subject. In the unmarked tense (‘P0’), if the event has happened or is happening, there is no modality marking, but if the event is yet to happen, the irrealis tonal marker is used (Robinson 2021: 44–45).

Similarly, in Chufie’, realis has been analysed as the unmarked (default) modality. The irrealis modality is marked with a /H/ tone following the subject of the clause (Hamm & Majenpoh 2024: 72).

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The marking of the realis/irrealis distinction is more complex in Ngiemboon, where both realis and irrealis are marked (Anderson 1983). The realis modality is marked with a polar tone before the verb, whereas the irrealis modality is marked with a polar tone both before and after the verb. Anderson offers the following underlying tone patterns for realis and irrealis modality in both /H/ and /L/ verbs, shown in Table 7. Note that we have marked the verb root tone in bold.

Table 7: Realis/irrealis underlying tone patterns in Ngiemboon (Anderson 1983: 70).

	Realis	Irrealis
/H/ verb root	L- H -Ø	L- H -P
/L/ verb root	H- L -Ø	H- L -P

Anderson notes that “P” indicates a polar tone, whose identity is the opposite of what follows it. The verbal prefixes for both realis and irrealis are polar and are represented as opposite of the tone that follows it, namely the verb root tone. Thus, if the verb root is /H/, the prefix is /L/, and in the same way, when the verb root is /L/, the prefix is /H/.

4.4 Tone in marking of mood

The declarative mood is the default, unmarked mood in GB languages. The other moods (interrogative, imperative and hortative) are often marked tonally (or intonationally) rather than segmentally.

In the interrogative mood, polar questions (“yes-no questions”) may have a sentence-final question marker that is segmental with inherent tone (Watters 2003: 249), but it is also common to find that polar questions are marked mainly by intonation. For example, in Wushi, a polar question can be viewed as a statement plus an “interrogative overlay”. This interrogative overlay consists of a moderate fall in pitch in the utterance-final syllable, with phonetic vowel lengthening in this syllable (Robinson 2021: 53).

Before discussing the imperative and hortative moods, it should be noted that many analyses of GB languages make a distinction between the imperative mood and the hortative mood. In many GB languages, the imperative mood is typically only used when addressing a single person and the singular subject pronoun is dropped. If multiple addressees are given a command, the hortative mood is used. The hortative mood, however, can be used with all the pronouns and

the pronouns are not dropped. The imperative and hortative moods also tend to have distinct forms. Often a command to a group of people uses the hortative mood form with the second person plural pronoun. Thus, imperative mood is used exclusively with a single addressee, and if any other participants are added, then a hortative form is used instead. There are, however, some languages where imperative is used for all 2nd person addressees, and hortative mood is used for other persons. This is an area where more research could be beneficial.

The reconstructed marker of the imperative in EGB languages is a /H/ tone following the verb (Elias et al. 1984: 37). This is corroborated by some descriptions of GB languages, seen in Table 8. The presence of this /H/ in non-EGB languages shows that it should likely be reconstructed for GB as a whole.

Table 8: Imperative suffixes

Language	Form	Source
Chufie' (EGB, Nun)	-H	Hamm & Majenpoh 2024
Chrambo (EGB, Nun)	-HL	Wright 2009
Chopechop (EGB, Nun)	-H	Robinson 2024
Supapya (EGB, Nun)	-H	Nashipu 2005
Medumba (EGB, Nun)	-á	Nganmou 1991
Meta (Momo)	-H/-V	Spreda 1990
Wushi (South Ring)	-H	Robinson 2022
Kenswei Nsei (South Ring)	-H	Ingle, pers. comm.
Kom (Central Ring)	-H	Shultz 1997
Babanki (Central Ring)	-H	Akumbu et al. 2020
Isu (West Ring)	-HL	Kießling 2011
Bafut (Ngemba)	-H	Mfonyam 1989
Limum (Nkambe)	Ø	Mfonyam 1989

In other Ngemba languages, the imperative is marked by a /HL/ tone sequence following the verb (Elias et al. 1984: 37). Similarly, in Chrambo, imperative marking has been analysed as a /HL/ tone sequence following the verb (Wright 2009: 59). However, the /L/ tone in this /HL/ sequence may actually be a boundary tone that is used when there is no object following the verb (see Hamm & Majenpoh 2024: 80).

In Wushi, the initial analysis of imperative marking proposed a /HL/ sequence following the verb (Robinson 2021: 58), but in subsequent work (Robinson 2022: 5), the imperative marking was re-analysed as a /H/ tone following the verb, with

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an utterance-final boundary /L/ tone that only occurs when the imperative verb form is found before a pause.

Interestingly, this /H/ imperative (IMP) marking after the verb, followed by the /L/ boundary tone (BT), can give rise to a “convex” surface tone pattern on the verb, i.e. [LHL], when a lexically /L/ verb is used, as illustrated in example (16):

- (16) Wushi (Robinson 2022: 6)
 [dɛ̃ɛ̃]
 /dɛ̃ɛ̃-H-L/
 teach-IMP-BT
 ‘Teach!’

Yip (2002a: 30) states that complex contour tones – tones that fall and then rise (“concave”) or that rise and then fall (“convex”) are rare in African languages. However, these complex contour tones on a single syllable are not uncommon in the GB languages. (The surface tone pattern would be identical if the verb in example (16) was replaced with a verb with a short vowel.)

The hortative mood can be loosely defined as a weak command. The hortative in GB languages is typically marked tonally with a /H/ tone preceding the verb. It may additionally be marked with a /H/ tone preceding the subject of the clause. A recent study (Beavon-Ham et al. 2024) showed that the Ring languages are more likely to only mark the hortative preceding the verb, whereas the EGB languages are more likely to mark it both preceding the verb and preceding the subject of the clause. This is illustrated in Table 9.

Two behaviours are seen for the /H/ hortative tone that precedes the verb. It can dock onto the verb as in Wushi in example (17), or it can dissociate the lexical tone of the verb which provides the condition for downstep, as in Mendenkye (EGB, Nun) in example (18).

- (17) Wushi (Robinson, fieldwork)
 [ɲwì pfǎ̀ɛ̃]
 /ɲwì H pfǎ̀ɛ̃/
 2PL HORT.hit
 ‘You should hit.’
- (18) Mendenkye (Langhout, personal communication)
 [pàpwèè gǎ̀tǎ móo]
 /pàpwèè H gǎ̀tǎ móo/
 people HORT.divide-PLUR things
 ‘People should divide things.’

Table 9: Hortative marking in GB languages that lack subject markers (Beavon-Ham et al. 2024).

Language	H-SBJ	H-verb
Mendenkye (Bangolan) (EGB, Nun)	-	+
Supapya (EGB, Nun)	+	+
Chrambo (EGB, Nun)	+	+
Chopechop (EGB, Nun)	+	+
Chufie' (EGB, Nun)	+	+
Shupamem (EGB, Nun)	-	+
Ngomba (Bamileke)	+	+
Wushi (South Ring)	-	+
Ngiemekohke (South Ring)	-	+
Kenswei Nsei (South Ring)	-	+

More research is needed on the hortative mood in other GB languages to understand how far-reaching these tentative conclusions are, or if there are other strategies for hortative marking.

It is clear from the previous sections on the verb phrase that tone plays an important role in the realization of all of the TAM categories. Even when segmental markers exist, they may only tell part of the story, since a floating tone may still be present, whose effects are only known in context.

5 **Tone in the clause & multi-clausal constructions**

Moving on to clause-level phenomena, we show below that tone is the distinguishing criterion for subject and/or object agreement in the verb phrase, and also marks conditional clauses as different from both other forms of subordinate clauses and coordinated independent clauses. The analysis of tonal data in conditional clauses is from very recent work and is somewhat tentative.

5.1 **Tone in subject and object agreement**

Many GB languages exhibit some form of concord between the subject and the verb via an agreement morpheme in the verb phrase. The form of this morpheme varies and can be a full CV, V, N, or a floating tone (Watters 2003). As a case in point, consider Bafut, where the tone of the subject marker (SM) is /L/ for classes

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1 and 9, and /H/ elsewhere.¹⁴ The subject marker is part of the verb phrase and indexes the subject by agreeing with the noun class of the noun that functions as the subject of the clause. However, it seems to be in process of segmental loss as it is not pronounced in normal speech (the top line of the Bafut examples below), but it is still present in careful speech (the second line in the examples below).¹⁵

- (19) Bafut (Tamanji 2009: 5)
 [bɪfɔ̃ bjá Ø kɪ ʃɪʔi tsəʔə dzá nʃɪʔi]
 /bɪ-fɔ̃ bj-á bí kɪ ʃɪʔi ɪ-tsəʔə dz-á N-ʃɪʔi/
 2-chief 2-the SM(2) P2 wash 8-dresses 8-the N-wash
 ‘The chiefs washed the dresses thoroughly.’
- (20) Bafut (Tamanji 2009: 101)
 [lɔ̃ŋɔ̃ wâ Ø kɪ kɔʔɔ ntáʔà]
 /lɔ̃ŋɔ̃ w-á à kɪ kɔʔɔ á n-táʔà/
 horse(1) 1-the SM(1) P2 climb PREP 1-mountain
 ‘The horse climbed up the mountain.’

In example (19), the subject marker is not pronounced in normal speech and there is no apparent change in the tone when comparing the surface and underlying forms. However, in example (20), we see that the /L/ tone from the subject marker is attached to the end of the word that precedes it. When the segments of the subject marker are not pronounced, the tones remain intact. Thus, in (19), the /H/ tone of the class 2 subject marker merges with the /H/ tone of the preceding word. But in (20), the /L/ tone of the class 1 subject marker docks leftward, producing a falling contour surface tone.

In some Nun languages, even the tonal agreement is lost, thus there is no evidence of subject agreement as in Chufie’ (Hamm & Majenpoh 2024) and Chrambo (Wright 2009).

Object agreement is less commonly documented in GB, but has been found in a handful of Bamileke and Nun languages: Ngiemboon (Anderson 1983), Yemba (Harro & Haynes 1991, Chen & Lehman 2021, Czuba 2022), Supapya (Nashipu 2005) and Chufie’ (Hamm & Majenpoh 2024). In all of these, the object marker is a floating tone except for Yemba where it can still be found as a vowel with a specific tone, the details of which depend on the class and type of onset of the noun. In rapid speech, however, it is often reduced to just a tonal morpheme (Chen & Lehman 2021).

¹⁴Bafut has CV and V forms of subject markers (Tamanji 2009: 101).

¹⁵“Speakers actually tend to omit the subjectival concord except in very slow and deliberate speech.” (Tamanji 2009: 100).

- (21) Yemba (adapted from [Chen & Lehman 2021](#): 915)
- a. à lè tóŋ-ó ↓sén
3SG P3 call-OM1 1.bird
‘He called the bird (some time ago).’
- b. à lè tóŋ-L m-↓bhá
3SG P3 call-OM9 9-dog
‘He called the dog (some time ago).’

Subject and object agreement offers an interesting view of present-day change in progress, where loss of segments lead to floating tonal morphemes in languages like Bafut and Yemba. In some Nun languages, the loss of the subject marker has fully taken place.

5.2 Tone in conditional marking

In recent years, some investigation of conditional marking has revealed the possibility of floating tonal morphemes in some Nun languages. There are other strategies for marking a conditional ‘if-clause’ with a segmental conditional morpheme, but these exist alongside floating tonal conditional morphemes.¹⁶

At this stage, analysis is not yet thorough enough to make definitive statements, but evidence points towards a /H/ that follows the subject in Chufie’, Supapya and Chrambo. [Table 10](#) illustrates the behaviour of the conditional /H/. In each case, the first word is a subject pronoun with underlying /L/ tone, and this is followed by a verb which is also underlyingly /L/.

Table 10: Conditional marking in Nun languages: Chufie’ (Hamm, personal fieldwork), Supapya (Hamm, personal fieldwork), Chrambo ([Wright 2011](#): 12).

Language	S	V	O	Translation
Chufie’	[ǎ	ń-zó ^u	↓sé ⁱ]	If he follows profit...
	/à H	N-zò ^u	sé ⁱ /	
Supapya	[tǐ	yúp	ŋgùp tǐ]	If we (dual) peel tree bark...
	/tì H	yùp	ŋgùp tí/	
Chrambo	[á	shíri	↓shíe]	If he turns corn...
	/à H	shìri	shíe/	

¹⁶It is possible that the different strategies encode different types of conditionals.

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In all three languages, the /H/ affects both the subject and the verb. In Chufie' and Supapya, there is a rising pitch on an underlyingly /L/ subject pronoun. In Chrambo, the conditional /H/ replaces the /L/ of the pronoun.

Concerning the verb perturbation, in all three languages the /L/ verb is realized with high pitch. More work is needed to give the precise characterization of the conditional tonal morpheme and to explain the processes involved here, but there is certainly evidence of tonal alternations in conjunction with conditional marking in these Nun languages. This has not yet been documented in other subgroups of GB.

6 Conclusion

The overwhelming evidence from research on GB tone is that the source of its tonal complexity are the two tones /H/ and /L/. Even in the languages where an underlying /M/ tone has been proposed in the synchronic analysis, this is analysed as historically derivable from /H/ and /L/ – see, for example, the description of a “phonemic mid tone” in some of the Ngemba languages and in Limbum (Mfonyam 1990: 22).

From a universal perspective, in a two-tone system, the low tone is often unmarked (Yip 2002b: 26, 41), however both tones are phonologically active in GB. The low tone, for example, is engaged in tone spreading – recall the low tone spreading from nominal prefix to root in Ngiemboon, described in Section 2.1.1 (cf. Smolders 2026 [this volume]). This means that GB languages do not have /H/ vs Ø tone systems (privative). Despite many languages having just /H/ and /L/ underlying tones, the GB languages are considered to have some of the most complex tone systems in Africa (Watters 2003: 236). The prevalence of floating tones (both lexical and grammatical tones that have remained after segments are lost) means that “any synchronic tonological analysis will have to involve reconstruction to some degree” (Kießling 2020: 1). Thus, for example, the reconstruction of disyllabic noun roots allows us to account for the four tone categories typically found synchronically in monosyllabic GB noun roots.

It is also worth pointing out that GB languages are not what Yip (2002b: 28) describes as “true contour tone languages”. In GB languages, a contour is never a single phonological object, but a sequence of tones – recall example (16) in Section 4.4.

Tone has a significant functional load in GB languages, both in terms of the lexicon and the grammar. The tonal complexities often become apparent only when examining data in the context of, for example, noun phrases and verb phrases.

Various tone processes are typical in GB languages: tone spreading, tone docking, tone merging and non-automatic downstep. Some GB analyses describe “double downstep”, where /LHL/ floating tones between two attached /H/ tones cause the register to lower twice.

There is also evidence of polar tones, boundary tones and underlyingly toneless morphemes in these languages. As already mentioned earlier in this section, floating tonal morphemes are widespread.

The TBU in GB languages is typically the syllable rather than the mora (see, for example, Robinson 2022: 9). Tones typically attach from left to right (see, for example, Anderson 1983 and Hamm & Majenpoh 2024: 4).

Despite the fact that consonants affecting tones – “depressor consonants” – are quite common in African languages (Yip 2002b: 33), there is little evidence of this phenomenon in the GB area (cf. Beavon-Ham & Pearce 2026 [this volume]). This is noteworthy given that depressor consonants often play a significant role in the related, but physically distant Narrow Bantu languages of East Africa (Maddieson 2004: 29–30).

We should also note that whereas some African languages are described as accentual, or somewhere between tonal and accentual (Yip 2002b: 139–140), there is no clear evidence of accentual systems operating in the GB languages, apart from stem-initial prominence which is widespread in GB (Franich 2024). This is likely due to the complexity of tonal phenomena in the GB area (Anderson 2001: 14).

Much research remains to be done on tone in GB. For example, there has been little analysis of the function of tone at the discourse level. Research on the relationship between intonation and tone is also lacking. More studies discussing the role of intonation in these highly tonal languages (see, for example, Kießling 2017) are needed.

There is also a need for more research into the critical interaction of tone processes in these languages. As tone is analysed in depth in more languages, we may be able to see whether typological patterns emerge regarding such interactions.

Abbreviations

AM	associative marker	CNS	consecutive
BT	boundary tone	EGB	Eastern Grassfields Bantu
C	consonant	GB	Grassfields Bantu
C1,	etc. class 1 (noun prefix)	H	high tone

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L	low tone (which falls)	SVO	subject, verb, object
L°	non-falling low tone	SM	subject marker
LOC	locative preposition	T̥	floating tone
M	mid tone	TBU	tone bearing unit
N	homorganic nasal	V	vowel
NPx	noun class prefix(es)	WGB	Western Grassfields Bantu
OM	object marker	↓	downstep
P	proto-	/.../	underlying representation
PLUR	pluractional	[...]	phonetic (surface)
PREP	preposition		representation
P0	unmarked tense	`	low tone
P1	immediate past tense	ˊ	high tone
P2	near/hesternal/medial past tense	-	mid tone
		ˋ	falling tone
P3	distant/remote past tense	ˊˊ	rising tone
REAL	realis modality		

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Chapter 15

Tonal systems in Benue-Congo and West Chadic A3 in Central Nigeria

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The paper describes the tonal systems of the Benue-Congo languages, excluding Bantoid and Bantu. Benue-Congo consists of four major branches, Kainji, Plateau, Jukunoid and Cross River. The Ukaan language is possibly also affiliated. Also included is a part of West Chadic, A3, since this appears to have been completely restructured by contact with its Benue-Congo neighbours. Tone systems are extremely variable, with Jukunoid including examples of some of the most levels globally, with five level tones. Only Cross River is lacking in such diverse systems. Especially in Plateau, there are examples of upstep, downstep and downdrift as well as extensive grammatical tone.

1 Introduction

Benue-Congo is the name for a large and complex group of languages, whose exact membership remains disputed. Originating with Westermann's (1927) *Benue-Cross-Fluss*, it took shape in Greenberg (1963), Williamson (1971) and De Wolf (1971). The name "Benue-Congo" was introduced by Greenberg (1963) who proposed a division into four branches, what would now be called Plateau, Jukunoid, Cross River, and Bantoid. Plateau was subsequently split into what is called Plateau today and Kainji (for which see Gerhardt 1989). Bantoid is not a coherent branch like the others, rather a nexus of individual families related to Bantu (cf. Grimm 2026 [this volume], Blench 2026 [this volume], Hamm et al. 2026 [this volume], and Connell 2026 [this volume]). For a period in the 1980s and 1990s,

Roger Blench. 2026. Tonal systems in Benue-Congo and West Chadic A3 in Central Nigeria. In Keith L. Snider & Virginia Beavon-Ham (eds.), *African tone phenomena*, 479–511. Berlin: Language Science Press. DOI: ??

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it was considered that all the languages in former “Eastern Kwa”, i.e. Yoruboid, Igboïd, Nupoid etc. were part of Benue-Congo, i.e. Western Benue-Congo. However, the evidence for this was never published and it seems easier to revert to the definition of Benue-Congo as in Greenberg’s original. With this in mind, Figure 1 provides a revised subclassification of Benue-Congo languages.

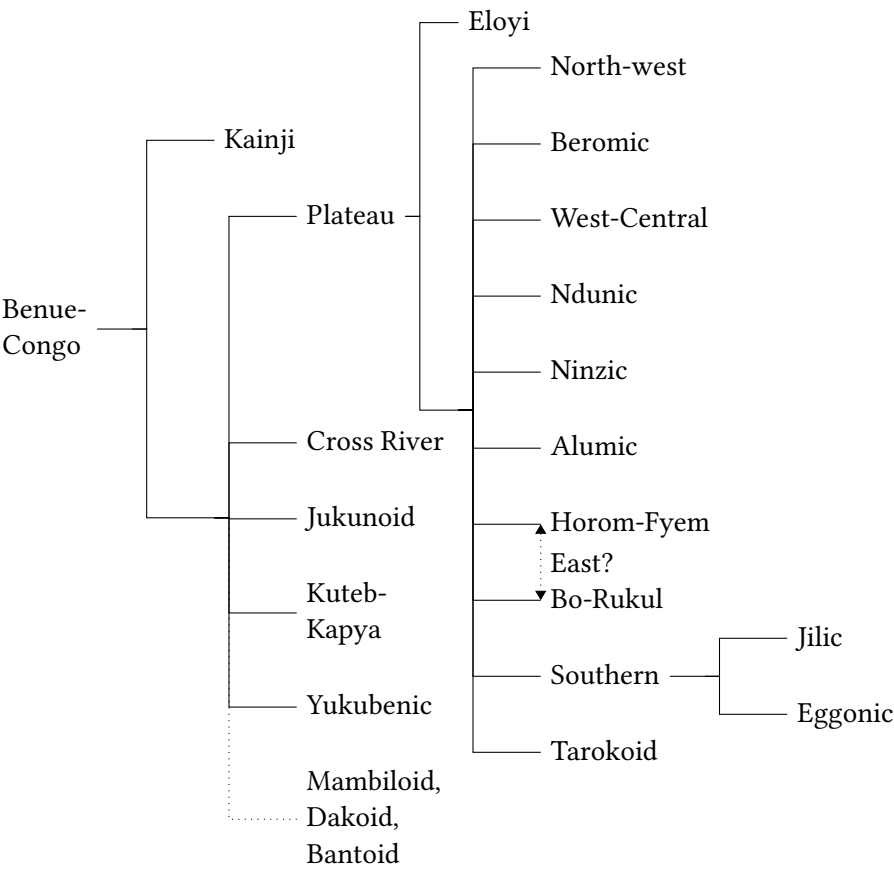


Figure 1: Revised subclassification of Benue-Congo languages

Benue-Congo is of considerable importance for the understanding of Bantoid, and thus Bantu. Strikingly the branches of Benue-Congo contain some of the most complex tone languages in Africa, so the chain of developments which led to the relatively simple tone systems in Bantu is an analytic problem of considerable interest.

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Tones are typically transcribed on vowels in African languages although analytically they are regarded as suprasegmentals and associated with structural units, such as morae. However, in Benue-Congo languages, it is possible for almost all sonorants to bear tones and in some rare examples, fricatives and stops, i.e. they are moraic. The most common non-vowels are the nasals, /n/, /m/, /ŋ/ and /ɲ/. Characteristically, these bear tones when in a homorganic relationship with a following stop. This can be at the beginning of a word or morpheme or more rarely in final position. Tone-bearing nasals do not usually have contour tones, and most often bear low tones, although there are always exceptions. Less common, but attested are tone-bearing liquids, /l/ and /r/. These are found in Kainji and Plateau languages in Central Nigeria and seem to arise from the assimilation of nasals to them. Thus an original $N + l \sim r \rightarrow \tilde{L} + l \sim r$. Rarer still are the fricatives and stops occurring in languages such as cLela [dri]¹, where the loss of a -V in a CV prefix, has shifted the tone leftwards onto the consonant (Dettweiler 2015). Similar processes have been noted in Nilo-Saharan languages, such as Lendu (Dimmendaal 1986).

Some tones arise from morphological processes and are thus distributed unevenly. An example of this is the superhigh tones which arise from tone-raising rules. For example, Boze [jer], an East Kainji language spoken north of Jos in Plateau State, Nigeria, has a rule where nominal plurals are formed by raising the stem tone one level. Singulars usually only exhibit three tone heights but High becomes Superhigh in plurals. Intriguingly, contour tones can also be subject to this rule, such that a High-Mid in a singular becomes the contour tone Superhigh-High in the plural (for which Unicode so far has no diacritic²). A small number of languages exhibit double contour tones, i.e. those where the tone makes two movements, while still being applying to a single mora. Examples of this are found in Angan (Plateau) and Mambila (Mambiloid). Perrin & Hill (1969) and Perrin (1974) describe the tonal system of Ba Mambila, which has four tone heights and multiple contour tones. These are notated with superscript numerals, and examples are presented of lexemes with three heights on a single mora, e.g. *ven*²¹⁴, ‘evil’.

This chapter discusses the tonal systems of Benue-Congo languages through analyses of sample languages in each of the four major branches listed above. It provides access to significant descriptions of tone where these exist, but it should

¹I have marked the ISO three-letter codes throughout the paper so that readers can more easily locate the languages discussed. However, this does not imply I endorse the divisions between speech forms they imply.

²It is worth adding that Unicode has so far not fully addressed the area of complex contour tones

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be noted that for some branches the evidence is very fragmentary and remains unpublished. A final section discusses the West Chadic A3 languages, which although not Benue-Congo, are geographically intertwined with them and have apparently been intensively restructured as a result of contact (cf. [Roberts 2026 \[this volume\]](#)). A final section covers the problems of developing orthographies which incorporate tone-marking and the strategies users of social media are developing to type tone languages on phones.

2 Kainji

2.1 Overview

The Kainji branch of Benue-Congo was historically divided into two families, East and West, which originated with Greenberg (1963) and Rowlands (1962) and persisted in later publications, e.g. [Gerhardt \(1989\)](#). The two sub-branches are geographically remote, so this division seemed reasonable. However, as [Blench \(2018b\)](#) pointed out, it is not supported by the linguistic evidence. The primary sub-branches of Kainji are Reshe and the Lake languages, and East Kainji consists of well-formed languages with strong resemblances to Bantu and other subfamilies of Kainji such as the Basa group.

2.2 West Kainji

West Kainji consists of some twenty-four languages or lects spoken in north-western Nigeria. Because of their extremely diverse lexicon and morphology, it is only comparatively recently that their unity has been recognised ([Gerhardt 1989](#) is where this division is first recognised in the standard publication). Typologically, they are difficult to characterise, but some of the more conservative branches have both nominal and verbal morphology highly reminiscent of Bantu (as indeed the *-tu* root for ‘person’). As elsewhere, languages with comparatively long words, such as the Kamuku and Kambari branches, have more restricted tone systems. The “Lake” languages, Laru [lan] and Lopa [lop] have lost all nominal morphology, probably under the influence of Busa, a neighbouring Mande language, and have correspondingly more elaborate tone systems (cf. [Konoshenko 2026 \[this volume\]](#)). Dedicated accounts of tone are few and tonal systems need to be extracted from more general grammatical descriptions. These include Hackett and Davey (2004) for C’Hungwere [nat], McGill (2009) for Cicipu [awc], Mort (2012) for TiCind [cdr], Dettweiler (2015) for C’Lela [dri] and Paterson (2019) for Ut-Ma’in [gel].

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TiCind is a Kamuku language spoken in Niger State, in northwest Nigeria. Mort (2012) describes the TiCind tonal system. TiCind has two level tones, high (H) and low (L). These are most clearly seen in disyllabic noun roots, which display all four expected patterns: HH, HL, LH, LL. TiCind has both lexical and grammatical tone (cf. Rolle & Bickmore 2026 [this volume]). Lexical tone contrasts are relatively rare, although they occur in noun roots. Many demonstratives and possessives form a minimal tone pair with a noun, such as [mì-ní] ‘NC10-water’ and [mí-ní] ‘NC10-1SG.POSS’. Tone is apparently lexicalised as a pattern on a morpheme rather than assigned to individual syllables. This can be seen in a language game characterised by reversed speech, since the tone melody across a morpheme remains the same even though the syllables have been reversed, showing that it exists independently from the syllables in the mind of speakers. An example of this is shown by nasality, which spreads within words from roots to affixes. When vowels are very short, notably in CV class markers, the nasality is not easy to hear, but is clearer when the word is reversed, for example;

/bè-ɾʷáʊǎ/ reversed: [ʊǎɾʷábé] NC4-snail

Automatic downstep occurs in Cinda, whereby a low tone triggers the lowering of the pitch of subsequent high and low tones, for example:

- (1) [bèràkwà ɛɛ bí-ní]
 /bè-ràkwà bí-ní/
 horse NC4-1SG.POSS
 ‘my horse’

When people are calling over a distance, downstep is apparently eliminated and the tones are pronounced at the same pitch throughout the utterance.

When tone spreading and automatic downstep occur, together they produce non-automatic downstep (cf. Smolders 2026 [this volume]). When a H tone spreads to the right, it can cause a L tone to be delinked and become floating (cf. Rolle et al. 2026 [this volume]). The L tone still causes downstep, but this is considered non-automatic because the L tone is not itself otherwise evident. We see this in (2). Here the second syllable of the noun [bágò] ‘quiver’ first undergoes H spreading which, in turn, is realized as a “super-high” tone, but only the second syllable undergoes upstep. The L tone of the first word that was dislodged by H spreading now occurs as a floating tone between the two words and causes the H tones of the second word to be downstepped (non-automatic).

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- (2) [bág^w ɔ ʔ í-ní]
/bágò ʔí-ní/
quiver NC2-1SG.POSS
'my quiver (of arrows)'

The super-high tone in (2) could also be considered an example of upstep. However, this analysis is problematic since the upstepped H is actually the same H tone as has spread from the the previous syllable. To enable an analysis of upstep, we would be required to say that, once the tone has spread, it divides and becomes two separate tones, the second of which is upstepped.

Verb roots are treated as toneless, but tense and aspect superimpose lexical tone patterns. These depend primarily on the number of syllables of the basic stem, i.e. the number of syllables of the root plus any extension suffix plus a final vowel. It is plausible that such patterns are common but not enough data is available to support this.

The Țt-Ma'in language [gel], spoken in northwest Nigeria (Paterson 2019), has a different tonal system when compared with TiCind (above) and Boze (below). Țt-Ma'in nouns have three contrastive level tones: high (H), mid (M), low (L) and apparently no contour tones. Each noun stem is presented with its noun class prefix; all noun class prefixes in citation form have a M tone. Table 1 shows an example where the noun class morphology is identical for all three singular/plural pairs. The schwa stands for what is an underspecified vowel.

Table 1: Minimal tone triplets for the noun stem /bah/ in Țt-Ma’in
(Source: Paterson 2019)

Stem Tone	Singular	Gloss	Plural	Gloss
H	ǎr-báh	‘C5-foot’	ǎt-báh	‘C6-feet’
M	ǎr-bāh	‘C5-bag’	ǎt-bāh	‘C6-bags’
L	ǎr-bàh	‘C5-lake’	ǎt-bàh	‘C6-lakes’

Stems always retain the same tone in singular and plural. Tone can also be used to distinguish stems in derivational processes, as in;

- (3) H āt-gáp 'C6-slapping'
 M ū-gāp 'C3-slap'
 L āt-gàp 'C6-hitting'

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C’Lela has a further reduced system, consisting of high (H), low (L) and falling (HL) (Dettweiler 2015). Falling [HL] tone occurs primarily in disyllabic noun roots, and its distribution is restricted as follows:

- 1. It is almost always in a word-final CV syllable where the V is phonetically long.
- 2. The preceding (penultimate) syllable is always low tone.
- 3. The penultimate syllable is almost always either a CVC syllable, or a CV syllable with a long vowel in the nucleus.

Examples illustrating this process are given in Table 2.

Table 2: Tone Pattern in C’Lela Noun Roots (Source: Dettweiler 2015)

(px-) L H L	Gloss
[ʔà.gâw]	‘bee-eater bird’
[də-ʔò.bô:]	‘squash’
[nà:.mâ:]	‘cow’
[tʃə-kà:.tâ:]	‘sandals’
[lòg.mô:]	‘elephant’
[wàr.kô:]	‘young child’

Given that Ut-Ma’in and C’Lela are closely related at the lexical level and share the common morphosyntactic feature of prefixes with an underspecified vowel, this suggests tone systems can rapidly diversify.

2.3 East Kainji

East Kainji consists of some thirty languages spoken north of Jos, Plateau and Bauchi States. Almost all have only a small number of speakers, and some are either moribund or extinct, so no useful information can be recovered about their tonal systems. Compared with West Kainji, which has undergone a wide variety of morphological changes that makes its branches look very diverse, the East Kainji languages for which data exist are comparatively similar. Figure 2 shows a tentative tree of East Kainji languages, marking those which have been confirmed as extinct.

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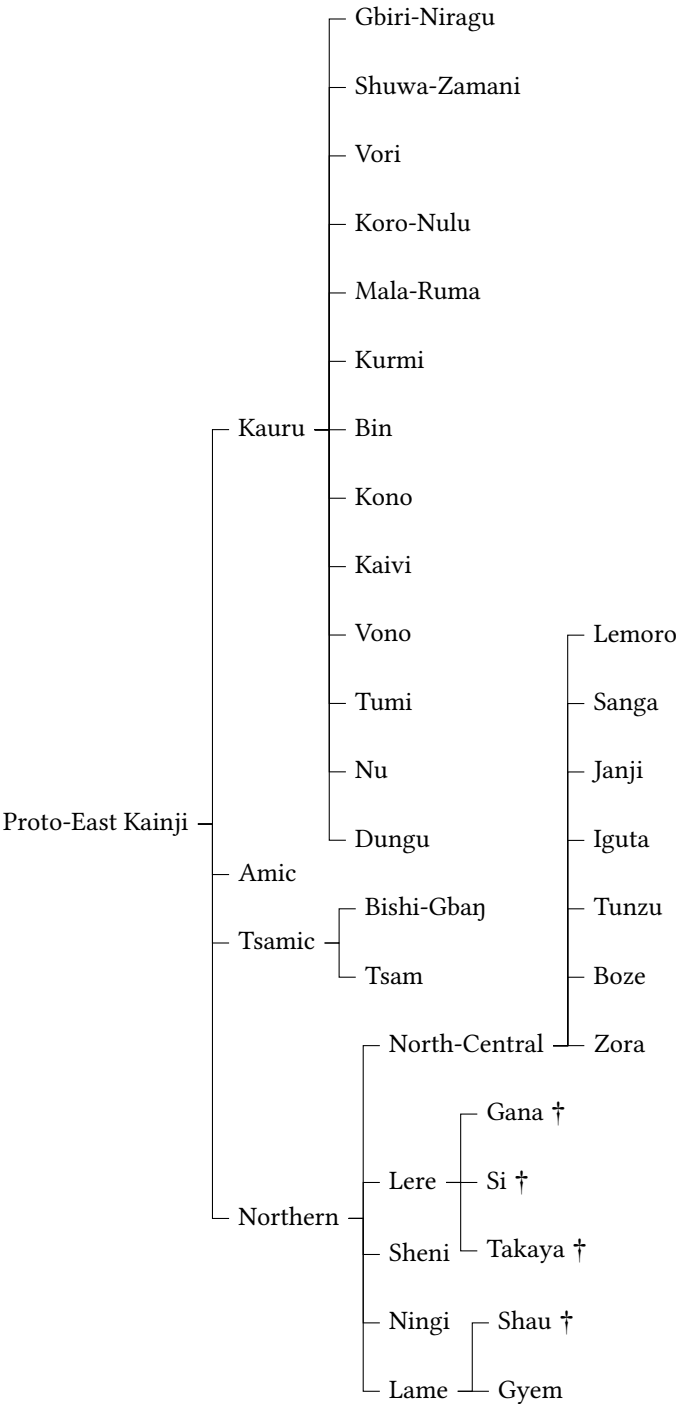


Figure 2: Tentative “tree” of East Kainji languages

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There is no full-length grammar of any East Kainji language, and much of the existing data occurs in survey reports where the tone transcription is impressionistic, to say the least. The author has studied the Boze language [jer] for some years so what follows derives from unpublished fieldwork. The Boze tone system is described to illustrate its elaboration in contrast to the much simpler systems in West Kainji.

Boze has three underlying level tones and rising and falling tones. However, a superhigh tone occurs on noun plurals where a tone-raising rule is applied such that a high becomes superhigh. This type of morphological conditioning is found in many other Central Nigerian languages (for Izere see [Blench 2000](#)). Some of the glide tones arise from long vowels and diphthongs, but others occur on single vowels. Surface tones are transcribed as in Table 3.

Table 3: Terminology and tone graphemes used in this paper

Terminology	Description	Grapheme
Superhigh	double acute accent above the syllable	˝
High	acute accent above the syllable	ˊ
Mid	macron	-
Low	grave accent over the syllable	ˋ
Superhigh-High		˝ˊ
Low-High	hachek over the syllable	ˊ̣
High-Low	circumflex over the syllable	ˆ
Low-Mid		⊠
Mid-High		⊡

This example shows a word where the tone-raising rule creates a superhigh tone;

(4) bēshēshēné ēshēshéně tree sp.

In word-medial position, resonants, typically /l/, /r/, /m/, /n/ and /ŋ/, can bear tones. /l/ is probably an original nasal which has assimilated and is realised as /ll/. The other resonants do not assimilate as shown in Table 4.

These tone-bearing resonants almost certainly derive from deletion of interconsonantal vowels. Their presence can be detected when speakers whistle the tones, even where the tones appear to be assimilated to the ear.

Boze is a language rich in ideophones; over a thousand have been recorded. In respect of tone, they show striking patterns of restriction. Most ideophones

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Table 4: Examples of tone-bearing resonants (Source: Personal data)

Boze	Gloss
cūllū	slimy; viscous; gluey
bētōrnō	boil or other lump on the skin
cōmpōrōŋ	describes s.o. tall and shapeless
ìcákùmbá	mumps
cōncōŋ	describes s.t. unsteady; wobbly; bouncy
bèrēñkō	cataract
àràŋmyē	liver

are reduplicated, CVCVCVCV, except those which have an exclamatory quality. Despite contour tones being common in other parts of the lexicon, Boze ideophones only ever have level tones. The tone of the ideophone is most commonly the same throughout, either mid or low, which correlates with persistence of the same vowel throughout the word. Words with tonal change constitute less than 5% of the sample, and typically non-reduplicated. Ideophones in this class also contain examples of variability of vowels. Table 5 shows examples of “typical” tone patterns in Boze ideophones, contrasted with those where there is internal tone change;

Table 5: Contrast in tone patterns in Boze ideophones (Source: Personal data)

Boze	Gloss
Level Tone Patterns	
bādānbādān	describes struggling with pain
gībīti	describes person with projecting forehead
bèndèlèŋ	describes suffering of a sick person
kparàrà	describes s.t. fried or roasted
Mobile Tone Patterns	
cārōŋ	describes s.t. soiled; spoiled; ugly; bad-looking
cə̀h̀kə̀lì	describes swift movement out of s.t.
fífíp	describes s.t. covered very well; describes door shut completely
gàgák	describes two things that are equal in strength

Even though tones are often mobile in other grammatical contexts in Boze, the tones of ideophones always remain static within an utterance.

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3 Plateau

The Plateau languages consist of between sixty and eighty languages spoken in central Nigeria, spreading from Lake Shiroro to the banks of the Benue River (Blench 2018a). Although most Plateau populations are small (2,000–10,000 speakers), there are probably more than a million speakers of Plateau languages, with the bulk of the numbers made up from large groups such as Berom and Eggon. Some Plateau languages, such as Sambe and Yangkam, are moribund and others are severely threatened, such as Ayu.

Plateau languages are notable for the complexity of both their segmental phonology and tonal systems. This arises from the erosion of the noun class affixes, which must once have been Bantu-like, i.e. with alternating prefixes showing alliterative concord with adjectives and other parts of speech. This is entirely lost as a productive strategy in most languages but in some cases it has been rebuilt. Tarokoid, for example, consists of five languages, four of which mark number by invariant suffixes or clitics. Tarok itself still has a restricted set of noun-class affix alternations, but since they bear no relation to other regional systems they appear to have been subsequently rebuilt. The consequence of these processes has been considerable variability within Plateau. Extreme cases like Rigwe [iri] has four level tones, glide tones and both upstep and downstep. In contrast, Zwall [sha] and Yangkam [bsx] have two level tones but the contrast is being lost. It is possible they can be treated as pitch accent languages, i.e. systems in which one syllable is more prominent than the other syllables in the same word. This is almost certainly due to bilingualism in Hausa exhibited by speakers.

Rigwe, spoken in Plateau State, Nigeria, some 30 km southwest of Jos, is a striking example of tonal complexity. It is generally classified as Central Plateau, and its nearest relatives are Tyapic and Izeric. Table 6 shows the tones which are distinguished for Rigwe.³

Rigwe has syllabic and non-syllabic nasals, hence the notation given above. Rigwe is a terrace-level language with a four-way contrast illustrated in Table 7.

The four tones occur in all major parts of speech. The table exemplifies a grammatical distinction marked only by tone, contrasting the high tone imperative with the low tone as the subject of declarative sentences.

A subset of Rigwe plurals have a restricted extra-low nasal prefix which is raised to mid, while an extra-low stem-tone becomes falling (Table 8).

³I would like to acknowledge the input of Daniel Gya, the translator of the Rigwe New Testament, for discussions on the tonal analysis.

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Table 6: Rigwe tones

High	´
Mid	Unmarked
Mid over a syllabic nasal	-
Low	`
Extra-Low	˘
Falling (High-Low)	ˆ
Falling (Mid-Low)	⊠
High-mid fall	⊠
Rising	˘ˆ
Downstep	⊠
Upstep	⊠

Table 7: Contrastive meanings for four tone levels (Source: Personal data)

Rigwe	Gloss
né	depart (imperative -the citation form)
²nē	sun
nè	depart (with a downstep when it has a subject)
ně	absence of rain [‘rain’ is <i>nzè</i> , a distinct lexeme]

Table 8: Extra-low nasal raising in Rigwe plurals (Source: Personal data)

Gloss	Singular	Plural
‘chair’	ñtɕù	ñtɕu⊠
‘chief’	ñgʷě	ñgʷe⊠
‘agama lizard’	ñdà	ñda⊠
‘scar’	ñmgbě	ñmgbɛ⊠
‘boyfriend’	ñtɕǎ	ñtɕa⊠

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However, the extra-low nasal prefix does not occur elsewhere in the language and must have arisen from a morphological process now-vanished. This points to the importance of the grammatical role of tone and reduced complexity of lexical tone.

Rising tone can result from the assimilation of low-high sequences in pronouns. Thus the 3rd person singular pronoun *àá* assimilates to the vowel it follows and the first vowel of *àá* is lost, leaving a floating tone, thereby creating a surface falling-rising tone. So;

- (5) *héri àá* → *héri`í*
 until he until he
- (6) *àá ní ruhú`ú hu*
 he did sickness he died
 'he became sick and died'

The perfective aspect in Rigwe is carried by the pronoun and reflected as downstep on the verb. The marker of perfective is a V slot added to the subject pronoun which carries HL tone, the L floating and triggering the downstep. Without downstep the aspect changes to incomplete. Contrast;

- (7) *àá ɛ́cé kaá*
 he farmed farm
 'He has farmed.'
- (8) *à cé kaá*
 he farms farm
 'He farms.'

A analogous contrast is seen between the perfective and the progressive;

- (9) *àá ɛ́sê j̀ɲʷa*
 he bought cow
 'He bought a cow.'
- (10) *à j̀i sê j̀ɲʷa*
 he AUX buy cow
 'He is buying a cow.'

In the progressive, the pronoun is shortened, leading to loss of downstep on the verb. In the case of 'buy', the stem vowel undergoes alternation in the habitual in comparison with the progressive.

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- (11) à sí jɪ^wa
 he buys cows
 ‘He buys cows.’

A similar rule applies to polysyllabic verbs;

- (12) ɪ^wé ʔʷɛlí èwèé rèwè
 you cheated little children
 ‘You cheated children.’
- (13) ɪ^wé ʃ^wɛlí èwèé rèwè
 you cheat little children
 ‘You cheat children.’

One interpretation could be that tonal variation on the verb emanates from its toneless condition in the lexicon. The tone would then be imposed by affixal material designating aspect.

The downstep at the end of a clause can denote a pause.

- (14) a. gà ñne ʃɪŋce ʒ^wóʒ^wó ʔné ré ñzi jì,
 all people Cinkye all went to go farm,
 ‘All the people of Cinkye have gone to farm,’
 b. ʔmédí ñne tà rí ñzi ʔʃí wé ʔrí
 but people down house us are just home
 ‘but the people in our house are at home.’

The downstep on *ʃí* in the incomplete second clause is an aspect harmony rule, which states that all clauses in a multi-clause sentence must show the same aspect marking. If the second clause is uttered in isolation, downstep is not present.⁴

Berom [bom] is spoken in the area adjacent to Rigwe, but is from a quite different branch of Plateau, has what can be regarded as a more typical tone system, three tone heights plus rising and falling. The Du dialect of Berom has been described by Bouquiaux (1970) but the present account is based on extensive work with the late Barnabas Dusu, the main translator of the Berom Bible, which is based on the significantly different Foron dialect. The phonology of Foron Berom is presented in Kuhn & Dusu (1985), but tone is only covered in brief. Table 9 shows examples of the five contrastive tones on a single vowel, /a/;

⁴A reviewer suggests an alternative interpretation is the lack of resetting the register to a full High, but it seems to me the loss of Downstep supports my analysis

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Table 9: The five contrastive tones in Berom

Berom	Gloss
á	it is
ā	he, she, it
â	is it?
ǎ	auxiliary preceding verb marking s.t. that has just happened
à	of, for

Contour tones are far more prevalent in pronouns and grammatical morphemes, although they occur in basic lexicon, nouns and verbs. They can be implicated in grammatical processes, though not in a systematic way. For example;

- (15) byók v.n. tickling
byök v. to tickle

Three-way height contrasts are common in nouns and verbs, for example;

- (16) bák ‘inner compound’
bāk ‘region’
bàk ‘sesame’

but contour tones do not typically form part of these sets.

Berom has a very reduced system of noun class affix alternation and the majority of nominal plurals are formed only by tone. Table 10 shows characteristic examples of tone plurals.

These do not form an obvious synchronic pattern and the tonal alternations most probably reflect a now-vanished system of prefixes. By contrast, Table 11 number marking by prefix alternation

The Berom tense/aspect system is intertwined with the pronominal system. Table 12 shows the independent subject pronouns. There is a marked division between singular and plural; the plural pronominal set is invariant (apart from contractions) and sentences must therefore be marked for tense and aspect.

The Berom singular subject pronouns have grammatical tones marking tense/aspect which may be realised on the subject pronoun) while the first and third person are subject to vowel raising, probably reflecting the tense/aspect distinctions.

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Table 10: Berom tone plurals (Source: Personal data)

Singular	Plural	Gloss
cùṇùn	cūṇùn	‘buffalo; bush-cow’
gūl	gùl	‘wind, cold’
jàrùm	jārùm	‘patas monkey’
jòrō	jōrō	‘incisor tooth of dog or cat’
lyàng	lyáng	‘scorpion’
pìm	pīm	‘brown house snake’
pyè	pyé	‘thing’
rìp	rīp	‘partridge’
shè	shē	‘bamboo’

Table 11: Berom number marking by prefix alternation

Singular	Plural
∅-	be-/pe-
∅-	ba-
∅-	ga-
∅-	ne-/n-/ŋ-/m-
∅-	re-
∅-	ra-
∅-	se-
kè-	ne-
nè/n-	be-
re-	ba-
wò-	be-

Table 12: Berom subject pronouns

No.	Singular	Plural
1	ma, me	wōt
2	hwa	yín, ’ín
3	a, e	yèn

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The forms are not precisely mapped between persons and the first person singular has more attested allomorphs than the other pronouns. Although there is a broad distinction between complete and incomplete aspect, the exact range of meaning can only be deduced from a range of example sentences.

In the case of the first person singular subject pronoun, there are two underlying forms, *ma* and *me*, each of which has three tonal allomorphs. The general distinction is that *ma* refers to all forms of past and completed action, while *me* expresses future, stative and intentionality. Within these two classes, tone marks subsets of TAM. Table 13 shows the allomorphs of the first person singular pronoun, arranged by tone-height.

Table 13: Allomorphs of the first person singular pronoun in Berom

Basic	Allomorphs	Senses
ma	má	completed past action, time-delimited negative past action, stative following previous action
	mā	recent past action, current stative
	mà	undelimited negative past action, conditional past
me	mé	future action, negative present
	mê	future intentionality, irrealis
	mē	present, current action

Although Berom has been written for more than a century, the significant meaning differences attributable to tone are not represented in the orthography, and readers’ ability to interpret texts is thereby compromised.

An even greater level of tonal complexity is exhibited by Angan [kci], a language spoken Southern Kaduna State, formerly known as Kamantan⁵. Although it has been classified with Gyong [Kagoma] it is more plausibly an isolate branch of Plateau. It is remarkable for the high diversity of nominal number marking strategies and the variety of segmental tones. Table 14 presents some examples of four tone levels in Angan, while Table 15 shows examples of Angan contour tones.

As Table 15 suggests, tone plays an important role in number marking. Table 16 shows examples where tone is used for noun plurals.

The patterns are far from consistent, but it is remarkable how many polar tones are part of the dataset, e.g. when the singular is rising, the plural will be falling

⁵Thanks to Paul Kura, a native speaker, for help with this language, which is quite endangered.

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Table 14: Four contrastive level tone heights in Angan (Source: Personal data)

Level	Marking	Angan	Gloss
Extra High	ǎ	[tʰǎ]	‘father’
High	á	[tʰé]	‘tree’
Mid	ā	[tʰwāŋ]	‘ashes’
Low	à	[tè]	‘relation’

Table 15: Angan contour tones (Source: Personal data)

F	â	Falling	High low	[tùl]	‘blacksmith’
R	ǎ	Rising	Low high	[ndžǔn]	‘friend’
MH	a↗	Mid Rising	Mid high	[tu↗]	‘blacksmiths’
ML	â	Mid Falling	Mid low	[dâzũ]	‘shame’
LHL	a↗	Rising Falling	Low high low	[mgbâ↗]	‘witch’
HLH	a↘	Falling Rising	High low high	[mgbâ↘]	‘witches’

Table 16: Nominal number marking via tone (Source: Personal data)

Angan	Singular	Plural
nose	ndʒa↗ŋ	ndʒâŋ
fly (generic)	nʒi↗ŋ	nʒîŋ
sling	ndʒêm	ndʒěm
friend	ndʒǔn	ndʒũn
saliva; spittle	nʦĩ	nʦî
knife	fâk	fʰâk
age-group	tâŋ	tǎŋ
hyena	mùrùm	múrùm
rat	pùtfūr	pútfūr
shoulder	fântfǝŋ	fântfǝŋ

(cf. Pulleyblank 2026 [this volume]). This undoubtedly reflects a prior system of prefixes which have been deleted or merged with the stems.

4 Jukunoid

Jukunoid consists of around twenty-three languages spoken in southeast Nigeria and in some isolated areas of adjacent Cameroun. The core monograph arguing for the coherence and classification of Jukunoid is Shimizu (1980b) which established a basic division between “Central Jukunoid” and “Yukuben-Kutep”. Geographically at least, these might be better conceptualised as “Northern” and “Southern” Jukunoid. The classification of Jukunoid remains debated. In her unpublished dissertation, Prischnegg (2008) argued that Oohum [=Yukuben] [ybl] and other poorly documented South Jukunoid lects would be better classified with Plateau, a view reflected in the Glottolog, which presents a “Yukubenic” branch of Benue-Congo⁶. The main descriptive monographs on Jukunoid languages which include analyses of the tonal systems are Shimizu (1980a) for Jukun, Storch (1999) for Hone, Koops (2009) for Kuteb and Richter (2014) for Mbembe.

Disregarding classification issues, it is the case that Jukunoid languages have some of the most complex tones within Benue-Congo, but also in Africa, competing with Omotic languages such as Yemsa (Wedekind 1990). Table 17 shows the tones marked by Shimizu, who has a somewhat idiosyncratic terminology. I have included in square brackets terms which would probably be used today.

Table 17: Jukunoid tones (Shimizu 1980b: 63)

High [Super-High]	´
Top [High]	-
Mid	unmarked
Low	`
Extra-Low	˘
Top-Low Fall [High-Low]	^
High-mid fall	⊠
Rising	ˆ
Downstep	⊞
Upstep	⊡

⁶This seems implausible to the present author

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These are still not entirely clear. High-mid fall would usually be represented as (̨) while Rising may be Low-Mid or Low-High. However, Shimizu was writing in pre-Unicode days when modern conventions had yet to be established. Languages such as Kporo [nza] and Ashuku [nza] appear to have five contrastive tone levels and four levels are common in neighbouring languages such as Icen [ich]. Shimizu takes the view that Proto-Jukunoid must have had three levels and that the additional tone heights are later developments.

For Jukun itself, the language of Wukari, [Shimizu \(1980a\)](#) establishes a three-way contrast for nouns;

- (17) -bá ‘dog’
 -bā ‘agama lizard’
 -bà ‘spleen’

but only a two way contrast for verbs;

- (18) t̄ā ‘to shoot’
 t̄à ‘to show’

with no examples of high tone verbs. The explanation for this is that high tone is reserved for tense/aspect marking in Jukun. [Shimizu \(1980a: 39\)](#) illustrates this contrast;

- (19) k̄ū bī
 ‘he came’
 (20) k̄ú bī
 ‘that he should come’

Mbembe is a Central Jukunoid language for which we have an in-depth grammatical description ([Richter 2014](#)). Mbembe has three register tone levels, High, Mid and Low, as well as contour tones. Tone in Mbembe is attached to a mora, usually vowels, but also syllabic nasals. It is used to mark number on nouns and to distinguish tense-aspect mood categories. Mbembe is relatively unusual in Benue-Congo in that singular/plural alternation in nouns is commonly achieved by changes in the stem vowel. This almost certainly explains the mobile nature of tones in number marking. Table 18 illustrates the combination of stem vowel alternation and tonal change in Mbembe noun plurals;

This is very much in contrast with South Jukunoid, where tones are typically static (see below). Floating tones, i.e. tones that are not attached to a segment,

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Table 18: Tonal change in Mbembe noun plurals (Source: Richter 2014)

Singular	Plural	Gloss
àdā	ādé	‘machete’
mbò	mbá	‘lizard’
ànè	ané	‘mountain’
àfè	áfé	‘necklace’

can mark grammatical categories. For example, a auxiliary floating high tone marks the subjunctive and may attaches either to the subject pronoun or to the subjunctive marker. As an example of marking mood through tonal change, Mbembe raises the tone on the pronominal element from mid to high (Table 19).

Table 19: Tonal change in Mbembe subjunctive marking (Source: Richter 2014)

Subject	Subjunctive	Example
Ñ	ń dò	I should go
ā	á dò	you SG should go
ē	é dò	he should go

Grammatical descriptions of South Jukunoid languages include Kuteb [kub] (Koops 2009) and Oohum [=Yukuben] [ybl] (Rennison 2012). Kuteb has four phonetic tones but only three are contrastive on lexical items, while the fourth tone is a super-high. Table 20 shows evidence for a three-way lexical contrast in Kuteb.

Table 20: Three-way lexical contrast in Kuteb (Source: Koops 2009)

L	Gloss	M	Gloss	H	Gloss
nyàŋ	‘be good’	nyāŋ	‘draw out’	nyán	‘only’
tsò	‘plant’	tsō	‘ascend’	tsó	‘show’
bàe	‘ignite’	bāe	‘stick to’	báe	‘write’
tùr	‘knead’	tūr	‘cut down’	túr	‘push’

Koops (2009) gives the following example of a phonetic super-high tone.

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- (21) m̄ nún̄ kūsǒg
I see house
'I saw a house.'

This is the result of a rule that steps up a high-tone noun stem when it follows a high-tone verb. This also applies to numerals following a head noun;

- (22) àndá ìtsǒŋ
women five
'five women'

Kuteb shows rare cases of falling and rising tones as in Table 21.

Table 21: Contour tones in Kuteb (Source: [Koops 2009](#))

Falling	Gloss	Rising	Gloss
kínûŋ	'bird'	kurúkûm	'toad'
kíkôg	'chest'	kícě	'bowl'
imbô	'chimpanzee'	gbămsa	'sickle'
ipâm	'pound (money)'	aměnjā	'brothers'

Phonetic falling tones, including Mid-Low, can arise from tone spreading from pronouns. In the following example, the low tone verb *sòm* becomes mid-low when following a mid tone pronoun;

- (23) nī sòm̄ iké
You PL sit here!

Three-tone systems with contour tones are found in the small languages spoke along the Nigeria-Cameroun border (e.g. [Kempf 2014](#) for Bezen and [Kempf & Prischneegg 2020](#) for Akum). In these languages, tone is very stable. For example, in Akum, tones in singular and plural nouns are the same and adjectives, despite having residual concord prefixes do not agree in tone with the head noun.

- (24) a. ikī úfī
'small head'
b. ākī ífī
'small heads'

5 Cross River

The Cross River languages were first established in Greenberg (1963) and included Upper Cross, Lower Cross, Ogoni, Central Delta and Bendi. Bendi languages are now generally treated as a branch of Bantoid (see Blench this volume). The remaining four groups are often referred to as Delta-Cross. Cross River languages are spoken mainly in southeast Nigeria with fragmentary populations in adjacent Cameroun. They total fifty-seven languages. Some isolated and now very endangered languages from the Central Delta branch are spoken encapsulated among Ijò speakers, which suggests they are the original inhabitants of the Niger Delta.

Descriptions of tone in Cross River languages are sporadic. Reconstructions exist for two of the four branches, Connell (1991) for Lower Cross and Sterk (n.d.) for Upper Cross. Otherwise we are reliant on sketches. For Lower Cross, almost all languages have consistently High, Low, Downstep (Connell 1991) and this can therefore be reconstructed to Proto-Lower Cross. A more nuanced account of an individual language is Obolo, e.g. Faraclas’ (1984) *Grammar of Obolo* and Rowland Oke (2003). The tones and their allotones are given as follows:

High	Level high tone	High-Low Fall
Low	Level low tone	Low-high rise
Extra high	Level extra-high tone	Extra high-High Fall

Unusually, Obolo permits words with only contour tones. For example;

- (25) a. sôsô
 ‘Please!’
 b. fǎfǎ
 ‘specially’

Tone also plays a role in tense/aspect marking. The tone on *nu*, ‘come’, changes from falling to low in the perfective.

- (26) ì-nû mé érè kèjí
 you-come IMP to here
 ‘you come here’

(27) î nù
 he come.PERF
 ‘he came’

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For Upper Cross, few of the texts analyse tone specifically. DuRop ([Kastelein 1994: 23](#)) has High and Low contrastive level tones, Rising, Falling and also rather rare downstep. DuRop has significant use of grammatical tone, as in the examples in [Table 22](#).

Table 22: Grammatical function of tones in DuRop (Source: [Kastelein 1994](#))

Function	Form	Gloss
Focus	ǒ-sìnì	‘he is <i>refusing</i> ’
	ó-sìnì	‘ <i>he</i> is refusing’
	ò-sìnì	‘he is refusing <i>x</i> ’
Negation	ǒ-sìní	‘he is <i>refusing</i> ’
	ó-sìní	‘he is not refusing’
Pluractional	kóm	‘build [a house]’
	kóóm	‘build [houses]’

Barnwell ([1969](#)) describes a similar system for Mbembe and Sterk’s transcriptions suggest he considered this could be applied to many languages in Upper Cross.

For Central Delta there is Gardner ([1980](#)) on Abuan [abn] and Kari ([2000](#)) on Ogbronuagum [ogu]. [Table 23](#) summarises the sources for Central Delta, which is fairly consistent and implies a system of High/Low/Falling/Downstep for the proto-language.

Table 23: Central Delta tones

Language	Levels	Reference
Abuan	H,L,F	Gardner (1980)
Ogbronuagum	H,L,D	Kari (2000)
Odual	H,L,D	Kari (2009)
Agholo	H,L,D,F	Isukul (n.d.)
Abureni	H,L,F	Etire (2015)

For Ogoni there is Bond ([2006](#)) on Eleme, Ikoro ([1996](#)) on Kana. [Table 24](#) tabulates the various accounts of Ogoni tone.

Kana is the subject of the most detailed description. There are four level tone heights, as exemplified by the minimal set ([Table 25](#)).

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Table 24: Ogoni tone levels

Language	Levels	Reference
Eleme	3	Bond (2006)
Tẹ̀	3	Nwí-Bàrì (2002)
Baan	3	Nakagawa (2000)
Kana	4	Ikoro (1996)
Gokana	4	Vopnu (1991)

Table 25: Kana contrastive tone heights (Source: Ikoro 1996)

Tone	Kana	Gloss
High	fá	‘intestine’
Raised	fa̋	‘vehicle’
Mid	fā	‘someone weeded’
Low	fà	‘to weed’

Kana makes key grammatical distinctions through tone alone (Table 26).

Table 26: Tone in Kana grammatical categories (Source: Ikoro 1996)

Noun	Gloss	Verb	Gloss
wààrà	‘donation’	wààrá	‘to donate’
ṣààʔǎ	‘change’	ṣààʔǎ	‘to repent’
yēb	‘dance’	yèb	‘to dance’

Ogoni languages are characterised by a loss of the noun-class system. Indeed, Williamson (1985) once characterised them as “Kwa” languages. The number of tone levels has increased, compared with other Cross River branches, to compensate for the loss of noun class marking.

6 West Chadic (A3)

It may seem unusual to include a branch of Chadic in a chapter on Benue-Congo. However, the A3 branch of West Chadic appears to have undergone major restructuring as a consequence of contact with Benue-Congo [probably Plateau]

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so that it has lost almost all its Chadic characteristics. West Chadic A includes Hausa (A1), Bole-Tangale (A3), A3 (Angas⁷) and Ron (A4). All of these except A3 have complex nominal and verbal morphology, but relatively simple tone systems. Hausa, for example, has High, Low and Falling. By contrast, A3 languages have lost almost all morphology, but have developed complex tone systems, marked by three level tones and a variable number of contour tones. In addition, they have a system of plural verbs, based on reduplication of the first syllable of the stem, which closely resembles neighbouring Plateau languages such as Berom and Izere. This indicates restructuring at the level of Proto-A3.

We have descriptions for Ngas [anc] (Jungraithmayr & Holubová 2016), Mwaghavul [sur] (Blench & Dawum n.d.), Mupun [sur] (Frajzyngier 1993), Tal [tal] (Blench & Dawum n.d.) and Goemai [ank] (Hellwig 2011). In many languages, minimal triplets are easily found, and it is reasonable to assume proto-A3 had three level tones. For Mwaghavul, the following examples provide evidence for a three-way contrast in level tones;

- (28) high kún ‘three’ kús ‘near’
mid kūn ‘be old’ kūs ‘to rub together’
low kùn ‘old age’ kùs ‘big’

Table 27 shows examples of minimal triplets in Tal.

Table 27: Minimal triplets in Tal (Source: Bulkaam & Blench n.d.)

High	Gloss	Mid	Gloss	Low	Gloss
tál	‘native speaker of Tal’	tāl	‘to greet’	tàl	‘greeting’
ḃél	‘flute’	ḃēl	‘to punch’	ḃèl	‘wisdom’
t’èi	‘tree sp.’	t’ēi	‘to lie down’	t’èi	‘lying’

In noun plus noun constructions, the tone of the first noun is lowered. This phenomenon is also noted for Ngas, called *Nomen regnans* in Jungraithmayr & Holubová (2016). The tone on the following noun does not appear to determine the pattern of lowering of the tone in the first noun. The following examples show the various possible patterns of tone-lowering.

- (29) High to Mid

⁷Although this formulation is found in major reference sources, it is locally disliked as privileging the Ngas language.

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- a. léé
‘item, thing’
- b. lēē lú
‘item of furniture’
- c. lēē pùk
‘soup ingredients’

(30) High to Low

- a. máár
‘farm’
- b. mààr màt
‘farmwork for in-law’
- c. lú
‘room’
- d. lù zók
‘store-room’

Mwaghavul uses tone to mark nominalisation, but not in a consistent way.

Table 28 shows a set of verbs where the noun is derived by tonal change.

Table 28: Tone-change marking nominalisation in Mwaghavul (Source: Personal data)

Verb	Gloss	Nominalisation	Gloss
sāā	‘to cut (pl.)’	sàà	‘cutting’
pīlāng	‘to thank’	píláng	‘thanks’
rīp	‘to itch’	rìp	‘irritant, irritation’
wām	‘to rot’	wām	‘rotten object’

Mwahgavul marks aspect by changes on verbal auxiliaries;

- (31) kí marks progressive action
- kì perfective marker

Thus;

- (32) wàghà kí jì
you PROG come
‘You are coming.’

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Versus;

- (33) Pànrét kɪ sò wàà
 Pànrét PERF go home
 ‘Pànrét has gone home.’

In her description of Goemai, Hellwig (2011) says; “Goemai is a tonal language with two level tones (high, low), two contour tones (falling, rising) plus a predictable downstep”. However, in her table (Hellwig 2011: 18) she assigns conventions for three tone levels on vowels, and the lexical items in the wordlist are so marked. Later she says “Phonetically, a mid tone (M) is also present, which arises through assimilatory processes ..., downdrift ... or downstep ...” (cf. Snider 2026 [this volume]). In practice, the same three tones of the northern languages are present.

Phonetic contour tones are heard in Chadic languages in sequences of two identical vowels, where the tones are dissimilar. However, there are clearly contour tones on short vowels. Tone transcriptions of A3 languages vary in their accuracy, but for those which have been the subject of direct elicitation, we can be fairly certain. Mwaghavul, for example, has HL contour tones on pronouns, which are not reflected in neighbouring languages. These most likely reflect assimilated auxiliaries, a pattern adopted from nearby Plateau languages via contact. More typically for A3 however, are tones which rise or fall by a single level, again characteristic of Plateau.

7 Conclusions

The chapter reviews the tonal systems of the Benue-Congo languages, specifically the Kainji, Plateau, Jukunoid and Cross River branches. The A3 branch of West Chadic is included, due to its radical restructuring as a result of contact with Benue-Congo. Given the large number of languages involved, and the fragmentary data on many of them, this can only provide a snapshot of better-described languages. However, it is worth underlining that the Central Nigerian area represents a hotspot for complex tonal systems, comparable to the Mande area in Cote d’Ivoire and the Grassfields of Cameroun. This situation almost certainly arises from the various pathways for the shortening of nominals with prefixes and CVCV roots. Segmental loss combined with the preservation of tonemes results both in complex contour tones and floating tones.

The detailed description of synchronic tonal systems, and revisiting the previous, rather sketchy, accounts of some languages is thus a priority. The rise of

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interest in the writing of minority languages, their increasing use in schools and in social media argues that greater attention should be given to marking tone in orthographies.

Abbreviations

C	Consonant
D	Downstep
F	Falling
H	High
L	Low
M	Mid
PERF	Perfective
PROG	Progressive
R	Rising
TAM	Tense/Aspect marking
V	Vowel

Acknowledgements

Many examples presented in this chapter derive either from fieldwork in Nigeria over the last four decades or unpublished materials kindly made available to me by their authors. I would particularly like to thank Michael Bulkaam, Timothy Terna, Raymond Dawum, Steve Dettweiler, Daniel Gya, the late John Nengel, Saleh Libisan, David Rowbory, Katherine Mort, Bruce Connell and Harald Hammarström for both data and discussions over many years. The Kay Williamson Educational Foundation funded much of the research reported here.

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Chapter 16

Tone in Mande languages

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This chapter describes the main tone properties of Mande, a West African language family comprising around 70 tone languages, with the most diverse tonologies among them ranging from two to five contrastive levels. The chapter covers (i) lexical contrasts and melodies; (ii) grammatical tones on Mande nouns and verbs; (iii) how Mande surface tone rules relate to the general trends in African tone processes; as well as (iv) boundary tones and (v) how tone phenomena align within the prosodic hierarchy in Mande. An interesting property of Mande is that High spread is not so common in this family and tends to be highly syntactically constrained, although syntactic constituency does not always coincide with prosodic phrasing. I also show that intonational phenomena in Mande are prone to contact influence which may lead to tone complexification.

1 Introduction

The Mande family comprises around 70 languages spoken in West Africa, mostly in Senegal, Gambia, Guinea-Bissau, Mali, Burkina-Faso, Guinea, Sierra Leone, Liberia and Ivory Coast in the West, but also in Benin and Nigeria in the East ([Hammarström et al. 2025](#)). The internal classification of the family is rather well established, with some controversies concerning the mid-taxa level ([Kastenholz 1997](#), [Vydrin 2009b, 2016b](#)). Here I follow Vydrin's ([2009b, 2016b](#)) classification as shown in Figure 1.

Mande is commonly viewed as belonging to the Niger-Congo phylum since Greenberg's ([1963](#)) classification; cf. also [Pozdniakov \(2018\)](#). I follow this assumption here as well, although some scholars point to insufficient historical-comparative evidence of Mande being part of Niger-Congo ([Dimmendaal 2011: 92](#)). Mande

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languages also share some features with other languages of the so-called Macro-Sudan Belt area, such as labio-velar stops, implosives, nasal vowels, multiple tone levels and SAuxOVX word order (Güldemann 2007).

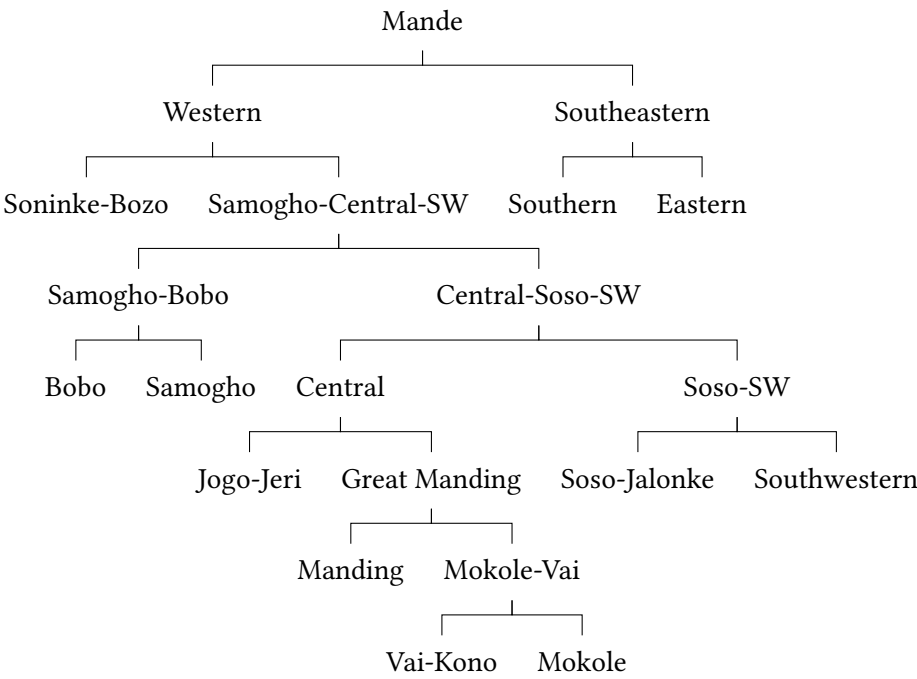


Figure 1: Mande internal family classification based on Vydrin (2009b, 2016b)

Unlike many other Niger-Congo branches, Mande family has relatively little segmental inflectional morphology, usually allowing for no more than one affixal marker per word form if any. However, tonal inflection and derivational morphology are quite abundant in Mande, and compounding is extremely productive (Green & Konoshenko 2022, Konoshenko 2022, Vydrin 2018). All Mande languages have a rigid SAux(O)VX structure. TAMP meanings are typically encoded by dedicated constructions involving the Aux, or predicative marker, and the verb, which appears either in its base form, or bearing segmental and/or tonal markers, cf. Section 3.

An initial survey of Mande tone systems was carried out by Vydrin (2003b), although he mostly focussed on lexical tone contrasts in individual languages. There has been a dramatic increase in field data since then, so the present

overview seeks to provide further insights based on Vydrin's (2003a) observations, especially in the domain of grammatical tones, surface rules and intonational phenomena. Yet, Mande tonologies have still been documented rather unevenly, the Manding and the Southern branches described in most detail, while only a few or just one language having a more detailed tonal description in other branches. Therefore, no robust generalisations about the family-internal frequency of specific tonal phenomena can be drawn at present and any tendencies formulated here should be taken with a grain of salt.

This chapter is organized as follows. Section 2 describes basic tonal contrasts attested in different Mande branches, as well as tonotactic constraints on word-internal tone sequences known as “melodies”. Section 3 provides a brief outline of major grammatical tone types in Mande (cf. Rolle & Bickmore 2026 [this volume]), and Section 4 summarizes the most recurrent surface rules. Section 5 outlines boundary tones and Section 6 shows how tonal phenomena operate at different layers of the prosodic hierarchy. I discuss the main findings in Section 7, and Section 8 concludes the chapter. The Appendix presents the languages discussed in the chapter, with their genetic affiliation and the source.

2 Lexical contrasts and tone melodies in Mande

2.1 Contrastive tones

All Mande languages are tonal and feature mainly level tones, ranging from two, e.g., in the Manding branch, to five, as in Eastern Dan (Southern). Examples (1–3) illustrate a binary L/H contrast in Niokolo Maninka, Susu and Kingi Soninke:¹

- (1) Niokolo Maninka (Manding): Low / High
 - a. ḡ ‘I’, ḡ ‘we’
 - b. ì ‘you’, í ‘they’ (Creissels 2013: 18–19)
- (2) Susu (Soso-Jalonke): Low / High
 - a. yèxéè ‘sheep’, yéxéé ‘fish’
 - b. tòó ‘today’, tóó ‘see’ (Touré 2004: 188, 192)
- (3) Kingi Soninke (Soninke-Bozo): Low / High

¹I use the following conventions to indicate tone on a given tone bearing unit: Superhigh or Top (SH) ṽ, High (H) ṡ, Mid (M) ṃ, Low (L) ṁ, and Extralow (XL) ṁ̇. Any contours are shown as sequences of these diacritics. Underlying tones are given in slash brackets //, grammatical tones in curly brackets {}, and surface tones in square brackets [], wherever applicable. A word boundary is indicated by #. Downstep is marked with an arrow ↓. Nasalization is marked under the vowel: ɔ̃.

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- a. *kàrá* ‘die’, *kára* ‘break’
- b. *qáwá* ‘be wet’, *qáwà* ‘gather’ (Creissels 2016: 29)

While both L and H are phonologically active in some languages having a binary tone contrast, e.g., in Guinean Kpelle (Southwestern; Konoshenko 2014b), it is not always symmetric. For example, in the Manding branch, L may be phonologically more active than H, as in Bamana and Kita Maninka, or H is more active in those languages that have undergone tone reversal, e.g., Odienne Jula and Niokolo Maninka (Creissels 2013, Creissels & Grégoire 1993, Green & Konoshenko 2022, Vydrin 2016a). Soninke (Soninke-Bozo) also features active L (Creissels 2016). For Kakabe (Mokole), Vydrina (2017: 351) postulates an underlying H vs. L vs. Ø contrast surfacing as H vs. L.

Examples (4–6) further illustrate Mande languages with more than two contrastive levels – three (Mano), four (Seenku, Kla-Dan) and five (Eastern Dan).

- (4) Mano (Southern): Low / Mid / High
 - a. *lè* ‘place’, *lē* copula, *lé* relative marker
 - b. *là* 3sg possessive marker, *lā* ‘save’, *lá* ‘annoy’ (Khachaturyan et al. 2022)
- (5) Seenku (Samogo): Extralow / Low / High / Superhigh
 - a. *sí* ‘terminalia sp.’, *sí* reciprocal marker, *sì* ‘first son’, *sĩ* ‘water jar’
 - b. *sǎn* ‘god’, *sá* ‘cry’, *sà* ‘second son’, *sǎn* ‘stream’ (McPherson 2019: 79)
- (6) Eastern Dan (Southern): Extralow / Low / Mid / High / Superhigh
 - kǎǎ* ‘scabies’, *káá* ‘2PL’ (negative present), *kāā* ‘2PL’ (prospective series), *kàà* ‘scratch’ (conjoint form), *kǎǎ* ‘reed’ (Vydrin 2021c)

Note, however, that even if a certain number of tone levels is contrastive in a language, their relative frequencies may be different in the lexicon. Thus, while Seenku (Samogo) clearly has four levels illustrated in (5), they are distributed unevenly in the existing language corpus, Low only being a peripheral tone: Superhigh (22.8%), High (14.2%), Low (1.2%), Extralow (22.7%); the rest accounting for contours (McPherson 2019: 90).

Tonemic *contours*, e.g., constituting independent units in tonal inventories rather than patterning as sequences of elementary level tones, e.g., as in many languages of South East Asia (Kirby & Brunelle 2017), are rather rare in Mande and are attested only in a handful of languages, e.g., Guro (Southern; Kuznetsova

2007), featuring a rising (LH) and a falling (HL) contour, and Eastern Dan (Southern; Vydin 2021c) with two falling contours (H-SL and M-SL). In such languages, contours are attested on light (C)V syllables and show no signs of internally complex structure. Many Mande languages allow contours on light syllables, but they are either not underlying, e.g., in Mwan (Southern; Perekhvalskaya & Vydin 2024), or they pattern as individual level tones in sandhi processes or in tone association rules, e.g., in Beng (Southern; Paperno 2014: 22–23) and Seenku (Samogo; McPherson 2019: 82–87), cf. also Section 4 for the discussion of Beng sandhi. Finally, there are languages, e.g., Goo (Southern; Aplonova & Vydin 2017: 459 and Mano (Southern; Khachaturyan 2015: 10), where surface contours are only possible on bimoraic structures, and they can be formed by all combinations of level tones.

Mande languages tend to have dense tonal specification, *toneless morphemes* are mostly attested in the Manding group (Vydin 2003b) and less common in other branches. Such morphemes are usually verbal suffixes or function words, i.e., predicative markers or postpositions. For example, in Kono (Southwestern), there is an enclitic postposition *ma* ‘on’ which copies the tone of its complement: /Héní ma/ → [Héní má] ‘on Heni’, /Pépèè ma/ → [Pépèè mà] ‘on Pepee’ (Konoshenko 2017: 88–89).

Underlying *floating* tones are quite common in Mande (cf. Rolle et al. 2026 [this volume]). They may belong to the lexical tonal specification of certain morphemes, e.g., in Dzuun (Samogo; Solomiac 2014) or Guinean Kpelle (Southwestern; Konoshenko 2014b), or function as independent referential markers, e.g., in Bamana (Manding; Vydin 2016a) or Soninke (Creissels 2016). Example (7a) illustrates a floating Low referential marker in Bamana.

(7) Bamana (Manding)

a. Mùsò‘ té yàn.

woman\REF NEG.be here

‘The woman is not here.’

b. Mùsò té yàn.

woman NEG.be here

‘There is no woman here.’ (Vydin 2016a: 90)

Tonal inversion has been reported for some Mande languages, e.g., Odienne Julia (Manding; Braconnier 1983) and Looma (Southwestern; Dwyer 1981).

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2.2 Tone melodies

Another seminal property of Mande languages is that they tend to impose various phonotactic constraints on tone sequences at the level of the *prosodic foot* (Green 2015, Vydrin 2003a, 2010, 2020, Leben 2002, 2003) or word, cf. Section 6. When it comes to lexical tones, this means that tonal elements in these languages tend to exhibit just a few recurrent foot- or word-level patterns, also referred to as *melodies* (Green 2015, Hyman 2011, Konoshenko 2014b, Sherwood 2020); although Mande languages having no systematic melodic restrictions are also attested, e.g., Mano (Southern; Khachaturyan et al. 2022). The underlying and surface realisations of those melodies are often quite different and depend on surface rules.

A further notable pattern within Mande is that monomorphemic structures may allow for a similar inventory of melodies with little regard to their syllable structure. This phenomenon, especially salient in Western Mande branch, laid the foundations of Autosegmental phonology after Leben's (1973) seminal description of Mende (Southwestern). Thus, in Mende, featuring a binary L/H contrast, the majority of monomorphemic words belong to one of five tone patterns: /H/, /L/, /HL/, /LH/, and /LHL/ regardless of the number of syllables they have (Leben 1973: 186, 1978). Kpelle and Kono within Southwestern Mande group feature similar inventories (Konoshenko 2014b, 2017, Welmers 1962). Here I illustrate this cross-word uniformity with less cited examples from Vai (Vai-Kono) having four basic melodies: /H/, /L/, /HL/ and /LH/:

(8) Vai (Vai-Kono, Welmers 1976: 29–30)

/H/	<i>tá</i>	'fire'	<i>sánjá</i>	'fence'
/L/	<i>bhù</i>	'gun'	<i>gbàsà</i>	'cassava'
/HL/	<i>bâ</i>	'mother'	<i>sám-bà</i>	'basket'
/LH/	<i>mǎ</i>	'person'	<i>sànjá</i>	'town'

In Bamana (Manding), 90% of the monomorphemic words bear one of *two* major (underlying) tone melodies, i.e., /H/ and /LH/. The latter melody can also be analysed as underlying /L/ with contextual [LH] realization; still the two analyses – /H/ vs. /LH/ or /H/ vs. /L/ – are mutually convertible (Vydrin 2020 and references therein):

(9) Bamana (Manding, Green 2015)

/H/	<i>bá</i>	'goat'	<i>bílén</i>	'still'	<i>kóró</i>	'underside'
/LH/	<i>bǎ</i>	'river'	<i>bílén</i>	'red'	<i>kàró</i>	'old'

Some languages featuring a simple binary L/H contrast have more complex rules for melody association depending on prosodic structure, e.g., light and heavy, final and non-final syllables as in Pakaawu Mandinka (Manding; [Creissels 2019](#): 18–21). Slightly simplifying, in this language, words with light final syllables can bear /HL/ underlying melodies but not /H/. In contrast, for items with heavy final syllables, /H/ is possible on a par with /HL/ (10).

- (10) Pakaawu Mandinka (Manding, [Creissels 2019](#): 20–21)
- | | | | | |
|------|--------------|-------------|--------------|----------------------|
| /HL/ | <i>ńtè</i> | ‘me’ | <i>náani</i> | ‘four’ |
| | <i>kúnùn</i> | ‘yesterday’ | <i>kánàa</i> | (prohibitive marker) |
| /H/ | <i>sínín</i> | ‘tomorrow’ | <i>túnáa</i> | ‘abroad’ |

Southern Mande languages, which have more complex level contrasts, generally impose fewer lexical tonotactic restrictions within a foot. However, even in this branch, tone combinations may not be entirely unpredictable, as I illustrate in (11) by the major tonotactic rules from Eastern Dan featuring five contrastive levels:

- (11) Eastern Dan (Southern)
- sequences of identical tones are allowed: *těě* ‘wind’, *tāā* ‘gift’, *piàà* ‘grill’
 - any initial tones combine with final Extra-Low: *bhǎǎn* ‘muslim’, *gbáǎ* ‘plantain sp.’, *gǎǎn* ‘man’, *kěǎ* ‘after’
 - initial mid can only combine with higher tones: *bhǎǎ* ‘bee sp.’, *bhǎǎ* ‘monkey trap’ ([Vydrin 2021b, 2021c](#): 12)

3 Grammatical tones

There are two most regularly recurring types of grammatical types in Mande: tonal head-marking in compounds ([Green & Konoshenko 2022](#)) and verbal tones making part of TAMP paradigms ([Konoshenko 2022](#)).

Mande compounds headed by nominal stems typically consist of a dependent stem (W1) and a head stem (W2; more precisely, it is the final word of the compound if there are more than two components). In many languages, with the exception of some Southern and Eastern Mande, the lexical tone of W2 is consistently neutralized, and sometimes W1 also surfaces with a modified tone. [Green & Konoshenko \(2022\)](#) identify two major types of W2 tone neutralization in Mande compounds. First, in the “Low” type, W2 lexical tone is replaced by the lowest tone in the inventory – Low as in (12) in Mano or Extralow as in Goo (13).

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(12) Mano (Southern)

- a. /sòṣ/ ‘amusement’ + /mī/ ‘person’ → [sòṣ#mì] ‘lover’
- b. /mèí/ ‘Maninka’ + /mī/ ‘person’ → [mèí#mì] ‘a Maninka’
(*Khachaturyan 2015*)

(13) Goo (Southern)

- a. /dí/ ‘witchcraft’ + /bhéén/ ‘person’ → [dí#bhéén] ‘sorcerer’
- b. /kùl/ ‘gun’ + /dàà/ ‘go up’ + /bhéén/ ‘person’ → [kùl#dàà#bhéén]
‘hunter (with a gun)’ (*Aplonova 2020*)

Example (13b) from Goo also shows a compound consisting of more than two elements, including a verb, which is also quite typical for Mande. In some languages of the Western branch, e.g., Vai, Susu and Southwestern Mande, the {L}reassignment on W2 is further complexified by H tone spread rule from W1, cf. Section 4.

Second, in the “Default” type, W2 is assigned a default, or unmarked tone, depending on the language tonology, cf. Section 2.1. The “Default” type is illustrated in (14) from Kakabe, where H is unmarked, and (15) from Niokolo Maninka, featuring unmarked L.

(14) Kakabe (Mokole)

- a. /nìngì/ ‘cow’ + /mùsù/ ‘woman’ → [nìngì#mùsù] ‘female cow’
- b. /dònsò/ ‘hunter’ + /kàyì/ ‘man’ → [dònsò#káyì] ‘male hunter’
(*Vydrina 2017*: 500)

(15) Niokolo Maninka (Manding)

- a. /fùlà/ ‘Fula’ + /mùsù/ ‘woman’ → [fùlà#mùsù] ‘Fula woman’
- b. /tìbààbù/ ‘European’ + /mùsù/ ‘woman’ → [tìbààbù#mùsù] ‘European woman’ (*Creissels 2013*: 27)

Grammatical tones are also abundant in Mande verbal morphology, as discussed in detail by Konoshenko (2022), cf. also Konoshenko (forthc.) for a more diachronically oriented summary. Examples (16a–b) feature two TAMP constructions from Guinean Kpelle (Southwestern). While the Perfect construction comprises a specific Perfect 1SG Aux and the verb *pá* ‘come’ with its lexical /H/ tone (16a), the Imperfective construction consists of an Imperfective 1SG Aux and the same verb *pá* ‘come’ bearing a grammatically specified replacive Low tone (16b).²

²In (16a–b), the surface tones are not identical to the underlying ones represented in the examples due to the operation of surface rules, i.e., H spread and contour simplification, cf. Section 4.

(16) Guinean Kpelle (Southwestern)

a. Perfect

ŋǎa pá b́é.

1SG.PRF come here

'I have come here.'

b. Imperfective

ŋǎ pá b́é.

1SG.IPFV come\L here

'I come here (often).' (author's own data)

Apart from nominal and verbal morphology, tones typically distinguish between different pronominal forms as shown in (1) from Niokolo Maninka (Manding), as well as person-number auxiliary forms in languages with higher degree of morphological fusion (Konoshenko 2020). Floating tones function as referential markers in some Manding languages (Vydrin 2016a), cf. (7). Finally, tonal changes accompany adjective reduplication in some Southern Mande languages, e.g., Eastern Dan (Vydrin 2021c).

4 Surface rules

So far, recurrent patterns in African tone rules have been mainly observed based on data from Kwa, Yoruboid, Igbooid, Chadic and Bantu languages (Hyman 1975, 2009, 2011, Hyman & Schuh 1974, Lionnet & Hyman 2018), while Mande tones have largely remained in the shadows with some rare exceptions (Creissels 1994). Therefore, the basic aim of this section is to compare Mande data against what has been suggested as general trends elsewhere in African languages.

One of the classical tonal process discussed in literature on African tones is *downdrift*, i.e., phonetic lowering of a H following a L tone in HLH sequences (Connell 2001, Yip 2002), sometimes also referred to as automatic downstep (cf. Snider 2026 [this volume]). It has been claimed that downdrift is most often found in languages with a binary L/H distinction (Creissels 1994: 217, Hyman 1975: 227–228, Yip 2002: 149). Mande languages generally follow this generalisation, since downdrift has been mainly reported for two-tone languages, e.g., Kingi Soninke (Soninke-Bozo; Creissels 2016), Kakabe (Mokole; Vydrina 2017) and Bamana (Manding; Vydrin 2016a); although some languages having a binary tone contrast lack downdrift, e.g., Vai (Vai-Kono; Welmers 1976: 29). Downdrift is usually not attested in languages with more complex tonal contrasts, yet it has been described for Dzuun (Samogo) featuring L/M/H contrast, where the second

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H is lowered in both HLH and HMH sequences (Solomiac 2014); as well as in Seenku (Samogo), contrasting four level tones, where downdrift affects Super-high tone and it is triggered by both Low and Extralow tones (McPherson 2019: 113–118).

Downstep refers to the phonetic lowering of a subsequent H triggered by a floating L, rather than a surface one, as in downdrift (Hyman & Schuh 1974, Yip 2002). This phenomenon is also quite common in Mande, mostly reported for two-tone languages, which typically have both downdrift and downstep, e.g., Mandinka (Manding; Creissels & Sambou 2013), Susu (Soso-Jalonke, Touré 1994: 50) and Guinean Kpelle (Southwestern; Konoshenko 2014b). However, Beng, a Southern Mande language with LMH contrast, also features downstep triggered by falling contours and affecting both H and M; but no downdrift is mentioned by Paperno (2014: 23). Seenku (Samogo) employs both downdrift and downstep despite having four contrastive tone levels (McPherson 2019).

Fifty years ago, Hyman & Schuh (1974) established that progressive tone spreading was among the most common tone rules in West African tone (non-Mande) languages, and thirty years later Hyman (2009: 7) made an argument that High tone tends to spread more often than other tones (cf. Smolders 2026 [this volume]). A closer look at Mande reveals that in these languages, *High spread* is indeed more common than L or M spread, but H spread itself is quite rare and tends to be morphosyntactically constrained.

In Section 3, we observed that W2 tone lowering is a grammatical tone operation attested in different Mande languages. In some of them, this lowering is further opacified by surface H tone spread from W1 to W2 (Green & Konoshenko 2022). High spread phenomena are attested in some other languages, e.g., Mende (Southwestern, Dwyer 1973), Susu (Soso-Jalonke, Green et al. 2013) and Mwan (Southern; Perekhvalskaya & Vydrin 2024), but they are in no way widespread in Mande. Furthermore, they are always limited to specific morphosyntactic or even lexical contexts. For example, in Mwan, H spread only occurs between a direct object and a subset of verbs in Perfective constructions, where the verb is marked by a grammatical {L} tone and a suffix, as well as between a complement and a postposition *dà* ‘under’ (Perekhvalskaya & Vydrin 2024). Thus, (17a) features High spread from the H-toned direct object onto the {L}-marked verb in Mwan, while in (17b) no spread occurs after the M-toned object.

(17) Mwan (Southern)

- a. [Jàdà á kpé fáá-dà]
 /Jàdà á kpé fàà-dà/
 lion 1SG.POSS duiker steal\L-PRF
 ‘A lion has stolen my duiker.’
- b. [Jàdà á bāā fàà-dà]
 /Jàdà á bāā fàà-dà/
 lion 1SG.POSS chicken steal\L-PRF
 ‘A lion has stolen my chicken.’ (Perekhvalskaya & Vydrin 2024)

Contour simplification via absorption or leveling is another common tone rule in African languages (Hyman & Schuh 1974), the most natural processes in absorption being /LH.H/ → [L.H] and /HL.L/ → [H.L] (Hyman 2009: 12). Note that here the term *contour* precisely refers to a case when non-identical tones are lexically or contextually associated with one TBU (usually a mora or a syllable). Except for a few languages, where contours are tonemic and consistently surface as such across different contexts, as in Guro and Eastern Dan discussed in Section 2.1, Mande languages tend to simplify surface contours. For example, in Beng (Southern), having a three-way L/M/H contrast, there are four contours: HL, ML, LH and MH, but they tend to be levelled in context. The rising LH tone becomes L before non-low tones including contours (18a), e.g., /dũ/ ‘knife’ – /dũ LH dō M/ → [dũ L dō M] ‘one knife’. Both falling tones are simplified before low including the rising LH (18b–c): /sĩ/ ‘oil palm’ – /sĩ ML è L/ → [sĩ M è L] ‘this is an oil palm’, /zô/ ‘mat’ – /zô HL è L/ → [zô H è L] ‘this is a mat’ (Paperno 2014: 20–23).

(18) Beng (Southern)

- a. LH → L / _ {H, M, MH, ML, HL}
 b. ML → M / _ {L, LH}
 c. HL → H / _ {L, LH} (Paperno 2014)

Beng rules suggest that, first, contour tones phonotactically behave as combinations of individual level elements (Section 2.1), and second, that contour simplification is a productive process in this language.

Similar simplification processes may take place in the foot domain, on a par with syllables. In Susu (Soso-Jalonke), HL contours are simplified both on CVV and CVCV structures, although there is a difference. Non-final HL always simplifies to H on CVV, the Low element either being absorbed by the following Low (19a) or downstepping the following High (19b); also note the neutralized final

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vowel length distinction. On CVCV structures, falling contour only flattens to High on CVCV before L tone (19c), but not before H (19d).³

(19) Susu (Soso-Jalonke)

- a. /yèxéè/ ‘sheep’ + /gbò/ ‘many’ → [yèxé gbò] ‘many sheep’
- b. /yèxéè/ ‘sheep’ + /síyà/ ‘few’ → [yèxé¹ síyà] ‘few sheep’
- c. /kònkórè/ ‘rooster’ + /gbò/ ‘few’ → [kònkórè gbò] ‘many roosters’
- d. /kònkórè/ ‘rooster’ + /síyà/ ‘few’ → [kònkórè síyà] ‘few roosters’
(Green et al. 2013)

Two more common rules are worth mentioning here. First, *tone plateauing*, i.e., dispreferred *HLH (Hyman 2011: 217–218), is not so common but attested in different Mande branches, e.g., /L-xL#L/ → [L#L] in Eastern Dan (Southern): /sòò/ ‘horse’ → [sòò wlìy] ‘under the horse’ (Vydrin 2021a: 57). In Pakaawu Mandinka (Manding), some HL#H sequences undergo the rule that Creissels (2019) calls *pont tonal* (tonal bridge): /HL#H/ → [HH#H], e.g., in the combinations of nouns having final light low-toned syllable with some modifiers (20):

(20) Pakaawu Mandinka (Manding)

- a. /bàsà/ ‘lizard’ + /dóo/ ‘a certain’ → [básá dóo] ‘a certain lizard’
- b. /súnjkútù/ ‘girl’ + /dóo/ ‘a certain’ → [súnjkútú dóo] ‘a certain girl’
- c. /tùbáábù/ ‘European’ + /dóo/ ‘a certain’ → [tùbáábú dóo] ‘a certain European’ (Creissels 2019: 36)

Finally, OCP-triggered *dissimilatory* tone rules, acting in the opposite way to plateauing, are also attested in Mande. Thus, in Wan (Southern), there is a rule assigning H to toneless units before either L or M (21a–b) or at the end of the phonological phrase, cf. Section 6, otherwise toneless units receive a default mid tone (21c).

(21) Wan (Southern)

- a. /ku blò/ → [kú blò] ‘long house’
- b. /ku dō/ → [kú dō] ‘one house’
- c. /ku tí/ → [kū tí] ‘a lot of houses’ (Nikitina 2019: 702)

Furthermore, in Wan, there is a dissimilation rule: /H#H/ → [H#L] shown in (22).

³Susu also has rising contour simplification /LH.H/ → [LL.H], cf. Green et al. 2013 for details.

(22) Wan (Southern)

- a. /t́ gó/ → [t́ gò] ‘in the swamp’
- b. /t́ kpò dō/ → [t́ kpò dō] ‘one river fish’ (Nikitina 2019: 705)

Similarly, in Eastern Dan (Southern), Mid tone is replaced by High after Mid, although this only concerns three deictic adverbs (Vydrin 2024). In Kakabe (Mokole), OCP accounts for H insertion (Vydrina 2017).

5 Tone and intonation

Intonation is one of the least studied phenomena in Mande prosody, as it is usually not covered in available descriptions, with some recent exceptions (McPherson 2019, Vydrin & Diané 2016, Vydrina 2017, Sherwood 2020). Apart from down-drift, or tonal declination discussed in Section 4, *boundary tones* are most prominent intonational phenomena in Mande. This section thus focusses on phrase-final prosody (Section 5.1), the intonation of polar questions (Section 5.2), and continuation intonation (Section 5.3). Focus is usually not marked via prosody in Mande languages, as in Bobo (Bobo, Sherwood 2020: 139–153), as they employ dedicated focus particles. However, emphasis or discourse prominence can be expressed via register raising or an increase in pitch span, cf. Vydrina (2017: 520–522) on Kakabe (Mokole).

5.1 Phrase-final prosody

Most Mande languages seem to lack dedicated postlexical prosody marking finality, the realization of affirmative utterances being sufficiently accounted for by lexical and grammatical tones as well as surface tone rules, on a par with down-drift (if any). However, a dedicated intonational-phrase final L% prosody has been postulated for Kakabe (Vydrina 2017: 527), cf. also examples (24a–b) below.

A typologically unusual property of Kakabe is that it also employs distinct boundary prosody to mark negation. Thus in Kakabe, the (↑)H%⁴ boundary tone, which primarily marks continuation (Section 5.3), also signals the end of negative affirmative utterances as shown in (23).

(23) Kakabe (Mokole)

⁴In the boundary (↑)H% tone, the arrow in parentheses marks an optional pitch level rise.

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Háráy má máá súúmáyè tòtìpá-H%
 DISC 1PL PFV.NEG fast.REF spoil
 ‘We don’t fail to fast.’ (Vydrina 2017: 542)

Vydrina (2017: 548–549) suggests that, since negative utterances lack completion and require further elaboration, (↑)H% boundary tone may have been extended from continuation to negative utterance marking due to the similarity of their roles in discourse.

A dedicated finality-related phenomenon is also attested in Mano (Southern), where High tone surfaces as Mid on the utterance-final tone bearing unit (24).

- (24) Mano (Southern)
 [āā kpó á sónō]
 /āā kpó á sónó/
 3SG.PRF>3SG put 3SG near
 ‘She put it near her.’ (Konoshenko et al. Forthcoming)

Bisyllabic morphemes like *sónó* ‘near’ (24) are arguably lexically associated with a single underlying /H/ melody, since grammatical tone replacement rules target the underlying lexical tone in Mano, e.g., Imperfective lowering: *pélé* ‘wash’ → *pèlè* ‘wash:IPFV’, cf. Section 3. However, final High lowering in (24) only affects H associated with the last TBU: /sónó##/ → [sónō##], [*sōnō##], hence it refers to the surface string of H tones. Therefore, while its phonological implementation is categorical (H → M), it operates as a late postlexical rule at the utterance level (Section 6).

5.2 Polar questions

High-pitched polar question marking is extremely common cross-linguistically, and yet, the opposite, i.e., lowering, strategy is widely attested in African languages (Rialland 2007, Salfner 2017). Interestingly enough, Mande languages reveal an *areal* split in intonational polar question marking, which does not perfectly correspond to the basic genetic split into Western and Southeastern branches (Vydrin 2009b), see Figure 1. Languages spoken in the Western part of the Mande area, most notably in Mali, Guinea, Sierra Leone and Senegal, e.g., Bamana (Manding), Kakabe (Mokole), Susu (Soso-Jalonke), Guinean Kpelle and Mende (Southwestern), tend to employ either a question particle or a typologically common utterance-final pitch rise in polar questions (Vydrin et al. 2017). According to Vydrin’s (2009b, 2016b) classification, all these languages belong to

the Western branch (Section 1). At the same time, languages spoken to the East, e.g., in Ivory Coast, Burkina Faso and Nigeria, employ final lengthening and/or lowering, or a dedicated particle in polar questions, but crucially, no rise. This holds for Guro, Beng (Southern), Boko and Samo (Eastern; [Rialland 2007](#), [Vydrin et al. 2017](#)), which are classified as Southeastern Mande, cf. Figure 1. Strikingly, no polar question rise is attested in several genetically Western, but geographically “Eastern” languages. Thus, Jogo (Jogo-Jeri), mostly spoken in Ghana, uses final lengthening ([Sapozhnikova 2017](#): 218), while Seenku (Samogo) and Bobo (Bobo), both spoken in Burkina Faso, employ final lowering ([McPherson 2019](#), [Sherwood 2020](#): 124–126); their polar question marking thus patterning similarly to other geographically “Eastern” languages.

The only two areally “Eastern” languages where polar question rises have been reported are Mano (Southern; [Khachaturyan 2015](#)) and Mwan (Southern; [Perekhvalskaya 2017](#)). Crucially, both languages are in contact with “Western” type languages, i.e., Guinean Kpelle (Southwestern) in case of Mano ([Konoshenko et al. Forthcoming](#)), and Jula (Manding) for Mwan ([Vydrin p.c.](#)).

5.3 Continuation intonation

Continuation intonation, i.e., a postlexical boundary prosody signalling non-finality or turn-keeping, has been mostly reported for Western Mande, and also for Tura (Southern), where, similarly to polar questions, final lengthening and lowering signal continuation ([Idiatov & Aplonova 2017](#)).

Among Western Mande, the most detailed description of continuation intonation has been provided for Kakabe (Mokole) by [Vydrina \(2017\)](#). In Kakabe, continuation is marked by several distinct Intonational phrase (IP) final tones. [Vydrina \(2017\)](#) distinguishes between three types of continuation rises in Kakabe outlined in Table 1.

Example (25) illustrates how these boundary tones appear in natural discourse; note the utterance-final L% discussed in Section 5.1.

Table 1: Continuation intonation in Kakabe ([Vydrina 2017](#): 527–579)

Boundary tone	Alignment	Function
↑%	End of the IP-final syllable	Polar questions, continuation
↑H%	IP-final syllable	Continuation, negation (cf. Section 5.1)
↑HL%	Two final moras	Continuation

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(25) Kakabe (Mokole)

a. final rise

mà nààlén Bààdû↑ kà b́ Bààdû↑

We came to Baadi, and when we came to Baadi,

kà nàà misíídè tɔ̀yàn↑ kà b́ misíídè tɔ̀↑

we came to the market place there, and we left the market place

má t̀kèètà Pámpá yàn nà-L%

and we arrive to here, to Pampa. (Vydрина 2017: 538)

b. final boundary H%

ba 'má lè má nɔ̀lén'dé nùn-H% mà kɔ̀lɔ̀nizélén-H%

You see, we were enslaved, we were colonized

tibáábòè̀nù nín fúlán dè b̀lò-L%

by white people and by Fulbe. (Vydрина 2017: 541)

c. final HL%

mà b́: t̀gú'ré lè ↑HL-kéla sàgòè ↑HL-máà, mà nì jée sò↑HL-r̀i

We do pounding, in the morning, we scoop water. (Vydрина 2017: 556)

Note that final rise ↑% and boundary ↑H% yield a similar rising output and both serve as continuation markers; yet their precise phonetic alignment and functional distribution are different. Final rise (↑%) does not interfere with lexical tones and is realized after them, but the surfacing of the boundary ↑H% depends on the lexical tone alignment (Vydрина 2017: 551–554). Functionally, only the final rise can mark polar questions, and ↑H% also marks negation, cf. Section 5.1.

To sum up, Kakabe shows that while similar prosodic elements can in principle mark polar questions and continuation, different strategies may also be used for the two discourse functions. Further evidence comes from Seenku, which employs final lowering for polar questions, but final rise to mark continuation (McPherson 2019: 126–127).

Continuation rise has also been attested in Guinean Kpelle (Southwestern; Konoshenko et al. Forthcoming, Sigova 2018), which is in intense contact with Guinean Mano (Southern) lacking this prosody (Khachaturyan 2020a,b, Khachaturyan & Konoshenko 2021). Konoshenko et al. (Forthcoming) demonstrate that Continuation rise can appear in multilingual Mano speech as a result of cross-linguistic influence from Kpelle.

6 Tone and prosodic hierarchy

The complex rules of lexical and postlexical tone assignment outlined in the previous sections tend to operate at different levels of the prosodic structure in

Mande languages. Thus, the minimal tone bearing unit tends to be a *mora*, e.g., in Susu (Soso-Jalonke; [Green et al. 2013](#)) and Kakabe (Mokole; [Vydrina 2017](#)), or a *syllable* as in Seenku (Samogo; [McPherson 2019](#)) and Soninke (Soninke-Bozo; [Creissels 2016](#)). The alignment of postlexical tones also tends to be mora-sensitive, e.g., in Guinean Kpelle (Southwestern), Mano (Southern; [Konoshenko et al. Forthcoming](#)) and Kakabe (Mokole; [Vydrina 2017](#)).

At a higher prosodic level, the *foot* is extremely prominent in Mande ([Vydrin 2010](#)); it has been discussed for Susu (Soso-Jalonke; [Green et al. 2013](#)), Guro (Southern; [Kuznetsova 2007](#)), Mano (Southern; [Khachaturyan 2015](#)), Bobo (Bobo; [Sherwood 2020](#)) and many other languages. Yet accounting for foot-level phenomena often sparks debates, see [Green \(2015\)](#) and [Vydrin \(2020\)](#) for two rather different approaches to defining foot in Bamana (Manding). While feet are both relevant for segments and tones, at the tonal level, the foot is the domain for tonotactic constraints, e.g., tone melodies (Section 2.2), and tone rules (Section 4), e.g., the /LH#H/ → [LL#H] tone settling rule operating within a foot in Bamana ([Green 2015](#), [Leben 2002, 2003](#)). Generally, the phonological foot tends to coincide with the *phonological word* in Mande, but specific configurations may also vary from language to language; and in some languages, e.g., Pakaawu Mandinka (Manding), there are no reasons to postulate foot as a prosodic domain ([Creissels 2019: 8](#)).

Phonological phrase, a prosodic domain dominating the word, is of direct relevance for tone sandhi. Thus, in Kakabe (Mokole), it is the domain of Tone Leveling, whereby phrase-final H tone(s) are deleted after a HL sequence: /HLH)_{PhP}/ → [HL)_{PhP}] ([Vydrina 2017: 418](#)), while in Wan (Southern), it is the domain of the dissimilation rule discussed in Section 4 ([Nikitina 2019](#)).

Phonological phrases tend to strongly correspond to intermediate syntactic constituents, e.g., combinations of a lexical head with its (internal) argument, as is the case in Wan (Southern; [Nikitina 2019](#)). Yet, syntactic and prosodic phrasing may still feature some divergences. Thus, in Guinean Kpelle (Southwestern), H spread occurs in various phonological phrases corresponding to syntactic constituents, e.g., direct object and verb (26a), but also in cases where there is no syntactic constituency, i.e., between a time adverb and an intransitive verb (26b).

(26) Guinean Kpelle (Southwestern)

a. direct object + verb

/béláá kàà)_{PhP}/ → [béláá káá]

sheep see\

‘saw a sheep’

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- b. time adverb + intransitive verb
 /(ŋé)_{PhP} (wéí pà)_{PhP}/ → [ŋé wéí pà]
 1SG.B yesterday come\L
 ‘I came yesterday.’ (Konoshenko 2014b)

Unlike in (26a) corresponding to a syntactic VP, a time adverb (26b) does not form an immediate constituent with the verb as it can surface at the beginning or at the very end of the clause. Hence in Guinean Kpelle, prosodic constituency is not directly predicted by syntax.

Phonological phrases group into an *intonational phrase*, which is commonly relevant for boundary phenomena. In Wan (Southern), a H tone may be assigned before H or L (Section 4) or at the end of the phonological phrase as shown in (27).

- (27) Wan (Southern)
 /(àà nɛ)_{IP}, (Pálē à ē)_{IP} / → [àà nɛ Pálē à ē]
 3SG.POSS child PN 3SG saw
 ‘His child, Pale saw him.’ (Nikitina 2019: 702)

In (27), the left-dislocated noun phrase ‘his child’ ending in a lexically toneless word is assigned a H, since it is located at the right edge of an intonational phrase. In Kakabe, the intonational phrase is the domain of H insertion (Vydrina 2017: 391) as well as of boundary tones discussed in Section 5.3, and in Guinean Kpelle, it is the domain of floating ^H regressive linking (Konoshenko 2014b) and of Continuation rise.

Finally, *utterance* is the highest prosodic unit. It is usually the domain of down-drift and finality-related prosodies. In Guinean Kpelle, utterance is the domain of contour simplification rule, cf. Section 4 (Konoshenko 2014b).

7 Discussion

All Mande languages are tonal, yet they have rather diverse tonologies correlating with foot structure, cf. Sections 2 and 6. Western Mande, with the exception of Samogo and Bobo languages, generally have a binary tone contrast and bisyllabic feet, while Southeastern branch features from three to five tone levels and a stronger tendency towards monosyllabicity, most likely due to contact, e.g., with Kru languages (cf. Russell & Sande 2026 [this volume]) which heavily influenced Southern Mande in particular (Vydrin 2009a). Konoshenko (2014a) also shows that more contrasts correlate with more grammatical tones in Mande, since both

phenomena likely appeared as a result of segmental restructuring via monosyllabification. However, Green & Konoshenko (2022) and Konoshenko (2022) argue that at least some grammatical tones, e.g., the head tone in compounds and verb tone in Perfective may be old and reconstructable at the proto-family level, cf. Section 3.

As for tone rules discussed in Section 4, Mande languages generally follow the African tonology trends outlined by Hyman & Schuh (1974) and Hyman (2009), although High tone spread is not so common in Mande and tends to be constrained to few syntactic contexts.

Section 5 demonstrates that Mande languages tend to have various postlexical phenomena operating at higher levels of the prosodic hierarchy (Section 6), and that postlexical tones are sensitive to contact and areal effects, e.g., the geographical patterning of polar question marking strategies (Section 5.2), and continuation rise attested in multilingual Mano speech under the influence of Kpelle (Section 5.3). On the one hand, this supports the idea that intonational phenomena are particularly prone to cross-linguistic influence (Mennen 2015). At the same time, it has been claimed (Hyman & Monaka 2011, Downing & Rialland 2017) that languages with more complex tone contrasts might impose more restrictions on boundary tones. While continuation rise has indeed been mostly reported for Western Mande having fewer tone contrasts, the development of this prosody in three-tone Mano suggests that that contact can lead to further complexification of non-simplex tonologies.

8 Conclusion

This chapter summarises basic tone patterns in Mande, i.e., its lexical and grammatical tones, surface rules and intonational phenomena at different layers of prosodic structure. For reasons of space, I have not covered some topics, e.g., prosodic prominence, segment-tone interaction as well as the interaction between lexical and boundary tones. Other limitations are due to the lack of detailed tone descriptions for many languages, e.g., from Eastern Mande branch. Hopefully, this will be overcome in the future, so that the present overview can be updated based on a larger empirical basis.

Abbreviations

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ALN	alienable	NEG	negative
B	basic auxiliary	PhP	phonological phrase
COP	copula	PN	personal name
DISC	discourse particle	POSS	possessive
H	high	PFV	perfective
IP	intonational phrase	PRF	perfect
L	low	REF	referential marker
M	mid	SG	singular

Funding and acknowledgments

This study has been supported by Kone Foundation, project “How indigenous languages change: A view from the heritage language perspective.” The author thanks the two anonymous reviewers for their valuable comments and corrections. All the remaining errors are my own.

Appendix A Languages discussed in this chapter

Language	Group	Source	Source type
Kingi Soninke	Soninke-Bozo	Creissels 2016	Article
Seenku	Samogo	McPherson 2019	Grammar
Dzuun	Samogo	Solomiac 2014	Grammar
Bobo	Bobo	Le Bris & Prost 1981	Dictionary with a grammar sketch
		Sherwood 2020	PhD thesis
Niokolo Maninka	Manding	Creissels 2013	Article
Bamana	Manding	Green 2015	Article
		Vydrin 2016a	Article
		Vydrin 2020	Article
Pakaawu Mandinka	Manding	Creissels 2019	Article
Mandinka	Manding	Creissels & Sambou 2013	Grammar
Vai	Vai-Kono	Welmers 1976	Grammar
Kakabe	Mokole	Vydrina 2017	PhD thesis
Susu	Soso-Jalonke	Touré 2004	Grammar
		Green et al. 2013	Article
Mende	Southwestern	Leben 1973	PhD thesis

		Leben 1978	Article
Liberian Kpelle	Southwestern	Welmers 1962	Article
Guinean Kpelle	Southwestern	Konoshenko 2014b	Article
Kono	Southwestern	Konoshenko 2017	Article
Mano	Southern	Khachaturyan 2015	Grammar
		Khachaturyan et al. 2022	Dictionary
Eastern Dan	Southern	Vydrin 2021b	Dictionary
		Vydrin 2021c	Grammar sketch
Goo	Southern	Aplonova 2020	Dictionary
			Grammar sketch
Wan	Southern	Nikitina 2019	Article
Beng	Southern	Paperno 2014	Grammar sketch
Mwan	Southern	Perekhvalskaya & Vydrin 2024	Presentation
Boko	Eastern	Prost 1976	Grammar

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Chapter 17

Tone in Kru languages

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Kru languages, spoken in West Africa, have highly complex tone systems, typically with four contrastive level tones as well as several contrastive contour tones. Grammatical tone, especially marking inflectional categories like aspect and negation, realized on subjects, is prevalent across the language family. Kru language tone systems have been particularly important for understanding tonal typology and how to best represent contour tones. This paper collates data and generalizations from existing descriptions of tone in Kru languages, and adds novel data where possible, to lay out the lexical tone systems, grammatical tone patterns, tonotactics, and segment/tone interactions found in Kru languages, ending with a note about the implications of Kru tonal patterns for linguistic typology and theory and directions for future research

This paper provides an overview of the tone systems in Kru languages. Kru languages each have at least three contrastive tone levels in addition to contrastive contour tones (§2). Grammatical tone (§3), tonal alternations (§4), and segment-tone interactions (§5) are all common in Kru tone systems. Because of their high numbers of contrastive tones and complex tonal interactions, Kru tone has come to bear on key questions of tonal analysis; we highlight some formal and typological implications of Kru tone in section (§6).

1 Background on Kru

Languages in the Kru family are spoken primarily in southern Liberia and southwestern Côte d'Ivoire. Kru languages are underdocumented and understudied,

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and a precise number of languages or speakers is unclear. Names for individual languages vary a great deal between sources (Singler 2020), and varieties which are linguistically quite different are often labeled the same for sociocultural and political reasons (Sande 2020a). We refer to 36 distinct varieties here, listed in Table 1 with ISO code where available. Labels for varieties come from descriptive sources, which are cited throughout and listed in the appendix in Table 8.

Kru is widely agreed to be within the larger Niger-Congo language family, though its position within Niger-Congo is debated. Westermann & Bryan (1952) list Kru as an isolate family, while other authors consider Kru to be grouped together with other families, including Kwa (Greenberg 1966), Gur (Voltaic) (Bennett & Sterk 1977), Gur and Adamawa (Williamson 1989, Williamson & Blench 2000), Mande (Vydrine 2004) and Niger-Kordofanian (Welmers 1977). Marchese (1979, 1983, 1989) questions the grouping of Kru with Kwa, and finds it most compelling to group Kru as a separate branch within Niger-Congo until sufficient evidence is found otherwise.

Some notable linguistic features of Kru languages include the predominance of monosyllabic words composed of CV, CVV and CLV syllable structures, harmony (typically ATR harmony, though height harmony and nasal harmony are also found), SAuxOV word order that often alternates with SVO order, the presence of portmanteau morphs that simultaneously function as pronominal subjects and tense/aspect/mood/polarity markers (often referred to as STAMP morphs as per Anderson 2016, a.o), the presence of three or more contrastive tone heights, and complex systems of both lexical and grammatical tone.

Internally, there has been consensus since Delafosse (1904) that there are two major sub-branches of Kru: Eastern and Western. These branches differ in a few notable ways: in terms of phonology, most Western Kru languages have contrastive nasal vowels but no contrastive central vowels, while most Eastern Kru languages have the opposite (Marchese 1989). With respect to word order, Eastern Kru show SAuxOXV word order, where the verb is always clause-final and adpositional phrases and adverbs surface, along with objects, between the auxiliary and verb. In most Western Kru languages, X elements such as adpositional phrases and adverbs surface after the verb: SAuxOVX (Marchese 1983).

Here we remain conservative in both our internal and external classification of Kru languages, adopting the widely accepted Eastern/Western split within Kru, as well as grouping a few established dialect clusters together in Table 1, but otherwise not assuming particular relationships among Kru languages or between Kru and other language families.¹

¹Three languages – Kuwaa, Aizi, and Seme – have previously been (tentatively) classified as

Table 1: Kru varieties included in our survey

Branch	Group	Variety	ISO
Western	Krumen	Bassa	bsq
		Dewoin	dee
		Guéré	gxx
		Nidrou	wec
		Nyabwa	nwb
		Krahn	krw
		Grebo	grj
		Jabo	–
		Klao	klu
		Wobé	wob
		Tépo	ted
		Plapo	ktj
		Piè	pye
Eastern	Godié	Kouya	kyf
		Godié	god
		Dadjriwalé	–
	Bété	Daloa Bété	bev
		Guibéroua Bété	bet
		Gbadi	btg
		–	–
	Dida-k ^w	Kpokolo	–
		Lakota Dida	dic
		Abou Dida	–
		Yocoboué Dida	gud
		Zaroko Dida	–
		Guébie	gie
		Gabɔgɔ	–
		Néyo	ney
	Dida-f	Vata	–
		Gakpa	–
		Gbeiwi	–
		Kagwé	–
		Laɓɔ	–
		Ménéhiri	–
		–	–
Unclear		Kuwaa	blh
		Aizi	ahi
		Seme	sif

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2 Lexical tone

All Kru languages make use of tone to differentiate lexical items. Each Kru language contrasts three or four level tones, which we represent here as L (low), m (mid/mid-low), M (mid-high), and H (high). When discussing a language that contrasts only three levels, we use L, m, and H. In IPA transcriptions, we use the following conventions, represented on the vowel [a] for consistency: L [à], m [ā] M [á], H [ǎ]. Contour tones, or multiple successive tone heights on a single syllable, are represented with sequences of L, m, M, and H (e.g. HL is a falling tone that starts high and ends low, and mH is a rising tone that starts mid-low and ends high). Contour tones are represented as tone macros on a single vowel such as HL [ǎ], Hm [ǎ̃], or mM [ǎ̃̃]. For languages where contours are produced on long vowels, contour tones are written over a sequence of two vowels: [ǎǎ] for a HL on a long vowel.²

Section 2.1 discusses the number of contrastive of level tones found across Kru languages, section 2.2 discusses the presence of contrastive contour tones, and section 2.3 discusses category-specific tone melodies.

2.1 Number of contrastive levels

Most Kru languages have four contrastive tone levels.³ This includes Eastern languages (Kouya, Daloa Bété, Guibéroua Bété, Gbadi, Kpokolo, Lakota Dida, Zaroko Dida, Guébie, Vata, Gbeiwi, Labo, Ménéhiri), Western languages (Guéré, Nyabwa, Krahn, Grebo, Jabo, Wobé, Nidrou), and languages whose classification is undetermined (Aizi, Seme). An example of a four-way minimal tone pair from Daloa Bété is given in Table 2.

Several Kru languages have three surface tone levels that are best treated as a four-way underlying contrast. For example, Luckau (1975) analyzes Bassa (Western), which has three surface levels, as a language with four underlying levels in which the m tone is phonetically realized as a Lm rise (Table 3).

Other Western languages with three surface tone levels that may be analyzed as four underlyingly include Klao, Tépo, Plapo, and Piè. In Klao, there are surface

Kru in the literature (Marchese 1976, 1983). More recent work suggests that Kuwaa and Aizi may be Western Kru languages (Vogler 2020), while Seme should not be considered Kru at all (Vogler 2015).

²We adopt the conventions and description of the source materials for each language presented here as to whether contour tones within a syllable are written on a single vowel or a long vowel.

³Some Kru languages have more than 4 distinct levels on the surface. For example, see the discussion of Guébie imperfective tone shift in Sande (2018).

Table 2: Four-way minimal tone pair in Daloa Bété (Zogbo 2004: p. 51)

	Tone	Word	Gloss
^a	H	sú	‘to dance with enthusiasm’
	M	sù	‘tree’
	m	sũ	‘to reduce to powder’
	L	sù	‘to carry’

^aIn all examples throughout, we use IPA symbols. In some cases, original sources did not use the IPA, in which case our choice of IPA symbol is based on the description of the symbols used in the original sources. For example, some Kru descriptive literature uses [ω] for a high, back, -ATR vowel, though we use the IPA symbol [ʊ] here.

Table 3: Four-way near-minimal tone pair in Bassa (Luckau 1975: p. 21)

Underlying	Surface	Word	Gloss
/H/	[H]	x ^w é	‘corpse’
/M/	[m]	x ^w ē	‘deer’
/m/	[Lm]	ɣ ^w ě	‘ape’
/L/	[L]	ɣ ^w è	‘to finish’

L, m, and H tone levels, but there must be two distinct types of mid tone due to their surface behavior – both mid tones have a lowering effect on a following H tone, but one of them lowers /H/ to [m] and the other lowers /H/ to [L] (Lightfoot 1974). Similarly, analyses of Tépo, Plapo and Piè consider there to be two types of L tone based on interactions with surrounding tones (Thalmann 2018).

In Godié (Eastern), there are three surface level contrasts [H m L], but nouns with final mid tones in isolation pattern in two different ways. One type of mid always surfaces as [m] while a second type of mid lowers to [L] in some contexts and causes lowering of the following tone in other contexts (Gratrix 1975: p. 108, 113). In Kagwé (Eastern), there are three surface level contrasts [H m L] but four underlying contrastive levels /H M m L/ (Koopman & Sportiche 1982). Surface mid tones in Kagwé pattern as two separate classes: Class A, analyzed as /M/, and Class B, analyzed as /m/. While both classes surface as [m] in isolation, the difference between the classes is visible in particular contexts, including in the plural and after certain nominal modifiers. Here, Class A nouns surface as [H], while Class B nouns surface as [m] (Table 4). Koopman & Sportiche also note that

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Class A nouns in Kagwé correspond to M-toned cognates in related languages like Vata, while Class B nouns in Kagwé correspond to m-toned cognates, indicating that the current three-way surface contrast of tone levels in Kagwé may have evolved from a four-way level contrast.

Table 4: Two classes of surface mid tone in Kagwé (Koopman & Sportiche 1982: p. 47–48)

Class	Gloss	Singular	Plural	‘his X’	‘this X’
A; /M/	‘spear’	lē	lí	̄ lé	màná lé
	‘child’	jō	jía	̄ jó	màná jó
B; /m/	‘hen’	ɲlū	ɲlī	̄ ɲlū	màná ɲlū
	‘bank’	kpā	kpī	̄ kpā	màná kpā

A smaller set of Kru languages are described as having only three contrastive and surface tone levels. These include the unclassified language Kuwaa, the Western language Dewoin, and the Eastern languages Dadjriwalé, Néyo, Gaḃɔgḃɔ, Abou Dida and Yocoboué Dida. Note that with the exception of Abou Dida, the tone systems of these languages are not described in enough detail to determine the number of underlying tones. It is possible that with additional data, at least some of these languages may be found to show evidence for four distinct underlying level tone contrasts like the rest of the Kru languages.

2.2 Contour tones

All Kru languages have at least one contrastive rising contour tone. Most languages also have at least one falling contour, but some Western languages lack phonemic falling contours entirely. Languages reported to lack falling contours include Aizi, Tépo, Plapo, Piè, Grebo, and Jabo. The fact that rising contours are more pervasive than falling contours in Kru is surprising given the claim that rising contours are more complex (and therefore typologically less common) than falling ones (Yip 2002: p. 27–30).

In some Kru languages, syllables that host contour tones must have long (bi-moraic) vowels; non-contrastive vowel lengthening on syllables hosting contour tones is typical. For example, Marchese (1983) states that in Godié (Eastern), in which vowel length is not contrastive, vowels lengthen to host contours as in [bèē] ‘groundnut’. In such languages, the tone-bearing unit can be analyzed as the mora, where two moras are required to realize two tones on a single syllable.

In other Kru languages, contours appear freely on short vowels, as in Wobé (Western) (Bearth & Link 1978, 1980), where 14 distinct tone melodies – four level tones and 10 distinct contours – contrast on CV syllables. Example minimal tone sets in Wobé are given in (1); S stands for a super-low tone, which only appears in falling contours in Wobé. In Wobé, with few restrictions, the same set of tone melodies is found on CV, CVV, and CVVV syllables (Bearth & Link 1980: p. 160).

(1) Minimal tone sets in Wobé (Bearth & Link 1980: p. 150–151)

	CV string	Tone melody	Gloss
a.	pa	LH	‘cocoon’
		LM	‘to carve (intrans.)’
		Lm	‘to carve’
		mM	‘to enter (intrans.)’
b.	kle	M	‘neck’
		L	‘lizard (species)’
		Lm	‘field’
		LS	‘monkey (species)’
c.	kpo	H	‘awalé (game)’
		m	‘hump’
		mH	‘hoe’

In several languages, certain contour tones are restricted to particular positions within the word. In Kuwaa (unclassified) and Wobé (Western), rising contours are reported to appear only in word-initial syllables, with falling contours only in morpheme-final syllables (Marchese 1983). In Guébie (Eastern), contour tones typically only appear word-finally, suggesting that tones are associated to tone-bearing units from left to right (Sande 2017).

One source of contour tones in Kru is alternations between CVLV and CLV strings, where L represents a sonorant consonant, typically /l/. For example, in Dadjriwalé (Eastern), some words are pronounced as CVLV in careful speech, but as CLV in fast speech (Gode 2006). Even if the first vowel is elided in fast speech, underlying tones remain, creating a surface contour on a single syllable. This is attested for words where the two vowels are identical (2a-b), but also for words where the vowels differ in quality (2c-d).

(2) Dadjriwalé C(V)LV words (Gode 2006: p. 64–65)

- a. [bòlɔ́] ~ [blɔ́] ‘path’
- b. [dábɔ́lɔ́] ~ [dábɔ́] ‘pigeon’

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- c. [gàlɛ̃] ~ [glɛ̃] ‘mask’
- d. [kòlɣ̃] ~ [klɣ̃] ‘jar’

In some Kru languages, surface CLV syllables can host contours, sometimes including contours or tone melodies that are not otherwise found on monosyllabic words (Gratrix 1975). The distribution of tones on surface CLV syllables can contribute to determining whether such strings are underlyingly /CLV/ or /CVLV/ (Sande 2024).

2.3 Category-specific tonal inventories

In most Kru languages, there are fewer tonal contrasts among verbs than nouns (see Smith 2011 on category-specific phonology across languages). In most cases, this is due to grammatical tone playing an important role in determining the surface tone of verbs. In many languages, including Klao, Krahn and Tépo in the West and Kpokolo and Abou Dida in the East, between two and six possible tonal melodies are attested on verbs, but many more on nouns. In Tépo (Thalmann 1980: p. 43), for instance, there are 12 tone melodies on CVCV nouns, but only 4 on CVCV verbs. In other languages, though, such as Grebo (Western) and Godié (Eastern), verbs and nouns have the same number of possible tonal contrasts.

In some Kru languages, ideophones and loan words show distinct tone patterns from words in other categories. For example, in Klao (Western) all nouns with a HM falling tone melody are English loanwords (Lightfoot 1974: p. 434), and in Godié (Eastern), borrowed nouns systematically get a HM tone melody (Gratrix 1975: p. 104). Other languages with loan-specific tone melodies include Guébie (Sande 2022b: p. 10), Grebo (Innes 1966: p. 18), Tépo (Thalmann 1980: p. 61), and Bassa (Bertkau 1975: p. 41).

3 Grammatical tone

All Kru languages make use of tone to mark grammatical meanings (cf. Rolle & Bickmore 2026 [this volume]). The specific meaning marked via tone, and the particular tonal alternations involved, differ from language to language, though certain tendencies recur across the family. Here we categorize grammatical tone patterns across Kru as derivational or inflectional morphology, where inflectional grammatical tone involves differentiating person and number, aspect, and negation, and derivational grammatical tone involves phenomena like compounding and nominalization. The examples provided in this section do not constitute an

exhaustive list of grammatical tone processes in Kru; rather, we focus on processes that are found across several Kru languages.

3.1 Derivational grammatical tone

Nearly all Kru languages are described as requiring a L or m tone melody on the second noun in a nominal compound or associative construction (Marchese 1983: p. 106). For example, in the Krumen (Western) cluster of languages – Tépo, Plapo, and Piè – a floating L is associated with the second member of a compound. This L tone does not override or replace the underlying tone melody of the noun, but rather interacts with it (cf. Rolle et al. 2026 [this volume]). For example, the floating L lowers H to m: /brī jú/ (m H) ‘cow offspring’ surfaces as [brī jū] (m m) (Thalmann 2018: p. 7–8). In Grebo and Jabo (Western) and Guébie (Eastern), the replacive melody on the second member of a compound has a /m/ tone no matter the underlying tone of the word, although regular tonotactic alternations may sometimes obscure this /m/ on the surface. Guébie examples are given in (3).

- (3) Guébie replacive m in nominal compounds (Sande 2022b: p. 27)

- a. pītò jō
in-law child
‘daughter-in-law’ (cf. /jó/ ‘child’)
- b. máná-dī-pō
meat-cut-AGT
‘butcher’ (cf. /dī/ ‘cut’, /-pō/ ‘AGT’)
- c. bītə wālī
house top
‘top of house’ (cf. /wālī/ ‘top’)

In several Eastern languages, tone lowering is attested in some contexts that are translated as noun-adjective constructions as well, though these may very well be better analyzed as nominal compounds. For instance, in Godié, after certain adjectives, like /núgλ/ ‘good’, the first syllable of the following noun surfaces with a lowered tone (4). Notably, this appears to be a stepwise alternation, where /H/ → [m] and /m/ → [L].

- (4) Tonal alternations after ‘good’ in Godié (Gratrix 1975: p. 101–102)
- a. /núgλ/ + /dú/ → [núgλ dū] ‘a good village’

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- b. /núgλ/ + /lói/ → [núgλ lōī] ‘a good bushbuck’
c. /núgλ/ + /kōlós/ → [núgλ kòlós] ‘good bamboo’

Nominalization constructions in Kru appear to be historically related to compounds; they often involve a nominalizing suffix or clitic that is clearly a reduced form of a common noun meaning ‘place’, ‘thing’, or ‘person’. In many cases, these nominalizing suffixes or clitics show the same tonal requirements as the second noun in a compound, as with the agentive nominalizer /-ɲɔ̃/, which has a mid tone in Guébie, just like the second member of a compound (3).

3.2 Inflectional grammatical tone

3.2.1 Person and number

In almost all Kru languages, first- and second-person pronouns differ only in tone. In the singular, first-person pronouns always have a higher tone than second-person ones, while the opposite is true for plural pronouns: first person plural is always lower-toned than second plural. An example pronominal inventory from Abou Dida (Eastern) is given in Table 5.

Table 5: Inventory of first- and second-person pronouns in Abou Dida (Miller 2005: p. 66)

	1SG	2SG	1PL	2PL
Subject	ī	ì	à	ā
Object	í	ì	à	á
Possessive	nā	nǎ	àṇā	āṇā
Emphatic	àmō	mǔ	àṇī	āṇī

Sometimes, the marker of person and number is solely tonal. For instance, in Grebo (Western), the first singular subject pronoun has two allomorphs, one segmental /ní/, and one which is just a High tone. When the pronoun has no segmental content, its High tone replaces the first tone in the verb’s lexical melody 5b.

- (5) Alternation between segmental and tonal 1sg subject pronoun in Grebo (Newman 1986: p. 184)
- a. ní kwidē-dà
1SG run-PST
‘I ran.’

- b. kwídē-dà
 1SG.run-PST
 ‘I ran.’

Similarly, in some Eastern languages, certain object pronouns may be realized as just tone. In Guébie, the second singular object pronoun has two allomorphs, one segmental /mē/⁴, and one which is just a mid tone (6).

- (6) Alternation between segmental and tonal 2SG object pronoun in Guébie (primary data)

- a. é né=mē jōkù-ní
 1SG NEG.FACT=2SG.ACC PART-see
 ‘I didn’t see you.’
- b. é nē jōkù-ní
 1SG NEG.FACT.2SG.ACC PART-see
 ‘I didn’t see you.’

3.2.2 Aspect

The basic aspectual distinction in Kru is perfective versus imperfective. Stative verbs tend to pattern morphologically with perfective verbs, and the two together (stative + perfective) are sometimes referred to as *factative* (Marchese 1986). In many Kru languages, this aspectual distinction is expressed by the tone of the verb, either alone or alongside other aspect marking. In Western Kru, the perfective tends to be unmarked, and the verb surfaces with its lexical tone. Marchese (1986: p. 33) cites Dewoin, Grebo, Nyabwa, Klao, Krahn, and Wobé as Western languages with an unmarked perfective. Krahn, for instance, has a three-way distinction in underlying verb tone, visible in the perfective (7).

- (7) Krahn unmarked perfective (Sauder 2000: p. 72)

- a. mónā pá ǎà sǎ wé
 Mona mend 3SG.POSS fishnet FOC
 ‘Mona fixed her fishing net.’ (cf. /pá/ ‘mend’)
- b. ǎ mū dé kwlá
 3PL go REL bush
 ‘They went to the bush.’ (cf. /mū/ ‘go’)

⁴This morpheme surfaces as [mē] in (6) due to vowel harmony with the +[ATR] vowel in the negative auxiliary.

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- c. 5 dbà tū kàú
 3SG fell tree dry
 ‘He cut down a dry tree.’ (cf. /dbà/ ‘fell’)

Several Eastern languages, including Gbadi, Néyo, Kouya, and Guébie, are also described as having unmarked perfectives. Elsewhere in Eastern Kru, the perfective is marked by Low tone. Verbs with underlying High and M tones surface with a falling (ML) contour (8-9), and verbs with underlying mid and Low tones surface as Low (10-11). Marchese (1986) speculates that the perfective in Proto-Kru was likely unmarked, and that the Low-tone perfective in Eastern Kru originates from what was historically a marker of perfect aspect.

- (8) Gbeiwi perfective (Charette 1982b: p. 250)
 à drà zàgà
 1PL kill.PFV agouti
 ‘We killed an agouti.’ (cf. /drá/ ‘kill’)
- (9) Kpokolo perfective (Vahoua 2018: p. 151)
 ɲàkómúní là kàràsó
 Gnakoméné call.PFV Krasso
 ‘Gnakoméné called Krasso.’ (cf. /là/ ‘call’)
- (10) Godié perfective (Marchese 1986: p. 39)
 5 lì tlē
 3SG eat.PFV snake
 ‘He ate a snake.’ (cf. /lī/ ‘eat’)
- (11) Lakota Dida perfective (Guéhoun 1993: p. 336)
 5 tì drāā dōkō
 3SG lose.PFV money PART
 ‘He lost the money.’ (cf. /tì/ ‘lose’)

In Western Kru, the imperfective is often marked by a suffix on the subject. However, in a few Western languages, the imperfective additionally involves a change in verb tone. In Krahn, for instance, underlyingly High and Mid verbs are unmarked in the imperfective (12a,b), while an underlyingly Low verb raises to Mid (12c).

- (12) Krahn imperfective (Sauder 2000: p. 73–74)
- a. 5 dé-ā pá sō
 3SG mother-IPFV mend fishnet
 ‘Her mother is mending a fishnet.’ (cf. /pá/ ‘mend’)

- b. $\tilde{a}$ mū dé gwlòó
 3PL.IPFV go REL town
 ‘They are going in town.’ (cf. /mū/ ‘go’)
- c. jībē-ā tlō ðòé klēí
 cat-IPFV chase.IPFV rat behind
 ‘The cat is chasing after the rat.’ (cf. /tlō/ ‘chase’)

In Eastern Kru, the imperfective tends to be marked by a change in verb tone towards the middle of the tonal inventory. Verbs with underlying High tones lower to M (13), and verbs with underlying Low tones raise to m (16), while underlying M and m tones surface as such (14-15).

- (13) Gbeiwi imperfective (Charette 1982b: p. 250)
 à drà zàgà
 1PL kill.IPFV agouti
 ‘We kill an agouti.’ (cf. /drà/ ‘kill’)
- (14) Kpokolo imperfective (Vahoua 2018: p. 141)
 jàkō là vāwā
 Djako call.IPFV Vahoua
 ‘Djako calls Vahoua.’ (cf. /là/ ‘call’)
- (15) Godié imperfective (Marchese 1986: p. 39)
 ō lī tlē
 3SG eat.IPFV snake
 ‘He eats snakes.’ (cf. /lī/ ‘eat’)
- (16) Lakota Dida imperfective (Guéhoun 1993: p. 308)
 ŋó-ā ɓlī mōgō
 woman-DEF limp.IPFV leg
 ‘The woman limps.’ (cf. /ɓlī/ ‘limp’)

In a handful of Eastern Kru languages, including Dadjriwalé, Godié, and Gaḃṵgḃṵ (all languages with a three-level contrast), imperfective aspect is marked by a replacive mid tone on the final syllable of the verb, regardless of the verb’s underlying tone. Sande (2018) reports that in Guébie, a 4-tone language, imperfective aspect and negative perfective aspect are realized via scalar tone lowering (H→M→m→L) of the verb (17a,b), unless the verb is already L, in which case the final tone of the subject raises one step (L→m→M→H→S, where S is superhigh), (17c,d).

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(17) Scalar tone shift in Guébie imperfective (Sande 2022b: p. 28–29)

- a. ǝ li bɛ̀
3SG eat.PFV thing
‘He ate something.’
- b. ǝ lī bɛ̀
3SG eat.IPFV thing
‘He is eating something.’
- c. ǝ̀cì pà
Djatchi run.PFV
‘Djatchi ran.’
- d. ǝ̀cī pà
Djatchi run.IPFV
‘Djatchi runs.’

The Guébie imperfective alternation only affects a single tone height, not an entire tone melody.

(18) Scalar tone shift in Guébie imperfective, continued (Sande 2018: p. 260–261)

- a. jú gbálá sí
boy climb.PFV trees
‘A boy climbed trees.’
- b. jú gbālá sí
boy climb.IPFV trees
‘A boy climbs trees.’
- c. ā ká dibò-ǝ bālā
1PL IRR plantain-PL harvest
‘We would harvest plantains.’
- d. ā bàlà dibò-ǝ
1PL harvest.IPFV plantain-PL
‘We harvest plantains.’

The data in (18a,b) indicate the independence of each level tone in a tone melody.⁵

⁵The examples in (18c,d) show that the Obligatory Contour Principle (OCP) (Leben 1973) is active in Guébie: the tone on both syllables of a disyllabic verb lowers if they are the same height, suggesting that the tone on both syllables is linked to a single tonal unit.

3.2.3 Negation

In many Eastern Kru languages, negation is expressed through a change in the tone of the final tone-bearing unit of the subject (Marchese 1986), perhaps best analyzed as a suffixal floating High tone. In Gabŭgbŭ, for instance, the final tone of the subject ‘Ozé’ is realized as a rising contour in negative contexts (19).

(19) Tonal negation marking in Gabŭgbŭ⁶ (Gnahore 2006: p. 20–21, 32–33)

- a. òzē ɓlū sākā
Ozé pound.IPFV rice
‘Ozé pounds rice.’
- b. òzě ɓlū sākā
Ozé.NEG pound.IPFV rice
‘Ozé doesn’t pound rice.’
- c. òzē ɓlù sākā
Ozé pound.PFV rice
‘Ozé pounded rice.’
- d. òzě sākā ɓlù
Ozé.NEG rice pound.PFV
‘Ozé didn’t pound rice.’

In Daloa Bété and Guibéroua Bété, negation is expressed tonally only when the subject is pronominal; with a non-pronominal subject, negation is expressed through a particle or auxiliary instead (20).

(20) Tonal negation in Guibéroua Bété on pronouns only (Werle & Gbalehi 1976: p. 175–177)

- a. òó nímá
3SG.NEG drink.IPFV
‘He doesn’t drink.’ (cf. /ó/ ‘3SG’)
- b. ná dībà ní lī kòkó
1SG.POSS father NEG eat.IPFV chicken
‘My father doesn’t eat chicken.’

Most languages, however, do not show the same restriction, as negation may be expressed through tone on any subject, whether pronominal or non-

⁶The descriptive source does not distinguish the transcription of different rising contours, so it is unclear whether the final rise on the subject nouns is produced as LH, as written, or mH.

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pronominal. In Guébie, for instance, negation is indicated through a High tone appended to the final tone-bearing unit of the subject, regardless of lexical category (21).⁷

(21) Subject-final tonal negation marking in Guébie (primary data)

- a. *ɲúú wōtō*
water.NEG be.cold
'The water is not cold.' (cf. /*ɲú*/ 'water')
- b. *ɲʷónó ɔ̌ pā cékè pōlō āné si*
woman 3SG sell.IPFV attiéké PART REL.NEG laugh
'The woman who sells attiéké is not laughing.' (cf. /*āné*/ 'REL')
- c. *sákà bòtògbā fà kō*
rice much.NEG there be
'There is not much rice.' (cf. /*bòtògbā*/ 'much')

In such languages, the form of a non-pronominal subject in negative contexts is typically created through the phonologically predictable addition of a -H or -M suffix. For pronominal subjects, however, forms in negation are typically unpredictable from a combination of separate tones. In Lakota Dida, for example, the 1SG subject pronoun and 'Dago' both carry a final mid tone in affirmative contexts (22a, c). In negative contexts, though, the 1SG pronoun surfaces with a LH rising contour, while the final syllable of the name 'Dago' surfaces with a mH rising contour (22b, d).

(22) Negation marking on different subjects in Lakota Dida (Guéhoun 1993: p. 73, 249)

- a. *ī bídō*
1SG wash.IPFV
'I bathe.'
- b. *ĩ bídò*
1SG.NEG wash
'I don't bathe.'
- c. *dàāgō lĩpĩ*
Dago speak.IPFV
'Dago speaks.'

⁷In Guébie, the negative H tone is added to the final TBU of the subject in all cases except when the final TBU is itself H, in which case the final vowel of the subject is lengthened to host the negative H as in (21a).

- d. dàāg55 līpì
 Dago.NEG speak
 ‘Dago doesn’t speak.’

4 Tonicactics

Most Kru languages show a preference against certain surface tonal sequences, resolved by various tonal alternations such as tone spreading and contour simplification (cf. [Smolders 2026 \[this volume\]](#)).

4.1 Tone spreading

Several Kru languages show tone patterns that could be analyzed as tone spreading. In multiple languages, tone spreading crosses word boundaries. For example, in Tépo (Western), there are two types of L tone (as mentioned in §2.1, labeled L_a and L_b). L_b both undergoes and triggers tone spreading, whereas L_a neither triggers nor undergoes tone spreading ([Thalmann 2018](#): p. 3). The example in (23a) shows that L_a (here represented by the verb /tò/ ‘buy’) is not a target of tone spreading, whereas in (23b) the L_b verb /hà/ ‘pick’ becomes HL after a H spreads from the preceding word. In (23c) we see that L_a does not spread its own tone, but in (23d) we see spreading from L_b in the same environment.

(23) Tépo tone spreading ([Thalmann 2018](#): p. 3)

- a. L_a : Not a target of spreading
 /bǒ tò būbuē/ → [bǒ tò būbuē] ‘I need to buy an orange’
- b. L_b : Target of spreading
 /bǒ hà būbuē/ → [bǒ hà būbuē] ‘I need to pick an orange’
- c. L_a : Not a trigger of spreading
 /tò kē būbuē/ → [tò kē būbuē] ‘buy an orange today’
- d. L_b : Trigger of spreading
 /hà kē būbuē/ → [hà kē būbuē] ‘pick an orange today’

In Bassa (Western), H tone spreads to the right ([Bertkau 1975](#)) (24a). Spreading is blocked, however, by a so-called fortis consonant (24b); see Section 5 for further discussion of segment-tone interactions, including a list of fortis consonants. In Bassa, HL contours also spread to the right, where an underlying HL at the end of

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one word results in a H on the original HL morpheme and a HL at the beginning of the next (24c);⁸ this process is also blocked by fortis consonants (24d).

(24) Bassa tone spreading (Bertkau 1975: p. 28)

- a. /ní kòð/ → [ní kòð] ‘my bucket’
- b. /ní dè/ → [ní dè] ‘my palm kernel’
- c. /sùkúù djù/ → [sùkúù djù] ‘school child’
- d. /sùkúù gbò/ → [sùkúù gbò] ‘school house’

In several Kru languages, some suffixes do not contribute a tone or tonal melody of their own; we refer to these as toneless suffixes. When a toneless suffix is added to a root or stem, the tone melody of the root or stem associates over the entire word, including the toneless suffix. For example, in Jabo (Western), a /MH/ contour on a noun, plus a toneless suffix, like the plural /-i/, surfaces as [M] on the noun and [H] on the suffix (Luckau 1975: p. 27). Likewise, a /mM/ noun plus a toneless suffix results in [m] on the noun and [M] on the suffix.⁹ Similar patterns are found with toneless suffixes in Guébie (Sande 2017, 2022b). In Klao (Western), it is often the case that an underlying contour is realized with the initial level on the word of origin and the second level on the following word (Lightfoot 1974: p. 428): /ɔ̃ kpā kô tĩ/ → [ɔ̃ kpā kó tĩ] ‘He held bananas’. Spreading does not occur onto a L or low rising contour, or if a rising contour is followed by an identical rising contour.

4.2 Contour simplification

In a few Western Kru languages, contour tones are simplified in the context of specific other tones. In Bassa, an underlying /mL/ contour surfaces as [L] after /L/ (Hobley 1964: p. 53), and in Tépo and Plapo, /LH/ surfaces as [L] before /L/ (Thalmann 2018: p. 6). Both of these alternations involve underlying sequences of LXL, where some tone X is unrealized when sandwiched between two L tones.

⁸An unrelated process of ‘lenis raising’ affects the initial tone on ‘school’ (Hobley 1964: p. 53). Additionally, contour simplification results in an underlying /Lm/ contour surfacing as [L] (Bertkau 1975: p. 36).

⁹Additionally, in Jabo, a /mH/ contour on a noun, plus a /M/ or /m/ suffix, results in a [M] noun and [H] suffix. Less transparently, a /MH/ noun plus a L suffix, like the yesterday past /-də/, results in a [H] noun and [L] suffix, and a /H/ noun plus a toneless suffix results in a [M] noun and [H] suffix.

4.3 Morpheme-specific tonal alternations

4.3.1 Reduplication

Reduplication is found in multiple distinct grammatical contexts in Kru languages. Sometimes these distinct grammatical uses of reduplication differ only in their tonal patterns. In Krumen (Western) languages, /H/ surfaces as [m] on the second copy in reciprocal reduplication, but there is no tone change in deverbal nominalizing reduplication. This construction-specific tonal alternation is analyzed as due to the presence of a floating L in reciprocal contexts (Thalmann 2018: p. 8). For example, in the Piè variety of Krumen, /lá/ ‘kill/ is reduplicated as [lí-lā] in reciprocal contexts but as [lí-lá] in deverbal nominalizing contexts (with vowel raising on the first copy). Similar facts hold in Guébie reciprocal reduplication and nominalizing reduplication contexts (Sande 2017, 2020b, 2022b).

4.3.2 Tonal alternations after subject pronouns

Tonal interactions at the boundary between a subject pronoun and verb or auxiliary are common across Kru languages. Such alternations are often morpheme-specific in that they are sensitive to the identity of the subject (pronominal versus non-pronominal subjects, or differences in person/number of the subject), and/or the tense, aspect, mood, or polarity of the clause. In the Dida-f (Eastern) languages Vata, Laḃ and Gbeiwi, for instance, the tone of a verb surfaces as identical to the tone of a subject pronoun in certain contexts, demonstrated here with data from Vata (25). This could be analyzed as tone spreading or assimilation.

- (25) Tonal assimilation at the subject-verb boundary in Vata (Koopman & Sportiche 1982: p. 56, 242)

- a. /n̩ l̩/
- [n̩ li]
- 2SG eat.PFV
- ‘You ate.’
- b. /ɔ́ lá jò/
- [ɔ́ lá jò]
- 3SG.NEG call.IPFV child
- ‘He doesn’t call a child.’

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- c. /ñ lē/
 [ñ lè]
 1SG eat.IPFV
 ‘I eat.’

Dissimilation is also found between a subject pronoun and verb in other contexts. For instance, an underlyingly /M/ verb surfaces as [H] after a /M/ subject pronoun in Vata (26). This tonal interaction does not take place at other boundaries, like the verb-object boundary, as evidenced by the lack of alternation between /M/ ‘call.IPFV’ and /M/ ‘child’ in (26).

- (26) Tonal dissimilation at the subject-verb boundary in Vata (Koopman & Sportiche 1982: p. 55)
 /ǝ lá jǝ/
 [ǝ lá jǝ]
 3SG call.IPFV child
 ‘He calls a child.’

Similar tonal assimilation at the same boundary is attested in Jabo, a Western language. In Jabo, an underlyingly /M/ verb surfaces as [H] after a /H/ subject pronoun (27a). Elsewhere, including after High-toned nominal subjects, an underlyingly /M/ verb surfaces as [M] (27b-c). Luckau analyzes this tonal distinction as owing to a difference in prosodic boundary: ‘a pronoun subject is separated from its verb by a phonological boundary of rank intermediate between morpheme boundary... and word boundary’ (Luckau 1975: p. 30).

- (27) Alternation of verb tone after different subjects in Jabo (Luckau 1975: p. 30)

- a. né dú-dà nǝ nē
 1SG pound-PST 3SG.ACC SCmpl
 ‘I pounded it.’
 b. dó dù-dà nǝ nē
 Dǝ pound-PST 3SG.ACC SCmpl
 ‘Dǝ pounded it.’
 c. ké dù-dà nǝ nē
 chief pound-PST 3SG.ACC SCmpl
 ‘The chief pounded it.’

In Abou Dida, Kagwé and Godié, all Eastern languages with three surface contrastive levels, certain verbs and auxiliaries alternate between [m] and [H] depending on the tone of the preceding subject. In Abou Dida, an underlyingly /H/ verb or auxiliary surfaces as [H] after a /m/ pronoun but as [m] after other pronouns and all non-pronominal subjects (Miller 2005: p. 68). In Kagwé, similarly, an underlyingly /M/ verb surfaces as [H] after an underlyingly /M/ subject pronoun, but [m] elsewhere (Koopman & Sportiche 1982: p. 49). In Godié, /H/ verbs surface as [m] after another /H/ tone (Gratrix 1975: p. 113).

The prevalence of morpheme-specific tonal alternations at the juncture between subject and verb leads to questions about the prosodic and morphosyntactic nature of this boundary in Kru. We encourage further investigation of this boundary, and alternations at this boundary, in future work.

4.4 Intonation and positional alternations

Certain tonal alternations in Kru languages are sensitive to position within a clause or utterance (§4.4.1). It is possible that these alternations occur as a result of interactions with intonation. Intonation remains extremely understudied in Kru, though a few languages are reported to show interactions with downstep or downdrift (§4.4.2), and stress or accent systems (§4.4.3).

4.4.1 Positional alternations

In several languages, certain tonal alternations are attested at sentence or utterance boundaries. For instance, in Klao (Western), a /ML/ contour is phonetically lowered in sentence-initial position and before /L/ (Lightfoot 1974: p. 427). In Guéré (Western), in non-final position, /HL/ is realized as [H], /mL/ as [m] or [mM], and /L/ as [m] (Fisher 1976: p. 103–104). There are several final-specific tonal alternations, usually resulting in a final lowered or falling tone. In Klao, in sentence-final position, a final L is realized as falling after another L (Lightfoot 1974: p. 430). In Nyabwa, L is realized as falling before pause (Bentinck 1978: p. 25). In Wobé, disyllabic words with a H-tone melody are realized as [H.M] before pause (Bearth & Link 1980: p. 31). And in Gbadi, /H/ is realized as [M] word-finally (Charette 1982a: p. 35).

4.4.2 Downstep

Kru languages are rarely described as displaying downstep or downdrift (Marchese 1983: p. 69), despite the prevalence of downstep in African tone languages in general (cf. Snider 2026 [this volume]). Downstep is reported for only a few

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Kru languages, including the Western languages Klao, Wobé, and Grebo, and the Eastern language Vata. In Klao, for instance, a mid tone is produced with lower pitch when it follows a Low tone (Lightfoot 1974: p. 428). Similarly, in Grebo, an underlyingly /M/ tone is lowered to [m] after a Low tone (Newman 1986: p. 180). In Vata, successive High tones are pronounced with progressively lower pitch, such that a High tone at the end of a phrase may be phonetically lower than a Low tone at the beginning of a phrase (Vogler 1987: p. 26). We hypothesize that the lack of downstep in most Kru languages is attributable to the already crowded nature of the tone system due to the high number of contrastive tone heights.

4.4.3 Stress

Two Western languages, Grebo and Jabo, are reported to have contrastive stress or accent in addition to tone (Sapir 1931, Innes 1966, Luckau 1975). In Grebo all disyllabic nouns and verbs bear one strong stress, in which the stress-bearing syllable is pronounced with ‘considerably greater force and tenseness’, and the onset consonant of the stressed syllable is ‘noticeably lengthened’ (Innes 1966: p. 19). Minimal pairs differing only in stress are attested in Grebo, including /'sónó/ ‘sting’ (n.) versus /só'nó/ ‘smell’ (n.) (Innes 1966: p. 19). To our knowledge, a stress or accent system has not been reported for any other Kru language.

5 Segment-tone interactions

Across Kru languages, tones interact with adjacent segments in various ways. That is, the presence of certain consonants restricts or alters the immediately following tone melody, and, perhaps surprisingly, in a few cases, the presence of a certain tone level or melody affects the realization of the preceding segment.

5.1 Depressor consonant effects

Some Kru languages show classic depressor consonant effects, where the pitch of a tone is lowered immediately following a voiced obstruent (cf. Beavon-Ham & Pearce 2026 [this volume]). Implosives pattern with sonorants (and voiceless consonants) in this respect; they are not depressors. For example, in Dewoin (Western), a L tone is realized as level low after voiced obstruents, but with a slight fall after other consonants. H and m tones surface as rising after voiced obstruents, but as level after other consonants (Marchese 1983: p. 67–68, citing Welmers). In Bassa (Western), a Low tone following a so-called lenis consonant – from the set /p t c k kp f s x^w bɔ̌ dʲ m n ɲ w/ – is realized as a fall, while a High

tone following a so-called fortis consonant – from the set /b d ʃ g gb v z ɲ m h ɣ^w/ – is realized as a rise (Bertkau 1975: p. 22).

In Guéré (Western), a voiced obstruent onset contributes an initial Low tone, resulting in a surface rise when the underlying tone is /H/, /M/ or /m/ (Table 6).

Table 6: Guéré depressor consonant effects (Bôle-Richard 2006: p. 15)

Underlying tone	/H/	/M/	/m/	/L/
Non-depressor	kó ~ kò ‘kaolin’	kpá ‘bone’	sṣ̃ ‘snail’	tà ‘chew’
Depressor	bṽ ‘swell’	zḙ ‘bump’	zṣ̃ ‘day’	zṣ̣ ‘year’

In Wobé and Nidrou, languages closely related to Guéré, the voicing contrast has been lost for initial obstruents. The tonal effect of depressor consonants remains, however: syllables which historically had voiced obstruent onsets now have initial voiceless obstruents and Low or rising contour tones. For instance, the Guéré words ‘snail’ /sṣ̃/ and ‘day’ /zṣ̃/, realized on the surface as represented in Table 6, form a minimal pair in terms of voicing. In Wobé, the voicing contrast has been lost, and the two words have instead been reinterpreted as a tonal minimal pair: /sṣ̃/ ‘day’ versus /sṣ̃/ ‘snail’ (Bôle-Richard 2006: p. 24). Kagwé, an Eastern language, has also undergone a similar process of a loss of voicing contrast with a corresponding reanalysis as tonal contrast (Koopman & Sportiche 1982: p. 58).

In other languages, voiced obstruents block certain tone processes. In Gbadi (Eastern), for example, the first tone of the second noun (the head) of a noun compound surfaces as mid after H: [sìpú jū], cat child, ‘kitten’, cf. /jú/ ‘child’. However, this process is blocked if the second noun of a compound begins with a voiced obstruent: [jú dù] (*[jú dū]), child village, ‘child’s village’ (Charette 1982a: p. 100). Voiced obstruents also block tone spreading in Kpokolo (Goprou 2017: p. 212) and Vata (Koopman & Sportiche 1982: p. 58).

5.2 Tones affecting segments

Traditionally, the literature on tone has assumed, or even explicitly stated, that “consonants affect tone but tone does not affect consonants” (Hyman & Schuh 1974: p. 108). Perhaps surprisingly, then, there are a few attested instances of

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tones affecting the realization of segments in Kru languages. In Kuwaa, for example, /kp/ surfaces as [gb] (there is no contrast between the two in this language) in L-toned syllables (Thompson 1976: p. 42). Additionally, Kaye & Charette (1981) describe a case of segmental deletion rules sensitive to tonal identity in Lakota Dida, as spoken in the village of Srikpalilié. When a verb stem has a m or L tone, the tone and its vocalic host are deleted in the presence of a M object enclitic: [ɲ ló] ‘I carry him’ (/lā=ɔ/ ‘carry=3SG.ACC’).¹⁰ However, a M or H tone on the verb stem requires maintaining the stem vowel and tone: [ɲ lóó] ‘I call him’ (from /lā=ɔ/ ‘call=3SG.ACC’) (Kaye & Charette 1981: p. 83).

5.3 Other restrictions on segment/tone co-occurrence

Other co-occurrence between certain tone melodies and certain segments are also described for Kru languages. In Klao, all syllables with the falling contour HL begin with ‘unvoiced velar stops’, while no syllables with the contour ML begin with an unvoiced velar stop (Lightfoot 1974: p. 434). In Wobé, [kp] appears only with non-low tones, while [gb] appears only with low tone and contours including a low tone (Bearth & Link 1980: p. 188). In Tépo, the velar stops [k] and [g] are in complementary distribution, with [k] appearing with high tones (level and high rising) and [g] appearing with low tones (mid, low and low rising) (Thalmann 1980: p. 7). In Nyabwa, [+high, +ATR] vowels rarely carry M tone, while [+high, -ATR] vowels rarely carry H tone (Bentinck 1978: p. 25). In Kpokolo, on verbs of the shape CVV, tone melodies are limited by the sonority of the onset consonant. Sonorant onsets (including /b/) can only co-occur with one of three possible melodies: /H/, /m/, or /mL/. Obstruent onsets, on the other hand, can co-occur with a larger set of melodies: /H/, /m/, /mL/, /Hm/, /HL/, /M/, or /Mm/ (Vahoua 2018: p. 34).

6 Implications

The large number of tone heights and distinct contours or melodies found in Kru languages is typologically unusual, and the tonal processes and interactions found in Kru languages bear on tonal typology. Additionally, the types of tone patterns and alternations found in Kru languages influence several debates in formal phonology, specifically having to do with representations of tone. This section discusses these implications, with reference to the specific tone patterns reported above.

¹⁰Surface [mM] contours do otherwise occur in Lakota Dida of Srikpalilié.

This section focuses on the implications of Kru tone for tonal representations; however, Kru tone data has also been leveraged to bear on formal debates such as whether morphophonology is best analyzed as involving items versus processes (Sande 2022a, 2018) and the size and nature of tonal domains (Sande 2018, 2019).

6.1 Tonal representations and processes

6.1.1 Tone features

There is a longstanding debate in the literature about whether tone – like segments – should be represented using binary distinctive features. Various sets of tonal features have been proposed over the years (Yip 1980, Clements 1983, Pulleyblank 1986, Snider 1999, Bao 1999). Hyman (2011) and Clements et al. (2011), though, conclude that tone should not be represented featurally, and there has instead been robust adoption of alternative representations such as tonal autosegments (Goldsmith 1976, Leben 1973). However, some recent work has again suggested that tones, at least in some languages, should be represented featurally (McPherson 2017, Meyase 2021). The tone feature system proposed by Yip (1980) and Pulleyblank (1986) is given in Table 28.

(28) Proposed features for 4-tone systems

	H	M	m	L
Upper	+	+	-	-
High/Raised	+	-	+	-

Most Kru languages have four underlying tone heights, and some have five distinct surface tone heights, posing an optimal testing ground for whether sets of (two) binary tone features can account for a range of tonal contrasts and alternations. Certain tonal patterns across Kru, like Kpokolo aspect marking, may be best handled using a representational tool like features. In Kpokolo (Eastern), there are six possible tone melodies on CV verbs: H, M, m, L, rising (Lm) and falling (mL) (Vahoua 2018). All tone melodies surface faithfully in the imperative (Table 7).¹¹ In imperfective and perfective aspects, however, verbs split into two classes: 1) H, M and rising, and 2) m, L, and falling. Class 1 tones surface with a ML contour in the perfective and as M in the imperfective. Class 2 tones surface as L in the perfective and as m in the imperfective.

¹¹While verbs of other shapes (e.g. CVV, CVCV) in Kpokolo may display additional melodies, all verbs pattern as described in Table 7, based on the stem-final tone.

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Table 7: Kpokolo tone classes (Vahoua 2018: 140–141, 153–154)

Class	1			2		
Underlying tone	/H/	/M/	Rising	/m/	/L/	Falling
Imperative	H	M	Lm	m	L	mL
perfective	ML	ML	ML	L	L	L
Imperfective	M	M	M	m	m	m

The behavior of the rising contour tone is particularly difficult for an account which treats the [Lm] rising tone as a combination of autosegmental /L/ + /m/: for instance, /m/ verbs surface as [m] in the imperfective, but /Lm/ verbs surface as [M].¹² On the other hand, the Kpokolo alternations could straightforwardly be accounted for with features like those in (29), where, in the imperfective, tones with the feature [+upper] surface as [M], and those with the feature [-upper] surface as [m].

(29) Tonal features to account for Kpokolo tone classes

	H	M	R	m	L	F
Upper	+	+	+	-	-	-
Raised	+	-	∅	+	-	∅
Contour	-	-	+	-	-	+

On the other hand, other tonal alternations across Kru appear to be scalar or step-wise in nature (see 4 and 17), and scalar or step-wise processes, especially across four levels, are difficult to account for in a principled way with a single set of features. Sande (2018) discusses why two binary tone features fail to account for the scalar tone shift in imperfective contexts in Guébie (17), where H→M→m→L. Namely, if a feature system like the one proposed by Yip (1980) and Pulleyblank (1986) in Table 28 is adopted, then the M and m tones share no features in common, so the change in a single feature cannot result in a change from M to m. One would have to have to propose a completely distinct set of rules, operations, or constraints to get from M to m than from H to M and m to L. Alternatively, adopting (scalar) autosegmental representations can more straightforwardly account for the pattern.

¹²[LM] does not appear to be a possible surface contour in Kpokolo, so an alternative analysis could be to consider [Lm] to be the phonetic realization of an underlying /LM/ contour.

Similarly, recall the alternation shown in (13-16), where /H/ and /M/ surface as [M] in the imperfective but /m/ and /L/ verbs surface as [m] in the imperfective in several Eastern Kru languages. Because m and M tones share no features in a binary feature system, there is no way in a featural representation system to unify the shift of /H/ and /M/ to [M] and the shift of /m/ and /L/ to [m] under a single rule or process, despite the fact that the two shifts are describable in the same way: they involve movement towards the middle of the tone system. This alternation could, instead, easily be modeled using scalar representations and constraints.

6.1.2 Contours as sequences of levels versus tonemic primes

Tone systems are often categorized into two types based on the behavior of contour tones. On one hand, *register* or *level* tone languages are the type most often found in Africa, where H and L tones are primitives and can combine to form rising and falling sequences. On the other hand, *contour* tone systems, often found in East and Southeast Asian languages, have combinations of rises and falls that pattern as complex units rather than as sequences of separable parts (Pike 1948, Yip 1989).

Some Kru languages demonstrably have contour tones that are best analyzed as sequences of level tones (see, for example, the Guébie data in 18); this is typologically unsurprising given that this type of tone system is common in African languages. However, at least in some Kru languages, tone systems seem to be best analyzed as contour-type systems, where contour tones are not decomposable into sequences of level tones, but rather pattern as a unit. In Grebo (Western), for instance, two types of disyllabic verbs display distinct tonal behavior. Some disyllabic verbs, like /jídí/ ‘steal’ behave as ‘monotonal’, while other disyllabic verbs, like /kwidē/ ‘run’ behave as ‘ditonal’ (Newman 1986: p. 184). The difference between these two types of verb is visible through interactions with adjacent material. For instance, one form of the first singular subject pronoun in Grebo is a floating H tone which replaces the first tone in the lexical verb melody (see 5). When a monosyllabic verb has an underlying contour tone, the entire contour is replaced (30a). A disyllabic verb like ‘steal’ patterns as though it has a single contour tone: both syllables of the verb are raised to High (30b). However, for a disyllabic verb like ‘run’, only the first syllable is raised to High (30c), patterning as if there are two separate level tones.

(30) Verb tone alternations after H-tone 1SG in Grebo (Newman 1986: p. 184)

a. /H/ + /lǎ/ → [lǎ] ‘1sg.kill’

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- b. /H/ + /jidí/ → [jidí] ‘1sg.steal’
- c. /H/ + /kwidē/ → [kwidē] ‘1sg.run’, *[kwidé]

Newman interprets the Grebo data, supported by an array of other tonal phenomena in the language, as evidence that a verb like ‘steal’ is underlyingly associated with a single rising contour toneme, which he labels /R/ for rising, realized across two syllables on the surface¹³, whereas a verb like ‘run’ is underlyingly associated with two separate tonemes – /M/ on the first syllable and /m/ on the second syllable. As such, both inseparable contours and contours which are sequences of two tones appear to co-exist within a single language.

Wobé (Western) is well-known for its high number of contrasting tonal melodies: 14 melodies in total, which includes 4 level tones and 10 contour tones (**Bearth & Link 1980**, **Singler 1984**). There is a debate in the literature as to whether contours in Wobé are best analyzed as inseparable tonemic primes or separable sequences of levels. **Bearth & Link (1980)**, motivated by differences between lexical tone melodies before pause, argue that contours are best treated as primes in Wobé. **Singler (1984)**, however, calls their analysis into question, arguing that Wobé contour tones are in fact decomposable into sequences of level tones. Further analytical work is needed on individual Kru languages to determine whether the trend across the family is that contours are best analyzed as tonemic primes versus sequences of level tones.

Just as **Hyman (2010)** describes Kuki-Thaadow, a Kuku-Chin (Tibeto-Burman) language of India and Myanmar, as “an African tone system in Southeast Asia” based on the behaviors of its contour tones as sequences of levels rather than tonemic primes, Kru languages like Grebo, Kpokolo and Wobé could be thought of as Southeast Asian-type tone systems in Africa. Perhaps one conclusion to draw from these facts is that these two categories of tonal systems are not limited to particular areas of the world, so tone systems are not best categorized based on the location in which the language is spoken, but by a set of linguistic properties. Particularly notable is the fact that tonemic contours and contours separable into level tones seem to be able to co-exist, at least in Grebo.

6.1.3 Summary of implications for tonal representations

In this section we have seen that the best abstract representation of tone depends on the nature of the tone system in question. Even tone systems in different languages within the same family, or individual tonal melodies within the same language, might be best analyzed using distinct tonal representational tools. While

¹³Similar phenomena, in which multiple surface syllables appear to be underlyingly associated with a single contour tone, are also reported for Kpokolo (**Vahoua 2018**) and Godié (**Gratrix 1975**).

some Kru tone systems may be best captured with tonal features, others are difficult or impossible to capture in featural approaches. Similarly, contour tones in some Kru languages may be best analyzed as tonemic primes, while contours in other Kru languages are clearly sequences of separable level tones, and Grebo may show the need for both representations of contours to co-exist in the same language.

7 Conclusions

Here, we have described the tonal inventory, basic tonal alternations, and grammatical tone properties in Kru languages. We have pointed out that Kru languages tend to have complex tone systems, with most showing evidence for four contrastive tone heights as well as several contrastive contour or multi-tone melodies. The complexity and nature of Kru tone systems reinforces (à la Hyman 2010, see the discussion in §6.1.2) that African tone languages are not always best analyzed as level-tone systems. Rather, there is greater diversity among the tonal properties of languages in a given area of the world than previously represented in most tonal typologies.

Kru tone phenomena have influenced what we know about the typology and representation of tone languages. We have seen clear examples of tones in Kru languages influencing segmental patterns, as described in §5.2, despite past claims that tones do not affect segments (Hyman & Schuh 1974). Additionally, several Kru languages have contrastive rising tones but not contrastive falling tones, which is surprising from a typological point of view because falling tones are more widely attested (Yip 2002: p. 27–30), see §2.2. Furthermore, as discussed in §6.1, different tonal representations are needed to formally best capture the tone systems of different Kru languages, suggesting that there is no single best way to represent tone across tone languages.

While Kru languages have taught us much about tonal typology and formal representation, there is still a lot of work to be done. Future work on tone correspondences across Kru languages will shed light on tonal reconstruction in Kru. We also suggest future work on intonation and its interactions within tone in Kru languages, which have been greatly understudied to this point. Additionally, we suggest a closer look at the nature of the boundary between subject and verb in Kru languages, where many grammatical tone and morpheme-specific tonal phenomena apply.

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Appendix

Abbreviations

		IRR	irrealis
		NEG	negative
		NOM	nominative
1	first person	PART	particle
2	second person	PL	plural
3	third person	PFV	perfective
ACC	accusative	POSS	possessive
AGT	agentive	PST	past
FOC	focus	REL	relative
IPFV	imperfective	SCMPL	sentence-completion particle
		SG	singular

Acknowledgements

Redacted

Contributions

Redacted

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Table 8: Kru varieties included in our survey with list of sources consulted

Variety	Source(s)
Bassa	Hobley (1964), Bertkau (1975), Luckau (1975)
Dewoin	Marchese (1983)
Guéré	Paradis (1983), Bôle-Richard (2006)
Nidrou	Bôle-Richard (2006)
Nyabwa	Bentinck (1978)
Krahn	Sauder (2000)
Grebo	Innes (1966), Luckau (1975), Newman (1986)
Jabo	Sapir (1931), Luckau (1975)
Klao	Lightfoot (1974)
Wobé	Bearth & Link (1978, 1980), Egner (1989), Bôle-Richard (2006)
Tépo	Thalmann (1980, 2018)
Plapo	Thalmann (2018)
Piè	Thalmann (2018)
Kouya	Saunders (2009)
Godié	Gratrix (1975), Marchese (1983), Egner (2015)
Dadjriwalé	Gode (2006)
Daloa Bété	Zogbo (1975), Zogbo (1979), Zogbo (2004)
Guibéroua Bété	Werle & Gbalehi (1976)
Gbadi	Bossé (1982), Charette (1982a)
Kpokolo	Goprou (2017), Vahoua (2018)
Lakota Dida	Kaye & Charette (1981), Guéhoun (1993)
Abou Dida	Miller (2005)
Yocoboué Dida	Masson (1992)
Zaroko Dida	Doua et al. (1979)
Guébie	Sande (2017, 2018, 2022b), original fieldwork
Gabɔgbɔ	Gnahore (2006)
Néyo	Grah (1983)
Vata	Vogler (1987), Kaye (1982), Koopman & Sportiche (1982)
Gakpa	Kaye (1982), Charette (1982b)
Gbeiwi	Kaye (1982), Charette (1982b), Tougbo (1998)
Kagwé	Koopman & Sportiche (1982), Charette (1982b)
Laɓɔ	Charette (1982b)
Ménéhiri	Kaye (1982), Charette (1982b)
Kuwaa	Ingemann & Thompson (1974), Thompson (1976)
Aizi	Hérault (1971)
Seme	Mous & Traore (2000), Toews (2015)

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Chapter 18

Tone in Northwestern Bantu

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Bantu tone systems are known for their range of tonal processes, particularly down-step, floating tones, and high-tone-spreading (HTS), and the prominence tone takes in encoding grammatical functions. One of Bantu's five major genealogical groups is Northwestern Bantu (NWB). This Bantu zone is highly heterogeneous, linguistically diverse, and cannot be defined as a uniform area of shared (tonal) features. Nevertheless, it stands out for the effects that final vowel loss and prosodic size restrictions have on the functional load of tone in an important subset of its clades (A31-40-50-60, A40-60-70, and A80-90, and to a certain extent A10-20-30). Languages in all Bantu A clades (and analytic West-Coastal Bantu languages such as Teke-Kukuya) use tone as sole exponent in marking tense-aspect-mood-polarity inflectional categories, whereas tone in the B clades of NWB only occurs as a co-exponent with a segmental morpheme. Tone in NWB also marks clause types, immediate-after-verb elements, and distinguishes phrase-final from phrase-medial verbs.¹

1 Introduction

Narrow Bantu languages constitute the largest subgroup of Niger-Congo, containing about 550 languages (Hammarstöm 2019, Philippson & Grollemund 2019), with the most recent updated inventory given by Hammarstöm (2019). Classifying this large family internally has been challenging for a long time, due to a lack of (sufficient) descriptive data. Therefore, Bantuists have traditionally used an areal-referential system for internal classification, originally based on Guthrie (1971). Maho (2009) updated and expanded this system to sixteen zones marked

¹I would like to thank the editors and two anonymous reviewers for their valuable feedback.

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by capital letters, each of which are divided into subgroups that are labeled by decimals. Individual languages thus receive a letter plus number code. For instance, Makaa has the label A83. This areal-referential system, however, often does not coincide with genealogical relations.

Recent large-scale phylogenetic analyses divide Bantu languages into four (Bostoen & Grégoire 2007)² or five (Grollemund et al. 2015, Koile et al. 2022) major genealogical groups, with Northwestern Bantu (NWB) as one of them. The authors disagree, however, on the details of exact clade groupings, number of splits, and proposed time depths. This also holds for the NWB group. Grollemund et al. (2015: 13297) propose thirteen sub-clades that form seven clusters. A simplified version is shown in Figure 2. Their NWB group notably includes the Jarawan clade with languages spoken in northeastern Nigeria, which are traditionally viewed as Bantoid, but not Narrow Bantu (cf. Blench 2026 [this volume]). In line with this traditional view, Koile et al. (2022) do not include Jarawan within NWB. At the same time, they also exclude Grollemund et al.'s (2015) clades B20 and B10-30 from NWB and group them with their “west-coastal” clade instead.

The differences in genealogical groupings point to a larger question about the typological cohesion of NWB languages, beyond lexicostatistic relations. Bantuists have long had a sense that NWB, also called “Forest Bantu” in the literature, is structurally different from the rest of Bantu, often referred to as “Savannah Bantu”. A particular point of interest concerns NWB’s relative older age, as this area around the Nigeria-Cameroon borderland constitutes the Bantu homeland from which the Bantu migration started around 5,000 years ago (Koile et al. 2022). Pakendorf et al. (2011: 54) characterize the NWB area as the Bantu zone with the highest linguistic diversity, while the other Bantu areas are thought of as more homogeneous. This is certainly also linked to NWB’s geographic location in contact with several other subgroups of the larger Benue-Congo family.

Whether NWB is diachronically more conservative or innovative is controversially discussed in the literature. Nurse & Philippson (2003: 5) note that

“[a]lthough they [NWB languages] are geographically closest to what we think was the Bantu cradleland, they are not thereby linguistically most similar to what we assume were the shapes and features of early Bantu. In fact, the opposite is true and we think that languages in eastern and southern Africa are in general more conservative than those in the northwest

²Bostoen & Grégoire’s (2007) classification is used by Hammarström et al. (2025) in *Glottolog* and contains the major groups “Bantu A-B10-B20-B30” (NWB in this paper), “Central-Western Bantu”, “East Bantu”, and “Mbam-Bubi”.

[...]. This is because the northwest languages form the eastern end of a geographical and typological area that embraces adjacent languages in West Africa, including many non-Bantu Niger-Congo languages.”

Güldemann (2011: 122–123) disagrees with this proposition of historical development, especially with respect to the morphological complexity of verb stems, assuming that the complex verb stems in Savannah Bantu are an innovation. He highlights, however, many features in NWB that are shared with languages of the “Macro Sudan Belt” and not found in the rest of Bantu, such as labial-velar consonants, possibly implosives (although they are also found in eastern Bantu languages), and the more analytic verb structure in the northwest.

It is important to note, however, that “for lack of indisputable innovations that delineate Bantu from these closest [Bantoid] relatives” (Pakendorf et al. 2011: 54), it is difficult to delimit NWB from non-Narrow Bantu languages of the area. Also within Bantu, it is not possible to define NWB as a uniform area of shared features that are found in all NWB languages but not outside of NWB. This is also true for tonal systems, as we will see. For instance, prosodic size restriction is a feature thought of as typical for the NWB area. It is, however, also found well beyond the narrow NWB zone of Grollemund et al. (2015) and Koile et al. (2022), and, at the same time, some NWB languages allow for larger prosodic units similar to Savannah Bantu. Due to such typological heterogeneity, the extension of NWB is often taken as wider with zones A and B at its core, but also including languages of zones C, and parts of D and H (e.g. Nurse & Philippson 2003, Nurse 2008, Di Garbo & Verkerk 2022).

In line with the phylogenetic studies and for the practical reason to make the scope of this study on tone more manageable, however, I take NWB in the narrow sense, only comprising the languages of zone A and B10–30 (Grollemund et al. 2015, Koile et al. 2022).³ These can be found under Glottocode *bant1295* for the group “Bantu A-B10-B20-B30” (Hammarström et al. 2025). This group comprises 75 languages, whose geographic distribution is represented in Figure 1. Figure 2 shows the languages and their genealogical groupings that I use as the core sample for this paper, while I also make more sporadic references to West-Coastal and East Bantu languages. The choice of sample languages was guided by considerations of balance between different subgroups and the availability of literature.

³Grollemund et al. (2015) show that also Jarawan and Grassfields Bantu are part of this language family. In accordance with Glottolog’s classification (Hammarström et al. 2025), I restrict myself to the Bantu proper languages, excluding Jarawan and Grassfields. The Grassfields languages and their tone systems are presented in Hamm et al. (2026 [this volume]).

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Generally, the B languages are underrepresented for a lack of description, particularly pertaining to information on the verbal complex and its tones.⁴

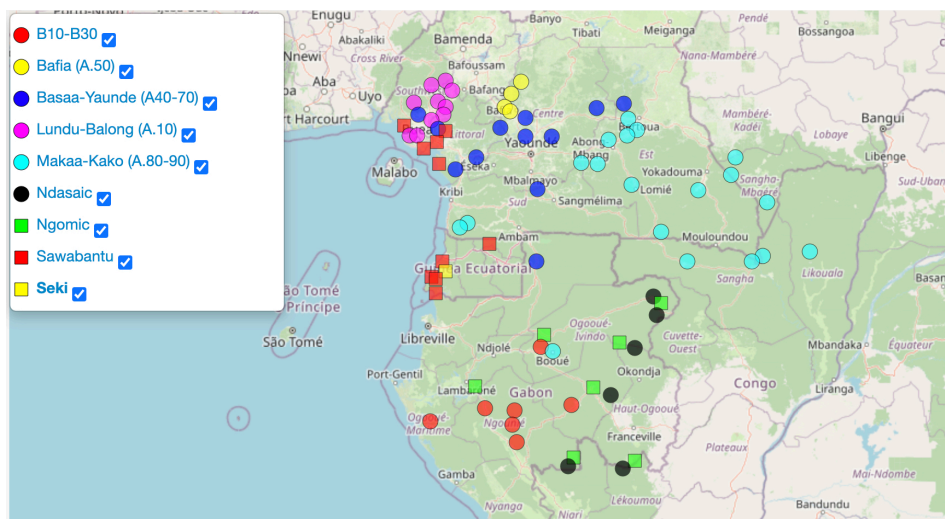


Figure 1: Map of the Northwestern Bantu languages
by *Glottolog*, Glottocode: bant1295 (Hammarström et al. 2025)

With respect to tone, Nurse (2008: 48) lists three features that he finds remarkable for NW (or Forest) Bantu, namely the widespread presence of floating tones, which reflects the loss of vowels, metatony, and the role played by tone in the encoding of tense and aspect categories (cf. Rolle et al. 2026 [this volume]). All three tonal features are significantly or, for metatony and tone marking tense and aspect, even exclusively found in the verb phrase. While tone as a lexical and phonological property is found in all areas of NWB languages, it seems most “active”, i.e. subject to change, in the verbal domain. Therefore, this paper has a heavy focus on the grammatical and syntactic functions of tone in the verb phrase. This paper is structured as following: In Section 2, I provide an overview of the tonal inventories and tonal processes found in NWB. This serves as preliminaries to the main part of this paper in Section 3, which outlines grammatical tones in the verb phrase, including tone in the encoding of tense, aspect, mood, and polarity (TAMP) categories, tone and its syntactic function in distinguishing clause types, metatony, and tone in the immediate-after-verb (IAV) position. I conclude in Section 4.

⁴The languages with an asterisk in Figure 2 are not represented in Grollemund et al.’s (2015) database. I classified them based on their affiliation to decimal groups, i.e. Balon with the other A10 languages and Gyeli with the other A80 languages.

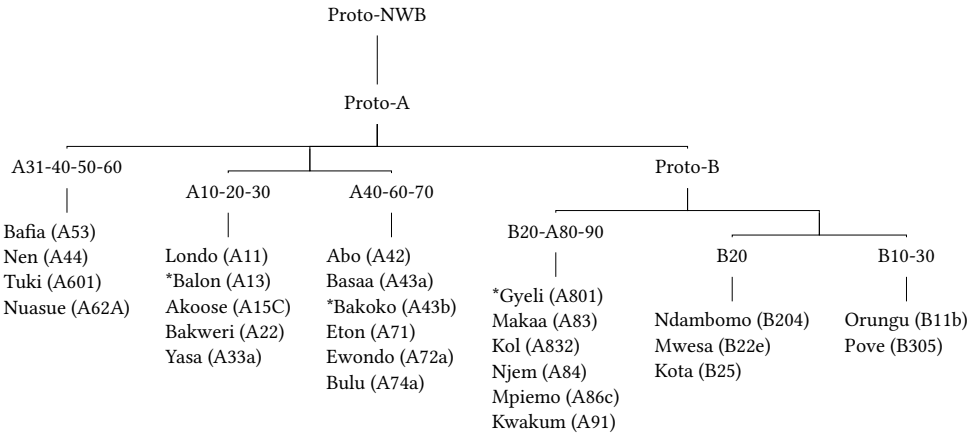


Figure 2: Genealogical groups of NWB languages, based on [Grolle-mund et al.'s \(2015\)](#) classification and [Grimm's \(2025b\)](#) diachronic dimension

2 Tone inventories and processes

Are tonal inventories and processes different in NWB from the rest of Bantu? To a large extent, the answer is no. Basic features of NWB tone systems, such as the opposition of two tone levels, the occurrence of contour and floating tones, and common processes such as downstep (cf. [Snider 2026 \[this volume\]](#)) and tone spreading (cf. [Smolders 2026 \[this volume\]](#)), are well attested in all of Bantu. NWB seems special, however, as many (but not all) languages make more than two phonemic distinctions, as shown in Table 1. The table gives an impression of tonal inventories across NWB, sorted by language subgroups and interpreting the sources as to whether tones and their patterns are thought of as phonemic or phonetic. In addition to H and L, NWB tonal inventories may have phonemic contour tones, downstep H, and/or toneless TBUs. Even the fact that, to my knowledge, every NWB language phonemically distinguishes H from L tones, is noteworthy, as this is not true for many Eastern Bantu languages, which may have privative instead of equipollent systems of phonemic H vs. toneless or, in rarer cases, L vs. toneless ([Hyman 2001a](#)). The exact distribution and geographic extension of equipollent vs. privative tone systems are, however, not well enough studied to tell whether there is a clear boundary.

Nine out of the represented eighteen languages are described with two phonemic tones: H vs. L. The other half contains additional phonemic tones that include a mid tone, which might be equivalent to a downstep H across languages of this

Table 1: Tone inventories in some NWB languages

Language ^a	Phonemic tones	Phonetic tones	Status of contours
Nen (A44)	H, L	H, L, downstep H	n/a
Bafia (A53)	H, L	H, L, M	sequence of H + L
Tuki (A601)	H, L, HL, LH	H, L, HL, LH and M ~ downstep H	phonemic
Balon (A13)	H, L	H, L, HL, LH, M	sequence of H + L
Bakweri (A22)	H, L, downstep H, ∅	H, L, HL, LH, M	sequence of H + L
Yasa (A33c)	H, L	H, L, HL, LH, super- high ^b	sequence H + L
Basaa (A43a)	H, L, HL, LH	H, L, HL, LH	phonemic
Eton (A71)	H, L, dissimilating H	H, L, HL, LH, down- step H	sequences of H + L
Bulu (A74a)	H, L	H, L, M, downstep H, upstep L	no information
Njem (A84)	H, L	H, L, downstep and up- step H, upstep L, HL, LH	sequence of H + L
Mpiemo (A86c)	H, L	H, L, M, HL, LH	sequence of H + L
Kwakum (A91)	H, L, ∅	H, L, downstep H, L°, HL, LH	sequence of H + L
Seki (B21)	H, L, HL	H, L, HL	phonemic
Ndambomo (B204)	H, L	H, L, HL, LH	sequence of H + L
Mwesa (B22e)	H, L, M	H, L, M, downstep H, HL, LH	phonetic
Kota (B25)	H, L	H, L	n/a
Orungu (B11b)	no information	H, L, downstep H, HL, downstep H-L	no information
Pove (B305)	H, L, downstep H	H, L, downstep H, ex- tra L, HL, LH	sequence of H + L

^aSources are as follows: Nen (Mous 2003: 286–288), Bafia (Guarisma 2003: 310–311), Tuki (Kongne Welaze 2004: 24–25), Balon (Kouoh Mboundja 2004: 91ff), Bakweri (Hawkinson 1986: 37ff), Yasa (Brown 2021: 38ff), Basaa (Hyman 2003: 260), Eton (Van de Velde 2008), Bulu (Ndi 2001: 9–10), Njem (Akumbu 2006: 53–54), Mpiemo (Festen 2008: 7–8), Kwakum (Njantcho Kouagang 2018: 51ff), Seki (Koumbi Mabiti 2025: 66–67), Ndambomo (Koumbi Mabiti 2025: 60–61), Mwesa (Mokrani 2016: 591), Kota (Magnana Ekoukou 2015: 95), Orungu (Ambourou 2007: 52ff), Pove (Mickala-Manfoumbi 1994: 115ff)

^bBrown (2021: 39) observes that “[o]nly falling contour tones are attested on short vowels. Rising contours are only found in vowel sequences.”

area, one or two contour tones, and rarely toneless TBUs. Phonetically, however, NWB languages have a much larger tonal repertoire. All languages but Kota are described as exhibiting other tones in addition to H and L, including contours and mid/downstep and in some cases also upstep L (Njem), super-high (Yasa), and extra L (Pove). Table 1 further illustrates that ten out of fourteen languages with contour tones, the contours are analyzed as sequences of level tones. Only in two cases (Tuki and Basaa) are they taken as phonemic, while in Seki, they have phonetic status, but it is not specified if that entails an analysis as sequence of level tones.

While contour and floating tones are not unique to NWB, their nature is special, as their development is often linked to final vowel loss and a resulting reduction in prosodic stem size. Both will become important also for understanding patterns of grammatical tone in the verbal complex discussed in Section 3. Patterns of final vowel loss and prosodic size constraints are, however, not homogeneous across NWB nor is the phenomenon restricted to the narrow NWB area, as shown in Table 2. Final vowel loss phenomena are represented by showing whether a language allows for consonant-final stems and whether a stem-final vowel has an inflectional function, as further explained in Table 4. If indicated as “no” in Table 2, stem-final vowels are lexical or absent altogether in the case of consonant-final stems. Prosodic size restrictions are reflected by the maximum number of verb extensions that can be added to a verb radical (see Section 3 for more details on the Bantu verb structure) and maximum number of syllables a verb stem can have. It should be noted, however, that the maximum number usually involves derivation through extension suffixes and thus does not reflect the most frequent pattern in verb stem length. For instance, while trisyllabic verb stems are attested in Gyeli, the most frequent type is the disyllabic stem, which accounts for 50% of the verbs, followed by monosyllabic stems (Grimm 2021: 95).⁵

Table 2 shows that NWB languages differ with respect to their degree of analytic vs. agglutinative morphological patterns.⁶ While the clades A31-40-50-60, A40-60-70, and A80-90 are more analytic with stem restrictions of maximally three syllables, clades B20 and B10-30 are clearly more agglutinative with up to

⁵It would be preferable to provide the most common stem size for each language. However, this information is rarely provided in descriptions and, similarly, authors do not always clearly state the maximal number of extension combinations and stem syllables. Numbers with a question mark in Table 2 indicate a lack of information and/or my uncertain interpretation of the data given in the literature.

⁶Derivation suffixes in the stem are taken as proxies for an analytic vs. agglutinative morphological type here, although I am aware that the discussion should include the status of the inflectional material in the verbal pre-stem. This, however, is beyond the scope of this paper.

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Table 2: Final vowel loss and prosodic size restriction in the wider NWB area

Language ^a ,	C-final stems	inflectional FV	Extensions	Verb stem syllables
North-Western Bantu				
Nen (A44)	yes	no	2	3
Bafia (A53)	yes	no	2	3
Tuki (A601)	no	no	1?	3
Balon (A13)	yes	no	1	3
Londo (A11)	no	yes	2–3	5
Bakweri (A22)	no	no	3	5
Yasa (A33c)	no	no	2	6
Basaa (A43a)	yes	no	2	3
Eton (A71)	yes	no	1	3
Bulu (A74a)	yes	no	1–2	3
Gyeli (801)	no	no	1–2	3
Njem (A84)	yes	no	1	3
Kwakum (A91)	yes	no	1	3
Mwesa (B22e)	no	yes?	2–3?	4
Kota (B25)	no	yes	?	4?
Orungu (B11b)	no	yes	3	4
Pove (B305)	no	yes	4	5
West-Coastal Bantu				
Teke-Kukuya (B77a)	no	yes	0	3
Tiene (B81)	no	(yes) ^b	1?	3
Sakata (C34)	no	yes	?	2
Bushong (C83)	yes	no	1	1–2

^aSources are as follows: Nen ([Mous 2003](#)), Bafia ([Guarisma 2003](#)), Tuki ([Kongne Welaze 2004](#)), Balon ([Kouoh Mboundja 2004](#)), Londo ([Kuperus 1985](#)), Bakweri ([Hawkinson 1986](#)), Yasa ([Brown 2021](#)), Basaa ([Hyman 2003](#)), Eton ([Van de Velde 2008](#)), Bulu ([Ndi 2001](#)), Gyeli ([Grimm 2021](#)), Njem ([Akumbu 2006](#)), Mpiemo ([Festen 2008](#)), Kwakum ([Njantcho Kouagang 2018](#)), Mwesa ([Mokrani 2016](#)), Kota ([Magnana Ekoukou 2015](#)), Orungu ([Ambouroue 2007](#)), Pove ([Mickala-Manfoumbi 1994](#)), Teke-Kukuya ([Li 2024](#)), Tiene ([Ellington 1977](#)), Sakata ([Ikamba 1987](#)), Bushong ([Vansina 1959](#))

^bAn inflectional final vowel -e in the past tense forms only applies to certain verb classes; some verbs end in a lexical vowel ([Ellington 1977](#): 73–76, 83).

four extension suffixes and 4-5 stem syllables. Clade A10-20-30 languages are predominantly agglutinative, but Balon (A13) clusters with the analytic languages. At the same time, analytic structures are also found in some languages of the West-Coastal Bantu area. These are, however, more similar to the B languages of NWB in that most of them seem to have an inflectional final vowel, whereas most Bantu A languages have lexical stem-final vowels (or consonant-final stems).

Having outlined the syllabic and morphological heterogeneity as an important influence on tone patterns in NWB, I now turn to more details on tonal features, including contour tones and toneless TBUs (Section 2.1),⁷ the distribution of tonal patterns by part-of-speech (Section 2.2), and tonal operations such as downstep, floating tones, and tone spreading (Section 2.3).

2.1 Contour tones and toneless TBUs

NWB is known to exhibit contour tones (see also Hyman 2001b and Hyman 2021: 49). Falling HL seem to be more frequent than rising LH tones, as stated for languages such as Balon (Kouoh Mboundja 2004: 103), Shiwa (Dougère 2007: 79–85), and Gyeli (Grimm 2021: 101). In Bekwel (Cheucle 2014: 350) and Seki (Koumbi Mabiti 2025: 66–67), only the falling contour occurs, but not the rising.

While surface contour tones are widely recognized in the area, their underlying phonological status as either one complex autosegment with the features [rising] and [falling] or multiple tonological entities linked to the same tone-bearing-unit is often not explicitly demonstrated in the descriptive literature. In most cases, contour tones are analyzed as sequences of level tones and only rarely as “real” contours. The decision between the two scenarios hinges upon what is considered to be the TBU (mora or syllable), the distributional patterns and minimal pairs found in the language, and considerations about the historical origins of contour tones. In Basaa (Hyman 2003: 260) and Gyeli (Grimm 2021: 101), for instance, contours are analyzed as phonemic units based on their minimal pairs contrasting with level tones.⁸ In comparison, Akumbu (2006: 55) shares Yip’s (2002: 133) observation that contour tones in African languages are fairly rare and often comprised of sequences of level tones in his analysis of Njem. He postulates that the syllable is the TBU, as light mono-moraic and heavy bi-moraic syllables can bear a similar number of tones.⁹ In Eton, Van de Velde (2008: 57–58)

⁷See also Hyman (2017a) for a Bantu-wide overview.

⁸For a detailed account on how to analytically distinguish contour pitch as sequences of level tones from phonemic units, see (Yip 2002: 50–52) and (Hyman 2011b: 208–209) for a general explanation, and (Grimm 2022: 477) for Gyeli.

⁹This, however, might be an argument for real contour tones, as one would want to propose that level tones associate with moras and form HL and/or LH sequences in bi-moraic syllables.

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views contour tones as sequences of level tones on prominent syllables. Everywhere else, they are the result of tonal sandhi.

The origins of contour tones for African languages more broadly are usually ascribed to (final) vowel loss with a surviving tone, as noted by Yip (1989: 151). Contours, however, did not yet exist in Proto-Bantu, which is reconstructed as opposing only H and L tones (Meeussen 1967: 84 and Hyman 2017a: 172). Diachronically, many NWB languages reduced bisyllabic to monosyllabic stems and a non-identical tone of the second syllable survived as a contour tone on the first syllable, whereas identical tones fused to the same tone. Cheucle (2014: 354, 506) concludes that the contour tones in monosyllables found in her sample of eight A80 languages correspond to the same tonal patterns found in disyllabic stems of Proto-Bantu.¹⁰ This is illustrated in Table 3 for some A80 languages. It should also be noted that final vowel loss does not automatically result in contour tones across all NWB languages. As Hyman (2003: 260) points out for Basaa, a final vowel is lost together with its tone and the remaining first vowel tonally corresponds to the level tone reconstructed for Proto-Bantu. This is also true for the Bekol examples in Table 3, although Bekol, just like Basaa, generally has contour tones (Cheucle 2014: 224).

Table 3: Historical development of contours in some A80 languages following final vowel loss (Cheucle 2014: 502, 505)

Proto-Bantu	Bekol (A832)	Konzime (A842)	Makaa (A83)	Translation
*bóngó	bwónɔ̃	é-bóó	bwóŋ	‘knee’
*gàngà	nkàŋ	nkàà	á-nkò	‘healer’
*bókò	mbwó	mbô	mbwô	‘hand’
*gòdó	nkù	nkǔ	nkùú	‘pig’

This historical development has repercussions on the nature and distribution of contour tones, which makes them look different from contours in the rest of Bantu: First, contours in the wider NW area frequently occur word-finally and especially on monosyllabic stems,¹¹ although not exclusively. In Tuki (A601),

¹⁰In terms of time depth, Cheucle (2014: 506) suggests that the emergence of contour tones was likely initiated at the Proto-Bantu-A80 phase for the languages in this group. I am not aware of any other literature that addresses the question of when contour tones may have developed after Proto-Bantu in NWB.

¹¹This phenomenon is found beyond the narrow NWB area. For instance, falling and rising tones occur in Ngwi (West-Coastal Bantu B861, DRC) on heavy CVC monosyllabic stems, whose vowels are phonetically lengthened to bimoraic syllables (Pacchiarotti & Bostoen 2024).

rising and falling contour tones are attested in monosyllabic stems, and on the first and second syllable of disyllabic stems, as shown in (1).

- (1) Tuki A601 (Kongne Welaze 2004: 24)
- a. ɲgô ‘paternal uncle, brother-in-law’
 - b. ɲgǔ ‘hen’
 - c. ì-nôní ‘bird’
 - d. mǎngá ‘mother’
 - e. wòtô ‘accept’

This diversity in stem positions where contours can occur contrasts with typical East Bantu systems where contours tend to occur on non-final syllables, as e.g. shown for Ganda in (2), which allows contours on CVV and closed syllables.

- (2) Ganda (East Bantu JE15, Uganda) (Dutcher & Paster 2008: 125)
- a. òkù-bòbbà ‘throb’
 - b. òkù-kóòkà ‘cry’

Second, many languages exhibit both falling and rising tones (see Table 1), as mentioned above. And third, NWB contours often occur on plain CV or monomoraic syllables, as seen for Tuki A601 in (1), whereas contours in eastern Bantu tend to be restricted to CVV and/or closed syllables, as in the Luganda case. In the wider NW area, however, contour tones can associate with a single TBU, squeezing a lot of tonal material together. In Teke-Kukuya (West-Coastal Bantu B77a, Republic of the Congo) a single TBU can host LH, HL, and even LHL, as shown in (3).

- (3) Teke-Kukuya (West-Coastal Bantu B77a, Republic of the Congo) (Paulian 1975: 130–131)
- a. mò-sǎ ‘weaving knot’
 - b. kǐ-kâ ‘to pick’
 - c. ndé bvɪ́ ‘he falls’

Some NWB languages are also described as having underlying toneless TBUs, in addition to phonemic H and L. This is the case in Basaa. According to Makasso (2012: 18), all bound morphemes are toneless. In Bakweri, verb syllables other than the initial are toneless (Hawkinson 1986: 159). And in Gyeli, second and third syllables in verb stems, CV noun class prefixes, the verbal plural particle

The uni-code key combination ‘@[for the LHL pattern in (3e) does not compile what it should. Hopefully, Sebastian can help with this.

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nga, and the “STAMP” portmanteau morpheme that encodes subject agreement, tense, aspect, mood, and polarity (Grimm 2021: 103). Arguments put forth to distinguish toneless from L TBUs include distributional asymmetries between nouns and verbs and the predictability which TBUs will surface L in isolation as well as behavior under high-tone-spreading (HTS). In both Bakweri Marlo & Odden (2019: 152) and Gyeli, H tones that attach to the right of multi-syllable verb stems spread leftwards up to, but not including the first syllable. This is taken as evidence that the radical is lexically specified for tone while the following TBUs are underlyingly toneless.¹²

2.2 Notes on the distribution of tonal patterns by part-of-speech

In this section, I present some basic patterns in the distribution of tone across nouns, verbs, and functional morphemes such as noun class prefixes and non-initial verb stem syllables, which are often related to verbal extensions. Given the size of the topic and the amount of variation, I will only be able to give some examples to list some broad tendencies.

Most descriptions do not seem to address questions of frequency and distribution of tonal (or syllabic) patterns compared across word classes. But those that do seem to come to similar conclusions: For the major lexical word classes, noun stems generally exhibit a more diverse range of tonal patterns than verb stems. For example, in Kwakum (A91), monosyllabic nouns can be, in order of descending frequency, L, H, L°, H'H, LH'H, LH, HL, HL°, and LHL. In comparison, monosyllabic verbs only display four tone patterns: H'H, L, H, L°. The same asymmetry is found in disyllabic stems, where nouns have fourteen different tonal patterns and verbs only eight (Njantcho Kouagang 2018: 55-58). Similarly, in Gyeli, noun stems exploit all kinds of possible combinations of level and contour tones, although with different frequency distributions, whereas verb stems are only tonally specified for the radical, i.e. the initial syllable, as either H or L (Grimm 2021: 99-102).

Finally, also Eton shows a similar difference of tone pattern complexity between nouns and verbs. For nouns, Van de Velde (2008: 70) remarks that

“[t]he tone scheme of basic stems is either high (43%), low (30%), high-low (9%) or low-high (18%), irrespective of the number of syllables. More complicated tone patterns, such as high-low-high, are not permitted.”

In contrast,

¹²See Grimm (2022: 485-487) for a detailed argument for an underspecification analysis.

“only the first structural tone of a verb stem can be high. Hence, there are no verb stems with a rising tone pattern and only monosyllabic stems can be entirely high. Contrary to noun stems, monosyllabic verb stems do not carry compound tones.” (Van de Velde 2008: 115)

Verb radicals lexically contrast H and L verb-initial syllables (Grimm 2025b) in almost all NWB languages. An exception is Orungu (B11b), which shows no lexical tone contrast in the verb stem. Instead, tone is solely assigned by the TAMP category tone encodes (Ambouroue 2007). Also Kota (B25) is exceptional in that a verb stem vowel depends on the subject agreement class, which seems to determine the tonal pattern of the following TAMP prefix (Magnana Ekoukou 2015: 157–158). When the TAMP prefix is L, as in the first and second person singular and agreement class 1, the verb stem tone is H. Vice versa, when the subject agreement class triggers an H TAMP prefix, the verb stem will be L.

Tone assignment in non-initial verb stem syllables shows yet more variation across NWB. As stated above, in some languages (Basaa, Bakweri, Gyeli), non-initial verb syllables are analyzed as underlyingly toneless and receive their tonal specification depending on the TAMP category. A different system of tonal polarity (cf. Pulleyblank 2026 [this volume]) is found in Bakweri (A22) where a polar tone, i.e. the opposite tone of the lexical verb root, is assigned to non-radical syllables (Hawkinson 1986: 56, 58). Extension suffixes in Pove (B305) are all lexically specified H (Mickala-Manfoumbi 1994: 426ff). In trisyllabic stems, the tone of the final syllable depends on the radical tone. If it is H, the verb will surface as H H ⁴H, if it is L, the pattern is L H L.

Another prominent class of functional bound morphemes concerns noun class prefixes. Noun class prefixes in Basaa are specified L, which is a floating L in the non-segmental prefixes of classes 1a, 3a, 7, 9, and 10 (Hyman 2003: 262). This is similar to Nen (A44), where all noun class prefixes are L, while all “independent pronominal prefixes” are H. In contrast, the “pronominal prefixes”¹³ and subject pronouns are L only in agreement classes 1 and 9 (Mous 2003: 300). This illustrates that tones of the same functional class, such as marking agreement, can be sensitive to both the part-of-speech the morpheme attaches to and the agreement class it encodes in some languages, a phenomenon that is common in Bantu and beyond.

¹³Mous (2003) uses the terms “independent pronominal prefixes” and “pronominal prefixes” to distinguish different agreement prefix patterns on different agreement targets. The former are used with emphatic demonstratives and pronominals, whereas the latter occur with, for instance, the connective and possessives.

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2.3 Tonal operations and floating tones

Tone rules and operations are abundant and diverse across NWB. I will therefore only present a few examples of the most prevalent processes, including downstep and tone spreading. Floating tones as tonal elements that are associated with tonal operations, such as attachment and spreading, will also be discussed. It should be noted that downstep, tone spreading, and floating tones are extremely common in all of Bantu and not a special feature of NWB.

First, downstep is widespread among all the NWB sample languages in Table 1. Its status as phonemic or phonetic, however, differs across languages.¹⁴ The conditions under which downstep occurs differ across languages. In Bantu overall, downstep is typically associated with Meeussen's rule as a special type of downstep, in which the second of two consecutive high tones (HH) is lowered to avoid sequences of H tones, relating to the Obligatory-Contour-Principle (OCP). In NWB, however, downstep seems instead more commonly triggered by (floating) L tones. In Akoose A15C, downstep results from two successive H tones that are separated by intervening L tones, creating a terracing effect (Hedinger 2008: 7). Downstep further occurs when an L tone between two H tones is lost, lowering the second H. And finally, an H on a nominal stem is downstepped if its L class prefix assimilates, for instance with a locative marker, as in (4). In this case, the H of the preceding element is realized on the class prefix and the L of the class prefix lowers the stem H.¹⁵

- (4) Akoose A15C (Hedinger 2008: 8)

á èdíb → [é^ldíb]

LOC water

'at the river'

In Eton (A71), downstepped H tones can never follow L tones (Van de Velde 2008: 51). Instead, downstep Hs are triggered by floating L tones if the floating L is preceded by an H (otherwise it is deleted), as illustrated in (5).¹⁶

¹⁴A downstep H tone is phonetically on a tone level between H and L. As descriptions show a high degree of variability in the information they provide on their analytic representations, it could even be the case that a reported M tone could be analyzed as a downstep H in certain languages.

¹⁵In Hedinger's (2008: 8) analysis, the L tone is "lost due to elision in the formation of 'at the river'."

¹⁶I generally strive for consistency across examples and their glossing. As authors vary in their conventions and to the extent to which they provide glosses in the first place, I took the liberty to add glosses where there were none and, at points, modify existing glosses to make them more consistent across this paper, to the best of my ability to interpret the data given in the sources.

- (5) Eton A71 (Van de Velde 2008: 58)

|à-ᵀtᵉ # ᵀ-dí| → /àtᵉ ˈdí/

1-PRS # INF-eat

‘S/he is eating.’

Second, tone spreading, especially high tone spreading (HTS) is a widespread tonal process in Bantu overall (Kisseberth & Odden 2003: 62) and also in NWB. For example, in Yasa (A33), H tones from the verb radical spread rightwards in verbs under suffixation (Brown 2021: 41), but stop before the final syllable, as shown in (6).

- (6) Yasa A33 (Brown 2021: 41–42)

a. bótá ‘hide (trans.)’ + -mà (positional) > bótá-mà ‘hide (intr.)’

b. nyángà ‘suckle’ + -ìyè (causative) > nyáng-ìyè ‘breastfeed’

c. báà ‘marry’ + -nà (reciprocal) > báá-nà ‘get married’

In Gyeli A80), HTS can occur both rightwards and leftwards. In the nominal domain, H tones spread rightwards onto CV noun class prefixes, as in the genitive construction in (7).

- (7) Gyeli A801 (Grimm 2022: 480)

ma-fwálá má be-túmbó → [mà-fwálá má bé-túmbó]

ma6-end 6:ATT be8-country

‘borders’

In contrast, H tones in the verbal domain attach on the right and spread leftwards across all toneless TBUs, which excludes the radical, as in (8).

- (8) Gyeli A801 (Grimm 2021: 110–111)

a videga-ᵀ → [à vîdégá]

1.PST1 turn-PST

‘S/he turned.’

Finally, floating tones are a common phenomenon in NWB. Similarly to contour tones, they are historically explained by vowel loss in combination with surviving tones. This is, for instance, the case in Kwakum A91 where Njantcho Kouagang (2018: 55) attributes both floating H and L tones to diachronic vowel loss.

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Floating tones are found in the verbal and the nominal domain. For instance, just like Basaa, non-segmental noun class prefixes of classes 9 and 10 in Bakweri consist of a floating L tone, which triggers downstep on the following H tone (Hawkinson 1986: 42). They only take effect when the noun is preceded by another H tone, as in (9).

- (9) Bakweri A22 (Hawkinson 1986: 42)

ó ①-*mbòà* → /ó mb'òà/

LOC 9-village

‘in the village’

Languages can possess both floating H and L tones, as in Eton (Van de Velde 2008: 55ff). In this language, floating H tones attach to the right of verbs with a leftwards effect. The floating H can either cause a rising tone if the preceding syllable is dominant or, if it is not, the floating H detaches and replaces the L (Van de Velde 2008: 55).

3 Grammatical tone in the verb phrase

Bantu languages, and especially NWB, are known for the significant role that tone plays in the verb phrase. The reason for this is that lexical tones interact and often change due to their grammatical functions and morphosyntactic environments. This phenomenon is discussed under a variety of terms in the literature, ranging from inflectional tone to melodic tone (Odden & Bickmore 2014, Marlo & Odden 2019) and grammatical tone (Rolle 2018) (cf. Rolle & Bickmore 2026 [this volume]). I use the term “grammatical tone” as a cover term for all these notions. Following Rolle (2018: 2, 6), I view grammatical tone as a tonological operation which functions as a specific type of non-concatenative morphology, and which is restricted to the context of a specific morpheme or construction. As such, it is only licensed in particular grammatical environments. A grammatical tone can either be the sole exponent to mark a grammatical category or construction type or be accompanied by a segmental co-exponent. In any case, there is no claim that grammatical tone is restricted to grammatical notions that are expressed by purely tonal means. Inflectional and melodic tone can be seen as subsets of grammatical tone. Inflectional tone refers to grammatical tones that occur in verb inflection to mark categories such as tense, aspect, mood, negation, and subject and object agreement. Melodic tone denotes the particular tonal operation of superimposing inflectional tones through tone attachment and spreading across the (macro-)stem, interacting with and potentially overriding lexical tones. Marlo &

Odden (2019: 152) describe inflectional melodic tone patterns as “composed of H tones which are assigned to locations such as the second stem mora, the penult or the ultima.” They go on to explain that positioning of L tones in melodic tone patterns is the exception, but found, for instance, in Mokpwe A22 in the NW area and in Fuliiru JD63 and Kamba E55 outside of NWB.¹⁷

In order to adequately explore Bantu tone, it is important to understand the structure of Bantu verbal predicates and their range of variation, as grammatical tones interact with segmental morphology and lexical tones. Bantu verbs in all zones are structurally divided into a pre-stem and a stem cluster, based on Meeussen’s (1967) reconstruction of the Proto-Bantu verb morphology, and illustrated by Güldemann (2022: 388) in Table 4. In the pre-stem, inflectional categories such as subject and object agreement are encoded as well as TAMP categories in two different positions. The stem starts with the lexical radical and can take derivational, valency-changing “extension” suffixes, e.g. causative, applicative, passive, and reciprocal morphemes. The final slot in the stem constitutes another TAMP-marking position. As shown previously in Table 2, many languages in the wider NWB area lack an inflectional final vowel and, instead, stem-final vowels are lexical.

Table 4: The structure of inflected verbs in Bantu, based on Güldemann (2022: 388)

Pre-stem cluster				Stem			
(Pre-initial)	Initial	(Post-initial) ⁺	(Pre-radical)	Radical	(Pre-final) ⁺	Final	(Post-final)
	subject		object	verb	derivation		participant
TAMP		TAMP			TAMP	TAMP	P

Notes: (...) optional, + possibly more than one, T = tense, A = aspect, M = mood, P = polarity

¹⁷ A reviewer commented that they had the impression that melodic tone is less widely attested in NWB than in Eastern Bantu and even that melodic tone does not exist in A40, A70, A80 and A90. My sense is that this impression may arise due to a better coverage of “melodic tone” phenomena in the literature on Eastern Bantu, which specifically uses this term. In contrast, grammatical tone phenomena in NWB are framed as floating or inflectional tone. At the same time, the effects of H tone docking and spreading may look different, as longer spreads are possible in the larger marcostems of East Bantu languages. They do, however, also exist in some of the more analytic NWB languages. In Gyeli A801, for instance, H tone docking and leftward spreading marks past tenses (Grimm 2021: 373).

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Many NWB languages, however, differ from this morphological template in important ways. As shown above in Table 2, languages in the clades A31-40-50-60, A40-60-70, and A80-90 are characterized by a prosodic size restrictions of the stem (Hyman & Inkelas 1997, Hyman 2004, 2010, Good 2022), limiting the stem to a maximum of three syllables. It is widely assumed that these NWB languages have undergone segmental and prosodic size erosion, leading to a reduction of the agglutinative verb structure proposed as the morphological structure in Proto-Bantu (Hyman 2011a), although this is controversially debated in Güldemann (2011). Prosodic constraints on the stem are also linked to a more analytic structure of the verbal complex, as summarized by Nurse (2007: 243): “a few Cameroonian Bantu languages in Zone A are analytic (A24, A43-44-45-46, A62, A83-84, maybe other A40 and A80 languages, A90, others?).” In what way NWB or Bantu A languages are analytic is so far under-explored. It seems, however, that the relation between the pre-stem and the stem is described as loose particularly in NWB languages with prosodic size restrictions, as for instance pointed out by Hyman (2003: 277) for Basaa.¹⁸ It is common that the subject agreement marker and a TAMP-encoding morpheme are analyzed as an independent portmanteau morpheme preceding the verb stem, as shown for a variety of NW languages in (10). Following Anderson (2011, 2015, 2016), I label these portmanteau morphemes in bold as “STAMP” morphs, as they often encode multiple functions at once, namely subject agreement, tense, aspect, mood, and polarity. Generally, the segmental part contributes to the subject agreement class information, while the tonal part often contributes to the TAMP information.

- (10) a. Balon A13 (Kouoh Mboundja 2004: 269)
 ǎ sák
 à-á sák
 1-PRS dance
 ‘He is dancing/he dances.’
- b. Kol A832 (Henson 2007: 183)
 ɲǎ kò bwògè kwàn
 ɲì=á kò bwòg kwàn
 1-PST2 go harvest 9.honey
 ‘S/he was in the process of collecting honey.’
- c. Kwakum A91 (Njantcho Kouagang 2018: 213)

¹⁸Descriptions of individual languages are split as to whether they represent the pre-stem and the stem as one or as separate units. The decision may come from Bantuist traditions to view, for instance, subject agreement as prefixes, but they are often not sufficiently argued for.

ní jǐdʒók gbààgbàà
 ⓘ-ní n-dʒókóⓁ gbààgbàà
 PRS-1SG PRS-hear 1.IDEO
 ‘I’m hearing claps of thunder.’

A more extreme analytic structure is presented for Nen (A44) by Mous (2003), who views not only the subject agreement marker as a free morpheme, but also the verb radical and the individual TAMP morphemes, such as the negative and future markers in (11).

- (11) Nen A44 (Mous 2003: 299)
- à só ɲò fám ó m’ímè
 1 NEG FUT leave LOC house
 ‘S/he won’t leave the house.’

There are, however, also different morphological accounts in other languages, representing the pre-stem and stem as one unit, as in (12).

- (12) a. Akoose A15C (Hedinger 2008: 181)
- sê-sólé ńmì mǒtò
 1PL-enter.PRF 6.SG.3POSS 6.car
 ‘We got into his car.’
- b. Yasa A33 (Brown 2021: 80)
- è-tóká-ndí jòòngà
 2SG-want-PRS 5.healing
 ‘Do you want to be healed?’
- c. Ewondo A72 (Redden 1980: 172)
- mǎ-ɲgá-yén bà-wàà bá-bē
 1SG-PST2-see 8-chimpanzee 8-two
 ‘I saw two chimpanzees.’

Irrespective of the analytic details of whether the verb complex consists of one or several words, it tends to form a tonal domain with grammatical tone operating in conjunction on both the pre-stem and the stem. This can be illustrated in Gyeli A801, which is analyzed as an analytic type in Grimm (2021, 2022), with a free STAMP morpheme preceding the verb. Tone combinations (and contrastive vowel lengthening) on the STAMP marker and the verb stem instantiate seven affirmative tense-mood categories as a basic system in simple predicates, as shown in Table 5.

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Table 5: Tone patterns on the STAMP morpheme and the verb stem encode tense in Gyeli A801

Basic distinction	TM category	STAMP	Verb stem		Gloss
NON-PST	PRES	yá	gyàgà	L	‘We buy.’
	INCH	yàá	gyàgà		‘We start to buy.’
	FUT	yàà	gyàgà		‘We will buy.’
PST	PST1	yà	gyàgá	H	‘We bought(recently).’
	PST2	yàá	gyàgá		‘We bought (a long time ago).’
other	IMP	yá	gyàgà	HL	‘Let’s buy!’
	SBJV	yá	gyàgáà		‘May we buy.’

While the overall order of functional and lexical morphemes outlined in Table 4 applies in NWB as well, many functional morphemes are solely tonal or contain an important tonal component, likely following segmental erosion, a process that is often used to explain the origin of contour tones (Section 2.3). This remarkable feature of the analytic NWB languages leads Nurse (2008: 48) to conclude that “tone carries a much larger relative role in Zone A languages.” Despite a lot of variation in the details across languages, some general tendencies can be observed. First, the locations of grammatical tone marking TAMP categories correspond to the locations of TAMP positions in the Bantu morphological verb template in Table 4, namely between the subject agreement marker and the verb radical¹⁹ and to the left of the verb stem. Tonal morphemes attach as floating tones to segmental hosts, either segmental functional morphemes, such as TAMP markers, or the lexical stem. I will provide more details in Section 3.1.

Second, Bantu languages, especially those in the NW, have a strong tendency to encode TAMP categories through a combination of segmental and tonal morphemes. Grimm (2022, 2025b) discusses the functional load of grammatical tone and the way that segmental and tonal morphemes share the work in TAMP-encoding, exhibiting three types of exponence: tone as the sole exponent, a segment as the sole exponent (possibly with a lexically specified tone that does not change and that does not affect the tonal pattern of any other element), and co-exponence of tone and segment. Grimm (2025b) shows that, historically, Proto-NW-Bantu only used the co-exponence type, while sole exponence

¹⁹In NWB languages, there is generally no object agreement on the verb, unlike in the template in Table 4. Instead, object pronouns appear as independent morphemes mostly after the verb, following the basic SVO word order. One exception are the B30 languages that van der Veen (2003: 389) describes as having both pre-radical object agreement markers and free object pronoun.

(tone-only and segment-only) is an innovation in the Bantu A clade. This may contribute to the impression that tone plays a relatively more important role in these languages.

Grammatical tone in the (NW) Bantu verb phrase is predominantly associated with its role in TAMP encoding, which I will explore further in Section 3.1. Tone has, however, also been demonstrated to function in other ways. It plays a role in distinguishing clause types, as discussed in Section 3.2. Another role of tone is examined in Section 3.3, illustrating how tone distinguishes phrase-final and phrase-medial verbs, a phenomenon that is known under the term of “metatony”. Finally, Section 3.4 explores grammatical tones that associate on elements following the verb and their functions.

3.1 TAMP encoding on the STAMP and verb stem

Tone is known to have an important grammatical function in the encoding of TAMP categories in all of Bantu. Many languages in the wider NW area, however, are remarkable in that tone is widely used as the only means to encode certain grammatical categories. “Tone as sole exponent” is one of two strategies in which grammatical tone appears. In this case, the tone’s functional load is said to be high, since the tone is the only exponent of a grammatical category. In Grimm’s (2025b: 22) sample, languages in all Bantu A clades (even the more agglutinative types in A10-20-30) use tone as sole exponent in TAMP marking. This pattern, shifts, however, in the A80-90 clade, where Kwakum A91 does not allow for tone as sole exponent and neither do the Bantu B clades of NWB. Tone as sole exponent, is, however, found in languages of West-Coastal Bantu that exhibit prosodic size restrictions, such as Teke-Kukuya (Li 2024: 109).

An extreme case of the functional load of tone is illustrated for Gyeli in Table 5 where tonal changes on the STAMP morpheme and the verb stem instantiate seven tense-mood categories. Another language with a high functional load of tone in TAMP marking is Basaa. Hyman (2003) shows that five out of fifteen TAMP categories are encoded by a sole tonal exponent, as shown for H and L tone verbs in Table 6. In the (distant) past 3, the narrative tense, and the singular imperative, an H tone attaches as a suffix and a prefix, respectively, to the verb stem. In the future prefect and the affirmative subjunctive, an H tone attaches to the STAMP morpheme so that underlyingly L agreement class STAMP markers, such as agreement class 1, surface H, as highlighted in bold in the table. These two categories further include tonal changes on the verb stem, showcasing that tonal morphemes can attach to the verb stem or both the STAMP morpheme and the verb stem in Basaa.

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Table 6: L and H STAMP morphemes in Basaa TAMP categories, exemplified with the verbs *jé* ‘eat’ and *lò* ‘come’ (based on Hyman 2003: 281 with modifications based on p.c. Emmanuel-Moselly Makasso)

TAMP category	Tense marking	L STAMP classes ^a		H STAMP classes ^b	
		Affirm.	Neg.	Affirm.	Neg.
Past 3	/-H/	à jé	à jé 'béeé	bá 'jé	bá 'jé 'béeé
		à lós	à lós 'béeé	bá lós	bá lós 'béeé
Narrative	/H-/	à jé	à jé 'béeé	bá jé	bá jé 'béeé
		à ló	à ló 'béeé	bá ló	bá ló 'béeé
Imperative (sg.)	/-H/	jé	jé 'bá'ánj		
		lós	lós 'bá'ánj		
Future perfect	/H≠L-...-H/	á 'jé	á 'jé 'béeé	bá 'jé	bá 'jé 'béeé
		á lós	á lós 'béeé	bá lós	bá lós 'béeé
Subjunctive	/H≠...-H/	á jé	à jé 'bá'ánj	bá jé	bá 'jé 'bá'ánj
		á ló'ó	à ló 'bá'ánj	bá ló'ó	bá ló 'bá'ánj

^aL STAMP classes: 1SG, 2SG, 1PL, 2PL, 1, 9

^bH STAMP classes: 2, 3, 4, 5, 6, 7, 8, 10, 19, 13

Impressionistically, it seems that a grammatical tone attaches more frequently to the verb or the verb and the STAMP morph instead of to the STAMP marker alone, although this needs to be investigated more systematically. The majority of TAMP categories are faithful to the STAMP morph’s underlying tone in Grimm’s (2025a) small sample of six NWB languages, whereas a change of the STAMP tone constitutes an exception restricted to certain TAMP categories. Examples of a grammatical tone attaching only to the subject agreement marker, however, is attested in, for instance, Nuasue, as shown in (13).

- (13) Nuasue A62A (Bébiné 2019: 461)
Ⓣ-ò-yòəs-in Ⓛ ò-ó → [úyòòsìn wò]
SBJV-1-look.at-APPL SIT 1-3SG.OBJ
‘May he look at him.’

In the second exponence strategy of grammatical tone, tone is a co-exponent of a TAMP category, together with a segmental morpheme. The crucial factor to distinguish a segment and tone co-exponent from a sole segmental exponent is that a tonal co-exponent causes a tonal change on a different tonal target than its segmental co-exponent. In (14a), the recent past form consists of a segmental

morpheme *a* that fuses with the subject agreement marker, while the H tone co-exponent affects the tonal pattern of the verb. Tuki has a similar strategy, but the segmental morpheme has a different segmental and tonal form and does not fuse with the subject agreement marker, as shown in (14b). Additionally, the H tone affecting the verb stem detaches and replaces an underlying L tone instead of causing an HL tone as in Bakoko. In Bulu, the *éloigné* [far] “aspect 3” causes downstep on a following H radical, as in (14c). What all these examples have in common is that the tonal change on the verb stem is not conditioned by a general phonological rule, but by the specific environment of the grammatical morpheme. For instance, in Bulu, not all H tone TAM prefixes trigger downstep on the following H tone, as seen with “aspect 4” *ngá* that does not affect the stem tone (Alexandre 1966: 109).

- (14) a. Co-exponent *a* @,²⁰ Bakoko A43b (Michel 2000: 58)
 mì a @ dĩŋ mǎn → [mà dĩŋ mǎn]
 1SG PST1 PST1 love 1.child
 ‘I loved the child.’
- b. Co-exponent *má* @, Tuki A601 (Kongne Welaze 2004: 78)
 à má @ tò: → [à má tò:]
 1 PST1 PST1 satiate
 ‘S/he ate his fill.’
- c. Co-exponent *ngá* @, Bulu A74 (Alexandre 1966: 109)
 èyòŋ mèn-ngá@-tǒó m-ǒŋǒ → [mèn-ngá-tǒó]
 when 1SG-ASP3-be 1-child
 ‘When I was a child,...’

Despite the many commonalities regarding the location of GT in the verb phrase and their functions of encoding TAMP categories, the examples presented on tone as sole and as co-exponent above show that NWB languages differ a lot in the details on how tone contributes to expressing the individual TAMP categories. From a form-to-function perspective, no particular tone pattern is associated with any particular TAMP category, except for the subjunctive which tends to involve an H STAMP morph. This, however, is a widespread feature reconstructed to Proto-Bantu (Nurse 2008: 233).²¹ Grimm (2025a: 252) comes to a

²⁰The morphological analysis of the recent past tense co-exponent follows Michel (2000), who explains “[t]he recent past tense marker consists of [a] and a floating H tone”.

²¹Nurse (2016: 69) states that the subjunctive in Bantu “is marked typically by -*é* and also a H on the SM. Suffixal -*é* is absent from northwest languages such as A40-50-60-70-80 (?), B20-30-50-60-70-82, C10-25-30-83, D30, H10-33. In most of these languages, one or more H on the verb suggests that *é* has been lost and its tone pattern kept, but this needs more checking.”

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similar conclusion and also observes that future tenses in some languages receive a special tone pattern on the STAMP morph. For instance, in Akoose, the STAMP morpheme retains its underlying lexical tone in all major tenses, except for in the future. Similarly, in Yasa and Gyeli, the future exhibits special features that deviate from the default case, although in these languages, the special pattern is not restricted to the future but also applies to other TAMP categories.

There are several factors that contribute to the variety of forms and the particular functions they encode. One factor is the underlying, synchronic tonal specification of the subject agreement markers across different agreement classes. In some languages, such as in Gyeli (Grimm 2021: 212), they are analyzed as toneless. In other languages, they have different tonal specifications, which are usually close to Proto-Bantu, with some variation. In Balon (A13), 1SG, 2SG and classes 1 and 9 are L while the rest is H (Kouoh Mboundja 2004: 256ff). In Pove (B305), all speech-act-participants (first and second person in the singular and plural) and agreement class 1 are L, but not class 9, while the rest is H (Mickala-Manfoumbi 1994: 420–421).

Another factor is the interaction of the underlying STAMP tone with the tone of certain TAMP categories, which is highly language-specific. It is common to find exceptions to a TAMP tone pattern in single agreement classes. In Gyeli (A801), the underlyingly toneless STAMP morphs receive their tonal specification through grammatical tone patterns on the STAMP and the verb stem, as shown above in Table 5. The first and second person singular and agreement class 1, however, show an exceptional pattern on the STAMP morph in a few TAMP categories. In the future, their tone is L: instead of HL (Grimm 2021: 372) and in the present negation, they are LH on a long vowel instead of H on a plain vowel (Grimm 2021: 392).

In Orungu (B11b), STAMP markers of all persons and agreement classes are H in all TAMP categories, except for agreement class 1, which is nearly always L, as shown for some TAMP categories in Table 7.²² In the past irrealis, however, the agreement class 1 STAMP marker surfaces H. Ambouroue (2007) shows that tone patterns are specific to individual TAMP categories, and each morphological verb slot (cf. Table 4) is subject to change by GT. The radical in Orungu has no L vs. H contrast—in contrast to all other languages in the sample. Basic TAMP categories are also encoded segmentally on the STAMP marker, a TAMP prefix, and the final vowel. The templatic tone structure in the Orungu verb inflection is thus rather different from Gyeli (see Table 5).

There is also variation in the underlying specification of tones in verb stems, contributing to the variation of tonal patterns in TAMP marking. While languages

²²Negative and non-indicative categories also have an L agreement class 1 STAMP marker.

Table 7: Tonal patterns in the Orungu B11b verb inflection of *-kol-* ‘buy’(Ambouroue 2007)

TAMP	AGR 1 ‘s/he’		AGR 2 ‘they’	
PRS	è-kól-à	‘s/he buys’	ʔí-k’ól-à	‘they buy’
	è-kól-íz-à	‘s/he sells’	ʔí-k’ól-íz-à	‘they sell’
PST perfect/ resultative	à-kól-i	‘s/he bought’	wá-kól-i	‘they bought’
	à-kól-i	‘s/he sold’	wá-kól-íz-i	‘they sold’
recent PST	à-kól-à	‘s/he bought’	wá-k’ól-à	‘they bought’
	à-kól-íz-à	‘s/he sold’	wá-k’ól-íz-à	‘they sold’
distant PST	à-yòl-i	‘s/he bought’	wá-yól-i	‘they bought’
	à-yòl-íz-i	‘s/he sold’	wá-yól-íz-i	‘they sold’
PST irrealis	á-yòl-è	‘had s/he bought’	wá-yòl-è	‘had they bought’
	á-yòl-íz-è	‘had s/he sold’	wá-yòl-íz-è	‘had they sold’
FUT	è-ḃé-k’ól-á	‘s/he will buy’	ʔí-ḃé-k’ól-á	‘they will buy’
	è-ḃé-k’ól-íz-á	‘s/he will sell’	ʔí-ḃé-k’ól-íz-á	‘they will sell’

Notes: The morphological template is STAMP-radical-(causative suffix *-íz-*)-final vowel.
Agreement class 2 is representative of all classes other than AGR 1.

generally have a distinction of L vs. H on the radical, they differ in their specifications for extension and final vowel suffixes (for the Bantu B languages and Londo A11 that have inflectional final vowels). In Gyeli (Grimm 2021: 103) and Basaa (Makasso 2014: 226), for instance, these are analyzed as underlyingly toneless. In Balon, they are L, but they differ depending on the number of syllables following the radical. In disyllabic stems, the H tone from a radical spreads to the extension or final vowel syllable. Therefore, disyllabic stems with an H radical have an H HL pattern. In contrast, L radicals surface L L, as shown in Table 8. This process does not apply to trisyllabic stems, whose non-initial syllables are always L.

In Bakweri, the stem-final lexical vowel carries a dissimilating tone: With H radicals, the final tone is L and, vice versa, with L radicals, the final vowel is H (Hawkinson 1986: 56), as shown for the minimal pairs in (15).

(15) Bakweri A22 (Hawkinson 1986)

- a. H-toned radicals
 - índà ‘gossip’
 - ókà ‘be sick’
 - ávà ‘rotate, turn’
- b. L-toned radicals

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Table 8: Tonal patterns of the Balon A13 verb stem (Kouoh Mboundja 2004: 232ff.)

Stem structure		H-toned radicals		L-toned radicals	
disyllabic	with stem-final vowel	yúfù	‘ask’	yòlò	‘laugh’
		kwálò	‘scratch’	nùkè	‘tremble’
		núnû	‘uproot’	nyàwò	‘let’
	with extension	fól-ìl	‘calm sb.’	yàn-ìl	‘forget’
		tél-ìl	‘open’	kèl-ìl	‘attach’
		lók-ân	‘drive’	kàk-ân	‘promise’
		búl-âm	‘observe’	bènd-âm	‘perch’
		pán-inì	‘scatter’	fèt-inì	‘imitate’
trisyllabic		nyúŋgw-èlè	‘tip over’	bàk-èlè	‘accuse’
		ká-y-isè	‘judge’	-	

indá ‘stay long’
òká ‘play’
àvá ‘sell’

Unlike in Balon, this pattern is not restricted to disyllabic stems, but applies to all non-initial syllables, as shown in (16).

(16) Bakweri A22 (Hawkinson 1986)

- a. H-toned radicals
kók-isè ‘punish’
ám-b-èlè ‘wait’
kóm-b-ènè ‘be near’
- b. L-toned radicals
làk-isé ‘forgive sb.’
kòv-élé ‘feed by hand’
kàk-éné ‘promise’

Finally, languages differ in the location and direction of floating tone attachment. Most commonly, they either attach before the stem and spread rightwards onto the stem, or they attach on the right and spread leftwards. In Eton, floating tones attach to both the left and the right of verb stems, as shown in (17).²³ Float-

²³I adjusted the glosses provided by Van de Velde (2008) to make annotation conventions more uniform across this paper. The <@g> in the gloss indicates an L tone with a segmental morpheme, which Van de Velde (2008) calls the “G-form”.

ing L tones attach to both the right and the left, whereas floating H tones only attach rightwards.

(17) Eton A71 (Van de Velde 2008: 57–58)

- a. L-attachment to following syllable
à-ᵀtᵉ ᵀ-dí → [àtᵉ 'dí]
1-PRS INF-eat
'She is eating.'
- b. L-attachment to preceding syllable
mà-kód<ᵀg>ò → [mà-kôdgò]
1SG-leave<G.PST>
'I left.'
- c. H-attachment to following syllable
mà-ᵀ-kpèlì → [màkpèlì]
1SG-PST-hurt.oneself
'I hurt myself [in the past].'

3.2 Tone indicating clause type

In addition to encoding TAMP categories, tone also distinguishes different clause types linked to subordination and relativization across Bantu at large, as described in Odden & Bickmore (2014: 9–10). This also applies in a variety of NWB languages, but is not a feature restricted to this area. Which clause types exactly are tonally marked, is language specific. In languages that have this type of syntactic tone, however, relative clauses and some other type(s) of dependent clauses trigger a tone change, often an H tone on the STAMP morph. In Akoose (A15C), Hedinger (2008: 102) explains that “[i]n specific grammatical environments, such as dependent clauses, relative clauses and questions, both affirmative and negative verb forms have distinct dependent forms.” The dependent forms are tonally different from the “independent” forms and are further subdivided into subject type and object type dependent clauses (Hedinger 2008: 105ff). that different clause types are marked by different tone patterns, as shown in (18).

(18) Akoose A15C (Hedinger 2008: 105–106)

- a. Independent type clause
mwǎn ěpiměé mbààngé
1.child throw.out.NEG.PRF cocoyam
'The child didn't throw out the cocoyams.'

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- b. Subject type dependent clause
 nzé ěpimè mbààngé?
 who throw.out eat.NEG.PRF cocoyam
 ‘Who didn’t throw out the cocoyams?’
- c. Object type dependent clause
 mbààngé cḥ mwăn ěpimèé
 cocoyam COP? 1.child throw.out.NEG.PRF
 ‘It is the cocoyams that the child didn’t throw out.’

Each TAMP category has different forms for each clause type. In some categories, such as the affirmative past and future perfective and imperfective, the present imperfective, and the perfect, the independent type and the subject dependent type are form-identical (Hedinger 2008: 104).

Also Bakweri (A22) marks relative clauses tonally in various TAMP categories through special tone patterns on the verb, as shown in (19).

(19) Bakweri A22 (Marlo & Odden 2014)

- a. Main clause pattern (constructed from Marlo & Odden’s (2014) data)
 à-mà-lá
 1-PST-eat
 ‘S/he ate.’
- b. Subject relative pattern (Marlo & Odden 2014: 304)
 à-mă-l’á
 1-PST-eat
 ‘which/the one person who ate’
- c. Object relative pattern (Marlo & Odden 2014: 303)
 á-zà-lá
 1-FUT-eat
 ‘(what/who) s/he won’t eat’

In addition to this, tones distinguish between subject and object relatives, as the contrast between (19b) and (19c) shows. With object relatives, all STAMP forms, irrespective of their underlying lexical tone, are marked through an H tone (Marlo & Odden 2014: 297). Tone marking subject relatives targets the tense marker *ma* and the verb final vowel, in contrast to the tone that marks object relatives and that targets the STAMP morpheme.²⁴

²⁴This presentation is oversimplified, as the details exceed the scope of this paper. There is more nuance to the interaction between relative marking and individual TAMP markers, as further discussed in Hawkinson (1986: 199–204).

Hawkinson (1986: 202–204) also notes an H tone on the STAMP marker in conjunction with a falling tone on the verb final to encode temporal (simultaneous and sequential) clauses, as in (20). She explains the H tone as a backgrounding device to mark the secondary event. Without the H tone on the STAMP, both events are primary.²⁵

(20) Bakweri A22 (Hawkinson 1986: 202)²⁶

- a. ná-tilà, à-làngà
1SG-write.SUB.FUT 1-read.FUT
‘When I will write, s/he will read.’
- b. nà-tilà, à-làngà
1SG-write.FUT 1-read.FUT
‘I will write, s/he will read.’

The LHL tone in (20a) does not come out as intended with the Unicode key combination @[.

In Yasa (A33), certain clause types trigger an H on the STAMP morpheme, including relative clauses (21a), a subset of temporal clauses (21b), and other subordinate constructions introduced by conjunctions (21c), as explained by Brown (2021: 50–51, 62). All STAMP morphemes in bold in (21) would surface L, which is their underlying base form, if they occurred in a main clause.²⁷

(21) Yasa (Brown 2021: 50–51, 62)

- a. à-bé mùmú [mé à ngwé ná-tóó]_{REL}
1-COP 1.man REL 1.CON 1SG.PRO 1SG-see.PST
‘He was the man that I saw.’
- b. bòlò nyàmà wávókwé wá-écé-cè mú [vá mú
together 10.animals 2.other 2-laugh-INCOMP 1.PRO when 1.PRO
á-yé púwà cóò]_{TEMP}
1-COP.PRS 9.back turned
‘The group of other animals laughed at him when he had his back turned.’

²⁵Syntactically, the difference between primary and secondary can be characterized as a difference between main clauses and dependent clauses.

²⁶(17a) should be a Grave plus acute plus grave accent, but unicode @ [doesn’t work. I am not sure whether -làngà ‘read’ should have an L tone here, as the verb is H elsewhere, but I stick with the tonal notation of the source.

²⁷Brown (2021) explains that other factors can also determine whether an L STAMP marker surfaces H, including certain past suffixes and the tonal pattern of the verb radical–H radicals prevent H on the STAMP. These conditions do not apply in (21).

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- c. èné é sèyá é ècèndò èvóókó [èné mú á-bòngá-ná nà
 thus LOC 9.path LOC 7.trip 7.one thus 1.PRO 1-meet-RECIP COM
 ìcìni]_{SUB}
 1.fly
 ‘Then on the path he met Fly.’

The syntactic function of tone on the STAMP morph to distinguish clause types seems mostly attested in the A10-20-30 clade of NWB, while it seems absent in most other subgroups. An exception is Kota in the B20 group, where tone is the sole exponent to encode the difference between the present in a main clause and in a relative clause, as shown in (22).

(22) Kota (B25) (*Magnana Ekoukou 2015*: 239)

- a. m-wánà àdʒá létè
 1-child eat.PRS rice
 ‘The child eats rice.’
 b. m-wánà àdʒá létè
 1-child eat.REL rice
 ‘the child that eats rice’

3.3 Metatony

The term “metatony” goes back to *Meeussen (1967)* and denotes a tonal change from a final L or non-H tone on verbs in phrase-final position to an H tone phrase-medially, as illustrated for Basaa in (23). The marking of phrase-final versus phrase-medial verbs is extremely widespread in Bantu, as I explain below, but historically, the tone was likely a co-exponent along with a segmental suffix. In the wider NW area, the segmental suffix was lost so that only a metatonic tone survives synchronically. The only exception I found is Balon A13 in (27) that allows a glimpse at a segmental co-exponent.

(23) Basaa A43 (*Makasso 2012*: 19)

- | | |
|---|-----------------------------|
| a. mààngé à n-tíŋ-il
mààngé à N-tíŋ-il
1.child 1 PST1-untie
‘The child untied.’ | (no metatony: “disjoint”) |
| b. mààngé à n-tíŋ-íŋl ń-kòò
mààngé à N-tíŋ-il ń-kòò
1-child 1 PST1-untie 3-rope
‘The child untied the rope.’ | (with metatony: “conjoint”) |

There are several points of contention with respect to the exact definition of metatony. Meeussen's (1967: 111) original notion of metatony only includes infinitival verb forms with the final vowel *-a* that change from L to H if followed by an object. A broader view of metatony, however, has been widely adopted (Nurse 2008, Hyman & Lionnet 2012), encompassing non-infinitival verb forms and post-verbal material other than objects. In this paper, I treat both cases as different forms of metatony. Nurse (2008: 204) mentions another type of H tone change that he suggests is related to metatony, namely instances where the metatonic H carries over onto the first tone-bearing unit of the complement. I treat these cases of tone on immediate-after-verb elements separately from metatony (Section 3.4), however, as evidence from Eton and Gyeli suggest that they are two distinct tones.

While some authors provide a strict phonological analysis of metatony (Hyman & Lionnet 2012), the phenomenon has also been related to notions of focus-marking and the “conjoint/disjoint distinction”, particularly in eastern and southern Bantu, e.g. (Nurse 2008, Costa & Kula 2008, Hyman 2017b).²⁸ (Hyman & Lionnet 2012) and Hyman (2017b) provide an excellent overview of the discussion of metatony in the literature and propositions for the historical origins of the metatonic H tone, which have been linked to the connective (genitive) **-á* morpheme (Angenot 1971) and to the initial **H* “augment” morpheme on the following noun phrase (Dimmendaal 1995, Schadeberg 1995).

Metatony is widespread in NWB, as also noted by Nurse (2008: 48), who states that metatony “occurs widely in Zones A and B and also in a few C languages, in D13 (Mituku), D27 (Bangubangu), and some H10.” In Grimm's (2025b) sample of twelve NWB languages, representing each clade of this family, metatony is attested in three quarters of the languages. There are four parameters according to which languages with metatony vary: a) the inflectional status of the verb (specific TAMP categories vs. infinitives), b) the sentence type in which the verb occurs, and c) the nature of post-verbal material that trigger metatony, and d) the function that metatony encodes, if any. In the following, I will illustrate these different parameters in one language, Gyeli, before describing the variation of each parameter across different NWB languages.

In Gyeli, the first syllable of a verb stem is lexically specified H or L,²⁹ whereas second and third syllables are underlyingly toneless. If the verb is in phrase-final

²⁸It should be noted, however, that the conjoint/disjoint distinction is not primarily marked by tone in many eastern and southern Bantu languages. Also, Hyman & Lionnet (2012: 181) convincingly argue that metatony in Abo is distinct from the Eastern Bantu type of conjoint/disjoint distinctions.

²⁹Monosyllabic H verbs surface as HL in isolation and in their uninflected form.

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position, the verb surfaces L, as shown in (24a). In certain TAMP categories, the phrase-medial verb takes a metatonic H tone that attaches to all toneless second and third syllables, whereas lexical L tones in monosyllabic verbs are replaced by an H tone, as shown in (24b).

(24) Gyeli A801 (based on Grimm 2021, 2022)

- a. mé nyê/dè/gyàgà/lùngèlè (phrase-final verb)
 mɛ-Ⓣ nyê/dè/gyàgà/lùngɛlɛ
 1SG-PRS see/eat/buy/aim.at
 ‘I see/eat/buy/aim at mangoes.’
- b. mé nyé/dé/gyàgá/lùngélé mántúà
 mɛ-Ⓣ nyê/dè/gyàgà/lùngɛlɛ-Ⓣ Ⓣma-ntúà
 1SG-PRS see/eat/buy/aim.at-R OBJ.LINKma6-mango
 (with H tone: realis)
 ‘I see/eat/buy/aim at mangoes.’
- c. mèè nyê/dè/gyàgà/lùngèlè mántúà
 mèè nyê/dè/gyàgà/lùngɛlɛ Ⓣma-ntúà
 1SG.FUT see/eat/buy/aim.at OBJ.LINKma6-mango
 (no H tone: irrealis)
 ‘I will see/eat/buy/aim at mangoes.’

This phrase-medial tone change only happens, however, in the present, the inchoative, and the recent and remote past in Gyeli (Grimm 2022: 488). In contrast, the metatonic H tone does not appear in the future, as shown in (24c), nor in the present negation and non-indicative imperative and subjunctive categories. In Gyeli, metatony is indeed conditioned by specific TAMP categories and not by the morphosyntactic material that follows the phrase-medial verb. Grimm (2021: 389) shows that in metatony TAMP categories any part of speech within the phrase triggers the metatonic H tone attachment to the verb, including nominal and pronominal objects, verbs (in complex predicates), adverbs, prepositions, and conjunctions. Functionally, the metatonic H tone divides the Gyeli tense categories into realis (with metatony) and irrealis (without metatony) categories.

The conditions under which metatony occurs in various NWB languages differ, among other factors, according to the inflectional status of the verb with respect to certain TAMP categories or the infinitival form that trigger metatony. In contrast to Gyeli where metatony occurs in the present, inchoative, and the two past tenses, metatony in Abo appears in the present, past, and perfect (Hyman & Lionnet 2012). In Pove B305, all TAMP categories except for the negative

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any kind of word. In Table 9, I expand Hyman & Lionnet’s (2012: 181) suggested typology of metatony. Originally, they only placed Lega, Songye, Abo, and Basaa, pointing out that the other logical possibilities (various word forms and before object only, only infinitive forms and before any word) likely also exist. And indeed, the combination of metatony with infinitives only but before any word is attested in Yasa (Brown 2021).

Table 9: Typology of metatony

	before object only	before any word
only infinitive forms	Lega (D25), Songye (L23)	Yasa (A33)
various verb forms		Abo (A42), Basaa (A43), Kpā? (A53), Gyeli (A801), Makaa (A83)

This typology further shows that NWB languages generally do not seem to place restrictions on the elements that follow a phrase-medial verb: the object-only type is only found outside of NWB. The category “before any word”, however, should be treated with caution. Although it is attested for these languages that other words than objects trigger metatony, only Abo, Basaa, and Gyeli seem to have been systematically investigated for the full range of parts of speech following a verb with metatony.

Finally, the functions that have been ascribed to metatony across NWB languages are extremely diverse, ranging from tense and mood distinctions to semantic differences pertaining to information structure or no obvious meaning at all. As metatony is conditioned by certain TAMP categories across languages, it is an obvious area to investigate potential functions. In some languages, there does not seem to be any common semantic denominator across the specific TAMP categories that exhibit metatony, e.g. present, past and perfect in Abo. In Gyeli, however, Grimm (2021, 2022) proposes that metatony marks a realis/irrealis distinction. And in Balon, metatony coincides with past marking: Kouoh Mboundja (2004) shows that the “disjoint” (phrase-final) form in (27a) not only takes an HL tone but also has the segmental suffix *-i*, which encodes a past form. In the “conjoint” (phrase-medial) form in (27b), the segmental suffix is deleted and the radical takes an H instead of an L tone.

- (27) Balon A13 (Kouoh Mboundja 2004: 279)
- a. “disjoint” (phrase-final) form

- m̩ bumb-î
 1SG grill-PST 2
 ‘I grilled.’
- b. “conjoint” (phrase-medial) form
 m̩ búmb swî
 1SG grill fish
 ‘I grilled fish.’

Guarisma (2003: 320–324) states that metatony expresses imperfective aspect categories in Kpā? (A53), including progressive, hypothetical, inceptive, anterior, and conclusive. It also occurs with the negative suffix *-bí*.

An often-cited function of metatony is its alleged focus-marking potential. Nurse (2008: 204) views metatony as a form of focus in the literature on Zone A, B, C, D10, and D20. He points out that metatony is mostly found in affirmative clauses in Kpa (A53), according to Guarisma (2003: 320–327), which he takes as a striking characteristic that links metatony to focus and that encodes a contrast between verb focus and post-verbal focus. Similarly, Costa & Kula (2008: 313) propose that “in Duala and Basaa, where a tonal distinction with respect to a following complement can still be seen, we have [immediate after verb] focus as opposed to initial focus, pointing to the fact that the tonal effects [...] are the indicator of focus (via prosodic structure).”

There seems to be growing consensus, however, that metatony generally does not serve a focus-marking function. One problem about focus-related claims is that the literature rarely examines metatony within the information structure system of the language, i.e. there is no discussion about how topic and focus are generally marked. For languages where this information is available, a focus-marking function can be ruled out. As Hyman (2017b: 109) puts it, “[i]n fact, it is not clear that metatony *sensu stricto* is ever related to focus.” He shows in the following that focus in Abo is encoded by the focus marker *ndí*, which changes its position depending on which constituent is in focus. As shown in (28), metatony occurs independently of focus-marking.

- (28) Abo A42 (Hyman 2017b: 110)
- a. “Neutral” focus
 m-ăn ă sɔ̃ŋsé má-sɔ̃ŋ m-é
 N1-child 1-PRS count ma6-tooth 6-3SG.POSS
 ‘The child is counting his teeth.’
- b. Subject focus

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m-ăn ndi ă sòŋsé má-sǎŋ m-é
 N1-child FOC 1-PRS count ma6-tooth 6-3SG.POSS
 ‘The CHILD is counting his teeth.’

c. Non-subject focus

m-ăn ă sòŋsé ndí má-sǎŋ m-é
 N1-child 1-PRS count FOC ma6-tooth 6-3SG.POSS
 ‘The child is counting HIS TEETH.’

Likewise, in Gyeli, the metatonic H tone surfaces on phrase-medial verbs in the relevant TAMP categories, no matter whether the object is in focus, as in (29a), or the subject in the cleft construction in (29b).

(29) Gyeli A801 (Grimm 2021: 522)

a. What do the children eat?

bwánò bá dé mántúà
 b-wánò ba-@ dè-@ @ma-ntúà
 ba2-child 2-PRS eat-R OBJ.LINKma6-mango
 ‘The children eat MANGOES.’

b. Who eats mangoes?

yî bwánò bá dé mántúà
 yî b-wánò ba-@ dè-@ @ma-ntúà
 7.COP ba2-child 2-PRS eat-R OBJ.LINKma6-mango
 ‘It’s the CHILDREN who eat mangoes.’

Also for Basaa, Makasso (2012: 15) argues that metatony encodes a conjoint/disjoint distinction, which he takes as a contrastive relation between the verb and other constituents, but crucially, it does not mark focus. In (30), the same sentence appears in the same past category. While (30a) constitutes one prosodic unit in which the verb takes a metatonic H, there is a prosodic break, marked by the comma, in (30b), and the verb surfaces non-H.

(30) Basaa A43 (Makasso 2012: 15)

a. “conjoint” form

bòr bá níló ɲgándàk
 2.man 2 PST1-come many
 ‘A lot of people came.’

b. “disjoint” form

bòr bá nlô, ngàndàk
 2.man 2 PST1-come many
 ‘A lot of people came.’

This conjoint/disjoint distinction points to a syntactic-semantic function, as also discussed in [Schadeberg \(1995: 176\)](#) and [Dimmendaal \(1995: 32\)](#). [Hadermann \(2005: 409\)](#) describes it as a suprasegmental phenomenon conditioned by syntax that provides evidence for the verb-complement relation. As such, there is a difference in interpretation between the two sentences in (30). While (30a) expresses the fact that many people came, (30b) indicates that there was a problem with the quantity. A detailed discussion of the literature on the conjoint/disjoint distinction exceeds the scope of this paper, particularly as it describes an alternation found in East Bantu, which seems to be distinct from related phenomena in NW and West-Coastal Bantu. The reader is referred to [Van der Wal \(2017\)](#) for an excellent overview of the conjoint/disjoint alternation and the question whether it is primarily related to focus or constituency.

In NWB, metatony has been described for some languages as not having any clear function. [Van de Velde \(2012\)](#) comes to this conclusion for Eton, stating that the metatonic tones have a morphological (tonal) form but no meaning, which he attributes to diachronic changes and erosion. Similarly, in Abo (A42), [Hyman & Lionnet \(2012\)](#) analyze metatony as one out of three phonological patterns across TAMP categories, which do not seem to map onto any clear function. As shown in Table 10, the metatony pattern that exhibits the characteristic phrase-medial tone change is found in the present, past, and perfect tenses. In contrast, verbs in the future are characterized by a final L, both phrase-finally and phrase-medially, while the stative, imperative, and subjunctive are marked by a final H in both positions.

As I will discuss in the next section, there seems to be a relation between metatonic tones on the verb stem and immediately following elements in the verb phrase. At least in some languages, a metatonic tone may carry over to a following word, typically an object noun.

3.4 Tones on immediate-after-verb elements

Grammatical tones on immediate-after-verb (IAV) elements occur in a range of NWB languages, although they seem generally less common than the case of metatony ([Grimm 2025b](#)). They are characterized in two different ways across NWB. First, IAV tones are viewed as an instantiation of metatony in some languages, either because the IAV is a carry-over from a metatonic H on the verb

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Table 10: Metatony as a phonological pattern in Abo A42 (Hyman 2017b: 109)

suffix tone	TAM	pre-pause	+ bitámbé ‘shoes’	Translation
-L ~ -H	present	ǎ pòŋò	ǎ pòŋó bitámbé	‘He is making shoes.’
	past	à póŋò	à pó↓ŋó bitámbé	‘He made shoes.’
	perfect	à má póŋò	à má pòŋó bitámbé	‘He has made shoes.’
-L	future	à káà pòŋò	à káà pòŋó bitámbé	‘He will make shoes.’
	stative	à pòŋó	à pòŋó bitámbé	‘He has made shoes.’
-H	imperative	pòŋó	pòŋó bitámbé	‘Make shoes!’
	subjunctive	sá pó↓ŋó	sá pó↓ŋó bitámbé	‘Let’s make shoes!’

or because the metatonic H is said to associate with an IAV element instead of the verb. The second type of analysis views metatony on a phrase-medial verb and an IAV tone as two distinct phenomena. In this case, the IAV tone that typically associates with an object, may have more in common with “tone-case” as discussed in Western Bantu (Kavari et al. 2012, Van der Wal 2015, Hyman 2017b). I will present examples of these different scenarios in the following.

In some languages, tonal patterns of IAV-elements depend on the preceding verb stem tone and may thus be linked to notions of metatony and the conjoint/disjoint distinction (Section 3.3). For example, in Basaa, an IAV element, such as an object noun prefix, takes an H tone only if it is preceded by a metatonic H tone (Hyman 2003: 280–281), as in (31). The distant past (past 3), future perfect, subjunctive and imperative are marked by a final H on the verb, but do not show metatony, as there is no tonal alternation between a phrase-medial and a phrase-final verb. In these non-metatonic tenses, the verb’s final H does not affect the following object, whose prefix surfaces L, as in (31c).

(31) Basaa A43 (Hyman 2003: 280–281)

- a. à bí nùŋùl
à bí nùŋul
1.PRS2 PRS2 sell
‘He sold.’
- b. à bí nùŋúl bí-sèl
à bí nùŋul @ bi-sèl
1.PRS2 PRS2 sell MET 8-basket
‘He sold baskets.’
- c. nùŋúl bì-sèl

nùŋul-@ bi-sèl
 sell-IMP 8-basket
 ‘Sell baskets!’

Makasso (2012: 18) clarifies that the metatonic H tone not only affects postverbal objects, but any adjacent post-verbal element, including nouns, verbs (infinitives), adverbs, and adjectives. In (32a), the H of the conjoint (metatony) form carries over to the comitative marker, whereas the comitative surfaces L in the disjoint (non-metatonic) case in (32b).

(32) Basaa A43 (Makasso 2012: 20)

- a. conjoint (metatony)
 à ń-gw^lés ní ńgùj
 1 PRS-like COM strength
 ‘She likes seriously.’
- b. disjoint (no metatony)
 à ń-gw^lés nì bèn
 1 PRS-like COM hate
 ‘She likes and hates.’

A similar situation is found in Abo (A42), according to Hyman & Lionnet (2012). Here, a metatonic H on the verb also raises a following overt L noun prefix, as shown in Table 10, and a floating L prefix, which is found with nouns and pronouns. In contrast to Basaa, however, metatony does not affect other IAV elements, such as demonstratives, prepositions, or particles. The reason for this is phonological, as these word classes do not come with a floating L.

Makaa (A83) represents a different type of an IAV tone related to metatony. Heath (2003: 343–344) explains that a “Macrostem H” either associates with the phrase-medial verb or a following object. This Macrostem H seems to correspond to metatony in that it only surfaces when the verb is not phrase-final and in that its occurrence is conditioned by certain TAMP categories. All TAMP categories trigger it, except for distant past (PST 2). The properties of the IAV element determine the location of attachment. If the verb precedes a noun of class 3 or 7, a preposition, pronoun, or verb, the metatonic (Macrostem H) associates leftward with the verb, as in (33a). This examples also illustrates the remarkable feature in Makaa that the metatonic H tone can occur multiple times in a phrase, namely after each non-final verb. If the verb precedes a noun of classes 1, 2, 4, 5, 6, 8, or 9, the metatonic H is realized on the object noun. In case the object noun has a CV prefix, as in classes 2, 4, 6, and 8, the metatonic H replaces the prefix L tone,

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as in (33b). If the noun does not have a CV prefix, as in classes 1, 5, and 9, the metatonic tone is realized on the first syllable of the stem, as in (33c).

(33) Makaa A83 (Heath 2003: 334–344)

- a. mǎ áǎmǎ nyǐngǎ gǔ gwòó
 mǎ áǎmǎ nyǐngǎ ⑤ gǔ ⑤ ⑤-gwòó
 1SG PST1 again(v.) MET pick MET 7-mushroom
 ‘I again picked a mushroom.’
- b. mǎ é cǎl mǎlǎndú
 mǎ é cǎl ⑤ mǎlǎndú
 1SG FUT1 cut.down MET 6-palm.tree
 ‘I will cut down the palm trees [today].’
- c. mǎ áǎmǎ díg kúùs
 mǎ áǎmǎ díg ⑤ kúùs
 1SG PST1 see MET 1.parrot
 ‘I saw the parrot.’

In all these cases, an IAV H tone is conditioned by metatony. Especially in Makaa, the metatonic tone behavior illustrates the fluidity and strong relation between the verb and an IAV element. There are, however, reported cases in NWB languages where an IAV tone is clearly distinct from the metatonic H. One such case is found in Gyeli (A801). As discussed in (24) in Section 3.3, metatony in Gyeli marks a realis/irrealis distinction. In addition to the metatonic H tone suffix that is always realized on the phrase-medial verb in certain TAMP categories, Gyeli has an independent grammatical tone following the verb. This H tone marks the nominal object closest to the verb. It can only attach to CV prefixes, as these are underlyingly toneless and thus able to host a floating tone—Gyeli strongly dis-prefers overriding lexical tones (Grimm 2022). The object-marking tone is independent of any particular TAMP categories. As (34b) shows, the H tone on the object prefix is not the result of H tone spreading from the verb or a co-occurrence with the metatonic tone, as the object-linking tone also attaches when the preceding verb ends in an L tone in a non-metatonic TAMP category. (34c) illustrates that only argument nominals carry this object-linking H tone, whereas nominal obliques do not.

(34) Gyeli A801, based on (Grimm 2022: 489–490)

- a. mé dé mántúà
 mɛ-Ⓜ dè-Ⓜ Ⓜma-ntúà
 1SG-PRS eat-R OBJ.LINKma6-mango
 ‘I eat mangoes.’
- b. mɛ̀ɛ̀ dè mántúà
 mɛ̀ɛ̀ dè Ⓜma-ntúà
 1SG.FUT eat OBJ.LINKma6-mango
 ‘I will eat mangoes.’
- c. mɛ̀ pàlé dè [màfú málálɛ̀]_{OBL}
 mɛ̀ pàlé dè ma-fú má-lálɛ̀
 1SG NEG.PST eat ma6-day 6-three
 ‘I haven’t eaten in three days.’

Van de Velde (2008, 2012) reports a similar phenomenon in Eton (A71) with two distinct tones: a non-final H suffix on the verb and a “link” tone that looks similar to the “object-linking H tone” in Gyeli (34b).³⁰ It occurs, however only after infinitives and, unlike Gyeli, does not seem to mark syntactic relations (Van de Velde 2008: 206). The link tone does not get realized if another element appears between the verb and the object, as in (35b), whereas in Gyeli it does (Grimm 2021: 484).

(35) Eton A71 (Van de Velde 2008: 206)

- a. àté kùl bíkòb bí tíd
 à-Ⓜté Ⓜ-kùl Ⓜ bì-kòb bí=tíd
 1-PRS INF-clean LT 8-skin 8.CON=animal
 ‘He cleans the skins.’
- b. àté kùl vè bíkòb bí tíd
 à-Ⓜté Ⓜ-kùl vè bì-kòb bí=tíd
 1-PRS INF-clean only 8-skin 8.CON=animal
 ‘He cleans only the skins.’

Van de Velde (2012) explores the question of whether the link tone could be related to focus-marking and clearly shows for Eton that it is not a focus-marking device: The link tone appears irrespective of which element in the sentence is in

³⁰Van de Velde (2012) proposes that the link tone is closer to what Meeussen (1967) calls metatony, although it has a different location. Given these different places of association and common patterns across NWB languages, I suggest viewing the non-final suffix as metatony and the link tone as a type of IAV tone.

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focus. The sentence in (36) can serve as a response to all the listed questions, which elicit object, subject, and predicate focus.

- (36) Eton A71 (Van de Velde 2012)
 [What will she buy at the market?]
 [What will she DO at the market?]
 [Will she buy MANIOC at the market? No,]
 [Will YOU buy cocoyams at the market? No,]
 èèy kùs mǎbàṇà
 à-èèj ①-kùs ② mǎ-bàṇà
 1-FUT INF-buy LT 6-cocoyam
 ‘She will buy cocoyams.’

Van de Velde (2012) concludes that neither the H suffix on phrase-medial verbs and the IAV link tone do not have an obvious function, i.e. they do not mark argumenthood, focus, nor do they signal a certain syntactic relation. As with the phenomenon of metatony, there is a huge of variation and differences in the details with respect to IAV tones. It is conceivable that distinct IAV tones, as in Gyeli and Eton, originated from metatony, with Makaa presenting an intermediary case of metatony being realized sometimes on the verb and sometimes on the object noun.

4 Conclusion

Enormous progress has been made in the description and understanding of NWB languages in the last decades. This language sub-family of Narrow Bantu has long been known for exhibiting different structures than the rest of Bantu. Several NWB clades are characterized by a more analytic verb structure, often including floating tones that attach to the pre-stem complex, including the STAMP morpheme, and the stem. Tonally, NWB languages are remarkable for their wealth of tone patterns, at least on the phonetic level, and the vast diversity of tonal processes, most prominently downstep, floating tones, and HTS. Tone in NWB is also so fascinating, as it is involved in a multitude of grammatical and syntactic functions, namely the encoding of TAMP categories and clause types, as well as metatony and immediate-after-verb tones that might be linked to argument marking, at least in some languages. It is clear that there is a lot more to be explored, especially in the verbal complex of Bantu B languages. More comparative studies and detailed descriptions of tonal systems in individual languages will shed light onto the historical development of grammatical tone in these languages.

Abbreviations

↓	downstep	MET	metatony
①	floating L tone	N	noun
②	floating H tone	NEG	negation
1-9	agreement class 1-9	NWB	Northwestern Bantu
Ø	Ø-noun class	OBJ	object
*	reconstructed form	OBJ.LINK	object linking H tone
1PL	first person plural	PB	Proto-Bantu
2PL	second person plural	PL	plural
1SG	first person singular	POSS	possessor pronoun
2SG	second person singular	PREP	preposition
ADV	adverb	PRO	pronoun
APPL	applicative	PROG	progressive
COM	comitative marker	PRF	perfect(ive)
COMP	complementizer	PRS	present
CON	connective marker	PST	past (not specified for recent or remote past)
CONJ	conjunction	PST1	recent past
COP	copula	PST2	remote past
FUT	future	R	realis mood
GT	grammatical tone	RECIP	reciprocal
H	high tone	REL	relative marker/clause
HL	falling contour tone	SBJV	subjunctive
HTS	high tone spreading	SP	subject prefix
FV	final vowel	SG	singular
IAV	immediate-after-verb	SIT	situational
IMP	imperative	STAMP	subject-tense-aspect-mood- polarity clitic
INCH	inchoative	SUB	subordinate
INCOMP	incompletive	TAMP	tense, aspect, mood, and polarity
INF	infinitive	TEMP	temporal clause
IRR	irrealis	TM	tense-mood
L	low tone	V	verb
le	<i>le</i> -noun class		
LOC	locative		
LT	link tone		
ma	<i>ma</i> -noun class		

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African tone phenomena

A vital resource for emerging and seasoned linguists alike, *African Tone Phenomena* serves as a handbook and inspiration for rigorous research into under-documented African tone systems. The volume features 18 peer-reviewed chapters authored by leading linguists. Each chapter prioritizes clear description, employing theoretical terminology only as needed to illuminate the phenomena being discussed.

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